Advanced Statistics and Machine Learning for Health Research

A Practical Course for Health Researchers

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June 11, 2026

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1. Advanced Statistics and Machine Learning for Health Research

Welcome

This course teaches advanced statistics and machine learning to people who work in health research. It assumes you know the basics — means, standard deviations, maybe a t-test — and takes you from there to the methods you actually see in current medical journals: splines, penalised regression, prediction models, Bayesian analysis, and dimensionality reduction.

Everything uses **clinical data**. Every exercise works in **R and Python**. Pick one or try both. Solutions for all exercises are in the Appendix.

1.1. What you will learn

By the end of this course, you will be able to:

- Model non-linear relationships properly instead of categorising continuous variables
- Build, validate, and report clinical prediction models to current standards (TRIPOD+AI)
- Use machine learning methods (random forests, gradient boosting, deep learning) and know when they help and when they do not
- Apply Bayesian methods including hierarchical models for multi-site studies
- Reduce high-dimensional data with PCA, t-SNE, and UMAP

1.2. Structure

Part	What it covers
Pre-Course	Probability, inference, regression refresher
I	Splines, penalised regression, survival analysis
II	Bayesian inference and hierarchical models
III	ML foundations, trees, ensembles, neural networks, ROC curves, calibration, decision curves
IV	PCA, UMAP, clustering
V	Clinical prediction models, validation, reporting
Advanced Toolkit	Causal inference, meta-analysis, journal-ready analysis

The **Pre-Course** material should be completed before Day 1 if you are taking this in person. If you are self-studying, work through everything in order.

The **Advanced Research Toolkit** is for after the course — use it when you are doing your own research and need guidance on causal inference, meta-analysis, or producing a journal-ready manuscript.

1.3. Am I ready?

Skim Chapters 1-3. If the material feels familiar, skip ahead to Part I. If it feels new, work through those chapters carefully first — they are the foundation everything else builds on.

Rough time estimates per chapter:

Chapter	Topic	Estimated time
0	Environment setup	1-2 hrs
1	Probability and distributions	2-3 hrs
2	Inference and estimation	2-3 hrs
3	Regression foundations	3-4 hrs
4	Splines and non-linear modelling	3-4 hrs
5	Penalised regression	2-3 hrs
6	Survival analysis	3-4 hrs
13	Bayesian inference	3-4 hrs
14	Applied Bayesian modelling	3-4 hrs
7	ML foundations	2-3 hrs
8	Trees and ensembles	2-3 hrs
8b	Introduction to neural networks	2-3 hrs
9	Model evaluation	2-3 hrs
15	Dimensionality reduction	2-3 hrs
16	Clustering	2-3 hrs
10	Building prediction models	3-4 hrs
11	Performance and validation	3-4 hrs
12	Reporting and TRIPOD+AI	2-3 hrs

1.4. Datasets

All exercises use real or realistically simulated clinical data:

- **Framingham Heart Study** — cardiovascular risk
- **Wisconsin Breast Cancer** — diagnostic classification
- **PBC (Mayo Clinic)** — liver disease survival
- **NHANES** — national health survey
- **Simulated meningitis** — based on Lopez-Ayala et al. (*BMJ* 2025)

Details in the Dataset Codebook.

1.5. Key sources

This course draws heavily on:

- Smits, van Kuijk & Wynants, *Improving Health Care with Clinical Prediction Models* (2026, CC BY 4.0)
- Van Calster et al., *Lancet Digital Health* (2025) — performance measures for prediction models
- Lopez-Ayala et al., *BMJ* (2025) — handling continuous variables and splines
- Collins et al., *BMJ* (2024) — TRIPOD+AI reporting guidelines
- Harrell, *Regression Modeling Strategies* (2015)
- McElreath, *Statistical Rethinking* (2020)

Full references appear at the bottom of each chapter.

1.6. Licence

Part I.

Pre-Course: Foundations

2. Setting Up Your Computing Environment

2.1. Why Two Languages?

Throughout this course, you will work with both **R** and **Python**. This is not an accident or an attempt to double your workload. In modern biostatistics, epidemiology, and health data science, both languages appear regularly in published research, collaborative projects, and industry applications. R has deep roots in classical statistics and has an extraordinary ecosystem of packages for survival analysis, Bayesian modeling, and clinical reporting. Python dominates in machine learning, deep learning, and large-scale data engineering. By learning both, you will be able to read and contribute to a wider range of projects, collaborate with more colleagues, and choose the best tool for each task.

You do not need to be an expert programmer to succeed in this course. We will introduce code gradually and explain every line. Think of R and Python as lab instruments: you will learn to use them by doing, and we will always prioritize understanding the statistical ideas over memorizing syntax.

> Which language should I pick?
>
> If you have no programming experience, **start with R and RStudio**. RStudio is designed for data analysis and is the standard in

biostatistics departments. You can always add Python later.
If you already know some Python, stick with Python. Both languages can do everything in this course.

2.2. Installing R and RStudio

2.2.1. What Are R and RStudio?

R is a programming language designed for statistical computing and graphics. It runs in a terminal or console, but working with raw R in a terminal is not the most pleasant experience. **RStudio** is an integrated development environment (IDE) that wraps around R, providing a code editor, a console, a file browser, a plot viewer, and much more in a single window. Think of R as the engine and RStudio as the dashboard and steering wheel.

2.2.2. Step 1: Download and Install R

1. Open your web browser and navigate to the Comprehensive R Archive Network (CRAN): https://cran.r-project.org/
2. You will see links for your operating system near the top of the page:

    - **Windows**: Click "Download R for Windows," then click "base," then click the link that says something like "Download R-4.x.x for Windows." Run the downloaded `.exe` installer and accept all default settings.
    - **macOS**: Click "Download R for macOS." Choose the appropriate version for your Mac. If you have an Apple Silicon Mac (M1, M2, M3, or M4 chip), download the `arm64` version. If you have an older Intel Mac, download the `x86_64` version. If you are unsure, click the Apple icon in the top-left corner of your screen,

choose "About This Mac," and look at the "Chip" or "Processor" field. Open the downloaded `.pkg` file and follow the installation prompts.
- **Linux**: Click "Download R for Linux," select your distribution (Ubuntu, Fedora, etc.), and follow the instructions. On Ubuntu, you can also install R from the terminal:

```
# Ubuntu/Debian
sudo apt update
sudo apt install r-base r-base-dev
```

3. Verify the installation by opening a terminal (or Command Prompt on Windows) and typing:

```
R --version
```

You should see output that includes the R version number (4.4 or later is recommended).

2.2.3. Step 2: Download and Install RStudio

1. Navigate to the Posit (formerly RStudio) download page: https://posit.co/download/rstudio-desktop/
2. The page will detect your operating system and suggest the correct installer. Click the download button.
3. Run the installer:

   - **Windows**: Run the `.exe` file and follow the prompts.
   - **macOS**: Open the `.dmg` file and drag RStudio to your Applications folder.
   - **Linux**: Install the `.deb` or `.rpm` package using your package manager.

4. Open RStudio. You should see four panels: the Source editor (top left), the Console (bottom left), the Environment/History pane (top right), and the Files/Plots/Packages/Help pane (bottom right). If R is installed correctly, you will see a welcome message in the Console that includes the R version number.

2.2.4. Step 3: A Quick Tour of RStudio

- **Console** (bottom left): Type R commands here and press Enter to run them immediately. Good for quick exploration.
- **Source Editor** (top left): Write and save R scripts (`.R` files) or Quarto documents (`.qmd` files). You can run lines or selections by pressing Ctrl+Enter (Cmd+Enter on Mac).
- **Environment** (top right): Shows all variables and data objects currently in memory.
- **Files/Plots/Packages/Help** (bottom right): Browse files on your computer, view plots, manage installed packages, and read documentation.

2.3. Installing Python and an Editor

2.3.1. What Are Python, JupyterLab, and VS Code?

Python is a general-purpose programming language widely used in data science and machine learning. Unlike R, Python was not designed exclusively for statistics, but its ecosystem of scientific libraries (NumPy, pandas, scikit-learn, etc.) makes it extremely powerful for data analysis.

You have two main choices for an editing environment:

- **JupyterLab**: A browser-based interactive environment where you write code in "notebooks" — documents that mix code, output, and

narrative text. Jupyter notebooks (`.ipynb` files) are very popular in data science.
- **VS Code (Visual Studio Code)**: A general-purpose code editor from Microsoft that supports Python (and R) through extensions. It can also run Jupyter notebooks directly inside the editor.

We recommend **VS Code** for this course because it handles both R and Python well and is widely used in industry. However, JupyterLab is perfectly fine if you prefer it.

2.3.2. Step 1: Install Python via Miniforge (Recommended)

We recommend installing Python through **Miniforge**, a lightweight distribution that includes the `conda` package manager. Conda makes it easy to install scientific packages and manage separate environments for different projects.

1. Download Miniforge from: https://conda-forge.org/miniforge/
2. Choose the installer for your operating system:
    - **Windows**: Download and run the `.exe` installer. Check the box that says "Add Miniforge to my PATH environment variable" during installation.
    - **macOS**: Download the `.sh` script. Open Terminal and run:

```bash
# Replace the filename with the one you downloaded
bash Miniforge3-MacOSX-arm64.sh
```

- **Linux**: Same approach as macOS — download the `.sh` script and run it in your terminal.

3. Close and reopen your terminal, then verify:

```
python --version
conda --version
```

You should see Python 3.11 or later and a conda version number.

2.3.3. Alternative: Install Python from python.org

If you prefer not to use conda, you can download Python directly from https://www.python.org/downloads/. Make sure to check the box "Add Python to PATH" during installation on Windows. You will then use `pip` instead of `conda` to install packages.

2.3.4. Step 2: Install VS Code

1. Download VS Code from https://code.visualstudio.com/
2. Install it using the standard process for your operating system.
3. Open VS Code, click the Extensions icon in the left sidebar (it looks like four small squares), and install the following extensions:

    - **Python** (by Microsoft) — provides Python language support, debugging, and Jupyter notebook integration.
    - **Jupyter** (by Microsoft) — adds full Jupyter notebook support inside VS Code.
    - **Quarto** (by Quarto) — optional, but useful if you want to edit `.qmd` files in VS Code.
    - **R** (by REditorSupport) — optional, provides R language support in VS Code.

2.3.5. Alternative: Install JupyterLab

If you prefer JupyterLab, install it after setting up Python:

```
conda install jupyterlab
# or, if using pip:
pip install jupyterlab
```

Launch it with:

```
jupyter lab
```

This will open JupyterLab in your web browser.

2.4. Installing Key R Packages

R packages extend the language with specialized tools. Think of them as apps you install on your phone — the base R system is the phone itself, and packages add new capabilities.

Open RStudio and run the following command in the Console. This will take several minutes the first time because it downloads and compiles many packages:

```
# Install all packages needed for this course
install.packages(c(
  "tidyverse",     # Data manipulation (dplyr, ggplot2, tidyr, readr, etc.)
  "rms",           # Regression Modeling Strategies (Frank Harrell's toolkit)
  "glmnet",        # Regularized regression (lasso, ridge, elastic net)
  "survival",      # Survival analysis (Cox models, Kaplan-Meier)
  "brms",          # Bayesian regression models via Stan
  "rstanarm",      # Bayesian applied regression modeling via Stan
  "ranger",        # Fast random forests
  "xgboost",       # Gradient boosted trees
  "mice",          # Multiple imputation by chained equations
```

```
  "dcurves",        # Decision curve analysis
  "pROC",           # ROC curve analysis
  "uwot",           # UMAP dimensionality reduction
  "cluster",        # Cluster analysis
  "gtsummary",      # Publication-ready summary tables
  "MatchIt",        # Propensity score matching
  "tidymodels"      # Unified machine learning framework
))
```

2.4.1. What Each Package Does (Brief Overview)

Table 2.1.: Key R packages for this course

Package	Purpose
`tidyverse`	A collection of packages for data wrangling and visualization. Includes `ggplot2` (plotting), `dplyr` (data manipulation), `tidyr` (reshaping), and more.
`rms`	Frank Harrell's Regression Modeling Strategies package. Provides tools for fitting and validating regression models with a focus on best practices.
`glmnet`	Fits penalized (regularized) generalized linear models, including lasso and ridge regression.
`survival`	The foundational package for survival (time-to-event) analysis in R.

Package	Purpose
`brms`	An interface to Stan for fitting Bayesian generalized linear mixed models using familiar R formula syntax.
`rstanarm`	Similar to `brms` but with pre-compiled Stan models for faster startup. Great for common Bayesian regression tasks.
`ranger`	A fast implementation of random forests, useful for both classification and regression.
`xgboost`	Extreme gradient boosting — one of the most powerful and popular machine learning algorithms.
`mice`	Handles missing data through multiple imputation, a principled approach to dealing with incomplete datasets.
`dcurves`	Implements decision curve analysis for evaluating clinical prediction models.
`pROC`	Computes and displays ROC curves and calculates the area under the curve (AUC).
`uwot`	Implements UMAP (Uniform Manifold Approximation and Projection) for dimensionality reduction.
`cluster`	Provides methods for cluster analysis including k-medoids (PAM), hierarchical clustering, and more.

Package	Purpose
gtsummary	Creates publication-quality summary and regression tables.
MatchIt	Implements propensity score matching and other matching methods for causal inference.
tidymodels	A unified framework for building, tuning, and evaluating machine learning models in R.

2.4.2. Verifying R Package Installation

After installation, verify that the key packages load without errors:

```
library(tidyverse)
library(rms)
library(glmnet)
library(survival)
library(ranger)
library(xgboost)
```

If you get an error like `there is no package called 'xyz'`, try reinstalling that specific package. If compilation errors occur (common on Linux), you may need to install system-level dependencies. RStudio will usually tell you what is missing.

2.5. Installing Key Python Packages

2.5.1. Using conda (Recommended)

Create a dedicated environment for this course. An environment is an isolated collection of packages that will not interfere with other Python projects on your machine:

```
# Create a new environment called 'mlstats' with Python 3.12
conda create -n mlstats python=3.12

# Activate the environment
conda activate mlstats

# Install all packages
conda install pandas numpy scikit-learn statsmodels matplotlib seaborn

# Some packages are best installed via pip even when using conda
pip install lifelines xgboost pymc bambi arviz umap-learn plotnine miceforest
```

2.5.2. Using pip Only

If you are not using conda:

```
# Create a virtual environment
python -m venv mlstats_env

# Activate it
# On macOS/Linux:
source mlstats_env/bin/activate
# On Windows:
mlstats_env\Scripts\activate
```

```
# Install packages
pip install pandas numpy scikit-learn statsmodels lifelines xgboost \
    pymc bambi arviz umap-learn matplotlib seaborn plotnine miceforest
```

2.5.3. What Each Python Package Does

Table 2.2.: Key Python packages for this course

Package	Purpose
`pandas`	The primary data manipulation library. DataFrames (tables) are the central data structure.
`numpy`	Numerical computing — arrays, linear algebra, random number generation.
`scikit-learn`	The standard machine learning library. Classification, regression, clustering, preprocessing, model evaluation.
`statsmodels`	Classical statistical models: linear regression, logistic regression, time series, and more. Provides p-values and confidence intervals that scikit-learn does not.
`lifelines`	Survival analysis in Python: Kaplan-Meier curves, Cox proportional hazards models, and more.
`xgboost`	Gradient boosted trees (same algorithm as the R version).

Package	Purpose
`pymc`	Bayesian statistical modeling and probabilistic programming.
`bambi`	BAyesian Model-Building Interface — a high-level interface to PyMC using R-style formulas.
`arviz`	Visualization and diagnostics for Bayesian models.
`umap-learn`	UMAP dimensionality reduction for Python.
`matplotlib`	The foundational plotting library for Python.
`seaborn`	Statistical data visualization built on top of matplotlib. Easier to use for common statistical plots.
`plotnine`	A Python implementation of R's ggplot2 grammar of graphics. If you like ggplot2, you will like plotnine.
`miceforest`	Multiple imputation using random forests in Python.

2.5.4. Verifying Python Package Installation

```
import pandas as pd
import numpy as np
import sklearn
import statsmodels
import matplotlib.pyplot as plt

print(f"pandas:       {pd.__version__}")
print(f"numpy:        {np.__version__}")
```

```
print(f"scikit-learn: {sklearn.__version__}")
print("All key packages loaded successfully!")
```

2.6. Hello World: Your First Analysis

Let us make sure everything works by loading a built-in dataset and creating a simple plot in both languages. We will use the classic `iris` dataset — measurements of petal and sepal dimensions for three species of iris flowers. While not a clinical dataset, it is available in both R and Python without any downloads, making it perfect for a quick test.

2.6.1. R

```
# Load the tidyverse (includes ggplot2 and dplyr)
library(tidyverse)

# The iris dataset is built into R
data(iris)

# Take a quick look at the first few rows
head(iris)

# Summary statistics
summary(iris)

# Create a scatter plot of Sepal Length vs. Petal Length, colored by Species
ggplot(iris, aes(x = Sepal.Length, y = Petal.Length, color = Species)) +
  geom_point(size = 2, alpha = 0.7) +
  labs(
    title = "Iris Dataset: Sepal Length vs. Petal Length",
```

```
    x = "Sepal Length (cm)",
    y = "Petal Length (cm)"
  ) +
  theme_minimal()
```

2.6.2. Python

```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris

# Load the iris dataset
iris_data = load_iris()
iris = pd.DataFrame(
    iris_data.data,
    columns=iris_data.feature_names
)
iris["species"] = pd.Categorical.from_codes(
    iris_data.target, iris_data.target_names
)

# Take a quick look
print(iris.head())
print(iris.describe())

# Create a scatter plot
plt.figure(figsize=(8, 5))
sns.scatterplot(
    data=iris,
    x="sepal length (cm)",
    y="petal length (cm)",
```

```
    hue="species",
    alpha=0.7,
    s=60
)
plt.title("Iris Dataset: Sepal Length vs. Petal Length")
plt.tight_layout()
plt.show()
```

If you see a colorful scatter plot with three clusters of points, congratulations — your setup is working.

2.7. How to Use the Course Materials

2.7.1. Navigating This Book

This course is delivered as a **Quarto book** — a collection of chapters that you can read in your web browser. Each chapter builds on previous ones, so we recommend reading them in order the first time through. However, you can always jump to a specific chapter using the table of contents in the left sidebar.

2.7.2. Code Blocks

Throughout the book, you will encounter code blocks like this:

```
# This is an R code block
x <- c(1, 2, 3, 4, 5)
mean(x)
```

Many chapters present code in **tabbed panels** labeled "R" and "Python." Click the tab for your preferred language. We encourage you to try both, but if you are short on time, pick the one you are more comfortable with and come back to the other later.

2.7.3. Exercises

Each chapter ends with exercises. They follow this pattern:

1. A description of the task inside a colored callout box.
2. **Starter code** that sets up the problem and leaves blanks for you to fill in.
3. A **Solution** hidden inside a collapsible box — try the exercise yourself before peeking!

2.7.4. Downloading Notebooks

For hands-on practice, you can download the exercises as standalone notebooks:

- **R users**: Look for `.qmd` or `.Rmd` files in the course repository that you can open in RStudio.
- **Python users**: Look for `.ipynb` (Jupyter notebook) files in the course repository that you can open in JupyterLab or VS Code.

The course repository is available on GitHub. Your instructor will provide the link.

2.7.5. Rendering Quarto Documents

If you want to render `.qmd` files yourself (to produce HTML or PDF output), you need to install Quarto:

1. Download Quarto from https://quarto.org/docs/get-started/
2. Install it following the instructions for your operating system.
3. In RStudio, you can render a `.qmd` file by clicking the "Render" button. In VS Code, use the Quarto extension's render command. From the terminal:

```
quarto render my_document.qmd
```

2.8. Troubleshooting Common Setup Issues

2.8.1. R and RStudio Issues

Problem: RStudio cannot find R. **Solution**: Make sure you installed R *before* installing RStudio. If you installed them in the wrong order, try reinstalling RStudio. On Windows, RStudio looks for R in standard installation locations. If you installed R to a non-standard path, go to Tools > Global Options > General and set the R version manually.

Problem: Package installation fails with a compilation error. **Solution**: Some R packages need to compile C++ or Fortran code. On Windows, install Rtools. On macOS, install the Xcode Command Line Tools by running `xcode-select --install` in Terminal. On Linux, install `build-essential` and `r-base-dev`.

Problem: `brms` or `rstanarm` fails to install. **Solution**: These packages depend on Stan, a probabilistic programming language that requires a C++ compiler. Follow the instructions above for installing compilation tools. On Windows, make sure Rtools is on your PATH. Installation can take 10–15 minutes — be patient.

Problem: Package loads but you get warnings about versions. **Solution**: Warnings (yellow text) are usually harmless — they often say things like "package was built under R version X.Y.Z." Errors (red text) are the ones that prevent code from running. If you get errors, try updating the package with `install.packages("package_name")`.

2.8.2. Python Issues

Problem: `python` command not found. **Solution**: On some systems, Python 3 is accessed via `python3` instead of `python`. Try `python3 --version`. If using conda, make sure you have activated your environment with `conda activate mlstats`.

Problem: `ModuleNotFoundError: No module named 'xyz'`. **Solution**: The package is not installed in your current environment. Make sure you have activated the correct conda environment (`conda activate mlstats`) or virtual environment before installing and importing packages.

Problem: `pip install pymc` fails with obscure errors. **Solution**: PyMC can be tricky to install because it depends on compiled numerical libraries. Using conda often resolves these issues: `conda install -c conda-forge pymc`. On Apple Silicon Macs, make sure you are using the ARM64 version of Miniforge.

Problem: Jupyter notebook does not show the correct Python kernel. **Solution**: Install the IPython kernel in your environment: `conda install ipykernel` or `pip install ipykernel`. Then register it: `python -m ipykernel install --user --name mlstats --display-name "ML Stats Course"`.

2.8.3. Common Error Messages

Error: `Error: object 'x' not found` **Solution**: You haven't run the earlier code blocks that create `x`. Go back and run all preceding code chunks in order. In RStudio, use "Run All Chunks Above" from the Run menu. In Jupyter, use "Run All Above" from the Cell menu.

Error: `Error in library(xxx): there is no package called 'xxx'` **Solution**: Install it first with `install.packages("xxx")` in R, or `pip install xxx` / `conda install xxx` in Python. Then try loading it again.

2.8.4. General Tips

- **Keep your software updated.** At the start of each semester, update R, RStudio, Python, and your packages.
- **Use separate environments.** Conda environments (Python) and `renv` (R) prevent package version conflicts between projects.
- **Read error messages carefully.** They usually tell you exactly what went wrong, even if the language is technical. Copy the last line of an error message into a search engine — someone else has almost certainly had the same problem.
- **Ask for help.** Post on the course discussion board with the full error message, your operating system, and what you were trying to do. Screenshots are helpful.

Exercise 0.1: Verify Your Setup

Confirm that both R and Python are working by completing the following tasks:

1. In R, load the `tidyverse` package and use `ggplot2` to create a histogram of the `Sepal.Width` column from the built-in `iris` dataset.
2. In Python, use `seaborn` to create a histogram of the `sepal width (cm)` column from scikit-learn's iris dataset.

2.9. R

```r
# Load tidyverse
library(tidyverse)

# Create a histogram of Sepal.Width from the iris dataset
# YOUR CODE HERE
```

2.10. Python

```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris

# Load iris and create a DataFrame
iris_data = load_iris()
iris = pd.DataFrame(iris_data.data, columns=iris_data.feature_names)

# Create a histogram of sepal width
# YOUR CODE HERE
```

Solution

2.11. R

```r
library(tidyverse)

ggplot(iris, aes(x = Sepal.Width)) +
  geom_histogram(binwidth = 0.2, fill = "steelblue", color = "white")
  labs(
    title = "Distribution of Sepal Width",
    x = "Sepal Width (cm)",
    y = "Count"
  ) +
  theme_minimal()
```

2.12. Python

```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris

iris_data = load_iris()
iris = pd.DataFrame(iris_data.data, columns=iris_data.feature_names)

plt.figure(figsize=(8, 5))
sns.histplot(iris["sepal width (cm)"], bins=15, kde=True, color="steel
plt.title("Distribution of Sepal Width")
plt.xlabel("Sepal Width (cm)")
plt.ylabel("Count")
plt.tight_layout()
plt.show()
```

Exercise 0.2: Explore a Clinical Dataset

Now let us work with something more relevant to health sciences. Both R and Python include the `mtcars` dataset (or we can simulate clinical-like data). In this exercise, create a scatter plot relating two variables and add a trend line.

2.13. R

```r
library(tidyverse)

# Let's simulate a small clinical dataset
set.seed(42)
n <- 200
clinical <- tibble(
  age = round(rnorm(n, mean = 55, sd = 12)),
  systolic_bp = round(100 + 0.8 * age + rnorm(n, sd = 10)),
  bmi = round(rnorm(n, mean = 27, sd = 5), 1)
)

# Create a scatter plot of age vs. systolic blood pressure with a trend line
# Hint: use geom_point() and geom_smooth(method = "lm")
# YOUR CODE HERE
```

2.14. Python

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# Simulate a small clinical dataset
np.random.seed(42)
n = 200
clinical = pd.DataFrame({
    "age": np.round(np.random.normal(55, 12, n)).astype(int),
})
clinical["systolic_bp"] = np.round(100 + 0.8 * clinical["age"] + np.random.
clinical["bmi"] = np.round(np.random.normal(27, 5, n), 1)

# Create a scatter plot of age vs. systolic blood pressure with a trend lin
# Hint: use sns.regplot() or sns.lmplot()
# YOUR CODE HERE
```

Solution

2.15. R

```r
library(tidyverse)

set.seed(42)
n <- 200
clinical <- tibble(
  age = round(rnorm(n, mean = 55, sd = 12)),
  systolic_bp = round(100 + 0.8 * age + rnorm(n, sd = 10)),
  bmi = round(rnorm(n, mean = 27, sd = 5), 1)
)

ggplot(clinical, aes(x = age, y = systolic_bp)) +
  geom_point(alpha = 0.5, color = "darkblue") +
  geom_smooth(method = "lm", color = "firebrick", se = TRUE) +
  labs(
    title = "Age vs. Systolic Blood Pressure",
    subtitle = "Simulated clinical data (n = 200)",
    x = "Age (years)",
    y = "Systolic Blood Pressure (mmHg)"
  ) +
  theme_minimal()
```

2.16. Python

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

np.random.seed(42)
n = 200
clinical = pd.DataFrame({
    "age": np.round(np.random.normal(55, 12, n)).astype(int),
})
clinical["systolic_bp"] = np.round(100 + 0.8 * clinical["age"] + np.ra
clinical["bmi"] = np.round(np.random.normal(27, 5, n), 1)

plt.figure(figsize=(8, 5))
sns.regplot(
    data=clinical, x="age", y="systolic_bp",
    scatter_kws={"alpha": 0.5, "color": "darkblue"},
    line_kws={"color": "firebrick"}
)
plt.title("Age vs. Systolic Blood Pressure\nSimulated clinical data (n
plt.xlabel("Age (years)")
plt.ylabel("Systolic Blood Pressure (mmHg)")
plt.tight_layout()
plt.show()
```

2.17. References and Further Reading

- Wickham, H., & Grolemund, G. (2017). *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. O'Reilly Media. Freely available at https://r4ds.had.co.nz/. The definitive introduction to the tidyverse ecosystem in R.

- VanderPlas, J. (2016). *Python Data Science Handbook: Essential Tools for Working with Data*. O'Reilly Media. Freely available at https://jakevdp.github.io/PythonDataScienceHandbook/. Covers NumPy, pandas, matplotlib, and scikit-learn in depth.
- McKinney, W. (2022). *Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter* (3rd ed.). O'Reilly Media. The authoritative guide to pandas, written by the creator of the library.
- Posit. (2024). *Quarto Documentation*. https://quarto.org/docs/guide/. The official guide to Quarto, the publishing system used for this course.
- Cetinkaya-Rundel, M., & Hardin, J. (2024). *Introduction to Modern Statistics* (2nd ed.). OpenIntro. Freely available at https://openintro-ims2.netlify.app/. An excellent, free statistics textbook that uses R.
- Harrell, F. E. (2015). *Regression Modeling Strategies: With Applications to Linear Models, Logistic and Ordinal Regression, and Survival Analysis* (2nd ed.). Springer. The companion text for the `rms` R package and a masterclass in regression modeling for health sciences.

3. Probability, Distributions, and the Language of Uncertainty

3.1. Why Probability Matters in Health Sciences

Medicine is a discipline of uncertainty. A patient presents with chest pain — is it a heart attack, acid reflux, or a pulled muscle? A new drug shows a 15% reduction in tumor size — is this a real effect or a fluke of sampling? A screening test comes back positive — what is the chance the patient actually has the disease?

Probability gives us a formal language for reasoning about uncertainty. It allows us to quantify how likely different outcomes are, how confident we should be in our conclusions, and how we should update our beliefs when new evidence arrives. Every statistical method you will learn in this course — from logistic regression to Bayesian modeling to machine learning — is built on the foundation of probability theory.

This chapter introduces the key ideas you need. We will move slowly, use concrete clinical examples, and build intuition before introducing formulas.

3.2. What Is Probability?

At its simplest, a **probability** is a number between 0 and 1 that represents how likely an event is to occur. A probability of 0 means the event is impossible; a probability of 1 means it is certain. A probability of 0.5 means the event is equally likely to happen or not happen.

But where do probabilities come from? There are two major perspectives.

3.2.1. The Frequentist Interpretation

The **frequentist** interpretation says that the probability of an event is the long-run relative frequency with which it occurs if we repeat the same process many times. For example:

- If we flip a fair coin many times, it will land heads about 50% of the time. We say $P(\text{heads}) = 0.5$.
- If a particular surgery has a 3% mortality rate, this means that out of every 100 patients who undergo the procedure (under similar conditions), roughly 3 will die.

This interpretation works well when we can imagine repeating an experiment. Coin flips, blood draws from a population, random assignment in a clinical trial — all of these fit naturally into the frequentist framework.

3.2.2. The Subjective (Bayesian) Interpretation

The **subjective** or **Bayesian** interpretation says that probability represents a *degree of belief*. It is a measure of how confident you are that something is true, given what you currently know.

- A cardiologist examining a 65-year-old male smoker with crushing chest pain might say: "I believe there is an 85% probability this is a myocardial infarction." This is not based on repeating the exact same patient encounter thousands of times — it is an informed judgment.
- Before seeing any data from a clinical trial, a researcher might believe there is a 30% chance a new drug is effective. After seeing positive results from the trial, they update that belief to 80%.

The Bayesian interpretation is powerful because it applies to one-time events and unique situations. You cannot repeat a patient's life, but you can still reason probabilistically about their prognosis.

3.2.3. Which Interpretation Is Correct?

Both are useful. In this course, we will use frequentist methods (hypothesis tests, confidence intervals, maximum likelihood) and Bayesian methods (posterior distributions, credible intervals, prior updating). Understanding both perspectives will make you a more versatile analyst. The key insight is that probability is a *tool for reasoning under uncertainty*, regardless of your philosophical starting point.

3.3. Random Variables

A **random variable** is a quantity whose value is determined by a random process. In clinical research, random variables are everywhere:

- The number of patients who respond to treatment (a count)
- A patient's systolic blood pressure (a measurement)
- Whether a tumor is malignant or benign (a category)
- Time from diagnosis to death (a duration)

We use capital letters like X, Y, or Z to denote random variables, and lowercase letters like x, y, z for specific observed values.

3.3.1. Discrete Random Variables

A **discrete** random variable takes on a countable number of values, typically whole numbers. Examples:

- The number of adverse events in a clinical trial: 0, 1, 2, 3, …

- The number of positive test results in a batch of 20 specimens: 0, 1, 2, …, 20
- A patient's pain score on a 0–10 scale: 0, 1, 2, …, 10

For a discrete random variable, we describe its behavior using a **probability mass function (PMF)**, which gives the probability of each possible value:

$$P(X = x)$$

All probabilities must be non-negative, and they must sum to 1:

$$\sum_{\text{all } x} P(X = x) = 1$$

3.3.2. Continuous Random Variables

A **continuous** random variable can take on any value within a range (including decimals). Examples:

- Body temperature: 36.2, 37.45, 38.1, …
- Serum creatinine level: any positive real number
- Time to event: any positive duration

For a continuous random variable, the probability of any *exact* value is technically zero (because there are infinitely many possible values). Instead, we talk about the probability of falling within a *range*, and we describe the distribution using a **probability density function (PDF)**, denoted $f(x)$. The probability that X falls between a and b is:

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx$$

This is the area under the density curve between a and b. The total area under the entire curve equals 1.

Do not worry if integrals make you nervous. The key idea is: for continuous variables, probability corresponds to *area under a curve*.

3.4. The Normal (Gaussian) Distribution

The **normal distribution** — also called the Gaussian distribution or the "bell curve" — is the most important probability distribution in statistics. It appears everywhere in nature and medicine.

3.4.1. Why Does It Matter?

1. **Many biological measurements are approximately normal.** Heights, weights, blood pressure readings, cholesterol levels, and many lab values follow roughly bell-shaped distributions in large populations.
2. **The Central Limit Theorem** (discussed later in this chapter) tells us that averages of measurements tend to follow a normal distribution, even when individual measurements do not.
3. **Many statistical methods assume normality** (or at least approximate normality) for valid inference.

3.4.2. Properties of the Normal Distribution

A normal distribution is completely described by two parameters:

- **Mean (μ)**: The center of the distribution. The peak of the bell curve.

- **Standard deviation (σ)**: A measure of spread. About 68% of values fall within one standard deviation of the mean, about 95% within two, and about 99.7% within three. This is sometimes called the **68-95-99.7 rule**.

The probability density function is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

You do not need to memorize this formula. What matters is understanding the shape and the role of μ and σ.

We write $X \sim \text{Normal}(\mu, \sigma^2)$ or $X \sim N(\mu, \sigma^2)$ to say that X follows a normal distribution with mean μ and variance σ^2.

3.4.3. Clinical Example: Blood Pressure

Systolic blood pressure in healthy adults is approximately normally distributed with a mean of about 120 mmHg and a standard deviation of about 15 mmHg. That is, $\text{SBP} \sim N(120, 15^2)$.

Using the 68-95-99.7 rule:

- About 68% of healthy adults have SBP between 105 and 135 mmHg (120 plus or minus 15).
- About 95% have SBP between 90 and 150 mmHg (120 plus or minus 30).
- About 99.7% have SBP between 75 and 165 mmHg (120 plus or minus 45).

If a patient has a systolic blood pressure of 160 mmHg, how unusual is that? We can answer this using **Z-scores**.

3.4.4. Z-Scores: Standardizing Values

A **Z-score** tells you how many standard deviations a value is from the mean:

$$Z = \frac{X - \mu}{\sigma}$$

For our patient with SBP = 160:

$$Z = \frac{160 - 120}{15} = \frac{40}{15} \approx 2.67$$

This patient's blood pressure is about 2.67 standard deviations above the mean. Using a standard normal table or a computer, we find that only about 0.4% of the healthy population would have a blood pressure this high or higher. This is strong evidence that this patient may have hypertension.

The **standard normal distribution** is a normal distribution with $\mu = 0$ and $\sigma = 1$. Any normal distribution can be converted to the standard normal by computing Z-scores.

Let us visualize the normal distribution and Z-scores:

3.4.5. R

```r
library(tidyverse)

# Parameters for systolic blood pressure
mu <- 120
sigma <- 15

# Create a sequence of values for the x-axis
```

```r
x <- seq(mu - 4 * sigma, mu + 4 * sigma, length.out = 300)
y <- dnorm(x, mean = mu, sd = sigma)

bp_data <- tibble(x = x, y = y)

ggplot(bp_data, aes(x = x, y = y)) +
  geom_line(color = "steelblue", linewidth = 1.2) +
  geom_area(data = bp_data |> filter(x >= 160),
            aes(x = x, y = y), fill = "firebrick", alpha = 0.4) +
  geom_vline(xintercept = 160, linetype = "dashed", color = "firebrick") +
  annotate("text", x = 168, y = 0.015,
           label = "Patient: 160 mmHg\n(Z = 2.67)", color = "firebrick",
           hjust = 0, size = 3.5) +
  labs(
    title = "Distribution of Systolic Blood Pressure in Healthy Adults",
    subtitle = paste0("Normal(", mu, ", ", sigma, "^2)"),
    x = "Systolic Blood Pressure (mmHg)",
    y = "Density"
  ) +
  theme_minimal()
```

3.4.6. Python

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

mu = 120
sigma = 15

x = np.linspace(mu - 4 * sigma, mu + 4 * sigma, 300)
y = stats.norm.pdf(x, loc=mu, scale=sigma)
```

```
fig, ax = plt.subplots(figsize=(9, 5))
ax.plot(x, y, color="steelblue", linewidth=1.5)

# Shade the area above 160
x_fill = x[x >= 160]
y_fill = stats.norm.pdf(x_fill, loc=mu, scale=sigma)
ax.fill_between(x_fill, y_fill, color="firebrick", alpha=0.4)

ax.axvline(160, linestyle="--", color="firebrick")
ax.text(162, 0.015, "Patient: 160 mmHg\n(Z = 2.67)",
        color="firebrick", fontsize=9)

ax.set_title("Distribution of Systolic Blood Pressure in Healthy Adults")
ax.set_xlabel("Systolic Blood Pressure (mmHg)")
ax.set_ylabel("Density")
plt.tight_layout()
plt.show()
```

3.5. The Binomial Distribution

The **binomial distribution** models the number of “successes” in a fixed number of independent trials, where each trial has the same probability of success. Despite the name, a “success” does not have to be a good thing — it is simply the outcome we are counting.

3.5.1. Parameters

- n: the number of trials (a fixed, known integer)
- p: the probability of success on each trial (between 0 and 1)

We write $X \sim \text{Binomial}(n, p)$, and the PMF is:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, 2, \ldots, n$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is the binomial coefficient ("n choose k"), which counts the number of ways to arrange k successes among n trials.

Mean: $E[X] = np$

Variance: $\text{Var}(X) = np(1-p)$

3.5.2. Clinical Examples

Disease prevalence. In a city, the prevalence of Type 2 diabetes among adults is 12%. If you randomly sample 50 adults, the number with diabetes follows a Binomial(50, 0.12) distribution. The expected number is $50 \times 0.12 = 6$.

Treatment response. A chemotherapy regimen achieves complete response in 35% of patients. If 20 patients are treated, the number who achieve complete response follows a Binomial(20, 0.35) distribution.

Key assumptions: The binomial distribution assumes that (1) each trial is independent (one patient's outcome does not affect another's), (2) the probability of success is the same for every trial, and (3) the number of trials is fixed in advance. In practice, these assumptions are sometimes approximate (e.g., disease probability may vary with age), but the binomial is still a useful starting model.

3.5.3. R

```r
library(tidyverse)

# Treatment response: 20 patients, 35% response rate
n <- 20
p <- 0.35

# Calculate probabilities for each possible outcome
outcomes <- tibble(
  k = 0:n,
  probability = dbinom(k, size = n, prob = p)
)

ggplot(outcomes, aes(x = k, y = probability)) +
  geom_col(fill = "steelblue", alpha = 0.8) +
  geom_vline(xintercept = n * p, linetype = "dashed", color = "firebrick") +
  annotate("text", x = n * p + 0.5, y = max(outcomes$probability),
           label = paste0("Expected value = ", n * p),
           hjust = 0, color = "firebrick") +
  labs(
    title = "Binomial Distribution: Treatment Response",
    subtitle = "n = 20 patients, p = 0.35 response rate",
    x = "Number of Responders",
    y = "Probability"
  ) +
  scale_x_continuous(breaks = 0:n) +
  theme_minimal()
```

3.5.4. Python

```python
import numpy as np
import matplotlib.pyplot as plt
```

```python
from scipy import stats

n = 20
p = 0.35

k = np.arange(0, n + 1)
probabilities = stats.binom.pmf(k, n, p)

fig, ax = plt.subplots(figsize=(9, 5))
ax.bar(k, probabilities, color="steelblue", alpha=0.8)
ax.axvline(n * p, linestyle="--", color="firebrick")
ax.text(n * p + 0.3, max(probabilities), f"Expected value = {n * p}",
        color="firebrick", fontsize=9)

ax.set_title("Binomial Distribution: Treatment Response")
ax.set_xlabel("Number of Responders")
ax.set_ylabel("Probability")
ax.set_xticks(k)
plt.tight_layout()
plt.show()
```

3.6. The Poisson Distribution

The **Poisson distribution** models the number of events occurring in a fixed interval of time or space, when events happen independently at a constant average rate. It is the go-to distribution for modeling rare events.

3.6.1. Parameters

- λ (lambda): the average number of events per interval. This is both the mean and the variance.

We write $X \sim \text{Poisson}(\lambda)$, and the PMF is:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \ldots$$

Mean: $E[X] = \lambda$

Variance: $\text{Var}(X) = \lambda$

The fact that the mean equals the variance is a distinctive property of the Poisson distribution. When you see count data where the variance is much larger than the mean, the Poisson may not be a good fit (a condition called **overdispersion**).

3.6.2. Clinical Examples

Adverse drug reactions. A hospital reports an average of 2.5 serious adverse drug reactions per month. If these events are independent, the number in any given month follows approximately a Poisson(2.5) distribution. The probability of zero serious reactions in a month is $P(X = 0) = e^{-2.5} \approx 0.082$, or about 8.2%.

Hospital admissions. An emergency department sees an average of 40 patients per shift. The number of arrivals per shift can be modeled with a Poisson(40) distribution (though in practice, arrivals may cluster, violating independence).

Rare disease incidence. If a rare genetic disorder affects 1 in 100,000 people per year and a city has 500,000 people, we expect about 5 cases per year. The annual count follows approximately a Poisson(5) distribution.

3.6.3. R

```
library(tidyverse)

# Adverse drug reactions: average 2.5 per month
lambda <- 2.5
k <- 0:12

adr_data <- tibble(
  k = k,
  probability = dpois(k, lambda = lambda)
)

ggplot(adr_data, aes(x = k, y = probability)) +
  geom_col(fill = "darkorange", alpha = 0.8) +
  geom_vline(xintercept = lambda, linetype = "dashed", color = "firebrick") +
  labs(
    title = "Poisson Distribution: Adverse Drug Reactions per Month",
    subtitle = expression(paste(lambda, " = 2.5 reactions/month")),
    x = "Number of Adverse Reactions",
    y = "Probability"
  ) +
  scale_x_continuous(breaks = k) +
  theme_minimal()
```

3.6.4. Python

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

lam = 2.5
```

```
k = np.arange(0, 13)
probs = stats.poisson.pmf(k, lam)

fig, ax = plt.subplots(figsize=(9, 5))
ax.bar(k, probs, color="darkorange", alpha=0.8)
ax.axvline(lam, linestyle="--", color="firebrick")
ax.set_title(f"Poisson Distribution: Adverse Drug Reactions per Month (lambda = {lam})")
ax.set_xlabel("Number of Adverse Reactions")
ax.set_ylabel("Probability")
ax.set_xticks(k)
plt.tight_layout()
plt.show()
```

3.7. The Central Limit Theorem

The **Central Limit Theorem (CLT)** is one of the most remarkable results in all of statistics. It states:

> If you take sufficiently large random samples from *any* population (regardless of the population's distribution), the distribution of the sample means will be approximately normal.

More precisely: if $X_1, X_2, \ldots, X_n$ are independent and identically distributed random variables with mean μ and standard deviation σ, then the sample mean $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ has a distribution that approaches $N(\mu, \sigma^2/n)$ as n grows large.

3.7.1. Why Is This Important?

The CLT is the reason the normal distribution is so central to statistics. It means that:

1. **We can use normal-based methods** (t-tests, confidence intervals, regression) even when the underlying data are not normally distributed, as long as our sample size is large enough.
2. **Sample means are more predictable than individual observations.** The standard deviation of the sample mean ($\sigma/\sqrt{n}$) shrinks as n grows, meaning larger samples give more precise estimates.

3.7.2. How Large Is "Large Enough"?

It depends on how non-normal the underlying distribution is:

- If the population is already roughly symmetric, $n = 15$–30 is often sufficient.
- If the population is heavily skewed (e.g., income, hospital length of stay), you may need $n = 50$ or more.
- For extremely skewed distributions, $n > 100$ may be needed.

3.7.3. Simulation Demonstration

Let us demonstrate the CLT visually. We will take samples from a **right-skewed** distribution (exponential, which looks nothing like a bell curve) and show that the distribution of sample means becomes increasingly normal as the sample size grows.

3.7.4. R

```
library(tidyverse)

set.seed(42)

# Parameters
```

```r
n_simulations <- 5000   # Number of samples to draw
sample_sizes <- c(1, 5, 30, 100)

# For each sample size, draw many samples and compute the mean of each
clt_data <- map_dfr(sample_sizes, function(n) {
  means <- replicate(n_simulations, mean(rexp(n, rate = 1)))
  tibble(
    sample_size = paste0("n = ", n),
    sample_mean = means
  )
})

# Convert to ordered factor so panels appear in the right order
clt_data$sample_size <- factor(clt_data$sample_size,
                                levels = paste0("n = ", sample_sizes))

ggplot(clt_data, aes(x = sample_mean)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 50, fill = "steelblue", alpha = 0.7) +
  geom_density(color = "firebrick", linewidth = 0.8) +
  facet_wrap(~sample_size, scales = "free") +
  labs(
    title = "Central Limit Theorem in Action",
    subtitle = "Sampling from an Exponential(1) distribution (mean = 1, right-skewed)",
    x = "Sample Mean",
    y = "Density"
  ) +
  theme_minimal()
```

3.7.5. Python

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(42)

n_simulations = 5000
sample_sizes = [1, 5, 30, 100]

fig, axes = plt.subplots(2, 2, figsize=(10, 8))
axes = axes.flatten()

for i, n in enumerate(sample_sizes):
    means = [np.mean(np.random.exponential(1, n)) for _ in range(n_simulatio
    axes[i].hist(means, bins=50, density=True, color="steelblue", alpha=0.7)
    axes[i].set_title(f"n = {n}")
    axes[i].set_xlabel("Sample Mean")
    axes[i].set_ylabel("Density")

fig.suptitle(
    "Central Limit Theorem in Action\n"
    "Sampling from an Exponential(1) distribution (right-skewed)",
    fontsize=13
)
plt.tight_layout()
plt.show()
```

Notice how the distribution of sample means changes:

- When $n = 1$, the distribution of "means" is just the original exponential distribution — heavily right-skewed.
- At $n = 5$, the skewness is already reduced.

- At $n = 30$, the distribution looks approximately bell-shaped.
- At $n = 100$, the distribution is very close to a perfect normal curve, and it is much narrower (less variable).

This is the CLT in action. It does not matter that we started with a skewed distribution — the means converge to normality.

3.8. Bayes' Theorem: Updating Beliefs with Evidence

3.8.1. The Idea

Bayes' theorem provides a mathematically precise way to update your beliefs when you receive new evidence. In clinical medicine, one of the most important applications is interpreting diagnostic tests.

Suppose a patient tests positive for a disease. What is the probability they actually have the disease? Your intuition might say "very high," but as we will see, the answer depends critically on how common the disease is in the population.

3.8.2. The Formula

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B)}$$

In the context of diagnostic testing:

- $P(\text{Disease} \mid \text{Positive test})$ = the probability the patient has the disease, given a positive test result. This is called the **positive predictive value (PPV)**.
- $P(\text{Positive test} \mid \text{Disease})$ = the probability of a positive test result if the patient truly has the disease. This is the test's **sensitivity** (true positive rate).

- $P(\text{Disease})$ = the probability of having the disease before the test, based on the population prevalence or clinical judgment. This is the **prior probability** or **pre-test probability**.
- $P(\text{Positive test})$ = the overall probability of a positive test result, accounting for both true positives and false positives.

We can expand the denominator using the **law of total probability**:

$$P(\text{Positive test}) = P(\text{Positive} \mid \text{Disease}) \times P(\text{Disease}) + P(\text{Positive} \mid \text{No disease}) \times P(\text{No diseas$$

Note that $P(\text{Positive} \mid \text{No disease})$ is the **false positive rate**, which equals $1 - \text{specificity}$.

3.8.3. Key Diagnostic Test Terminology

Table 3.1.: Diagnostic test performance metrics

Term	Definition	Formula
Sensitivity	Probability of a positive test given disease is present	$P(+ \mid D)$
Specificity	Probability of a negative test given disease is absent	$P(- \mid \bar{D})$
Positive Predictive Value (PPV)	Probability of disease given a positive test	$P(D \mid +)$
Negative Predictive Value (NPV)	Probability of no disease given a negative test	$P(\bar{D} \mid -)$
Prevalence	Proportion of the population with the disease	$P(D)$

3.8.4. Clinical Example: HIV Screening

Let us work through a concrete example. Consider an HIV screening test with the following characteristics:

- **Sensitivity**: 99.7% (the test correctly identifies 99.7% of people who have HIV)
- **Specificity**: 99.5% (the test correctly identifies 99.5% of people who do not have HIV)
- **Prevalence**: 0.3% (the proportion of the general population with HIV)

These numbers look excellent. But what happens when a randomly selected person tests positive?

Using Bayes' theorem:

$$PPV = P(D \mid +) = \frac{P(+ \mid D) \times P(D)}{P(+ \mid D) \times P(D) + P(+ \mid \bar{D}) \times P(\bar{D})}$$

$$= \frac{0.997 \times 0.003}{0.997 \times 0.003 + 0.005 \times 0.997}$$

$$= \frac{0.002991}{0.002991 + 0.004985}$$

$$= \frac{0.002991}{0.007976} \approx 0.375$$

The positive predictive value is only about **37.5%**! This means that if a person from the general population tests positive, there is only about a 37.5% chance they actually have HIV. More than 6 out of 10 positive results are false positives.

This counterintuitive result arises because the disease is rare. Even with a highly accurate test, the large number of healthy people (99.7% of the population) generates many false positives that overwhelm the true positives from the small number of infected people.

This is why HIV screening programs use **confirmatory testing**: a second, different test is performed on all positive results. It is also why screening is more effective in high-prevalence populations (e.g., among people with known risk factors).

3.8.5. The Role of Prevalence

The PPV depends heavily on prevalence. Let us compute the PPV across a range of prevalence values:

3.8.6. R

```r
library(tidyverse)

sensitivity <- 0.997
specificity <- 0.995

prevalence <- seq(0.001, 0.30, by = 0.001)

ppv <- (sensitivity * prevalence) /
  (sensitivity * prevalence + (1 - specificity) * (1 - prevalence))

npv <- (specificity * (1 - prevalence)) /
  (specificity * (1 - prevalence) + (1 - sensitivity) * prevalence)

test_data <- tibble(
  prevalence = prevalence,
```

```r
  PPV = ppv,
  NPV = npv
) |>
  pivot_longer(cols = c(PPV, NPV), names_to = "metric", values_to = "value")

ggplot(test_data, aes(x = prevalence, y = value, color = metric)) +
  geom_line(linewidth = 1.2) +
  scale_x_continuous(labels = scales::percent_format()) +
  scale_y_continuous(labels = scales::percent_format(), limits = c(0, 1)) +
  labs(
    title = "How Prevalence Affects PPV and NPV",
    subtitle = "Sensitivity = 99.7%, Specificity = 99.5%",
    x = "Disease Prevalence",
    y = "Predictive Value",
    color = "Metric"
  ) +
  theme_minimal() +
  scale_color_manual(values = c("PPV" = "steelblue", "NPV" = "firebrick"))
```

3.8.7. Python

```python
import numpy as np
import matplotlib.pyplot as plt

sensitivity = 0.997
specificity = 0.995

prevalence = np.arange(0.001, 0.301, 0.001)

ppv = (sensitivity * prevalence) / (
    sensitivity * prevalence + (1 - specificity) * (1 - prevalence)
)
```

```
npv = (specificity * (1 - prevalence)) / (
    specificity * (1 - prevalence) + (1 - sensitivity) * prevalence
)

fig, ax = plt.subplots(figsize=(9, 5))
ax.plot(prevalence * 100, ppv * 100, color="steelblue", linewidth=1.5, label=
ax.plot(prevalence * 100, npv * 100, color="firebrick", linewidth=1.5, label=
ax.set_xlabel("Disease Prevalence (%)")
ax.set_ylabel("Predictive Value (%)")
ax.set_title("How Prevalence Affects PPV and NPV\nSensitivity = 99.7%, Speci
ax.legend()
ax.set_ylim(0, 105)
plt.tight_layout()
plt.show()
```

The plot shows a crucial pattern: when prevalence is very low, even an excellent test produces many false positives (low PPV). As prevalence increases, PPV rises. Conversely, NPV is very high when prevalence is low (a negative result is very reassuring) but decreases as disease becomes more common.

Clinical takeaway: Always consider the pre-test probability (prevalence in the relevant population) when interpreting a diagnostic test result. A positive mammogram in a 25-year-old woman with no risk factors means something very different from the same result in a 60-year-old woman with a family history of breast cancer.

3.9. Exercises

Exercise 1.1: Z-Scores and the Normal Distribution

A laboratory reports that fasting blood glucose levels in healthy adults follow a normal distribution with a mean of 90 mg/dL and a standard deviation of 10 mg/dL.

1. What is the Z-score for a patient with a fasting glucose of 115 mg/dL?
2. What proportion of healthy adults have a fasting glucose above 115 mg/dL?
3. What fasting glucose value corresponds to the 95th percentile?
4. Plot the normal distribution and shade the area above 115 mg/dL.

3.10. R

```r
mu <- 90
sigma <- 10
patient_value <- 115

# 1. Calculate the Z-score
z_score <- # YOUR CODE HERE

# 2. Proportion above 115 (upper tail probability)
prop_above <- # YOUR CODE HERE (hint: use pnorm())

# 3. 95th percentile
percentile_95 <- # YOUR CODE HERE (hint: use qnorm())

# 4. Plot (modify the blood pressure example from the chapter)
# YOUR CODE HERE
```

3.11. Python

```
from scipy import stats
import numpy as np
import matplotlib.pyplot as plt

mu = 90
sigma = 10
patient_value = 115

# 1. Calculate the Z-score
z_score = # YOUR CODE HERE

# 2. Proportion above 115
prop_above = # YOUR CODE HERE (hint: use stats.norm.sf() or 1 - stats.norm.cdf())

# 3. 95th percentile
percentile_95 = # YOUR CODE HERE (hint: use stats.norm.ppf())

# 4. Plot
# YOUR CODE HERE
```

Solution

3.12. R

```r
library(tidyverse)

mu <- 90
sigma <- 10
patient_value <- 115

# 1. Z-score
z_score <- (patient_value - mu) / sigma
cat("Z-score:", z_score, "\n")
# Z-score: 2.5

# 2. Proportion above 115
prop_above <- 1 - pnorm(patient_value, mean = mu, sd = sigma)
# Equivalently: pnorm(patient_value, mean = mu, sd = sigma, lower.tail
cat("Proportion above 115:", round(prop_above, 4), "\n")
# About 0.0062 or 0.62%

# 3. 95th percentile
percentile_95 <- qnorm(0.95, mean = mu, sd = sigma)
cat("95th percentile:", round(percentile_95, 1), "mg/dL\n")
# About 106.4 mg/dL

# 4. Plot
x <- seq(mu - 4 * sigma, mu + 4 * sigma, length.out = 300)
y <- dnorm(x, mean = mu, sd = sigma)
plot_data <- tibble(x = x, y = y)

ggplot(plot_data, aes(x = x, y = y)) +
  geom_line(color = "steelblue", linewidth = 1.2) +
  geom_area(data = plot_data |> filter(x >= patient_value),
            fill = "firebrick", alpha = 0.4) +
  geom_vline(xintercept = patient_value, linetype = "dashed", color =
  annotate("text", x = 118, y = 0.02,
           label = paste0("Glucose = ", patient_value, " mg/dL\nZ = ",
           color = "firebrick", hjust = 0) +
  labs(
    title = "Distribution of Fasting Blood Glucose",
    x = "Fasting Blood Glucose (mg/dL)", y = "Density"
  ) +
  theme_minimal()
```

3.13. Python

```python
from scipy import stats
import numpy as np
import matplotlib.pyplot as plt

mu = 90
sigma = 10
patient_value = 115

# 1. Z-score
z_score = (patient_value - mu) / sigma
print(f"Z-score: {z_score}")
# Z-score: 2.5

# 2. Proportion above 115
prop_above = 1 - stats.norm.cdf(patient_value, loc=mu, scale=sigma)
# Or equivalently: stats.norm.sf(patient_value, loc=mu, scale=sigma)
print(f"Proportion above 115: {prop_above:.4f}")
# About 0.0062 or 0.62%

# 3. 95th percentile
percentile_95 = stats.norm.ppf(0.95, loc=mu, scale=sigma)
print(f"95th percentile: {percentile_95:.1f} mg/dL")
# About 106.4 mg/dL

# 4. Plot
x = np.linspace(mu - 4 * sigma, mu + 4 * sigma, 300)
y = stats.norm.pdf(x, loc=mu, scale=sigma)

fig, ax = plt.subplots(figsize=(9, 5))
ax.plot(x, y, color="steelblue", linewidth=1.5)
x_fill = x[x >= patient_value]
y_fill = stats.norm.pdf(x_fill, loc=mu, scale=sigma)
ax.fill_between(x_fill, y_fill, color="firebrick", alpha=0.4)
ax.axvline(patient_value, linestyle="--", color="firebrick")
ax.text(117, 0.02, f"Glucose = {patient_value} mg/dL\nZ = {z_score}",
        color="firebrick")
ax.set_title("Distribution of Fasting Blood Glucose")
ax.set_xlabel("Fasting Blood Glucose (mg/dL)")
ax.set_ylabel("Density")
plt.tight_layout()
plt.show()
```

Exercise 1.2: Binomial Distribution — Vaccine Efficacy

A COVID-19 vaccine has an efficacy of 90%, meaning that among people exposed to the virus, a vaccinated person has a 90% chance of NOT getting symptomatic disease (equivalently, a 10% chance of a breakthrough infection).
In a group of 25 vaccinated individuals who are all exposed to the virus:

1. What is the expected number of breakthrough infections?
2. What is the probability of zero breakthrough infections?
3. What is the probability of 5 or more breakthrough infections?
4. Plot the full probability distribution.

3.14. R

```r
n <- 25
p <- 0.10  # probability of breakthrough infection

# 1. Expected number
expected <- # YOUR CODE HERE

# 2. P(X = 0)
prob_zero <- # YOUR CODE HERE (hint: dbinom())

# 3. P(X >= 5)
prob_five_or_more <- # YOUR CODE HERE (hint: use pbinom with lower.tail)

# 4. Plot the distribution
# YOUR CODE HERE
```

3.15. Python

```
from scipy import stats
import numpy as np
import matplotlib.pyplot as plt

n = 25
p = 0.10

# 1. Expected number
expected = # YOUR CODE HERE

# 2. P(X = 0)
prob_zero = # YOUR CODE HERE (hint: stats.binom.pmf())

# 3. P(X >= 5)
prob_five_or_more = # YOUR CODE HERE (hint: stats.binom.sf())

# 4. Plot
# YOUR CODE HERE
```

Solution

3.16. R

```r
library(tidyverse)

n <- 25
p <- 0.10

# 1. Expected number
expected <- n * p
cat("Expected breakthrough infections:", expected, "\n")
# 2.5

# 2. P(X = 0)
prob_zero <- dbinom(0, size = n, prob = p)
cat("P(X = 0):", round(prob_zero, 4), "\n")
# About 0.0718

# 3. P(X >= 5)
prob_five_or_more <- 1 - pbinom(4, size = n, prob = p)
# Equivalently: pbinom(4, size = n, prob = p, lower.tail = FALSE)
cat("P(X >= 5):", round(prob_five_or_more, 4), "\n")
# About 0.0980

# 4. Plot
outcomes <- tibble(
  k = 0:n,
  probability = dbinom(k, size = n, prob = p)
)

ggplot(outcomes, aes(x = k, y = probability)) +
  geom_col(aes(fill = k >= 5), alpha = 0.8) +
  scale_fill_manual(values = c("FALSE" = "steelblue", "TRUE" = "firebrick"),
                    guide = "none") +
  geom_vline(xintercept = expected, linetype = "dashed") +
  labs(
    title = "Breakthrough Infections in 25 Vaccinated Individuals",
    subtitle = "Binomial(25, 0.10); red bars = 5 or more infections",
    x = "Number of Breakthrough Infections", y = "Probability"
  ) +
  scale_x_continuous(breaks = seq(0, 25, 2)) +
  theme_minimal()
```

3.17. Python

```python
from scipy import stats
import numpy as np
import matplotlib.pyplot as plt

n = 25
p = 0.10

# 1. Expected number
expected = n * p
print(f"Expected breakthrough infections: {expected}")
# 2.5

# 2. P(X = 0)
prob_zero = stats.binom.pmf(0, n, p)
print(f"P(X = 0): {prob_zero:.4f}")
# About 0.0718

# 3. P(X >= 5)
prob_five_or_more = stats.binom.sf(4, n, p)  # sf = survival function = 1 - cdf
print(f"P(X >= 5): {prob_five_or_more:.4f}")
# About 0.0980

# 4. Plot
k = np.arange(0, n + 1)
probs = stats.binom.pmf(k, n, p)
colors = ["firebrick" if ki >= 5 else "steelblue" for ki in k]

fig, ax = plt.subplots(figsize=(9, 5))
ax.bar(k, probs, color=colors, alpha=0.8)
ax.axvline(expected, linestyle="--", color="black")
ax.set_title("Breakthrough Infections in 25 Vaccinated Individuals\nBinomial(25, 0.10
ax.set_xlabel("Number of Breakthrough Infections")
ax.set_ylabel("Probability")
ax.set_xticks(range(0, 26, 2))
plt.tight_layout()
plt.show()
```

Exercise 1.3: Poisson Distribution — Emergency Department Visits

An emergency department receives an average of 8 trauma cases per day.

1. What is the probability of receiving exactly 8 trauma cases tomorrow?
2. What is the probability of receiving 12 or more?
3. On how many days per year (365 days) would you expect 0 trauma cases?
4. Create a bar plot of the Poisson(8) distribution from 0 to 20.

3.18. R

```r
lambda <- 8

# 1. P(X = 8)
prob_exactly_8 <- # YOUR CODE HERE (hint: dpois())

# 2. P(X >= 12)
prob_12_or_more <- # YOUR CODE HERE (hint: ppois with lower.tail)

# 3. Expected days with 0 cases in a year
prob_zero <- # YOUR CODE HERE
expected_zero_days <- # YOUR CODE HERE

# 4. Plot
# YOUR CODE HERE
```

3.19. Python

```
from scipy import stats
import numpy as np
import matplotlib.pyplot as plt

lam = 8

# 1. P(X = 8)
prob_exactly_8 = # YOUR CODE HERE (hint: stats.poisson.pmf())

# 2. P(X >= 12)
prob_12_or_more = # YOUR CODE HERE (hint: stats.poisson.sf())

# 3. Expected days with 0 cases
prob_zero = # YOUR CODE HERE
expected_zero_days = # YOUR CODE HERE

# 4. Plot
# YOUR CODE HERE
```

Solution

3.20. R

```
library(tidyverse)

lambda <- 8

# 1. P(X = 8)
prob_exactly_8 <- dpois(8, lambda = lambda)
cat("P(X = 8):", round(prob_exactly_8, 4), "\n")
# About 0.1396

# 2. P(X >= 12)
prob_12_or_more <- ppois(11, lambda = lambda, lower.tail = FALSE)
cat("P(X >= 12):", round(prob_12_or_more, 4), "\n")
# About 0.1121

# 3. Expected days with 0 cases
prob_zero <- dpois(0, lambda = lambda)
expected_zero_days <- 365 * prob_zero
cat("P(X = 0):", format(prob_zero, scientific = TRUE), "\n")
cat("Expected zero-case days per year:", round(expected_zero_days, 3),
# P(0) is very small (~0.000335), so about 0.12 days/year

# 4. Plot
k <- 0:20
poisson_data <- tibble(
  k = k,
  probability = dpois(k, lambda = lambda)
)

ggplot(poisson_data, aes(x = k, y = probability)) +
  geom_col(fill = "darkorange", alpha = 0.8) +
  geom_vline(xintercept = lambda, linetype = "dashed", color = "firebr
  labs(
    title = "Poisson Distribution: Trauma Cases per Day",
    subtitle = expression(paste(lambda, " = 8")),
    x = "Number of Trauma Cases", y = "Probability"
  ) +
  scale_x_continuous(breaks = k) +
  theme_minimal()
```

3.21. Python

```python
from scipy import stats
import numpy as np
import matplotlib.pyplot as plt

lam = 8

# 1. P(X = 8)
prob_exactly_8 = stats.poisson.pmf(8, lam)
print(f"P(X = 8): {prob_exactly_8:.4f}")
# About 0.1396

# 2. P(X >= 12)
prob_12_or_more = stats.poisson.sf(11, lam)
print(f"P(X >= 12): {prob_12_or_more:.4f}")
# About 0.1121

# 3. Expected days with 0 cases
prob_zero = stats.poisson.pmf(0, lam)
expected_zero_days = 365 * prob_zero
print(f"P(X = 0): {prob_zero:.6f}")
print(f"Expected zero-case days per year: {expected_zero_days:.3f}")
# Very small, about 0.12 days per year

# 4. Plot
k = np.arange(0, 21)
probs = stats.poisson.pmf(k, lam)

fig, ax = plt.subplots(figsize=(9, 5))
ax.bar(k, probs, color="darkorange", alpha=0.8)
ax.axvline(lam, linestyle="--", color="firebrick")
ax.set_title(f"Poisson Distribution: Trauma Cases per Day (lambda = {l
ax.set_xlabel("Number of Trauma Cases")
ax.set_ylabel("Probability")
ax.set_xticks(k)
plt.tight_layout()
plt.show()
```

Exercise 1.4: Bayes' Theorem — Interpreting a Cancer Screening Test

A breast cancer screening mammogram has the following characteristics:

- Sensitivity: 87% (detects 87% of actual cancers)
- Specificity: 95% (correctly clears 95% of women without cancer)
- Prevalence of breast cancer in women aged 50–59 undergoing screening: 2%

1. Calculate the positive predictive value (PPV): if a woman in this age group has a positive mammogram, what is the probability she actually has breast cancer?
2. Calculate the negative predictive value (NPV).
3. How does the PPV change if we consider a high-risk population where prevalence is 10%?
4. Create a plot showing PPV as a function of prevalence (from 0.1% to 30%).

3.22. R

```r
sensitivity <- 0.87
specificity <- 0.95
prevalence <- 0.02

# 1. PPV
ppv <- # YOUR CODE HERE

# 2. NPV
npv <- # YOUR CODE HERE

# 3. PPV with prevalence = 10%
ppv_high_risk <- # YOUR CODE HERE

# 4. Plot PPV vs. prevalence
# YOUR CODE HERE
```

3.23. Python

```
import numpy as np
import matplotlib.pyplot as plt

sensitivity = 0.87
specificity = 0.95
prevalence = 0.02

# 1. PPV
ppv = # YOUR CODE HERE

# 2. NPV
npv = # YOUR CODE HERE

# 3. PPV with prevalence = 10%
ppv_high_risk = # YOUR CODE HERE

# 4. Plot PPV vs. prevalence
# YOUR CODE HERE
```

Solution

3.24. R

```r
library(tidyverse)

sensitivity <- 0.87
specificity <- 0.95
prevalence <- 0.02

# 1. PPV
ppv <- (sensitivity * prevalence) /
  (sensitivity * prevalence + (1 - specificity) * (1 - prevalence))
cat("PPV (prevalence = 2%):", round(ppv, 4), "\n")
# About 0.2623 or 26.2%

# 2. NPV
npv <- (specificity * (1 - prevalence)) /
  (specificity * (1 - prevalence) + (1 - sensitivity) * prevalence)
cat("NPV (prevalence = 2%):", round(npv, 4), "\n")
# About 0.9972 or 99.7%

# 3. PPV with high-risk prevalence
prev_high <- 0.10
ppv_high_risk <- (sensitivity * prev_high) /
  (sensitivity * prev_high + (1 - specificity) * (1 - prev_high))
cat("PPV (prevalence = 10%):", round(ppv_high_risk, 4), "\n")
# About 0.6588 or 65.9%

# 4. Plot
prev_range <- seq(0.001, 0.30, by = 0.001)
ppv_curve <- (sensitivity * prev_range) /
  (sensitivity * prev_range + (1 - specificity) * (1 - prev_range))

plot_data <- tibble(prevalence = prev_range, PPV = ppv_curve)

ggplot(plot_data, aes(x = prevalence, y = PPV)) +
  geom_line(color = "steelblue", linewidth = 1.2) +
  geom_point(data = tibble(prevalence = c(0.02, 0.10),
                           PPV = c(ppv, ppv_high_risk)),
             color = "firebrick", size = 3) +
  annotate("text", x = 0.04, y = ppv - 0.03,
           label = paste0("General pop (2%): PPV = ", round(ppv * 100,
           hjust = 0, color = "firebrick", size = 3.5) +
  annotate("text", x = 0.12, y = ppv_high_risk - 0.03,
           label = paste0("High risk (10%): PPV = ", round(ppv_high_ri
           hjust = 0, color = "firebrick", size = 3.5) +
  scale_x_continuous(labels = scales::percent_format()) +
  scale_y_continuous(labels = scales::percent_format(), limits = c(0,
  labs(
    title = "Mammography PPV Depends on Prevalence",
    subtitle = "Sensitivity = 87%, Specificity = 95%",
    x = "Prevalence",
```

3.25. Python

```python
import numpy as np
import matplotlib.pyplot as plt

sensitivity = 0.87
specificity = 0.95
prevalence = 0.02

# 1. PPV
ppv = (sensitivity * prevalence) / (
    sensitivity * prevalence + (1 - specificity) * (1 - prevalence)
)
print(f"PPV (prevalence = 2%): {ppv:.4f}")
# About 0.2623 or 26.2%

# 2. NPV
npv = (specificity * (1 - prevalence)) / (
    specificity * (1 - prevalence) + (1 - sensitivity) * prevalence
)
print(f"NPV (prevalence = 2%): {npv:.4f}")
# About 0.9972 or 99.7%

# 3. PPV with high-risk prevalence
prev_high = 0.10
ppv_high = (sensitivity * prev_high) / (
    sensitivity * prev_high + (1 - specificity) * (1 - prev_high)
)
print(f"PPV (prevalence = 10%): {ppv_high:.4f}")
# About 0.6588 or 65.9%

# 4. Plot
prev_range = np.arange(0.001, 0.301, 0.001)
ppv_curve = (sensitivity * prev_range) / (
    sensitivity * prev_range + (1 - specificity) * (1 - prev_range)
)

fig, ax = plt.subplots(figsize=(9, 5))
ax.plot(prev_range * 100, ppv_curve * 100, color="steelblue", linewidt
ax.plot([2, 10], [ppv * 100, ppv_high * 100], "o", color="firebrick",
ax.annotate(f"General pop (2%): PPV = {ppv*100:.1f}%",
            xy=(2, ppv*100), xytext=(5, ppv*100 - 5),
            color="firebrick", fontsize=9,
            arrowprops=dict(arrowstyle="->", color="firebrick"))
ax.annotate(f"High risk (10%): PPV = {ppv_high*100:.1f}%",
            xy=(10, ppv_high*100), xytext=(13, ppv_high*100 - 5),
            color="firebrick", fontsize=9,
            arrowprops=dict(arrowstyle="->", color="firebrick"))
ax.set_xlabel("Prevalence (%)")
ax.set_ylabel("Positive Predictive Value (%)")
ax.set_title("Mammography PPV Depends on Prevalence\nSensitivity = 87%
```

Exercise 1.5: Central Limit Theorem — Hands-On Simulation

Demonstrate the Central Limit Theorem yourself by simulating sample means from a non-normal distribution.

1. Generate 10,000 individual observations from a **Poisson distribution** with $\lambda = 3$ (this distribution is right-skewed). Plot a histogram.
2. Now, for each of the following sample sizes (n = 5, 15, 50, 200), draw 5,000 random samples of that size, compute the mean of each sample, and plot the distribution of sample means.
3. Overlay a normal curve on the $n = 50$ histogram to show how well the normal approximation fits.

3.26. R

```
library(tidyverse)

set.seed(123)
lambda <- 3

# 1. Histogram of raw Poisson data
raw_data <- rpois(10000, lambda = lambda)
# YOUR CODE HERE: plot a histogram

# 2. CLT simulation for different sample sizes
sample_sizes <- c(5, 15, 50, 200)
# YOUR CODE HERE: for each n, draw 5000 samples, compute means, plot

# 3. Overlay normal curve on n = 50
# YOUR CODE HERE
```

3.27. Python

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

np.random.seed(123)
lam = 3

# 1. Histogram of raw Poisson data
raw_data = np.random.poisson(lam, 10000)
# YOUR CODE HERE: plot a histogram

# 2. CLT simulation
sample_sizes = [5, 15, 50, 200]
# YOUR CODE HERE

# 3. Overlay normal curve
# YOUR CODE HERE
```

Solution

3.28. R

```r
library(tidyverse)

set.seed(123)
lambda <- 3
n_sim <- 5000

# 1. Raw Poisson data
raw_data <- rpois(10000, lambda = lambda)
ggplot(tibble(x = raw_data), aes(x = x)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white", alpha = 0.8) +
  labs(title = "Raw Poisson(3) Data (right-skewed)",
       x = "Value", y = "Count") +
  theme_minimal()

# 2. CLT simulation
sample_sizes <- c(5, 15, 50, 200)

clt_data <- map_dfr(sample_sizes, function(n) {
  means <- replicate(n_sim, mean(rpois(n, lambda = lambda)))
  tibble(n = paste0("n = ", n), sample_mean = means)
})

clt_data$n <- factor(clt_data$n, levels = paste0("n = ", sample_sizes))

ggplot(clt_data, aes(x = sample_mean)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 40, fill = "steelblue", alpha = 0.7) +
  facet_wrap(~n, scales = "free_y") +
  labs(title = "CLT: Distribution of Sample Means from Poisson(3)",
       x = "Sample Mean", y = "Density") +
  theme_minimal()

# 3. Overlay normal curve on n = 50
n50_means <- replicate(n_sim, mean(rpois(50, lambda = lambda)))
theoretical_sd <- sqrt(lambda / 50)

ggplot(tibble(x = n50_means), aes(x = x)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 40, fill = "steelblue", alpha = 0.7) +
  stat_function(fun = dnorm, args = list(mean = lambda, sd = theoretical_sd),
                color = "firebrick", linewidth = 1.2) +
  labs(title = "Sample Means (n=50) with Normal Approximation Overlay",
       subtitle = paste0("Theoretical: N(", lambda, ", ", round(theoretical_sd^2, 4),
       x = "Sample Mean", y = "Density") +
  theme_minimal()
```

3.29. Python

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

np.random.seed(123)
lam = 3
n_sim = 5000

# 1. Raw Poisson data
raw_data = np.random.poisson(lam, 10000)
fig, ax = plt.subplots(figsize=(8, 4))
ax.hist(raw_data, bins=range(0, 15), color="steelblue", alpha=0.8, edgecolor="white")
ax.set_title("Raw Poisson(3) Data (right-skewed)")
ax.set_xlabel("Value")
ax.set_ylabel("Count")
plt.tight_layout()
plt.show()

# 2. CLT simulation
sample_sizes = [5, 15, 50, 200]
fig, axes = plt.subplots(2, 2, figsize=(10, 8))
axes = axes.flatten()

for i, n in enumerate(sample_sizes):
    means = [np.mean(np.random.poisson(lam, n)) for _ in range(n_sim)]
    axes[i].hist(means, bins=40, density=True, color="steelblue", alpha=0.7)
    axes[i].set_title(f"n = {n}")
    axes[i].set_xlabel("Sample Mean")
    axes[i].set_ylabel("Density")

fig.suptitle("CLT: Distribution of Sample Means from Poisson(3)", fontsize=13)
plt.tight_layout()
plt.show()

# 3. Overlay normal curve on n = 50
n50_means = [np.mean(np.random.poisson(lam, 50)) for _ in range(n_sim)]
theoretical_sd = np.sqrt(lam / 50)

fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(n50_means, bins=40, density=True, color="steelblue", alpha=0.7, label="Simula
x_range = np.linspace(min(n50_means), max(n50_means), 200)
ax.plot(x_range, stats.norm.pdf(x_range, loc=lam, scale=theoretical_sd),
        color="firebrick", linewidth=1.5, label="Normal approx.")
ax.set_title("Sample Means (n=50) with Normal Approximation Overlay")
ax.set_xlabel("Sample Mean")
ax.set_ylabel("Density")
ax.legend()
plt.tight_layout()
plt.show()
```

3.30. Chapter Summary

In this chapter, we covered the foundational language of probability and statistics:

- **Probability** quantifies uncertainty. The frequentist interpretation focuses on long-run frequencies; the Bayesian interpretation treats probability as a degree of belief. Both are useful.
- **Random variables** can be discrete (countable values) or continuous (any value in a range).
- The **normal distribution** is the most important distribution in statistics, characterized by its mean and standard deviation. Z-scores standardize values relative to the distribution.
- The **binomial distribution** models the number of successes in a fixed number of independent trials — useful for modeling treatment response rates, disease prevalence, and more.
- The **Poisson distribution** models counts of rare events occurring at a constant rate — useful for adverse events, hospital admissions, and rare disease incidence.
- The **Central Limit Theorem** tells us that sample means are approximately normally distributed regardless of the underlying population distribution, as long as the sample size is large enough. This justifies the widespread use of normal-based statistical methods.
- **Bayes' theorem** provides a principled way to update probabilities in light of new evidence. In diagnostic testing, it reveals that the predictive value of a test depends critically on disease prevalence, not just sensitivity and specificity.

These concepts underpin everything that follows in this course. In the next chapter, we will build on this foundation to discuss statistical inference — how to draw conclusions about populations from samples.

3.31. References and Further Reading

- Wasserman, L. (2004). *All of Statistics: A Concise Course in Statistical Inference*. Springer. A rigorous but accessible introduction to probability and statistics. Chapters 1–3 cover the material in this chapter at a more mathematical level.
- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). *Bayesian Data Analysis* (3rd ed.). CRC Press. The definitive textbook on Bayesian statistics. Chapter 1 provides an excellent introduction to Bayesian thinking.
- Harrell, F. E. (2015). *Regression Modeling Strategies: With Applications to Linear Models, Logistic and Ordinal Regression, and Survival Analysis* (2nd ed.). Springer. Chapter 2 discusses the role of probability and distributions in the context of regression modeling for health sciences.
- Altman, D. G., & Bland, J. M. (1994). "Diagnostic tests 1: Sensitivity and specificity." *BMJ*, 308(6943), 1552. A classic short paper explaining sensitivity, specificity, and predictive values for a clinical audience.
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- Gigerenzer, G., Gaissmaier, W., Kurz-Milcke, E., Schwartz, L. M., & Woloshin, S. (2007). "Helping doctors and patients make sense of health statistics." *Psychological Science in the Public Interest*, 8(2), 53–96. An excellent discussion of how to communicate probabilistic information about diagnostic tests and treatments.
- Cetinkaya-Rundel, M., & Hardin, J. (2024). *Introduction to Modern Statistics* (2nd ed.). OpenIntro. Freely available at https://openintro-

ims2.netlify.app/. Chapters 3–4 provide an accessible introduction to probability and distributions with many worked examples.

- Rice, J. A. (2007). *Mathematical Statistics and Data Analysis* (3rd ed.). Thomson Brooks/Cole. A thorough treatment of probability distributions for students comfortable with calculus.
- Motulsky, H. (2018). *Intuitive Biostatistics: A Nonmathematical Guide to Statistical Thinking* (4th ed.). Oxford University Press. A superb introduction to biostatistics for medical professionals, with clear explanations of probability, distributions, and Bayesian reasoning.

4. Statistical Inference: What We Can and Cannot Conclude

4.1. Introduction

Imagine you run a clinical trial comparing a new antihypertensive drug to placebo. After 12 weeks, the treatment group's mean systolic blood pressure dropped by 8.3 mmHg more than the placebo group. But how confident should you be in that number? Would you see the same result if you repeated the trial? Could the difference be explained by chance alone? And even if the effect is "real," is 8.3 mmHg enough to matter for patient health?

These are the questions that **statistical inference** answers. Inference is the bridge between the data you *observed* (your sample) and the truth you *want to know* (the population). This chapter will equip you to cross that bridge carefully, avoiding the many pitfalls that trip up even experienced researchers.

We will use a single running example throughout: a randomized controlled trial (RCT) of **Drug X** versus placebo for systolic blood pressure reduction. This trial enrolled 200 participants (100 per arm), measured blood pressure at baseline and 12 weeks, and found a mean difference of 8.3 mmHg (SD = 14.2 mmHg) favoring Drug X.

4.2. Point Estimates and Sampling Variability

4.2.1. What Is a Point Estimate?

A **point estimate** is a single number calculated from your sample that serves as your best guess for an unknown population parameter. Common examples:

What you want to know	Point estimate
Population mean blood pressure reduction	Sample mean ($\bar{x}$)
Population proportion who respond to treatment	Sample proportion ($\hat{p}$)
Population odds ratio for treatment effect	Sample odds ratio ($\widehat{OR}$)

In our trial, the point estimate for the mean difference in blood pressure reduction is **8.3 mmHg**. This is a single number – our best guess – but it comes with uncertainty.

4.2.2. Sampling Variability

If we ran the same trial again with 200 new participants from the same population, we would almost certainly get a *different* point estimate. Maybe 7.1 mmHg, or 9.8 mmHg, or 6.5 mmHg. This natural fluctuation from sample to sample is called **sampling variability**.

The key insight: **every point estimate is "wrong" in the sense that it almost never equals the exact population parameter.** The question is *how wrong* it might be.

4.2.3. The Standard Error

The **standard error (SE)** quantifies how much a point estimate is expected to vary across repeated samples. For a sample mean:

$$SE(\bar{x}) = \frac{s}{\sqrt{n}}$$

where s is the sample standard deviation and n is the sample size.

For the difference between two independent means:

$$SE(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

In our trial, assuming equal SDs of 14.2 mmHg in each group:

$$SE = \sqrt{\frac{14.2^2}{100} + \frac{14.2^2}{100}} = \sqrt{2.016 + 2.016} = \sqrt{4.033} \approx 2.01 \text{ mmHg}$$

This tells us: if we repeated this trial many times, the sample mean difference would typically vary by about 2 mmHg from trial to trial.

> **!** Important
>
> **Standard deviation vs. standard error:** The standard deviation (SD = 14.2) describes variability among *individual patients*. The standard error (SE = 2.01) describes variability among *sample means*. They answer different questions. The SD is always larger than the SE because means are less variable than individual observations.

4.3. Confidence Intervals: What They Actually Mean

4.3.1. Constructing a 95% Confidence Interval

A **95% confidence interval (CI)** for our mean difference is:

$$\bar{x} \pm t^* \times SE = 8.3 \pm 1.97 \times 2.01 = (4.34,\ 12.26) \text{ mmHg}$$

where t^* is the critical value from the t-distribution (approximately 1.97 for large samples).

4.3.2. The Correct Interpretation

Here is one of the most commonly misunderstood concepts in statistics.

> ⚠ Common Misconception
>
> **Wrong:** "There is a 95% probability that the true mean difference lies between 4.34 and 12.26 mmHg."
> **Right:** "If we repeated this trial many times and computed a 95% CI each time, approximately 95% of those intervals would contain the true mean difference."

Why does this distinction matter? The true population parameter is a fixed (unknown) number – it does not have a probability of being "in" or "out" of an interval. The *interval* is the random quantity, not the parameter. Each time you run a new study, you get a new interval. Some of those intervals capture the truth; some do not. A 95% CI means that the *procedure* works 95% of the time.

In practice, a useful way to think about it: the confidence interval gives you a range of plausible values for the population parameter, given the data you observed. Values inside the interval are reasonably compatible with your data; values outside are not.

4.3.3. What the Width Tells You

The width of a confidence interval reflects **precision**. A narrow CI means you have a precise estimate; a wide CI means substantial uncertainty. Width depends on:

1. **Sample size** – larger n gives narrower CIs
2. **Variability** – less variable data gives narrower CIs
3. **Confidence level** – 99% CIs are wider than 95% CIs (more confidence requires more hedging)

In our trial, the CI of (4.34, 12.26) is moderately wide – the true effect could be as small as about 4 mmHg or as large as 12 mmHg. This matters clinically: a 4 mmHg reduction and a 12 mmHg reduction may lead to very different treatment decisions.

4.4. P-values: What They Are, What They Are Not

4.4.1. Definition

A **p-value** is the probability of observing a test statistic as extreme as (or more extreme than) the one you calculated, *assuming the null hypothesis is true*.

For our trial, the null hypothesis is $H_0 : \mu_{\text{Drug X}} - \mu_{\text{Placebo}} = 0$ (no difference). The test statistic is:

$$t = \frac{8.3 - 0}{2.01} = 4.13$$

With 198 degrees of freedom, $p < 0.001$.

4.4.2. What a P-value Is

- The p-value measures the **compatibility between your data and the null hypothesis**
- A small p-value means your data would be unlikely if the null hypothesis were true
- It quantifies evidence against H_0, not evidence *for* any particular alternative

4.4.3. What a P-value Is NOT

The American Statistical Association published a landmark statement in 2016 (Wasserstein & Lazar, 2016) because p-values are so widely misinterpreted. Here are the key points:

P-value Misconceptions – From the ASA Statement

1. **A p-value is NOT the probability that the null hypothesis is true.** A p-value of 0.03 does *not* mean there is a 3% chance the drug doesn't work.
2. **A p-value is NOT the probability that the result was produced by chance.** It is calculated *assuming* chance is the only explanation.
3. **Statistical significance does NOT mean clinical or practical importance.** A tiny, meaningless effect can have $p < 0.001$ with a large enough sample.
4. **A large p-value does NOT mean the null hypothesis is true.** It means you failed to find strong evidence against it – absence of evidence is not evidence of absence.
5. **The p-value does NOT measure the size of an effect.** It

mixes effect size with sample size. Always report the effect size and confidence interval alongside the p-value.

4.4.4. The "Statistical Significance" Problem in Medicine

The threshold of $p < 0.05$ is deeply entrenched in medical research. But there is nothing magical about 0.05. It is an arbitrary convention, originally proposed as a convenient rule of thumb by Ronald Fisher, not as a rigid decision boundary.

Problems with dichotomizing at $p = 0.05$:

- A result with $p = 0.049$ is treated as fundamentally different from $p = 0.051$, even though they are essentially identical in evidential strength.
- Journals preferentially publish "significant" results, creating **publication bias**.
- Researchers may unconsciously (or consciously) engage in **p-hacking** – trying multiple analyses until one crosses the 0.05 threshold.
- In large clinical databases, nearly everything is "statistically significant" because of enormous sample sizes, regardless of clinical relevance.

A better approach: Report the point estimate, confidence interval, and p-value. Interpret them together. Judge results by their clinical importance, not just their p-value. As the ASA statement concludes: "No single index should substitute for scientific reasoning."

4.5. Effect Sizes That Matter Clinically

4.5.1. Why Effect Sizes Matter

Statistical significance tells you whether an effect is likely to be non-zero. It says nothing about whether the effect is *large enough to matter*. In medicine, this distinction is critical: a drug that lowers blood pressure by 0.5 mmHg might achieve $p < 0.001$ in a trial of 50,000 people, but no clinician would prescribe it based on that effect.

4.5.2. Cohen's d (Standardized Mean Difference)

Cohen's d expresses the difference between two means in standard deviation units:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}}$$

For our trial: $d = 8.3/14.2 = 0.58$, which is conventionally considered a "medium" effect.

Cohen's d	Interpretation
0.2	Small
0.5	Medium
0.8	Large

> **Note**
>
> These benchmarks (from Cohen, 1988) are rough guidelines, not rigid rules. A "small" effect by Cohen's standards might be highly clinically meaningful (e.g., a small reduction in mortality), while a "large" effect

> might be trivial in another context. Always interpret effect sizes in the clinical context.

4.5.3. Odds Ratios and Risk Ratios

For binary outcomes (e.g., death, hospitalization, disease incidence), effect sizes are typically reported as:

Risk Ratio (Relative Risk):

$$RR = \frac{P(\text{event} \mid \text{treatment})}{P(\text{event} \mid \text{control})}$$

An RR of 0.75 means the treatment group had 75% the risk of the control group, i.e., a 25% relative risk reduction.

Odds Ratio:

$$OR = \frac{\text{odds of event in treatment}}{\text{odds of event in control}} = \frac{p_1/(1-p_1)}{p_2/(1-p_2)}$$

Odds ratios are the natural output of logistic regression. When the outcome is rare ($< 10\%$), the OR approximates the RR. When the outcome is common, the OR exaggerates the effect compared to the RR.

Risk Difference (Absolute Risk Reduction):

$$RD = P(\text{event} \mid \text{control}) - P(\text{event} \mid \text{treatment})$$

If 15% of control patients are hospitalized vs. 10% of treatment patients, the RD = 0.05 (or 5 percentage points).

4.5.4. Number Needed to Treat (NNT)

The NNT is perhaps the most clinician-friendly effect size:

$$NNT = \frac{1}{RD} = \frac{1}{|p_1 - p_2|}$$

Using our example: $NNT = 1/0.05 = 20$. This means you need to treat 20 patients to prevent one additional hospitalization. A low NNT is desirable. For reference:

- Statins for secondary prevention of MI: NNT around 30-80
- Aspirin for acute MI: NNT around 40
- Antibiotics for strep pharyngitis to prevent rheumatic fever: NNT around 4,000

4.5.5. Statistically Significant but Clinically Meaningless

Consider a large pragmatic trial (n = 20,000) of a new diabetes drug that lowers HbA1c by 0.1% compared to the current standard of care, with $p = 0.002$. Is this clinically meaningful?

Almost certainly not. A 0.1% reduction in HbA1c is below the threshold most endocrinologists consider clinically relevant (typically 0.3-0.5%). The p-value is tiny because the sample size is enormous – even a trivial true effect becomes "significant" with enough data.

This is why you should **always** report and interpret effect sizes (with confidence intervals), not just p-values.

4.6. Type I and Type II Errors

4.6.1. The Two Kinds of Mistakes

When making a decision based on a hypothesis test, there are two ways to be wrong:

	H_0 is true (no effect)	H_0 is false (real effect)
Reject H_0	Type I error (α)	Correct (Power)
Fail to reject H_0	Correct	Type II error (β)

- **Type I error (false positive):** You conclude the drug works, but it actually doesn't. Probability = α (typically set at 0.05).
- **Type II error (false negative):** You conclude the drug doesn't work, but it actually does. Probability = β.

4.6.2. Statistical Power

Power $= 1 - \beta =$ the probability of correctly detecting a real effect.

Power depends on:

1. **Effect size** – larger effects are easier to detect
2. **Sample size** – more data provides more power
3. **Significance level (α)** – higher α gives more power (but more false positives)
4. **Variability** – less noise makes signals easier to detect

In clinical trials, we typically aim for **80% power** (i.e., $\beta = 0.20$). This means we accept a 20% chance of missing a true effect – a remarkably high false-negative rate that is often underappreciated.

Clinical Implication

Many clinical trials are **underpowered** – they have too few participants to detect realistic effect sizes. A "negative" trial (one that fails to reach $p < 0.05$) does not necessarily mean the treatment is ineffective. It may simply mean the trial was too small. Always check the confidence interval: if it includes both clinically meaningful and null effects, the trial was inconclusive, not negative.

4.6.3. A Power Calculation Example

Suppose you are planning a trial of Drug X and expect a 5 mmHg difference (SD = 15 mmHg). How many participants per group do you need for 80% power at $\alpha = 0.05$?

$$n = \frac{2(z_{\alpha/2} + z_{\beta})^2\sigma^2}{\Delta^2} = \frac{2(1.96 + 0.84)^2 \times 15^2}{5^2} = \frac{2 \times 7.84 \times 225}{25} \approx 141 \text{ per group}$$

You need approximately 141 participants per group, or 282 total.

4.7. Multiple Testing

4.7.1. The Problem

When you perform multiple statistical tests, the probability of at least one false positive increases rapidly. If you test 20 independent hypotheses at $\alpha = 0.05$, the probability of at least one Type I error is:

$$1 - (1 - 0.05)^{20} = 1 - 0.95^{20} = 1 - 0.358 = 0.642$$

That is a **64.2% chance** of at least one false positive. This is a massive problem in:

- **Genomics/GWAS:** Testing millions of genetic variants simultaneously
- **Clinical trials with multiple endpoints:** Primary, secondary, and exploratory outcomes
- **Subgroup analyses:** Testing treatment effects in men vs. women, old vs. young, etc.

4.7.2. Bonferroni Correction

The simplest fix: divide α by the number of tests.

$$\alpha_{\text{adjusted}} = \frac{\alpha}{m}$$

where m is the number of tests. If you run 20 tests, you require $p < 0.05/20 = 0.0025$ for each test.

Pros: Simple, controls the **family-wise error rate** (FWER) – the probability of *any* false positives.

Cons: Very conservative. As m grows, it becomes extremely difficult to detect real effects. With 1 million SNPs in a GWAS, you need $p < 5 \times 10^{-8}$.

4.7.3. False Discovery Rate (FDR) Control

The **Benjamini-Hochberg procedure** (1995) controls the **false discovery rate** – the expected proportion of rejected hypotheses that are false positives. This is less stringent than Bonferroni and more appropriate when you expect many true positives among your tests.

Steps:

1. Rank all m p-values from smallest to largest: $p_{(1)} \leq p_{(2)} \leq ... \leq p_{(m)}$
2. Find the largest k such that $p_{(k)} \leq \frac{k}{m} \times q$, where q is the desired FDR (e.g., 0.05)
3. Reject all hypotheses with $p \leq p_{(k)}$

In genomics, FDR control at $q = 0.05$ means: among all the genes we declare "significant," we expect no more than 5% to be false positives. This is often a more useful guarantee than Bonferroni's promise of *no* false positives.

4.7.4. When Does Multiple Testing Matter?

Scenario	Correction needed?
Pre-specified primary endpoint in a clinical trial	No (single test)
One primary + several secondary endpoints	Yes
Subgroup analyses	Yes (or treat as exploratory)
Genome-wide association study	Yes (millions of tests)
Exploratory data analysis	No formal correction, but be transparent

4.8. Exercises

Exercise 2.1: Confidence Interval Simulation

Simulate 100 trials, each with n=50 per group, true mean difference = 5, SD = 10. Compute a 95% CI for each trial. What proportion of CIs contain the true value of 5?

4.9. R

```r
set.seed(42)
n_sims <- 100
n_per_group <- 50
true_diff <- 5
sd_val <- 10

contains_true <- numeric(n_sims)

for (i in 1:n_sims) {
  group1 <- rnorm(n_per_group, mean = 0, sd = sd_val)
  group2 <- rnorm(n_per_group, mean = true_diff, sd = sd_val)

  test_result <- t.test(group2, group1)
  ci <- test_result$conf.int

  contains_true[i] <- (ci[1] <= true_diff & ci[2] >= true_diff)
}

cat("Proportion of CIs containing true value:", mean(contains_true), "\n")

# Bonus: visualize the CIs
library(ggplot2)
ci_data <- data.frame(
  sim = 1:n_sims,
  lower = numeric(n_sims),
  upper = numeric(n_sims),
  contains = logical(n_sims)
)

set.seed(42)
for (i in 1:n_sims) {
  group1 <- rnorm(n_per_group, mean = 0, sd = sd_val)
  group2 <- rnorm(n_per_group, mean = true_diff, sd = sd_val)
  test_result <- t.test(group2, group1)
  ci_data$lower[i] <- test_result$conf.int[1]
  ci_data$upper[i] <- test_result$conf.int[2]
  ci_data$contains[i] <- (ci_data$lower[i] <= true_diff & ci_data$upper[i]
}

ggplot(ci_data, aes(x = sim, y = (lower + upper) / 2, color = contains)) +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.3) +
  geom_hline(yintercept = true_diff, linetype = "dashed", color = "blue") +
  scale_color_manual(values = c("red", "darkgreen")) +
  labs(x = "Simulation", y = "Mean Difference",
       title = "95% Confidence Intervals from 100 Simulated Trials") +
  theme_minimal()
```

4.10. Python

```
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

np.random.seed(42)
n_sims = 100
n_per_group = 50
true_diff = 5
sd_val = 10

contains_true = []
lower_bounds = []
upper_bounds = []

for i in range(n_sims):
    group1 = np.random.normal(0, sd_val, n_per_group)
    group2 = np.random.normal(true_diff, sd_val, n_per_group)

    diff = np.mean(group2) - np.mean(group1)
    se = np.sqrt(np.var(group1, ddof=1)/n_per_group + np.var(group2, ddof=1
    t_crit = stats.t.ppf(0.975, df=n_per_group*2 - 2)

    lower = diff - t_crit * se
    upper = diff + t_crit * se

    lower_bounds.append(lower)
    upper_bounds.append(upper)
    contains_true.append(lower <= true_diff <= upper)

print(f"Proportion of CIs containing true value: {np.mean(contains_true):.2

# Visualize
fig, ax = plt.subplots(figsize=(10, 6))
for i in range(n_sims):
    color = "green" if contains_true[i] else "red"
    ax.plot([i, i], [lower_bounds[i], upper_bounds[i]], color=color, linewi

ax.axhline(y=true_diff, color="blue", linestyle="--", label="True differenc
ax.set_xlabel("Simulation")
ax.set_ylabel("Mean Difference")
ax.set_title("95% Confidence Intervals from 100 Simulated Trials")
ax.legend()
plt.tight_layout()
plt.show()
```

Solution

You should find that approximately 95 out of 100 CIs contain the true value of 5. The exact number will vary due to randomness, but it should be close to 95. The red intervals in the plot are the ones that "missed" – they do not contain the true value. This is the correct interpretation of a 95% CI: the *procedure* captures the truth 95% of the time.

Exercise 2.2: P-value Interpretation

A clinical trial reports: "The new drug reduced 30-day mortality from 12% to 9% (p = 0.04)." For each statement below, indicate whether it is TRUE or FALSE and explain why.

1. There is a 96% probability that the drug truly reduces mortality.
2. There is a 4% probability the result is due to chance.
3. If the drug had no effect, we would see a difference this large or larger about 4% of the time.
4. The drug reduces mortality by 3 percentage points.

4.11. R

```r
# This is a conceptual exercise, but let's verify the numbers:
p_control <- 0.12
p_treatment <- 0.09

risk_difference <- p_control - p_treatment
risk_ratio <- p_treatment / p_control
nnt <- 1 / risk_difference

cat("Risk Difference:", risk_difference, "\n")
cat("Risk Ratio:", round(risk_ratio, 3), "\n")
cat("NNT:", round(nnt, 1), "\n")
```

4.12. Python

```python
# This is a conceptual exercise, but let's verify the numbers:
p_control = 0.12
p_treatment = 0.09

risk_difference = p_control - p_treatment
risk_ratio = p_treatment / p_control
nnt = 1 / risk_difference

print(f"Risk Difference: {risk_difference}")
print(f"Risk Ratio: {risk_ratio:.3f}")
print(f"NNT: {nnt:.1f}")
```

Solution

1. **FALSE.** The p-value does not give the probability that the drug works. It does not tell you the probability of any hypothesis being true or false.

2. **FALSE.** The p-value is not the probability that the result is "due to chance." It is calculated *assuming* chance is the only explanation.
3. **TRUE.** This is the correct definition of a p-value: if the null hypothesis (no difference) were true, we would observe a result this extreme or more extreme approximately 4% of the time.
4. **TRUE (as a point estimate).** The absolute risk reduction is 12% - 9% = 3 percentage points. The NNT = 1/0.03 = 33.3, meaning you need to treat about 34 patients to prevent one death. Whether this is clinically meaningful depends on the drug's cost, side effects, and the patient population.

Exercise 2.3: Multiple Testing Correction

You are analyzing gene expression data and have tested 1,000 genes for differential expression between a treatment and control group. You find 60 genes with $p < 0.05$. Apply both Bonferroni correction and BH-FDR correction. How many genes remain significant under each approach?

4.13. R

```r
# Simulate 1000 p-values: 950 null (uniform), 50 truly differentially expre
set.seed(123)
n_genes <- 1000
n_true <- 50

# Null p-values are uniformly distributed
p_null <- runif(n_genes - n_true, 0, 1)

# True effects: p-values will tend to be small
p_true <- rbeta(n_true, 1, 20)  # skewed toward 0

p_values <- c(p_null, p_true)

# How many nominally significant?
cat("Nominally significant (p < 0.05):", sum(p_values < 0.05), "\n")

# Bonferroni correction
p_bonferroni <- p.adjust(p_values, method = "bonferroni")
cat("Significant after Bonferroni:", sum(p_bonferroni < 0.05), "\n")

# BH-FDR correction
p_fdr <- p.adjust(p_values, method = "BH")
cat("Significant after BH-FDR:", sum(p_fdr < 0.05), "\n")

# Compare
cat("\nBonferroni is much more conservative.\n")
cat("FDR retains more discoveries while controlling the false discovery rat
```

4.14. Python

```
import numpy as np
from statsmodels.stats.multitest import multipletests

np.random.seed(123)
n_genes = 1000
n_true = 50

# Null p-values (uniform) and true effect p-values (small)
p_null = np.random.uniform(0, 1, n_genes - n_true)
p_true = np.random.beta(1, 20, n_true)

p_values = np.concatenate([p_null, p_true])

print(f"Nominally significant (p < 0.05): {np.sum(p_values < 0.05)}")

# Bonferroni correction
reject_bonf, pvals_bonf, _, _ = multipletests(p_values, alpha=0.05, method='bonferroni')
print(f"Significant after Bonferroni: {np.sum(reject_bonf)}")

# BH-FDR correction
reject_fdr, pvals_fdr, _, _ = multipletests(p_values, alpha=0.05, method='fdr_bh')
print(f"Significant after BH-FDR: {np.sum(reject_fdr)}")

print("\nBonferroni is much more conservative.")
print("FDR retains more discoveries while controlling the false discovery rate.")
```

Solution

With the simulated data (seed = 123), you should see approximately:

- **Nominally significant:** ~60-110 genes (includes both true and false positives)
- **Bonferroni:** ~15-30 genes (very conservative, few false positives but many missed true effects)
- **BH-FDR:** ~30-50 genes (moderate, good balance of sensitivity and false positive control)

The Bonferroni correction controls the family-wise error rate (probability of *any* false positives), while BH-FDR controls the expected proportion of false positives among rejections. In genomics, where you expect many true positives, FDR is usually more appropriate because Bonferroni's stringency causes you to miss too many real discoveries.

Exercise 2.4: Power Analysis

You are planning a trial to test whether a lifestyle intervention reduces HbA1c in type 2 diabetes patients. You expect a 0.4% reduction (SD = 1.2%). Compute the required sample size per group for 80% power at alpha = 0.05. Then recompute for 90% power. How does the sample size change?

4.15. R

```r
# Using the power.t.test function
# 80% power
result_80 <- power.t.test(
  delta = 0.4,       # expected difference
  sd = 1.2,          # standard deviation
  sig.level = 0.05,
  power = 0.80,
  type = "two.sample",
  alternative = "two.sided"
)
cat("Sample size per group (80% power):", ceiling(result_80$n), "\n")

# 90% power
result_90 <- power.t.test(
  delta = 0.4,
  sd = 1.2,
  sig.level = 0.05,
  power = 0.90,
  type = "two.sample",
  alternative = "two.sided"
)
cat("Sample size per group (90% power):", ceiling(result_90$n), "\n")

cat("\nIncreasing power from 80% to 90% requires about",
    round((ceiling(result_90$n) / ceiling(result_80$n) - 1) * 100),
    "% more participants per group.\n")
```

4.16. Python

```
from statsmodels.stats.power import TTestIndPower

analysis = TTestIndPower()

# Cohen's d = delta / sd = 0.4 / 1.2
effect_size = 0.4 / 1.2

# 80% power
n_80 = analysis.solve_power(effect_size=effect_size, alpha=0.05, power=0.80
                            alternative='two-sided')
print(f"Sample size per group (80% power): {int(np.ceil(n_80))}")

# 90% power
n_90 = analysis.solve_power(effect_size=effect_size, alpha=0.05, power=0.90
                            alternative='two-sided')
print(f"Sample size per group (90% power): {int(np.ceil(n_90))}")

print(f"\nIncreasing from 80% to 90% power requires about "
      f"{int(round((n_90/n_80 - 1) * 100))}% more participants per group.")
```

Solution

With an effect size of 0.4% and SD of 1.2% (Cohen's d = 0.33):

- **80% power:** approximately 143 per group (286 total)
- **90% power:** approximately 191 per group (382 total)

Going from 80% to 90% power requires roughly 33% more participants. This illustrates the diminishing returns of increasing power – each additional percentage point of power costs more participants. Most trials settle for 80% power as a reasonable compromise.

4.17. Key Takeaways

1. **Point estimates are always uncertain.** Report confidence intervals, not just point estimates.
2. **Confidence intervals describe the precision of your estimate.** They tell you the range of plausible values for the population parameter.
3. **P-values measure compatibility with the null hypothesis.** They are not the probability that the null is true, nor the probability of a chance finding.
4. **Statistical significance is not clinical significance.** Always report and interpret effect sizes.
5. **NNT is the most clinician-friendly effect size** for binary outcomes.
6. **Underpowered studies produce inconclusive results,** not evidence that a treatment is ineffective.
7. **Multiple testing inflates false positives.** Use Bonferroni (conservative) or FDR (less conservative) corrections as appropriate.

4.18. References and Further Reading

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5. Regression: The Workhorse of Clinical Research

5.1. Introduction

If you read clinical research papers, you will encounter regression in nearly every one. It is by far the most commonly used statistical method in medicine. Whether researchers are adjusting for confounders in an observational study, building a risk prediction model, or estimating the effect of a treatment, regression is the tool they reach for.

This chapter covers the foundations: linear regression for continuous outcomes and logistic regression for binary outcomes. We will work through clinical examples in detail, discuss assumptions and diagnostics, and build your intuition for interpreting regression output – a skill you will use throughout your career.

5.2. Why Regression?

Regression serves two distinct (but related) purposes in clinical research:

1. **Estimating adjusted effects.** In an observational study comparing smokers to non-smokers, smokers are also older, more likely to be male, and more likely to drink alcohol. Simple comparisons confound the smoking effect with these other variables. Regression lets you

estimate the effect of smoking *while holding age, sex, and alcohol constant* – what we call "adjusting for confounders."

2. **Predicting outcomes.** A cardiologist wants to estimate a patient's 10-year risk of coronary heart disease (CHD) based on their age, sex, smoking status, blood pressure, and cholesterol. Regression can combine these predictors into a single risk score. The Framingham Risk Score, one of the most widely used tools in medicine, is essentially a logistic regression model.

These two goals require different approaches to model building and evaluation, as we will see in later chapters. For now, we focus on the mechanics.

5.3. Simple Linear Regression

5.3.1. The Setup

Suppose we have data from 150 adults and want to understand the relationship between **age** (in years) and **systolic blood pressure** (SBP, in mmHg). We suspect that blood pressure increases with age.

Simple linear regression fits a straight line through the data:

$$\text{SBP}_i = \beta_0 + \beta_1 \times \text{Age}_i + \epsilon_i$$

where:

- β_0 is the **intercept** – the predicted SBP when Age = 0 (often not directly meaningful)
- β_1 is the **slope** – the predicted change in SBP for each one-year increase in age
- ϵ_i is the **residual** – the difference between the observed and predicted SBP for person i

We assume: $\epsilon_i \sim N(0, \sigma^2)$ – the residuals are normally distributed with mean zero and constant variance.

5.3.2. Fitting the Line

The method of **ordinary least squares (OLS)** finds the values of β_0 and β_1 that minimize the sum of squared residuals:

$$\min_{\beta_0, \beta_1} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

This has a closed-form solution:

$$\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

5.3.3. Interpreting the Output

Suppose our regression yields:

$$\widehat{\text{SBP}} = 90.5 + 0.65 \times \text{Age}$$

Intercept (90.5 mmHg): The predicted SBP when Age = 0. This is a mathematical necessity, not a clinical statement – no one in our study is zero years old. The intercept anchors the line but is rarely interpreted on its own.

Slope (0.65 mmHg/year): For each additional year of age, systolic blood pressure is predicted to increase by 0.65 mmHg, on average. This is the key quantity of interest. A 10-year age difference corresponds to a predicted 6.5 mmHg difference in SBP.

> **!** Important
>
> **Correlation is not causation.** The slope tells you about the *association* between age and blood pressure. It does not mean that aging *causes* blood pressure to rise by exactly 0.65 mmHg per year. There may be confounders (e.g., weight gain, decreased physical activity, dietary changes) that both increase with age and raise blood pressure. To estimate causal effects, you need additional assumptions or study designs (e.g., randomized trials, instrumental variables).

5.3.4. Hypothesis Testing for the Slope

The standard test asks: is β_1 significantly different from zero?

$$H_0 : \beta_1 = 0 \quad \text{(no linear relationship)}$$
$$H_A : \beta_1 \neq 0$$

The test statistic is: $t = \hat{\beta}_1 / SE(\hat{\beta}_1)$

If the output gives $\hat{\beta}_1 = 0.65$, $SE = 0.08$, then $t = 0.65/0.08 = 8.13$, which is highly significant ($p < 0.001$). But remember from Chapter 2: report the confidence interval for β_1 as well. Here: $0.65 \pm 1.96 \times 0.08 = (0.49, 0.81)$. We are quite confident that each year of age is associated with a 0.49 to 0.81 mmHg increase in SBP.

5.4. Multiple Linear Regression

5.4.1. Adding Predictors

In practice, we rarely care about just one predictor. Let us add **sex** (0 = female, 1 = male) and **BMI** (kg/m^2) to our model:

$$\text{SBP}_i = \beta_0 + \beta_1 \times \text{Age}_i + \beta_2 \times \text{Male}_i + \beta_3 \times \text{BMI}_i + \epsilon_i$$

5.4.2. Interpreting "Adjusted" Coefficients

Suppose the output is:

Variable	Coefficient	SE	p-value
Intercept	72.3	6.1	< 0.001
Age	0.58	0.07	< 0.001
Male	4.2	1.8	0.021
BMI	0.92	0.21	< 0.001

Each coefficient is interpreted as the effect of that variable **holding all other variables constant** (also called "adjusting for" or "controlling for" the other variables):

- **Age (0.58):** Each additional year of age is associated with a 0.58 mmHg higher SBP, *after adjusting for sex and BMI*. Notice this is slightly lower than the 0.65 from simple regression – some of the apparent age effect was actually due to BMI increasing with age.
- **Male (4.2):** Males have, on average, 4.2 mmHg higher SBP than females, *after adjusting for age and BMI*.
- **BMI (0.92):** Each 1 kg/m^2 increase in BMI is associated with a 0.92 mmHg higher SBP, *after adjusting for age and sex*.

> **Note**
>
> The phrase "after adjusting for" is critical. It means: if we compare two people who are the same age and sex, but differ by 1 unit of BMI, we predict that the person with the higher BMI has 0.92 mmHg higher

> SBP. This is the power of multiple regression – it lets you isolate the association with one variable while holding others constant.

5.5. Assumptions of Linear Regression

Linear regression relies on four key assumptions, sometimes remembered by the acronym **LINE**:

5.5.1. 1. Linearity

The relationship between each predictor and the outcome must be approximately linear (after accounting for other predictors). A curved relationship will produce biased estimates and poor predictions.

How to check: Plot residuals vs. each predictor, or residuals vs. fitted values. Look for systematic patterns (curves, U-shapes). If you see curvature, consider adding polynomial terms (e.g., Age^2), using splines, or transforming the variable.

5.5.2. 2. Independence

Observations must be independent of each other. This is violated when you have:

- Repeated measures on the same patient
- Patients clustered within hospitals
- Time-series data

Violations do not bias coefficients but severely affect standard errors, p-values, and confidence intervals. If you have non-independent data, you need methods like mixed models or GEE (covered in later chapters).

5.5.3. 3. Normality of Residuals

The residuals (ϵ_i) should be approximately normally distributed. This matters most for small samples. With large samples (n > 30-50 per group), the Central Limit Theorem makes inference robust to non-normality.

How to check: Q-Q plot (residuals plotted against theoretical normal quantiles). Points should fall approximately on a straight line.

5.5.4. 4. Equal Variance (Homoscedasticity)

The spread of residuals should be roughly constant across all levels of the predicted values. If residuals "fan out" as predicted values increase (heteroscedasticity), standard errors are biased.

How to check: Plot residuals vs. fitted values. The spread should be roughly uniform. If it fans out, consider transforming the outcome (e.g., log transformation) or using robust standard errors.

5.5.5. Diagnostic Plots

The standard set of diagnostic plots includes:

1. **Residuals vs. Fitted** – checks linearity and homoscedasticity
2. **Q-Q Plot** – checks normality of residuals
3. **Scale-Location** – another check for homoscedasticity
4. **Residuals vs. Leverage** – identifies influential observations (Cook's distance)

> Practical Advice
>
> No real dataset perfectly satisfies all assumptions. The question is whether violations are severe enough to invalidate your conclusions.

> Linear regression is reasonably robust to moderate violations, especially with large samples. However, extreme non-linearity or strong heteroscedasticity can seriously distort your results.

5.6. Logistic Regression for Binary Outcomes

5.6.1. Why Not Linear Regression for Binary Outcomes?

Many outcomes in medicine are binary: alive/dead, disease/no disease, hospitalized/not. You might think you could code the outcome as 0/1 and use linear regression. But this causes problems:

- Predicted values can fall below 0 or above 1, which are impossible for probabilities
- The residuals are not normally distributed
- The variance is not constant (it depends on the predicted probability)

Logistic regression solves these problems by modeling the **log-odds** (logit) of the outcome:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots$$

where $p = P(Y = 1)$ is the probability of the event.

5.6.2. The Logit Link

The **logit** function transforms a probability (bounded between 0 and 1) into a value that can range from $-\infty$ to $+\infty$:

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$$

The inverse (the **logistic function**) converts back:

$$p = \frac{e^{\beta_0+\beta_1 x_1+\dots}}{1+e^{\beta_0+\beta_1 x_1+\dots}} = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\dots)}}$$

This ensures predicted probabilities always fall between 0 and 1.

5.6.3. Interpreting Coefficients as Odds Ratios

In logistic regression, the coefficients are on the log-odds scale, which is not intuitive. We exponentiate them to get **odds ratios (OR)**:

$$OR = e^{\beta}$$

5.6.4. Clinical Example: Predicting 10-year CHD Risk

Let us build a model predicting 10-year coronary heart disease (CHD) risk from age, sex, smoking status, total cholesterol, and systolic blood pressure. This is similar to the Framingham Risk Score.

Suppose our logistic regression output is:

Variable	Coefficient (β)	SE	OR (e^{β})	95% CI for OR	p-value
Intercept	-8.45	0.92	–	–	< 0.001
Age (per year)	0.065	0.012	1.067	(1.042, 1.093)	< 0.001

Variable	Coefficient (β)	SE	OR (e^{β})	95% CI for OR	p-value
Male	0.72	0.19	2.054	(1.414, 2.985)	< 0.001
Current smoker	0.58	0.21	1.786	(1.182, 2.698)	0.006
Total cholesterol (per 10 mg/dL)	0.11	0.04	1.116	(1.032, 1.207)	0.006
SBP (per 10 mmHg)	0.18	0.05	1.197	(1.085, 1.321)	< 0.001

Interpretation:

- **Age (OR = 1.067):** Each additional year of age is associated with 6.7% higher odds of CHD, adjusted for sex, smoking, cholesterol, and SBP.
- **Male (OR = 2.054):** Males have approximately twice the odds of developing CHD compared to females, after adjustment.
- **Current smoker (OR = 1.786):** Current smokers have about 79% higher odds of CHD compared to non-smokers, after adjustment.
- **Total cholesterol (OR = 1.116 per 10 mg/dL):** Each 10 mg/dL increase in total cholesterol is associated with 11.6% higher odds of CHD.
- **SBP (OR = 1.197 per 10 mmHg):** Each 10 mmHg increase in SBP is associated with about 20% higher odds of CHD.

5.6.5. Predicted Probabilities

You can use the model to compute individual risk. For a 55-year-old male smoker with cholesterol of 240 mg/dL and SBP of 150 mmHg:

$$\text{logit}(p) = -8.45 + 0.065(55) + 0.72(1) + 0.58(1) + 0.11(24) + 0.18(15)$$

$$= -8.45 + 3.575 + 0.72 + 0.58 + 2.64 + 2.70 = 1.765$$

$$p = \frac{e^{1.765}}{1 + e^{1.765}} = \frac{5.84}{6.84} = 0.854$$

Wait – that is an 85.4% predicted probability of CHD, which seems too high. This is because our coefficients are illustrative; real Framingham coefficients would produce more calibrated predictions. The important point is the *method*: plug in patient values, compute the linear predictor, and convert to a probability using the logistic function.

> **Note**
>
> **Odds ratios vs. risk ratios:** Remember from Chapter 2 that odds ratios exaggerate the effect when the outcome is common. If the baseline risk is 5%, an OR of 2.0 corresponds to a risk ratio of approximately 1.9. But if the baseline risk is 40%, an OR of 2.0 corresponds to a risk ratio of only about 1.4. In logistic regression, you always get ORs; converting to risk ratios requires additional steps. For common outcomes, consider using log-binomial regression or Poisson regression with robust standard errors to obtain risk ratios directly.

5.7. Model Fit

5.7.1. R-squared (Linear Regression)

R^2 is the proportion of variance in the outcome explained by the predictors:

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

- $R^2 = 0$: the model explains none of the variance (no better than the mean)
- $R^2 = 1$: the model explains all the variance (perfect prediction)

In clinical research, R^2 values are often modest (0.10 - 0.40) because human biology is variable. An R^2 of 0.25 means your model explains 25% of the variance in blood pressure – the remaining 75% is due to factors not in your model, measurement error, and inherent biological variability.

Adjusted R^2 penalizes for the number of predictors, preventing the illusion of improvement from adding noise variables:

$$R^2_{\text{adj}} = 1 - \frac{(1 - R^2)(n-1)}{n-k-1}$$

where k is the number of predictors. Always report adjusted R^2 when comparing models with different numbers of predictors.

5.7.2. AIC (Akaike Information Criterion)

AIC balances model fit against complexity:

$$AIC = -2\log(L) + 2k$$

where L is the likelihood and k is the number of parameters. Lower AIC is better. AIC is useful for comparing non-nested models (models that are not subsets of each other). Unlike R^2, AIC has no absolute interpretation – it is only meaningful relative to other models fit to the same data.

5.7.3. Deviance (Logistic Regression)

In logistic regression, we do not have a traditional R^2. Instead, we use **deviance**, which plays a similar role to the sum of squares in linear regression:

$$\text{Deviance} = -2\log(L)$$

A **likelihood ratio test** compares the deviance of your model to a null model (intercept only). Pseudo-R^2 measures (e.g., Nagelkerke, McFadden) attempt to provide an R^2-like summary, but they do not have the same clean interpretation as linear R^2.

5.8. Categorical Predictors and Dummy Variables

5.8.1. The Problem

Regression requires numeric predictors. How do you include a variable like **race** (White, Black, Hispanic, Asian, Other)?

5.8.2. Dummy (Indicator) Variables

Create $k-1$ binary variables for a categorical predictor with k levels. One level is chosen as the **reference category**:

Patient	White	Black	Hispanic	Asian
White patient	1	0	0	0
Black patient	0	1	0	0
Hispanic patient	0	0	1	0
Asian patient	0	0	0	1
Other patient	0	0	0	0

Here, "Other" is the reference category (all dummies = 0). Each coefficient represents the difference between that group and the reference group.

> ⚠ Warning
>
> The choice of reference category changes the coefficients but NOT the model's predictions or overall fit. It only changes which comparisons are directly reported. Choose a reference category that makes clinical sense – often the most common group or the control/standard-of-care group.

5.8.3. Interpreting Dummy Variable Coefficients

If the coefficient for Black (vs. Other) is 3.8, it means: "On average, Black patients have 3.8 mmHg higher SBP than Other patients, holding all other variables constant."

Most statistical software handles this automatically. In R, factors are automatically converted to dummy variables. In Python (statsmodels or scikit-learn), you may need to create them explicitly or use formula notation.

5.9. Interaction Terms

5.9.1. When Effects Depend on Each Other

An **interaction** occurs when the effect of one predictor depends on the level of another predictor. For example:

- The effect of a drug might differ between men and women
- The effect of age on blood pressure might be steeper for smokers than non-smokers
- The effect of exercise on weight loss might depend on baseline BMI

5.9.2. Specifying an Interaction

To test whether the age-SBP relationship differs by sex:

$$\text{SBP} = \beta_0 + \beta_1\text{Age} + \beta_2\text{Male} + \beta_3(\text{Age} \times \text{Male}) + \epsilon$$

The interaction term β_3 captures *how much the slope of Age differs for males vs. females.*

5.9.3. Interpreting Interaction Terms

Suppose we get:

Variable	Coefficient
Intercept	85.0
Age	0.50
Male	-5.0
Age x Male	0.20

For **females** (Male = 0):

$$\text{SBP} = 85.0 + 0.50 \times \text{Age}$$

For **males** (Male = 1):

$$\begin{aligned}\text{SBP} &= 85.0 + 0.50 \times \text{Age} + (-5.0)(1) + 0.20 \times \text{Age}(1) \\ &= 80.0 + 0.70 \times \text{Age}\end{aligned}$$

So the age slope is 0.50 mmHg/year for females but 0.70 mmHg/year for males. The interaction coefficient (0.20) is the *difference in slopes*. Males' blood pressure increases more steeply with age than females'.

> **!** Important
>
> **Main effects rule:** When you include an interaction term, you must also include both "main effects" (the individual terms). Never include Age × Male without also including Age and Male separately. Dropping main effects changes the interpretation of the interaction term in misleading ways.

> When to Test for Interactions
>
> Do not test every possible interaction – this leads to multiple testing problems and overfitting. Only test interactions that are:
>
> 1. **Pre-specified** based on clinical reasoning (e.g., "We hypothesize the drug works differently in men vs. women")
> 2. **Scientifically plausible** (there is a biological mechanism for effect modification)
> 3. **Adequately powered** (interaction tests require much larger samples than main effect tests – roughly 4x the sample size)

5.10. Exercises

Exercise 3.1: Simple Linear Regression

Using simulated clinical data, fit a simple linear regression of systolic blood pressure on age. Interpret the slope and intercept. Create a scatterplot with the regression line.

5.11. R

```r
# Simulate clinical data
set.seed(42)
n <- 150
age <- round(runif(n, 30, 75))
sbp <- 85 + 0.6 * age + rnorm(n, 0, 12)

clinical_data <- data.frame(age = age, sbp = sbp)

# Fit the model
model <- lm(sbp ~ age, data = clinical_data)
summary(model)

# Interpret
cat("\nIntercept:", coef(model)[1], "mmHg\n")
cat("Slope:", coef(model)[2], "mmHg per year of age\n")
cat("R-squared:", summary(model)$r.squared, "\n")

# Confidence interval for the slope
confint(model, "age", level = 0.95)

# Scatterplot with regression line
library(ggplot2)
ggplot(clinical_data, aes(x = age, y = sbp)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = TRUE, color = "blue") +
  labs(x = "Age (years)", y = "Systolic Blood Pressure (mmHg)",
       title = "Age vs. Systolic Blood Pressure") +
  theme_minimal()
```

5.12. Python

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
import matplotlib.pyplot as plt

np.random.seed(42)
n = 150
age = np.random.uniform(30, 75, n).round()
sbp = 85 + 0.6 * age + np.random.normal(0, 12, n)

clinical_data = pd.DataFrame({'age': age, 'sbp': sbp})

# Fit the model
X = sm.add_constant(clinical_data['age'])
model = sm.OLS(clinical_data['sbp'], X).fit()
print(model.summary())

print(f"\nIntercept: {model.params['const']:.2f} mmHg")
print(f"Slope: {model.params['age']:.3f} mmHg per year of age")
print(f"R-squared: {model.rsquared:.3f}")
print(f"95% CI for slope: ({model.conf_int().loc['age', 0]:.3f}, "
      f"{model.conf_int().loc['age', 1]:.3f})")

# Scatterplot with regression line
fig, ax = plt.subplots(figsize=(8, 6))
ax.scatter(age, sbp, alpha=0.5)
age_range = np.linspace(30, 75, 100)
predicted = model.params['const'] + model.params['age'] * age_range
ax.plot(age_range, predicted, 'b-', linewidth=2, label='Regression line')
ax.set_xlabel('Age (years)')
ax.set_ylabel('Systolic Blood Pressure (mmHg)')
ax.set_title('Age vs. Systolic Blood Pressure')
ax.legend()
plt.tight_layout()
plt.show()
```

Solution

You should find a slope of approximately 0.6 mmHg/year (close to the true value used in simulation), an intercept around 85, and R^2 around 0.40-0.50. The interpretation: each additional year of age is associated with about 0.6 mmHg higher systolic blood pressure. The intercept (approximately 85) represents the predicted SBP at age 0, which is an extrapolation and not clinically meaningful. The 95% CI for the slope should be entirely positive, confirming the association.

Exercise 3.2: Multiple Regression and Confounding

Add BMI and sex to the model from Exercise 3.1. Compare the age coefficient before and after adjustment. Does it change? What does this tell you about confounding?

5.13. R

```r
set.seed(42)
n <- 150
age <- round(runif(n, 30, 75))
sex <- rbinom(n, 1, 0.5)  # 1 = male
# BMI increases slightly with age (confounding)
bmi <- 22 + 0.08 * age + 2 * sex + rnorm(n, 0, 3)
# SBP depends on age, sex, AND BMI
sbp <- 70 + 0.45 * age + 4 * sex + 1.0 * bmi + rnorm(n, 0, 10)

df <- data.frame(age = age, sex = factor(sex, labels = c("Female", "Male"))
                 bmi = bmi, sbp = sbp)

# Unadjusted model (age only)
model_unadj <- lm(sbp ~ age, data = df)
cat("=== Unadjusted Model ===\n")
cat("Age coefficient:", coef(model_unadj)["age"], "\n")
cat("R-squared:", summary(model_unadj)$r.squared, "\n\n")

# Adjusted model (age + sex + BMI)
model_adj <- lm(sbp ~ age + sex + bmi, data = df)
cat("=== Adjusted Model ===\n")
summary(model_adj)
cat("\nAge coefficient (unadjusted):", round(coef(model_unadj)["age"], 3),
cat("Age coefficient (adjusted):", round(coef(model_adj)["age"], 3), "\n")
cat("Change:", round(coef(model_unadj)["age"] - coef(model_adj)["age"], 3),
```

5.14. Python

```python
import numpy as np
import pandas as pd
import statsmodels.api as sm
import statsmodels.formula.api as smf

np.random.seed(42)
n = 150
age = np.random.uniform(30, 75, n).round()
sex = np.random.binomial(1, 0.5, n)
# BMI increases slightly with age (confounding)
bmi = 22 + 0.08 * age + 2 * sex + np.random.normal(0, 3, n)
# SBP depends on age, sex, AND BMI
sbp = 70 + 0.45 * age + 4 * sex + 1.0 * bmi + np.random.normal(0, 10, n)

df = pd.DataFrame({'age': age, 'sex': sex, 'bmi': bmi, 'sbp': sbp})

# Unadjusted model
model_unadj = smf.ols('sbp ~ age', data=df).fit()
print("=== Unadjusted Model ===")
print(f"Age coefficient: {model_unadj.params['age']:.3f}")
print(f"R-squared: {model_unadj.rsquared:.3f}\n")

# Adjusted model
model_adj = smf.ols('sbp ~ age + C(sex) + bmi', data=df).fit()
print("=== Adjusted Model ===")
print(model_adj.summary())
print(f"\nAge coefficient (unadjusted): {model_unadj.params['age']:.3f}")
print(f"Age coefficient (adjusted): {model_adj.params['age']:.3f}")
print(f"Change: {model_unadj.params['age'] - model_adj.params['age']:.3f}")
```

Solution

The unadjusted age coefficient should be notably larger than the adjusted coefficient. In the simulation, the true age effect is 0.45, but the unadjusted estimate is inflated because BMI increases with age (confounding path: Age -> BMI -> SBP). After adjusting for BMI and sex, the age coefficient moves closer to the true value of 0.45. This demonstrates **confounding**: the unadjusted age coefficient captured not only the direct effect of age but also the indirect effect through BMI. Adjusted R^2 should also be substantially higher in the multiple regression model.

Exercise 3.3: Logistic Regression

Using simulated data, fit a logistic regression predicting 10-year CHD risk from age, sex, smoking, and cholesterol. Compute and interpret the odds ratios.

5.15. R

```r
set.seed(42)
n <- 500

age <- round(runif(n, 35, 75))
male <- rbinom(n, 1, 0.5)
smoker <- rbinom(n, 1, 0.25)
cholesterol <- round(rnorm(n, 220, 40))

# Generate CHD outcome (binary) based on logistic model
log_odds <- -7 + 0.06 * age + 0.5 * male + 0.4 * smoker + 0.008 * cholesterol
prob_chd <- plogis(log_odds)  # inverse logit
chd <- rbinom(n, 1, prob_chd)

df <- data.frame(age, male = factor(male), smoker = factor(smoker),
                 cholesterol, chd)

cat("CHD prevalence:", mean(chd), "\n\n")

# Fit logistic regression
model <- glm(chd ~ age + male + smoker + cholesterol,
             data = df, family = binomial)
summary(model)

# Odds ratios with 95% CI
or_table <- data.frame(
  OR = exp(coef(model)),
  Lower = exp(confint(model))[, 1],
  Upper = exp(confint(model))[, 2]
)
cat("\nOdds Ratios:\n")
print(round(or_table[-1, ], 3))  # Exclude intercept

# Predicted probability for a specific patient
new_patient <- data.frame(age = 60, male = factor(1), smoker = factor(1),
                          cholesterol = 260)
pred_prob <- predict(model, newdata = new_patient, type = "response")
cat("\nPredicted CHD probability for 60-year-old male smoker with",
    "cholesterol 260:", round(pred_prob, 3), "\n")
```

5.16. Python

```
import numpy as np
import pandas as pd
import statsmodels.formula.api as smf
from scipy.special import expit

np.random.seed(42)
n = 500

age = np.random.uniform(35, 75, n).round()
male = np.random.binomial(1, 0.5, n)
smoker = np.random.binomial(1, 0.25, n)
cholesterol = np.random.normal(220, 40, n).round()

# Generate CHD outcome
log_odds = -7 + 0.06 * age + 0.5 * male + 0.4 * smoker + 0.008 * cholesterol
prob_chd = expit(log_odds)
chd = np.random.binomial(1, prob_chd)

df = pd.DataFrame({
    'age': age, 'male': male, 'smoker': smoker,
    'cholesterol': cholesterol, 'chd': chd
})

print(f"CHD prevalence: {chd.mean():.3f}\n")

# Fit logistic regression
model = smf.logit('chd ~ age + male + smoker + cholesterol', data=df).fit()
print(model.summary())

# Odds ratios with 95% CI
params = model.params[1:]  # exclude intercept
conf = model.conf_int().iloc[1:]
or_table = pd.DataFrame({
    'OR': np.exp(params),
    'Lower 95% CI': np.exp(conf[0]),
    'Upper 95% CI': np.exp(conf[1])
})
print("\nOdds Ratios:")
print(or_table.round(3))

# Predicted probability for a specific patient
new_patient = pd.DataFrame({
    'age': [60], 'male': [1], 'smoker': [1], 'cholesterol': [260]
})
pred_prob = model.predict(new_patient)
print(f"\nPredicted CHD probability for 60yo male smoker, chol=260: {pred_prob.values[0]:.
```

Solution

The logistic regression should recover coefficients close to the true values used in simulation (age: 0.06, male: 0.5, smoker: 0.4, cholesterol: 0.008). The odds ratios should be approximately:

- Age (per year): OR around 1.06 – each year of age increases CHD odds by about 6%
- Male: OR around 1.65 – males have about 65% higher odds of CHD
- Smoker: OR around 1.50 – smokers have about 50% higher odds of CHD
- Cholesterol (per mg/dL): OR around 1.008 – each 1 mg/dL increase in cholesterol increases CHD odds by 0.8% (or about 8% per 10 mg/dL)

The predicted probability for a high-risk patient (60-year-old male smoker, cholesterol 260) should be noticeably higher than the population average.

Exercise 3.4: Regression Diagnostics

Using the linear regression model from Exercise 3.1, create diagnostic plots to assess model assumptions. Identify any potential violations.

5.17. R

```r
set.seed(42)
n <- 150
age <- round(runif(n, 30, 75))
sbp <- 85 + 0.6 * age + rnorm(n, 0, 12)
df <- data.frame(age = age, sbp = sbp)

model <- lm(sbp ~ age, data = df)

# Standard diagnostic plots (2x2)
par(mfrow = c(2, 2))
plot(model)
par(mfrow = c(1, 1))

# Or using ggplot2 for prettier diagnostics
library(ggplot2)
library(patchwork)

diag_data <- data.frame(
  fitted = fitted(model),
  residuals = resid(model),
  std_residuals = rstandard(model)
)

p1 <- ggplot(diag_data, aes(x = fitted, y = residuals)) +
  geom_point(alpha = 0.5) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  geom_smooth(se = FALSE, color = "blue") +
  labs(title = "Residuals vs Fitted", x = "Fitted Values", y = "Residuals") +
  theme_minimal()

p2 <- ggplot(diag_data, aes(sample = std_residuals)) +
  stat_qq() +
  stat_qq_line(color = "red") +
  labs(title = "Normal Q-Q Plot", x = "Theoretical Quantiles",
       y = "Standardized Residuals") +
  theme_minimal()

p1 + p2
```

5.18. Python

```python
import numpy as np
import pandas as pd
import statsmodels.api as sm
import statsmodels.formula.api as smf
import matplotlib.pyplot as plt
from scipy import stats

np.random.seed(42)
n = 150
age = np.random.uniform(30, 75, n).round()
sbp = 85 + 0.6 * age + np.random.normal(0, 12, n)
df = pd.DataFrame({'age': age, 'sbp': sbp})

model = smf.ols('sbp ~ age', data=df).fit()

fig, axes = plt.subplots(2, 2, figsize=(12, 10))

# 1. Residuals vs Fitted
axes[0, 0].scatter(model.fittedvalues, model.resid, alpha=0.5)
axes[0, 0].axhline(y=0, color='red', linestyle='--')
axes[0, 0].set_xlabel('Fitted Values')
axes[0, 0].set_ylabel('Residuals')
axes[0, 0].set_title('Residuals vs Fitted')

# 2. Q-Q Plot
stats.probplot(model.resid, dist="norm", plot=axes[0, 1])
axes[0, 1].set_title('Normal Q-Q Plot')

# 3. Scale-Location
std_resid = np.sqrt(np.abs(model.get_influence().resid_studentized_internal))
axes[1, 0].scatter(model.fittedvalues, std_resid, alpha=0.5)
axes[1, 0].set_xlabel('Fitted Values')
axes[1, 0].set_ylabel('sqrt(|Standardized Residuals|)')
axes[1, 0].set_title('Scale-Location')

# 4. Residuals vs Leverage
sm.graphics.influence_plot(model, ax=axes[1, 1], criterion="cooks")
axes[1, 1].set_title('Residuals vs Leverage')

plt.tight_layout()
plt.show()

# Formal test for normality
shapiro_stat, shapiro_p = stats.shapiro(model.resid)
print(f"Shapiro-Wilk test for normality: W={shapiro_stat:.4f}, p={shapiro_p:.4f}")
```

Solution

Since the data were simulated from a correctly specified linear model with normal errors, the diagnostic plots should look well-behaved:

1. **Residuals vs. Fitted:** Points should be randomly scattered around zero with no systematic pattern. No evidence of non-linearity or heteroscedasticity.
2. **Q-Q Plot:** Points should fall approximately along the diagonal line, confirming normality.
3. **Scale-Location:** The spread of residuals should be roughly constant across fitted values.
4. **Residuals vs. Leverage:** No points should have both high leverage and large residuals (high Cook's distance).

In real clinical data, you will rarely see such clean diagnostics. Common violations include: non-linearity (curved pattern in residuals vs. fitted), heteroscedasticity (fanning out of residuals), and heavy tails (deviations at the ends of the Q-Q plot).

Exercise 3.5: Interaction Terms

Test whether the effect of age on SBP differs between males and females by adding an interaction term. Interpret the interaction coefficient and create a visualization.

5.19. R

```r
set.seed(42)
n <- 300
age <- round(runif(n, 30, 75))
sex <- factor(rbinom(n, 1, 0.5), labels = c("Female", "Male"))

# True interaction: steeper age slope for males
sbp <- 80 + 0.45 * age +
       ifelse(sex == "Male", -8 + 0.25 * age, 0) +
       rnorm(n, 0, 10)

df <- data.frame(age = age, sex = sex, sbp = sbp)

# Model WITHOUT interaction
model_no_int <- lm(sbp ~ age + sex, data = df)
cat("=== Model without interaction ===\n")
summary(model_no_int)

# Model WITH interaction
model_int <- lm(sbp ~ age * sex, data = df)
cat("\n=== Model with interaction ===\n")
summary(model_int)

# Is the interaction significant?
anova(model_no_int, model_int)

# Visualization
library(ggplot2)
ggplot(df, aes(x = age, y = sbp, color = sex)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(x = "Age (years)", y = "Systolic Blood Pressure (mmHg)",
       title = "Age-SBP Relationship by Sex",
       subtitle = "Steeper slope for males indicates an interaction") +
  theme_minimal() +
  scale_color_manual(values = c("Female" = "coral", "Male" = "steelblue"))
```

5.20. Python

```python
import numpy as np
import pandas as pd
import statsmodels.formula.api as smf
import matplotlib.pyplot as plt

np.random.seed(42)
n = 300
age = np.random.uniform(30, 75, n).round()
sex = np.random.binomial(1, 0.5, n)  # 1 = male

# True interaction: steeper age slope for males
sbp = 80 + 0.45 * age + (-8 + 0.25 * age) * sex + np.random.normal(0, 10, n)

df = pd.DataFrame({'age': age, 'sex': sex, 'sbp': sbp})

# Model WITHOUT interaction
model_no_int = smf.ols('sbp ~ age + C(sex)', data=df).fit()
print("=== Model without interaction ===")
print(model_no_int.summary())

# Model WITH interaction
model_int = smf.ols('sbp ~ age * C(sex)', data=df).fit()
print("\n=== Model with interaction ===")
print(model_int.summary())

# Compare models (likelihood ratio test)
from scipy.stats import chi2
lr_stat = -2 * (model_no_int.llf - model_int.llf)
lr_p = chi2.sf(lr_stat, df=1)
print(f"\nLikelihood ratio test: chi2={lr_stat:.2f}, p={lr_p:.4f}")

# Visualization
fig, ax = plt.subplots(figsize=(8, 6))
for s, label, color in [(0, 'Female', 'coral'), (1, 'Male', 'steelblue')]:
    mask = df['sex'] == s
    ax.scatter(df.loc[mask, 'age'], df.loc[mask, 'sbp'],
               alpha=0.3, color=color, label=label)
    age_range = np.linspace(30, 75, 100)
    pred = model_int.params['Intercept'] + model_int.params['age'] * age_range
    if s == 1:
        pred += model_int.params['C(sex)[T.1]'] + model_int.params['age:C(sex)[T.1]'] * ag
    ax.plot(age_range, pred, color=color, linewidth=2)

ax.set_xlabel('Age (years)')
ax.set_ylabel('Systolic Blood Pressure (mmHg)')
ax.set_title('Age-SBP Relationship by Sex')
ax.legend()
plt.tight_layout()
plt.show()
```

Solution

The interaction term should be statistically significant and close to the true value of 0.25. Interpretation:

- **Female age slope:** approximately 0.45 mmHg/year (the main effect of age)
- **Male age slope:** approximately 0.45 + 0.25 = 0.70 mmHg/year (main effect + interaction)
- **Interaction coefficient (~0.25):** Males' blood pressure increases 0.25 mmHg/year *faster* than females' blood pressure

The visualization should show two regression lines with noticeably different slopes – the male line is steeper. The lines may cross at some point, which reflects the negative main effect of male sex combined with the positive interaction.

This has clinical implications: if the age-SBP relationship truly differs by sex, then risk prediction models should account for this interaction rather than assuming a uniform age effect.

5.21. Key Takeaways

1. **Simple linear regression** fits a line; the slope is the predicted change in the outcome per unit increase in the predictor.
2. **Multiple regression coefficients** are "adjusted" – they represent the effect of one variable holding others constant.
3. **Always check assumptions** with diagnostic plots. Violations can invalidate your inferences.
4. **Logistic regression** is for binary outcomes. Exponentiate coefficients to get odds ratios.

5. **Categorical predictors** require dummy variables. The coefficients represent differences from the reference category.
6. **Interaction terms** allow effects to differ across subgroups. Only test interactions that are pre-specified and scientifically plausible.
7. **Model fit statistics** (R^2, AIC, deviance) help you evaluate and compare models, but no single metric tells the whole story.

5.22. References and Further Reading

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Part II.

Advanced Statistical Methods

6. Modelling Non-Linear Relationships: Splines, Fractional Polynomials, and Why Categorisation Fails

> **Learning Objectives**
>
> By the end of this chapter, you will be able to:
>
> 1. Explain why categorising continuous variables leads to information loss and biased estimates
> 2. Recognise when a linear term is insufficient for modelling a continuous predictor
> 3. Describe fractional polynomials and their role in flexible modelling
> 4. Understand how splines work, including the concepts of knots, cubic splines, and restricted cubic splines
> 5. Implement and interpret restricted cubic spline models in both R and Python
> 6. Compare linear, categorical, and spline models using appropriate fit statistics
> 7. Apply the practical recommendations from Lopez-Ayala et al. (2025) in your own analyses

> **What is different about this chapter**
>
> This chapter introduces tools you probably haven't seen before. The pace is faster than the previous chapters, and the concepts (splines, knot placement) may feel unfamiliar. Budget extra time here.
> **Why it matters for your research:** If you build a prediction model or run a regression without properly handling continuous variables, your results will be less accurate — or wrong. Systematic reviews show that most published clinical prediction models categorise continuous variables inappropriately. This chapter teaches you the right way.

6.1. Introduction

Imagine you are studying the relationship between body mass index (BMI) and mortality. You have data on thousands of patients, and you need to build a regression model. What do you do with BMI?

If you are like most researchers in clinical medicine, you probably split BMI into categories: underweight (<18.5), normal (18.5–24.9), overweight (25–29.9), and obese (30+). You then include these categories as dummy variables in your regression model. This approach feels natural — it mirrors how we think clinically, and the results are easy to present in a table.

This is almost always the wrong approach.

Categorising continuous variables is one of the most common and most damaging analytic practices in clinical research. It throws away information, introduces bias, reduces statistical power, and can lead to completely wrong conclusions about the shape of relationships between predictors and outcomes.

This chapter draws heavily on a landmark tutorial by Lopez-Ayala et al. (2025), published in the *BMJ* in 2025, which provides a comprehensive guide to modelling continuous variables correctly. We will follow their framework closely, supplementing it with additional clinical examples and implementation guidance.

6.2. The Problem with Categorising Continuous Variables

6.2.1. What Happens When You Categorise

When you convert a continuous variable into categories, you make three implicit assumptions, all of which are almost certainly wrong:

1. **Within each category, the relationship is flat.** A patient with a BMI of 18.6 is treated identically to a patient with a BMI of 24.8 — despite a difference of over 6 kg/m^2.
2. **At the category boundary, there is a sudden jump.** A patient with a BMI of 24.9 is treated as fundamentally different from a patient with a BMI of 25.0 — despite a difference of only 0.1 kg/m^2.
3. **The cut-points are meaningful.** The chosen boundaries (often based on clinical convention or percentiles) capture the true biology of the relationship.

None of these assumptions hold for most biological relationships. Biology is smooth, not stepped.

6.2.2. The Consequences Are Serious

Lopez-Ayala et al. (2025) enumerate several consequences of categorisation:

- **Loss of statistical power.** Categorising a continuous variable into two groups is equivalent to discarding roughly one-third of your data (Royston et al. 2006). Even with more categories, substantial information is lost.

- **Biased effect estimates.** Because the relationship within each category is forced to be flat, the estimated effects are systematically wrong. The direction and magnitude of the bias depend on where you place the cut-points.

- **Residual confounding.** When a categorised variable is used as a confounder in an adjusted model, the adjustment is incomplete. The true confounding effect varies smoothly, but your model only adjusts in crude steps. This is sometimes called *residual confounding due to categorisation*.

- **Arbitrary cut-points change conclusions.** Different choices of cut-points can lead to different — even contradictory — conclusions from the same data. This is a form of researcher degrees of freedom that inflates false positive rates.

- **Loss of predictive accuracy.** Prediction models that categorise continuous predictors perform worse than those that model them appropriately.

> ⚠ A Common But Flawed Justification
>
> Researchers often justify categorisation by saying "it's easier to interpret" or "clinicians think in categories." While clinical decision-making does involve thresholds (e.g., treat if blood pressure > 140 mmHg), the *modelling* of relationships should reflect reality, not convenience. You can always convert a continuous model's output into clinically actionable categories *after* modelling — but you cannot recover the information lost by categorising *before* modelling.

6.2.3. A Visual Demonstration

Consider the true relationship between a biomarker (say, C-reactive protein) and the log-odds of a diagnosis. Suppose the true relationship is U-shaped: both very low and very high values are associated with increased risk.

If you categorise CRP into tertiles (low, medium, high), you will estimate a *monotonically increasing* relationship — completely missing the U-shape. The low-risk group is contaminated by some high-risk individuals at the low end, and the comparison between groups obscures the smooth curve.

```
library(ggplot2)

set.seed(42)
x <- seq(0, 10, length.out = 300)
# True U-shaped relationship
y_true <- 0.3 * (x - 5)^2 - 2

df <- data.frame(x = x, y_true = y_true)

# Categorise into tertiles
df$category <- cut(df$x, breaks = quantile(df$x, probs = c(0, 1/3, 2/3, 1)),
                   include.lowest = TRUE, labels = c("Low", "Medium", "High"))
cat_means <- aggregate(y_true ~ category, data = df, FUN = mean)

df$y_cat <- cat_means$y_true[match(df$category, cat_means$category)]

ggplot(df, aes(x = x)) +
  geom_line(aes(y = y_true), linewidth = 1.2, colour = "#2c3e50") +
  geom_step(aes(y = y_cat), linewidth = 1, colour = "#e74c3c", linetype = "dashed") +
  labs(x = "Biomarker Level", y = "Log-Odds of Outcome",
       subtitle = "Black curve = true relationship; Red steps = categorised estimate") +
  theme_minimal(base_size = 14) +
  theme(plot.subtitle = element_text(size = 11))
```

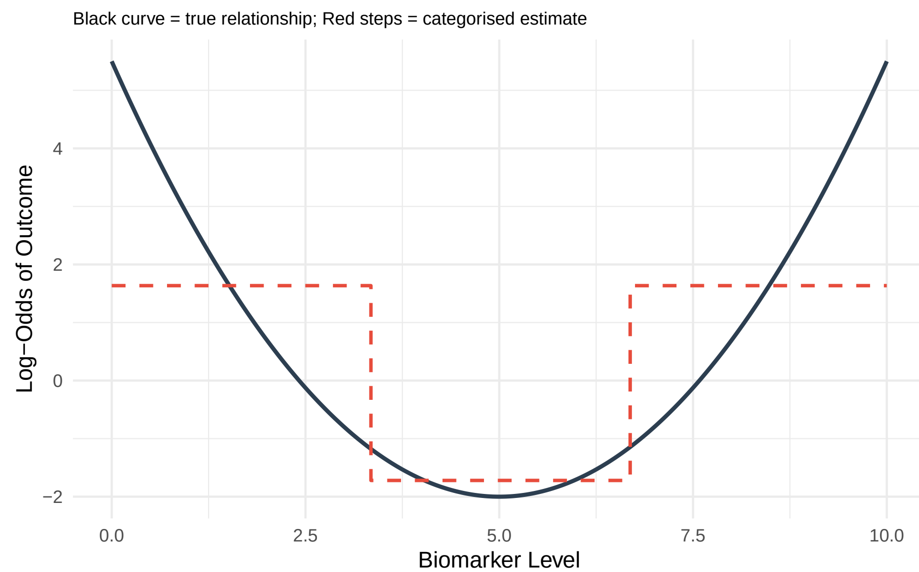


Figure 6.1.: True non-linear relationship (solid curve) versus what categorisation into tertiles recovers (stepped line). Categorisation misses the U-shape entirely.

6.3. When Linear Terms Are Not Enough

6.3.1. The Linearity Assumption

In standard logistic or linear regression, including a continuous predictor x as a simple linear term assumes:

$$\text{logit}(p) = \beta_0 + \beta_1 x$$

This means that each one-unit increase in x is associated with the same change in the log-odds, regardless of where on the x scale you start. For

BMI, this would mean that going from 20 to 21 has the same effect as going from 35 to 36.

For many clinical variables, this assumption is **wrong**:

- **BMI and mortality**: U-shaped or J-shaped. Both very low and very high BMI are associated with increased mortality.
- **Alcohol consumption and cardiovascular risk**: J-shaped. Moderate consumption may be associated with lower risk than either abstinence or heavy drinking (though this remains debated).
- **Age and many outcomes**: Often non-linear, with steeper increases at older ages.
- **Blood pressure and stroke risk**: The relationship steepens at higher pressures.
- **CSF glucose and bacterial meningitis**: A threshold effect where low values are strongly predictive but the relationship flattens at higher values (we will explore this in detail below).

6.3.2. Detecting Non-Linearity

Before choosing a modelling strategy, it helps to assess whether non-linearity is present:

1. **Visual inspection.** Plot the predictor against the outcome using a loess (locally estimated scatterplot smoothing) curve or a generalised additive model (GAM) smooth. If the relationship is not a straight line, you need a non-linear term.

2. **Likelihood ratio test.** Fit a model with a linear term and a model with a non-linear term (e.g., a spline). Compare them using a likelihood ratio test. A significant result indicates that the non-linear model fits substantially better.

3. **AIC/BIC comparison.** The Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) are single-number summaries that balance *how well a model fits* against *how many parameters it uses*. Each starts from the model's log-likelihood (a measure of fit) and then adds a penalty for every extra parameter, so a more complex model is only rewarded if it improves the fit by more than its added complexity "costs." This guards against the temptation to keep adding flexibility just to chase the data. **Lower values indicate a better fit-versus-complexity balance**, so you compare the linear and non-linear models and prefer the one with the smaller AIC (or BIC). BIC penalises extra parameters more heavily than AIC, so it tends to favour simpler models, especially in large samples. The values are only meaningful *relative to each other* on the same dataset — an AIC of 480 is not "good" or "bad" in isolation, it is only lower or higher than a competing model's.

4. **Residual plots.** Plot residuals against the predictor. A systematic pattern (e.g., a curve) suggests non-linearity.

> Practical Rule
>
> As Lopez-Ayala et al. (2025) recommend: when in doubt, use a non-linear term. The cost of modelling a truly linear relationship with a spline is minimal (you lose very little power), but the cost of forcing a non-linear relationship into a linear term can be severe (biased estimates, wrong conclusions).

6.4. Fractional Polynomials

6.4.1. The Idea

Fractional polynomials (FPs), developed by Royston and Altman (1994), extend conventional polynomials by allowing *fractional* and *negative* powers. Instead of being restricted to $x, x^2, x^3, \ldots$, fractional polynomials choose from the set of powers:

$$\{-2, -1, -0.5, 0, 0.5, 1, 2, 3\}$$

where x^0 is defined as $\log(x)$.

A **first-degree fractional polynomial** (FP1) has the form:

$$\beta_0 + \beta_1 x^{p_1}$$

A **second-degree fractional polynomial** (FP2) has the form:

$$\beta_0 + \beta_1 x^{p_1} + \beta_2 x^{p_2}$$

where p_1 and p_2 are selected from the allowed power set. If $p_1 = p_2$, the second term becomes $x^{p_1} \log(x)$ (the so-called "repeated powers" convention).

6.4.2. How FPs Work

The algorithm searches through all combinations of powers from the allowed set and selects the combination that best fits the data (by maximum likelihood or AIC). For an FP2, there are 36 possible combinations of powers, making the search computationally straightforward. Where does 36 come from? The allowed power set has 8 values. An FP2 chooses two powers, and the order does not matter, so there are $\binom{8}{2} = 28$ pairs of *distinct* powers, plus 8 cases where the two powers are *equal* (the "repeated powers" form $x^p \log x$ described above) — giving $28 + 8 = 36$ candidate models in total. The algorithm simply fits all 36 and keeps the best.

The key advantage of fractional polynomials is that they can capture a wide range of curve shapes — monotone increasing, monotone decreasing, U-shaped, J-shaped, S-shaped — with only one or two parameters beyond the linear term. (A relationship is **monotone** if it moves in a single direction across the whole range of the predictor: *monotone increasing* means the outcome only ever rises as the predictor rises, *monotone decreasing* means it only ever falls — in neither case does the curve reverse direction, even if its steepness changes.)

6.4.3. Strengths and Limitations

Strengths:

- Parsimonious: only 1–2 additional parameters
- Can capture many common curve shapes
- Well-established methodology with formal model selection procedures (the FP selection algorithm, also called MFP for multivariable fractional polynomials)
- Particularly useful when sample size is modest

Limitations:

- Behaviour at the extremes of the predictor range can be erratic (the curves are governed by global functions)
- Cannot capture complex local features (e.g., a bump in the middle of the range)
- The set of allowed shapes, while broad, is still constrained

Lopez-Ayala et al. (2025) note that fractional polynomials and splines are *complementary* approaches, and in many cases they give similar results. However, for most applications in clinical research, **restricted cubic splines have become the recommended default** due to their greater flexibility and better behaviour at the boundaries.

6.5. Splines: The Recommended Approach

6.5.1. What is a spline, in plain language?

Think of a spline as a flexible ruler. A straight line forces the same slope everywhere. A spline lets the curve bend at specific points (called **knots**), producing a smooth curve that follows the data's actual shape. The result is a single smooth function — not separate lines stitched together.

You have already seen the problem a spline solves: in Chapter 5, we used a straight line to model the relationship between age and blood pressure. That works if the relationship is genuinely linear. But many clinical relationships are not — BMI and mortality is U-shaped, age and surgical risk accelerates in the elderly. A spline captures these patterns.

6.5.2. What Is a Spline?

A spline is a piecewise polynomial function — a curve made up of polynomial segments joined smoothly at specific points called **knots**. The idea is simple: rather than fitting one global function to the entire range of a predictor, you fit separate polynomial segments in different intervals and then connect them smoothly.

Think of it like bending a flexible ruler: the ruler is smooth everywhere, but you can create complex curves by bending it at specific points.

6.5.3. Knots

Knots are the points along the predictor's range where the polynomial segments connect. The number and placement of knots control the flexibility of the spline:

- **More knots** = more flexible curve (can capture more complex shapes)
- **Fewer knots** = smoother, more constrained curve

How to choose knot locations:

The standard recommendation, following Harrell (2015), is to place knots at **fixed percentiles** of the predictor's distribution:

Number of Knots	Percentile Locations
3	10th, 50th, 90th
4	5th, 35th, 65th, 95th
5	5th, 27.5th, 50th, 72.5th, 95th

This data-driven placement ensures that knots are spread across the range of observed data, with more knots where there are more observations.

6.5.4. Cubic Splines

A **cubic spline** builds the curve out of short cubic (third-degree) pieces, one between each pair of neighbouring knots, and glues them together so that the joins are invisible. "Cubic" simply means each piece is a curve of the form $a + bx + cx^2 + dx^3$ — flexible enough to bend once or twice, which is usually all you need over a short interval. The cleverness is in *how* the pieces are joined: at every knot, neighbouring pieces are forced to meet at the same height, with the same slope, and with the same curvature (rate of change of slope). In calculus terms, the curve has **continuous first and second derivatives** at each knot. The practical consequence is that the eye sees one smooth curve, with no kinks or sudden changes in bend where the pieces meet.

The price of this flexibility is paid in **degrees of freedom** — roughly, the number of free parameters the model has to estimate from the data. A cubic spline with k knots uses $k + 3$ parameters: the cubic shape of the first segment, plus one extra parameter for each additional knot. More knots mean a more flexible curve but more parameters to estimate, and in small samples that flexibility can be expensive. This cost is exactly what restricted cubic splines, described next, are designed to reduce.

6.5.5. Restricted Cubic Splines (RCS)

A **restricted cubic spline** (also called a **natural cubic spline**) adds one more constraint: the function is forced to be **linear** beyond the outermost knots (i.e., in the tails of the distribution).

This constraint is motivated by the fact that there are typically few observations in the tails, so we have little information to estimate complex curves there. Concretely, "linear beyond the outer knots" means that below the first knot and above the last knot the curve is a **straight line** — it contains no squared (x^2) or cubic (x^3) terms in those regions. Those higher-order terms are precisely what let a curve bend, so removing them in the data-sparse tails stops the model from inventing dramatic bends where it has almost no data to justify them. Inside the knots, the full cubic flexibility is retained.

Forcing linearity in the tails:

- Prevents wild oscillations at the extremes (an unrestricted cubic spline can swing violently just past the last data points, because the squared and cubic terms are unconstrained there)
- Reduces the number of parameters (an RCS with k knots uses $k-1$ parameters, versus $k+3$ for an unrestricted cubic spline — the four "saved" parameters are the squared and cubic terms removed from the two tails)
- Makes extrapolation more sensible (extending a straight line beyond the data is far more defensible than extending a cubic, which could shoot off to implausible values)

The degrees-of-freedom saving is the practical headline: a 4-knot RCS costs only 3 parameters ($4-1$), so it adds just **2 parameters beyond a simple linear term** while still capturing most biologically plausible curve shapes. That is why it is such an economical default.

> **!** The Recommended Default
>
> Lopez-Ayala et al. (2025) and Harrell (2015) recommend **restricted cubic splines with 3 to 5 knots** as the default approach for modelling continuous predictors. With 4 knots (3 degrees of freedom), you can capture the vast majority of biologically plausible curve shapes. Use 5 knots (4 df) if you suspect a particularly complex relationship and have adequate sample size.
> An RCS with 3 knots (2 df) can capture monotone and simple U/J-shaped curves. This is a good starting point when sample size is limited or when you do not expect a complex relationship.

6.5.6. Why Restricted Cubic Splines?

To summarise the case for RCS:

1. **Flexibility.** They can capture a wide variety of non-linear shapes without requiring you to specify the shape in advance.
2. **Parsimony.** With $k - 1$ parameters for k knots, they are relatively economical. A 4-knot RCS adds only 2 parameters beyond a linear term.
3. **Good boundary behaviour.** The linearity constraint in the tails prevents overfitting where data are sparse.
4. **Interpretability through plots.** While the individual coefficients of a spline are not directly interpretable, the resulting curve is highly interpretable when plotted.
5. **Well supported by software.** The `rms` package in R and various Python libraries make implementation straightforward.

6.5.7. Interpreting Spline Models

This is a critical point that many researchers initially find confusing:

You cannot interpret the individual regression coefficients of a spline model. An RCS with 4 knots produces 3 coefficients (e.g., `rcs(bmi, 4)`, `rcs(bmi, 4)'`, `rcs(bmi, 4)''`), but these individual numbers are meaningless in isolation. They define the *shape* of the curve jointly.

Instead, you interpret spline models through:

1. **Plots.** Plot the predicted outcome (or log-odds, or hazard ratio) as a function of the predictor. This shows you the entire shape of the relationship.

2. **Specific contrasts.** You can calculate the predicted difference in outcome between any two values of the predictor (e.g., the odds ratio for BMI = 35 vs BMI = 25).

3. **Overall tests.** Two distinct questions can be asked, and they use different degrees of freedom. The first is *"does this predictor matter at all?"* — a **chunk test** (Wald or likelihood ratio test) of all the spline terms together, with $k-1$ degrees of freedom (the total number of parameters the RCS spends on the predictor). The second is *"is the relationship actually curved, or would a straight line do?"* — the **non-linearity test**, with $k-2$ degrees of freedom.

   Why $k-2$? Of the $k-1$ parameters the spline uses, exactly **one** describes the overall linear trend (the straight-line component). The remaining $k-2$ parameters are the "extra" terms that allow the curve to bend away from that straight line. Testing whether *those* $k-2$ terms are jointly zero is therefore a direct test of non-linearity: if they are not needed, the relationship is adequately linear; if they are, a straight line is too simple. So for a 4-knot RCS ($k=4$): the overall test has 3 df, and the non-linearity test has 2 df.

```
library(rms)
library(ggplot2)

set.seed(123)
n <- 500
x <- runif(n, 20, 45)
# True non-linear relationship (a gentle U-shape in BMI). The coefficients
# are kept modest so the outcome is not near-separated --- otherwise the
# fitted log-odds and their confidence band explode and the curve becomes
# invisible against a huge y-axis.
logit_p <- -2 + 0.02 * (x - 30)^2
p <- plogis(logit_p)
y <- rbinom(n, 1, p)

dd <- datadist(x)
options(datadist = "dd")
```

```r
fit <- lrm(y ~ rcs(x, 4))
knot_locs <- attr(rcs(x, 4), "parms")

# Prediction across the range of BMI
x_pred <- seq(20, 45, length.out = 200)
pred <- as.data.frame(Predict(fit, x = x_pred))

# Zoom the y-axis to the fitted curve and its confidence band so the
# U-shape is clearly visible.
y_lo <- min(pred$lower)
y_hi <- max(pred$upper)

ggplot(pred, aes(x = x, y = yhat)) +
  geom_ribbon(aes(ymin = lower, ymax = upper), alpha = 0.2, fill = "#3498db") +
  geom_line(linewidth = 1.2, colour = "#2c3e50") +
  geom_point(data = data.frame(x = knot_locs, y = y_lo),
             aes(x = x, y = y), shape = 17, size = 3, colour = "#e74c3c") +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
  coord_cartesian(ylim = c(y_lo, y_hi)) +
  labs(x = "BMI (kg/m\u00b2)", y = "Log-Odds of Outcome",
       subtitle = "RCS with 4 knots; triangles mark knot positions") +
  theme_minimal(base_size = 14)
```

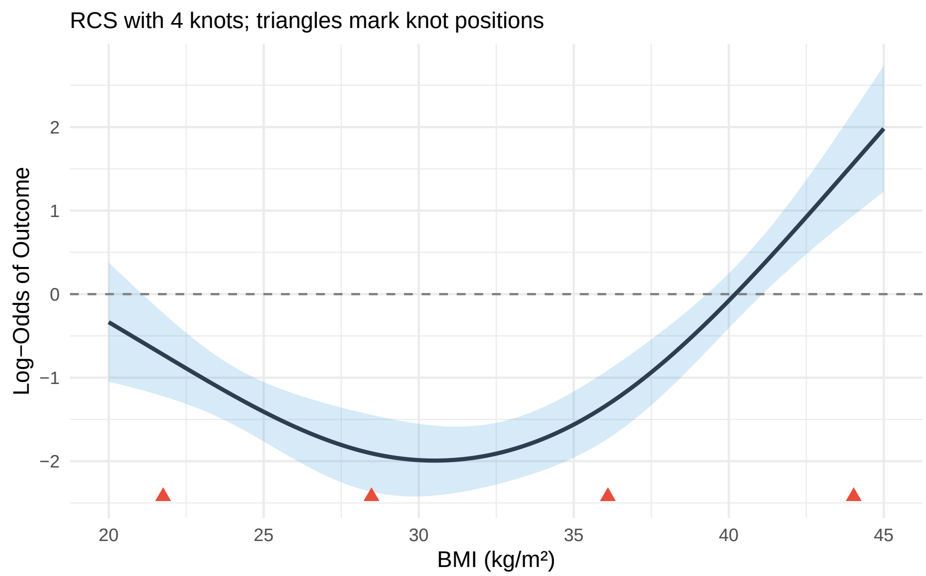


Figure 6.2.: Illustration of an RCS(4) fit. Knot locations are marked with triangles. The relationship is linear beyond the outer knots and flexibly curved between them.

6.6. Clinical Example: CSF Glucose and Bacterial Meningitis

This example is adapted from Lopez-Ayala et al. (2025), who use it to illustrate the full workflow for modelling a continuous predictor.

6.6.1. The Clinical Question

In patients presenting with suspected meningitis, cerebrospinal fluid (CSF) glucose is one of several laboratory values used to distinguish bacterial from viral meningitis. Low CSF glucose is a hallmark of bacterial infection, as bacteria consume glucose. But what is the precise shape of this relationship? And is it appropriate to use a simple cut-point (e.g., CSF glucose < 2.2 mmol/L)?

6.6.2. Step-by-Step Analysis

Step 1: Visualise the raw relationship.

Before fitting any model, plot the outcome against the predictor using a smoothing function (loess or GAM). This gives you a preliminary sense of the shape.

```
library(ggplot2)
library(rms)

set.seed(2025)
n <- 400
csf_glucose <- rlnorm(n, meanlog = log(3), sdlog = 0.6)
csf_glucose <- pmin(csf_glucose, 10)

# True relationship: steep drop in probability below ~2, then flat
logit_p <- 3 - 4 * pmax(0, 2.5 - csf_glucose) - 0.3 * csf_glucose
p_bact <- plogis(logit_p)
bacterial <- rbinom(n, 1, p_bact)

df_csf <- data.frame(csf_glucose = csf_glucose, bacterial = bacterial)

ggplot(df_csf, aes(x = csf_glucose, y = bacterial)) +
  geom_jitter(height = 0.05, alpha = 0.3, width = 0.05) +
  geom_smooth(method = "loess", se = TRUE, colour = "#2c3e50") +
  labs(x = "CSF Glucose (mmol/L)", y = "Probability of Bacterial Meningitis",
```

```
    subtitle = "Loess smooth suggests a non-linear (threshold-like) relati
theme_minimal(base_size = 14)
```

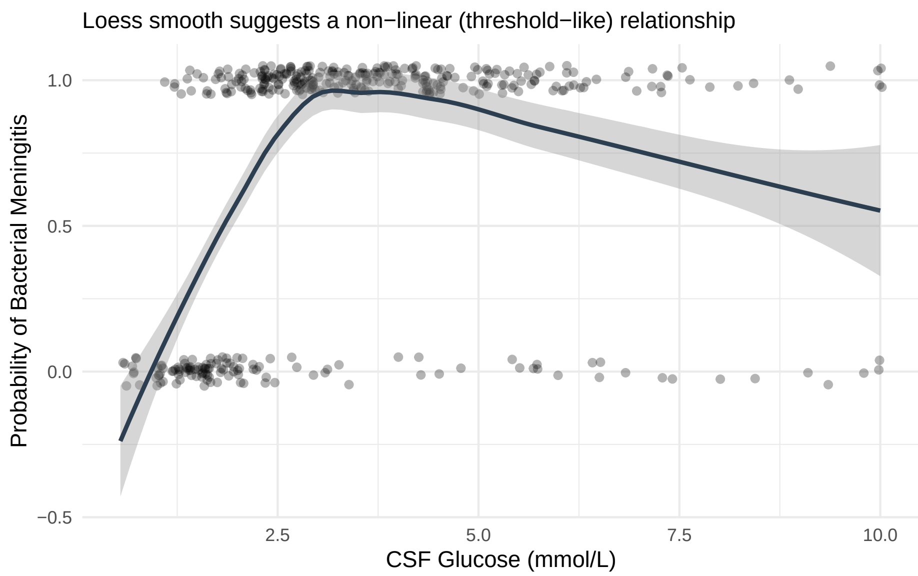


Figure 6.3.: Simulated relationship between CSF glucose and probability of bacterial meningitis, illustrating the non-linear (threshold-like) pattern.

Step 2: Fit competing models.

We fit three models and compare them:

- **Model 1 (Linear):** `bacterial ~ csf_glucose`
- **Model 2 (Categorised):** `bacterial ~ I(csf_glucose < 2.2)` (a common clinical cut-point)
- **Model 3 (RCS):** `bacterial ~ rcs(csf_glucose, 4)`

```r
library(rms)

dd_csf <- datadist(df_csf)
options(datadist = "dd_csf")

# Model 1: Linear
fit_linear <- lrm(bacterial ~ csf_glucose, data = df_csf)

# Model 2: Categorised
df_csf$glucose_low <- as.numeric(df_csf$csf_glucose < 2.2)
fit_cat <- lrm(bacterial ~ glucose_low, data = df_csf)

# Model 3: RCS with 4 knots
fit_rcs <- lrm(bacterial ~ rcs(csf_glucose, 4), data = df_csf)

# Compare AIC
aic_vals <- c(
  Linear = AIC(fit_linear),
  Categorised = AIC(fit_cat),
  RCS_4knots = AIC(fit_rcs)
)

cat("AIC Comparison:\n")
print(round(aic_vals, 1))
cat("\nLower AIC = better fit (penalised for complexity)\n")
```

```
AIC Comparison:
     Linear Categorised  RCS_4knots
      456.3       366.7       339.0

Lower AIC = better fit (penalised for complexity)
```

Step 3: Test for non-linearity.

```
# The anova() function in rms tests the overall effect and non-linearity sepa
cat("ANOVA for RCS model:\n")
print(anova(fit_rcs))
```

```
ANOVA for RCS model:
                Wald Statistics          Response: bacterial

 Factor      Chi-Square d.f. P
 csf_glucose 91.23      3    <.0001
  Nonlinear  83.01      2    <.0001
 TOTAL       91.23      3    <.0001
```

The ANOVA output from `rms` provides two key tests:

- **Total effect** (all spline terms combined): tests whether CSF glucose has *any* association with the outcome
- **Nonlinear component** (spline terms beyond the linear): tests whether the relationship departs from linearity

If the non-linear component is significant, this confirms that a linear term is insufficient.

Step 4: Plot the fitted curves from all three models.

```
x_seq <- seq(min(df_csf$csf_glucose), max(df_csf$csf_glucose), length.out = 2

# Predictions from each model
pred_linear <- predict(fit_linear, newdata = data.frame(csf_glucose = x_seq)
                       type = "fitted")
pred_cat <- predict(fit_cat,
                    newdata = data.frame(glucose_low = as.numeric(x_seq < 2.2
                    type = "fitted")
```

```r
pred_rcs_obj <- Predict(fit_rcs, csf_glucose = x_seq)

plot_df <- data.frame(
  csf_glucose = rep(x_seq, 3),
  probability = c(pred_linear, pred_cat, as.data.frame(pred_rcs_obj)$yhat),
  Model = rep(c("Linear", "Categorised (<2.2)", "RCS (4 knots)"), each = length(x_seq))
)

# Convert RCS from log-odds to probability for fair comparison
plot_df$probability[plot_df$Model == "RCS (4 knots)"] <-
  plogis(plot_df$probability[plot_df$Model == "RCS (4 knots)"])

ggplot(plot_df, aes(x = csf_glucose, y = probability, colour = Model)) +
  geom_line(linewidth = 1.1) +
  scale_colour_manual(values = c("Linear" = "#3498db",
                                 "Categorised (<2.2)" = "#e74c3c",
                                 "RCS (4 knots)" = "#2c3e50")) +
  labs(x = "CSF Glucose (mmol/L)", y = "Predicted Probability of Bacterial Meningitis") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "bottom")
```

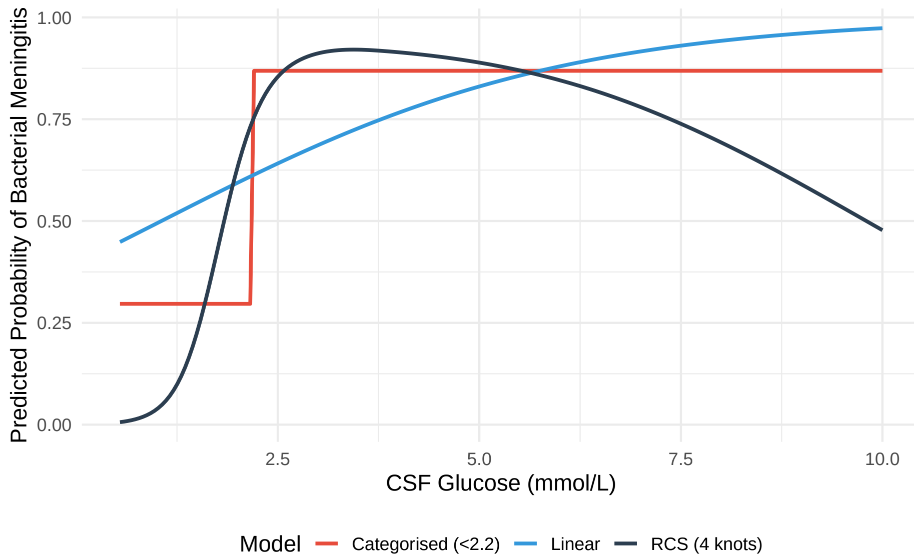


Figure 6.4.: Comparison of linear (blue), categorised (red), and RCS (black) model fits for the CSF glucose–bacterial meningitis relationship.

6.6.3. What This Example Teaches Us

1. The **linear model** misses the threshold-like nature of the relationship. It estimates a constant decrease in probability per unit increase in glucose, which underestimates the risk at very low glucose levels and overestimates it at intermediate levels.

2. The **categorised model** captures the broad direction (low glucose = higher risk) but loses all nuance. It cannot tell you that a glucose of 0.5 mmol/L carries a very different risk than a glucose of 2.0 mmol/L — both are simply "low."

3. The **RCS model** captures the steep decline in risk at low glucose levels and the flattening at higher levels, providing the most accurate and informative picture.

6.7. Practical Recommendations

Lopez-Ayala et al. (2025) provide a practical framework (their Box 2) that we adapt here:

> Workflow for Modelling Continuous Predictors
>
> 1. **Keep continuous variables continuous.** Do not categorise unless there is a strong *a priori* biological or clinical reason (and even then, think twice).
> 2. **Start by visualising** the predictor–outcome relationship using a smoother (loess, GAM, or a spline with many knots).
> 3. **Fit a restricted cubic spline** with 3–5 knots as the default modelling strategy. Use 3 knots when sample size is small or the relationship is expected to be simple; 4–5 knots when sample size is adequate and the relationship may be complex.
> 4. **Test for non-linearity** using a likelihood ratio test or the Wald test from `anova(fit)` in `rms`. If non-linearity is not significant, you *may* simplify to a linear term — but the spline model will give very similar results anyway, so simplification is optional.
> 5. **Present results as plots** showing the predicted outcome across the range of the predictor, with confidence intervals. Supplement with specific contrasts (e.g., odds ratio for value A vs value B) as needed.

6. **Do not report individual spline coefficients** in tables. They are not interpretable. Instead, report the overall test for the predictor and any non-linearity test, along with a figure.
7. **Use AIC or likelihood ratio tests** to compare models (linear vs spline, different numbers of knots).

6.8. Implementation

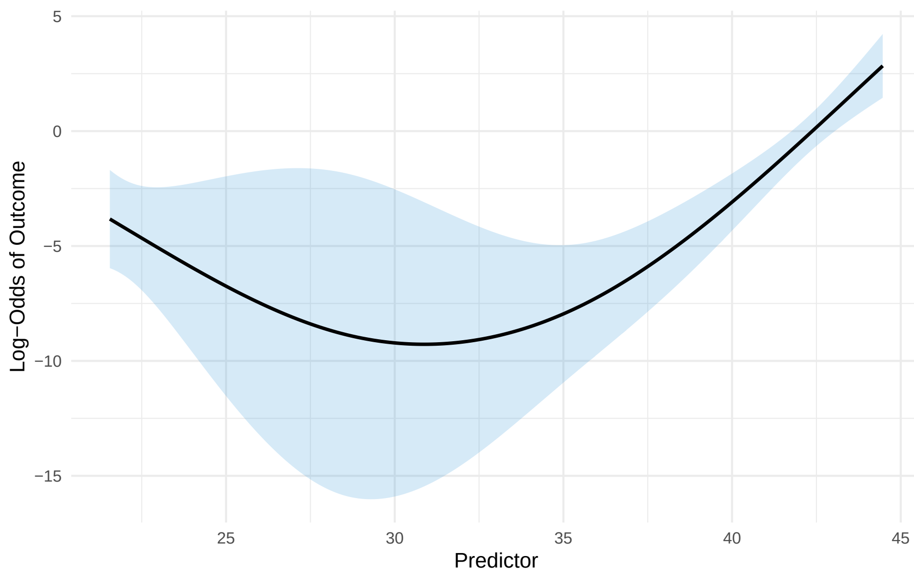
The `rms` package by Frank Harrell is the gold standard for this in R, and `patsy` + `statsmodels` give the equivalent capability in Python. The walkthroughs below are **fully self-contained and runnable**: each one first simulates a small example dataset (a continuous `predictor`, two `confounder`s, and a binary `outcome`) so you can run the code as-is, then fits and plots a restricted cubic spline. To use them on your own study, replace the simulated `dat` with your own data frame and keep the rest unchanged.

6.8.0.1. R

The key function is `rcs(variable, number_of_knots)` inside an `rms` model fit such as `lrm()`.

```
library(rms)
library(ggplot2)

# --- Simulated example data (replace this block with your own data) ---
set.seed(1)
n <- 400
predictor   <- runif(n, 20, 45)
confounder1 <- rnorm(n)
```

```r
confounder2 <- rbinom(n, 1, 0.4)
logit_p <- -8 + 0.05 * (predictor - 30)^2 + 0.3 * confounder1 + 0.5 * confounder2
dat <- data.frame(
  outcome = rbinom(n, 1, plogis(logit_p)),
  predictor = predictor,
  confounder1 = confounder1,
  confounder2 = confounder2
)

# Step 1: Tell rms about the data distribution (required for Predict/summary)
dd <- datadist(dat)
options(datadist = "dd")

# Step 2: Fit a logistic regression with the predictor modelled as an RCS(4)
fit <- lrm(outcome ~ rcs(predictor, 4) + confounder1 + confounder2, data = dat)

# Step 3: Examine the model
print(fit)         # Overall model summary
anova(fit)         # Tests for each predictor (total effect + non-linearity)

# Step 4: Plot the fitted relationship across the range of the predictor
pred_df <- as.data.frame(Predict(fit, predictor))
ggplot(pred_df, aes(x = predictor, y = yhat)) +
  geom_ribbon(aes(ymin = lower, ymax = upper), alpha = 0.2, fill = "#3498db") +
  geom_line(linewidth = 1) +
  labs(x = "Predictor", y = "Log-Odds of Outcome") +
  theme_minimal(base_size = 13)
```




```
# Step 5: A specific contrast --- log-odds ratio for predictor 30 vs 25
summary(fit, predictor = c(25, 30))
```

```
Logistic Regression Model

lrm(formula = outcome ~ rcs(predictor, 4) + confounder1 + confounder2,
    data = dat)

                       Model Likelihood      Discrimination    Rank Discrim.
                             Ratio Test             Indexes          Indexes
Obs              400    LR chi2      193.30      R2       0.775    C       0.982
 0               357    d.f.              5      R2(5,400)0.375    Dxy     0.964
 1                43    Pr(> chi2) <0.0001    R2(5,115.1)0.805    gamma   0.964
max |deriv| 7e-06                                Brier    0.028    tau-
a   0.185
```

```
            Coef    S.E.    Wald Z Pr(>|Z|)
Intercept   15.1163 16.5458  0.91  0.3609
predictor   -0.8784  0.7545 -1.16  0.2443
predictor'   1.9043  2.7241  0.70  0.4845
predictor'' -1.9463  6.4234 -0.30  0.7619
confounder1  0.2734  0.2730  1.00  0.3165
confounder2  0.1893  0.5931  0.32  0.7496

                Wald Statistics          Response: outcome

 Factor       Chi-Square d.f. P
 predictor    47.13      3    <.0001
  Nonlinear   24.85      2    <.0001
 confounder1   1.00      1    0.3165
 confounder2   0.10      1    0.7496
 TOTAL        47.73      5    <.0001
             Effects              Response : outcome

 Factor       Low     High     Diff.  Effect    S.E.    Lower 0.95 Upper 0.95
 predictor    25.0000 30.00000 5.0000 -2.469800 1.18720 -
4.7967000 -0.14295
  Odds Ratio 25.0000 30.00000 5.0000  0.084601      NA  0.0082572  0.86679
 confounder1 -0.6428  0.68494 1.3277  0.363050 0.36245 -
0.3473400  1.07340
  Odds Ratio -0.6428  0.68494 1.3277  1.437700      NA  0.7065600  2.92540
 confounder2  0.0000  1.00000 1.0000  0.189310 0.59313 -
0.9732000  1.35180
  Odds Ratio  0.0000  1.00000 1.0000  1.208400      NA  0.3778700  3.86450
```

6.8.0.2. Python

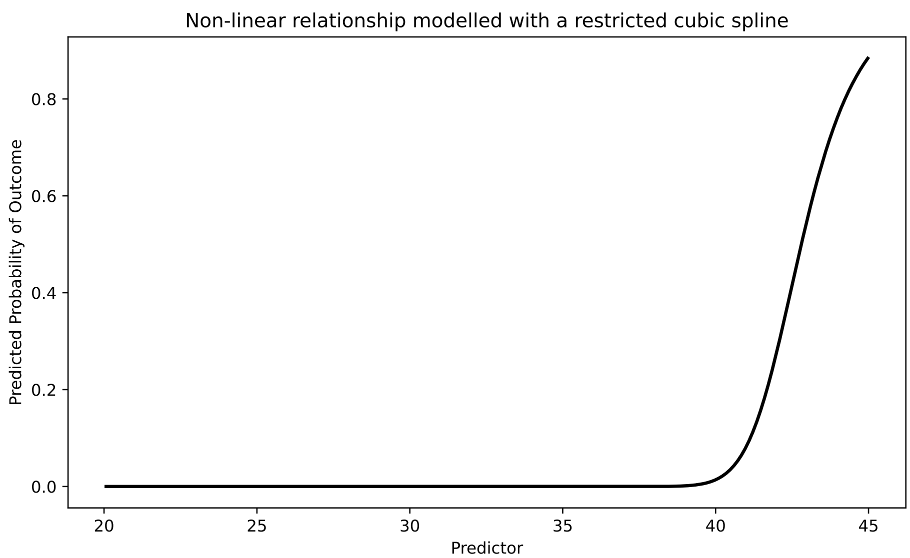
`patsy`'s `cr()` builds a natural (restricted) cubic spline basis; `df` equals the number of knots minus one, so `df=3` corresponds to a 4-knot RCS.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.api as sm
import statsmodels.formula.api as smf
from scipy.special import expit

# --- Simulated example data (replace this block with your own data) ---
rng = np.random.default_rng(1)
n = 400
predictor = rng.uniform(20, 45, n)
confounder1 = rng.normal(size=n)
confounder2 = rng.binomial(1, 0.4, n)
logit_p = -8 + 0.05 * (predictor - 30) ** 2 + 0.3 * confounder1 + 0.5 * confo
dat = pd.DataFrame({
    "outcome": rng.binomial(1, expit(logit_p)),
    "predictor": predictor,
    "confounder1": confounder1,
    "confounder2": confounder2,
})

# Step 1: Fit using the formula interface. patsy's cr() builds a natural
# (restricted) cubic spline basis; df=3 corresponds to a 4-knot RCS. Using th
# formula API means statsmodels remembers the spline knots and reapplies them
# automatically when we predict on new data below.
model = smf.glm(
    "outcome ~ cr(predictor, df=3) + confounder1 + confounder2",
    data=dat, family=sm.families.Binomial(),
)
result = model.fit()
print(result.summary())

# Step 2: Plot the fitted relationship across the range of the predictor
```

```
# (holding the confounders fixed at reference values).
x_range = np.linspace(dat["predictor"].min(), dat["predictor"].max(), 200)
grid = pd.DataFrame({"predictor": x_range, "confounder1": 0.0, "confounder2": 0})
pred_prob = result.predict(grid)

plt.figure(figsize=(8, 5))
plt.plot(x_range, pred_prob, "k-", linewidth=2)
plt.xlabel("Predictor")
plt.ylabel("Predicted Probability of Outcome")
plt.title("Non-linear relationship modelled with a restricted cubic spline")
plt.tight_layout()
plt.show()
```

```
                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:                outcome   No. Observations:                  400
Model:                            GLM   Df Residuals:                      395
Model Family:                Binomial   Df Model:                            4
Link Function:                  Logit   Scale:                          1.0000
Method:                          IRLS   Log-Likelihood:                -
28.986
Date:                Wed, 10 Jun 2026   Deviance:                       57.973
Time:                        16:33:27   Pearson chi2:                     74.3
No. Iterations:                   100   Pseudo R-squ. (CS):             0.3538
Covariance Type:            nonrobust
==========================================================================================
                             coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
Intercept              -5.303e+12   9.85e+13     -0.054      0.957   -
1.98e+14    1.88e+14
cr(predictor, df=3)[0]  5.303e+12   9.85e+13      0.054      0.957   -
1.88e+14    1.98e+14
```

```
cr(predictor, df=3)[1]  5.303e+12   9.85e+13      0.054      0.957   -
1.88e+14    1.98e+14
cr(predictor, df=3)[2]  5.303e+12   9.85e+13      0.054      0.957   -
1.88e+14    1.98e+14
confounder1                0.9267      0.335      2.770      0.006       0.27
confounder2                1.2288      0.705      1.743      0.081      -
0.153       2.610
==========================================================================
```



6.9. Comparison with Other Approaches

The table below summarises the main options. As a quick reminder of the two flexible approaches: **fractional polynomials** (introduced earlier in this chapter) model the predictor using a small number of non-integer powers (such as $\sqrt{x}$, $1/x$, or $\log x$) chosen to fit the data, giving a smooth global curve; **restricted cubic splines** instead stitch together local cubic pieces between knots and force straight lines in the tails. Both avoid the pitfalls of categorisation, but they bend the curve in different ways.

Table 6.1.: Comparison of approaches for modelling continuous variables

Approach	Pros	Cons
Linear term	Simple, 1 df	Assumes linearity — often wrong
Categorisation	"Easy" to present	Information loss, bias, arbitrary cut-points
Quadratic polynomial	Can capture U-shapes, 2 df	Symmetric, poor boundary behaviour
Fractional polynomials	Flexible, parsimonious	Global functions; limited local flexibility
Restricted cubic splines	Flexible, good boundary behaviour, well-supported	Coefficients not directly interpretable
GAM (smoothing splines)	Very flexible, automatic smoothing	Can overfit; penalty selection needed

For most clinical research, **restricted cubic splines with 3–5 knots** represent the best balance of flexibility, parsimony, and practical usability.

6.10. Exercises

6.10.1. Exercise 1: Recognising the Problem

Consider a study of the association between systolic blood pressure (SBP) and stroke risk. The researchers categorise SBP into: normal (<120), elevated (120–129), Stage 1 hypertension (130–139), and Stage 2 hypertension (>=140).

Questions:

a) What assumptions does this categorisation impose on the SBP–stroke relationship?
b) A patient with SBP of 131 mmHg is classified identically to a patient with SBP of 139 mmHg. Why is this problematic?
c) Suggest a better modelling approach and explain how you would implement it.

6.10.2. Exercise 2: Fitting and Interpreting Spline Models

Using the built-in `PBC` dataset (primary biliary cholangitis) from the `survival` package in R, model the relationship between serum bilirubin and transplant status (a binary outcome: transplanted or not).

6.10.2.1. R

```
library(rms)
library(ggplot2)

# Load the PBC data
data(pbc, package = "survival")
pbc <- pbc[!is.na(pbc$trt), ]  # Remove incomplete cases

# Create a binary outcome: transplanted (status == 1) vs not
pbc$transplant <- as.numeric(pbc$status == 1)

# Set up datadist
dd <- datadist(pbc)
options(datadist = "dd")

# Task 1: Fit a logistic regression with bilirubin as a linear term
```

```r
fit_linear <- lrm(transplant ~ bili, data = pbc)

# Task 2: Fit a logistic regression with bilirubin modelled using RCS (4 knots)
fit_rcs <- lrm(transplant ~ rcs(bili, 4), data = pbc)

# Task 3: Test for non-linearity
anova(fit_rcs)

# Task 4: Compare AIC
AIC(fit_linear)
AIC(fit_rcs)

# Task 5: Plot the relationship
p <- Predict(fit_rcs, bili)
plot(p, xlab = "Serum Bilirubin (mg/dL)",
     ylab = "Log-Odds of Transplant")

# Task 6: What does the plot tell you about the bilirubin-transplant
#          relationship? Is it linear?
```

6.10.2.2. Python

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.api as sm
from patsy import dmatrix

# Simulate data similar to PBC bilirubin-transplant relationship
np.random.seed(42)
n = 300
bili = np.random.exponential(scale=2, size=n)
```

```
bili = np.clip(bili, 0.3, 25)
logit_p = -3 + 0.1 * bili + 0.005 * bili**2  # Non-linear true relationship
prob = 1 / (1 + np.exp(-logit_p))
transplant = np.random.binomial(1, prob)

df = pd.DataFrame({'bili': bili, 'transplant': transplant})

# Task 1: Fit logistic regression with linear bilirubin
X_linear = sm.add_constant(df[['bili']])
fit_linear = sm.GLM(df['transplant'], X_linear,
                     family=sm.families.Binomial()).fit()
print("Linear model:")
print(fit_linear.summary())

# Task 2: Create spline basis and fit logistic regression with splines
spline_basis = dmatrix("cr(bili, df=3)", data=df, return_type='dataframe')
fit_spline = sm.GLM(df['transplant'], spline_basis,
                      family=sm.families.Binomial()).fit()
print("\nSpline model:")
print(fit_spline.summary())

# Task 3: Compare AIC
print(f"\nAIC Linear: {fit_linear.aic:.1f}")
print(f"AIC Spline: {fit_spline.aic:.1f}")

# Task 4: Plot the predicted log-odds across bilirubin range
bili_range = np.linspace(df['bili'].min(), df['bili'].max(), 200)
spline_pred = dmatrix("cr(bili_range, df=3)",
                       {"bili_range": bili_range},
                       return_type='dataframe')
pred_prob = fit_spline.predict(spline_pred)

plt.figure(figsize=(8, 5))
```

```
plt.plot(bili_range, pred_prob, 'k-', linewidth=2)
plt.xlabel('Serum Bilirubin (mg/dL)')
plt.ylabel('Predicted Probability of Transplant')
plt.title('Non-linear effect of bilirubin on transplant (RCS)')
plt.tight_layout()
plt.show()
```

6.10.3. Exercise 3: Categorisation vs Splines Head-to-Head

Simulate data with a known non-linear (U-shaped) relationship and compare the performance of categorised and spline models.

6.10.3.1. R

```
library(rms)
library(ggplot2)

set.seed(2025)
n <- 1000
x <- rnorm(n, mean = 50, sd = 10)

# True U-shaped relationship
logit_p <- -5 + 0.004 * (x - 50)^2
y <- rbinom(n, 1, plogis(logit_p))

sim_data <- data.frame(x = x, y = y)
dd <- datadist(sim_data)
options(datadist = "dd")

# Model 1: Categorise into quartiles
sim_data$x_cat <- cut(sim_data$x,
```

```
                        breaks = quantile(sim_data$x, c(0, 0.25, 0.5, 0.75, 1)
                        include.lowest = TRUE)
fit_cat <- lrm(y ~ x_cat, data = sim_data)

# Model 2: Linear term
fit_lin <- lrm(y ~ x, data = sim_data)

# Model 3: RCS with 5 knots
fit_rcs <- lrm(y ~ rcs(x, 5), data = sim_data)

# Questions:
# a) Compare AIC of the three models. Which fits best?
# b) Plot all three fitted curves on the same graph along with the
#    true logit function. Which model best recovers the truth?
# c) What happens if you change the quartile cut-points to tertiles?
#    Does it affect the categorised model's conclusions?

cat("AIC values:\n")
cat("Categorised:", AIC(fit_cat), "\n")
cat("Linear:     ", AIC(fit_lin), "\n")
cat("RCS(5):     ", AIC(fit_rcs), "\n")
```

6.10.3.2. Python

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.api as sm
from patsy import dmatrix
from scipy.special import expit

np.random.seed(2025)
```

```
n = 1000
x = np.random.normal(50, 10, size=n)

# True U-shaped relationship
logit_p = -5 + 0.004 * (x - 50)**2
y = np.random.binomial(1, expit(logit_p))

df = pd.DataFrame({'x': x, 'y': y})

# Model 1: Categorise into quartiles
df['x_cat'] = pd.qcut(df['x'], q=4, labels=['Q1', 'Q2', 'Q3', 'Q4'])
X_cat = pd.get_dummies(df['x_cat'], drop_first=True).astype(float)
X_cat = sm.add_constant(X_cat)
fit_cat = sm.GLM(df['y'], X_cat, family=sm.families.Binomial()).fit()

# Model 2: Linear
X_lin = sm.add_constant(df[['x']])
fit_lin = sm.GLM(df['y'], X_lin, family=sm.families.Binomial()).fit()

# Model 3: RCS
X_rcs = dmatrix("cr(x, df=4)", data=df, return_type='dataframe')
fit_rcs = sm.GLM(df['y'], X_rcs, family=sm.families.Binomial()).fit()

print(f"AIC Categorised: {fit_cat.aic:.1f}")
print(f"AIC Linear:      {fit_lin.aic:.1f}")
print(f"AIC RCS:         {fit_rcs.aic:.1f}")

# Plot all three fitted curves
x_range = np.linspace(df['x'].min(), df['x'].max(), 200)

# True curve
true_logit = -5 + 0.004 * (x_range - 50)**2
true_prob = expit(true_logit)
```

```
# RCS predictions
X_rcs_pred = dmatrix("cr(x_range, df=4)", {"x_range": x_range},
                      return_type='dataframe')
rcs_prob = fit_rcs.predict(X_rcs_pred)

# Linear predictions
X_lin_pred = sm.add_constant(pd.DataFrame({'x': x_range}))
lin_prob = fit_lin.predict(X_lin_pred)

plt.figure(figsize=(10, 6))
plt.plot(x_range, true_prob, 'k--', linewidth=2, label='True relationship')
plt.plot(x_range, rcs_prob, 'b-', linewidth=2, label='RCS')
plt.plot(x_range, lin_prob, 'r-', linewidth=1.5, label='Linear')
plt.xlabel('Predictor (x)')
plt.ylabel('Predicted Probability')
plt.legend()
plt.title('Comparing modelling strategies for a U-shaped relationship')
plt.tight_layout()
plt.show()
```

6.11. Summary

Key Concept	Details
Do not categorise	Categorising continuous variables loses information, introduces bias, and reduces power
Linear terms	Assume a constant effect per unit change; often wrong for biological variables

Key Concept	Details
Fractional polynomials	Use non-integer powers for flexible global curves; parsimonious but limited locally
Splines	Piecewise polynomials joined at knots; flexible and locally adaptive
Restricted cubic splines	Splines forced to be linear in the tails; recommended default with 3–5 knots
Interpretation	Use plots and contrasts, not individual coefficients
Testing	Overall association test ($k - 1$ df); non-linearity test ($k - 2$ df)

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7. Penalised Regression: Ridge, LASSO, and Elastic Net

ℹ Learning Objectives

By the end of this chapter, you will be able to:

1. Explain why standard regression fails when the number of predictors is large relative to sample size
2. Describe the bias-variance tradeoff and how penalised regression exploits it
3. Distinguish between Ridge (L2), LASSO (L1), and Elastic Net penalties
4. Explain what the tuning parameters lambda and alpha control
5. Use cross-validation to select optimal penalty parameters
6. Interpret regularisation path plots (coefficient traces)
7. Implement penalised regression in both R (`glmnet`) and Python (`sklearn`)
8. Choose the appropriate method for a given clinical research problem

7.1. Introduction: Why Standard Regression Is Not Enough

Throughout the earlier chapters, we have used standard regression — ordinary least squares (OLS) for continuous outcomes, logistic regression for binary outcomes, and Cox regression for *time-to-event* data (outcomes measured as the time until an event such as death or relapse occurs; also called survival data, covered in the survival analysis chapter). These methods work well when the number of predictors is small relative to the sample size and the predictors are not highly correlated.

Two pieces of notation recur throughout this chapter: we write p for the **number of predictors** (variables) and n for the **number of observations** (patients). So "$p > n$" is shorthand for *more predictors than patients* — the genomics-style setting where standard regression breaks down.

But modern clinical research increasingly involves situations where these assumptions break down:

- **Genomics and proteomics.** A study might measure expression levels of 20,000 genes in 200 patients. There are 100 times more predictors than observations.
- **Radiomic features.** A CT scan can be summarised by hundreds of texture and shape features. A typical study might have 300 features and 150 patients.
- **Electronic health records.** A prediction model might draw on hundreds of laboratory values, medications, diagnoses, and demographic variables.
- **Multi-centre clinical prediction.** Even with a modest number of clinical predictors, small samples in some centres can lead to unstable estimates.

In all these situations, standard regression either **fails entirely** (when $p > n$) or produces **unreliable, overfitted** models (when p is close to n).

Penalised regression — also known as regularised regression or shrinkage methods — solves this problem by adding a penalty to the regression coefficients, pulling them toward zero. This trades a small amount of **bias** for a large reduction in **variance**, typically producing models that predict much better on new data. (If those two terms are unfamiliar, do not worry — the bias-variance tradeoff is exactly what the next section unpacks; for now, read "bias" as *systematic error* and "variance" as *instability from one sample to the next*.)

7.2. The Problem: Overfitting and Instability

7.2.1. What Happens with Too Many Predictors

Consider a logistic regression model for predicting 30-day mortality after cardiac surgery. You have 200 patients (40 deaths) and want to use 50 candidate predictors (age, comorbidities, lab values, surgical variables, etc.).

With standard logistic regression, this model is in serious trouble. A common rule of thumb (events per variable, or EPV) suggests you need at least 10–20 events per predictor (Steyerberg 2019). With 40 events and 50 predictors, your EPV is 0.8 — far below the minimum.

What goes wrong?

1. **Overfitting.** The model fits the training data very well — perhaps perfectly — but performs terribly on new patients. It has memorised the noise in the training data rather than learning the true signal.

2. **Unstable coefficients.** Small changes in the data (adding or removing a few patients) cause large changes in the estimated coefficients. Some coefficients may be absurdly large.

3. **Separation.** In logistic regression, if a predictor perfectly separates the outcomes (e.g., all patients with a certain lab value died), the maximum likelihood estimate for that coefficient is infinite.
4. **Non-convergence.** When $p > n$, the standard regression algorithm simply does not converge — there is no unique solution.

7.2.2. A Demonstration

To make the problem concrete, the simulation below creates a dataset where we *know* the truth: 100 patients, 40 candidate predictors, but only the first 5 actually influence the outcome (the other 35 are pure noise with a true coefficient of zero). We then fit a standard logistic regression and compare the coefficients it estimates (blue bars) against the true values (red dots). If the method were working well, the blue bars should sit near the red dots — near zero for the 35 noise predictors. Instead, because there are far too many predictors for the available information, many coefficients are estimated as large positive or negative values for variables that have *no real effect at all*. That is overfitting made visible: the model is fitting the noise.

7.2.2.1. R

```
library(ggplot2)

set.seed(42)
n <- 100
p <- 40

# True model: only 5 predictors matter
X <- matrix(rnorm(n * p), n, p)
colnames(X) <- paste0("X", 1:p)
```

```
true_beta <- c(rep(1, 5), rep(0, p - 5))
logit_p <- X %*% true_beta
y <- rbinom(n, 1, plogis(logit_p))

# Fit standard logistic regression (suppress warnings about non-convergence)
fit_standard <- suppressWarnings(
  glm(y ~ X, family = binomial())
)

coef_df <- data.frame(
  predictor = paste0("X", 1:p),
  estimated = coef(fit_standard)[-1],  # Remove intercept
  true = true_beta
)

ggplot(coef_df, aes(x = reorder(predictor, -abs(estimated)))) +
  geom_col(aes(y = estimated), fill = "#3498db", alpha = 0.7) +
  geom_point(aes(y = true), colour = "#e74c3c", size = 3) +
  coord_flip() +
  labs(x = "Predictor", y = "Coefficient Value",
       subtitle = "Blue bars = estimated; Red dots = true values. Many coefficients are wild
  theme_minimal(base_size = 12)
```

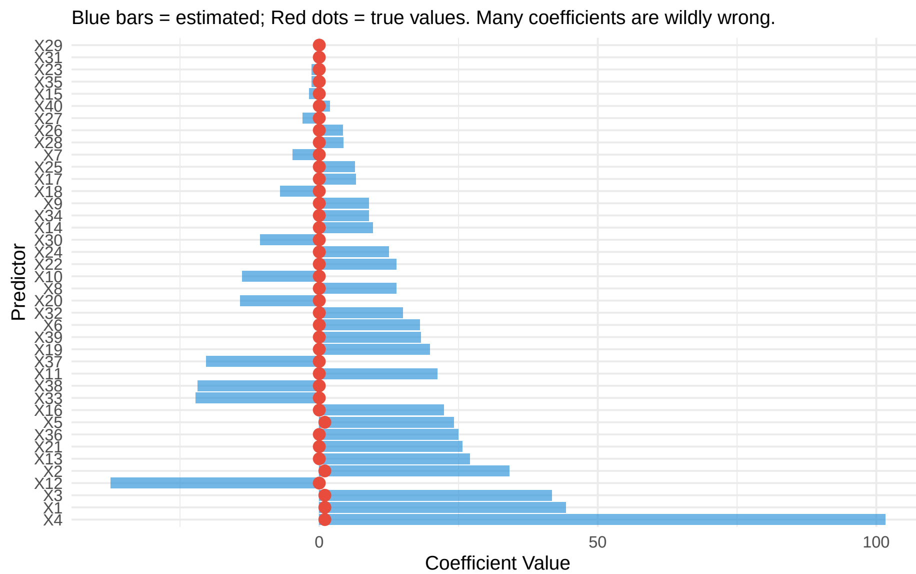


Figure 7.1.: Simulated example: standard logistic regression with 40 predictors and 100 observations. Many coefficients are wildly inflated. True coefficients (most are zero) are shown in red.

7.2.2.2. Python

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LogisticRegression

rng = np.random.default_rng(42)
n, p = 100, 40

# True model: only the first 5 predictors matter
X = rng.normal(size=(n, p))
```

```python
true_beta = np.array([1.0] * 5 + [0.0] * (p - 5))
y = rng.binomial(1, 1 / (1 + np.exp(-(X @ true_beta))))

# Standard (essentially unpenalised) logistic regression
fit = LogisticRegression(penalty=None, max_iter=10000).fit(X, y)
estimated = fit.coef_[0]

order = np.argsort(np.abs(estimated))
plt.figure(figsize=(7, 8))
plt.barh(range(p), estimated[order], color="#3498db", alpha=0.7, label="Estimated")
plt.plot(true_beta[order], range(p), "o", color="#e74c3c", label="True")
plt.yticks(range(p), [f"X{j+1}" for j in order], fontsize=7)
plt.xlabel("Coefficient value")
plt.title("Many coefficients are wildly wrong (blue) vs the truth (red)")
plt.legend(loc="best")
plt.tight_layout()
plt.show()
```

```
([<matplotlib.axis.YTick object at 0x13345d060>, <matplotlib.axis.YTick object at 0x13345c9a
```

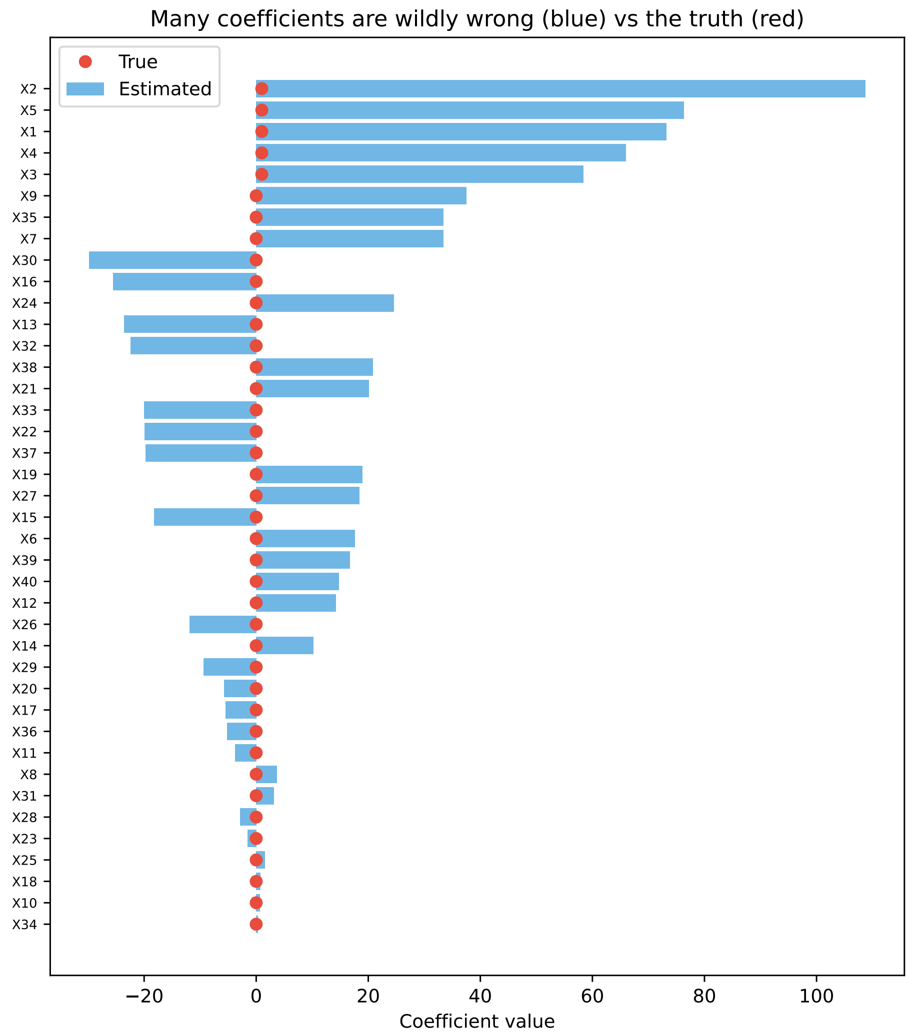


Figure 7.2.: The same demonstration in Python: standard logistic regression with 40 predictors and 100 observations. Estimated coefficients (blue) bear little resemblance to the true values (red dots).

7.3. The Bias-Variance Tradeoff

The fundamental insight behind penalised regression is the **bias-variance tradeoff**.

7.3.1. Prediction Error Decomposition

The expected prediction error of a model can be decomposed into three components:

$$\text{Expected Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Noise}$$

- **Bias** is the systematic error introduced by simplifying assumptions. A model that is too simple (e.g., a linear model for a non-linear relationship) has high bias.
- **Variance** is the sensitivity of the model to the specific training data. A model that is too complex (e.g., 50 predictors with 40 events) has high variance.
- **Irreducible noise** is the inherent randomness in the outcome that no model can capture.

7.3.2. The Tradeoff

Standard regression (OLS, maximum likelihood) is **unbiased** — meaning that if you could repeat your study many times on fresh samples and average the coefficient estimates, that average would land on the true value. There is no systematic tendency to over- or under-estimate. (This is a guaranteed mathematical property of least squares and maximum likelihood under the usual modelling assumptions; it is *why* these methods are the default starting point.) But notice what "unbiased on average" does *not* promise: that any *single* estimate from your *one* actual dataset is close to the truth. In high-dimensional or small-sample settings, the estimates scatter enormously from sample to sample — they are correct on average but wildly imprecise in any given sample. That scatter is the variance, and it is what penalisation attacks.

Penalised regression deliberately introduces a small amount of bias — shrinking coefficients toward zero — in exchange for a dramatic reduction in variance. The net effect is a model with **lower total prediction error** despite being biased.

Think of it this way: if the true coefficient is 0.5, an unbiased estimator might give you estimates of -2.3, 3.1, 0.8, -1.5, etc. across different samples (unbiased on average, but highly variable). A biased estimator that consistently gives you 0.3 is much closer to the truth in any given sample, even though it is systematically too low.

The figure below makes this tradeoff visual. The horizontal axis is the penalty strength λ, plotted so that the penalty *increases* as you move to the right (equivalently, model complexity *decreases* to the right). Watch the three curves as the penalty grows: **bias²** (red) rises, because shrinking coefficients toward zero introduces systematic error; **variance** (blue) falls, because a more constrained model is less sensitive to the particular sample; and the **total error** (dark line), which is their sum plus irreducible noise, is U-shaped. The dashed vertical line marks the sweet spot — the penalty that minimises total error. Too little penalty (far left) and variance dominates; too much (far right) and bias dominates. Cross-validation, introduced later, is simply an automatic way to find that dashed line from data.

```
library(ggplot2)

lambda <- seq(0.01, 5, length.out = 200)
bias2 <- 2 * (1 - exp(-0.5 * lambda))
variance <- 3 * exp(-lambda)
total <- bias2 + variance + 0.5

df_bv <- data.frame(
```

```
  lambda = rep(lambda, 3),
  error = c(bias2, variance, total),
  component = rep(c("Bias²", "Variance", "Total Error"), each = length(lambda))
)

ggplot(df_bv, aes(x = lambda, y = error, colour = component)) +
  geom_line(linewidth = 1.2) +
  scale_colour_manual(values = c("Bias²" = "#e74c3c", "Variance" = "#3498db",
                                  "Total Error" = "#2c3e50")) +
  geom_vline(xintercept = lambda[which.min(total)], linetype = "dashed", colour = "grey40")
  annotate("text", x = lambda[which.min(total)] + 0.3, y = max(total) * 0.9,
           label = "Optimal\npenalty", size = 3.5) +
  labs(x = "Penalty Strength (lambda)", y = "Error",
       colour = "Component") +
  scale_x_reverse() +
  theme_minimal(base_size = 14) +
  theme(legend.position = "bottom")
```

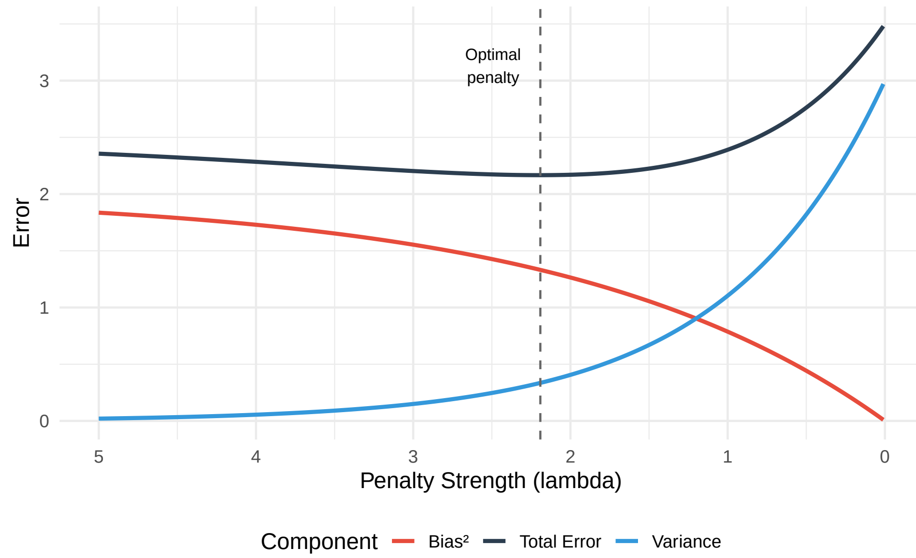


Figure 7.3.: The bias-variance tradeoff. As model complexity increases (or penalty decreases), bias decreases but variance increases. The sweet spot minimises total error.

7.4. The Penalised Regression Framework

7.4.1. The General Idea

In standard regression, we estimate coefficients by minimising a loss function (e.g., residual sum of squares for linear regression, negative log-likelihood for logistic regression):

$$\hat{\beta} = \arg\min_{\beta} \{\text{Loss}(\beta)\}$$

In penalised regression, we add a **penalty term** that grows with the size of the coefficients:

$$\hat{\beta} = \arg\min_{\beta} \{\text{Loss}(\beta) + \lambda \cdot \text{Penalty}(\beta)\}$$

The penalty term discourages large coefficients. The tuning parameter $\lambda \geq 0$ controls the strength of the penalty:

- $\lambda = 0$: no penalty, equivalent to standard regression
- $\lambda \to \infty$: all coefficients shrunk to zero (null model)
- Intermediate λ: the sweet spot, chosen by cross-validation

We will lean on **cross-validation** repeatedly to choose λ, so here is the idea in one sentence (it gets a full treatment in Section 7.8): cross-validation repeatedly hides part of the data, fits the model on the rest, and checks how well it predicts the hidden part — giving an honest estimate of out-of-sample performance for each candidate λ, so we can pick the value that predicts best on data the model has not seen.

The three main methods differ only in the form of the penalty.

! Standardisation Is Essential

Before applying penalised regression, all predictors must be **standardised**. Standardising a variable means two simple steps: subtract its mean (so the new variable is centred at zero) and divide by its standard deviation (so a one-unit change now means "one standard deviation"). For a predictor x_j with mean $\bar{x}_j$ and standard deviation s_j, each value becomes

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}.$$

After this, every predictor is on the same footing: mean zero, standard deviation one, no units.

Why does this matter here specifically? The penalty adds up the sizes of the coefficients ($\sum \beta_j^2$ for ridge, $\sum |\beta_j|$ for LASSO) and treats every coefficient the same way. But a coefficient's size depends on the units of its predictor. Measure a drug dose in milligrams and the coefficient is small; measure the *same* dose in kilograms and the coefficient is a million times larger — even though nothing about the science changed. Because the penalty only "sees" coefficient size, it would shrink the kilogram version far more aggressively than the milligram version. Standardising removes this arbitrariness, so the penalty reflects each predictor's genuine contribution rather than the accident of its measurement scale.
Both `glmnet` (R) and `sklearn` (Python) standardise internally by default and then report coefficients back on the original scale, so you usually do not have to do this by hand — but you should know it is happening, because it explains why penalised coefficients are not directly comparable to ordinary regression coefficients unless you account for it.

7.5. Ridge Regression (L2 Penalty)

7.5.1. Definition

Ridge regression, introduced by Hoerl and Kennard (1970), adds the **squared sum of coefficients** (L2 norm) as the penalty:

$$\hat{\beta}_{\text{Ridge}} = \arg\min_{\beta} \left\{ \underbrace{\sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2}_{\text{(1) how badly the model fits the data}} + \underbrace{\lambda \sum_{j=1}^{p} \beta_j^2}_{\text{(2) penalty for large coefficients}} \right\}$$

The equation looks dense, but it is saying something simple. Read it as a tug-of-war between two quantities the model tries to keep small at the same time:

- **Term (1) — misfit.** This is the ordinary regression criterion: for each patient i, take the gap between the observed outcome y_i and the model's prediction, square it, and add up over all patients. Smaller means the model fits the data better. On its own, minimising this term is just standard regression.
- **Term (2) — penalty.** This adds up the squared coefficients and multiplies by λ. It grows whenever coefficients get large, so it pushes them back toward zero. λ is the dial that sets how strong that pushback is.

"$\arg\min_\beta$" just means *find the set of coefficients β that makes the whole expression as small as possible.* So ridge regression looks for coefficients that fit the data well **and** stay modest in size, with λ deciding how much weight to give each goal. Set $\lambda = 0$ and the penalty vanishes, recovering ordinary regression; make λ very large and the coefficients are squeezed close to zero.

Note that the **intercept** β_0 is typically **not penalised** — only the predictor coefficients $\beta_1, \ldots, \beta_p$ are shrunk, because the intercept just sets the baseline level of the outcome and shrinking it would have no useful effect.

7.5.2. Key Properties

Ridge has three properties worth understanding. The first is the one that matters most in practice; the other two explain *why* ridge is so well-behaved.

1. **It shrinks every coefficient toward zero, but never all the way to zero.** No matter how strong the penalty, ridge makes coefficients smaller without ever switching any off completely — they get tiny,

but stay non-zero. The practical upshot is that ridge keeps *all* of your predictors in the model.

This is the place to define a term used throughout the chapter. **Variable selection** means automatically deciding which predictors to *keep* and which to *drop entirely* from the model — in effect, setting some coefficients to exactly zero so those variables disappear. It is what you would otherwise do by hand when you trim a model down to a short, usable list of predictors. **Ridge does not do variable selection**: because its coefficients never hit exactly zero, every predictor stays in, just with a reduced weight. (The LASSO, covered next, *does* perform variable selection — this is the single biggest difference between the two methods, so keep the contrast in mind.)

2. **It has a closed-form solution.** "Closed-form" means the answer can be computed directly with a single formula, rather than by an iterative search. For ridge that formula is:

$$\hat{\beta}_{\text{Ridge}} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$$

You do not need to manipulate this by hand — the software does. The point to take away is the role of the $\lambda\mathbf{I}$ piece: adding it guarantees the matrix can always be inverted (i.e., the equation always has a unique, stable solution), even when there are more predictors than patients ($p > n$) or when predictors are highly correlated — exactly the cases where ordinary regression fails. This is the mathematical reason ridge "rescues" problems that would otherwise have no stable answer.

3. **It handles multicollinearity gracefully.** *Multicollinearity* is when predictors carry overlapping information — systolic and diastolic blood pressure, or several related inflammatory markers. Ordinary regression reacts badly to this: it cannot decide how to split the shared effect between the correlated variables, so it produces wildly unstable coefficients that can even come out with the wrong sign.

Ridge calms this down by pulling correlated predictors toward a shared, modest value instead of letting them swing against each other.

7.5.3. When to Use Ridge

- You believe **many predictors** contribute to the outcome (none is truly zero)
- Predictors are **highly correlated** (multicollinearity)
- You care about **prediction accuracy** more than variable selection
- Example: predicting ICU mortality using dozens of physiological measurements, most of which carry some information

7.6. LASSO Regression (L1 Penalty)

7.6.1. Definition

The LASSO (Least Absolute Shrinkage and Selection Operator), introduced by Tibshirani (1996), uses the **sum of absolute values** (L1 norm) as the penalty:

$$\hat{\beta}_{\text{LASSO}} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

7.6.2. Key Properties

1. **Shrinks some coefficients to exactly zero.** This is the defining feature of the LASSO. When λ is large enough, some coefficients are forced to zero, effectively removing those predictors from the model. The LASSO thus performs **automatic variable selection**.

2. **No closed-form solution.** Unlike Ridge, the LASSO requires iterative optimisation algorithms (coordinate descent is the standard approach in `glmnet`).

3. **Selects at most n predictors** (when $p > n$). The LASSO can select at most $\min(n, p)$ predictors. When you have far more predictors than observations, this is a useful property.

4. **Struggles with correlated predictors.** When two predictors are highly correlated, the LASSO tends to pick one and drop the other arbitrarily. This can lead to **unstable variable selection** — which predictor is kept can change across different samples.

7.6.3. Why Sparsity Matters Clinically

In clinical research, we often want to identify the **most important predictors** from a larger set of candidates. This is common in:

- **Clinical prediction model development.** A parsimonious model with 5–10 predictors is more practical than one with 50.
- **Biomarker discovery.** From a panel of 100 candidate biomarkers, which ones are truly associated with the outcome?
- **Risk factor identification.** Among many potential lifestyle, genetic, and environmental factors, which are the strongest?

The LASSO's ability to set coefficients exactly to zero makes it a natural tool for these tasks.

> ⚠ LASSO Is Not a Substitute for Domain Knowledge
>
> While the LASSO performs automatic variable selection, the results should always be interpreted in light of clinical knowledge. A variable dropped by the LASSO is not necessarily unimportant — it may be correlated with another retained variable, or the sample may be too

small to detect its effect. Similarly, a variable retained by the LASSO is not necessarily causal.
Use the LASSO as a tool to *complement*, not replace, clinical reasoning about which variables matter.

7.7. Elastic Net: The Best of Both Worlds

7.7.1. Definition

The Elastic Net, proposed by Zou and Hastie (2005), combines both L1 and L2 penalties:

$$\hat{\beta}_{\text{EN}} = \arg\min_{\beta} \left\{ \text{Loss}(\beta) + \lambda \left[\alpha \sum_{j=1}^{p} |\beta_j| + \frac{(1-\alpha)}{2} \sum_{j=1}^{p} \beta_j^2 \right] \right\}$$

The parameter $\alpha \in [0, 1]$ controls the **mix** of L1 and L2:

- $\alpha = 1$: pure LASSO
- $\alpha = 0$: pure Ridge
- $0 < \alpha < 1$: Elastic Net (a blend)

7.7.2. Why Combine?

The Elastic Net addresses the LASSO's limitations while retaining its strengths:

1. **Correlated predictors.** When predictors are correlated, the LASSO picks one and drops the rest. The Elastic Net tends to **keep correlated predictors together** (the "grouping effect"), either including all of them or excluding all of them. This is often more scientifically reasonable.

2. **More than n predictors.** The LASSO selects at most n predictors. The Elastic Net can select more, which is useful in genomics where thousands of genes may be relevant.

3. **Stability.** The Elastic Net generally produces more stable results than the LASSO across different samples.

7.7.3. Choosing Alpha

In practice, α is often chosen by **cross-validation** alongside λ. Cross-validation (defined fully in Section 7.8) is the standard tool for choosing *any* tuning parameter: it scores each candidate value by how well the resulting model predicts data it was not trained on. Here we have two dials to set — α (the L1/L2 mix) and λ (the overall strength) — so we search over a grid of both:

- Try $\alpha \in \{0.1, 0.25, 0.5, 0.75, 0.9, 1.0\}$
- For each α, use cross-validation to find the optimal λ (this is the inner search `cv.glmnet` does automatically)
- Select the (α, λ) combination with the best cross-validated performance

In practice this is a short loop over the α values, calling cross-validation once per α and keeping the best pair — exactly the pattern shown in the R implementation walkthrough later in this chapter. If you would rather not search at all, $\alpha = 0.5$ is a reasonable default that balances L1 and L2 equally and works well in many clinical datasets.

7.8. Choosing Lambda: Cross-Validation

7.8.1. The Cross-Validation Procedure

The tuning parameter λ controls the penalty strength. We cannot estimate it from the data using the same approach as the regression coefficients — it must be chosen externally. **K-fold cross-validation** is the standard approach:

1. Divide the data into K folds (typically $K = 10$).
2. For each value of λ on a grid:

    a. For each fold, fit the model on the other $K - 1$ folds and predict the held-out fold.
    b. Calculate the prediction error (e.g., mean squared error, deviance, or misclassification rate).

3. Average the prediction error across folds for each λ.
4. Select the λ that minimises the average prediction error.

7.8.2. Lambda.min vs Lambda.1se

Cross-validation typically identifies two candidate values of λ:

- **lambda.min**: the value of λ that minimises the cross-validated error. This gives the model with the best predictive performance.

- **lambda.1se**: the largest λ whose cross-validated error is within one standard error of the minimum. This gives a **simpler, more parsimonious model** that is nearly as good as the best.

Which Lambda to Use?

In clinical prediction modelling, `lambda.min` is often preferred when predictive accuracy is the primary goal. In biomarker discovery or variable selection, `lambda.1se` is often preferred because it gives a more parsimonious model with fewer selected variables.
Steyerberg (2019) recommends reporting results for both and discussing the trade-off between parsimony and performance.

```r
library(glmnet)

set.seed(42)
n <- 200
p <- 30
X <- matrix(rnorm(n * p), n, p)
true_beta <- c(rep(1.5, 5), rep(0, 25))
y <- rbinom(n, 1, plogis(X %*% true_beta))

cv_fit <- cv.glmnet(X, y, family = "binomial", alpha = 1, nfolds = 10)
plot(cv_fit)
```

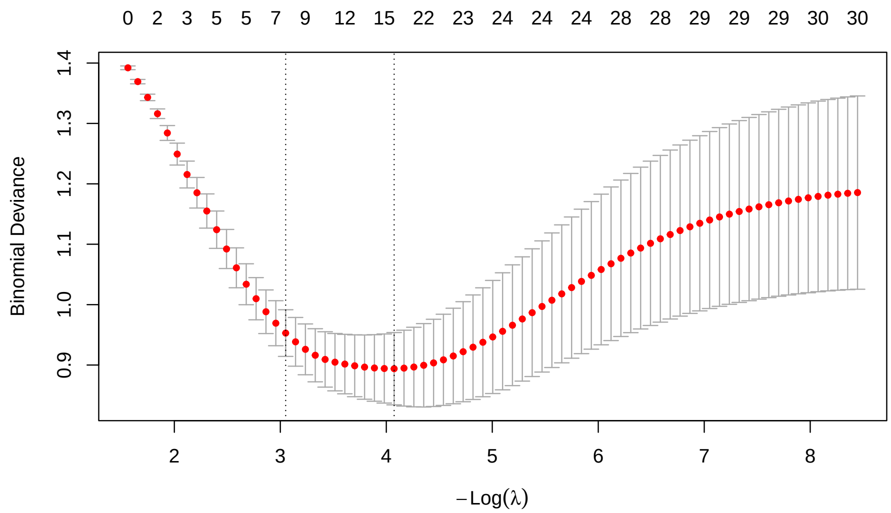


Figure 7.4.: Cross-validation curve for LASSO. The x-axis is -log(lambda) (so the penalty decreases left to right), the y-axis is cross-validated deviance. Vertical lines mark lambda.1se (left) and lambda.min (right).

How to read this plot. This is the standard `glmnet` cross-validation plot, and it repays a careful look. One orientation point first: current versions of `glmnet` label the x-axis $-\log(\lambda)$ (negative log lambda), so the penalty *decreases* from left to right. This is the opposite layout to a plain $\log(\lambda)$ axis, so read it carefully:

- The **x-axis** is $-\log(\lambda)$. Because of the minus sign, the penalty is *strongest* on the **left** (large λ, a heavily shrunk model) and *weakest* on the **right** (small λ, a complex model close to ordinary regression). Each red dot is one candidate λ.
- The **y-axis** is the cross-validated **deviance** — a measure of prediction error (lower is better; see the deviance explanation later in this

chapter). Each red dot is the average error across the 10 folds for that λ, and the **vertical bars** through it show ± 1 standard error of that average, i.e. how uncertain the estimate is.

- The numbers along the **top** count how many predictors remain in the model (have non-zero coefficients) at each λ — notice they *grow* from left to right (few on the strongly-penalised left, all of them on the weakly-penalised right) as the penalty releases more coefficients from zero.
- The two **dashed vertical lines** mark the two recommended choices. Because larger λ is now on the left, `lambda.1se` (the largest λ within one standard error of the best — the simpler, more parsimonious model) is the **left** line, and `lambda.min` (the λ with the lowest average error) is the **right** line.

So the plot is really a picture of the bias-variance tradeoff from earlier, drawn from real cross-validated data: error is high when the penalty is too strong (over-shrinking, far left) *and* when it is too weak (overfitting, far right), with a basin of good values in between that the two dashed lines bracket.

7.9. Regularisation Paths

A **regularisation path** (also called a coefficient trace or coefficient path) shows how each coefficient changes as λ varies from large (strong penalty) to small (weak penalty).

The code below fits a LASSO over the *whole range* of λ in one call (`glmnet` does this automatically — you do not specify a single λ), then plots every coefficient as a line. The two dashed vertical lines reuse the `lambda.min` and `lambda.1se` values found by cross-validation in the previous section, so you can see which coefficients survive at each recommended penalty. On the $-\log(\lambda)$ axis the penalty relaxes as you move left to right, so you watch predictors switch on one by one from left (strong penalty, all near zero) to right (weak penalty, all active).

```r
library(glmnet)

fit_lasso <- glmnet(X, y, family = "binomial", alpha = 1)
# glmnet 5.0 draws this plot on a -log(lambda) axis, so the reference
# lines are placed at -log(lambda.min) and -log(lambda.1se) to match.
plot(fit_lasso, xvar = "lambda", label = TRUE)
abline(v = -log(cv_fit$lambda.min), lty = 2, col = "blue")
abline(v = -log(cv_fit$lambda.1se), lty = 2, col = "red")
legend("topleft", legend = c("lambda.min", "lambda.1se"),
       col = c("blue", "red"), lty = 2, cex = 0.8)
```

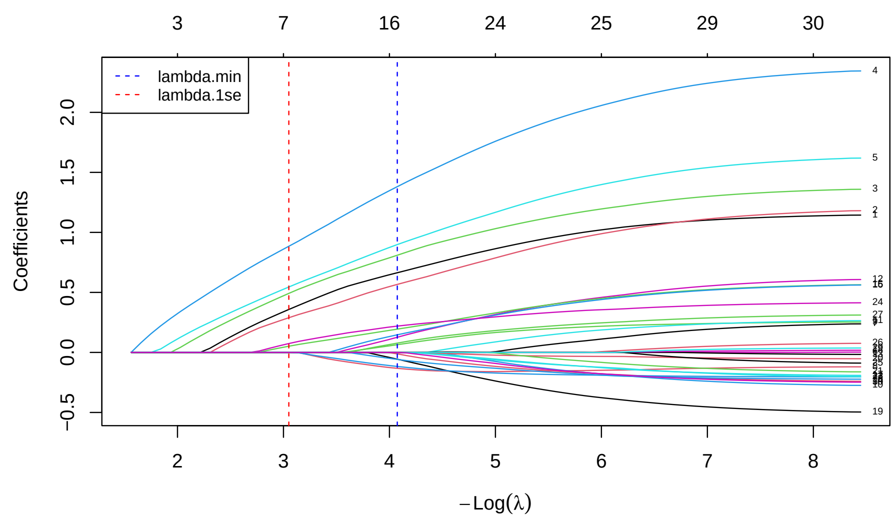


Figure 7.5.: LASSO regularisation path. Each line is one coefficient. As lambda decreases (left to right), more coefficients become non-zero. The five truly non-zero coefficients (shown as they enter the model) are recovered first.

7.9.1. How to Read a Regularisation Path

First, the x-axis. Current `glmnet` (version 5.0 and later) labels it $-\log(\lambda)$ — *negative* log lambda. The minus sign matters: it means the penalty *decreases* as you move from left to right. So large penalties sit on the **left** and small penalties on the **right** — the mirror image of a plain $\log(\lambda)$ axis. This is the same axis as the cross-validation plot in the previous section, so the two figures line up. (If you have seen older `glmnet` figures with a $\log(\lambda)$ axis, the left–right orientation there is reversed; always read the axis label.)

With that settled, here is how to read the curves:

- **Left side** (large λ, small $-\log\lambda$): strong penalty, all coefficients squeezed to near zero.
- **Right side** (small λ, large $-\log\lambda$): weak penalty, coefficients approach their ordinary (unpenalised) regression values.
- **Order of entry**: predictors whose lines peel away from zero **earliest — i.e. furthest to the left, while the penalty is still strong** — are the most important, because they survive the heaviest shrinkage.
- **Sign and magnitude**: the sign and eventual magnitude of each line reveal the direction and strength of that predictor's association.

The two dashed vertical lines mark the cross-validated `lambda.min` and `lambda.1se`, so you can read off which coefficients are non-zero at each recommended penalty. This plot is extremely useful for understanding the relative importance of predictors and the stability of the model across different penalty strengths.

7.10. Clinical Example: Breast Cancer Diagnosis from Image Features

7.10.1. The Dataset

The Original Wisconsin Breast Cancer dataset (the `BreastCancer` data in the `mlbench` package, used in the code below) contains 699 biopsy samples scored on **9 cytological attributes**, each graded 1–10 by a pathologist: clump thickness, uniformity of cell size, uniformity of cell shape, marginal adhesion, single epithelial cell size, bare nuclei, bland chromatin, normal nucleoli, and mitoses. The outcome is the diagnosis: malignant or benign.

With 9 predictors and several hundred observations, this is not a $p > n$ problem, but it is still a useful setting for penalised regression because many of the cytological features are highly correlated (cell-size and cell-shape uniformity, for instance, tend to move together). Correlated predictors are exactly where the three methods behave differently, which is what makes the comparison below instructive.

> **What is "deviance"?**
>
> The code below reports model performance as **deviance**, a term that recurs whenever we fit logistic (and other generalised linear) models, so it is worth pinning down. Deviance is a measure of **how poorly a model fits**, built from the likelihood: it is defined as $-2\times$ the log-likelihood of the fitted model. Lower deviance means a better fit. For ordinary linear regression, deviance reduces to the residual sum of squares, so you can think of it as the natural extension of "squared prediction error" to logistic regression and other models where squared error does not directly apply. When we report **cross-validated deviance**, we mean the deviance computed on held-out folds — so it measures how well the model predicts *new* data, not how well it memorised the training data. It is the default error measure `glmnet` minimises when choosing λ for a binary outcome.

The worked example below is shown in both R (`glmnet`) and Python (`scikit-learn`). Both fit LASSO, Ridge, and Elastic Net to the same kind of problem and compare the resulting coefficients; the datasets shipped with each language differ slightly (R's `mlbench` carries the 9-attribute Original Wisconsin data; scikit-learn ships the 30-feature Diagnostic version), so the exact numbers differ, but the *pattern* is identical and is the point of the example.

7.10.1.1. R

```r
library(glmnet)
library(ggplot2)
```

```
# Load the dataset (9 cytological attributes, Original Wisconsin data)
# install.packages("mlbench")
data(BreastCancer, package = "mlbench")

# Prepare the data
bc <- BreastCancer[complete.cases(BreastCancer), ]
bc$diagnosis <- ifelse(bc$Class == "malignant", 1, 0)

# Extract numeric predictors (the 9 cytological attributes, columns 2-10)
X <- as.matrix(apply(bc[, 2:10], 2, as.numeric))
y <- bc$diagnosis

# Fit LASSO with cross-validation
set.seed(42)
cv_lasso <- cv.glmnet(X, y, family = "binomial", alpha = 1, nfolds = 10)

cat("Lambda.min:", round(cv_lasso$lambda.min, 4), "\n")
cat("Lambda.1se:", round(cv_lasso$lambda.1se, 4), "\n\n")

# Coefficients at lambda.1se (sparser model)
coef_1se <- coef(cv_lasso, s = "lambda.1se")
cat("Coefficients at lambda.1se:\n")
print(round(coef_1se[coef_1se[, 1] != 0, , drop = FALSE], 4))

# Coefficients at lambda.min (best predictive model)
coef_min <- coef(cv_lasso, s = "lambda.min")
cat("\nCoefficients at lambda.min:\n")
print(round(coef_min[coef_min[, 1] != 0, , drop = FALSE], 4))

# Compare LASSO, Ridge, and Elastic Net
set.seed(42)
cv_ridge <- cv.glmnet(X, y, family = "binomial", alpha = 0, nfolds = 10)
cv_enet <- cv.glmnet(X, y, family = "binomial", alpha = 0.5, nfolds = 10)
```

```r
cat("\n--- Cross-Validated Deviance (at lambda.min) ---\n")
cat("Ridge:       ", round(min(cv_ridge$cvm), 4), "\n")
cat("Elastic Net:", round(min(cv_enet$cvm), 4), "\n")
cat("LASSO:       ", round(min(cv_lasso$cvm), 4), "\n")
```

```
Lambda.min: 0.0021
Lambda.1se: 0.0166

Coefficients at lambda.1se:
10 x 1 sparse Matrix of class "dgCMatrix"
                lambda.1se
(Intercept)        -6.2171
Cl.thickness        0.3250
Cell.size           0.1183
Cell.shape          0.2039
Marg.adhesion       0.1165
Epith.c.size        0.0615
Bare.nuclei         0.2941
Bl.cromatin         0.2267
Normal.nucleoli     0.1274
Mitoses             0.0040

Coefficients at lambda.min:
10 x 1 sparse Matrix of class "dgCMatrix"
                lambda.min
(Intercept)        -8.9990
Cl.thickness        0.4795
Cell.size           0.0208
Cell.shape          0.3009
Marg.adhesion       0.2722
Epith.c.size        0.0878
Bare.nuclei         0.3597
```

```
Bl.cromatin         0.3871
Normal.nucleoli     0.1901
Mitoses             0.3298

--- Cross-Validated Deviance (at lambda.min) ---
Ridge:       0.2064
Elastic Net: 0.1796
LASSO:       0.1799
```

```r
# Extract coefficients at lambda.min for each method
coef_ridge <- as.vector(coef(cv_ridge, s = "lambda.min"))[-1]
coef_enet <- as.vector(coef(cv_enet, s = "lambda.min"))[-1]
coef_lasso <- as.vector(coef(cv_lasso, s = "lambda.min"))[-1]

pred_names <- colnames(X)

comp_df <- data.frame(
  predictor = rep(pred_names, 3),
  coefficient = c(coef_ridge, coef_enet, coef_lasso),
  method = rep(c("Ridge", "Elastic Net", "LASSO"), each = length(pred_names))
)

ggplot(comp_df, aes(x = predictor, y = coefficient, fill = method)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  scale_fill_manual(values = c("Ridge" = "#3498db", "Elastic Net" = "#2ecc71",
                               "LASSO" = "#e74c3c")) +
  coord_flip() +
  labs(x = "Predictor", y = "Coefficient (standardised)",
       fill = "Method") +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom")
```

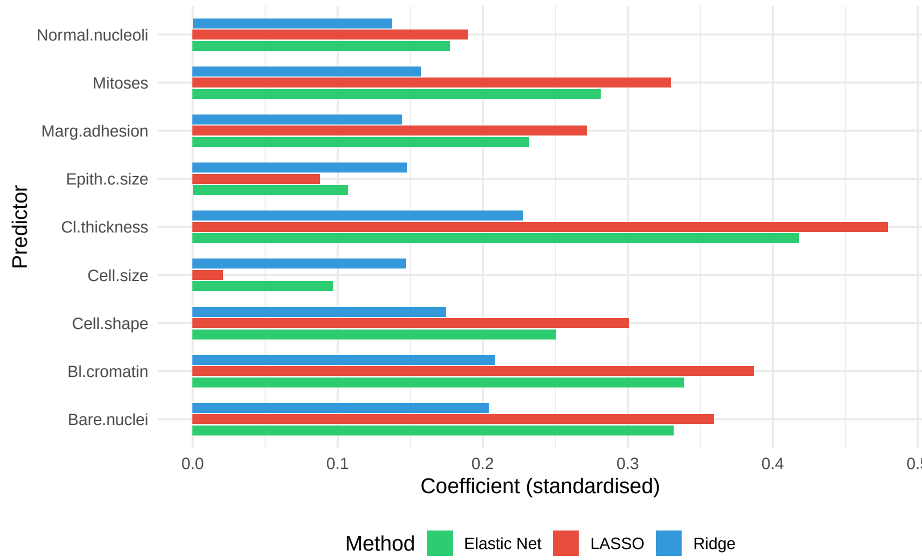


Figure 7.6.: Coefficient estimates from Ridge, Elastic Net, and LASSO for the breast cancer dataset (R). Ridge retains all predictors; LASSO sets some to zero; Elastic Net is intermediate.

7.10.1.2. Python

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_breast_cancer
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegressionCV

# scikit-learn ships the 30-feature Diagnostic version of the dataset
data = load_breast_cancer()
```

```
X = StandardScaler().fit_transform(data.data)
y = data.target  # 1 = benign, 0 = malignant in sklearn's coding
feature_names = data.feature_names

# Fit LASSO, Ridge, and Elastic Net, each tuned by cross-validation
# In sklearn, C = 1/lambda, so it tunes C; penalty selects the norm
common = dict(Cs=20, cv=10, scoring="neg_log_loss",
              max_iter=10000, random_state=42)
lasso = LogisticRegressionCV(penalty="l1", solver="saga", **common).fit(X, y)
ridge = LogisticRegressionCV(penalty="l2", solver="lbfgs", **common).fit(X, y)
enet = LogisticRegressionCV(penalty="elasticnet", solver="saga",
                            l1_ratios=[0.5], **common).fit(X, y)

n_lasso = int(np.sum(np.abs(lasso.coef_[0]) > 1e-6))
n_ridge = int(np.sum(np.abs(ridge.coef_[0]) > 1e-6))
n_enet = int(np.sum(np.abs(enet.coef_[0]) > 1e-6))
print(f"Non-zero coefficients (of {X.shape[1]}):")
print(f"  Ridge:       {n_ridge}")
print(f"  Elastic Net: {n_enet}")
print(f"  LASSO:       {n_lasso}")
```

```
Non-zero coefficients (of 30):
  Ridge:       30
  Elastic Net: 28
  LASSO:       19
```

```
# Show the first 10 features for legibility
k = 10
idx = np.arange(k)
width = 0.27

plt.figure(figsize=(9, 6))
```

```
plt.barh(idx - width, ridge.coef_[0][:k], height=width, label="Ridge", color=
plt.barh(idx, enet.coef_[0][:k], height=width, label="Elastic Net", color="#2
plt.barh(idx + width, lasso.coef_[0][:k], height=width, label="LASSO", color=
plt.yticks(idx, [n.replace(" ", "\n") for n in feature_names[:k]], fontsize=8
plt.xlabel("Coefficient (standardised)")
plt.legend(loc="best")
plt.tight_layout()
plt.show()
```

```
([<matplotlib.axis.YTick object at 0x13985a860>, <matplotlib.axis.YTick objec
```

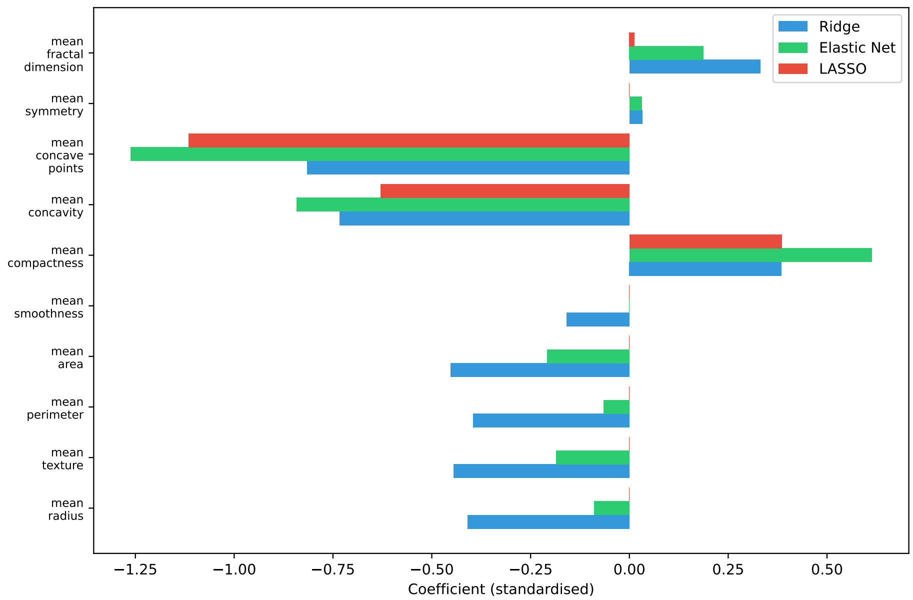


Figure 7.7.: Coefficient estimates from Ridge, Elastic Net, and LASSO for the breast cancer dataset (Python). Ridge retains all predictors; LASSO sets many to zero; Elastic Net is intermediate.

7.10.2. Interpretation

Several patterns emerge, and they are visible directly in the coefficient figure above — compare the three coloured bars for each predictor (the counts below describe the R fit on the 9-attribute data; the Python fit shows the same qualitative pattern on its 30 features):

1. **Ridge** retains all predictors: every blue bar is non-zero, with the weights spread across correlated features rather than concentrated on one.
2. **LASSO** selects a subset — many red bars sit at exactly zero (those predictors are dropped), leaving only the features that carry the most unique information.
3. **Elastic Net** behaves intermediately: more green bars are non-zero than red (LASSO) ones, but some are still shrunk to zero — visibly between the other two.
4. The cross-validated deviance (printed above) is similar across all three methods, confirming that penalised regression is robust to the choice of penalty type in this dataset.

7.11. When to Use Which Method

The following decision framework can guide your choice:

7.11.1. Summary Table

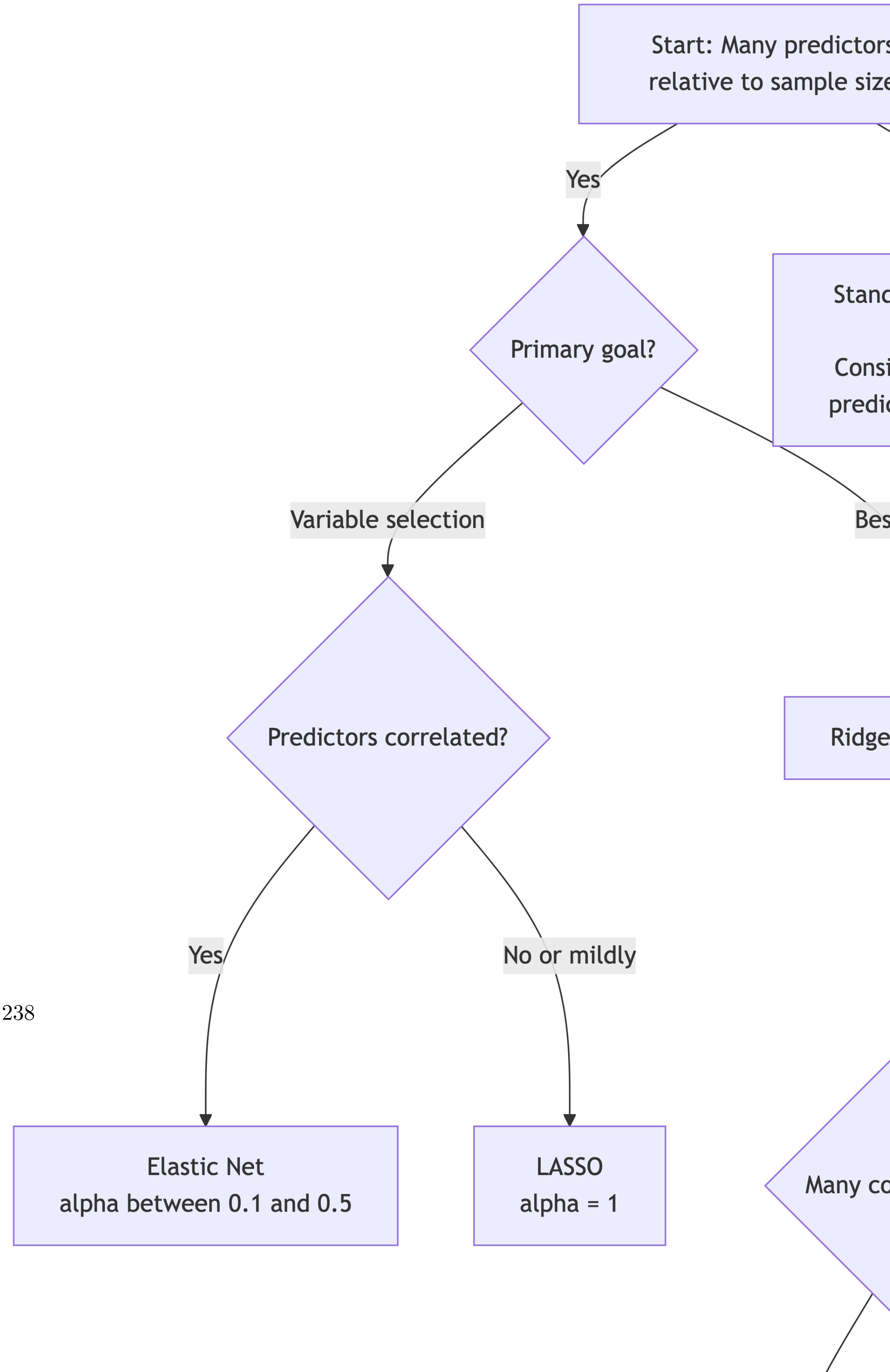

Start: Many predictors
relative to sample size
Yes
Primary goal?
Variable selection
Predictors correlated?
Yes
No or mildly
Elastic Net
alpha between 0.1 and 0.5
LASSO
alpha = 1

Table 7.1.: When to use Ridge, LASSO, or Elastic Net

Scenario	Recommended Method	Rationale
Few predictors, large sample	Standard regression (or Ridge for mild shrinkage)	Penalisation not needed, but mild shrinkage can still improve predictions
Many predictors, want variable selection	LASSO	Sets unimportant predictors to exactly zero
Many correlated predictors, want variable selection	Elastic Net	Groups correlated predictors together
Many correlated predictors, want best prediction	Ridge	Keeps all predictors, handles collinearity well
Genomics / high-dimensional ($p >> n$)	Elastic Net or LASSO	Need sparsity; Elastic Net more stable
Clinical prediction model	Ridge or Elastic Net	Stability and prediction are key

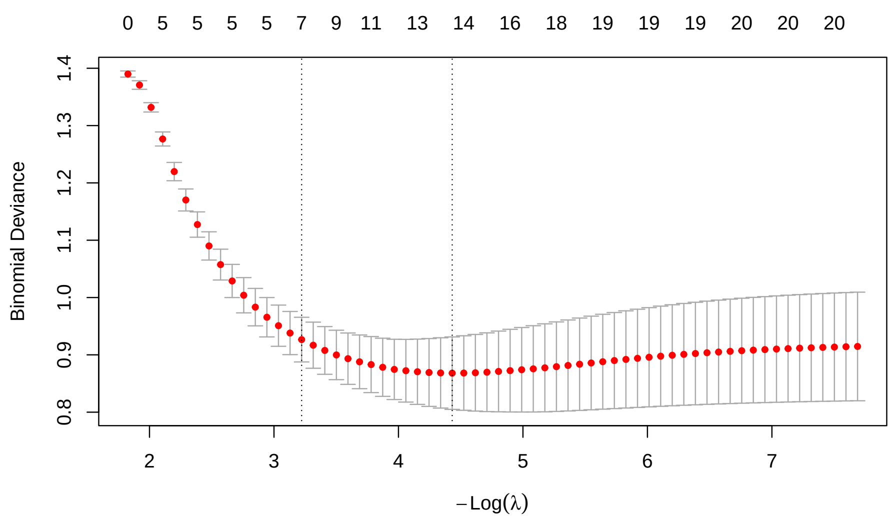
7.12. Implementation

The two walkthroughs below are self-contained: each simulates a small labelled dataset, splits it into training and test sets, then fits, tunes, and uses a penalised model. To apply them to your own study, replace the simulated-data block with your own predictors and outcome.

7.12.0.1. R: Using `glmnet`

The `glmnet` package is the standard implementation for penalised regression in R. It is fast (it uses the coordinate-descent algorithm of Friedman, Hastie, and Tibshirani (2010)), handles logistic and Cox regression as well as linear, and has excellent cross-validation support. For the underlying theory, the standard reference is *The Elements of Statistical Learning* (Hastie et al. 2009). Note that in `glmnet`, `X` must be a **matrix** (not a data frame) and `y` a vector — forgetting this is the most common cause of errors.

```r
library(glmnet)

# --- Step 0: Simulated example data (replace with your own) ---
set.seed(42)
n <- 300; p <- 20
X_all <- matrix(rnorm(n * p), n, p)
colnames(X_all) <- paste0("x", 1:p)
true_beta <- c(rep(1.2, 5), rep(0, p - 5))   # only first 5 predictors matter
y_all <- rbinom(n, 1, plogis(X_all %*% true_beta))

# Train/test split (75% / 25%)
train <- sample(seq_len(n), size = 0.75 * n)
X <- X_all[train, ];  y <- y_all[train]
X_test <- X_all[-train, ];  y_test <- y_all[-train]

# --- Step 1: Fit LASSO with cross-validation ---
# alpha = 1 for LASSO, 0 for Ridge, 0-1 for Elastic Net
# family = "gaussian" (linear), "binomial" (logistic), "cox" (survival)
cv_fit <- cv.glmnet(X, y, family = "binomial", alpha = 1, nfolds = 10)

# --- Step 2: Examine the CV curve (deviance vs log lambda) ---
plot(cv_fit)
```




```r
# --- Step 3: Extract coefficients at the two recommended lambdas ---
cat("Non-zero coefficients at lambda.1se (parsimonious model):\n")
coef_1se <- coef(cv_fit, s = "lambda.1se")
print(round(coef_1se[coef_1se[, 1] != 0, , drop = FALSE], 3))

# --- Step 4: Predict on the held-out test set ---
predictions <- predict(cv_fit, newx = X_test, s = "lambda.min",
                       type = "response")   # predicted probabilities
cat("\nFirst few test-set predicted probabilities:\n")
print(round(head(as.vector(predictions)), 3))

# --- Step 5: Search over alpha for the best Elastic Net mix ---
alphas <- c(0, 0.25, 0.5, 0.75, 1)
min_errors <- sapply(alphas, function(a) {
  min(cv.glmnet(X, y, family = "binomial", alpha = a, nfolds = 10)$cvm)
})
```

```r
best_alpha <- alphas[which.min(min_errors)]
cat("\nBest alpha (0 = Ridge, 1 = LASSO):", best_alpha, "\n")
```

```
Non-zero coefficients at lambda.1se (parsimonious model):
8 x 1 sparse Matrix of class "dgCMatrix"
            lambda.1se
(Intercept)      0.267
x1               0.619
x2               0.613
x3               0.662
x4               0.462
x5               0.525
x11              0.051
x14             -0.014

First few test-set predicted probabilities:
[1] 0.722 0.960 0.961 0.147 0.330 0.620

Best alpha (0 = Ridge, 1 = LASSO): 1
```

7.12.0.2. Python: Using `scikit-learn`

In scikit-learn the regularisation strength is parameterised as `C = 1/lambda` (so *larger* `C` means *weaker* penalty), and `LogisticRegression` does **not** standardise automatically — you must scale the predictors yourself with `StandardScaler`.

```python
import numpy as np
from sklearn.linear_model import LogisticRegression, LogisticRegressionCV
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
```

```
# --- Step 0: Simulated example data (replace with your own) ---
rng = np.random.default_rng(42)
n, p = 300, 20
X_all = rng.normal(size=(n, p))
true_beta = np.array([1.2] * 5 + [0.0] * (p - 5))   # only first 5 matter
prob = 1 / (1 + np.exp(-(X_all @ true_beta)))
y_all = rng.binomial(1, prob)

X_train, X_test, y_train, y_test = train_test_split(
    X_all, y_all, test_size=0.25, random_state=42)

# Standardise (fit the scaler on training data only, then apply to both)
scaler = StandardScaler().fit(X_train)
X_train_s = scaler.transform(X_train)
X_test_s = scaler.transform(X_test)

# --- Step 1: LASSO (L1) tuned by cross-validation ---
lasso_cv = LogisticRegressionCV(
    penalty="l1", solver="saga", Cs=30, cv=10,
    scoring="neg_log_loss", max_iter=10000, random_state=42
).fit(X_train_s, y_train)

print(f"Best C (= 1/lambda): {lasso_cv.C_[0]:.4f}")
print("Non-zero coefficients:")
for j, coef in enumerate(lasso_cv.coef_[0]):
    if abs(coef) > 1e-6:
        print(f"  x{j+1}: {coef:.3f}")

# --- Step 2: Predict on the held-out test set ---
test_probs = lasso_cv.predict_proba(X_test_s)[:, 1]
print("\nFirst few test-set predicted probabilities:")
print(np.round(test_probs[:6], 3))
```

```
# --- Step 3: Ridge and Elastic Net for comparison ---
ridge_cv = LogisticRegressionCV(
    penalty="l2", solver="lbfgs", Cs=30, cv=10,
    scoring="neg_log_loss", max_iter=10000, random_state=42
).fit(X_train_s, y_train)
enet_cv = LogisticRegressionCV(
    penalty="elasticnet", solver="saga", Cs=30, cv=10,
    l1_ratios=[0.1, 0.5, 0.9], scoring="neg_log_loss",
    max_iter=10000, random_state=42
).fit(X_train_s, y_train)
print(f"\nElastic Net best l1_ratio (alpha): {enet_cv.l1_ratio_[0]:.2f}")

# --- Step 4: Regularisation path for the LASSO ---
Cs = np.logspace(-3, 2, 60)
coefs = []
for C in Cs:
    m = LogisticRegression(penalty="l1", C=C, solver="saga", max_iter=10000)
    m.fit(X_train_s, y_train)
    coefs.append(m.coef_[0])
coefs = np.array(coefs)

plt.figure(figsize=(9, 6))
for j in range(p):
    plt.plot(np.log(1.0 / Cs), coefs[:, j])
plt.xlabel("log(lambda)")
plt.ylabel("Coefficient")
plt.title("LASSO Regularisation Path")
plt.tight_layout()
plt.show()
```

```
Best C (= 1/lambda): 0.3857
Non-zero coefficients:
  x1: 0.904
```

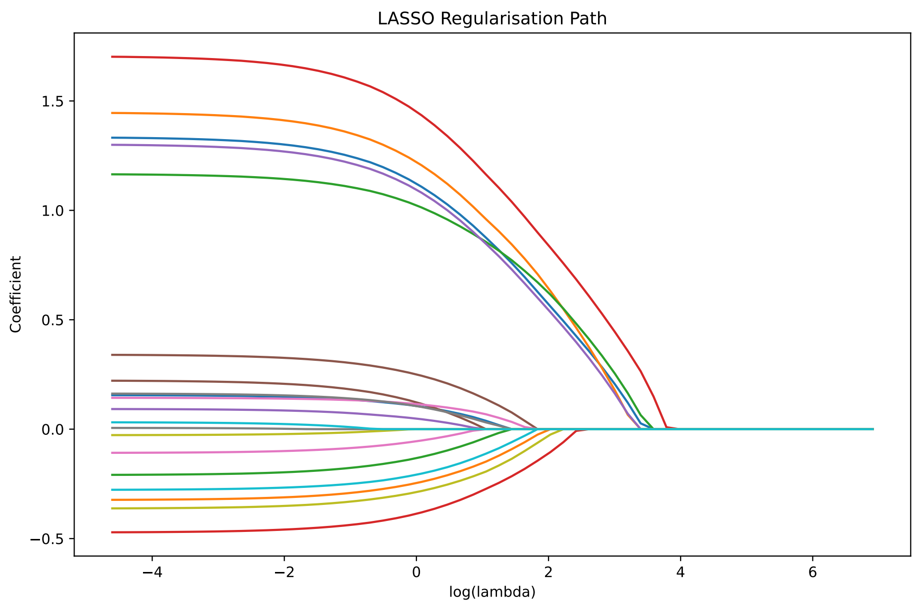
```
  x2: 0.989
  x3: 0.875
  x4: 1.197
  x5: 0.875
  x6: 0.151
  x7: -0.000
  x9: -0.205
  x10: -0.124
  x11: 0.048
  x12: -0.160
  x13: -0.050
  x14: -0.287
  x15: 0.001
  x16: 0.015
  x17: 0.075
  x18: 0.039

First few test-set predicted probabilities:
[0.498 0.069 0.449 0.957 0.47  0.582]

Elastic Net best l1_ratio (alpha): 0.90
LogisticRegression(C=0.001, max_iter=10000, penalty='l1', solver='saga')
LogisticRegression(C=0.0012154742500762859, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.0014773776525985112, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.001795714494371641, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.0021826447283974874, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.0026529484644318944, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.0032245905452963947, max_iter=10000, penalty='l1',
                   solver='saga')
```

```
LogisticRegression(C=0.003919406774847221, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.004763938010401341, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.005790443980602483, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.007038135554931555, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.008554672535565685, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.0103979841848149, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.012638482029342977, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.015361749466718281, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.018671810912919196, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.022695105366946685, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.027585316176291827, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.033529241492495566, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.040753929658717755, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.0495353520895917, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.060208944933361284, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.07318242219076174, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.08895134973108232, max_iter=10000, penalty='l1',
                   solver='saga')
```

```
LogisticRegression(C=0.10811807510766078, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.13141473626117553, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.1597312280060254, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.19414919457438817, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.23598334667821938, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.2868316813342009, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.34863652276780843, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.4237587160604063, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.5150678076168121, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.6260516572014815, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.7609496685459876, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=0.9249147277217336, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=1.1242100350620863, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=1.3664483492953243, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=1.6608827826277148, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=2.018760254679039, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=2.4537511066398165, max_iter=10000, penalty='l1',
                   solver='saga')
```

```
LogisticRegression(C=2.982471286216888, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=3.6251170499885315, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=4.406236427773573, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=5.3556669177068965, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=6.509675230458164, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=7.912342618981318, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=9.617248711152964, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=11.689518164985776, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=14.208308325339209, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=17.269832906594324, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=20.991037201085547, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=25.514065200312874, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=31.011689265747755, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=37.69390975388364, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=45.81597669054491, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=55.68813990945267, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=67.68750009458527, max_iter=10000, penalty='l1',
                   solver='saga')
```

```
LogisticRegression(C=82.27241341700457, max_iter=10000, penalty='l1',
                   solver='saga')
LogisticRegression(C=100.0, max_iter=10000, penalty='l1', solver='saga')
```



7.13. Common Pitfalls

Pitfalls to Avoid

1. **Forgetting to standardise.** If you do not standardise predictors, the penalty will disproportionately affect predictors with larger scales. `glmnet` and `sklearn` handle this internally, but be aware when interpreting coefficients — they are on the

standardised scale unless you back-transform.

2. **Interpreting LASSO-selected variables as "important."** A variable dropped by the LASSO may be correlated with a retained variable. It is not necessarily unimportant — it is just redundant *given the other predictors in the model.*

3. **Reporting p-values for LASSO-selected variables.** This is subtle and frequently done wrong, so it is worth spelling out. A common workflow is: run the LASSO, see which variables survive, then refit an *ordinary* (unpenalised) regression on just those survivors and report its p-values. Those p-values are **invalid** — they are far too small, and the confidence intervals far too narrow. The reason is **double-dipping**: the data were already used once to *choose* the variables (the LASSO only kept the ones that looked strongest in this particular sample), and then the *same* data are used again to *test* them. Standard p-values assume the variables were specified in advance, independently of the data; but here the selection step has cherry-picked the most promising-looking variables, so of course they look significant when re-tested on the same data. A variable that survived partly by chance will appear "significant" even when its true effect is zero, inflating the false-positive rate. The honest treatment of this problem — doing valid inference *after* a data-driven selection step — is called **post-selection inference** and is an active research area; practical tools include sample-splitting (select on one half, test on the other) and dedicated methods such as the `selectiveInference` R package. The safe default for this course: treat LASSO output as a *list of selected predictors and a prediction model*, not as a source of p-values.

4. **Using the same data for tuning and evaluation.** The cross-validation used to select λ should be considered part of

model building. To honestly assess model performance, you need a **separate** test set or nested cross-validation.

5. **Ignoring the 1se rule.** Researchers often default to `lambda.min`, but `lambda.1se` frequently gives a much simpler model with negligibly worse performance. Always consider both.

6. **Applying penalised regression when standard regression suffices.** If you have 10 predictors and 1000 observations, standard regression is fine. Penalisation will not hurt (much), but it adds complexity without clear benefit unless you are specifically developing a prediction model.

7.14. Penalised Regression in Prediction Modelling

In the context of clinical prediction models — which we discuss in later chapters — penalised regression plays a specific role:

1. **Shrinkage corrects for optimism.** Prediction models developed on finite samples are always overfitted to some degree. The apparent performance (e.g., C-statistic, calibration) is too optimistic. Penalisation provides a principled way to shrink coefficients, correcting for this optimism *during* model development rather than *after*.

2. **Alternative to backward selection.** Traditional backward stepwise selection is known to inflate Type I error rates, produce unstable models, and overestimate effect sizes. LASSO and Elastic Net provide a more principled approach to variable selection with better statistical properties.

3. **Uniform shrinkage vs differential shrinkage.** A simple post-hoc shrinkage factor (as recommended by Steyerberg (2019)) applies the same shrinkage to all coefficients. Penalised regression allows

differential shrinkage — strong effects are shrunk less than weak effects — which is more efficient.

7.15. Exercises

7.15.1. Exercise 1: Conceptual Questions

a) A colleague tells you: "I used LASSO and it removed age from my model predicting mortality. This means age is not a risk factor for death." Explain why this interpretation is incorrect.

b) You fit a Ridge regression model to predict length of hospital stay using 20 clinical variables. The cross-validated RMSE is 2.1 days. You then fit a standard linear regression on the same data and get RMSE = 1.8 days. Your colleague says the standard model is better because it has lower RMSE. What is wrong with this comparison?

c) Explain in one sentence why the LASSO produces exact zeros but Ridge does not.

7.15.2. Exercise 2: Predicting Diabetes Onset

Using the Pima Indians Diabetes dataset (available in R and Python), build penalised regression models to predict diabetes onset.

7.15.2.1. R

```
library(glmnet)
library(mlbench)

# Load data
```

```r
data(PimaIndiansDiabetes2, package = "mlbench")
pima <- na.omit(PimaIndiansDiabetes2)

# Prepare data
X <- as.matrix(pima[, 1:8])
y <- ifelse(pima$diabetes == "pos", 1, 0)

# Task 1: Fit a LASSO model with 10-fold cross-validation
set.seed(123)
cv_lasso <- cv.glmnet(X, y, family = "binomial", alpha = 1, nfolds = 10)

# Task 2: Plot the cross-validation curve
plot(cv_lasso)

# Task 3: Which variables are selected at lambda.1se?
coef(cv_lasso, s = "lambda.1se")

# Task 4: Fit Ridge and Elastic Net (alpha = 0.5) models
cv_ridge <- cv.glmnet(X, y, family = "binomial", alpha = 0, nfolds = 10)
cv_enet <- cv.glmnet(X, y, family = "binomial", alpha = 0.5, nfolds = 10)

# Task 5: Compare the three methods:
#   - How many variables does each select (at lambda.1se)?
#   - What is the cross-validated deviance for each?
#   - Plot the regularisation path for the LASSO model

# Task 6: Which method would you recommend for this problem, and why?
```

7.15.2.2. Python

```python
import numpy as np
import pandas as pd
```

```
from sklearn.linear_model import LogisticRegressionCV
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import cross_val_score
from sklearn.datasets import load_diabetes  # Note: this is regression
import matplotlib.pyplot as plt

# Load the Pima Indians Diabetes dataset
# Available via: https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-
# Or use the following URL:
url = ("https://raw.githubusercontent.com/jbrownlee/Datasets/master/"
       "pima-indians-diabetes.data.csv")
columns = ['pregnancies', 'glucose', 'blood_pressure', 'skin_thickness',
           'insulin', 'bmi', 'diabetes_pedigree', 'age', 'outcome']
pima = pd.read_csv(url, names=columns)

# Remove rows with zero values in columns where zero is not meaningful
cols_no_zero = ['glucose', 'blood_pressure', 'skin_thickness',
                'insulin', 'bmi']
pima[cols_no_zero] = pima[cols_no_zero].replace(0, np.nan)
pima = pima.dropna()

X = pima[columns[:-1]].values
y = pima['outcome'].values

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Task 1: Fit LASSO with cross-validation
lasso_cv = LogisticRegressionCV(
    penalty='l1', solver='saga', Cs=50, cv=10,
    scoring='neg_log_loss', max_iter=10000, random_state=42
)
lasso_cv.fit(X_scaled, y)
```

```
print("LASSO selected features:")
for name, coef in zip(columns[:-1], lasso_cv.coef_[0]):
    if abs(coef) > 1e-6:
        print(f"  {name}: {coef:.4f}")

# Task 2: Fit Ridge with cross-validation
ridge_cv = LogisticRegressionCV(
    penalty='l2', solver='lbfgs', Cs=50, cv=10,
    scoring='neg_log_loss', max_iter=10000, random_state=42
)
ridge_cv.fit(X_scaled, y)

# Task 3: Fit Elastic Net with cross-validation
enet_cv = LogisticRegressionCV(
    penalty='elasticnet', solver='saga', Cs=50, cv=10,
    l1_ratios=[0.25, 0.5, 0.75], scoring='neg_log_loss',
    max_iter=10000, random_state=42
)
enet_cv.fit(X_scaled, y)

# Task 4: Compare cross-validated scores
for name, model in [('LASSO', lasso_cv), ('Ridge', ridge_cv),
                     ('Elastic Net', enet_cv)]:
    scores = cross_val_score(model, X_scaled, y, cv=10,
                             scoring='neg_log_loss')
    print(f"{name}: Mean CV log-loss = {-scores.mean():.4f} "
          f"(+/- {scores.std():.4f})")

# Task 5: Plot regularisation path for LASSO
Cs = np.logspace(-2, 2, 80)
coefs = []
for C in Cs:
    model = LogisticRegressionCV(penalty='l1', solver='saga', Cs=[C],
```

```
                                      cv=5, max_iter=10000, random_state=42)
    model.fit(X_scaled, y)
    coefs.append(model.coef_[0])

coefs = np.array(coefs)
plt.figure(figsize=(10, 6))
for j, name in enumerate(columns[:-1]):
    plt.plot(np.log10(1/Cs), coefs[:, j], label=name)
plt.xlabel('log10(lambda)')
plt.ylabel('Coefficient')
plt.title('LASSO Regularisation Path - Pima Diabetes')
plt.legend(loc='best')
plt.tight_layout()
plt.show()
```

7.15.3. Exercise 3: The Effect of Correlation

This exercise demonstrates how LASSO and Ridge handle correlated predictors differently.

7.15.3.1. R

```r
library(glmnet)
library(MASS)

set.seed(2025)
n <- 200

# Create 4 correlated predictors (correlation ~ 0.9)
Sigma <- matrix(0.9, 4, 4)
diag(Sigma) <- 1
```

```
X_corr <- mvrnorm(n, mu = rep(0, 4), Sigma = Sigma)

# Add 6 independent predictors (noise)
X_indep <- matrix(rnorm(n * 6), n, 6)
X <- cbind(X_corr, X_indep)
colnames(X) <- c(paste0("Corr_", 1:4), paste0("Noise_", 1:6))

# True model: all 4 correlated predictors have effect = 0.5
true_beta <- c(rep(0.5, 4), rep(0, 6))
y <- rbinom(n, 1, plogis(X %*% true_beta))

# Task 1: Fit LASSO and examine which variables are selected
cv_lasso <- cv.glmnet(X, y, family = "binomial", alpha = 1)
cat("LASSO coefficients (lambda.1se):\n")
print(round(as.matrix(coef(cv_lasso, s = "lambda.1se")), 3))

# Task 2: Fit Ridge and examine coefficients
cv_ridge <- cv.glmnet(X, y, family = "binomial", alpha = 0)
cat("\nRidge coefficients (lambda.1se):\n")
print(round(as.matrix(coef(cv_ridge, s = "lambda.1se")), 3))

# Task 3: Fit Elastic Net (alpha = 0.5)
cv_enet <- cv.glmnet(X, y, family = "binomial", alpha = 0.5)
cat("\nElastic Net coefficients (lambda.1se):\n")
print(round(as.matrix(coef(cv_enet, s = "lambda.1se")), 3))

# Questions:
# a) Does LASSO select all 4 correlated predictors, or does it pick
#    just one or two? Why?
# b) How does Ridge handle the correlated predictors differently?
# c) Does Elastic Net improve on the LASSO in terms of selecting the
#    correlated predictors?
# d) Run the analysis 10 times with different seeds. How stable is the
```

```
#    LASSO variable selection compared to Ridge and Elastic Net?
```

7.15.3.2. Python

```python
import numpy as np
import pandas as pd
from sklearn.linear_model import LogisticRegressionCV
from sklearn.preprocessing import StandardScaler
from scipy.special import expit

np.random.seed(2025)
n = 200

# Create 4 correlated predictors (correlation ~ 0.9)
mean = np.zeros(4)
cov = np.full((4, 4), 0.9)
np.fill_diagonal(cov, 1.0)
X_corr = np.random.multivariate_normal(mean, cov, size=n)

# Add 6 independent noise predictors
X_indep = np.random.randn(n, 6)
X = np.hstack([X_corr, X_indep])

feature_names = [f"Corr_{i+1}" for i in range(4)] + \
                [f"Noise_{i+1}" for i in range(6)]

# True model: all 4 correlated predictors contribute
true_beta = np.array([0.5]*4 + [0.0]*6)
y = np.random.binomial(1, expit(X @ true_beta))

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
# Task 1: LASSO
lasso = LogisticRegressionCV(penalty='l1', solver='saga', Cs=50,
                             cv=10, max_iter=10000, random_state=42)
lasso.fit(X_scaled, y)
print("LASSO coefficients:")
for name, coef in zip(feature_names, lasso.coef_[0]):
    print(f"  {name}: {coef:.4f}")

# Task 2: Ridge
ridge = LogisticRegressionCV(penalty='l2', solver='lbfgs', Cs=50,
                             cv=10, max_iter=10000, random_state=42)
ridge.fit(X_scaled, y)
print("\nRidge coefficients:")
for name, coef in zip(feature_names, ridge.coef_[0]):
    print(f"  {name}: {coef:.4f}")

# Task 3: Elastic Net
enet = LogisticRegressionCV(penalty='elasticnet', solver='saga', Cs=50,
                            cv=10, l1_ratios=[0.5], max_iter=10000,
                            random_state=42)
enet.fit(X_scaled, y)
print("\nElastic Net coefficients:")
for name, coef in zip(feature_names, enet.coef_[0]):
    print(f"  {name}: {coef:.4f}")

# Task 4: Stability analysis - run 10 times with different seeds
print("\n--- Stability Analysis (LASSO) ---")
for seed in range(10):
    np.random.seed(seed)
    idx = np.random.choice(n, n, replace=True)  # Bootstrap sample
    lasso_boot = LogisticRegressionCV(penalty='l1', solver='saga',
                                      Cs=20, cv=5, max_iter=10000,
                                      random_state=42)
```

```
    lasso_boot.fit(X_scaled[idx], y[idx])
    selected = [feature_names[j] for j in range(10)
                if abs(lasso_boot.coef_[0][j]) > 1e-6]
    print(f"  Seed {seed}: {selected}")
```

7.16. Summary

Concept	Details
The problem	Standard regression fails or overfits when p is large relative to n
Bias-variance tradeoff	Adding bias (shrinkage) reduces variance, lowering total prediction error
Ridge (L2)	Shrinks all coefficients toward zero; never exactly zero; handles collinearity
LASSO (L1)	Shrinks some coefficients to exactly zero; automatic variable selection
Elastic Net	Combines L1 and L2; groups correlated predictors; alpha controls the mix
Lambda	Penalty strength; chosen by cross-validation
lambda.min	Best predictive performance
lambda.1se	Most parsimonious model within 1 SE of the best
Standardise first	Essential so the penalty treats all predictors equally
Post-selection inference	P-values after LASSO selection are invalid without special methods

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8. Survival Analysis: Time-to-Event Data in Medicine

8.1. Why Ordinary Regression Fails for Time-to-Event Data

In medicine, we frequently ask questions like: *How long until a patient relapses after chemotherapy? How long does a hip replacement last before revision surgery is needed? What is the median survival time after a Stage IV lung cancer diagnosis?*

These are **time-to-event** questions. The outcome is not simply "did the event happen?" (which would be logistic regression) nor is it a continuous measurement at a fixed time point (which would be linear regression). The outcome is *how long* until something happens.

You might think: "Why not just use linear regression with time as the outcome?" There is a fundamental problem — **censoring**.

8.1.1. The Censoring Problem

Imagine a clinical trial that follows 100 patients for 5 years after a cancer diagnosis. At the end of the study:

- 40 patients have died (we know their exact survival time).
- 30 patients are still alive at year 5 (we know they survived *at least* 5 years, but we do not know when they will die).

- 20 patients moved away or were lost to follow-up at various time points.
- 10 patients withdrew from the study.

For the last 60 patients, we have **incomplete information**. We know they survived *at least* until the last time we observed them, but not how much longer. This is **right censoring** — the event of interest (death) may occur to the right of (after) our last observation.

If we simply excluded these 60 patients, we would introduce severe bias — we would systematically underestimate survival times because the long-survivors are disproportionately censored. If we treated their last-observed time as their event time, we would also underestimate survival. Survival analysis methods were developed precisely to handle this problem.

8.1.2. Types of Censoring

- **Right censoring** (most common): The event has not yet occurred when observation ends. The true event time is to the right of the censoring time.
- **Left censoring**: The event occurred *before* observation began (e.g., a patient already had the disease at enrollment, but we do not know when it started).
- **Interval censoring**: The event is known to have occurred within an interval but the exact time is unknown (e.g., a tumor detected at a scheduled 6-month scan — it appeared sometime between the last scan and this one).

In this chapter, we focus primarily on right censoring, which dominates clinical research.

8.2. The Kaplan-Meier Estimator

The **Kaplan-Meier (KM) estimator** is the workhorse of survival analysis. It provides a **non-parametric** estimate of the survival function $S(t) = P(T > t)$, the probability of surviving beyond time t.

> **"Non-parametric" — what it means here**
>
> A **non-parametric** method makes no assumption about the *shape* of the underlying distribution. A *parametric* survival model (e.g., a Weibull or exponential model) assumes the survival curve follows a specific mathematical formula governed by a few parameters; you are essentially fitting a smooth, pre-specified curve to the data. The Kaplan-Meier estimator does the opposite: it lets the data speak for themselves, building the survival curve directly from the observed event times as a series of steps, without assuming any particular form. The advantage is flexibility and honesty — the curve reflects exactly what happened in your cohort. The cost is that it cannot extrapolate beyond the observed follow-up and uses the data less efficiently than a correctly-specified parametric model.

8.2.1. How It Works

The KM estimator calculates survival probability at each time point where an event occurs:

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

where:

- t_i is a time at which at least one event occurred,

- d_i is the number of events at time t_i,
- n_i is the number of individuals at risk just before time t_i (alive and uncensored).

The intuition (for a clinical audience). Forget the product symbol for a moment and think of survival as running a gauntlet. At each moment when a death occurs, ask a simple question: *of the patients still alive and being followed right now, what fraction made it through this instant?* That fraction is $\left(1 - \frac{d_i}{n_i}\right)$ — the number who died (d_i) out of those still at risk (n_i), subtracted from one. To survive *to* time t, a patient must clear every one of these hurdles in sequence, so we **multiply** the individual survival fractions together (that is what the $\prod$ symbol means — a running product). This is exactly how you would compute the chance of passing several independent checkpoints: multiply the chance of passing each.

A worked snippet makes it concrete. Suppose 100 patients are at risk, 10 die on day 30, then of the 90 remaining 9 die on day 60. The survival probability after day 30 is $1 - 10/100 = 0.90$; after day 60 it is $0.90 \times (1 - 9/90) = 0.90 \times 0.90 = 0.81$. The curve only steps *down* at the moments events occur, and stays flat in between — which is why KM curves look like staircases.

The key insight is that censored individuals contribute to the **risk set** (n_i) up until the moment they are censored, and then they quietly drop out (they reduce n_i for later time points but never count as an event d_i). They are not discarded — they contribute all the information they can, namely "this patient was still event-free up to here."

8.2.2. Clinical Example: Primary Biliary Cholangitis (PBC)

We will use the classic PBC dataset from the Mayo Clinic, which is included in R's `survival` package. This dataset contains 418 patients with primary biliary cholangitis (PBC), a chronic liver disease. Patients were followed from enrollment until death, transplant, or censoring.

! Packages you need to load first

The R code in this chapter relies on three packages. If you are running it yourself, install them once and then load them at the top of your script — the code will error otherwise (for example, the `%>%` pipe comes from `dplyr`/`tidyverse`, and `ggsurvplot()` comes from `survminer`):

```
# Run once if not already installed:
# install.packages(c("survival", "survminer", "tidyverse"))

library(survival)   # core survival analysis: Surv(), survfit(), coxph(), the pbc data
library(survminer)  # ggsurvplot() for publication-quality Kaplan-Meier plots
library(dplyr)      # the %>% pipe and data wrangling (loaded via tidyverse too)
```

`survminer` in particular is a separate package from `survival` and is easy to forget — without it, `ggsurvplot()` will not be found.

8.2.2.1. R

```
# Load the packages used below (see the note above for installation)
library(survival)
library(survminer)
library(dplyr)

# Load the PBC dataset
data(pbc, package = "survival")

# Prepare data - status: 0 = censored, 1 = transplant, 2 = dead
# For now, treat death as the event and censor everything else
pbc_clean <- pbc %>%
  filter(!is.na(trt)) %>%
```

```r
  mutate(
    status_binary = ifelse(status == 2, 1, 0),
    time_years = time / 365.25,
    trt_label = factor(trt, labels = c("D-penicillamine", "Placebo"))
  )

cat("N =", nrow(pbc_clean), "\n")
cat("Events (deaths):", sum(pbc_clean$status_binary), "\n")
cat("Censored:", sum(pbc_clean$status_binary == 0), "\n")

# Fit overall KM curve
km_overall <- survfit(Surv(time_years, status_binary) ~ 1, data = pbc_clean)
print(km_overall)
```

```r
# Plot KM curve with confidence intervals
ggsurvplot(
  km_overall,
  data = pbc_clean,
  conf.int = TRUE,
  risk.table = TRUE,
  xlab = "Time (years)",
  ylab = "Survival probability",
  title = "Overall Survival in PBC Patients",
  ggtheme = theme_minimal()
)
```

8.2.2.2. Python

```python
import pandas as pd
import numpy as np
from lifelines import KaplanMeierFitter
from lifelines.statistics import logrank_test
```

```
import matplotlib.pyplot as plt

# Load PBC data (from R via CSV or use lifelines datasets)
# We'll replicate the data preparation
from lifelines.datasets import load_rossi  # placeholder - in practice, load PBC
# For demonstration, we use the PBC data exported from R or a similar dataset
# Here we simulate the structure:
import subprocess

# In practice, you would load the PBC data. We use lifelines' built-in loader:
try:
    pbc = pd.read_csv("data/pbc.csv")
except FileNotFoundError:
    # Create from R's dataset structure for demonstration
    from lifelines.datasets import load_regression_dataset
    print("Note: Using lifelines example data for demonstration.")
    print("In practice, load the PBC dataset from the survival R package.")

# Kaplan-Meier estimation
kmf = KaplanMeierFitter()

# Example with generic time-to-event data
np.random.seed(42)
n = 312
time_years = np.random.exponential(scale=8, size=n)
event_observed = np.random.binomial(1, 0.45, size=n)

kmf.fit(time_years, event_observed=event_observed, label="PBC Patients")

print(kmf.summary.head(10))
print(f"\nMedian survival time: {kmf.median_survival_time_:.2f} years")
```

```
fig, ax = plt.subplots(figsize=(8, 5))
kmf.plot_survival_function(ax=ax, ci_show=True)
ax.set_xlabel("Time (years)")
ax.set_ylabel("Survival probability")
ax.set_title("Kaplan-Meier Survival Estimate")
plt.tight_layout()
plt.show()
```

8.2.3. Interpreting the KM Curve

Key features to examine:

1. **Median survival time**: the time at which $\hat{S}(t) = 0.5$ (where the curve crosses the 50% line). If the curve never drops to 50%, the median is undefined.
2. **Survival at specific time points**: e.g., "5-year survival is 65%."
3. **Shape of the curve**: A steep early drop suggests high early mortality. A long flat tail suggests most survivors do well long-term.
4. **Confidence bands**: Wider bands mean more uncertainty, often because fewer patients remain at risk.
5. **Censoring marks** (tick marks on the curve): Heavy censoring near the tail means the late estimates are unreliable.

8.2.4. Confidence Intervals for the KM Estimator

The survival curve is an *estimate* from a finite sample, so it comes with uncertainty — the confidence band you see around a KM curve. The pointwise confidence interval for $\hat{S}(t)$ is computed via **Greenwood's formula**, which estimates the variance (the "wobble") of $\hat{S}(t)$:

$$\widehat{\text{Var}}[\hat{S}(t)] = \hat{S}(t)^2 \sum_{t_i \le t} \frac{d_i}{n_i(n_i - d_i)}$$

You do not need to compute this by hand — `survfit()` does it for you — but the formula carries one genuinely useful intuition, which is the only thing worth taking away from it. Look at each term in the sum: it grows when n_i (the number still at risk) is **small**. Early in follow-up, n_i is large and each term is tiny, so the band is tight. Late in follow-up, after many patients have died or been censored, n_i shrinks, each term balloons, and the band fans out. **That is why the confidence band on every KM curve is narrow on the left and wide on the right**: the tail of the curve is estimated from only a handful of remaining patients, so it is far less reliable. When you read a KM plot, treat the wide right-hand tail with caution.

A 95% confidence interval is then approximately $\hat{S}(t) \pm 1.96\sqrt{\widehat{\text{Var}}[\hat{S}(t)]}$. In practice, software uses a log-log transformation of this interval because it guarantees the limits stay within the valid [0, 1] range for a probability (a plain $\pm$ interval can stray below 0 or above 1 in the tails).

8.3. The Log-Rank Test

The **log-rank test** compares survival curves between two or more groups. It is the most widely used test for this purpose.

The null hypothesis is that survival is identical across groups: $H_0 : S_1(t) = S_2(t)$ for all t.

The test statistic compares the observed number of events in each group to the number expected under the null hypothesis, summed over all event times. It is most powerful when the **proportional hazards assumption** holds — that is, when one group has a consistently higher (or lower) hazard than the other across time.

8.3.0.1. R

```
# Compare survival between treatment groups
km_by_trt <- survfit(Surv(time_years, status_binary) ~ trt_label, data = pbc_

# Log-rank test
logrank <- survdiff(Surv(time_years, status_binary) ~ trt_label, data = pbc_c
print(logrank)
```

```
ggsurvplot(
  km_by_trt,
  data = pbc_clean,
  pval = TRUE,
  conf.int = TRUE,
  risk.table = TRUE,
  xlab = "Time (years)",
  ylab = "Survival probability",
  legend.title = "Treatment",
  palette = c("#E7B800", "#2E9FDF"),
  ggtheme = theme_minimal()
)
```

8.3.0.2. Python

```
from lifelines import KaplanMeierFitter
from lifelines.statistics import logrank_test
import numpy as np

np.random.seed(42)

# Simulate two treatment groups
```

```
n_per_group = 156
time_trt = np.random.exponential(scale=8.5, size=n_per_group)
event_trt = np.random.binomial(1, 0.42, size=n_per_group)

time_placebo = np.random.exponential(scale=7.5, size=n_per_group)
event_placebo = np.random.binomial(1, 0.48, size=n_per_group)

# Log-rank test
results = logrank_test(time_trt, time_placebo, event_trt, event_placebo)
print(f"Log-rank test statistic: {results.test_statistic:.3f}")
print(f"p-value: {results.p_value:.4f}")

# Plot both curves
fig, ax = plt.subplots(figsize=(8, 5))
kmf_trt = KaplanMeierFitter()
kmf_trt.fit(time_trt, event_trt, label="D-penicillamine")
kmf_trt.plot_survival_function(ax=ax)

kmf_placebo = KaplanMeierFitter()
kmf_placebo.fit(time_placebo, event_placebo, label="Placebo")
kmf_placebo.plot_survival_function(ax=ax)

ax.set_xlabel("Time (years)")
ax.set_ylabel("Survival probability")
ax.set_title("Survival by Treatment Group")
plt.tight_layout()
plt.show()
```

8.3.1. Reading the Log-Rank Output

The `survdiff()` output in R looks like a small table with one row per group, plus a chi-squared statistic and p-value at the bottom. Here is what each piece tells you and how to report it:

- `N` — the number of patients in each group.
- `Observed` — how many events (deaths) actually occurred in each group.
- `Expected` — how many events you would have expected in each group *if survival were identical across groups* (the null hypothesis). This is the key comparison: it accounts for how long each group was followed and how many were at risk over time.
- **Observed vs Expected** — the heart of the test. If a group has *more* observed deaths than expected, it is doing worse than the average; *fewer* than expected, better. The test asks whether the gap between observed and expected, accumulated across all event times, is larger than chance would produce.
- **`Chisq` and `p`** — the test statistic and its p-value. A small p-value (conventionally < 0.05) means the survival curves differ more than would be expected by chance; you reject the hypothesis that the two groups have identical survival.

For the PBC treatment comparison above, the relevant question is whether D-penicillamine changed survival versus placebo. A **non-significant** p-value (as is the case in this dataset) means there is no evidence the drug altered survival — which matches the historical conclusion that D-penicillamine is not effective in PBC. Report it as, for example: *"There was no significant difference in survival between D-penicillamine and placebo (log-rank p = 0.75)."* Crucially, a non-significant log-rank test is **not** proof the treatments are equivalent — it may simply reflect insufficient events to detect a difference.

The `ggsurvplot()` figure shows the same comparison visually: two KM curves with the log-rank p-value annotated (`pval = TRUE`). If the curves sit almost on top of each other, that is the visual counterpart of a non-significant test.

When the Log-Rank Test is Weak

The log-rank test has low power when survival curves cross (e.g., one treatment is better early but worse later) — the early and late differences cancel out, hiding a real effect. In such cases, consider the **weighted log-rank test** (e.g., Fleming-Harrington, which can up-weight early or late differences) or the **restricted mean survival time (RMST)** approach, which compares the average event-free time up to a chosen horizon and does not rely on proportional hazards.

8.4. Cox Proportional Hazards Model

The **Cox proportional hazards (PH) model** is the most widely used regression model for survival data. Introduced by Sir David Cox in 1972, it models the **hazard function** — the instantaneous rate of the event occurring at time t, given survival to that point.

What is a "hazard", in plain terms?

The word *hazard* trips up most clinicians on first encounter, because it is not a probability and not a risk. Think of it as a **speedometer for risk**: at any instant t (among patients who have survived to that instant), it measures *how fast* events are occurring right now. Like a car's speed, it is a *rate* (events per unit time), it applies to *this moment* rather than the whole journey, and it can change from moment to moment.
Contrast it with quantities you already use:

- A **risk** (or cumulative probability) is like the total distance travelled — "35% of patients have died **by** 5 years." It accumulates over a period.
- A **risk ratio / relative risk (RR)** compares those totals

between two groups over a fixed period — "the treated group's 5-year risk is half the control group's."
- A **hazard** is the instantaneous speed, and a **hazard ratio (HR)** compares the *speeds* of two groups at any given moment.

Why bother with a speed rather than a simple risk ratio? Because censoring and varying follow-up make "total risk by time t" hard to compare directly — patients are observed for different lengths of time. The hazard sidesteps this by asking, at each instant, only about the patients actually still at risk at that instant. The Cox model is built on this instantaneous quantity:

$$h(t|X) = h_0(t)\exp(\beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p)$$

where:

- $h(t|X)$ is the hazard at time t for an individual with covariates X,
- $h_0(t)$ is the **baseline hazard** (an unspecified function of time),
- $\beta_1, \dots, \beta_p$ are regression coefficients,
- $\exp(\beta_j)$ is the **hazard ratio** for a one-unit increase in X_j.

8.4.1. Key Properties

1. **Semi-parametric**: The baseline hazard $h_0(t)$ is left unspecified — the model makes no assumption about the shape of the hazard over time. Only the relative effect of covariates is estimated.
2. **Proportional hazards assumption**: The ratio of hazards for any two individuals is constant over time. If patient A has twice the hazard of patient B at time 1, they also have twice the hazard at time 5.

3. **Hazard ratios**: HR $= \exp(\beta)$. An HR > 1 means increased hazard (worse prognosis); HR < 1 means decreased hazard (better prognosis); HR $= 1$ means no effect.

8.4.2. Fitting the Cox Model

8.4.2.1. R

```r
# Fit Cox model with multiple predictors
cox_model <- coxph(
  Surv(time_years, status_binary) ~ age + sex + albumin + bili + edema,
  data = pbc_clean
)

summary(cox_model)
```

```r
# Forest plot of hazard ratios
ggforest(cox_model, data = pbc_clean)
```

8.4.2.2. Python

```python
from lifelines import CoxPHFitter
import pandas as pd
import numpy as np

# Create a simulated clinical dataset
np.random.seed(42)
n = 312
df = pd.DataFrame({
    'time_years': np.random.exponential(scale=8, size=n),
    'event': np.random.binomial(1, 0.45, size=n),
```

```
    'age': np.random.normal(50, 10, size=n),
    'sex': np.random.binomial(1, 0.5, size=n),
    'albumin': np.random.normal(3.5, 0.5, size=n),
    'bili': np.random.exponential(scale=3, size=n),
    'edema': np.random.choice([0, 0.5, 1], size=n, p=[0.75, 0.15, 0.10])
})

cph = CoxPHFitter()
cph.fit(df, duration_col='time_years', event_col='event',
        formula='age + sex + albumin + bili + edema')

cph.print_summary()
```

```
fig, ax = plt.subplots(figsize=(8, 5))
cph.plot(ax=ax)
ax.set_title("Cox PH Model - Hazard Ratios")
plt.tight_layout()
plt.show()
```

8.4.3. Interpreting Hazard Ratios

Consider a Cox model result where bilirubin has $\hat{\beta} = 0.15$ (HR = 1.16, 95% CI: 1.10–1.23, $p < 0.001$).

Interpretation: "For each 1 mg/dL increase in serum bilirubin, the hazard of death increases by 16% (HR = 1.16, 95% CI: 1.10–1.23), adjusting for age, sex, albumin, and edema."

> ⚠ Hazard Ratios Are Not Risk Ratios
>
> A hazard ratio of 2.0 does **not** mean "twice the risk of death." It means the instantaneous rate of death is twice as high at any given

moment. Over long follow-up, the cumulative risk difference depends on the baseline hazard. Be precise in your language when reporting results.

8.5. Checking the Proportional Hazards Assumption

The proportional hazards assumption is critical. If it is violated, the hazard ratios from the Cox model do not have a stable interpretation — the "effect" of a covariate depends on when you look.

8.5.1. Schoenfeld Residuals

The most common diagnostic tool is the **Schoenfeld residual test**. The intuition is simple: if a covariate's effect (its hazard ratio) is truly constant over time — which is exactly what proportional hazards requires — then a plot of its residuals against time should be a flat, horizontal cloud with no trend. If instead the residuals drift up or down over time, the covariate's effect is *changing* over time, and the PH assumption is violated for that covariate.

8.5.1.1. R

```r
# Test proportional hazards assumption
ph_test <- cox.zph(cox_model)
print(ph_test)
```

```r
# Plot Schoenfeld residuals
par(mfrow = c(2, 3))
plot(ph_test)
```

8.5.1.2. Python

```
# Check proportional hazards assumption in lifelines
cph.check_assumptions(df, p_value_threshold=0.05, show_plots=True)
```

What to look for in the output. The `cox.zph()` table in R gives one row per covariate plus a `GLOBAL` row, each with a chi-squared statistic and a **p-value**. Read it as a series of hypothesis tests where the null is "this covariate satisfies proportional hazards":

- A **large p-value (e.g., > 0.05)** is the *good* outcome here — it means no evidence against proportional hazards, so the assumption is reasonable for that covariate.
- A **small p-value (< 0.05)** flags a covariate whose effect changes over time — a PH violation you need to address (see the next section).
- The `GLOBAL` row tests all covariates jointly; if it is non-significant, the model as a whole is broadly consistent with PH.

Then back up the numbers with the **Schoenfeld residual plots**: for each covariate you want the smoothed line to be roughly **flat/horizontal**. A clear upward or downward slope is the visual signature of a violation — for example, a covariate whose effect is strong early but fades with time will show a declining trend. A useful habit: do not rely on the p-value alone (with large samples even trivial deviations become "significant"); always eyeball the plot to judge whether the trend is practically meaningful, not just statistically detectable.

8.5.2. What To Do When PH Is Violated

If the PH assumption is violated for a covariate, you have several options:

1. **Stratify** on the offending variable: `coxph(Surv(time, event) ~ x1 + strata(x2), data = df)`.

Stratification is the most common fix and deserves a proper explanation, because what it does is subtle. Recall that the Cox model assumes a single shared **baseline hazard** $h_0(t)$ that every patient's hazard is a multiple of. When you `strata(x2)`, you tell the model: *do not assume one common baseline — let each level of* `x2` *have its own, separately-shaped baseline hazard*. So if you stratify on, say, hospital site or disease stage, patients in stage 1 follow one baseline time-course and stage-4 patients another, and the model no longer forces their hazards to stay in fixed proportion.

The important trade-off: a stratified variable **drops out of the results table** — you no longer get a hazard ratio for it, because you are no longer modelling its effect, just absorbing it into the baseline. So stratify on variables you want to *adjust for* but are not interested in estimating (nuisance variables like centre, or a strong confounder that violates PH). Keep the variables you actually want HRs for as ordinary covariates. The other covariates (`x1` above) still get hazard ratios, now estimated within strata. A practical note: stratification works best for categorical variables with a few levels; a continuous variable must be binned into categories first, which loses information.

2. **Include a time interaction**: allow the effect to change over time by adding a term such as `x:time` (or `tt()` in `survival`). This estimates *how* the hazard ratio drifts — useful when the time-dependence is itself of interest.

3. **Use time-varying covariates** (see below) when the covariate's *value* (not just its effect) changes during follow-up.

4. **Use an alternative model**: accelerated failure time (AFT) models or fully parametric models (Weibull, log-normal), which do not rest on proportional hazards at all.

8.6. Time-Varying Covariates

In many clinical settings, covariates change over the course of follow-up. A patient's lab values, treatment status, or disease stage may evolve. The standard Cox model assumes covariates are fixed at baseline, but the model can be extended to handle **time-varying covariates**.

The idea is to split each patient's follow-up into intervals. Within each interval, the covariate values are fixed, but they can change between intervals. In R, this is accomplished using the `tmerge()` function from the `survival` package or by manually creating a "counting process" data format with start and stop times:

```
# Conceptual example - counting process format
# PatientID  start  stop  event  lab_value
#     1        0      3     0     2.1
#     1        3      7     0     3.4
#     1        7     10     1     5.8    <- event at time 10
```

This is an advanced topic. The key takeaway is that the Cox model is flexible enough to accommodate covariates that change over time, but the data preparation is more complex and requires careful thought about when measurements were taken.

8.7. Competing Risks

Standard survival analysis assumes that the event of interest is the only possible event, or that censoring is **non-informative** (i.e., censored patients have the same future risk as those who remain under observation).

Competing risks arise when another event prevents the event of interest from ever being observed. The defining feature is that the competing event *removes the patient from being at risk for your event altogether* — it is not just a temporary loss to follow-up. Classic clinical examples:

- Studying time to heart transplant rejection, but the patient dies of infection first.
- Studying time to cancer recurrence, but the patient dies of cardiovascular disease.
- Studying time to hospital readmission, but the patient dies before readmission.

Why this matters clinically. The instinctive fix — treat the competing event (e.g., death) as just another censoring — is **wrong**, and in a specific, important direction: it **overestimates** the probability of your event of interest. The reason is intuitive once stated. Ordinary censoring assumes the censored patient *could still have the event later* and is simply unobserved (e.g., they moved away). But a patient who has died of another cause can **never** experience cancer recurrence — they are not "out there, still at risk." By treating them as censored, the standard Kaplan-Meier "1 − S(t)" silently pretends they might still recur, and inflates the recurrence probability. In older patients or any setting with substantial mortality from other causes, this bias can be large and clinically misleading — you would quote a recurrence risk higher than any real patient faces. This is why, whenever a meaningful fraction of patients can experience a competing event, you should report the **cumulative incidence function** (which correctly accounts for competing events) rather than 1 minus the Kaplan-Meier estimate.

8.7.1. Two Approaches

There are two established ways to handle competing risks, and the distinction between them is one of the most commonly confused points in clinical survival analysis. They answer **different questions**, so the "right" one depends on what you want to know.

1. **Cause-specific hazard models.** Fit a separate Cox model for each event type, treating the competing events as censored. Each model estimates the **rate of that specific event among patients still alive and event-free** at each instant. This answers an *etiological* question: *what drives the rate of this particular event?* A hazard ratio here describes the effect of a covariate on the biological process leading to that event, holding the at-risk population fixed. Because competing events are censored, the cause-specific hazard ratios should **not** be read directly as effects on the *probability* of the event over time.

2. **Fine-Gray subdistribution hazard model.** Rather than censoring patients who experience the competing event, the Fine-Gray model **keeps them in the risk set** (conceptually, they remain "unable to have the event" forever). It models the **subdistribution hazard**, which links directly to the **cumulative incidence function** — the actual probability of experiencing the event by time t *in the real world, where competing events also happen.* This answers a *prognostic / predictive* question: *what is the chance this patient actually experiences the event by 5 years, given they might die of something else first?*

The crucial consequence: a covariate can increase the cause-specific hazard of an event yet *decrease* its cumulative incidence (because it also raises the competing mortality), so the two models can even point in opposite directions. Neither is "wrong" — they answer different questions.

Rule of Thumb for Competing Risks

If you are interested in **etiology** (what causes the event, mechanistically?), use **cause-specific hazards**. If you are interested in **prediction** (what is the absolute probability of the event by time t?), use the **Fine-Gray model** and report **cumulative incidence functions** rather than 1 minus KM. For a clinical prediction model, Fine-Gray (or directly modelling the cumulative incidence) is usually what you want.

For a clinically-oriented, practical treatment of competing risks — including worked examples of both approaches and guidance on which to choose — see the prediction-modelling textbook by Smits, van Kuijk, and Wynants (Smits et al. 2026). Accessible methodological introductions are given by Austin, Lee, and Fine (Austin et al. 2016), and the original subdistribution-hazard method is due to Fine and Gray (Fine and Gray 1999). In R, the `tidycmprsk` and `cmprsk` packages fit Fine-Gray models and estimate cumulative incidence; in Python, `scikit-survival` and `lifelines` provide competing-risks tools.

8.8. Full Clinical Example: PBC Prognostic Model

Let us build a complete survival model for the PBC dataset, following the steps a clinical researcher would take.

Kaplan-Meier vs Cox — when and why to use each

Before reading the model output, it helps to be crystal clear on how the two main tools differ, because they are often confused:

	Kaplan-Meier (+ log-rank)	**Cox proportional hazards**
What it gives you	A survival *curve* for a whole group (or a few groups)	The *effect* of each predictor on the hazard, as a hazard ratio

Handles covariates?	Only categorical group comparisons (e.g., treatment arm, stage)	Multiple predictors at once, including continuous ones, **adjusted for each other**
Output to report	Median survival, x-year survival %, log-rank p-value	Hazard ratios with 95% CIs and p-values
Analogy	Like comparing mean outcomes between groups (a *t*-test / ANOVA of survival)	Like multivariable regression for survival

In short: reach for **Kaplan-Meier** to *describe and visualise* survival and to compare a small number of groups; reach for the **Cox model** when you need to *adjust for several variables simultaneously* and quantify each one's independent effect. They are complementary — a typical paper shows KM curves for the headline comparison and a Cox model for the adjusted analysis.

The worked example below follows the standard sequence: explore the data, fit the multivariable Cox model, assess how well it discriminates, and use it to predict survival for individual patients.

8.8.0.1. R

```
# Step 1: Explore the data
pbc_model <- pbc_clean %>%
  select(time_years, status_binary, age, sex, albumin, bili,
         protime, stage, edema, trt_label) %>%
  drop_na()
```

```r
cat("Complete cases:", nrow(pbc_model), "\n")
cat("Events:", sum(pbc_model$status_binary), "\n")
cat("Median follow-up:", median(pbc_model$time_years), "years\n")

# Step 2: Fit multivariable Cox model
cox_full <- coxph(
  Surv(time_years, status_binary) ~ age + sex + albumin + log(bili) +
    protime + stage + edema + trt_label,
  data = pbc_model
)
summary(cox_full)
```

```r
# Step 3: Assess model discrimination
cat("Concordance (C-index):", summary(cox_full)$concordance[1], "\n")
cat("This means the model correctly ranks survival times",
    round(summary(cox_full)$concordance[1] * 100, 1),
    "% of the time.\n")
```

```r
# Step 4: Predicted survival for new patients
new_patients <- data.frame(
  age = c(40, 65),
  sex = c("f", "f"),
  albumin = c(3.8, 2.5),
  bili = c(1.0, 10.0),
  protime = c(10.5, 13.0),
  stage = c(2, 4),
  edema = c(0, 1),
  trt_label = c("Placebo", "Placebo")
)

pred_surv <- survfit(cox_full, newdata = new_patients)

plot(pred_surv, col = c("blue", "red"), lwd = 2,
```

```
     xlab = "Time (years)", ylab = "Survival probability",
     main = "Predicted Survival: Low-risk vs High-risk Patient")
legend("topright", c("Low risk", "High risk"),
       col = c("blue", "red"), lwd = 2)
```

8.8.0.2. Python

```
from lifelines import CoxPHFitter, KaplanMeierFitter
import pandas as pd
import numpy as np

np.random.seed(42)
n = 276

# Simulate a PBC-like dataset
pbc_sim = pd.DataFrame({
    'time_years': np.abs(np.random.exponential(7, n) +
                         np.random.normal(0, 1, n)),
    'event': np.random.binomial(1, 0.45, n),
    'age': np.random.normal(50, 10, n),
    'sex': np.random.binomial(1, 0.12, n),  # mostly female in PBC
    'albumin': np.random.normal(3.5, 0.4, n),
    'log_bili': np.random.normal(0.5, 1.0, n),
    'protime': np.random.normal(11, 1, n),
    'stage': np.random.choice([1, 2, 3, 4], n, p=[0.05, 0.25, 0.40, 0.30]),
    'edema': np.random.choice([0, 0.5, 1], n, p=[0.75, 0.15, 0.10]),
    'treatment': np.random.binomial(1, 0.5, n)
})

# Fit Cox model
cph = CoxPHFitter()
cph.fit(pbc_sim, duration_col='time_years', event_col='event',
```

```
        formula='age + sex + albumin + log_bili + protime + stage + edema + treatment')

cph.print_summary()
print(f"\nConcordance index: {cph.concordance_index_:.3f}")
```

```
# Predict survival for specific patient profiles
low_risk = pd.DataFrame({
    'age': [40], 'sex': [0], 'albumin': [3.8], 'log_bili': [0.0],
    'protime': [10.5], 'stage': [2], 'edema': [0], 'treatment': [0]
})
high_risk = pd.DataFrame({
    'age': [65], 'sex': [0], 'albumin': [2.5], 'log_bili': [2.3],
    'protime': [13.0], 'stage': [4], 'edema': [1], 'treatment': [0]
})

fig, ax = plt.subplots(figsize=(8, 5))
cph.predict_survival_function(low_risk).plot(ax=ax, label='Low risk', color='blue')
cph.predict_survival_function(high_risk).plot(ax=ax, label='High risk', color='red')
ax.set_xlabel("Time (years)")
ax.set_ylabel("Survival probability")
ax.set_title("Predicted Survival: Low-risk vs High-risk Patient")
ax.legend()
plt.tight_layout()
plt.show()
```

8.8.1. What the Outputs Are Telling You

It is easy to run this code and be confronted with a wall of numbers. Here is what to actually look at, in order:

1. **The `summary(cox_full)` / `print_summary()` table** is the heart of it. For each predictor, focus on three columns:

- `exp(coef)` **— the hazard ratio.** This is the headline number. HR > 1 = higher hazard (worse survival) per unit increase in that predictor; HR < 1 = protective. For example, an HR of 2.7 for `log(bili)` means each one-unit rise in log-bilirubin nearly triples the instantaneous hazard, holding the other predictors fixed.
- **The 95% confidence interval** (`lower .95`, `upper .95`). If it excludes 1, the effect is statistically significant; its width tells you how precisely the effect is estimated.
- `Pr(>|z|)` **— the p-value.** Which predictors are independently associated with survival after adjustment? You are *learning*: which factors carry prognostic information, in which direction, and how strongly. In the PBC model you should see that higher bilirubin, older age, lower albumin, and worse edema/stage all predict shorter survival — which matches the clinical picture of advanced liver disease.

2. **The concordance (C-index)** summarises *discrimination* — how well the model ranks who dies sooner. 0.5 is a coin-flip; 1.0 is perfect. A value around 0.8 (as here) means that, given two patients, the model correctly identifies the one who dies first about 80% of the time. This is the single best one-number summary of predictive performance.

3. **The predicted survival curves** turn the model into something a clinician can use: plug in a patient's covariates and read off their estimated survival probability over time. The low-risk vs high-risk curves should be visibly separated — the *visual* proof that the predictors above translate into meaningfully different prognoses. This is also where the link back to Kaplan-Meier becomes clear: a KM curve describes an *observed group*, whereas these Cox-predicted curves are *individualised* and *adjusted* for the full covariate profile.

If you observe nothing else, observe these three things: **direction and size of each hazard ratio, whether its CI excludes 1, and the C-index.**

Together they answer "what matters, how much, and how well does the whole model predict?"

8.9. Exercises

8.9.1. Exercise 1: Kaplan-Meier Curves by Disease Stage

Using the PBC dataset, create Kaplan-Meier curves stratified by disease stage (1–4). Perform a log-rank test to compare survival across stages. Report the median survival time for each stage.

8.9.1.1. R Starter Code

```
library(survival)   # Surv(), survfit(), survdiff()
library(survminer)  # ggsurvplot()

# Fit KM by stage
km_stage <- survfit(Surv(time_years, status_binary) ~ stage, data = pbc_clean)

# Your code: plot the curves and perform the log-rank test
# Hint: use ggsurvplot() with pval = TRUE
# Hint: use survdiff() for the formal test
```

8.9.1.2. Python Starter Code

```
# Fit KM for each stage and plot
# Your code here
# Hint: loop over stages, fit KaplanMeierFitter for each
# Use logrank_test() for pairwise comparisons
```

8.9.2. Exercise 2: Build and Evaluate a Cox Model

Build a Cox proportional hazards model for PBC survival using age, bilirubin (log-transformed), albumin, prothrombin time, and edema as predictors.

1. Report the hazard ratios and 95% confidence intervals.
2. Check the proportional hazards assumption. Which covariates, if any, violate it?
3. Calculate the concordance index. How well does the model discriminate?
4. Predict the 5-year survival probability for a 55-year-old woman with bilirubin = 3 mg/dL, albumin = 3.2 g/dL, prothrombin time = 11 seconds, and no edema.

8.9.2.1. R Starter Code

```
library(survival)   # Surv(), coxph(), cox.zph()

# Your code here
# cox_model <- coxph(Surv(time_years, status_binary) ~ ..., data = pbc_clean)
# summary(cox_model)
# cox.zph(cox_model)
```

8.9.2.2. Python Starter Code

```
# Your code here
# cph = CoxPHFitter()
# cph.fit(...)
# cph.check_assumptions(...)
```

8.9.3. Exercise 3: Competing Risks Thinking

Consider a study of time to kidney transplant rejection in 500 recipients. During follow-up, some patients die before experiencing rejection (a competing risk).

1. Explain why treating death as censoring would bias the results. What direction would the bias go?
2. Which approach would you use if your goal were to estimate the probability of rejection within 2 years — cause-specific hazards or Fine-Gray? Why?
3. (Bonus) In R, use the `cmprsk` or `tidycmprsk` package to fit a Fine-Gray model. In Python, explore `lifelines` or `scikit-survival` for competing risk methods.

8.10. Summary

Concept	What It Does	Key Assumption
Kaplan-Meier	Non-parametric survival estimate	Non-informative censoring
Log-rank test	Compares survival curves	Proportional hazards (most powerful)
Cox PH model	Regression for survival data	Proportional hazards, non-informative censoring
Fine-Gray model	Competing risks regression	Models subdistribution hazard

Survival analysis is fundamental to clinical research. The methods in this chapter — Kaplan-Meier estimation, log-rank testing, and Cox regression — form the backbone of nearly every clinical trial and prognostic study. The proportional hazards assumption must always be checked, and competing risks should be considered whenever another event can prevent observation of the event of interest.

8.11. References and Further Reading

The foundational paper on proportional hazards regression is Cox (1972), "Regression Models and Life-Tables," published in the *Journal of the Royal Statistical Society, Series B*, which introduced the semi-parametric model that now bears his name. This paper is one of the most cited in all of statistics.

For a comprehensive and accessible introduction to survival analysis in the context of this course, see Smits et al. (2026), which covers Kaplan-Meier estimation, the Cox model, and competing risks with modern clinical examples. Their treatment of model diagnostics and the proportional hazards assumption is particularly practical.

Bradburn et al. provide an excellent tutorial series on survival analysis for medical researchers, published in the *British Journal of Cancer*. Their series covers Kaplan-Meier methods, the Cox model, and advanced topics in a format accessible to clinicians without extensive statistical training.

For hands-on implementation, the `survival` and `survminer` packages in R are the standard tools. Therneau and Grambsch's *Modeling Survival Data: Extending the Cox Model* (Springer, 2000) is the definitive reference for the R `survival` package. In Python, the `lifelines` library by Cameron Davidson-Pilon provides an excellent API for survival analysis, and its online documentation includes worked clinical examples.

For competing risks specifically, Fine and Gray (1999), "A Proportional Hazards Model for the Subdistribution of a Competing Risk," in the *Journal of the American Statistical Association*, is the key methodological reference. Austin and Fine (2017) provide practical guidance on when and how to use competing risk methods in clinical research.

Part III.

Bayesian Methods

9. Bayesian Inference: A Different Way to Think About Evidence

> **Prerequisites**
>
> This chapter assumes you are comfortable with: (1) probability and Bayes' theorem (Chapter 3), (2) basic R or Python programming (Chapter 2), and (3) the regression concepts from Chapter 5. If Bayes' theorem feels unfamiliar, review Section 3.8 before continuing.

9.1. Introduction

You have now completed the core statistical methods that form the backbone of most clinical research: regression, splines, penalised regression, and survival analysis. All of these methods share a common philosophical framework — they are **frequentist**. In this part of the course, we step back to explore a fundamentally different way of thinking about evidence.

Every statistical method you have learned so far in this course has been **frequentist**. In frequentist statistics, probability refers to long-run frequencies: "if we repeated this experiment infinitely many times, the parameter would fall in our 95% confidence interval 95% of the time." That is a mathematically precise statement, but it is not the question most clinicians actually want answered. What clinicians want to know is: *given the data I have in front of me, what is the probability that this treatment works?*

Bayesian inference answers that question directly. It treats probability as a measure of **uncertainty** — a degree of belief — rather than a long-run frequency. This chapter introduces the Bayesian framework from first principles, shows you how it works with a medical example you already understand (diagnostic testing), and then builds toward the computational machinery (MCMC) that makes modern Bayesian analysis possible.

> **Why now?**
>
> Bayesian methods are increasingly required in clinical research. The US FDA released updated guidance in 2026 on the use of Bayesian statistics in medical device trials, and Bayesian adaptive trial designs are now standard in oncology. Understanding this framework is no longer optional for quantitative researchers in medicine and public health.

> **Notation used in this chapter**
>
> - $P(A \mid B)$ means "the probability of A, given that B is true"
> - θ (theta) is the parameter we want to estimate (e.g., a treatment effect)
> - D stands for "data" — the observations we have collected
> - $\propto$ means "is proportional to"

9.2. Two Philosophies of Probability

9.2.1. The Frequentist View

In the frequentist framework:

- **Parameters are fixed** but unknown constants.
- **Data are random** — they vary from sample to sample.

- **Probability** describes the long-run frequency of events across hypothetical replications.
- A 95% confidence interval means: if we repeated this study infinitely, 95% of the resulting intervals would contain the true parameter.

The frequentist approach does **not** assign a probability to the parameter itself. You cannot say "there is a 95% probability that the true effect lies in this interval" — even though nearly everyone, including many researchers, interprets confidence intervals that way.

9.2.2. The Bayesian View

In the Bayesian framework:

- **Parameters are uncertain** and described by probability distributions.
- **Data are fixed** — once observed, they do not change.
- **Probability** quantifies our degree of belief or uncertainty about unknown quantities.
- A 95% credible interval means exactly what it sounds like: there is a 95% probability that the parameter lies in this interval, given the data and the model.

Table 9.1.: Comparison of frequentist and Bayesian frameworks

Feature	Frequentist	Bayesian
What is probability?	Long-run frequency	Degree of belief
Parameters	Fixed, unknown	Random variables with distributions
Data	Random (varies across replications)	Fixed (observed once)
Inference	P-values, confidence intervals	Posterior distributions, credible intervals

Feature	Frequentist	Bayesian
Prior information	Not formally incorporated	Formally incorporated via priors

9.3. Bayes' Theorem: You Already Know This

Bayes' theorem is the mathematical engine of Bayesian inference. You have seen it before in epidemiology, even if it was not called by that name.

$$P(\theta \mid D) = \frac{P(D \mid \theta)\, P(\theta)}{P(D)}$$

Where:

- $P(\theta \mid D)$ is the **posterior** — our updated belief about the parameter θ after seeing data D.
- $P(D \mid \theta)$ is the **likelihood** — how probable the observed data are under a particular value of θ.
- $P(\theta)$ is the **prior** — our belief about θ before seeing the data.
- $P(D)$ is the **marginal likelihood** (or evidence) — a normalising constant that ensures the posterior integrates to 1.

In words: **Posterior** $\propto$ **Likelihood** $\times$ **Prior**.

9.3.1. A Medical Example: COVID-19 Rapid Testing

Suppose a new rapid antigen test for COVID-19 has:

- **Sensitivity** = 85% (probability of a positive test given truly infected)
- **Specificity** = 98% (probability of a negative test given truly uninfected)

A patient in your emergency department tests positive. What is the probability they actually have COVID-19?

This depends on the **prevalence** (the prior probability of disease). Let us work through two scenarios.

Scenario 1: High prevalence (20% — during a winter surge)

$$P(\text{COVID} \mid +) = \frac{P(+ \mid \text{COVID}) \times P(\text{COVID})}{P(+)}$$

$$= \frac{0.85 \times 0.20}{(0.85 \times 0.20) + (0.02 \times 0.80)} = \frac{0.17}{0.17 + 0.016} = \frac{0.17}{0.186} \approx 0.914$$

A positive test in a high-prevalence setting gives a 91.4% posterior probability of disease.

Scenario 2: Low prevalence (1% — summer, low community transmission)

$$P(\text{COVID} \mid +) = \frac{0.85 \times 0.01}{(0.85 \times 0.01) + (0.02 \times 0.99)} = \frac{0.0085}{0.0085 + 0.0198} = \frac{0.0085}{0.0283} \approx 0.300$$

The same positive test now gives only a 30% posterior probability. The **prior matters**.

This is Bayesian reasoning in its simplest form: your conclusion depends on what you knew before you saw the evidence. Every Bayesian analysis extends this logic to continuous parameters and complex models.

9.4. Prior Distributions

The prior distribution encodes what we know (or do not know) about a parameter before collecting data. Choosing an appropriate prior is one of the most important — and most debated — steps in Bayesian analysis.

9.4.1. Types of Priors

Uninformative (flat) priors assign equal probability to all values. For example, a uniform distribution over $(-\infty, +\infty)$ for a regression coefficient. These are sometimes called "objective" priors because they appear to let the data speak for themselves.

- In practice, truly flat priors are improper (they do not integrate to 1) and can cause computational problems.
- They are rarely truly uninformative — a flat prior on a log-odds scale is not flat on the probability scale.

Weakly informative priors gently constrain parameters to reasonable ranges without being strongly opinionated. For example, a $\text{Normal}(0, 10)$ prior on a log-odds regression coefficient says: "I think the coefficient is probably somewhere between –20 and +20 on the log-odds scale, but I am not very sure."

- These are the **recommended default** for most applied work.
- They prevent the model from exploring absurd parameter values while remaining flexible.
- Andrew Gelman's recommendation: use priors that rule out implausible values but remain vague within the plausible range.

Informative priors incorporate genuine prior knowledge. For example, if previous meta-analyses show that a drug reduces mortality by 10–30%, you might use a prior centred on that range.

- These are especially valuable in small-sample settings (paediatric trials, rare diseases).
- They must be justified and subjected to sensitivity analysis.

The three types are easiest to grasp visually. The figure below shows the same idea — a prior belief about a response rate θ between 0 and 1 — expressed three ways: a *flat* prior that treats every rate as equally plausible, a *weakly informative* prior that gently favours the middle while staying open-minded, and an *informative* prior that encodes a specific prior belief (here, a rate around 25%). Think of it as a dial for *how strong an opinion you bring before seeing the data*: flat = no opinion, weakly informative = a loose hunch, informative = a confident expectation from previous evidence.

9.4.1.1. R

```
library(tidyverse)

theta <- seq(0, 1, length.out = 400)

prior_types <- tibble(
  type = factor(c("Uninformative\n(flat): Beta(1,1)",
                  "Weakly informative:\nBeta(2,2)",
                  "Informative:\nBeta(10,30)"),
                levels = c("Uninformative\n(flat): Beta(1,1)",
                           "Weakly informative:\nBeta(2,2)",
                           "Informative:\nBeta(10,30)")),
  a = c(1, 2, 10),
  b = c(1, 2, 30)
)

prior_curves <- prior_types %>%
  rowwise() %>%
```

```
  reframe(type = type, theta = theta, density = dbeta(theta, a, b))

ggplot(prior_curves, aes(theta, density)) +
  geom_area(fill = "#3498db", alpha = 0.25) +
  geom_line(linewidth = 1, colour = "#2c3e50") +
  facet_wrap(~ type, scales = "free_y") +
  labs(x = expression(theta ~ "(response rate)"), y = "Prior plausibility",
       title = "Three flavours of prior belief") +
  theme_minimal(base_size = 13)
```

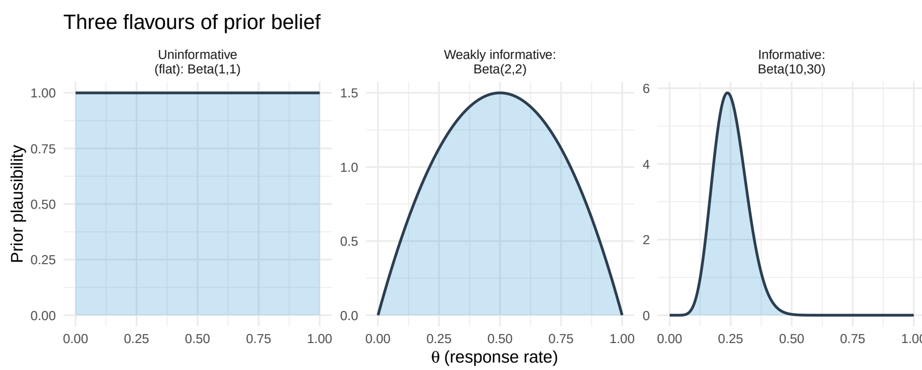

Figure 9.1.: Three types of prior belief about a response rate, from no opinion (flat) to a confident prior expectation (informative). The x-axis is the response rate; the height shows how plausible the prior considers each value.

9.4.1.2. Python

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import beta
```

```
theta = np.linspace(0, 1, 400)
priors = [
    ("Uninformative (flat): Beta(1,1)", 1, 1),
    ("Weakly informative: Beta(2,2)", 2, 2),
    ("Informative: Beta(10,30)", 10, 30),
]

fig, axes = plt.subplots(1, 3, figsize=(12, 3.5))
for ax, (name, a, b) in zip(axes, priors):
    d = beta.pdf(theta, a, b)
    ax.fill_between(theta, d, alpha=0.4, color="steelblue")
    ax.plot(theta, d, color="steelblue", linewidth=1.5)
    ax.set_title(name, fontsize=10)
    ax.set_xlabel(r"$\theta$ (response rate)")
    ax.set_ylabel("Prior plausibility")
fig.suptitle("Three flavours of prior belief", fontsize=12, y=1.05)
plt.tight_layout()
plt.show()
```

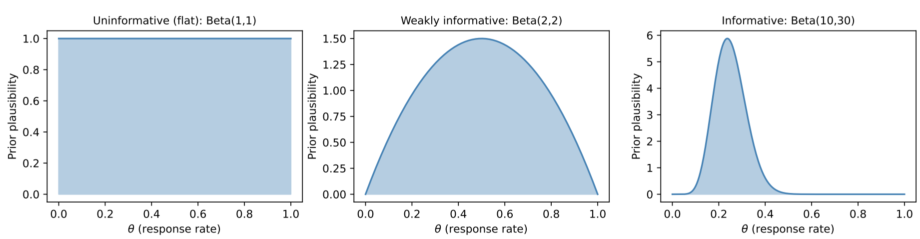

Figure 9.2.: Three types of prior belief about a response rate, from no opinion (flat) to a confident prior expectation (informative).

9.4.2. The Beta Distribution as a Conjugate Prior

Do not be put off by the name — the Beta distribution is simply **a flexible curve for describing a belief about a percentage** (a number between 0 and 1, such as a response rate or a complication rate). By tuning its two knobs, you can make it flat, gently humped, or sharply peaked anywhere on the 0–1 scale, which is exactly what we need to express priors like the three above.

The reason it is the go-to prior for proportions is a convenient mathematical accident called **conjugacy**: when you start with a Beta prior and observe binomial data (successes out of trials), the updated belief — the posterior — is *also* a Beta distribution. The practical payoff is enormous: you do not need any simulation or heavy computation to update your belief, just simple arithmetic. (We will meet problems later where this trick is not available and we *do* need computation — that is what MCMC, below, is for.)

A concrete clinical setup. Suppose a new treatment is given to **20 patients and 8 respond**. We want to estimate the true underlying response rate, θ. A natural first guess is simply $8/20 = 40\%$, but with only 20 patients that estimate is shaky, and we may also have prior expectations about the response rate from earlier studies. Bayesian updating lets us combine the two. We will carry this "8 responses in 20 patients" example through the next few sections.

The Beta distribution is parameterised by two shape parameters, α and β. Loosely, you can read α as a count of "prior successes" and β as "prior failures" — so larger values mean a more confident (narrower) prior:

- $\text{Beta}(1, 1)$ — uniform on [0, 1], an uninformative prior (no opinion).
- $\text{Beta}(2, 2)$ — a gentle hump centred at 0.5, weakly informative.
- $\text{Beta}(10, 30)$ — centred at 0.25, moderately informative (e.g., prior belief that the response rate is around 25%).

The updating rule is beautifully simple. If the prior is $\text{Beta}(\alpha_0, \beta_0)$ and we observe y successes in n trials, the posterior is:

$$\theta \mid y \sim \text{Beta}(\alpha_0 + y,\ \beta_0 + n - y)$$

In words: **add your observed successes to α and your observed failures to β.** For our example ($y = 8$ responses, $n - y = 12$ non-responses), a weakly-informative $\text{Beta}(2, 2)$ prior becomes a $\text{Beta}(2+8,\ 2+12) = \text{Beta}(10, 14)$ posterior — the data have been folded directly into the belief.

9.4.3. R

```r
library(tidyverse)

# Observed data: 8 successes out of 20 trials
y <- 8; n <- 20

# Define three priors (factor levels fix the facet order:
# uninformative -> weakly informative -> informative)
prior_levels <- c("Uninformative: Beta(1,1)",
                  "Weakly informative: Beta(2,2)",
                  "Informative: Beta(10,30)")
priors <- tibble(
  prior_name = factor(prior_levels, levels = prior_levels),
  alpha0 = c(1, 2, 10),
  beta0  = c(1, 2, 30)
)

theta <- seq(0, 1, length.out = 500)

# Build plotting data
```

```
plot_data <- priors %>%
  rowwise() %>%
  reframe(
    prior_name = prior_name,
    theta = theta,
    Prior     = dbeta(theta, alpha0, beta0),
    Posterior = dbeta(theta, alpha0 + y, beta0 + n - y)
  ) %>%
  pivot_longer(cols = c(Prior, Posterior),
               names_to = "Distribution", values_to = "Density") %>%
  mutate(prior_name = factor(prior_name, levels = prior_levels))

ggplot(plot_data, aes(x = theta, y = Density, colour = Distribution,
                      linetype = Distribution)) +
  geom_line(linewidth = 1) +
  facet_wrap(~ prior_name, scales = "free_y") +
  scale_colour_manual(values = c("Prior" = "steelblue", "Posterior" = "firebr
  labs(x = expression(theta ~ "(response rate)"),
       y = "Density",
       title = "Effect of Prior Choice on the Posterior",
       subtitle = "Data: 8/20 patients responded") +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom")
```

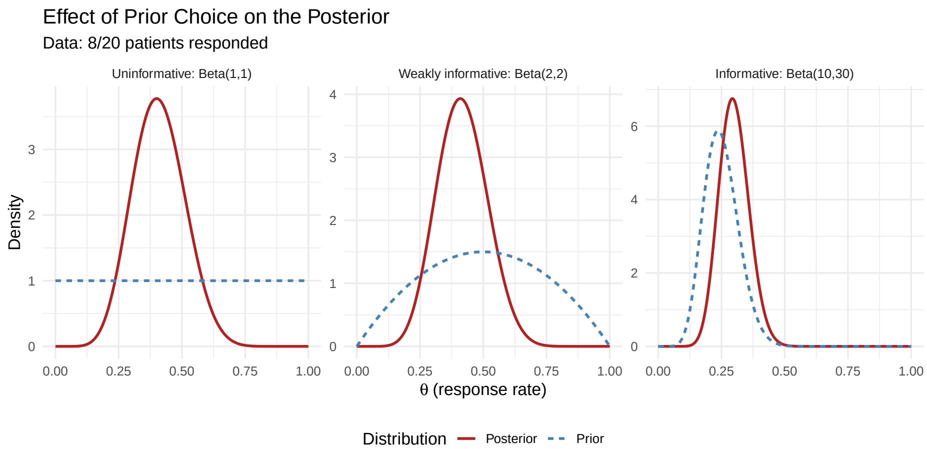

Figure 9.3.: Three Beta priors and their posteriors after observing 8 responses in 20 patients.

9.4.4. Python

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import beta

y, n = 8, 20
theta = np.linspace(0, 1, 500)

priors = [
    ("Uninformative: Beta(1,1)", 1, 1),
    ("Weakly informative: Beta(2,2)", 2, 2),
    ("Informative: Beta(10,30)", 10, 30),
]

fig, axes = plt.subplots(1, 3, figsize=(14, 4), sharey=False)
```

```
for ax, (name, a0, b0) in zip(axes, priors):
    prior_density = beta.pdf(theta, a0, b0)
    post_density  = beta.pdf(theta, a0 + y, b0 + n - y)
    ax.plot(theta, prior_density, label="Prior", color="steelblue", linewidtl
    ax.plot(theta, post_density, label="Posterior", color="firebrick", linew:
    ax.set_title(name, fontsize=10)
    ax.set_xlabel(r"$\theta$ (response rate)")
    ax.set_ylabel("Density")
    ax.legend(fontsize=9)

fig.suptitle("Effect of Prior Choice on the Posterior\nData: 8/20 patients re
             fontsize=12, y=1.04)
plt.tight_layout()
plt.show()
```

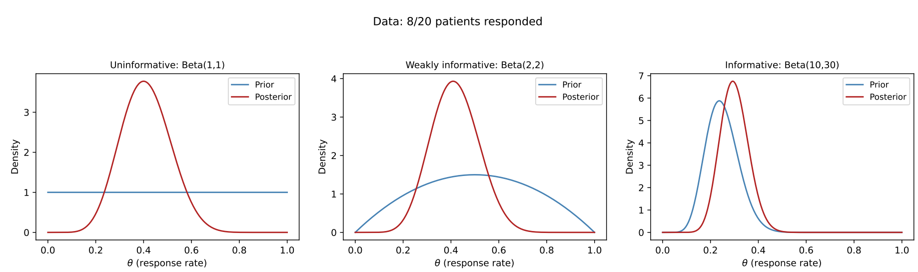

Figure 9.4.: Three Beta priors and their posteriors after observing 8 responses in 20 patients.

Notice what happens with only **20 patients** — a small sample. The choice of prior visibly matters: with the uninformative Beta$(1, 1)$ and weakly-informative Beta$(2, 2)$ priors, the posterior sits close to the observed rate of $8/20 = 0.40$, but with the informative Beta$(10, 30)$ prior the posterior is clearly **pulled down toward the prior's expectation of 0.25**. With this little data, the prior is not overwhelmed — it leaves a real fingerprint on the answer.

This cuts both ways, and it is the key practical lesson. When data are scarce, the prior carries real weight, so it must be chosen and justified carefully (and you should always check how much the conclusions shift under different reasonable priors — a *sensitivity analysis*). It is only as the sample size grows *much larger* that the likelihood dominates and the posterior becomes insensitive to the prior — the regime where "the data overwhelm the prior." Twenty patients is firmly in the small-sample regime, which is exactly why Bayesian methods, with a sensible prior, are so useful here in the first place.

9.5. The Likelihood

The likelihood answers a simple question: *for each possible value of the true response rate θ, how well does it explain the data we actually saw?* Written $P(D \mid \theta)$, it is the same quantity that appears in maximum likelihood estimation (MLE) — but in the Bayesian framework we *combine* it with the prior rather than maximising it on its own.

For our example of 8 responses in 20 patients, the likelihood is given by the **binomial formula**:

$$L(\theta) = \underbrace{\binom{20}{8}}_{\substack{\text{number of ways}\\ \text{8 of 20 could respond}}} \underbrace{\theta^8}_{\substack{\text{8 patients}\\ \text{responded}}} \underbrace{(1-\theta)^{12}}_{\substack{\text{12 did}\\ \text{not respond}}}$$

You will likely not have met this formula before, and you do not need to memorise it — but the idea behind it is intuitive. If the true response rate were θ, then the chance that any one patient responds is θ and the chance they do not is $(1-\theta)$. To get *8 responders and 12 non-responders*, we multiply θ eight times and $(1-\theta)$ twelve times; the $\binom{20}{8}$ term simply counts the many different orders in which those 8 responses could have occurred among the 20 patients. Plug in a candidate rate — say $\theta = 0.4$ — and the formula tells you how likely "8 out of 20" is under that rate.

The likelihood is largest at $\hat{\theta}_{\text{MLE}} = 8/20 = 0.40$ — exactly the observed proportion, which makes sense: the rate that best explains "8 of 20" is 40%. The Bayesian posterior will sit near this value but be pulled slightly toward the prior — a sensible compromise between what the data say and what we believed beforehand, known as **shrinkage**.

9.6. The Posterior Distribution

The posterior is the complete answer to a Bayesian analysis. It is not a single number; it is a full probability distribution over the parameter. From the posterior you can extract:

- **Point estimates**: the posterior mean, median, or mode.
- **Credible intervals**: ranges containing the parameter with a specified probability.
- **Probabilities of hypotheses**: e.g., $P(\theta > 0.3 \mid \text{data})$.

9.6.1. Credible Intervals vs Confidence Intervals

This distinction is critical and frequently confused.

A 95% frequentist confidence interval means: if the experiment were repeated many times, 95% of the computed intervals would contain the true parameter. It does **not** mean there is a 95% probability the true parameter is in *this* interval.

A 95% Bayesian credible interval means: given the data and the model, there is a 95% probability that the parameter lies in this interval.

The Bayesian credible interval says what most people *think* a confidence interval says. This interpretability is one of the major practical advantages of Bayesian inference.

9.6.2. R

```r
# Posterior from weakly informative prior Beta(2,2) + data (8/20)
alpha_post <- 2 + 8
beta_post  <- 2 + 12

# 95% credible interval (equal-tailed)
ci_95 <- qbeta(c(0.025, 0.975), alpha_post, beta_post)
cat("Posterior mean:", alpha_post / (alpha_post + beta_post), "\n")
cat("95% credible interval:", round(ci_95, 3), "\n")

# Probability that response rate exceeds 30%
prob_gt_30 <- 1 - pbeta(0.30, alpha_post, beta_post)
cat("P(theta > 0.30 | data):", round(prob_gt_30, 3), "\n")
```

```
Posterior mean: 0.4166667
95% credible interval: 0.232 0.615
P(theta > 0.30 | data): 0.88
```

9.6.3. Python

```python
from scipy.stats import beta

alpha_post = 2 + 8
beta_post  = 2 + 12

ci_95 = beta.ppf([0.025, 0.975], alpha_post, beta_post)
```

```
print(f"Posterior mean: {alpha_post / (alpha_post + beta_post):.3f}")
print(f"95% credible interval: [{ci_95[0]:.3f}, {ci_95[1]:.3f}]")

prob_gt_30 = 1 - beta.cdf(0.30, alpha_post, beta_post)
print(f"P(theta > 0.30 | data): {prob_gt_30:.3f}")
```

```
Posterior mean: 0.417
95% credible interval: [0.232, 0.615]
P(theta > 0.30 | data): 0.880
```

9.7. Markov Chain Monte Carlo (MCMC)

Before the mechanics, some context — because this is where Bayesian analysis starts to feel unfamiliar to most clinicians. So far, our "8 of 20" example had a tidy shortcut: because the Beta prior pairs perfectly with the binomial likelihood (the conjugacy trick from earlier), we could write the posterior down with simple arithmetic and no computer simulation. That shortcut is the exception, not the rule.

9.7.1. Why We Need It

The Beta-Binomial example above had a **neat closed-form answer** — we could compute the posterior with pen and paper because the Beta prior is conjugate to the binomial likelihood. But that convenience disappears as soon as a model becomes realistic. A clinical prediction model with age, sex, blood pressure, cholesterol, and a dozen other predictors has many parameters and no conjugate shortcut, so there is **no formula** for the posterior — you cannot simply write it down.

This is the practical problem MCMC solves, and it is worth stating plainly because it is genuinely new territory: instead of computing the posterior with a formula, we **let the computer draw thousands of samples from it** and then describe those samples (their average, their spread, their 95% range). It is the same logic as estimating a population by taking a large random sample — except here we are "sampling" from a probability distribution we cannot see directly. The key idea: rather than evaluating $P(\theta \mid D)$ everywhere, we construct a guided random walk through parameter space that, over many steps, spends time in each region *in proportion to its posterior probability.* Collect those visited points and you have a picture of the posterior.

You will never do this by hand, and you rarely even write the sampler yourself — modern software does it for you (see below). The most widely used algorithm today is the **No-U-Turn Sampler (NUTS)**, a refinement of Hamiltonian Monte Carlo (HMC), and it is the default in Stan, brms, rstanarm, and PyMC. What you *do* need to understand as a user is (1) what the samples represent and (2) how to check the sampler worked — which is what the next two sections cover.

9.7.2. How It Works Conceptually

The algorithm is a guided exploration of parameter space. Picture a hiker trying to map out where a mountain range is tallest, but in fog — they can only feel whether a step takes them uphill or downhill. The "height" here is the **posterior density**: how plausible a parameter value is, given the prior and the data combined. The sampler walks around, preferring to spend time on the high ground (plausible values) but occasionally wandering into the lowlands, and records every spot it stands on. Concretely:

1. **Start** at some initial parameter value θ_0.
2. **Propose** a nearby candidate value θ^*.
3. **Compare the posterior density** at θ^* versus the current spot θ_0 — i.e. ask "is this candidate more or less plausible, given prior + data?"
4. **Accept or reject**: if θ^* is more plausible, move there; if less plausible, move there only *sometimes* (with a probability equal to how much

less plausible it is). This "sometimes accept a worse spot" rule is what stops the chain getting stuck on one peak and lets it map the whole posterior.

5. **Repeat** thousands of times, writing down every value visited.

The crucial payoff is in step 4: because the walk lingers on high-density regions in proportion to their plausibility, the **collection of recorded values is itself a sample from the posterior**. You never computed the posterior formula — you reconstructed it from where the sampler spent its time. Pile those recorded values into a histogram and it traces out the posterior distribution, as the figure below shows. We typically run several independent chains (e.g., 4) from different starting points and discard each chain's initial **"warm-up"** period, during which it is still finding its way to the high-density region.

9.7.2.1. R

```
library(tidyverse)

# Target posterior for the 8/20 example with a Beta(2,2) prior: Beta(10, 14)
a_post <- 10; b_post <- 14

# Draw samples from the posterior (in practice MCMC would generate these;
# here we sample directly to illustrate what the accumulating draws look like
set.seed(1)
draws <- rbeta(4000, a_post, b_post)

theta <- seq(0, 1, length.out = 300)
true_curve <- tibble(theta = theta, density = dbeta(theta, a_post, b_post))

stages <- c(50, 500, 4000)
hist_data <- map_dfr(stages, function(k) {
  tibble(n_samples = paste0("After ", k, " samples"), theta = draws[1:k])
```

```
}) %>%
  mutate(n_samples = factor(n_samples,
           levels = paste0("After ", stages, " samples")))

ggplot(hist_data, aes(theta)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30,
                 fill = "grey70", colour = "white") +
  geom_line(data = expand_grid(n_samples = factor(paste0("After ", stages, " samples"),
                                levels = paste0("After ", stages, " samples")),
                              true_curve),
            aes(theta, density), colour = "firebrick", linewidth = 1) +
  facet_wrap(~ n_samples) +
  labs(x = expression(theta ~ "(response rate)"), y = "Density",
       title = "MCMC samples accumulating into the posterior") +
  theme_minimal(base_size = 12)
```

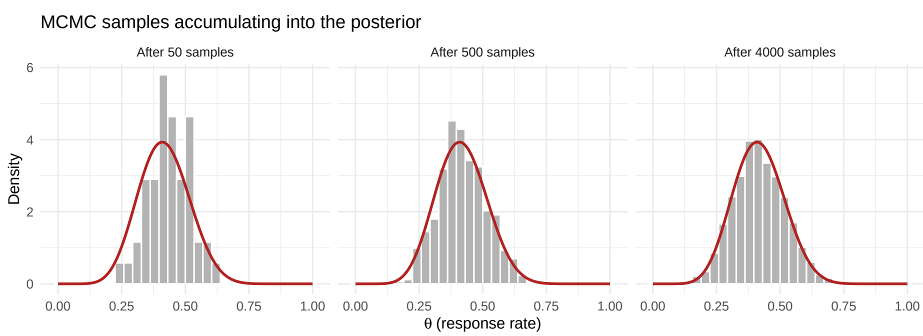

Figure 9.5.: How MCMC reconstructs a posterior. Each grey histogram is the collection of values the sampler has visited so far; as more samples accumulate (left to right), the histogram fills in the true posterior density (red curve) it was never given as a formula.

9.7.2.2. Python

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import beta

a_post, b_post = 10, 14
rng = np.random.default_rng(1)
draws = rng.beta(a_post, b_post, size=4000)
theta = np.linspace(0, 1, 300)
true_curve = beta.pdf(theta, a_post, b_post)

stages = [50, 500, 4000]
fig, axes = plt.subplots(1, 3, figsize=(12, 3.6), sharex=True)
for ax, k in zip(axes, stages):
    ax.hist(draws[:k], bins=30, density=True, color="grey", alpha=0.6,
            edgecolor="white")
    ax.plot(theta, true_curve, color="firebrick", linewidth=1.5)
    ax.set_title(f"After {k} samples", fontsize=10)
    ax.set_xlabel(r"$\theta$ (response rate)")
axes[0].set_ylabel("Density")
fig.suptitle("MCMC samples accumulating into the posterior", fontsize=12, y=
plt.tight_layout()
plt.show()
```

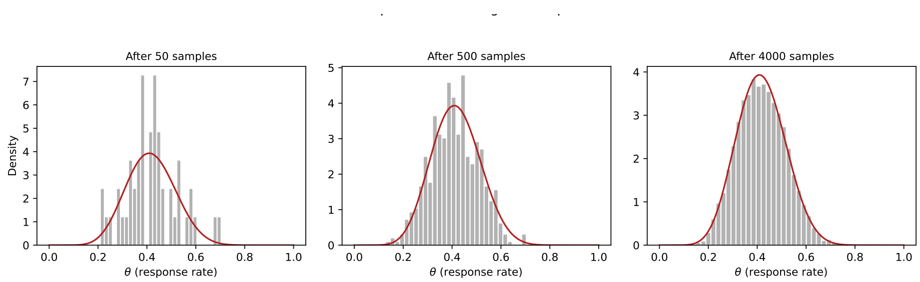

Figure 9.6.: How MCMC reconstructs a posterior. As more samples accumulate (left to right), the histogram of visited values fills in the true posterior density (red curve).

9.7.3. Diagnosing Convergence

MCMC hands you an answer, but you should not trust it blindly — the sampler can fail to explore the posterior properly, in which case the summaries are wrong. Fortunately, checking is quick: the software prints these diagnostics automatically, and you mainly need to recognise a green light from a red flag. There are three to know, and all three must look good before you believe the results. This is unfamiliar territory if you come from ordinary regression (where there is nothing analogous to "did the fitting algorithm converge?"), so here is each one in plain terms.

Trace plots — "did the chains explore freely?" A trace plot draws each chain's sampled value against the iteration number. You want it to look like a **"fuzzy caterpillar"**: a dense, flat, fuzzy band where all the chains overlap and rapidly zig-zag across the same range, with no drift up or down. That tells you the chains settled into the same region and explored it thoroughly. **Red flags**: a chain that wanders off on its own, drifts steadily in one direction, or gets "stuck" flat for long stretches — any of these means it has not converged. (The earlier figure contrasts good and poor mixing.)

$\hat{R}$ **(R-hat) — "do the separate chains agree?"** We run several chains from different starting points; if they have all found the same posterior, they should look interchangeable. $\hat{R}$ measures exactly that, by comparing the spread *within* each chain to the spread *between* chains. If they agree, $\hat{R} \approx 1.0$. **Rule of thumb**: $\hat{R} < 1.01$ is reassuring; values above ~1.05 signal a real problem — the chains disagree, so do not trust the estimates. The fix is usually to run the chains longer or reconsider the model.

Effective sample size (ESS) — "how much independent information do I really have?" Consecutive MCMC draws are correlated (each step is near the last), so 4,000 draws carry *less* information than 4,000 truly independent ones — they might be worth only, say, 1,000. ESS estimates that equivalent number of independent draws. A small ESS means your reported posterior mean and 95% interval are themselves noisy. **Rule of thumb**: aim for ESS in the high hundreds or more (400) for both the central estimate and the interval limits; if it is low, run more iterations.

In practice you glance at all three after every fit: trace plots fuzzy and overlapping, every $\hat{R} \approx 1.00$, and ESS comfortably in the hundreds. Only then do you read off the posterior.

9.7.4. R

```r
set.seed(42)

# Simulate a well-behaved chain (approximately normal draws)
good_chain <- cumsum(rnorm(2000, 0, 0.1)) * 0
good_chain[1] <- rnorm(1, 5, 1)
for (i in 2:2000) {
  good_chain[i] <- good_chain[i-1] + rnorm(1, 0, 0.3)
  # Metropolis-like: pull toward mean of 5
  good_chain[i] <- good_chain[i] + 0.1 * (5 - good_chain[i])
}
```

```r
# Simulate a poorly-mixing chain (high autocorrelation)
bad_chain <- numeric(2000)
bad_chain[1] <- 2
for (i in 2:2000) {
  bad_chain[i] <- bad_chain[i-1] + rnorm(1, 0, 0.02)
}

par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
plot(good_chain, type = "l", col = "steelblue",
     main = "Good mixing", xlab = "Iteration", ylab = expression(theta),
     cex.main = 1.1)
plot(bad_chain, type = "l", col = "firebrick",
     main = "Poor mixing", xlab = "Iteration", ylab = expression(theta),
     cex.main = 1.1)
```

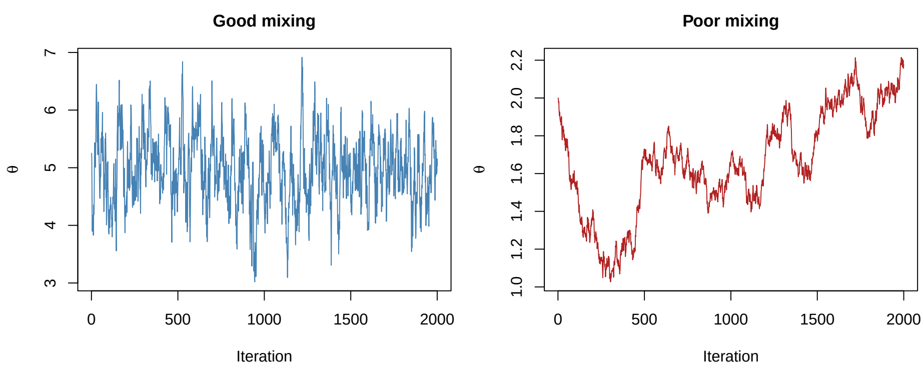

Figure 9.7.: Simulated MCMC trace plots: a well-behaved chain (left) and a poorly-mixing chain (right).

9.7.5. Python

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(42)

# Good chain: rapidly mixing
good_chain = np.zeros(2000)
good_chain[0] = np.random.normal(5, 1)
for i in range(1, 2000):
    good_chain[i] = good_chain[i-1] + np.random.normal(0, 0.3)
    good_chain[i] += 0.1 * (5 - good_chain[i])

# Bad chain: high autocorrelation, slow drift
bad_chain = np.zeros(2000)
bad_chain[0] = 2.0
for i in range(1, 2000):
    bad_chain[i] = bad_chain[i-1] + np.random.normal(0, 0.02)

fig, axes = plt.subplots(1, 2, figsize=(12, 4))
axes[0].plot(good_chain, color="steelblue", linewidth=0.5)
axes[0].set_title("Good mixing")
axes[0].set_xlabel("Iteration")
axes[0].set_ylabel(r"$\theta$")

axes[1].plot(bad_chain, color="firebrick", linewidth=0.5)
axes[1].set_title("Poor mixing")
axes[1].set_xlabel("Iteration")
axes[1].set_ylabel(r"$\theta$")

plt.tight_layout()
plt.show()
```

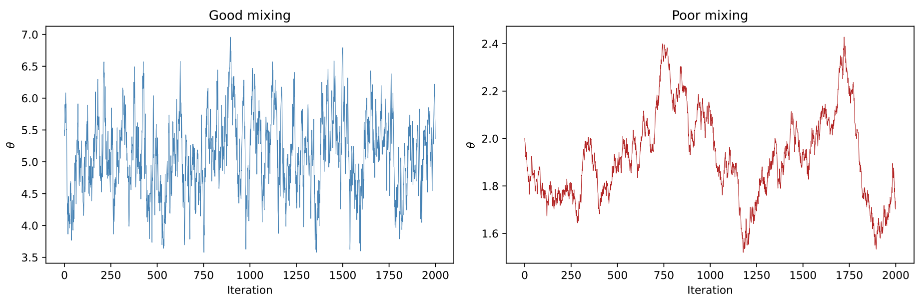

Figure 9.8.: Simulated MCMC trace plots: a well-behaved chain (left) and a poorly-mixing chain (right).

9.7.6. A first taste of Bayesian software

You do not need to write MCMC samplers yourself. Modern software handles this entirely. Here is what fitting a Bayesian logistic regression looks like in practice, using the Framingham heart-study data (10-year coronary heart disease as the outcome):

9.8. R

```
library(rstanarm)

# Load the data first (CSV included with the course materials)
framingham <- read.csv("data/framingham.csv")

fit <- stan_glm(chd_10yr ~ age + sex + smoking,
                data = framingham,
                family = binomial(),
                prior = normal(0, 2.5),  # weakly informative
```

```
                chains = 4, iter = 2000, seed = 42)
summary(fit)
```

9.9. Python

```python
import pandas as pd
import bambi as bmb

# Load the data first (CSV included with the course materials)
framingham = pd.read_csv("data/framingham.csv")

model = bmb.Model("chd_10yr ~ age + sex + smoking",
                   data=framingham, family="bernoulli")
results = model.fit(draws=2000, chains=4, random_seed=42)
results.summary()
```

The next chapter covers this in full detail, including how to check your model and interpret the output.

9.10. When Bayesian Methods Add Value

Bayesian inference is not always necessary, but there are situations where it provides genuine advantages over frequentist methods:

Small samples. When data are scarce (rare diseases, paediatric populations, early-phase trials), informative priors allow you to borrow strength from previous studies. A frequentist analysis of 12 patients yields wide confidence intervals and little power; a Bayesian analysis that incorporates prior evidence can produce more precise and clinically useful estimates.

Prior evidence is available. If there are five previous randomised trials of a drug, it would be wasteful to ignore that evidence when analysing a sixth. Bayesian methods provide a principled framework for incorporating it.

Rare events. When estimating the probability of a rare adverse event (e.g., 0 events in 200 patients), the frequentist MLE is zero — clearly an underestimate. A Bayesian analysis with an appropriate prior yields a plausible non-zero estimate.

Sequential updating. In adaptive clinical trials, data are analysed repeatedly as they accumulate. Bayesian methods handle this naturally: today's posterior becomes tomorrow's prior. Frequentist methods require complex corrections for multiple looks at the data.

Direct probability statements. Clinicians want to know "what is the probability that this treatment is effective?" Bayesian methods answer that question directly: $P(\text{treatment effect} > 0 \mid \text{data})$.

> **! Bayesian methods are not a magic fix**
>
> Bayesian inference does not rescue a poorly designed study, does not eliminate bias, and does not make small samples equivalent to large ones. The prior helps, but it also introduces subjectivity. Sensitivity analyses showing how results change under different priors are essential.

9.11. Practical Example: Estimating a Surgical Complication Rate

A hospital performs a new minimally invasive cardiac procedure. In the first 15 cases, 2 patients experience a major complication. The national average for the traditional open procedure is approximately 12%. What can we say about this hospital's complication rate?

9.11.1. R

```
# Data
y <- 2   # complications
n <- 15  # procedures

# Prior: weakly informative, centred near the national average of 12%
# Beta(3, 22) has mean 3/25 = 0.12 and is moderately concentrated
alpha_prior <- 3
beta_prior  <- 22

# Posterior
alpha_post <- alpha_prior + y
beta_post  <- beta_prior + n - y

theta <- seq(0, 0.5, length.out = 500)

plot_df <- tibble(
  theta = theta,
  Prior      = dbeta(theta, alpha_prior, beta_prior),
  Likelihood = dbeta(theta, y + 1, n - y + 1),  # normalised for visualisatio
  Posterior = dbeta(theta, alpha_post, beta_post)
) %>%
  pivot_longer(-theta, names_to = "Component", values_to = "Density")

ggplot(plot_df, aes(x = theta, y = Density, colour = Component)) +
  geom_line(linewidth = 1.2) +
  scale_colour_manual(values = c("Prior" = "steelblue",
                                  "Likelihood" = "grey50",
```

```r
                                    "Posterior" = "firebrick")) +
  labs(x = "Complication rate",
       y = "Density",
       title = "Bayesian Estimation of Surgical Complication Rate",
       subtitle = "2 complications in 15 procedures; Prior: Beta(3, 22)") +
  geom_vline(xintercept = 0.12, linetype = "dashed", colour = "grey40") +
  annotate("text", x = 0.12, y = max(dbeta(theta, alpha_post, beta_post)) * 0.55,
           label = "National average = 12%", angle = 90, vjust = -0.4,
           size = 3.3, colour = "grey30") +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom")

# Posterior summaries
cat("Posterior mean:", round(alpha_post / (alpha_post + beta_post), 3), "\n")
cat("95% credible interval:",
    round(qbeta(c(0.025, 0.975), alpha_post, beta_post), 3), "\n")
cat("P(rate > 0.20 | data):",
    round(1 - pbeta(0.20, alpha_post, beta_post), 3), "\n")
```

```
Posterior mean: 0.125
95% credible interval: 0.043 0.242
P(rate > 0.20 | data): 0.087
```

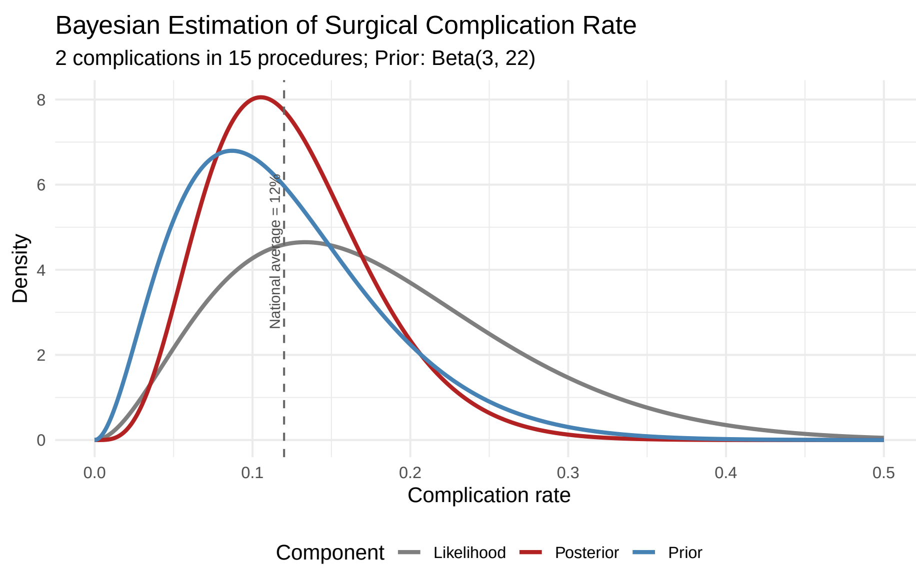

Figure 9.9.: Posterior distribution for the complication rate of a new surgical procedure.

9.11.2. Python

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import beta

y, n = 2, 15
alpha_prior, beta_prior = 3, 22
alpha_post = alpha_prior + y
beta_post_ = beta_prior + n - y

theta = np.linspace(0, 0.5, 500)
```

```
fig, ax = plt.subplots(figsize=(8, 5))
ax.plot(theta, beta.pdf(theta, alpha_prior, beta_prior),
        label="Prior", color="steelblue", linewidth=1.5)
ax.plot(theta, beta.pdf(theta, y + 1, n - y + 1),
        label="Likelihood (normalised)", color="grey", linewidth=1.5)
ax.plot(theta, beta.pdf(theta, alpha_post, beta_post_),
        label="Posterior", color="firebrick", linewidth=1.5)
ax.axvline(0.12, linestyle="--", color="grey", linewidth=1)
ax.text(0.12, ax.get_ylim()[1] * 0.55, "National average = 12%", rotation=90,
        ha="right", va="center", fontsize=8, color="grey")
ax.set_xlabel("Complication rate")
ax.set_ylabel("Density")
ax.set_title("Bayesian Estimation of Surgical Complication Rate\n"
             "2 complications in 15 procedures; Prior: Beta(3, 22)")
ax.legend()
plt.tight_layout()
plt.show()

ci = beta.ppf([0.025, 0.975], alpha_post, beta_post_)
print(f"Posterior mean: {alpha_post / (alpha_post + beta_post_):.3f}")
print(f"95% credible interval: [{ci[0]:.3f}, {ci[1]:.3f}]")
print(f"P(rate > 0.20 | data): {1 - beta.cdf(0.20, alpha_post, beta_post_):.3f}")
```

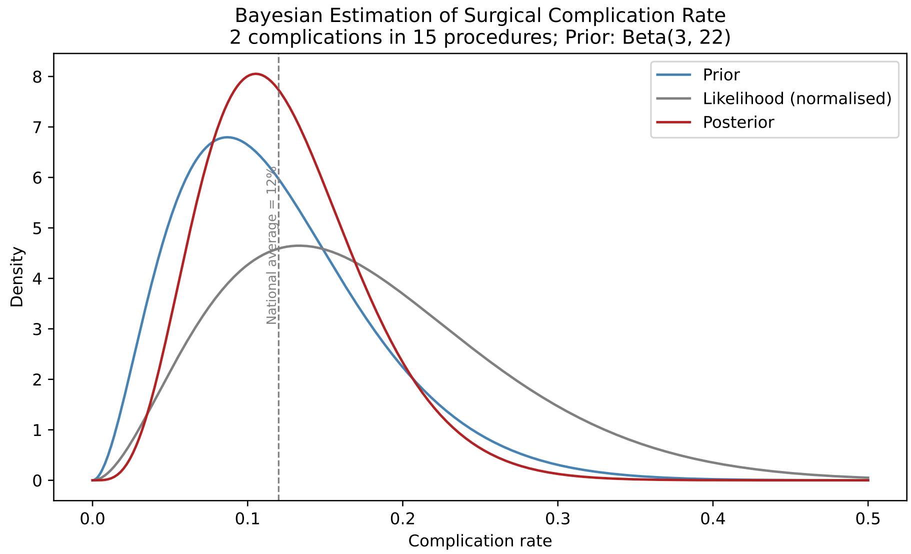

Figure 9.10.: Posterior distribution for the complication rate of a new surgical procedure.

```
Posterior mean: 0.125
95% credible interval: [0.043, 0.242]
P(rate > 0.20 | data): 0.087
```

The posterior mean is pulled between the MLE ($2/15 = 0.133$) and the prior mean (0.12), settling near 0.125. With only 15 observations, the prior meaningfully influences the result — and that is appropriate, because we have genuine prior information about surgical complication rates.

9.12. Exercises

9.12.1. Exercise 1: Diagnostic Test Updating

A screening mammogram has a sensitivity of 90% and a specificity of 92%. Among women aged 50–59, the prevalence of breast cancer is approximately 2%.

a. Calculate the posterior probability of breast cancer given a positive mammogram using Bayes' theorem.
b. If the patient then receives a confirmatory ultrasound (sensitivity 95%, specificity 85%), use the posterior from (a) as the new prior and calculate the updated posterior probability after a positive ultrasound.
c. What does this sequential updating illustrate about the Bayesian framework?

9.12.2. R

```r
# Part (a): Mammogram
sens_mammo <- 0.90
spec_mammo <- 0.92
prevalence <- 0.02

post_mammo <- (sens_mammo * prevalence) /
  (sens_mammo * prevalence + (1 - spec_mammo) * (1 - prevalence))
cat("P(cancer | positive mammogram):", round(post_mammo, 4), "\n")

# Part (b): Ultrasound, using posterior from (a) as new prior
sens_us <- 0.95
spec_us <- 0.85
prior_us <- post_mammo
```

```
post_us <- (sens_us * prior_us) /
  (sens_us * prior_us + (1 - spec_us) * (1 - prior_us))
cat("P(cancer | positive mammogram AND positive ultrasound):",
    round(post_us, 4), "\n")
```

9.12.3. Python

```
# Part (a)
sens_mammo, spec_mammo, prevalence = 0.90, 0.92, 0.02
post_mammo = (sens_mammo * prevalence) / \
    (sens_mammo * prevalence + (1 - spec_mammo) * (1 - prevalence))
print(f"P(cancer | positive mammogram): {post_mammo:.4f}")

# Part (b)
sens_us, spec_us = 0.95, 0.85
post_us = (sens_us * post_mammo) / \
    (sens_us * post_mammo + (1 - spec_us) * (1 - post_mammo))
print(f"P(cancer | pos mammo AND pos ultrasound): {post_us:.4f}")
```

9.12.4. Exercise 2: Prior Sensitivity Analysis

A Phase II trial of a new antibiotic enrols 30 patients with resistant urinary tract infections. Twenty-one patients achieve clinical cure.

a. Using three different priors — Beta(1,1), Beta(5,5), and Beta(20,10) — compute the posterior mean and 95% credible interval for the cure rate.
b. Plot all three posteriors on the same graph.
c. Which prior has the most influence on the posterior? Why?

9.12.5. R

```r
library(tidyverse)  # tibble(), bind_rows(), ggplot()

y <- 21; n <- 30
priors <- list(
  list(name = "Beta(1,1)", a = 1, b = 1),
  list(name = "Beta(5,5)", a = 5, b = 5),
  list(name = "Beta(20,10)", a = 20, b = 10)
)

theta <- seq(0, 1, length.out = 500)
plot_df <- tibble()

for (p in priors) {
  a_post <- p$a + y
  b_post <- p$b + n - y
  ci <- qbeta(c(0.025, 0.975), a_post, b_post)
  cat(sprintf("%s -> Posterior mean: %.3f, 95%% CI: [%.3f, %.3f]\n",
              p$name, a_post / (a_post + b_post), ci[1], ci[2]))
  plot_df <- bind_rows(plot_df,
    tibble(theta = theta, Density = dbeta(theta, a_post, b_post),
           Prior = p$name))
}

ggplot(plot_df, aes(x = theta, y = Density, colour = Prior)) +
  geom_line(linewidth = 1.2) +
  labs(x = "Cure rate", y = "Density",
       title = "Posterior Distributions Under Different Priors",
       subtitle = "21/30 patients cured") +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom")
```

9.12.6. Python

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import beta

y, n = 21, 30
priors = [("Beta(1,1)", 1, 1), ("Beta(5,5)", 5, 5), ("Beta(20,10)", 20, 10)]
theta = np.linspace(0, 1, 500)

fig, ax = plt.subplots(figsize=(8, 5))
for name, a0, b0 in priors:
    a_post, b_post = a0 + y, b0 + n - y
    ci = beta.ppf([0.025, 0.975], a_post, b_post)
    mean_post = a_post / (a_post + b_post)
    print(f"{name} -> Mean: {mean_post:.3f}, 95% CI: [{ci[0]:.3f}, {ci[1]:.3
    ax.plot(theta, beta.pdf(theta, a_post, b_post), label=name, linewidth=1.

ax.set_xlabel("Cure rate")
ax.set_ylabel("Density")
ax.set_title("Posterior Distributions Under Different Priors\n21/30 patients
ax.legend()
plt.tight_layout()
plt.show()
```

9.12.7. Exercise 3: Interpreting MCMC Diagnostics

You fit a Bayesian logistic regression model and obtain the following diagnostics for the treatment effect parameter:

- Chain 1 R-hat: 1.08
- Chain 2 R-hat: 1.08

- Bulk ESS: 150
- Tail ESS: 85
- The trace plot shows two chains that appear to be exploring different regions of the parameter space.

a. Is there evidence of convergence problems? Explain which diagnostics are concerning and why.
b. What steps would you take to fix these problems?
c. Would you trust the posterior summaries from this model? Why or why not?

9.13. Summary

Bayesian inference provides a coherent framework for updating beliefs in light of evidence. Its core components — the prior, the likelihood, and the posterior — are connected by Bayes' theorem. While simple conjugate models (like Beta-Binomial) can be solved analytically, most real-world models require MCMC sampling. The key practical advantages of Bayesian methods include direct probability statements, principled incorporation of prior knowledge, and natural handling of sequential data — all of which are particularly valuable in clinical research.

In the next chapter, we will move from theory to practice: fitting Bayesian regression and hierarchical models using modern software.

Key Takeaways

1. Bayesian inference treats parameters as uncertain and data as fixed — the reverse of frequentist thinking.
2. The posterior is proportional to the likelihood times the prior.
3. Credible intervals have the intuitive interpretation people mistakenly attribute to confidence intervals.
4. MCMC allows us to sample from complex posteriors; always

check convergence with trace plots, R-hat, and ESS.

5. Bayesian methods shine when samples are small, prior evidence exists, or direct probability statements are needed.

9.14. References and Further Reading

The foundational textbook for applied Bayesian analysis is McElreath's *Statistical Rethinking* (2nd edition, 2020), which teaches Bayesian inference through simulation and coding rather than mathematical derivation. It is highly recommended for students coming from a non-statistical background. For a more comprehensive and mathematically rigorous treatment, Gelman, Carlin, Stern, Dunson, Vehtari, and Rubin's *Bayesian Data Analysis* (3rd edition, 2013), commonly known as BDA3, remains the standard reference; a free PDF is available from the authors' website. Kruschke's *Doing Bayesian Data Analysis* (2nd edition, 2015) offers a gentle introduction with extensive worked examples in R.

For the clinical and regulatory context, the FDA's 2010 guidance document *Guidance for the Use of Bayesian Statistics in Medical Device Clinical Trials* provides a framework for incorporating Bayesian methods in regulatory submissions, and updated guidance released in 2026 reflects the growing acceptance of Bayesian adaptive designs. Spiegelhalter, Abrams, and Myles' *Bayesian Approaches to Clinical Trials and Health-Care Evaluation* (2004) is particularly relevant for public health students, covering prior elicitation, adaptive designs, and cost-effectiveness analysis. The Stan Development Team's documentation (mc-stan.org) includes excellent case studies and practical tutorials that complement any of the above textbooks.

10. Applied Bayesian Modelling: Regression, Hierarchical Models, and Clinical Applications

10.1. Introduction

The previous chapter introduced the conceptual foundations of Bayesian inference: priors, likelihoods, posteriors, and MCMC. This chapter puts those ideas to work. We will fit Bayesian regression models, learn the critical model-checking steps that distinguish careful Bayesian analysis from careless button-pressing, and then tackle the most powerful application of the Bayesian framework in clinical research: **hierarchical (multilevel) models**.

Hierarchical models deserve special attention because they solve a problem that arises constantly in medicine. Clinical data are almost always grouped: patients within hospitals, measurements within patients, sites within multi-centre trials. Frequentist mixed-effects models handle this, but the Bayesian hierarchical framework does it more flexibly and with clearer interpretation. By the end of this chapter, you will understand partial pooling, shrinkage, and why these ideas matter for clinical decision-making.

10.2. Bayesian Linear Regression

10.2.1. How It Differs from Frequentist Regression

In frequentist ordinary least squares (OLS), we obtain point estimates of regression coefficients and standard errors. A 95% confidence interval is a statement about long-run coverage.

In Bayesian linear regression, each coefficient has a **full posterior distribution**. We obtain not just an estimate and an interval, but the entire shape of our uncertainty. This is the same linear model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi} + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2)$$

Reading this equation in words: the outcome for patient i (say, their blood pressure, y_i) is built up from a baseline value plus a contribution from each predictor, plus some leftover noise.

- y_i is the outcome we want to model for patient i.
- β_0 (the **intercept**) is the predicted outcome when every predictor equals zero — a baseline level.
- $\beta_1, \dots, \beta_p$ (the **slopes** or coefficients) say how much the outcome changes for a one-unit increase in each predictor $x_1, \dots, x_p$. For example, $\beta_1 = 0.5$ for age means the outcome rises by 0.5 units for each additional year.
- $x_{1i}, \dots, x_{pi}$ are the predictor values measured on patient i (age, BMI, and so on).
- ϵ_i is the **residual** — the part of patient i's outcome that the predictors do not explain. We assume it is random noise centred on zero with standard deviation σ, which captures how much patients scatter around the line.

The symbol $\sim$ Normal$(0, \sigma^2)$ simply means "is drawn from a bell-shaped (normal) distribution centred at 0 with spread σ." This is the same linear model you would fit with `lm()`; nothing about the structure changes.

What *is* new in the Bayesian version is that we place priors on all parameters: $\beta_0, \beta_1, \ldots, \beta_p, \sigma$. A prior is just a starting belief about plausible values before seeing the data. The MCMC sampler then combines those priors with the data and generates draws from the joint posterior $P(\beta_0, \beta_1, \ldots, \beta_p, \sigma \mid \mathbf{y})$ — our updated belief about every parameter once the data have been taken into account.

10.2.2. Clinical Example: Predicting Systolic Blood Pressure

We will model systolic blood pressure (SBP) as a function of age, BMI, and whether the patient is on antihypertensive medication.

> ❗ Load your packages first
>
> Every R example in this chapter (and the exercises) assumes you have loaded the **tidyverse** — this is where functions like `tibble()`, `mutate()`, the pipe `%>%`, and `ggplot()` come from. If you skip it, R will not recognise these functions. We show `library(tidyverse)` at the top of each code block as a reminder; you only need to run it once per session. The Bayesian models additionally need `library(brms)`, and some plots need `library(bayesplot)`.

10.2.3. R

```
library(tidyverse)  # provides tibble(), mutate(), %>%, ggplot()
library(brms)       # provides brm() for Bayesian models

# Simulate clinical data
```

```
set.seed(123)
n <- 200
bp_data <- tibble(
  age = round(rnorm(n, 55, 12)),
  bmi = round(rnorm(n, 28, 5), 1),
  on_meds = rbinom(n, 1, 0.4),
  sbp = round(90 + 0.5 * age + 0.8 * bmi - 8 * on_meds + rnorm(n, 0, 12))
)

# Fit Bayesian linear regression with weakly informative priors
fit_bp <- brm(
  sbp ~ age + bmi + on_meds,
  data = bp_data,
  family = gaussian(),
  prior = c(
    prior(normal(0, 20), class = "b"),          # weakly informative for slopes
    prior(normal(120, 30), class = "Intercept"), # centred on typical SBP
    prior(exponential(0.1), class = "sigma")     # positive scale parameter
  ),
  chains = 4,
  iter = 2000,
  warmup = 1000,
  seed = 42,
  silent = 2,
  refresh = 0
)

summary(fit_bp)
```

```
library(tidyverse)
library(bayesplot)  # provides mcmc_areas() for posterior plots
mcmc_areas(fit_bp, pars = c("b_age", "b_bmi", "b_on_meds"),
           prob = 0.95, prob_outer = 0.99) +
```

```
  labs(title = "Posterior Distributions of Regression Coefficients",
       subtitle = "Shaded region = 95% credible interval") +
  theme_minimal(base_size = 13)
```

10.2.4. Python

```python
import numpy as np
import pandas as pd
import pymc as pm
import arviz as az

np.random.seed(123)
n = 200
bp_data = pd.DataFrame({
    'age': np.round(np.random.normal(55, 12, n)),
    'bmi': np.round(np.random.normal(28, 5, n), 1),
    'on_meds': np.random.binomial(1, 0.4, n)
})
bp_data['sbp'] = np.round(
    90 + 0.5 * bp_data['age'] + 0.8 * bp_data['bmi']
    - 8 * bp_data['on_meds'] + np.random.normal(0, 12, n)
)

with pm.Model() as bp_model:
    # Priors
    intercept = pm.Normal("Intercept", mu=120, sigma=30)
    b_age     = pm.Normal("b_age", mu=0, sigma=20)
    b_bmi     = pm.Normal("b_bmi", mu=0, sigma=20)
    b_meds    = pm.Normal("b_on_meds", mu=0, sigma=20)
    sigma     = pm.Exponential("sigma", lam=0.1)

    # Linear model
```

```
    mu = intercept + b_age * bp_data['age'].values + \
         b_bmi * bp_data['bmi'].values + \
         b_meds * bp_data['on_meds'].values

    # Likelihood
    y_obs = pm.Normal("sbp", mu=mu, sigma=sigma, observed=bp_data['sbp'].valu

    # Sample
    trace_bp = pm.sample(1000, tune=1000, chains=4, random_seed=42,
                         progressbar=False)

print(az.summary(trace_bp, var_names=["Intercept", "b_age", "b_bmi",
                                      "b_on_meds", "sigma"]))
```

```
az.plot_forest(trace_bp, var_names=["b_age", "b_bmi", "b_on_meds"],
               combined=True, hdi_prob=0.95, figsize=(8, 4))
import matplotlib.pyplot as plt
plt.title("Posterior Distributions of Regression Coefficients\n95% HDI")
plt.tight_layout()
plt.show()
```

The output shows the **posterior mean**, **standard deviation** (the Bayesian analogue of the standard error), and the **95% credible interval** for each coefficient. Notice that the on-medication effect has a credible interval that is entirely below zero, providing direct evidence that medication lowers SBP.

> **i** What are `chains`, `iter`, and `warmup`?
>
> These three arguments control the MCMC sampler and appear in every `brm()` call in this chapter, so it is worth pausing on them.
>
> - **`chains = 4`** runs the sampler four separate times from different

starting points. If all four chains end up exploring the same region, that is reassuring evidence the sampler has found the right answer (this is what the $\hat{R}$ diagnostic checks).

- `iter = 2000` is the total number of steps each chain takes.
- `warmup = 1000` (sometimes called *burn-in*) is the number of *initial* steps that are **thrown away**. When a chain starts, it begins at an arbitrary point that may be far from the posterior, and it also needs time to tune its internal step size. Those early, unreliable samples would distort our estimates, so we discard them and keep only what comes after.

So `iter = 2000` with `warmup = 1000` means each chain takes 2000 steps but we *keep* only the last 1000 — giving $4 \times 1000 = 4000$ usable posterior draws. A useful analogy: warm-up is like letting a car engine idle until it reaches operating temperature before you start the timed lap. The warm-up laps still happen; you simply do not record them.

10.3. Bayesian Logistic Regression

For binary outcomes (e.g., 30-day readmission, treatment response), we use Bayesian logistic regression. The model is identical to the frequentist version except that all parameters receive priors:

$$\text{logit}(P(y_i = 1)) = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi}$$

10.3.1. R

```
library(tidyverse)
library(brms)
```

```
# Simulate readmission data
set.seed(456)
n <- 300
readmit_data <- tibble(
  age = round(rnorm(n, 65, 10)),
  charlson = rpois(n, 2),
  los = rpois(n, 5) + 1,  # length of stay in days
  readmit_30 = rbinom(n, 1,
    plogis(-3 + 0.02 * round(rnorm(n, 65, 10)) +
             0.3 * rpois(n, 2) + 0.1 * (rpois(n, 5) + 1)))
)

fit_readmit <- brm(
  readmit_30 ~ age + charlson + los,
  data = readmit_data,
  family = bernoulli(link = "logit"),
  prior = c(
    prior(normal(0, 2.5), class = "b"),       # weakly informative on log-odd
    prior(normal(0, 5), class = "Intercept")
  ),
  chains = 4,
  iter = 2000,
  warmup = 1000,
  seed = 42,
  silent = 2,
  refresh = 0
)

# Posterior summary on the odds ratio scale
posterior_samples <- as_draws_df(fit_readmit)
or_summary <- posterior_samples %>%
  transmute(
    OR_age = exp(b_age),
```

```
    OR_charlson = exp(b_charlson),
    OR_los = exp(b_los)
  ) %>%
  summarise(across(everything(),
    list(mean = mean,
         q2.5 = ~quantile(., 0.025),
         q97.5 = ~quantile(., 0.975))))

cat("Posterior odds ratios (mean [95% CrI]):\n")
cat(sprintf("  Age:      %.3f [%.3f, %.3f]\n",
    or_summary$OR_age_mean, or_summary$OR_age_q2.5, or_summary$OR_age_q97.5))
cat(sprintf("  Charlson: %.3f [%.3f, %.3f]\n",
    or_summary$OR_charlson_mean, or_summary$OR_charlson_q2.5,
    or_summary$OR_charlson_q97.5))
cat(sprintf("  LOS:      %.3f [%.3f, %.3f]\n",
    or_summary$OR_los_mean, or_summary$OR_los_q2.5, or_summary$OR_los_q97.5))
```

10.3.2. Python

```
import numpy as np
import pandas as pd
import pymc as pm
import arviz as az

np.random.seed(456)
n = 300
readmit_data = pd.DataFrame({
    'age': np.round(np.random.normal(65, 10, n)),
    'charlson': np.random.poisson(2, n),
    'los': np.random.poisson(5, n) + 1
})
```

```python
from scipy.special import expit
lp = -3 + 0.02 * readmit_data['age'] + 0.3 * readmit_data['charlson'] + \
     0.1 * readmit_data['los']
readmit_data['readmit_30'] = np.random.binomial(1, expit(lp))

with pm.Model() as readmit_model:
    intercept = pm.Normal("Intercept", mu=0, sigma=5)
    b_age     = pm.Normal("b_age", mu=0, sigma=2.5)
    b_charl   = pm.Normal("b_charlson", mu=0, sigma=2.5)
    b_los     = pm.Normal("b_los", mu=0, sigma=2.5)

    logit_p = intercept + b_age * readmit_data['age'].values + \
              b_charl * readmit_data['charlson'].values + \
              b_los * readmit_data['los'].values

    y_obs = pm.Bernoulli("readmit", logit_p=logit_p,
                          observed=readmit_data['readmit_30'].values)
    trace_readmit = pm.sample(1000, tune=1000, chains=4, random_seed=42,
                               progressbar=False)

# Compute odds ratios from posterior
posterior = az.extract(trace_readmit)
for var in ["b_age", "b_charlson", "b_los"]:
    or_vals = np.exp(posterior[var].values)
    print(f"OR {var}: {or_vals.mean():.3f} "
          f"[{np.percentile(or_vals, 2.5):.3f}, "
          f"{np.percentile(or_vals, 97.5):.3f}]")
```

10.4. Prior Predictive Checks

A prior predictive check asks: **before seeing any data, does the model generate plausible outcomes?** This is a critical step that is often skipped. If your priors allow the model to predict systolic blood pressures of 500 mmHg or negative values, your priors are too vague.

The procedure:

1. Sample parameter values from the priors (not the posterior).
2. Simulate data from the model using those parameter values.
3. Plot the simulated data and check whether it covers a plausible range.

10.4.1. R

```r
library(tidyverse)
library(brms)

# Generate prior predictive samples
pp_check_prior <- brm(
  sbp ~ age + bmi + on_meds,
  data = bp_data,
  family = gaussian(),
  prior = c(
    prior(normal(0, 20), class = "b"),
    prior(normal(120, 30), class = "Intercept"),
    prior(exponential(0.1), class = "sigma")
  ),
  sample_prior = "only",
  chains = 4,
  iter = 2000,
  warmup = 1000,
  seed = 42,
  silent = 2,
  refresh = 0
)
```

```
pp_samples <- posterior_predict(pp_check_prior)

# Plot density of prior predictive samples vs observed data
tibble(
  simulated = as.vector(pp_samples[sample(nrow(pp_samples), 50), ]),
  type = "Prior predictive"
) %>%
  ggplot(aes(x = simulated)) +
  geom_density(fill = "steelblue", alpha = 0.3) +
  geom_density(data = bp_data, aes(x = sbp), fill = "firebrick", alpha = 0.3)
  labs(x = "Systolic Blood Pressure (mmHg)",
       y = "Density",
       title = "Prior Predictive Check",
       subtitle = "Blue = simulated from priors; Red = observed data") +
  xlim(-100, 350) +
  theme_minimal(base_size = 13)
```

10.4.2. Python

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(42)
n_sim = 2000

# Simulate from priors
intercepts = np.random.normal(120, 30, n_sim)
b_ages     = np.random.normal(0, 20, n_sim)
b_bmis     = np.random.normal(0, 20, n_sim)
b_meds_arr = np.random.normal(0, 20, n_sim)
sigmas     = np.random.exponential(10, n_sim)
```

```
# Pick a "typical" patient: age=55, bmi=28, on_meds=0
sim_sbp = intercepts + b_ages * 55 + b_bmis * 28 + b_meds_arr * 0
sim_sbp += np.random.normal(0, sigmas)

fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(sim_sbp, bins=80, density=True, alpha=0.4, color="steelblue",
        label="Prior predictive", range=(-500, 800))
ax.hist(bp_data['sbp'].values, bins=30, density=True, alpha=0.4,
        color="firebrick", label="Observed data")
ax.set_xlabel("Systolic Blood Pressure (mmHg)")
ax.set_ylabel("Density")
ax.set_title("Prior Predictive Check\nBlue = simulated from priors; Red = observed")
ax.set_xlim(-200, 400)
ax.legend()
plt.tight_layout()
plt.show()
```

How to read this plot. The blue curve is the range of blood pressures the model considers possible *based on the priors alone*, before it has seen a single patient. The red curve is the real data, shown only for comparison. What we are looking for is a sanity check: the blue curve should comfortably cover the clinically realistic range (roughly 80–200 mmHg) without putting serious weight on impossible values such as negative pressures or readings above 400 mmHg. Here the blue curve is wide — it does not yet know the typical SBP — but it is centred in a sensible place and is not absurd, so these priors are acceptable. It is meant to be broader than the data; the data will sharpen it later.

If the prior predictive distribution is wildly broader than any plausible clinical range, consider tightening the priors. If it is too narrow and excludes plausible values, widen them. This iterative process happens **before** fitting the model to data.

10.5. Posterior Predictive Checks

After fitting, we ask: **does the fitted model generate data that look like the observed data?** This is the Bayesian equivalent of residual diagnostics.

The procedure:

1. Draw parameter values from the posterior.
2. Simulate new data from the model using those posterior draws.
3. Compare the distribution of simulated data to the observed data.

If the model is well specified, the simulated data should be nearly indistinguishable from the real data in terms of summary statistics (mean, variance, range, shape).

How to read this plot. Each light blue line is one fake dataset the *fitted* model generates; the dark line is the observed data. Unlike the prior predictive check, here the model has already learned from the data, so we expect a close match. If the dark line sits comfortably within the spread of blue lines — same centre, same width, same overall shape — the model is reproducing the data well. If the dark line falls outside the blue cloud (for example, the data are skewed but the model only produces symmetric curves), the model is misspecified and the likelihood or priors need revisiting.

10.5.1. R

```
library(tidyverse)
library(brms)

pp_check(fit_bp, type = "dens_overlay", ndraws = 100) +
  labs(title = "Posterior Predictive Check",
```

```
      subtitle = "Light blue = simulated from posterior; Dark = observed") +
  theme_minimal(base_size = 13)
```

10.5.2. Python

```
import pymc as pm
import arviz as az
import matplotlib.pyplot as plt

# Using the previously fitted model
with bp_model:
    ppc = pm.sample_posterior_predictive(trace_bp, random_seed=42,
                                          progressbar=False)

az.plot_ppc(az.from_pymc3(posterior_predictive=ppc,
                           observed_data={"sbp": bp_data['sbp'].values}),
            num_pp_samples=100, figsize=(8, 5))
plt.title("Posterior Predictive Check")
plt.tight_layout()
plt.show()
```

10.6. Bayesian Hierarchical (Multilevel) Models

10.6.1. Why They Matter

Consider a multi-site clinical trial testing a new antihypertensive drug across 12 hospitals. The treatment effect might vary from site to site due to differences in patient populations, clinical protocols, or local practice patterns. How should we estimate the treatment effect at each site?

There are three naive approaches:

1. **Complete pooling**: ignore the site structure and estimate a single treatment effect. This assumes all sites are identical — clearly wrong.
2. **No pooling**: estimate a separate treatment effect at each site. This ignores that the sites are studying the same drug. Small sites will have wildly imprecise estimates.
3. **Partial pooling (hierarchical model)**: the Bayesian approach. Each site has its own treatment effect, but those effects are drawn from a shared distribution. Sites with small samples are **pulled toward the overall mean**, while sites with large samples retain their individual estimates.

This pulling toward the grand mean is called **shrinkage**, and it is one of the most powerful ideas in applied statistics.

10.6.2. The Model Structure

A Bayesian hierarchical model for treatment effects across J hospitals:

$$y_{ij} = \beta_0 + \beta_1 \text{treatment}_{ij} + u_j + \epsilon_{ij}$$

Reading the equation: y_{ij} is the outcome for patient i in hospital j. The first two terms, $\beta_0 + \beta_1\text{treatment}_{ij}$, are the *average* picture across all hospitals — the same intercept and treatment effect we would get from an ordinary regression. The new piece is u_j, a small "nudge" that is specific to hospital j. It captures how that hospital departs from the average. The terms are:

- β_0 is the overall baseline outcome (the **fixed intercept**, shared by all hospitals).
- β_1 is the **average treatment effect** across all hospitals.
- u_j is the **random effect** for hospital j: how much hospital j's own treatment effect sits above or below the average β_1. A hospital

with $u_j = +2$ responds 2 units more than the typical site; one with $u_j = -2$ responds 2 units less.
- ϵ_{ij} is the residual error for an individual patient — the usual unexplained patient-to-patient noise.

What does "random effect" mean? Instead of estimating each hospital's deviation u_j in complete isolation, we assume the deviations themselves come from a shared distribution:

$$u_j \sim \text{Normal}(0, \tau^2).$$

This single line is what makes the model "hierarchical." It says the hospital effects scatter around zero (zero meaning "exactly average") with a spread of τ. Because all the u_j are tied together by this common distribution, information is shared across hospitals: a hospital with very few patients borrows strength from the others rather than relying only on its own noisy data. That borrowing is exactly the partial pooling and shrinkage we explore below.

The role of τ. The parameter τ is the **between-hospital standard deviation** — it answers the clinically important question "how much do hospitals actually differ?"

- If τ is close to **zero**, the hospitals are nearly identical, every u_j is tiny, and the model collapses toward a single shared treatment effect (complete pooling).
- If τ is **large**, hospitals differ substantially, and each site keeps much of its own estimate (closer to no pooling).

Crucially, τ is not fixed in advance; it is **estimated from the data**, so the model learns how much pooling is warranted rather than assuming it.

In the Bayesian framework, τ also gets a prior. Because τ is a standard deviation it cannot be negative, so we use a prior defined only on positive values. Two common choices are the **half-normal** and the **half-Cauchy**

— the right-hand halves of the ordinary normal and Cauchy distributions. Both place most of their weight near zero (gently regularising toward "hospitals are similar") while still permitting larger values. The difference is in the tail: the half-Cauchy has a much heavier tail, so it is more willing to allow a large between-hospital spread if the data demand it. This makes the half-Cauchy a popular, mildly conservative default for group-level standard deviations, especially when the number of groups is small.

10.6.3. R

```
library(tidyverse)

tau_grid <- seq(0, 20, length.out = 400)

prior_df <- bind_rows(
  tibble(tau = tau_grid,
         density = 2 * dnorm(tau_grid, 0, 5),    # half-normal(0, 5)
         prior = "Half-Normal(0, 5)"),
  tibble(tau = tau_grid,
         density = 2 * dcauchy(tau_grid, 0, 5),  # half-Cauchy(0, 5)
         prior = "Half-Cauchy(0, 5)")
)

ggplot(prior_df, aes(x = tau, y = density, colour = prior)) +
  geom_line(linewidth = 1) +
  scale_colour_manual(values = c("Half-Normal(0, 5)" = "steelblue",
                                  "Half-Cauchy(0, 5)" = "firebrick")) +
  labs(x = expression(tau~"(between-hospital SD)"),
       y = "Prior density",
       colour = "Prior",
       title = "Priors for a Group-Level Standard Deviation",
       subtitle = "Both favour small tau; the half-Cauchy keeps a heavier ta
```

```
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom")
```

10.6.4. Python

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

tau_grid = np.linspace(0, 20, 400)
half_normal = 2 * stats.norm.pdf(tau_grid, 0, 5)    # half-normal(0, 5)
half_cauchy = 2 * stats.cauchy.pdf(tau_grid, 0, 5)  # half-Cauchy(0, 5)

fig, ax = plt.subplots(figsize=(6.5, 4))
ax.plot(tau_grid, half_normal, color="steelblue", linewidth=2,
        label="Half-Normal(0, 5)")
ax.plot(tau_grid, half_cauchy, color="firebrick", linewidth=2,
        label="Half-Cauchy(0, 5)")
ax.set_xlabel(r"$\tau$ (between-hospital SD)")
ax.set_ylabel("Prior density")
ax.set_title("Priors for a Group-Level Standard Deviation\n"
             "Both favour small tau; the half-Cauchy keeps a heavier tail")
ax.legend()
plt.tight_layout()
plt.show()
```

10.6.5. Clinical Example: Treatment Effects Across Hospitals

10.6.6. R

```
library(tidyverse)  # tibble(), map2_dfr(), %>%, ggplot()
library(brms)

# Simulate multi-site trial data
set.seed(789)
n_hospitals <- 12
patients_per_hospital <- c(15, 20, 25, 30, 35, 40, 50, 60, 80, 100, 120, 150)
true_grand_effect <- -8  # mmHg reduction in SBP
tau_true <- 3            # between-hospital SD

hospital_effects <- rnorm(n_hospitals, 0, tau_true)

trial_data <- map2_dfr(1:n_hospitals, patients_per_hospital, function(j, nj) {
  tibble(
    hospital = factor(j),
    treatment = rep(0:1, length.out = nj),
    sbp = 140 + (true_grand_effect + hospital_effects[j]) * treatment +
      rnorm(nj, 0, 15)
  )
})

# Fit hierarchical model
fit_hier <- brm(
  sbp ~ treatment + (treatment | hospital),
  data = trial_data,
  prior = c(
    prior(normal(0, 20), class = "b"),
    prior(normal(140, 30), class = "Intercept"),
    prior(cauchy(0, 5), class = "sd"),
```

```
    prior(exponential(0.1), class = "sigma")
  ),
  chains = 4,
  iter = 2000,
  warmup = 1000,
  seed = 42,
  silent = 2,
  refresh = 0,
  control = list(adapt_delta = 0.95)
)

# Extract hospital-specific treatment effects (partial pooling)
ranefs <- ranef(fit_hier)$hospital[, , "treatment"]
grand_mean <- fixef(fit_hier)["treatment", "Estimate"]

# No-pooling estimates (separate OLS per hospital)
no_pool <- trial_data %>%
  group_by(hospital) %>%
  summarise(no_pool_est = coef(lm(sbp ~ treatment))[2],
            n = n(), .groups = "drop") %>%
  mutate(
    hospital_num = as.numeric(hospital),
    partial_pool_est = grand_mean + ranefs[, "Estimate"],
    # Label each hospital with its sample size so the reader can see
    # which sites are small (and therefore shrink most)
    hospital_label = paste0("Hospital ", hospital, " (n = ", n, ")")
  )

# Plot shrinkage
ggplot(no_pool, aes(y = reorder(hospital_label, n))) +
  geom_point(aes(x = no_pool_est, colour = "No pooling"), size = 3) +
  geom_point(aes(x = partial_pool_est, colour = "Partial pooling"), size = 3) +
  geom_vline(xintercept = grand_mean, linetype = "dashed", colour = "grey40") +
```

```
  annotate("text", x = grand_mean + 0.5, y = 12.5,
           label = paste0("Grand mean = ", round(grand_mean, 1)),
           hjust = 0, size = 3.5) +
  geom_segment(aes(x = no_pool_est, xend = partial_pool_est,
                   yend = reorder(hospital_label, n)),
               arrow = arrow(length = unit(0.15, "cm")),
               colour = "grey60") +
  scale_colour_manual(values = c("No pooling" = "steelblue",
                                  "Partial pooling" = "firebrick")) +
  labs(x = "Treatment Effect (mmHg change in SBP)",
       y = "Hospital (ordered by sample size)",
       colour = "Estimate Type",
       title = "Shrinkage in a Hierarchical Model",
       subtitle = "Arrows show how partial pooling pulls estimates toward the
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom")
```

10.6.7. Python

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

np.random.seed(789)
n_hospitals = 12
patients = [15, 20, 25, 30, 35, 40, 50, 60, 80, 100, 120, 150]
grand_effect = -8
tau = 3

hospital_effects = np.random.normal(0, tau, n_hospitals)

rows = []
```

```
for j in range(n_hospitals):
    nj = patients[j]
    trt = np.tile([0, 1], nj // 2 + 1)[:nj]
    sbp = 140 + (grand_effect + hospital_effects[j]) * trt + \
          np.random.normal(0, 15, nj)
    for i in range(nj):
        rows.append({'hospital': j, 'treatment': trt[i], 'sbp': sbp[i]})

trial_data = pd.DataFrame(rows)

# No-pooling estimates
no_pool = []
for j in range(n_hospitals):
    df_j = trial_data[trial_data['hospital'] == j]
    trt_mean = df_j[df_j['treatment'] == 1]['sbp'].mean()
    ctl_mean = df_j[df_j['treatment'] == 0]['sbp'].mean()
    no_pool.append(trt_mean - ctl_mean)

no_pool = np.array(no_pool)
grand_mean_est = np.mean(no_pool)

# Simulate partial pooling (shrinkage toward grand mean)
# Shrinkage factor depends on sample size: larger sites shrink less
shrinkage_factor = np.array([15 / (15 + nj) for nj in patients])
partial_pool = grand_mean_est + (1 - shrinkage_factor) * (no_pool - grand_mean_est)

# Plot
fig, ax = plt.subplots(figsize=(10, 6))
order = np.argsort(patients)
y_pos = np.arange(n_hospitals)

for i, idx in enumerate(order):
    ax.plot([no_pool[idx], partial_pool[idx]], [i, i],
```

```
            color="grey", linewidth=1, zorder=1)
    ax.annotate("", xy=(partial_pool[idx], i), xytext=(no_pool[idx], i),
                arrowprops=dict(arrowstyle="->", color="grey"))

ax.scatter(no_pool[order], y_pos, color="steelblue", s=60, zorder=2,
           label="No pooling")
ax.scatter(partial_pool[order], y_pos, color="firebrick", s=60, zorder=2,
           label="Partial pooling")
ax.axvline(grand_mean_est, linestyle="--", color="grey", linewidth=1)
ax.set_yticks(y_pos)
ax.set_yticklabels([f"Hospital {order[i]+1}\n(n={patients[order[i]]})"
                     for i in range(n_hospitals)], fontsize=9)
ax.set_xlabel("Treatment Effect (mmHg)")
ax.set_title("Shrinkage in a Hierarchical Model\n"
             "Arrows show estimates pulled toward the grand mean")
ax.legend()
plt.tight_layout()
plt.show()
```

10.6.8. Understanding Shrinkage

The figure above illustrates the key behaviour of hierarchical models:

- **Small hospitals** (few patients) are pulled strongly toward the grand mean. Their individual data are noisy, so the model relies more on the shared information from other sites.
- **Large hospitals** (many patients) retain estimates close to their no-pooling values. Their data are informative enough to override the group-level information.
- **The grand mean** itself is estimated from all sites, so every site contributes to it.

This is **partial pooling** — a principled compromise between treating all sites as identical (complete pooling) and treating them as completely independent (no pooling). It is particularly valuable when some sites have very few patients, where no-pooling estimates would be unreliable.

> **Shrinkage is not bias**
>
> Shrinkage toward the grand mean might seem like it introduces bias. In a narrow sense it does: individual site estimates are biased toward the mean. But this bias is traded for reduced variance, and the overall mean squared error is lower. This is the bias-variance trade-off that we discussed in earlier chapters, now appearing naturally in the Bayesian framework.

10.6.9. Varying Slopes

The model above includes a varying (random) slope for treatment by hospital. This means we are estimating not just "does the treatment effect vary across hospitals?" but also "by how much?" The between-hospital standard deviation τ directly quantifies this heterogeneity. If the 95% credible interval for τ excludes zero, there is evidence of meaningful variation across sites.

10.7. The Practical Bayesian Workflow

A complete Bayesian analysis follows a structured workflow:

10.7.1. Step 1: Specify the Model

- Write down the likelihood (data-generating process).
- Choose priors for all parameters.

- Justify your choices: are the priors weakly informative? Are they based on prior studies?

You don't have to write a likelihood from scratch

"Specify the likelihood" sounds like it requires deriving probability formulas by hand. In practice, with `brms` (R) or `bambi`/PyMC (Python) you almost never do. The likelihood is determined by the **type of outcome** you are modelling, and you select it with a single `family =` argument. The mapping is the same one you already use for choosing a regression in everyday practice:

Your outcome	Example	`family` to use	Underlying likelihood
Continuous, roughly symmetric	Blood pressure, BMI	`gaussian()`	Normal
Binary (yes/no)	30-day readmission, mortality	`bernoulli()`	Bernoulli (logistic)
Count	Number of admissions	`poisson()`	Poisson
Time-to-event	Survival time	`cox()` / `weibull()`	Survival

So the practical question is not "what is the mathematical form of the likelihood?" but the familiar clinical question "**what kind of measurement is my outcome?**" Pick the matching family, and the software writes the likelihood for you. The equations in this chapter are there so you understand *what the software is doing* — not because you need to type them out.

10.7.2. Step 2: Prior Predictive Check

- Simulate data from the priors alone.
- Verify that the implied range of outcomes is clinically plausible.
- Iterate if necessary: tighten or widen priors.

10.7.3. Step 3: Fit the Model

- Run MCMC (typically 4 chains, 1000–2000 post-warmup iterations each).
- Check for warnings (divergent transitions, max treedepth warnings).

10.7.4. Step 4: Diagnose Convergence

- Inspect trace plots: should look like “hairy caterpillars”.
- Check $\widehat{R} < 1.01$ for all parameters.
- Check ESS > 400 for both bulk and tail.
- If any diagnostic fails, do **not** interpret the results. Fix the model first.

10.7.5. Step 5: Posterior Predictive Check

- Simulate data from the fitted posterior.
- Compare to observed data: distributions, summary statistics, patterns.
- If the model is badly misspecified, revise the likelihood or priors.

10.7.6. Step 6: Summarise and Report

- Report posterior means (or medians), credible intervals, and relevant posterior probabilities.
- Include sensitivity analyses: how do results change under alternative priors?
- Report MCMC diagnostics alongside the results.

Reporting Template

"We estimated the treatment effect using a Bayesian [model type] with [prior specification]. The posterior mean treatment effect was X (95% credible interval: [L, U]). The posterior probability that the treatment effect exceeds the clinically meaningful threshold of Y was Z%. All chains converged ($\hat{R} < 1.01$, bulk ESS > [value], tail ESS > [value])."

10.8. Implementation: Software Choices

10.8.1. R Ecosystem

brms (Bayesian Regression Models using Stan) is the recommended package for applied Bayesian analysis in R. It uses the familiar R formula syntax and compiles models in Stan behind the scenes. Essentially, anything you can fit with `lme4` can be fit with `brms`, but with priors and full posterior inference.

rstanarm provides pre-compiled Stan models for common regression types. It is faster to start (no compilation time) but less flexible than brms.

Both produce Stan code that you can inspect, which is excellent for learning.

10.8.2. Python Ecosystem

PyMC (version 5+) is the most mature probabilistic programming library in Python. It uses the JAX or NumPy backends and provides NUTS sampling out of the box.

Bambi (BAyesian Model-Building Interface) is the Python analogue of brms: it provides a formula-based interface on top of PyMC, making it easy to specify complex models without writing raw PyMC code.

10.8.3. R (brms)

```r
library(brms)

# Specify and fit
fit <- brm(
  outcome ~ predictor1 + predictor2 + (1 | group),
  data = my_data,
  family = gaussian(),
  prior = c(
    prior(normal(0, 10), class = "b"),
    prior(normal(0, 20), class = "Intercept"),
    prior(cauchy(0, 5), class = "sd"),
    prior(exponential(0.1), class = "sigma")
  ),
  chains = 4, iter = 2000, warmup = 1000, seed = 42
)

# Diagnose
plot(fit)                   # trace plots
summary(fit)                # R-hat, ESS
pp_check(fit)               # posterior predictive check
conditional_effects(fit)    # visualise effects
```

```
# Posterior probability of meaningful effect
hypothesis(fit, "predictor1 > 0.5")
```

10.8.4. Python (bambi)

```python
import bambi as bmb
import arviz as az

# Specify and fit
model = bmb.Model("outcome ~ predictor1 + predictor2 + (1 | group)",
                  data=my_data, family="gaussian")
results = model.fit(draws=1000, tune=1000, chains=4, random_seed=42)

# Diagnose
az.plot_trace(results)        # trace plots
print(az.summary(results))    # R-hat, ESS
az.plot_ppc(results)          # posterior predictive check

# Posterior probability
posterior = az.extract(results)
prob = (posterior['predictor1'] > 0.5).mean()
print(f"P(predictor1 > 0.5 | data) = {prob:.3f}")
```

10.9. Exercises

10.9.1. Exercise 1: Bayesian Logistic Regression for ICU Mortality

A dataset contains 500 ICU admissions with the following variables: age, APACHE II score, mechanical ventilation status (yes/no), and 28-day mortality (outcome).

a. Simulate a dataset with plausible parameter values.
b. Fit a Bayesian logistic regression model using weakly informative priors.
c. Perform a prior predictive check: do the priors imply plausible mortality rates?
d. Report posterior odds ratios with 95% credible intervals.
e. Calculate $P(\text{OR}_{\text{APACHE}} > 1.10 \mid \text{data})$ — the probability that each unit increase in APACHE score increases the odds of death by more than 10%.

10.9.2. R

```
library(tidyverse)
library(brms)

set.seed(101)
n <- 500
icu_data <- tibble(
  age = round(rnorm(n, 62, 15)),
  apache = round(rnorm(n, 18, 7)),
  ventilated = rbinom(n, 1, 0.35),
  mortality = rbinom(n, 1,
    plogis(-4.5 + 0.02 * round(rnorm(n, 62, 15)) +
             0.12 * round(rnorm(n, 18, 7)) + 0.8 * rbinom(n, 1, 0.35)))
)

fit_icu <- brm(
  mortality ~ age + apache + ventilated,
  data = icu_data,
  family = bernoulli(),
  prior = c(
```

```
    prior(normal(0, 2.5), class = "b"),
    prior(normal(0, 5), class = "Intercept")
  ),
  chains = 4, iter = 2000, warmup = 1000, seed = 42,
  silent = 2, refresh = 0
)

# Posterior OR for APACHE
post <- as_draws_df(fit_icu)
or_apache <- exp(post$b_apache)
cat(sprintf("OR APACHE: %.3f [%.3f, %.3f]\n",
    mean(or_apache), quantile(or_apache, 0.025), quantile(or_apache, 0.975)))
cat(sprintf("P(OR > 1.10 | data): %.3f\n", mean(or_apache > 1.10)))
```

10.9.3. Python

```
import numpy as np
import pandas as pd
import pymc as pm
import arviz as az
from scipy.special import expit

np.random.seed(101)
n = 500
icu_data = pd.DataFrame({
    'age': np.round(np.random.normal(62, 15, n)),
    'apache': np.round(np.random.normal(18, 7, n)),
    'ventilated': np.random.binomial(1, 0.35, n)
})
lp = -4.5 + 0.02 * icu_data['age'] + 0.12 * icu_data['apache'] + \
     0.8 * icu_data['ventilated']
icu_data['mortality'] = np.random.binomial(1, expit(lp))
```

```python
with pm.Model() as icu_model:
    intercept = pm.Normal("Intercept", 0, 5)
    b_age = pm.Normal("b_age", 0, 2.5)
    b_apache = pm.Normal("b_apache", 0, 2.5)
    b_vent = pm.Normal("b_ventilated", 0, 2.5)

    logit_p = intercept + b_age * icu_data['age'].values + \
              b_apache * icu_data['apache'].values + \
              b_vent * icu_data['ventilated'].values

    y = pm.Bernoulli("mort", logit_p=logit_p,
                     observed=icu_data['mortality'].values)
    trace = pm.sample(1000, tune=1000, chains=4, random_seed=42,
                      progressbar=False)

post = az.extract(trace)
or_apache = np.exp(post['b_apache'].values)
print(f"OR APACHE: {or_apache.mean():.3f} "
      f"[{np.percentile(or_apache, 2.5):.3f}, "
      f"{np.percentile(or_apache, 97.5):.3f}]")
print(f"P(OR > 1.10 | data): {(or_apache > 1.10).mean():.3f}")
```

10.9.4. Exercise 2: Hierarchical Model for Multi-Site Drug Trial

Twelve hospitals participate in a trial comparing a new statin to standard care. The primary outcome is change in LDL cholesterol (mg/dL) at 12 weeks. Hospitals vary in patient volume from 20 to 200 patients.

a. Simulate data where the true average treatment effect is –25 mg/dL with between-hospital SD of 5 mg/dL.
b. Fit a Bayesian hierarchical model with random intercepts and random slopes for treatment by hospital.

c. Create a shrinkage plot showing how hospital-specific estimates are pulled toward the grand mean.
d. Compare the hierarchical model estimates to separate hospital-by-hospital OLS regressions. Which approach is more appropriate and why?

10.9.5. Exercise 3: Prior Sensitivity for Rare Events

A new surgical technique is tested in 40 patients. Zero patients experience a major adverse event.

a. Compute the posterior distribution for the adverse event rate using three priors: Beta(1,1), Beta(0.5, 0.5) (Jeffreys prior), and Beta(1, 9) (informative: prior belief ~10% rate).
b. Report the posterior mean and 95% upper credible bound for each prior.
c. A frequentist would report the point estimate as $0/40 = 0$. Explain why the Bayesian estimates are more useful for regulatory decision-making.

10.10. Summary

Bayesian regression models extend familiar linear and logistic regression by placing priors on parameters and yielding full posterior distributions rather than point estimates. The practical workflow — specify, check priors, fit, diagnose, check posteriors, report — ensures rigorous and transparent analysis. Hierarchical models, with their partial pooling and shrinkage properties, are especially powerful for multi-site clinical studies where borrowing information across groups improves estimation. Modern software (brms in R, PyMC/bambi in Python) makes these methods accessible to applied researchers.

Key Takeaways

1. Bayesian regression yields full posterior distributions for all coefficients, not just point estimates.
2. Prior and posterior predictive checks are essential quality-control steps.
3. Hierarchical models perform partial pooling: small groups shrink toward the grand mean, large groups retain their individual estimates.
4. Shrinkage reduces mean squared error by trading a small amount of bias for a large reduction in variance.
5. Always report MCMC diagnostics ($\hat{R}$, ESS, trace plots) alongside posterior summaries.

10.11. References and Further Reading

The primary reference for the brms package is Buerkner's 2017 paper "brms: An R Package for Bayesian Multilevel Models Using Stan" published in the *Journal of Statistical Software*, which provides a thorough overview of the formula syntax, prior specification, and model families available. For the hierarchical modelling concepts presented in this chapter, Gelman and Hill's *Data Analysis Using Regression and Multilevel/Hierarchical Models* (2006) remains the classic applied reference, and Gelman, Carlin, Stern, Dunson, Vehtari, and Rubin's *Bayesian Data Analysis* (3rd edition, 2013) provides the theoretical foundations.

McElreath's *Statistical Rethinking* (2nd edition, 2020) devotes several excellent chapters to multilevel models with a focus on intuition and simulation. For the clinical applications of hierarchical Bayesian models, the FDA's 2010 guidance on Bayesian statistics in medical device trials and its 2026 update provide the regulatory perspective. Berry, Carlin, Lee, and Muller's *Bayesian Adaptive Methods for Clinical Trials* (2010) covers the design and analysis of adaptive trials using Bayesian methods, including hierarchical borrowing across subgroups and historical controls.

On the Python side, the PyMC documentation (pymc.io) includes a growing library of case studies, and Osvaldo Martin's *Bayesian Analysis with Python* (3rd edition, 2024) provides a practical guide to PyMC for applied researchers. The ArviZ documentation (arviz-devs.github.io/arviz) covers posterior diagnostics and visualisation in detail. For the concept of shrinkage and partial pooling, Efron and Morris's classic 1977 paper "Stein's Paradox in Statistics" in *Scientific American* remains a wonderfully accessible introduction, showing that shrinkage estimators outperform individual estimates even in simple settings.

Part IV.

Introduction to Machine Learning

11. What Is Machine Learning? Core Concepts for Clinical Researchers

11.1. What Machine Learning Is (and Is Not)

If you have worked through the regression and Bayesian chapters, you already understand the core of machine learning — you just did not call it that. This part of the course focuses on **supervised** machine learning: algorithms that learn to predict an outcome from labelled examples. We also introduce the vocabulary, workflow, and evaluation framework that the ML community uses, which differs in emphasis from the statistical tradition you have been working in so far.

Machine learning (ML) is a collection of algorithms that learn patterns from data to make predictions or discover structure — without being explicitly programmed with rules. That is the textbook definition. Here is a more practical one for clinical researchers:

Machine learning is pattern recognition at scale. Given enough examples of inputs and outputs, an ML algorithm finds the mapping between them. A logistic regression that predicts 30-day readmission from 5 clinical variables is, technically, machine learning. An algorithm that reads chest X-rays and flags pneumonia is also machine learning. The difference is one of complexity and scale, not of kind.

11.1.1. Common Misconceptions

Let us address several misconceptions head-on:

1. **"ML is always better than regression."** False. For many clinical prediction tasks with modest sample sizes and well-understood relationships, logistic or Cox regression performs just as well as or better than complex ML models — and is far more interpretable. A 2019 systematic review by Christodoulou et al. in the *Journal of Clinical Epidemiology* found that ML models did not consistently outperform logistic regression for clinical prediction.

2. **"ML finds truth in data automatically."** False. ML finds patterns — including spurious ones. Without careful validation, feature engineering, and domain knowledge, ML models can learn noise, artifacts, and biases in the data.

3. **"ML doesn't need assumptions."** Misleading. ML models may not require linearity or normality assumptions, but they make other assumptions: that the training data is representative of future data, that the features capture the relevant information, that the data is not systematically biased.

4. **"More data always helps."** Mostly true, but not always. More biased data produces a more confidently wrong model. Data quality matters at least as much as data quantity.

5. **"Deep learning is always the best approach."** False. For tabular clinical data (the kind in most EHR studies), gradient-boosted trees often outperform deep learning. Deep learning excels with images, text, and time series.

11.1.2. How is ML different from what I already know?

If you have worked through the regression chapters, you already know machine learning — you just did not call it that. Logistic regression is a machine learning algorithm. The difference is mainly one of emphasis:

- **Traditional statistics** focuses on understanding relationships (what is the effect of smoking on lung cancer risk, adjusted for age?).
- **Machine learning** focuses on prediction accuracy (given these 50 variables, what is the probability this patient will be readmitted within 30 days?).

The tools overlap substantially. The new concepts in this chapter — cross-validation, regularisation, ensemble methods — are techniques for improving prediction performance when you have many variables and complex relationships.

11.2. Supervised vs Unsupervised vs Reinforcement Learning

11.2.1. Supervised Learning

In **supervised learning**, the algorithm learns from labeled examples: input-output pairs where the "right answer" is known.

- **Classification**: The output is a category. *Is this skin lesion malignant or benign? Will this patient be readmitted within 30 days?*
- **Regression** (in the ML sense): The output is a continuous value. *What will this patient's HbA1c be in 6 months? What is the expected length of stay?*

Most clinical prediction models — logistic regression, random forests, neural networks for diagnosis — are supervised learning.

11.2.2. Unsupervised Learning

In **unsupervised learning**, there are no labels. The algorithm discovers structure in the data on its own.

- **Clustering**: Group patients into subgroups based on similarity. *Are there distinct phenotypes of sepsis patients?*
- **Dimensionality reduction**: Compress high-dimensional data into fewer dimensions for visualization or preprocessing. *Can we summarize 50 lab values into a few meaningful components?*
- **Anomaly detection**: Identify unusual observations. *Which patients have lab value patterns that are outliers?*

We cover dimensionality reduction and clustering in detail later in the course. This part focuses on **supervised learning** — the prediction-oriented methods that build on the regression foundations you already know.

11.2.3. Reinforcement Learning

In **reinforcement learning (RL)**, an agent learns by trial and error, receiving rewards or penalties for actions. While RL has exciting potential in medicine (e.g., optimizing treatment policies for sepsis management or dynamic treatment regimes), it requires large amounts of interaction data and is rarely used in standard clinical research. We mention it for completeness but will not cover it further in this course.

> **Where Does Traditional Statistics Fit?**
>
> Linear regression, logistic regression, and Cox regression are all forms of supervised learning. The distinction between "statistics" and "machine learning" is largely cultural and historical. Statistics emphasizes inference (understanding relationships, testing hypotheses). ML em-

phasizes prediction (making accurate forecasts on new data). The methods overlap substantially.

11.3. The Bias-Variance Tradeoff

This is arguably the single most important concept in machine learning. Every predictive model navigates a tension between two sources of error:

- **Bias**: Error from oversimplifying the model. A model with high bias misses important patterns. It *underfits* the data.
- **Variance**: Error from making the model too flexible. A model with high variance captures noise as if it were signal. It *overfits* the data.

The total prediction error can be decomposed as:

$$\text{Expected Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Noise}$$

11.3.1. A Clinical Analogy

Think of a diagnostic criterion:

- **Too specific** (high bias, low variance): A criterion that requires 5 out of 5 symptoms to diagnose a disease. It misses many true cases (underfitting — the rule is too rigid) but rarely triggers a false alarm.
- **Too sensitive** (low bias, high variance): A criterion that diagnoses the disease if *any* 1 of 5 symptoms is present. It catches nearly every true case but also flags many healthy individuals (overfitting to noise).

The optimal diagnostic criterion balances sensitivity and specificity. Similarly, the optimal ML model balances bias and variance.

11.3.2. Visualizing the Tradeoff

> You have already seen bias and variance
>
> In Chapter 4, you modelled non-linear relationships. A straight line through a U-shaped relationship has **high bias** — it systematically misses the true pattern. A spline with 20 knots that follows every wiggle in the training data has **high variance** — it captures noise and will perform poorly on new data. The bias-variance tradeoff is the formal name for this tension you have already experienced.

In practice, we cannot directly observe bias and variance separately. We detect the tradeoff by comparing **training error** (how well the model fits the data it learned from) and **test error** (how well it performs on new data). When training error is low but test error is high, the model is overfitting.

11.4. Training, Validation, and Test Sets

To evaluate how well a model will perform on new, unseen patients, we **must** evaluate it on data it has never seen during training. This is the most important methodological principle in ML.

11.4.1. The Three-Way Split

Set	Purpose	Typical Size
Training set	Fit the model (learn parameters)	60–70%
Validation set	Tune hyperparameters, select among models	15–20%

Set	Purpose	Typical Size
Test set	Final, unbiased performance estimate	15–20%

The **test set must never be used for any decision** — not for feature selection, not for hyperparameter tuning, not for choosing between models. It is opened exactly once, at the very end, to report final performance. If you use the test set to make modeling decisions, it becomes a validation set, and your reported performance will be optimistically biased.

> ⚠ The Cardinal Sin of Machine Learning
>
> Using the test set during model development — and then reporting performance on that same test set — is the ML equivalent of p-hacking. It produces overoptimistic results that will not replicate. In clinical ML, this can mean deploying a model that performs worse in practice than expected, with potential patient harm.

11.4.2. When Data Is Limited

In clinical research, we often have hundreds (not millions) of patients. A single 70/15/15 split wastes data and produces unstable estimates. The solution is **cross-validation**.

11.5. Cross-Validation

Cross-validation (CV) is a resampling strategy that uses all the data for both training and validation, by rotating which portion is held out.

11.5.1. k-Fold Cross-Validation

The most common approach:

1. Randomly split the data into k equally sized **folds** (typically $k = 5$ or $k = 10$).
2. For each fold $i = 1, \ldots, k$:
   - Train the model on all folds except fold i.
   - Evaluate on fold i.
3. Average the k performance estimates.

This gives a more stable and less biased estimate of performance than a single train/test split.

11.5.2. Variants

Plain k-fold works well when the outcome is balanced and every row is an independent patient. Clinical data often violate those assumptions, so several variants exist. Each one fixes a specific problem you are likely to meet:

- **Stratified k-fold**: Ensures each fold has the same proportion of events/classes as the full dataset. *Why it matters:* with a rare outcome (say a 5% readmission rate and 10 folds), an ordinary random split could easily produce a fold with almost no events — giving a meaningless performance estimate for that fold. Stratification guarantees every fold contains a fair share of events. **Use it by default for any classification problem.**
- **Repeated k-fold**: Repeat the entire k-fold procedure m times with different random splits, then average. *Why it matters:* a single 10-fold split is itself somewhat random — shuffle the data differently and you get a slightly different number. Repeating (e.g. 10-fold repeated

5 times = 50 estimates) averages out that luck-of-the-draw and gives a more stable estimate.

- **Leave-one-out (LOO)**: $k = n$ (each fold is a single observation). Nearly unbiased but high variance and computationally expensive. Useful only for very small datasets.
- **Grouped/clustered CV**: When observations are not independent (e.g., multiple visits per patient, or patients clustered within hospitals), entire groups must be kept together — all of a patient's rows go into the *same* fold. *Why it matters:* if the same patient appears in both the training and validation folds, the model can "recognise" them at validation time and post an inflated score that will not hold up on genuinely new patients. This is one of the most common and most damaging mistakes in clinical ML. **Never split a patient's data across training and validation sets.**

11.5.2.1. R

```r
library(tidymodels)

# Simulate clinical data
set.seed(42)
n <- 500
clin_data <- tibble(
  age = rnorm(n, 60, 12),
  bmi = rnorm(n, 28, 5),
  sbp = rnorm(n, 135, 20),
  glucose = rnorm(n, 110, 30),
  smoking = rbinom(n, 1, 0.25),
  readmit = factor(rbinom(n, 1, 0.15), levels = c("0", "1"),
                   labels = c("No", "Yes"))
)
```

```r
# Create 10-fold CV with stratification
folds <- vfold_cv(clin_data, v = 10, strata = readmit, repeats = 5)
cat("Number of resamples:", nrow(folds), "\n")
cat("Training size per fold:", nrow(training(folds$splits[[1]])), "\n")
cat("Validation size per fold:", nrow(testing(folds$splits[[1]])), "\n")

# Fit logistic regression with cross-validation using tidymodels
log_spec <- logistic_reg() %>%
  set_engine("glm")

log_recipe <- recipe(readmit ~ ., data = clin_data)

log_wf <- workflow() %>%
  add_model(log_spec) %>%
  add_recipe(log_recipe)

cv_results <- fit_resamples(log_wf, resamples = folds,
                            metrics = metric_set(roc_auc, accuracy))

collect_metrics(cv_results)
```

11.5.2.2. Python

```python
import numpy as np
import pandas as pd
from sklearn.model_selection import (StratifiedKFold, RepeatedStratifiedKFol
                                     cross_val_score)
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline

np.random.seed(42)
```

```python
n = 500

X = pd.DataFrame({
    'age': np.random.normal(60, 12, n),
    'bmi': np.random.normal(28, 5, n),
    'sbp': np.random.normal(135, 20, n),
    'glucose': np.random.normal(110, 30, n),
    'smoking': np.random.binomial(1, 0.25, n)
})
y = np.random.binomial(1, 0.15, n)

# 10-fold stratified CV, repeated 5 times
cv = RepeatedStratifiedKFold(n_splits=10, n_repeats=5, random_state=42)

# Pipeline: scale features, then logistic regression
pipe = make_pipeline(StandardScaler(), LogisticRegression(max_iter=1000))

scores = cross_val_score(pipe, X, y, cv=cv, scoring='roc_auc')
print(f"Mean AUC: {scores.mean():.3f} (+/- {scores.std():.3f})")
print(f"Number of CV iterations: {len(scores)}")
```

What the code is showing. This is a worked example of stratified, repeated cross-validation on simulated patient data. Walking through the output:

- It builds 10 folds and repeats the whole process 5 times, so it reports **50 train/validate cycles** (`Number of resamples: 50`). Each cycle trains the model on 90% of patients and checks it on the held-out 10%, with stratification keeping the event rate steady across folds.
- The headline result is the **mean AUC together with its spread** (mean ± standard deviation). The mean is your best single estimate of how well the model discriminates; the spread tells you how much that estimate wobbles depending on which patients land in which

fold. A tight spread means the estimate is trustworthy; a wide spread is a warning that performance is fragile, often a sign of too little data.

The point of the exercise is not the specific number (the data are random here, so the AUC will sit near 0.5) but the *workflow*: this is the template you reuse for real data, simply swapping in your own dataset and model.

11.6. Feature Engineering in Clinical Data

Feature engineering is the process of transforming raw variables into features that are more useful for modeling. In clinical ML, this often means incorporating domain knowledge.

11.6.1. Examples of Clinical Feature Engineering

Raw Data	Engineered Feature	Rationale
Date of birth + visit date	Age at visit	More meaningful than raw dates
Systolic BP, Diastolic BP	Mean arterial pressure, pulse pressure	Physiologically meaningful composites
Serum creatinine, age, sex, race	eGFR (CKD-EPI equation)	Standard clinical measure of kidney function
Multiple lab values over time	Rate of change (slope), variability (SD)	Trajectory matters more than single values
ICD-10 codes	Charlson comorbidity index	Summarizes comorbidity burden

Raw Data	Engineered Feature	Rationale
Free-text clinical notes	Extracted symptoms via NLP	Unlocks unstructured data
Medication list	Binary flags for drug classes	Captures treatment intent

Domain Knowledge Is Your Superpower

Clinical researchers have a massive advantage in ML: you understand the data. A computer scientist might feed raw creatinine into a model; you know to compute eGFR. A data scientist might treat blood pressure as two independent numbers; you know that pulse pressure has specific physiological meaning. Feature engineering is where clinical expertise directly improves model performance.

11.7. Feature Selection

With electronic health record data, you might have hundreds or thousands of candidate features. Including all of them risks overfitting and reduces interpretability. **Feature selection** identifies the most informative subset.

11.7.1. Three Approaches

1. Filter methods: Rank features by some statistical criterion (correlation with outcome, mutual information, chi-squared test) *before* fitting any model. Fast but ignores feature interactions.

2. Wrapper methods: Iteratively fit models with different feature subsets and select the best-performing set. Examples: forward selection, backward elimination, recursive feature elimination (RFE). More accurate but computationally expensive.

3. Embedded methods: Feature selection happens as part of model fitting. Examples: LASSO regression (L1 penalty shrinks some coefficients to zero), random forest variable importance, elastic net. Often the best practical choice.

11.7.1.1. R

```
library(tidymodels)

# LASSO for feature selection
set.seed(42)
n <- 400
df_feat <- tibble(
  age = rnorm(n, 60, 12),
  bmi = rnorm(n, 28, 5),
  sbp = rnorm(n, 135, 20),
  glucose = rnorm(n, 110, 30),
  smoking = rbinom(n, 1, 0.25),
  noise1 = rnorm(n),  # irrelevant
  noise2 = rnorm(n),  # irrelevant
  noise3 = rnorm(n),  # irrelevant
  outcome = factor(rbinom(n, 1, plogis(-3 + 0.02 * rnorm(n, 60, 12) +
                                       0.05 * rnorm(n, 28, 5))),
                   labels = c("No", "Yes"))
)

# LASSO logistic regression
lasso_spec <- logistic_reg(penalty = 0.01, mixture = 1) %>%
  set_engine("glmnet")
```

```r
lasso_recipe <- recipe(outcome ~ ., data = df_feat) %>%
  step_normalize(all_numeric_predictors())

lasso_wf <- workflow() %>%
  add_model(lasso_spec) %>%
  add_recipe(lasso_recipe)

lasso_fit <- fit(lasso_wf, data = df_feat)

# Extract coefficients
tidy(lasso_fit) %>%
  filter(term != "(Intercept)") %>%
  arrange(desc(abs(estimate)))
```

11.7.1.2. Python

```python
from sklearn.linear_model import LogisticRegression
from sklearn.feature_selection import RFE
from sklearn.preprocessing import StandardScaler
import pandas as pd
import numpy as np

np.random.seed(42)
n = 400
X = pd.DataFrame({
    'age': np.random.normal(60, 12, n),
    'bmi': np.random.normal(28, 5, n),
    'sbp': np.random.normal(135, 20, n),
    'glucose': np.random.normal(110, 30, n),
    'smoking': np.random.binomial(1, 0.25, n),
    'noise1': np.random.normal(0, 1, n),
    'noise2': np.random.normal(0, 1, n),
```

```
    'noise3': np.random.normal(0, 1, n)
})
y = np.random.binomial(1, 0.15, n)

# LASSO feature selection
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

lasso = LogisticRegression(penalty='l1', solver='saga', C=1.0, max_iter=5000)
lasso.fit(X_scaled, y)

coef_df = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': lasso.coef_[0]
}).sort_values('Coefficient', key=abs, ascending=False)

print("LASSO Coefficients (features with 0 coefficient are excluded):")
print(coef_df.to_string(index=False))

# Recursive Feature Elimination
rfe = RFE(LogisticRegression(max_iter=5000), n_features_to_select=4)
rfe.fit(X_scaled, y)
selected = X.columns[rfe.support_]
print(f"\nRFE selected features: {list(selected)}")
```

What the code is telling you. Both examples are a small experiment: we built a dataset where `age` and `bmi` genuinely relate to the outcome, while `noise1`, `noise2`, and `noise3` are pure random noise that we added on purpose. A good feature-selection method should *keep the real predictors and discard the noise*. That is exactly what you should see in the output:

- The **LASSO** table lists each predictor with its coefficient, sorted by size. The L1 penalty shrinks unhelpful coefficients all the way

to **exactly zero** — so the noise variables drop out (coefficient = 0) while age and BMI survive with non-zero coefficients. Reading the table top-to-bottom tells you which variables the model considers most important.
- The **RFE** line prints the handful of features it chose to keep; the noise variables should not be among them.

The clinical point is reassurance: when you face a spreadsheet with dozens of candidate variables, these methods let the data tell you which ones carry signal, rather than you guessing or keeping everything (which invites overfitting). If a variable you *expected* to matter gets dropped, that is itself informative and worth investigating.

11.8. Support Vector Machines

Support vector machines (SVMs) are a supervised learning method that finds the optimal decision boundary between classes.

> **i** Why a clinician should care about SVMs
>
> SVMs are not just a theoretical curiosity — they shine in a situation that is increasingly common in clinical research: **many measurements on relatively few patients**. Think of a study with gene-expression levels for 20,000 genes measured on 80 tumour samples, or hundreds of radiomic features extracted from a few hundred scans. Classical logistic regression breaks down when there are more predictors than patients, but an SVM is specifically built to draw a sensible boundary in these "wide" datasets and tends to resist overfitting even when the number of features is enormous.
> Concrete clinical uses where SVMs have been successful include classifying tumour subtypes from gene-expression panels, distinguishing disease from healthy tissue on imaging-derived features, and flagging

> abnormal ECG or EEG patterns. The practical takeaway: when you have a binary question, a modest number of patients, and a large pile of measurements per patient, an SVM is a strong first thing to try.

11.8.1. The Intuition: Maximum Margin

Imagine plotting patients in a 2D space where the x-axis is age and the y-axis is a biomarker level. Some patients have the disease (red dots), others do not (blue dots). Many possible lines could separate the two groups. SVM finds the line (or hyperplane, in higher dimensions) that **maximizes the margin** — the distance between the boundary and the nearest points from each class (Figure 11.1).

The nearest points are called **support vectors** — they "support" (define) the boundary. Moving any other point does not change the boundary. This makes SVMs robust to outliers that are far from the boundary. Clinically, this is reassuring: the decision rule hinges on the genuinely borderline patients (the hard cases near the boundary), not on the clear-cut ones far from it.

11.8.2. The Kernel Trick

What if the classes are not linearly separable? Consider a disease where risk is high both for *very low* and *very high* values of a biomarker, but low in the middle. No straight line can separate "diseased" from "healthy" in that picture (Figure 11.2, left). The **kernel trick** maps the data into a higher-dimensional space where a linear boundary *can* separate the classes (Figure 11.2, right) — without ever having to compute those new coordinates explicitly.

Common kernels include:

- **Linear**: No transformation (works when data is linearly separable).

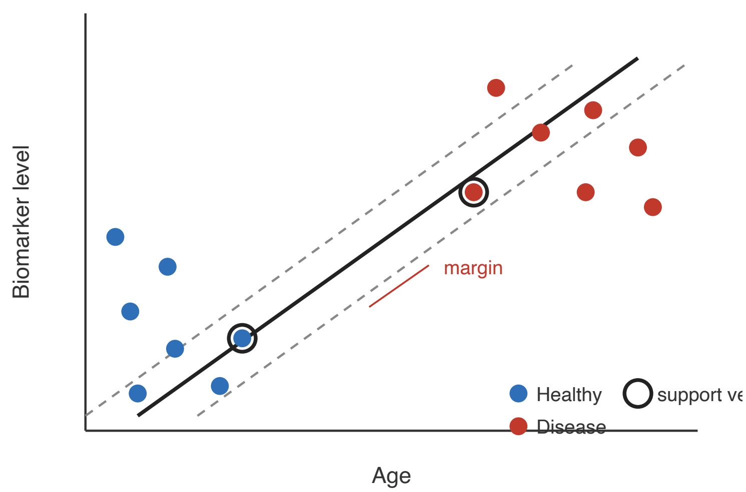

Figure 11.1.: A support vector machine chooses the separating line with the widest possible margin (the gap between the dashed lines). Only the points sitting on the margin — the circled **support vectors** — determine where the boundary goes; the other patients could be moved freely without changing it. A wide margin is what gives the model its robustness: a new patient has to fall well inside the wrong territory before being misclassified.

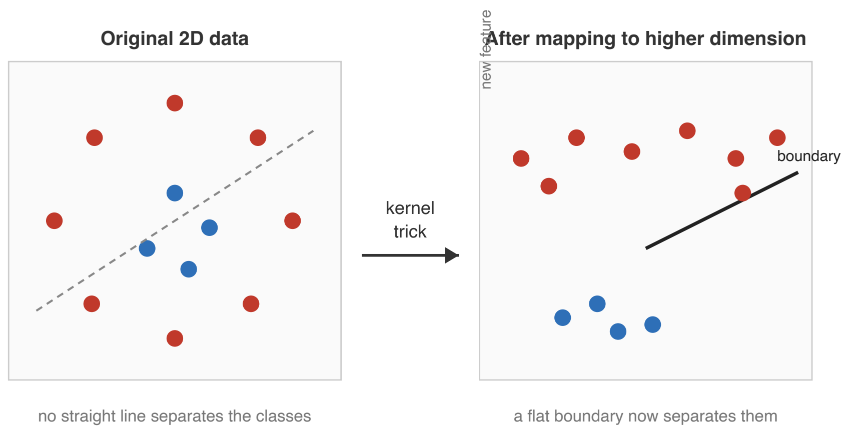

Figure 11.2.: Left: the diseased patients (red) surround the healthy ones (blue), so no straight line works in the original two measurements. Right: the kernel trick effectively adds a new dimension (here, roughly "distance from the centre"), lifting the red points up and the blue points down so that a simple flat boundary now separates them. The curved boundary we see back in the original 2D plot is the shadow of this flat boundary in the higher-dimensional space.

- **Radial basis function (RBF)**: Maps to infinite-dimensional space. Very flexible, works well in many settings.
- **Polynomial**: Maps to polynomial feature space.

You do not need to understand the mathematics of kernels to use SVMs effectively. The practical implication is: if a linear SVM does not work well, try an RBF kernel.

11.8.3. When SVMs Are Useful

SVMs work well for:

- Binary classification with moderate sample sizes
- High-dimensional data (many features relative to observations)
- Clear margin of separation between classes

SVMs are less ideal for:

- Very large datasets (training can be slow)
- Multi-class problems (require one-vs-one or one-vs-rest schemes)
- Interpretability (the model is a "black box" — no simple coefficients)

11.8.3.1. R

```
library(tidymodels)
library(kernlab)

set.seed(42)
n <- 300
svm_data <- tibble(
  x1 = rnorm(n),
  x2 = rnorm(n),
  class = factor(ifelse(x1^2 + x2^2 + rnorm(n, 0, 0.3) > 1.5,
```

```
                        "Disease", "Healthy"))
)

# Fit SVM with RBF kernel
svm_spec <- svm_rbf(cost = 1, rbf_sigma = 0.5) %>%
  set_engine("kernlab") %>%
  set_mode("classification")

svm_fit <- svm_spec %>% fit(class ~ ., data = svm_data)

# Evaluate with cross-validation
svm_wf <- workflow() %>%
  add_model(svm_spec) %>%
  add_formula(class ~ .)

folds <- vfold_cv(svm_data, v = 5, strata = class)
svm_cv <- fit_resamples(svm_wf, resamples = folds,
                        metrics = metric_set(roc_auc, accuracy))
collect_metrics(svm_cv)
```

11.8.3.2. Python

```
from sklearn.svm import SVC
from sklearn.model_selection import cross_val_score, StratifiedKFold
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline
import numpy as np

np.random.seed(42)
n = 300
x1 = np.random.normal(0, 1, n)
x2 = np.random.normal(0, 1, n)
```

```
X = np.column_stack([x1, x2])
y = (x1**2 + x2**2 + np.random.normal(0, 0.3, n) > 1.5).astype(int)

# SVM with RBF kernel
pipe = make_pipeline(StandardScaler(), SVC(kernel='rbf', C=1.0, gamma=0.5))

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
scores = cross_val_score(pipe, X, y, cv=cv, scoring='accuracy')
print(f"SVM accuracy: {scores.mean():.3f} (+/- {scores.std():.3f})")

auc_scores = cross_val_score(
    make_pipeline(StandardScaler(), SVC(kernel='rbf', C=1.0, gamma=0.5,
                                         probability=True)),
    X, y, cv=cv, scoring='roc_auc'
)
print(f"SVM AUC: {auc_scores.mean():.3f} (+/- {auc_scores.std():.3f})")
```

What the code is doing. We deliberately simulate a dataset where the disease cases form a *ring* around the healthy cases (`x1^2 + x2^2 > 1.5`) — exactly the non-linearly-separable situation from Figure 11.2 that a straight-line classifier cannot handle. We then fit an SVM with an **RBF kernel** and estimate its performance with 5-fold cross-validation. The printed accuracy and AUC (both should land well above 0.5, the no-information level) confirm that the RBF kernel successfully learned the curved boundary. The lesson for practice: the same two lines of code would have failed with a linear kernel here; switching to RBF is what makes it work.

11.9. ML vs Traditional Statistics: What Is Actually Different?

This is a question clinical researchers often ask, and the honest answer is: less than you think.

Aspect	Traditional Statistics	Machine Learning
Primary goal	Inference (understand relationships)	Prediction (forecast accurately)
Model choice	Guided by theory and assumptions	Guided by cross-validated performance
Feature selection	Guided by domain knowledge, parsimony	Data-driven, automated
Evaluation	p-values, confidence intervals	AUC, accuracy, calibration, cross-validation
Interpretability	Usually high (coefficients)	Varies (logistic regression = high; deep learning = low)
Sample size	Can work with small samples	Often needs more data for complex models
Assumptions	Explicit (linearity, normality, etc.)	Implicit (representative data, stationarity)

The biggest practical difference is in **workflow**. Statistical modeling typically follows a hypothesis-driven process: specify a model based on theory, fit it, check assumptions, interpret coefficients. ML follows a more empirical process: try many models, tune them, evaluate on held-out data, pick the best performer.

Neither approach is inherently superior. For a clinical trial with 200 patients and 5 pre-specified predictors, logistic regression is perfectly appropriate and more interpretable than a random forest. For a hospital system with 500,000 patient records, 1,000 candidate features, and the goal of predicting ICU transfer, a gradient-boosted tree with cross-validated tuning may outperform logistic regression.

11.10. Exercises

11.10.1. Exercise 1: Cross-Validation Experiment

Using the simulated clinical dataset below, compare logistic regression and SVM (RBF kernel) using 10-fold stratified cross-validation. Report AUC for both models. Which performs better? Why?

11.10.1.1. R Starter Code

```
library(tidyverse)  # tibble()
library(tidymodels) # vfold_cv(), logistic_reg(), svm_rbf(), etc.

set.seed(123)
n <- 600
ex_data <- tibble(
  age = rnorm(n, 65, 10),
  creatinine = rlnorm(n, 0, 0.5),
  hemoglobin = rnorm(n, 12, 2),
  platelets = rnorm(n, 250, 70),
  wbc = rlnorm(n, 2, 0.4),
  icu = factor(rbinom(n, 1, plogis(-4 + 0.03 * rnorm(n, 65, 10) +
                                    0.5 * rlnorm(n, 0, 0.5))),
               labels = c("No", "Yes"))
)
```

```
# Your code: set up tidymodels workflows for logistic_reg and svm_rbf
# Use vfold_cv with strata = icu
# Compare using roc_auc
```

11.10.1.2. Python Starter Code

```python
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.model_selection import cross_val_score, StratifiedKFold
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline

np.random.seed(123)
n = 600
# Your code: create X and y
# Compare LogisticRegression vs SVC using cross_val_score with scoring='roc_a
```

11.10.2. Exercise 2: Feature Engineering Challenge

You have the following raw variables for a diabetes prediction model:

- `height_cm`, `weight_kg`
- `sbp`, `dbp` (systolic and diastolic blood pressure)
- `fasting_glucose`, `hba1c`
- `age`, `sex`
- `waist_circumference`, `hip_circumference`
- `total_cholesterol`, `hdl`, `ldl`, `triglycerides`

1. List at least 5 clinically meaningful engineered features you could create from these variables. For each, explain *why* it is clinically meaningful.

2. Implement the feature engineering in R (using `recipes`) or Python (using pandas/sklearn).
3. Discuss which original features might become redundant after engineering.

11.10.3. Exercise 3: The Bias-Variance Tradeoff in Practice

Using polynomial regression on a simulated dataset:

1. Fit polynomials of degree 1, 3, 5, 10, and 20 to a training set.
2. Plot the fitted curves overlaid on the data.
3. Compute training error and test error for each degree.
4. Identify the degree that minimizes test error. Explain this in terms of the bias-variance tradeoff.

11.10.3.1. R Starter Code

```r
set.seed(42)
x_train <- sort(runif(50, 0, 10))
y_train <- sin(x_train) + rnorm(50, 0, 0.3)
x_test <- sort(runif(200, 0, 10))
y_test <- sin(x_test) + rnorm(200, 0, 0.3)

# Your code: fit poly(x, degree) for degree in c(1, 3, 5, 10, 20)
# Compute RMSE on train and test for each
# Plot the results
```

11.10.3.2. Python Starter Code

```
import numpy as np
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

np.random.seed(42)
x_train = np.sort(np.random.uniform(0, 10, 50))
y_train = np.sin(x_train) + np.random.normal(0, 0.3, 50)
x_test = np.sort(np.random.uniform(0, 10, 200))
y_test = np.sin(x_test) + np.random.normal(0, 0.3, 200)

# Your code: loop over degrees [1, 3, 5, 10, 20]
# Fit PolynomialFeatures + LinearRegression
# Compute train and test RMSE
```

11.11. Summary

This chapter introduced the foundational concepts of machine learning that every clinical researcher should understand before applying ML methods:

- **ML is pattern recognition**, not magic. It requires careful validation and domain knowledge.
- **Supervised learning** (prediction from labeled data) is the most common ML paradigm in clinical research.
- The **bias-variance tradeoff** governs all model selection: too simple = underfitting, too complex = overfitting.
- **Training/validation/test splits** and **cross-validation** are essential for honest performance evaluation.
- **Feature engineering** — where clinical expertise meets data science — is often more important than model choice.
- **Feature selection** (filter, wrapper, embedded) prevents overfitting in high-dimensional settings.

- **SVMs** find maximum-margin boundaries and can handle non-linear problems via the kernel trick.
- **ML and statistics are complementary**, not competing. The best clinical researchers use both.

11.12. References and Further Reading

For an integrated treatment of machine learning and statistical methods tailored to clinical and public health researchers, see Smits et al. (2026), particularly Chapter 9, which covers the supervised learning concepts and cross-validation approaches discussed here with additional worked medical examples.

Boehmke and Greenwell's *Hands-On Machine Learning with R* (HOML) provides an excellent practical introduction to ML methods in R, using the tidymodels framework. It covers the bias-variance tradeoff, cross-validation, and feature engineering with clear code examples and is freely available online.

Lgatto's *Introduction to Machine Learning* (IntroML) is a concise, well-organized online resource that covers the core concepts without excessive mathematical detail, making it a good supplement for students new to the field.

Christodoulou et al. (2019), "A Systematic Review Shows No Performance Benefit of Machine Learning over Logistic Regression for Clinical Prediction Models," published in the *Journal of Clinical Epidemiology*, provides important context for when ML methods do and do not outperform traditional approaches — a message every clinical researcher should internalize before defaulting to complex methods.

For support vector machines, the original formulation by Vapnik (1995) in *The Nature of Statistical Learning Theory* (Springer) is the theoretical foundation, though Hastie, Tibshirani, and Friedman's *The Elements of*

Statistical Learning (Springer, 2nd edition, 2009) provides a more accessible treatment in Chapters 9 and 12. James et al., *An Introduction to Statistical Learning* (ISLR, 2nd edition, 2021), covers SVMs in Chapter 9 at an introductory level with R lab exercises.

For a thoughtful perspective on the relationship between statistics and machine learning, Breiman's (2001) paper "Statistical Modeling: The Two Cultures" in *Statistical Science* remains essential reading.

12. Decision Trees, Random Forests, and Gradient Boosting

Of all the machine learning methods in this book, the tree-based family is the one clinicians tend to adopt fastest — because it mirrors how clinical reasoning already works. A doctor triaging chest pain mentally runs through a series of questions ("Is the troponin elevated? Are there ECG changes? Is the patient over 65?"), and each answer narrows down the likely diagnosis. That branching logic *is* a decision tree. This chapter starts from that intuition and builds up to the methods that win most clinical prediction competitions on tabular data: random forests and gradient boosting, which combine hundreds of trees into a single, far more accurate model.

Why invest in these methods rather than stopping at logistic regression? Because they automatically capture two things that are pervasive in medicine and awkward for standard regression: **non-linear effects** (risk that rises sharply only past a threshold) and **interactions** (a lab value that matters only in diabetic patients). You do not have to specify these by hand — the trees discover them. The price is interpretability, which is why the chapter spends time on how to read a tree and how to inspect variable importance.

12.1. Decision Trees (CART)

A **decision tree** is perhaps the most intuitive machine learning model. It makes predictions by asking a series of yes/no questions about the input features, splitting the data at each step into increasingly homogeneous groups. The result looks like an inverted tree — hence the name.

Why start with CART? CART (Classification And Regression Trees) is the foundation for everything else in this chapter — random forests and gradient boosting are both just clever ways of combining many CART trees. But a single tree is also valuable in its own right for clinical work: it produces a rule you can literally read off and apply at the bedside without a computer, and it handles a mix of continuous and categorical predictors, missing values, and interactions with no special preprocessing. When you need a model that a clinical team will actually trust and use, a small decision tree is often the most practical choice. Figure 12.1 shows the kind of transparent rule a tree produces.

Figure 12.1.: A simple, readable decision tree for 30-day readmission risk. To use it, start at the top and follow the answers to your patient's questions down to a leaf, which gives the predicted risk group. This is the kind of transparent rule that makes trees appealing at the bedside.

12.1.1. How Trees Split

The algorithm works top-down, **greedily** selecting the best split at each node. "Greedy" means it takes the single best step available right now without looking ahead — like a clinician ordering the one test that best separates the likely diagnoses at this moment, then reassessing. Concretely:

1. **Consider every feature and every possible split point.** For a continuous variable like age, the tree tries every threshold ("age <

60?", "age < 61?", …); for a binary variable like *has diabetes*, there is only one possible split. So at the very first node the algorithm might evaluate thousands of candidate questions.

2. **Score each candidate split by how much it improves the "purity" of the two resulting groups.** A split is good if it sends mostly-readmitted patients down one branch and mostly-not-readmitted patients down the other. "Purity" is just a number measuring how one-sided a group is (defined precisely in the next section).
3. **Choose the single split that improves purity the most**, and use it to divide the patients into two child nodes.
4. **Repeat recursively** on each child node — each becomes a new little problem with its own best question — until a **stopping criterion** is met: the node is too small to split further (`minsplit`), the tree has reached its maximum allowed depth (`maxdepth`), or no remaining split improves purity enough to be worthwhile.

A worked picture: at the root, the algorithm might find that "prior admissions 2?" best separates readmitted from non-readmitted patients. Among the patients who answered "yes," it then searches again and might pick "has heart failure?"; among those who answered "no," it might instead pick "age 75?". Different branches can ask about different variables — this is precisely how trees capture interactions automatically.

12.1.2. Splitting Criteria

"Purity" needs to be turned into a number the computer can compare. **Impurity** is simply a score for *how mixed* a group of patients is: it is **0 when everyone in the group has the same outcome** (all readmitted, or all not) and **largest when the group is a 50/50 coin-flip**. A split is judged good when it produces child groups that are *purer* (lower impurity) than the parent. Two formulas are used to measure this; do not be put off by the notation — each one is just a one-line recipe.

In the formulas below, imagine a group of patients in which a fraction p_1 have outcome 1 (readmitted), p_2 have outcome 2, and so on, across K possible outcomes. For the binary readmission example there are just two: p_{yes} and $p_{\text{no}} = 1 - p_{\text{yes}}$.

Gini impurity — the chance two randomly drawn patients from the group have *different* outcomes:

$$G = \sum_{k=1}^{K} p_k(1 - p_k) = 1 - \sum_{k=1}^{K} p_k^2$$

Read it as: take each outcome's proportion, and you are summing how "spread out" the group is across outcomes. Gini is 0 when a node is pure (all one class) and maximised when classes are equally mixed. For a binary problem the maximum is 0.5, reached at $p = 0.5$.

Worked example. A node of 100 patients with 30 readmitted and 70 not has $p_{\text{yes}} = 0.30$, $p_{\text{no}} = 0.70$, so $G = 1-(0.30^2+0.70^2) = 1-(0.09+0.49) = 0.42$ — fairly impure. If a split sends 40 patients to a child node that is 35 readmitted / 5 not ($p = 0.875$), that child has $G = 1-(0.875^2+0.125^2) = 0.22$ — much purer. The tree favours splits that drive these child Gini values down.

Entropy (information gain) — an alternative impurity measure borrowed from information theory:

$$H = -\sum_{k=1}^{K} p_k \log_2(p_k)$$

It behaves almost identically to Gini: 0 when the node is pure, and maximised (at $\log_2 K$) when outcomes are evenly split. The reduction in entropy achieved by a split is called the **information gain**.

In practice, Gini and entropy produce very similar trees, so the choice rarely matters. Gini is the default in most implementations because it is slightly faster to compute (no logarithm). **You do not need to compute these by hand** — the software does it for every candidate split. The reason to understand them is to know what the tree is optimising: at each step it is hunting for the question that most cleanly separates the outcomes.

For **regression trees** (continuous outcomes, e.g. predicting length of stay), there are no classes to count, so impurity is instead the **variance** within the node. A split is good when it produces child nodes whose values are tightly clustered — equivalently, when it gives the largest **reduction in residual sum of squares**.

12.1.3. Clinical Example: Hospital Readmission

Let us build a decision tree to predict 30-day hospital readmission — a common quality metric tied to reimbursement penalties in the United States.

12.1.3.1. R

```
# Simulate hospital readmission data
set.seed(42)
n <- 1000

readmit_data <- tibble(
  age = rnorm(n, 68, 12),
  length_of_stay = rpois(n, 5) + 1,
  num_comorbidities = rpois(n, 3),
  prior_admissions = rpois(n, 1),
  discharge_hemoglobin = rnorm(n, 11, 2),
  discharge_creatinine = rlnorm(n, 0.2, 0.5),
  has_diabetes = rbinom(n, 1, 0.35),
```

```r
  has_chf = rbinom(n, 1, 0.25),
  readmitted = factor(
    rbinom(n, 1, plogis(-3 + 0.02 * (rnorm(n, 68, 12) - 68) +
                        0.15 * rpois(n, 1) +
                        0.1 * rpois(n, 3) +
                        0.3 * rbinom(n, 1, 0.25) -
                        0.1 * rnorm(n, 11, 2))),
    labels = c("No", "Yes")
  )
)

cat("Readmission rate:", mean(readmit_data$readmitted == "Yes"), "\n")
cat("N =", nrow(readmit_data), "\n")
```

```r
# Fit a decision tree
tree_model <- rpart(
  readmitted ~ age + length_of_stay + num_comorbidities +
    prior_admissions + discharge_hemoglobin +
    discharge_creatinine + has_diabetes + has_chf,
  data = readmit_data,
  method = "class",
  control = rpart.control(cp = 0.01, maxdepth = 5, minsplit = 20)
)

# Visualize
rpart.plot(tree_model, type = 4, extra = 106, under = TRUE,
           box.palette = "RdYlGn", roundint = FALSE,
           main = "Decision Tree: 30-Day Readmission")
```

12.1.3.2. Python

```
import numpy as np
import pandas as pd
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.model_selection import cross_val_score, StratifiedKFold
import matplotlib.pyplot as plt

np.random.seed(42)
n = 1000

df = pd.DataFrame({
    'age': np.random.normal(68, 12, n),
    'length_of_stay': np.random.poisson(5, n) + 1,
    'num_comorbidities': np.random.poisson(3, n),
    'prior_admissions': np.random.poisson(1, n),
    'discharge_hemoglobin': np.random.normal(11, 2, n),
    'discharge_creatinine': np.random.lognormal(0.2, 0.5, n),
    'has_diabetes': np.random.binomial(1, 0.35, n),
    'has_chf': np.random.binomial(1, 0.25, n)
})
y = np.random.binomial(1, 0.18, n)

feature_names = list(df.columns)

# Fit decision tree
tree = DecisionTreeClassifier(max_depth=5, min_samples_split=20,
                               min_samples_leaf=10, random_state=42)
tree.fit(df, y)

fig, ax = plt.subplots(figsize=(16, 8))
plot_tree(tree, feature_names=feature_names, class_names=['No', 'Yes'],
          filled=True, rounded=True, ax=ax, fontsize=8, max_depth=3)
```

```
ax.set_title("Decision Tree: 30-Day Readmission")
plt.tight_layout()
plt.show()
```

12.1.4. Reading a Decision Tree

Each node in the tree displays:

- The **splitting rule** (e.g., "age >= 72")
- The **predicted class** (the majority class in that node)
- The **class probabilities** (e.g. 0.34 readmitted / 0.66 not)
- The **percentage of observations** that reach this node

To classify a new patient, start at the root and follow the branches based on the patient's feature values until you reach a terminal node (leaf). The prediction is the majority class of that leaf, and the leaf's class probability is the patient's estimated risk.

Worked example. Suppose a 78-year-old patient has had no prior admissions and no heart failure. Starting at the root of Figure 12.1, "prior admissions 2?" is **No**, so we go right to "age 75?", which is **Yes**, landing in the *moderate risk* leaf (~28% readmission risk). A different patient with three prior admissions and heart failure follows the left branches into the *high risk* leaf (~62%). Notice two clinically useful properties: the prediction is reached by at most a handful of questions, and you can explain *exactly why* any patient received their risk estimate — a transparency that "black box" models cannot offer. The flip side is that a single tree's boundaries are coarse (everyone in a leaf gets the same risk) and can shift noticeably if the data change slightly, which is the motivation for the ensemble methods later in this chapter.

12.1.5. Pruning: Controlling Tree Complexity

Why pruning matters. A tree left to grow without limits will keep asking questions until each leaf contains just one or two patients — effectively building a private rule for every individual in the training set. It will look *perfect* on the data it was built from and then fail badly on new patients, because it has memorised noise (one patient's quirky lab value) instead of learning generalisable patterns. This is **overfitting**, and for a clinical tool it is dangerous: a model that boasts 99% accuracy in development but collapses in the next clinic is worse than no model, because people trusted it.

Pruning is the cure: it deliberately cuts the tree back to a smaller, simpler set of rules that capture the real signal and ignore the noise. A pruned tree usually performs slightly *worse* on the training data but *better* on new patients — which is the only performance that matters. Think of it as the bias–variance tradeoff (from the previous chapter) made tangible: a few extra mistakes in development buy a large gain in reliability. The goal is the smallest tree whose performance is statistically indistinguishable from the bigger one.

Two strategies:

1. **Pre-pruning** (early stopping): Set constraints *before* growing the tree — maximum depth, minimum samples per leaf, minimum information gain to split. Simple but can stop too early.

2. **Post-pruning** (cost-complexity pruning): Grow a full tree, then prune branches that provide minimal improvement. The **complexity parameter (cp)** in R's `rpart` controls this: higher cp = more pruning. Select cp via cross-validation.

12.1.5.1. R

```
# View the CP table
printcp(tree_model)

# Plot CV error vs cp
plotcp(tree_model)

# Prune to optimal cp
optimal_cp <- tree_model$cptable[which.min(tree_model$cptable[, "xerror"]), "
pruned_tree <- prune(tree_model, cp = optimal_cp)

cat("Optimal cp:", optimal_cp, "\n")
cat("Number of terminal nodes (full):", sum(tree_model$frame$var == "<leaf>")
cat("Number of terminal nodes (pruned):", sum(pruned_tree$frame$var == "<lea
```

12.1.5.2. Python

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import cross_val_score
import numpy as np

# Compare trees of different depths
depths = [2, 3, 5, 7, 10, 15, None]
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

results = []
for d in depths:
    t = DecisionTreeClassifier(max_depth=d, random_state=42)
    scores = cross_val_score(t, df, y, cv=cv, scoring='roc_auc')
    depth_label = str(d) if d is not None else "None (full)"
    results.append({'max_depth': depth_label,
```

```
                    'mean_auc': scores.mean(),
                    'std_auc': scores.std()})
    print(f"max_depth={depth_label:>6s}: AUC = {scores.mean():.3f} (+/- {scores.std():.3f})'

best = max(results, key=lambda x: x['mean_auc'])
print(f"\nBest: max_depth={best['max_depth']} with AUC={best['mean_auc']:.3f}")
```

What the code is showing. Both versions answer the same question — *how big should the tree be?* — by measuring performance on held-out data rather than the training data:

- The **R** version uses cost-complexity pruning. `printcp()` prints a table where each row is a candidate tree size, and the key column is `xerror` (the cross-validated error). As the tree grows, `xerror` first falls (the tree is learning real structure) and then rises again (it has started memorising noise). `plotcp()` shows this as a U-shape. The code picks the complexity parameter `cp` at the bottom of the U and prunes to it; the final two lines print how many leaves the full tree had versus the smaller pruned tree — you should see the pruned tree is considerably smaller.
- The **Python** version makes the same point more directly by sweeping `max_depth` from 2 up to "no limit" and printing the cross-validated AUC for each. Reading down the list, AUC climbs, peaks at a modest depth, then stops improving (or worsens) as the tree gets deeper — and the script reports the depth that won. The deepest tree is *not* the best.

The shared lesson: deeper is not better. The right-sized tree is the one that does best on patients it has never seen, and cross-validation is how you find it.

12.2. Why Single Trees Are Unstable

Decision trees have a critical weakness: **high variance**. Small changes in the training data can produce a completely different tree. This instability means that:

- Predictions are unreliable for individual trees.
- Results may not replicate across datasets.
- Performance on new data is often poor.

The solution is to combine many trees into an **ensemble**. Two main strategies exist: **bagging** (reduce variance by averaging) and **boosting** (reduce bias by learning sequentially).

12.3. Bagging: Bootstrap Aggregating

Bagging (Bootstrap AGGregatING), introduced by Leo Breiman in 1996, reduces variance by:

1. Drawing B bootstrap samples (random samples with replacement) from the training data.
2. Fitting a separate decision tree to each bootstrap sample.
3. Averaging predictions (regression) or taking a majority vote (classification) across all B trees.

By averaging many high-variance, low-bias trees, bagging dramatically reduces the overall variance without increasing bias. The key insight: the variance of an average of B identically distributed random variables decreases with B (provided they are not perfectly correlated).

12.4. Random Forests

Random forests extend bagging with one additional trick: at each split, instead of considering all features, the algorithm randomly selects a **subset of features** (typically $\sqrt{p}$ for classification or $p/3$ for regression, where p is the total number of features).

This feature randomization **decorrelates the trees**. In bagging without feature randomization, if one feature is very strong, most trees will split on it first, making the trees correlated. Random forests force trees to consider different features, producing more diverse trees whose average is more stable.

12.4.1. Key Hyperparameters

Parameter	What It Controls	Typical Default
`num.trees` / `n_estimators`	Number of trees in the forest	500–2000
`mtry` / `max_features`	Features considered at each split	$\sqrt{p}$ (classification), $p/3$ (regression)
`min.node.size` / `min_samples_leaf`	Minimum observations in a leaf	1 (classification), 5 (regression)
`max.depth`	Maximum tree depth	Unlimited (let trees grow fully)

12.4.2. Out-of-Bag (OOB) Error

Each bootstrap sample includes about 63% of the original observations (due to sampling with replacement). The remaining 37% — the **out-of-bag** observations — were not used to build that particular tree. Random forests use these OOB observations as a built-in validation set: each observation is predicted by the trees that did *not* include it in their bootstrap sample.

The OOB error is a nearly unbiased estimate of the test error — no separate validation set required. This is one of the most elegant features of random forests.

12.4.3. Fitting a Random Forest

12.4.3.1. R

```
library(ranger)

# Fit random forest
set.seed(42)
rf_model <- ranger(
  readmitted ~ age + length_of_stay + num_comorbidities +
    prior_admissions + discharge_hemoglobin +
    discharge_creatinine + has_diabetes + has_chf,
  data = readmit_data,
  num.trees = 500,
  mtry = 3,
  min.node.size = 10,
  importance = "impurity",
  probability = TRUE,   # predict probabilities
  seed = 42
)

cat("OOB prediction error:", rf_model$prediction.error, "\n")
```

```
# Variable importance
importance_df <- tibble(
```

```
  Variable = names(rf_model$variable.importance),
  Importance = rf_model$variable.importance
) %>%
  arrange(desc(Importance))

ggplot(importance_df, aes(x = reorder(Variable, Importance), y = Importance)) +
  geom_col(fill = "#2E86AB") +
  coord_flip() +
  labs(x = NULL, y = "Importance (Gini impurity reduction)",
       title = "Random Forest Variable Importance") +
  theme_minimal(base_size = 14)
```

12.4.3.2. Python

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import cross_val_score, StratifiedKFold
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

np.random.seed(42)

rf = RandomForestClassifier(
    n_estimators=500,
    max_features='sqrt',
    min_samples_leaf=10,
    oob_score=True,
    random_state=42,
    n_jobs=-1
)
rf.fit(df, y)
```

```
print(f"OOB accuracy: {rf.oob_score_:.3f}")

# Cross-validated AUC
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
auc_scores = cross_val_score(rf, df, y, cv=cv, scoring='roc_auc')
print(f"CV AUC: {auc_scores.mean():.3f} (+/- {auc_scores.std():.3f})")
```

```
importances = pd.Series(rf.feature_importances_, index=feature_names)
importances = importances.sort_values(ascending=True)

fig, ax = plt.subplots(figsize=(8, 5))
importances.plot(kind='barh', ax=ax, color='#2E86AB')
ax.set_xlabel("Importance (Gini impurity reduction)")
ax.set_title("Random Forest Variable Importance")
plt.tight_layout()
plt.show()
```

12.4.4. Variable Importance: Two Measures

1. **Impurity-based importance** (default): Total reduction in Gini impurity (or variance) from splits on that feature, summed across all trees. Fast but can be biased toward high-cardinality features.
2. **Permutation importance**: For each feature, randomly shuffle its values and measure how much the OOB (or test) error increases. More reliable but slower. This is the recommended approach for inference.

> ⚠ Variable Importance Is Not Causal
>
> A feature with high importance in a random forest is *predictive* of the outcome, not necessarily *causally* related to it. A correlated proxy

(e.g., number of medications as a proxy for disease severity) may appear important even though intervening on it would have no effect. Do not confuse prediction with causation.

12.5. Gradient Boosting

While bagging reduces variance by averaging independent trees, **gradient boosting** reduces bias by building trees *sequentially*, where each new tree corrects the mistakes of the previous ensemble.

12.5.1. The Intuition: Learning from Mistakes

1. Fit a simple tree to the data.
2. Compute the **residuals** (errors) from this tree.
3. Fit a new tree to predict these residuals.
4. Add this new tree to the ensemble (scaled by a learning rate).
5. Repeat steps 2–4 for B iterations.

Each new tree focuses on the observations the current ensemble gets wrong. Over many iterations, the ensemble becomes increasingly accurate.

12.5.2. The Mathematics (Brief)

Gradient boosting minimizes a loss function $L(y, \hat{y})$ by gradient descent in function space. At iteration m:

$$\hat{f}_m(x) = \hat{f}_{m-1}(x) + \eta \cdot h_m(x)$$

where $h_m(x)$ is a tree fitted to the negative gradient of the loss (the "pseudo-residuals") and η is the **learning rate** (also called the shrinkage parameter).

12.5.3. Key Hyperparameters

Parameter	What It Controls	Typical Range
`n_estimators` / `nrounds`	Number of boosting iterations	100–5000
`learning_rate` / `eta`	Step size for each tree	0.01–0.3
`max_depth`	Depth of each tree	3–8 (shallow trees!)
`subsample`	Fraction of data used per tree	0.5–1.0
`colsample_bytree`	Fraction of features per tree	0.5–1.0
`min_child_weight`	Minimum sum of instance weights in a leaf	1–10
`reg_alpha` (L1) / `reg_lambda` (L2)	Regularization	0–10

> The Learning Rate / Number of Trees Tradeoff
>
> A smaller learning rate requires more trees but generally produces better results. A common strategy: set the learning rate to 0.01–0.05, then tune the number of trees via early stopping (stop when validation error stops improving).

12.5.4. XGBoost and LightGBM

XGBoost (eXtreme Gradient Boosting) by Chen and Guestrin (2016) added regularization, efficient computation, and handling of missing values to gradient boosting, making it the dominant algorithm for tabular data competitions and many clinical prediction tasks.

LightGBM by Microsoft further improves efficiency with gradient-based one-side sampling (GOSS) and exclusive feature bundling. It is often faster than XGBoost on large datasets.

12.5.4.1. R

```r
library(xgboost)

# Prepare data for xgboost (requires numeric matrix)
X_train <- readmit_data %>%
  select(age, length_of_stay, num_comorbidities, prior_admissions,
         discharge_hemoglobin, discharge_creatinine, has_diabetes, has_chf) %>%
  as.matrix()

y_train <- as.numeric(readmit_data$readmitted == "Yes")

dtrain <- xgb.DMatrix(data = X_train, label = y_train)

# Fit XGBoost with cross-validation to find optimal nrounds
set.seed(42)
xgb_cv <- xgb.cv(
  params = list(
    objective = "binary:logistic",
    eval_metric = "auc",
    eta = 0.05,
    max_depth = 4,
```

```
    subsample = 0.8,
    colsample_bytree = 0.8
  ),
  data = dtrain,
  nrounds = 1000,
  nfold = 5,
  early_stopping_rounds = 50,
  verbose = 0
)

cat("Best iteration:", xgb_cv$best_iteration, "\n")
cat("Best CV AUC:", max(xgb_cv$evaluation_log$test_auc_mean), "\n")
```

```
# Fit final model with optimal nrounds
xgb_model <- xgb.train(
  params = list(
    objective = "binary:logistic",
    eval_metric = "auc",
    eta = 0.05,
    max_depth = 4,
    subsample = 0.8,
    colsample_bytree = 0.8
  ),
  data = dtrain,
  nrounds = xgb_cv$best_iteration,
  verbose = 0
)

# Variable importance
importance_matrix <- xgb.importance(model = xgb_model)
xgb.plot.importance(importance_matrix, top_n = 8,
                    main = "XGBoost Variable Importance")
```

12.5.4.2. Python

```
import xgboost as xgb
from sklearn.model_selection import cross_val_score, StratifiedKFold
import numpy as np

np.random.seed(42)

xgb_model = xgb.XGBClassifier(
    n_estimators=1000,
    learning_rate=0.05,
    max_depth=4,
    subsample=0.8,
    colsample_bytree=0.8,
    eval_metric='auc',
    early_stopping_rounds=50,
    random_state=42,
    use_label_encoder=False
)

# Cross-validation
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
scores = cross_val_score(
    xgb.XGBClassifier(
        n_estimators=500, learning_rate=0.05, max_depth=4,
        subsample=0.8, colsample_bytree=0.8,
        random_state=42, use_label_encoder=False, eval_metric='logloss'
    ),
    df, y, cv=cv, scoring='roc_auc'
)
print(f"XGBoost CV AUC: {scores.mean():.3f} (+/- {scores.std():.3f})")
```

```
# Fit and plot importance
xgb_final = xgb.XGBClassifier(
    n_estimators=500, learning_rate=0.05, max_depth=4,
    subsample=0.8, colsample_bytree=0.8,
    random_state=42, use_label_encoder=False, eval_metric='logloss'
)
xgb_final.fit(df, y)

fig, ax = plt.subplots(figsize=(8, 5))
xgb.plot_importance(xgb_final, ax=ax, importance_type='gain',
                    title='XGBoost Feature Importance (Gain)')
plt.tight_layout()
plt.show()
```

12.6. Hyperparameter Tuning

Let us start from the very beginning, because this idea trips up almost everyone the first time.

When you fit a random forest, there are two completely different kinds of "settings" involved. The first kind, the model **learns by itself** from the data — for a tree, that means the actual questions at each split ("prior admissions 2?"). You never choose these; the algorithm finds them. The second kind, **you must choose before training even starts** — for example, how many trees to grow, or how deep each tree may go. These pre-set dials are called **hyperparameters**, and the model cannot learn them from the data because they govern *how the learning itself happens*.

> **i** Parameters vs. hyperparameters (the one distinction to remember)
>
> - A **parameter** is something the model estimates from the data (a regression coefficient, a tree's split rules). You do not set these.
> - A **hyperparameter** is a knob *you* set beforehand that controls the learning process (number of trees, tree depth, learning rate). The model cannot figure these out on its own.
>
> An analogy: baking bread, the *recipe quantities the dough develops into* (how the crumb sets) are like parameters — they emerge from the process. The **oven temperature and baking time** are hyperparameters — you dial them in beforehand, and getting them wrong ruins the loaf no matter how good the dough.

Why bother tuning them? Every library ships with default hyperparameter values, and they are reasonable starting points — but they are generic, not tailored to *your* dataset. The best settings depend on how much data you have, how noisy it is, and how many predictors there are. Tuning simply means **trying several combinations of settings and keeping the one that performs best on held-out data** (via the cross-validation you met earlier). Done well, tuning often buys a meaningful bump in performance for free; done carelessly (tuning on the test set), it produces a model that looks great and then disappoints.

The hyperparameters you will see in the code below, in plain language:

Hyperparameter (R / Python)	What it controls	Rule of thumb
`trees` / `n_estimators`	How many trees are in the forest	More is safer (just slower); a few hundred is usually plenty

Hyperparameter (R / Python)	What it controls	Rule of thumb
`mtry` / `max_features`	How many predictors each split is allowed to consider	Smaller values make trees more different from each other, which can help
`min_n` / `min_samples_leaf`	The smallest group of patients allowed in a leaf	Larger values = simpler, smoother trees that overfit less
`max_depth`	How many questions deep a tree may grow	Shallower = simpler; limits overfitting

You do not need to memorise these — the point is that each one trades flexibility against the risk of overfitting, and tuning finds the sweet spot for your data.

There are two common strategies for searching through the possible combinations.

12.6.1. Grid Search

Grid search means you write down a list of candidate values for each hyperparameter, and the computer tries *every possible combination*. If you list 4 values for `mtry` and 5 values for `min_n`, grid search trains and evaluates all $4 \times 5 = 20$ combinations and reports which did best. It is thorough and easy to reason about, but the cost explodes as you add hyperparameters: 4 settings each for just five hyperparameters is already $4^5 = 1024$ models to fit — and each one is itself fit five times under 5-fold cross-validation. This multiplicative blow-up is grid search's main drawback.

12.6.2. Random Search

Random search instead samples a fixed number of combinations *at random* from the ranges you specify (say, 20 combinations, regardless of how many hyperparameters there are). Bergstra and Bengio (2012) showed this is usually *more* efficient than grid search, and the reason is intuitive: in most problems only one or two hyperparameters really matter, but you do not know in advance which. A grid wastes much of its budget trying many values of the unimportant dials while testing only a few values of the important one. Random search, for the same number of model fits, ends up trying *more distinct values of every dial* — so it is more likely to stumble onto a good value of whichever one actually matters. **In practice, start with random search**; it gives you most of the benefit for a fraction of the compute.

12.6.2.1. R

```r
library(tidymodels)

# Define the model with tunable parameters
rf_spec <- rand_forest(
  trees = 500,
  mtry = tune(),
  min_n = tune()
) %>%
  set_engine("ranger") %>%
  set_mode("classification")

# Create recipe
rf_recipe <- recipe(readmitted ~ ., data = readmit_data)

# Workflow
```

```r
rf_wf <- workflow() %>%
  add_model(rf_spec) %>%
  add_recipe(rf_recipe)

# Define tuning grid
rf_grid <- grid_random(
  mtry(range = c(2, 7)),
  min_n(range = c(5, 30)),
  size = 20
)

# Cross-validation
set.seed(42)
folds <- vfold_cv(readmit_data, v = 5, strata = readmitted)

# Tune (this may take a moment)
rf_tune_results <- tune_grid(
  rf_wf,
  resamples = folds,
  grid = rf_grid,
  metrics = metric_set(roc_auc)
)

# Best parameters
show_best(rf_tune_results, metric = "roc_auc", n = 5)
```

```r
# Select best and finalize
best_params <- select_best(rf_tune_results, metric = "roc_auc")
cat("Best mtry:", best_params$mtry, "\n")
cat("Best min_n:", best_params$min_n, "\n")

final_wf <- finalize_workflow(rf_wf, best_params)
final_fit <- fit(final_wf, data = readmit_data)
```

12.6.2.2. Python

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import RandomizedSearchCV, StratifiedKFold
import numpy as np

np.random.seed(42)

# Define parameter distributions
param_dist = {
    'n_estimators': [100, 200, 500],
    'max_features': ['sqrt', 'log2', 3, 5],
    'min_samples_leaf': [5, 10, 15, 20, 30],
    'max_depth': [5, 10, 15, 20, None]
}

rf = RandomForestClassifier(random_state=42, n_jobs=-1)
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

# Random search
search = RandomizedSearchCV(
    rf, param_dist, n_iter=20, cv=cv, scoring='roc_auc',
    random_state=42, n_jobs=-1, verbose=0
)
search.fit(df, y)

print(f"Best AUC: {search.best_score_:.3f}")
print(f"Best parameters: {search.best_params_}")
```

What the code is doing, step by step. Both versions automate the "try many settings, keep the best" recipe. Reading the R version line by line:

1. `rand_forest(... mtry = tune(), min_n = tune())` — we declare a random forest but, instead of fixing `mtry` and `min_n`, we mark them with `tune()`, which is a placeholder meaning *"leave this blank for now; we will search for a good value."*
2. The `recipe` and `workflow` just bundle the model with the outcome and predictors — think of the workflow as a sealed envelope containing everything needed to fit the model.
3. `grid_random(...)` builds the list of 20 random combinations of `mtry` and `min_n` to try (these are the candidate "oven settings").
4. `vfold_cv(...)` creates the 5-fold cross-validation splits. This is the crucial part: **every candidate setting is judged on held-out patients, not on the data it was trained on.** That is what stops us from simply picking the most complex, overfit model.
5. `tune_grid(...)` does the heavy lifting — it fits the forest for each of the 20 settings, across all 5 folds, and records the cross-validated AUC for each.
6. `show_best(...)` then prints the top settings.

What the output shows. `show_best` returns a small table: each row is one hyperparameter combination, with its mean cross-validated AUC and the spread across folds. You read it top-down — the first row is the winning combination. `select_best` grabs that winner, and `finalize_workflow` + `fit` retrain the forest on *all* the data using those best settings, giving you the final model. The Python `RandomizedSearchCV` does exactly the same thing in one object, printing `Best AUC` and the `Best parameters` dictionary at the end. The specific numbers are not the lesson (the data here are simulated); the lesson is the *workflow*: define the dials to search, score each candidate honestly on held-out folds, and keep the winner.

! The cardinal rule of tuning

Tuning must happen **inside** cross-validation, using only the training data — never tune by looking at your final test set. If you try hundreds of settings and pick the one that scores best on the test set, that test score is no longer an honest estimate of future performance: you have, in effect, let the test set leak into model development. Keep a truly untouched test set (or external validation data) for the *final* check, after all tuning is done.

12.7. Comparing Methods: When to Use What

Method	Strengths	Weaknesses	Best For
Single tree	Interpretable, handles non-linearity	Unstable, overfits	Exploration, simple rules
Random forest	Robust, low tuning needed, OOB error	Less interpretable than single tree	General-purpose prediction
Gradient boosting	Often highest accuracy, flexible	More tuning, can overfit, slower	Maximizing prediction accuracy
Logistic regression	Interpretable, well-understood inference	Assumes linearity (without transforms)	Inference, small samples, reporting ORs

Practical Advice for Clinical Prediction Models

Start with logistic regression as your baseline. If you need better prediction and have enough data (typically hundreds of events), try a random forest. If you want to squeeze out the last bit of performance and are willing to tune carefully, use XGBoost. Always compare against the logistic regression baseline — if the improvement is marginal, prefer the simpler model.

12.8. Clinical Example: Readmission Risk Stratification

Let us bring everything together by comparing all three methods on the hospital readmission dataset.

12.8.0.1. R

```
library(tidymodels)

set.seed(42)
folds <- vfold_cv(readmit_data, v = 10, strata = readmitted)

# Logistic regression
lr_spec <- logistic_reg() %>% set_engine("glm")
lr_wf <- workflow() %>%
  add_model(lr_spec) %>%
  add_recipe(recipe(readmitted ~ ., data = readmit_data))
lr_res <- fit_resamples(lr_wf, resamples = folds,
                        metrics = metric_set(roc_auc))

# Random forest
rf_spec <- rand_forest(trees = 500, mtry = 3, min_n = 10) %>%
  set_engine("ranger") %>%
  set_mode("classification")
rf_wf <- workflow() %>%
  add_model(rf_spec) %>%
  add_recipe(recipe(readmitted ~ ., data = readmit_data))
```

```r
rf_res <- fit_resamples(rf_wf, resamples = folds,
                        metrics = metric_set(roc_auc))

# Boosted trees (via xgboost engine in tidymodels)
xgb_spec <- boost_tree(trees = 500, tree_depth = 4, learn_rate = 0.05,
                        min_n = 10) %>%
  set_engine("xgboost") %>%
  set_mode("classification")
xgb_wf <- workflow() %>%
  add_model(xgb_spec) %>%
  add_recipe(recipe(readmitted ~ ., data = readmit_data))
xgb_res <- fit_resamples(xgb_wf, resamples = folds,
                         metrics = metric_set(roc_auc))

# Collect and compare results
bind_rows(
  collect_metrics(lr_res) %>% mutate(model = "Logistic Regression"),
  collect_metrics(rf_res) %>% mutate(model = "Random Forest"),
  collect_metrics(xgb_res) %>% mutate(model = "XGBoost")
) %>%
  select(model, .metric, mean, std_err) %>%
  arrange(desc(mean))
```

12.8.0.2. Python

```python
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from sklearn.model_selection import cross_val_score, StratifiedKFold
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline
import numpy as np
import pandas as pd
```

```
np.random.seed(42)
cv = StratifiedKFold(n_splits=10, shuffle=True, random_state=42)

models = {
    'Logistic Regression': make_pipeline(
        StandardScaler(), LogisticRegression(max_iter=1000)
    ),
    'Random Forest': RandomForestClassifier(
        n_estimators=500, max_features='sqrt', min_samples_leaf=10,
        random_state=42, n_jobs=-1
    ),
    'Gradient Boosting': GradientBoostingClassifier(
        n_estimators=500, learning_rate=0.05, max_depth=4,
        subsample=0.8, random_state=42
    )
}

results = []
for name, model in models.items():
    scores = cross_val_score(model, df, y, cv=cv, scoring='roc_auc')
    results.append({
        'Model': name,
        'Mean AUC': f"{scores.mean():.3f}",
        'Std': f"{scores.std():.3f}"
    })
    print(f"{name:25s}: AUC = {scores.mean():.3f} (+/- {scores.std():.3f})")

print(pd.DataFrame(results).to_string(index=False))
```

12.9. Exercises

12.9.1. Exercise 1: Build and Prune a Classification Tree

Using the readmission dataset (or a dataset of your choice):

1. Fit a full, unpruned classification tree. How many terminal nodes does it have?
2. Use cross-validation to find the optimal complexity parameter (cp in R) or maximum depth (in Python).
3. Prune the tree and plot it. How does it compare to the full tree?
4. What are the top 3 splitting variables? Do they make clinical sense?

12.9.1.1. R Starter Code

```r
library(rpart)      # rpart(), rpart.control()
library(rpart.plot) # rpart.plot() for visualising the tree

# Fit a full tree (set cp very low to avoid early pruning)
full_tree <- rpart(readmitted ~ ., data = readmit_data, method = "class",
                   control = rpart.control(cp = 0.001))

# Your code:
# 1. Count terminal nodes
# 2. Use printcp() and plotcp() to find optimal cp
# 3. Prune and plot
# 4. Examine variable importance: full_tree$variable.importance
```

12.9.1.2. Python Starter Code

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import cross_val_score

# Your code:
# 1. Fit trees with different max_depth values
# 2. Cross-validate each
# 3. Plot the optimal tree
# 4. Examine feature_importances_
```

12.9.2. Exercise 2: Random Forest vs XGBoost Tuning Challenge

1. Split the readmission data into 80% training and 20% test sets.
2. Using the training set with 5-fold cross-validation:
   a. Tune a random forest (try different `mtry`/`max_features` and `min_n`/`min_samples_leaf` values).
   b. Tune an XGBoost model (try different `learning_rate`, `max_depth`, and `n_estimators` values).
3. Select the best hyperparameters for each model.
4. Evaluate both models on the held-out test set. Which performs better?
5. Create variable importance plots for both models. Do they agree on the most important features?

12.9.2.1. R Starter Code

```
library(tidymodels) # initial_split(), training(), testing(), tune_grid()

# Split data
```

```r
set.seed(42)
split <- initial_split(readmit_data, prop = 0.8, strata = readmitted)
train <- training(split)
test <- testing(split)

# Your code: define models with tune(), create grids, run tune_grid()
# Finalize models, predict on test set, compare
```

12.9.2.2. Python Starter Code

```python
from sklearn.model_selection import train_test_split, RandomizedSearchCV
from sklearn.ensemble import RandomForestClassifier
import xgboost as xgb

X_train, X_test, y_train, y_test = train_test_split(
    df, y, test_size=0.2, stratify=y, random_state=42
)

# Your code: define param_dist for RF and XGBoost
# Run RandomizedSearchCV for each
# Evaluate on test set with roc_auc_score
```

12.9.3. Exercise 3: Interpreting a Clinical Prediction Model

You have built a gradient-boosted model for 30-day hospital readmission. Your hospital administration asks you to present the results.

1. Write a non-technical summary (3–4 sentences) explaining what the model does and how well it performs, suitable for a hospital quality committee.

2. The model identifies "number of prior admissions" and "discharge creatinine" as the two most important predictors. Explain to a clinical audience what this means and *does not* mean (recall the caution about variable importance and causation).
3. How would you propose to validate this model before deploying it in clinical practice? What could go wrong if you skip external validation?

12.10. Summary

Method	How It Works	Key Advantage	Key Risk
Decision tree	Sequential yes/no splits	Highly interpretable	Overfits, unstable
Random forest	Average of many decorrelated trees	Robust, minimal tuning	Less interpretable
Gradient boosting	Sequential trees correcting errors	Often best accuracy	Can overfit, many hyperparameters

The progression from single trees to random forests to gradient boosting follows a clear logic:

- **Single trees** are interpretable but unstable and prone to overfitting.
- **Random forests** reduce variance by averaging many decorrelated trees, with the OOB error providing a built-in validation estimate.
- **Gradient boosting** reduces both bias and variance through sequential learning, often achieving the best predictive accuracy at the cost of more tuning.

For clinical prediction models, always compare against a logistic regression baseline. Use cross-validation for honest evaluation, tune hyperparameters systematically, and remember that the most complex model is not always the best choice.

12.11. References and Further Reading

The foundational paper on random forests is Breiman (2001), "Random Forests," published in *Machine Learning*. This paper introduced the algorithm, the concept of out-of-bag error estimation, and permutation-based variable importance. It remains one of the most cited papers in machine learning.

For XGBoost, the key reference is Chen and Guestrin (2016), "XGBoost: A Scalable Tree Boosting System," presented at KDD 2016. This paper describes the algorithmic innovations (regularized objective, approximate split finding, sparsity-aware computation) that made XGBoost the dominant method for tabular data.

Smits et al. (2026) provide an accessible treatment of tree-based methods in the context of clinical research, including practical guidance on when ensemble methods offer meaningful advantages over logistic regression and how to interpret variable importance in clinical terms.

For a thorough treatment of the CART algorithm, see Breiman, Friedman, Olshen, and Stone's *Classification and Regression Trees* (1984, Wadsworth). Although the original text is somewhat dated, the core ideas remain current.

Hastie, Tibshirani, and Friedman's *The Elements of Statistical Learning* (Springer, 2nd edition, 2009) covers bagging (Section 8.7), random forests (Chapter 15), and boosting (Chapter 10) with mathematical rigor. James et al.'s *An Introduction to Statistical Learning* (ISLR, 2nd edition, 2021) provides a gentler introduction in Chapters 8 and covers tree-based methods with R lab exercises.

Boehmke and Greenwell's *Hands-On Machine Learning with R* dedicates several chapters to tree-based methods with practical tidymodels code, and is an excellent companion for the R exercises in this chapter.

For the gradient boosting framework, Friedman (2001), "Greedy Function Approximation: A Gradient Boosting Machine," in *Annals of Statistics*, is the foundational statistical treatment. The paper introduced the general gradient boosting framework that XGBoost and LightGBM build upon.

Regarding clinical applications and best practices, the TRIPOD statement (Collins et al., 2015) provides guidelines for reporting clinical prediction models, including those based on machine learning. Any readmission model (or similar clinical prediction tool) developed with these methods should be reported following TRIPOD guidelines and externally validated before clinical deployment.

13. Introduction to Neural Networks

Deep learning is the technology behind most of the headline-grabbing medical AI of the last decade — systems that read chest X-rays, grade diabetic retinopathy, or flag arrhythmias on an ECG. This chapter is a practical, clinician-facing tour of what neural networks are, how they learn, and — crucially — *when they are and are not the right tool* for a health research question. The code in this chapter is shown for illustration and is set to `eval: false`, so the rendered page displays the code and our explanation of what it would produce, rather than live output. The aim is not to make you a deep learning engineer, but to let you read a deep learning paper critically, talk to a collaborator who builds these models, and recognise when a far simpler method would serve your patients just as well.

13.1. When Does Deep Learning Make Sense?

In the previous chapter, you learned that gradient-boosted trees are powerful, flexible, and remarkably effective for clinical prediction on structured data. A natural question follows: when should you reach for something more complex?

The short answer: **deep learning excels when your data has spatial, sequential, or linguistic structure that tabular methods cannot exploit.** If your data lives in a spreadsheet — rows of patients, columns of lab values — gradient-boosted trees will usually match or beat a neural network, with less effort and better interpretability. But if your data is a chest X-ray, a clinical note, an ECG tracing, or a pathology slide, deep learning is the tool that unlocked performance previously thought impossible.

13.1.1. Common Misconceptions

1. **"Deep learning is always better than traditional ML."** False. For tabular clinical data, a 2022 NeurIPS benchmark by Grinsztajn et al. showed that tree-based models consistently outperform neural networks on typical tabular datasets. A 2026 clinical benchmark confirmed that TabPFN — a transformer designed specifically for tabular data — exceeded the best traditional ML model in only 17% of clinical prediction tasks. Start with logistic regression or XGBoost; reach for deep learning only when the data type demands it.

2. **"Deep learning requires millions of examples."** Misleading. Transfer learning — starting from a model pretrained on millions of images and fine-tuning on your hundreds — has made deep learning practical even with small clinical datasets. Many successful medical imaging studies use fewer than 5,000 labelled examples.

3. **"Neural networks are uninterpretable black boxes."** Partially true, but increasingly addressable. Techniques such as Grad-CAM (which highlights the image regions driving a prediction) and attention visualization (which shows which words or time steps a model focuses on) provide clinically meaningful explanations. They are not as clean as a regression coefficient, but they are far from opaque.

4. **"I need a GPU cluster to do deep learning."** Not necessarily. Transfer learning with pretrained models can be done on a single consumer GPU or even in the cloud (Google Colab offers free GPU access). Training a model from scratch on large datasets is another matter entirely.

13.2. How Neural Networks Learn

A neural network is, at its core, a series of matrix multiplications followed by non-linear transformations. If you have worked through logistic regression, you already understand the basic idea: take a weighted sum of inputs and pass them through a sigmoid function to produce a probability. A neural network does exactly the same thing — but stacks many such layers on top of each other, allowing it to learn increasingly abstract representations.

13.2.1. The Building Blocks

Neurons (nodes): Each neuron computes a weighted sum of its inputs, adds a bias, and applies a non-linear function (called an **activation function**):

$$a = f\left(\sum_{i=1}^{n} w_i x_i + b\right)$$

where x_i are the inputs, w_i are the weights, b is the bias, and f is the activation function. In clinical terms, this is exactly the structure of a risk score: each input (say, age, creatinine, blood pressure) is multiplied by a weight that says how much it matters, the weighted contributions are added up, an intercept (the bias b) is added, and the total is squashed into a usable range by f. A single neuron with a sigmoid activation *is* a logistic regression. The power of a network comes from wiring thousands of these simple units together so the later ones can act on features the earlier ones discovered.

Layers: Neurons are organized into layers:

- **Input layer**: receives the raw data (pixel values, lab values, word embeddings).

- **Hidden layers**: intermediate layers that learn increasingly abstract features. A network with many hidden layers is called "deep" — hence "deep learning."
- **Output layer**: produces the final prediction (a probability, a class label, a continuous value).

Activation functions: The non-linear functions that give neural networks their power. Without them, stacking layers would be pointless — the composition of linear functions is still linear. Common choices include:

- **ReLU** (Rectified Linear Unit): $f(x) = \max(0, x)$. The default for hidden layers. Simple, fast, and effective.
- **Sigmoid**: $f(x) = \frac{1}{1+e^{-x}}$. Used in the output layer for binary classification. You already know this from logistic regression.
- **Softmax**: Generalizes sigmoid to multiple classes. Used in the output layer for multi-class classification.

13.2.2. Training: Gradient Descent and Backpropagation

Neural networks learn by adjusting their weights to minimize a **loss function** — a measure of how wrong the predictions are. The process works as follows:

1. **Forward pass**: Feed data through the network to produce a prediction.
2. **Compute loss**: Compare the prediction to the true label using a loss function (cross-entropy for classification, mean squared error for regression).
3. **Backward pass (backpropagation)**: Compute the gradient of the loss with respect to every weight in the network, using the chain rule of calculus.
4. **Update weights**: Adjust each weight in the direction that reduces the loss, scaled by a **learning rate** η:

$$w \leftarrow w - \eta \frac{\partial \mathcal{L}}{\partial w}$$

Read the update rule in plain language: the gradient $\partial \mathcal{L}/\partial w$ is the slope of the error with respect to one weight — it tells you which direction makes the predictions *worse*, so we step in the opposite direction (hence the minus sign). The learning rate η is the size of that step. A useful mental picture is walking downhill in fog: the gradient is the steepest direction underfoot, and η is how big a stride you take before re-checking. This process repeats over many **epochs** (complete passes through the training data). The learning rate is critical: too large, and the model oscillates; too small, and it converges painfully slowly.

> You Already Know This Pattern
>
> Gradient descent is not unique to deep learning. It is the same optimization strategy used in logistic regression and many other statistical models. The difference is scale: a logistic regression with 10 predictors has 11 parameters; a deep neural network may have millions.

13.2.3. Regularization: Preventing Overfitting

Deep networks have enormous capacity and will happily memorize the training data if you let them. The strategies for preventing this parallel what you learned in Chapter 7, but with some additions:

- **Dropout**: During training, randomly "turn off" a fraction of neurons in each layer. This prevents co-adaptation and acts as an implicit ensemble. Typical dropout rates range from 0.2 to 0.5.
- **Weight decay (L2 regularization)**: Penalize large weights, exactly as in ridge regression.

- **Early stopping**: Monitor performance on a validation set and stop training when performance begins to deteriorate.
- **Data augmentation**: Artificially expand the training set by applying random transformations (rotations, flips, crops, colour jitter). Especially important in medical imaging, where labelled data is scarce.

13.3. Clinical Example: Neural Network for Readmission Prediction

To connect these concepts to what you already know, let us apply a simple neural network to the hospital readmission data from Chapter 12. This is *not* a scenario where deep learning is the right tool — gradient-boosted trees will likely perform as well or better on tabular data — but it illustrates the mechanics.

13.3.0.1. R

```
library(keras3)

# Simulate readmission data (same structure as Chapter 8)
set.seed(42)
n <- 1000

readmit_data <- tibble(
  age = rnorm(n, 68, 12),
  length_of_stay = rpois(n, 5) + 1,
  num_comorbidities = rpois(n, 3),
  prior_admissions = rpois(n, 1),
  discharge_hgb = rnorm(n, 11, 2),
  discharge_creatinine = rlnorm(n, 0.2, 0.5),
```

```
  has_diabetes = rbinom(n, 1, 0.35),
  has_chf = rbinom(n, 1, 0.25)
)

readmit_prob <- plogis(-3 + 0.02 * (readmit_data$age - 68) +
                        0.15 * readmit_data$prior_admissions +
                        0.1 * readmit_data$num_comorbidities +
                        0.3 * readmit_data$has_chf -
                        0.1 * readmit_data$discharge_hgb)
readmit_data$readmitted <- rbinom(n, 1, readmit_prob)

# Prepare data: scale features, split into train/test
x <- readmit_data %>% select(-readmitted) %>% as.matrix()
y <- readmit_data$readmitted

# Standardize
x_mean <- apply(x, 2, mean)
x_sd   <- apply(x, 2, sd)
x_scaled <- scale(x, center = x_mean, scale = x_sd)

# Train/test split
set.seed(123)
train_idx <- sample(n, 800)
x_train <- x_scaled[train_idx, ]
x_test  <- x_scaled[-train_idx, ]
y_train <- y[train_idx]
y_test  <- y[-train_idx]

# Define a simple feedforward neural network
model <- keras_model_sequential(input_shape = ncol(x_train)) %>%
  layer_dense(units = 32, activation = "relu") %>%
  layer_dropout(rate = 0.3) %>%
  layer_dense(units = 16, activation = "relu") %>%
```

```
  layer_dropout(rate = 0.3) %>%
  layer_dense(units = 1, activation = "sigmoid")

model %>% compile(
  optimizer = optimizer_adam(learning_rate = 0.001),
  loss = "binary_crossentropy",
  metrics = "AUC"
)

# Train with early stopping
history <- model %>% fit(
  x_train, y_train,
  epochs = 50,
  batch_size = 32,
  validation_split = 0.2,
  callbacks = list(
    callback_early_stopping(
      patience = 5, restore_best_weights = TRUE
    )
  ),
  verbose = 0
)

# Evaluate on test set
results <- model %>% evaluate(x_test, y_test, verbose = 0)
cat("Test loss:", round(results[[1]], 3), "\n")
cat("Test AUC:", round(results[[2]], 3), "\n")
```

13.3.0.2. Python

```
import numpy as np
import tensorflow as tf
```

```python
from tensorflow.keras import layers, models, callbacks
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# Simulate readmission data (same structure as Chapter 8)
np.random.seed(42)
n = 1000

age = np.random.normal(68, 12, n)
length_of_stay = np.random.poisson(5, n) + 1
num_comorbidities = np.random.poisson(3, n)
prior_admissions = np.random.poisson(1, n)
discharge_hgb = np.random.normal(11, 2, n)
discharge_creatinine = np.random.lognormal(0.2, 0.5, n)
has_diabetes = np.random.binomial(1, 0.35, n)
has_chf = np.random.binomial(1, 0.25, n)

X = np.column_stack([age, length_of_stay, num_comorbidities,
                     prior_admissions, discharge_hgb,
                     discharge_creatinine, has_diabetes, has_chf])

prob = 1 / (1 + np.exp(-(-3 + 0.02 * (age - 68) +
                          0.15 * prior_admissions +
                          0.1 * num_comorbidities +
                          0.3 * has_chf -
                          0.1 * discharge_hgb)))
y = np.random.binomial(1, prob)

# Split and scale
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=123
)
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Define a simple feedforward neural network
model = models.Sequential([
    layers.Dense(32, activation="relu", input_shape=(X_train.shape[1],)),
    layers.Dropout(0.3),
    layers.Dense(16, activation="relu"),
    layers.Dropout(0.3),
    layers.Dense(1, activation="sigmoid")
])

model.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=0.001),
    loss="binary_crossentropy",
    metrics=[tf.keras.metrics.AUC(name="auc")]
)

# Train with early stopping
history = model.fit(
    X_train, y_train,
    epochs=50,
    batch_size=32,
    validation_split=0.2,
    callbacks=[
        callbacks.EarlyStopping(
            patience=5, restore_best_weights=True
        )
    ],
    verbose=0
)

loss, auc = model.evaluate(X_test, y_test, verbose=0)
```

```
print(f"Test loss: {loss:.3f}")
print(f"Test AUC:  {auc:.3f}")
```

What the code is doing. This walks through the full life-cycle of a neural network on tabular data, end to end. First we simulate 1,000 patients and a true readmission risk that depends on age, prior admissions, comorbidities, heart failure, and discharge haemoglobin — so we *know* the right answer and can judge the model. We then standardise every feature (subtract the mean, divide by the standard deviation) so that no variable dominates simply because it is measured on a larger scale, and split the data into training and held-out test sets. The architecture is a small stack: an input layer, two hidden layers (32 then 16 neurons) with ReLU activations and 30% dropout between them to curb overfitting, and a single sigmoid output that emits a readmission probability. `compile` chooses the optimiser (Adam), the loss (binary cross-entropy, the same quantity logistic regression minimises), and the metric to watch (AUC). `fit` trains for up to 50 epochs but, thanks to early stopping with `patience = 5`, halts once the validation AUC stops improving and rewinds to the best weights. Because the chapter is `eval: false`, no numbers are printed here, but the final two lines would report the test-set loss and AUC — the AUC being the headline: the probability the model ranks a randomly chosen readmitted patient above a randomly chosen non-readmitted one. Expect something in the 0.65–0.75 range for data this noisy.

> **Compare This to Chapter 8**
>
> Run the XGBoost model from Chapter 12 on the same data and compare the AUC. You will likely find that XGBoost matches or exceeds the neural network — with less code, less tuning, and immediate access to variable importance. This is the norm for tabular clinical data.

13.4. Key Architectures for Clinical Research

You do not need to understand every layer and parameter to use deep learning effectively. But you do need to understand *which architecture suits which data type* and *why*.

13.4.1. Convolutional Neural Networks (CNNs): Learning from Images

CNNs learn spatial hierarchies of features from images. Early layers detect edges and textures; deeper layers combine these into complex patterns (tumour boundaries, retinal vessel structures, skin lesion shapes).

Instead of connecting every neuron to every input, CNNs use **convolutional filters** — small windows that slide across the image, detecting local patterns. A 3×3 filter might learn to detect a horizontal edge; stacking many filters across many layers builds up a rich feature hierarchy.

Landmark clinical applications:

Application	Key Study	Architecture	Performance
Chest X-ray diagnosis	Rajpurkar et al., 2017	DenseNet-121	Radiologist-level pneumonia detection
Diabetic retinopathy	Gulshan et al., *JAMA* 2016	Inception-v3	AUC 0.991 for referable DR
Skin cancer classification	Esteva et al., *Nature* 2017	Inception-v3	Dermatologist-level performance

Application	Key Study	Architecture	Performance
Pathology	Campanella et al., *Nature Medicine* 2019	ResNet	Weakly supervised cancer detection in whole-slide images

From CNNs to Vision Transformers

Classic CNN architectures (ResNet, DenseNet) dominated medical imaging from 2015 to 2021. Since then, **Vision Transformers (ViTs)** have emerged as strong alternatives in research. ViTs split images into patches, treat each patch as a "token" (analogous to a word in NLP), and use self-attention to model relationships between patches. However, there is an "architectural gap" between research and clinical deployment: as of early 2026, nearly all FDA-cleared radiology AI devices still use CNNs, not transformers or foundation models (*Lancet Digital Health*, 2026). In the research literature, CNNs and ViTs achieve comparable performance on many tasks, and hybrid architectures are increasingly common.

13.4.2. Recurrent Networks and Transformers: Learning from Sequences

Recurrent Neural Networks (RNNs) were designed for sequential data — time series, text, and any data where order matters. The network maintains a **hidden state** updated at each time step, allowing it to retain information from earlier in the sequence.

Long Short-Term Memory (LSTM) networks solve the tendency of standard RNNs to forget long-range dependencies. They use gating mechanisms to control what information to retain, forget, and output at each step. LSTMs were the dominant architecture for clinical time series from roughly 2015 to 2022.

Transformers have largely superseded RNNs and LSTMs. Instead of processing sequences one element at a time, transformers use **self-attention** to relate every element to every other element simultaneously. This parallelism makes them faster to train, and the attention mechanism captures long-range dependencies more effectively. Medformer (NeurIPS 2024) is the current state-of-the-art transformer architecture specifically designed for medical time series classification.

Clinical applications of sequential architectures:

Data Type	Example	Current Preferred Architecture
ECG tracings	Arrhythmia detection, digital biomarkers	Transformers
EEG signals	Seizure detection, ICU monitoring	Transformers with attention-based interpretability
ICU vital signs	Mortality prediction, clinical deterioration	Transformers, temporal CNNs
Wearable sensor data	Activity recognition, gait analysis	Temporal CNNs, hybrid models

13.4.3. Large Language Models: Learning from Clinical Text

The transformer architecture is also the foundation of **large language models (LLMs)**. In clinical research, LLMs and their smaller predecessors have been applied to:

- **Named entity recognition**: Identifying diseases, medications, procedures, and lab values in clinical notes. Models like ClinicalBERT and PubMedBERT (~110M parameters) remain workhorses for structured extraction.
- **Information extraction**: Pulling structured data from pathology and radiology reports.
- **Report generation**: Drafting radiology or pathology reports from imaging studies.
- **Clinical question answering**: Med-PaLM 2 achieved 86.5% on MedQA; Med-Gemini reached 91.1%. The MedHELM benchmark (*Nature Medicine*, 2025) tested 9 frontier LLMs across 121 medical tasks, finding scores of 0.73–0.85 for clinical note generation but only 0.56–0.72 for clinical decision support.

> ⚠ LLMs Are Not Clinical Decision-Making Tools (Yet)
>
> LLMs can hallucinate — generating plausible-sounding but factually incorrect medical information. They lack the ability to reason causally about individual patients. Current evidence supports their use as *assistants* (drafting notes, extracting structured data, literature search) rather than as autonomous clinical decision-makers. Always verify LLM outputs against primary sources.

13.4.4. Deep Learning for Survival Analysis

If you worked through Chapter 8, you know that time-to-event data requires special handling. Deep learning has extended survival analysis beyond the

Cox model:

Model	Approach	Key Feature
DeepSurv (Katzman et al., 2018)	Neural network within Cox PH framework	Learns non-linear risk functions
DeepHit (Lee et al., 2018)	Custom loss; no parametric assumptions	Handles competing risks directly
SurvTRACE (2024)	Transformer-based	Models competing events with attention
DySurv (*JAMIA*, 2025)	Conditional variational autoencoder	Dynamic risk from longitudinal EHR data

For most clinical survival analysis, Cox regression and random survival forests remain the appropriate starting point. Deep survival models become relevant with high-dimensional inputs (imaging, genomics) or complex temporal patterns.

13.5. Transfer Learning and Foundation Models

Transfer learning is the single most important practical technique in deep learning for health research. The idea is simple: take a model that has already learned useful features from a large dataset, and adapt it to your specific task with a much smaller dataset.

13.5.1. How Transfer Learning Works

1. **Start with a pretrained model**: A CNN trained on ImageNet (14 million natural images) has learned to detect edges, textures, shapes,

and objects. These features transfer surprisingly well to medical images.
2. **Replace the output layer**: Swap the final classification head with one that predicts your clinical outcome.
3. **Fine-tune**: Train the modified model on your clinical dataset. You can freeze the pretrained layers (fast, works with very small datasets) or fine-tune all layers with a small learning rate (better performance, requires more data).

13.5.2. Foundation Models in Medicine

The field is moving beyond ImageNet toward **domain-specific foundation models** pretrained on medical data:

Model	Domain	Training Data	Published
MedSAM	Image segmentation	1.57M image-mask pairs, 10 modalities	*Nature Communications*, 2024
Biomed-CLIP	Vision-language	15M biomedical image-text pairs	Microsoft, 2023
RET-Found	Ophthalmology	1.6M retinal images, self-supervised	*Nature*, 2023
UNI	Pathology	100K+ whole-slide images	*Nature Medicine*, 2024
Virchow	Pathology	1.5M whole-slide images	Paige/Microsoft, 2024

Practical Advice for Clinical Researchers

If you want to apply deep learning to medical images, do not train from scratch. Use a pretrained foundation model (MedSAM for segmentation, BiomedCLIP for classification) and fine-tune on your labelled data. With even a few hundred annotated images, you can achieve strong performance. This is the practical path for most clinical research groups.

13.5.3. Transfer Learning in Practice

The following example shows the complete workflow for fine-tuning a pretrained ResNet50 for binary image classification. In practice, you would replace the data loading step with your own clinical images.

13.5.3.1. R

```
library(keras3)

# Load pretrained ResNet50 (trained on ImageNet, without classification head)
base_model <- application_resnet50(
  weights = "imagenet",
  include_top = FALSE,
  input_shape = c(224, 224, 3)
)
freeze_weights(base_model)

# Add new classification head for binary outcome
model <- keras_model_sequential(input_shape = c(224, 224, 3)) %>%
  base_model() %>%
  layer_global_average_pooling_2d() %>%
  layer_dropout(rate = 0.3) %>%
  layer_dense(units = 1, activation = "sigmoid")

model %>% compile(
```

```
  optimizer = optimizer_adam(learning_rate = 1e-4),
  loss = "binary_crossentropy",
  metrics = "AUC"
)

# Data augmentation (rotation, flipping, shifting)
train_datagen <- image_data_generator(
  rescale = 1/255,
  rotation_range = 20,
  horizontal_flip = TRUE,
  validation_split = 0.2
)

# Point to your image directory organized as: images/class_0/ and images/class_1/
train_gen <- flow_images_from_directory(
  "path/to/images/", train_datagen,
  target_size = c(224, 224), batch_size = 32,
  class_mode = "binary", subset = "training"
)

val_gen <- flow_images_from_directory(
  "path/to/images/", train_datagen,
  target_size = c(224, 224), batch_size = 32,
  class_mode = "binary", subset = "validation"
)

# Fine-tune with early stopping
history <- model %>% fit(
  train_gen, epochs = 20,
  validation_data = val_gen,
  callbacks = list(
    callback_early_stopping(
      patience = 3, restore_best_weights = TRUE
```

```
    )
  )
)
```

13.5.3.2. Python

```python
import tensorflow as tf
from tensorflow.keras.applications import ResNet50
from tensorflow.keras import layers, models, callbacks
from tensorflow.keras.preprocessing.image import ImageDataGenerator

# Load pretrained ResNet50 (trained on ImageNet, without classification head)
base_model = ResNet50(weights="imagenet", include_top=False,
                      input_shape=(224, 224, 3))
base_model.trainable = False

# Add new classification head for binary outcome
model = models.Sequential([
    base_model,
    layers.GlobalAveragePooling2D(),
    layers.Dropout(0.3),
    layers.Dense(1, activation="sigmoid")
])

model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=1e-4),
              loss="binary_crossentropy",
              metrics=[tf.keras.metrics.AUC(name="auc")])

# Data augmentation (rotation, flipping, shifting)
train_datagen = ImageDataGenerator(
    rescale=1.0/255, rotation_range=20,
    horizontal_flip=True, validation_split=0.2
```

```
)

# Point to your image directory organized as: images/class_0/ and images/class_1/
train_gen = train_datagen.flow_from_directory(
    "path/to/images/", target_size=(224, 224),
    batch_size=32, class_mode="binary", subset="training"
)

val_gen = train_datagen.flow_from_directory(
    "path/to/images/", target_size=(224, 224),
    batch_size=32, class_mode="binary", subset="validation"
)

# Fine-tune with early stopping
history = model.fit(
    train_gen, epochs=20, validation_data=val_gen,
    callbacks=[
        callbacks.EarlyStopping(
            patience=3, restore_best_weights=True
        )
    ]
)
```

What the code is doing. This is the workhorse recipe for medical imaging in a typical research group. Rather than training a network from scratch (which would need tens of thousands of labelled images), we load ResNet50 with its ImageNet weights already in place — a model that has *already* learned to detect edges, textures, and shapes from 14 million natural photographs. We discard its original 1,000-category classifier (`include_top = FALSE`) and freeze its weights so they are not disturbed. We then bolt on a fresh head — a pooling layer, dropout, and a single sigmoid neuron — which is the only part that learns from our clinical images. The `image_data_generator` performs *data augmentation*: each epoch it shows the network slightly rotated and flipped versions of the same images, artificially enlarging a small dataset and teaching the model to ignore irrelevant orientation. The `flow_images_from_directory` lines expect your images sorted into one folder per class. There is no output to read because the code is illustrative (`eval: false`) and points at a placeholder path; the practical takeaway is the *pattern* — freeze a pretrained backbone, train a small new head, augment aggressively — which lets you reach strong performance with only a few hundred annotated scans.

13.6. Practical Considerations

13.6.1. When to Use (and Not Use) Deep Learning

Scenario	Recommendation
Tabular EHR data, <50 features	Logistic regression or XGBoost
Tabular data combined with images or text	DL for the unstructured component; consider multimodal fusion
Medical images (X-ray, CT, MRI, pathology)	DL with transfer learning
Clinical notes, radiology reports	Transformer-based models
ECG, EEG, ICU time series	DL is appropriate; consider transformers or temporal CNNs
Survival analysis, standard covariates	Cox regression or random survival forests first

13.6.2. Data Requirements

Approach	Typical Data Needed
Training from scratch	Tens of thousands of labelled examples
Transfer learning (fine-tuning)	Hundreds to low thousands
Foundation model adaptation	As few as 50–100 examples for simple tasks
Self-supervised pretraining	Large amounts of *unlabelled* data

13.6.3. Compute

Approach	Hardware
Fine-tuning a pretrained model	Single GPU; Google Colab (free) works
Training a moderate model from scratch	1–4 GPUs; cloud instance ~$2–4/hour
Training a foundation model	GPU cluster; not realistic for most research groups
Running inference	CPU is often sufficient

13.6.4. Reporting Deep Learning Studies

If you publish research involving deep learning, several reporting guidelines apply:

- **TRIPOD+AI** (Collins et al., *BMJ* 2024): 27-item checklist for prediction models using regression or ML. Applies to all prediction model studies.

- **CLAIM** (Mongan, Moy, and Kahn, *Radiology: AI* 2020): Checklist for AI in Medical Imaging. Covers study design, data, model, evaluation, and discussion.
- **CONSORT-AI / SPIRIT-AI** (Liu et al. and Rivera et al., *Nature Medicine* 2020): Extensions for reporting randomised trials and protocols involving AI.
- **MINIMAR** (Hernandez-Boussard et al., *JAMIA* 2020): Minimum information for medical AI reporting.

13.6.5. Regulatory Context

As of early 2026, the FDA has authorized over 1,350 AI-enabled medical devices, with 76% in radiology. Nearly all are Class II devices cleared via the 510(k) pathway. The EU AI Act, which entered into force in August 2024, classifies AI-enabled medical devices as high-risk, with full compliance obligations from August 2026. If you are developing a model intended for clinical deployment, regulatory awareness is essential from the outset.

13.7. Challenges and Limitations

⚠ The External Validation Problem

A systematic review of 86 deep learning algorithms in radiology found that **81% exhibited decreased accuracy on external datasets**, with nearly a quarter experiencing a substantial drop of 0.10 or greater in AUC. Domain shift — differences in patient demographics, imaging equipment, acquisition protocols, and disease prevalence between development and deployment sites — remains the greatest obstacle to clinical translation.

Fairness and bias: A 2024 study in *Nature Medicine* demonstrated that even if a model is optimized for fairness at a single site, fairness does not transfer to out-of-distribution datasets. Subgroup analysis across age, sex, and ethnicity is essential.

Interpretability: A systematic review of 67 studies (2019–2024) found that the addition of explainability methods (Grad-CAM, saliency maps) provided no statistically significant improvement in diagnostic accuracy beyond the AI prediction itself. Interpretability tools are valuable for debugging and trust-building, but they are not a substitute for rigorous external validation.

Temporal degradation: Clinical AI models can degrade over time as clinical practice, coding conventions, and patient populations change. Post-deployment monitoring is essential but rarely implemented.

13.8. Exercises

13.8.1. Exercise 1: Neural Network vs XGBoost on Tabular Data

Using the readmission dataset from this chapter, fit both a neural network and an XGBoost model. Compare their performance using 5-fold cross-validated AUC.

13.8.1.1. R Starter Code

```r
library(tidyverse)  # tibble()
library(keras3)
library(tidymodels)
library(xgboost)

# Use the readmit_data from above (or re-simulate it)
```

```
set.seed(42)
n <- 1000

readmit_data <- tibble(
  age = rnorm(n, 68, 12),
  length_of_stay = rpois(n, 5) + 1,
  num_comorbidities = rpois(n, 3),
  prior_admissions = rpois(n, 1),
  discharge_hgb = rnorm(n, 11, 2),
  discharge_creatinine = rlnorm(n, 0.2, 0.5),
  has_diabetes = rbinom(n, 1, 0.35),
  has_chf = rbinom(n, 1, 0.25)
)

readmit_prob <- plogis(-3 + 0.02 * (readmit_data$age - 68) +
                         0.15 * readmit_data$prior_admissions +
                         0.1 * readmit_data$num_comorbidities +
                         0.3 * readmit_data$has_chf -
                         0.1 * readmit_data$discharge_hgb)
readmit_data$readmitted <- factor(rbinom(n, 1, readmit_prob),
                                   labels = c("No", "Yes"))

# Your code:
# 1. Set up a tidymodels XGBoost workflow with 5-fold CV
# 2. Train a Keras neural network with 5-fold CV (use a loop over folds)
# 3. Compare AUC from both approaches
# 4. Which model performs better? Does this surprise you?
```

13.8.1.2. Python Starter Code

```
import numpy as np
from sklearn.model_selection import StratifiedKFold, cross_val_score
```

```python
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline
import xgboost as xgb
import tensorflow as tf
from tensorflow.keras import layers, models, callbacks
from sklearn.metrics import roc_auc_score

np.random.seed(42)
n = 1000

# Re-simulate readmission data (same as above)
# Your code:
# 1. Run 5-fold CV with XGBClassifier using cross_val_score
# 2. Run 5-fold CV with a Keras model (manual loop)
# 3. Compare mean AUC across folds
# 4. Which model performs better? Does this surprise you?
```

13.8.2. Exercise 2: Architecture Matching

For each of the following clinical tasks, identify (a) the most appropriate deep learning architecture, (b) whether deep learning is likely to outperform gradient-boosted trees, and (c) the most relevant reporting guideline. Write 2–3 sentences justifying each answer.

1. Predicting 30-day mortality from 15 structured EHR variables (age, labs, vitals, comorbidities).
2. Classifying skin lesions as benign or malignant from dermoscopy images.
3. Detecting atrial fibrillation from 12-lead ECG tracings.
4. Extracting medication names from unstructured discharge summaries.
5. Predicting length of stay from a combination of structured EHR data and a chest X-ray at admission.

13.8.3. Exercise 3: Critical Appraisal of a Deep Learning Study

Find a recent (2024 or later) paper that applies deep learning to a clinical task in your area of interest. Evaluate it against the CLAIM or TRIPOD+AI checklist:

1. Was the model externally validated? If so, how did performance compare to internal validation?
2. Were subgroup analyses reported (by age, sex, ethnicity)?
3. Was the model compared to a simpler baseline (e.g., logistic regression)?
4. Were the training data, code, and model weights made available?
5. Based on your assessment, how close is this model to clinical deployment? What would you want to see before trusting it with patient care?

13.9. Summary

Concept	Key Takeaway
When to use deep learning	Images, text, time series, and sequences — not tabular data
Neural network basics	Weighted sums + non-linear activations, stacked in layers
CNNs	Learn spatial features from images via convolutional filters
Transformers	Use self-attention to model relationships in sequences; dominant for NLP and increasingly for imaging
LLMs in medicine	Strong for extraction and generation; not reliable for autonomous decision-making
Transfer learning	Start from pretrained weights; fine-tune on your data

Concept	Key Takeaway
Foundation models	Domain-specific pretrained models (MedSAM, RETFound, UNI) dramatically reduce data requirements
External validation	81% of radiology DL models degrade on external data
Reporting	Use TRIPOD+AI, CLAIM, CONSORT-AI, or MINIMAR as appropriate

13.10. References and Further Reading

Goodfellow IJ, Bengio Y, Courville A. *Deep Learning.* MIT Press, 2016. Freely available at deeplearningbook.org. The definitive theoretical reference. Chapters 6 (deep feedforward networks), 9 (CNNs), and 10 (RNNs) are directly relevant.

Zhang A, Lipton ZC, Li M, Smola AJ. *Dive into Deep Learning.* Cambridge University Press, 2023. Freely available at d2l.ai. An interactive textbook with executable code in PyTorch, TensorFlow, and JAX. More practical and accessible than Goodfellow et al. Adopted at over 500 universities.

Howard J, Gugger S. *Deep Learning for Coders with fastai and PyTorch.* O'Reilly, 2020. Companion to the free fast.ai course. Top-down, coding-first approach ideal for researchers who want to apply deep learning without years of theory.

Grinsztajn L, Oyallon E, Varoquaux G. Why do tree-based models still outperform deep learning on typical tabular data? *NeurIPS* 2022. The benchmark study demonstrating that XGBoost consistently outperforms neural networks on tabular datasets. Essential reading before choosing deep learning for structured clinical data.

Ma J, He Y, Li F, et al. Segment anything in medical images. *Nature Communications* 2024;15:654. The MedSAM paper: a foundation model for universal medical image segmentation trained on 1.57 million image-mask pairs across 10 modalities.

Zhou Y, Chia MA, Wagner SK, et al. A foundation model for generalizable disease detection from retinal images. *Nature* 2023. RETFound: self-supervised pretraining on 1.6 million retinal images, with strong downstream performance from minimal fine-tuning.

Collins GS, et al. TRIPOD+AI statement. *BMJ* 2024;385:e078378. The current reporting standard for prediction models using regression or machine learning.

Bedi S, Cui H, Fuentes M, et al. MedHELM: Holistic Evaluation of Large Language Models for Medical Tasks. *Nature Medicine* 2025. The most comprehensive LLM evaluation for clinical tasks, testing 9 frontier models across 121 tasks.

Wiegrebe S, et al. Deep learning for survival analysis: a review. *Artificial Intelligence Review*, Springer, 2024. Comprehensive taxonomy covering DeepSurv, DeepHit, neural ODEs, and transformer-based approaches.

For hands-on practice, Stanford CS231n (computer vision) and CS224n (NLP) offer free lecture materials. MIT 6.S191 (Introduction to Deep Learning) at introtodeeplearning.com provides accessible video lectures updated annually.

14. Evaluating Models: Beyond Accuracy

14.1. Introduction

You have built a model. It produces predictions. But how do you know if those predictions are any good? The answer is not as simple as asking "how often is it right?" In clinical medicine, the consequences of different types of errors are rarely symmetric, the outcomes we care about are often rare, and the population we apply a model to may look nothing like the population we trained it on. This chapter introduces the core concepts and tools for evaluating the performance of classification models, with a focus on clinical prediction.

We will move beyond the seductive simplicity of "accuracy" and learn to ask better questions: Does the model separate sick from well? Are the predicted probabilities trustworthy? Would using this model actually help patients? And does it help all patients equally?

14.2. The Accuracy Trap

Consider the following scenario. You are developing a model to predict a rare but serious condition — say, pancreatic cancer — in a primary care population. The prevalence is approximately 0.1%. You build a classifier and proudly report 99.9% accuracy. Your colleague, less impressed, points out that a model which simply predicts “no cancer” for every single patient would also achieve 99.9% accuracy. It would miss every true case, which is the entire point of the exercise, but its accuracy would be nearly perfect.

This is the **accuracy paradox**: when outcomes are imbalanced, accuracy becomes meaningless because it is dominated by the majority class. In clinical medicine, outcomes are almost always imbalanced. Most people do not have the disease you are screening for. Most surgical patients do not die within 30 days. Most pregnancies do not result in pre-eclampsia. A metric that ignores the structure of errors is not useful for clinical decision-making.

> **!** The Golden Rule of Model Evaluation
>
> Never report accuracy alone. Always examine how the model performs separately for those with and without the outcome.

14.3. Sensitivity and Specificity

The most fundamental decomposition of model performance separates its behaviour in two groups: those who truly have the condition and those who do not.

Sensitivity (also called recall or the true positive rate) answers: among everyone who truly has the disease, what proportion did the model correctly identify?

$$\text{Sensitivity} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Specificity (the true negative rate) answers: among everyone who is truly disease-free, what proportion did the model correctly identify as negative?

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Positives}}$$

These two metrics are properties of the **test itself** and do not depend on the prevalence of the disease in the population. This makes them useful for describing a test's intrinsic discriminative ability.

14.3.1. Clinical Context: Screening vs Confirmation

The relative importance of sensitivity and specificity depends on the clinical purpose:

- **Screening tests** (e.g., mammography for breast cancer, the PHQ-2 for depression) prioritise **high sensitivity**. The goal is to catch as many true cases as possible. We accept more false positives because the next step is a confirmatory test, not treatment. Missing a case (false negative) is the costly error.

- **Confirmatory tests** (e.g., biopsy for cancer, Western blot for HIV) prioritise **high specificity**. The goal is to be sure that a positive result is real, because a positive typically triggers treatment, which may carry risks. A false positive leading to unnecessary surgery or chemotherapy is the costly error.

Memory Aid

SnN**out**: a test with high **Sen**sitivity, when **N**egative, rules **out** the disease. **Sp**P**in**: a test with high **Sp**ecificity, when **P**ositive, rules **in** the disease.

14.4. Predictive Values: When Prevalence Matters

While sensitivity and specificity describe the test, clinicians and patients need to answer a different question: given my test result, what is the probability I actually have the disease?

Positive Predictive Value (PPV): among everyone who tested positive, what proportion truly has the disease?

$$\text{PPV} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Negative Predictive Value (NPV): among everyone who tested negative, what proportion is truly disease-free?

$$\text{NPV} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Negatives}}$$

Critically, PPV and NPV depend on the **prevalence** of the disease in the population being tested. A test with 95% sensitivity and 95% specificity will have very different PPVs depending on context:

Table 14.1.: PPV and NPV for a test with 95% sensitivity and 95% specificity at different prevalence levels.

Prevalence	PPV	NPV
50%	95.0%	95.0%
10%	67.9%	99.4%
1%	16.1%	99.9%
0.1%	1.9%	100%

At 0.1% prevalence, even with an excellent test, fewer than 2% of positive results are true positives. This is why mass screening for rare diseases

generates so many false alarms, and why testing should be targeted to higher-risk populations whenever possible.

14.4.1. Exercise: Bayes' Theorem in Practice

14.4.1.1. R

```
# Calculate PPV using Bayes' theorem
# PPV = (Sensitivity * Prevalence) /
#        (Sensitivity * Prevalence + (1 - Specificity) * (1 - Prevalence))

calculate_ppv <- function(sensitivity, specificity, prevalence) {
  numerator <- sensitivity * prevalence
  denominator <- numerator + (1 - specificity) * (1 - prevalence)
  return(numerator / denominator)
}

# Example: HIV rapid test (sensitivity = 99.7%, specificity = 99.5%)
# In a general population (prevalence ~ 0.4%)
ppv_general <- calculate_ppv(0.997, 0.995, 0.004)
cat("PPV in general population:", round(ppv_general * 100, 1), "%\n")

# In an STI clinic population (prevalence ~ 5%)
ppv_clinic <- calculate_ppv(0.997, 0.995, 0.05)
cat("PPV in STI clinic:", round(ppv_clinic * 100, 1), "%\n")

# In a population with known exposure (prevalence ~ 30%)
ppv_exposed <- calculate_ppv(0.997, 0.995, 0.30)
cat("PPV with known exposure:", round(ppv_exposed * 100, 1), "%\n")

# Plot PPV across a range of prevalences
prevalences <- seq(0.001, 0.5, by = 0.001)
```

```r
ppvs <- sapply(prevalences, function(p) calculate_ppv(0.997, 0.995, p))

plot(prevalences * 100, ppvs * 100,
     type = "l", lwd = 2, col = "steelblue",
     xlab = "Prevalence (%)", ylab = "Positive Predictive Value (%)",
     main = "PPV depends heavily on prevalence",
     las = 1)
abline(h = 50, lty = 2, col = "grey50")
```

14.4.1.2. Python

```python
import numpy as np
import matplotlib.pyplot as plt

def calculate_ppv(sensitivity, specificity, prevalence):
    """Calculate PPV using Bayes' theorem."""
    numerator = sensitivity * prevalence
    denominator = numerator + (1 - specificity) * (1 - prevalence)
    return numerator / denominator

# Example: HIV rapid test (sensitivity = 99.7%, specificity = 99.5%)
# In a general population (prevalence ~ 0.4%)
ppv_general = calculate_ppv(0.997, 0.995, 0.004)
print(f"PPV in general population: {ppv_general*100:.1f}%")

# In an STI clinic population (prevalence ~ 5%)
ppv_clinic = calculate_ppv(0.997, 0.995, 0.05)
print(f"PPV in STI clinic: {ppv_clinic*100:.1f}%")

# In a population with known exposure (prevalence ~ 30%)
ppv_exposed = calculate_ppv(0.997, 0.995, 0.30)
print(f"PPV with known exposure: {ppv_exposed*100:.1f}%")
```

```
# Plot PPV across a range of prevalences
prevalences = np.linspace(0.001, 0.5, 500)
ppvs = [calculate_ppv(0.997, 0.995, p) for p in prevalences]

plt.figure(figsize=(8, 5))
plt.plot(prevalences * 100, np.array(ppvs) * 100, lw=2, color="steelblue")
plt.axhline(y=50, linestyle="--", color="grey", alpha=0.7)
plt.xlabel("Prevalence (%)")
plt.ylabel("Positive Predictive Value (%)")
plt.title("PPV depends heavily on prevalence")
plt.tight_layout()
plt.show()
```

What the code is showing. This puts the prevalence lesson on its feet with a real test: an HIV rapid test with near-perfect sensitivity (99.7%) and specificity (99.5%). The `calculate_ppv` function is just Bayes' theorem written out — it asks "of everyone who tests positive, what fraction are truly positive?" The three worked examples show the same excellent test producing wildly different answers depending on *who* you test: a positive result means roughly a 44% chance of true infection in the general population (prevalence 0.4%), about 91% in an STI clinic (5%), and around 99% after known exposure (30%). The plot then sweeps prevalence from near zero to 50% and traces the PPV: a steep climb at low prevalence that flattens out, with the dashed line marking where a positive result becomes more likely true than false. The clinical message a health researcher should take away is that the same test can be trustworthy or near-useless depending entirely on the pre-test probability of the patient in front of you — which is why we target testing to higher-risk groups.

14.5. The Confusion Matrix

A **confusion matrix** is the fundamental bookkeeping device for classification evaluation. For a binary outcome, it is a 2x2 table that cross-classifies predictions against truth:

	Condition Positive	Condition Negative
Predicted Positive	True Positive (TP)	False Positive (FP)
Predicted Negative	False Negative (FN)	True Negative (TN)

Every metric discussed in this chapter can be derived from this table. The crucial insight is that a single model can produce **different** confusion matrices depending on the **threshold** you choose for classifying a predicted probability as "positive" or "negative."

14.5.1. Building and Inspecting Confusion Matrices

14.5.1.1. R

```
library(caret)

# Simulate predicted probabilities and true outcomes
set.seed(42)
n <- 1000
true_outcome <- rbinom(n, 1, 0.15)  # 15% prevalence
# Simulate a moderately good model
pred_prob <- plogis(rnorm(n, mean = -1 + 2 * true_outcome, sd = 1.2))

# Confusion matrix at threshold = 0.5
pred_class_50 <- ifelse(pred_prob >= 0.5, 1, 0)
cm_50 <- confusionMatrix(factor(pred_class_50), factor(true_outcome), positi
```

```
print(cm_50)

# Confusion matrix at threshold = 0.2 (more sensitive)
pred_class_20 <- ifelse(pred_prob >= 0.2, 1, 0)
cm_20 <- confusionMatrix(factor(pred_class_20), factor(true_outcome), positive = "1")
print(cm_20)

cat("\nAt threshold 0.5:\n")
cat("  Sensitivity:", round(cm_50$byClass["Sensitivity"], 3), "\n")
cat("  Specificity:", round(cm_50$byClass["Specificity"], 3), "\n")

cat("\nAt threshold 0.2:\n")
cat("  Sensitivity:", round(cm_20$byClass["Sensitivity"], 3), "\n")
cat("  Specificity:", round(cm_20$byClass["Specificity"], 3), "\n")
```

14.5.1.2. Python

```
import numpy as np
from sklearn.metrics import confusion_matrix, classification_report
from scipy.special import expit

np.random.seed(42)
n = 1000
true_outcome = np.random.binomial(1, 0.15, n)  # 15% prevalence
# Simulate a moderately good model
pred_prob = expit(np.random.normal(-1 + 2 * true_outcome, 1.2))

# Confusion matrix at threshold = 0.5
pred_class_50 = (pred_prob >= 0.5).astype(int)
print("=== Threshold = 0.5 ===")
print(confusion_matrix(true_outcome, pred_class_50))
print(classification_report(true_outcome, pred_class_50, target_names=["Negative", "Positive
```

```
# Confusion matrix at threshold = 0.2 (more sensitive)
pred_class_20 = (pred_prob >= 0.2).astype(int)
print("=== Threshold = 0.2 ===")
print(confusion_matrix(true_outcome, pred_class_20))
print(classification_report(true_outcome, pred_class_20, target_names=["Negat
```

What the code is showing. This demonstrates the single most important fact about a confusion matrix: it changes when you move the threshold, even though the model itself never changes. We simulate 1,000 patients at 15% prevalence and a moderately good model, then build two confusion matrices from the *same* predicted probabilities — one classifying anyone above 0.5 as positive, the other lowering the bar to 0.2. The printed tables and the sensitivity/specificity lines tell the story: at the strict 0.5 threshold the model is cautious, so specificity is high but it misses many true cases (low sensitivity); dropping to 0.2 catches far more true cases (sensitivity rises) at the cost of more false alarms (specificity falls). Reading these outputs side by side is the intuition behind every ROC curve that follows: there is no single "right" confusion matrix, only the one that matches the clinical cost of missing a case versus raising a false alarm.

14.6. ROC Curves

The **Receiver Operating Characteristic (ROC) curve** is perhaps the most widely used graphical tool for evaluating classification models. It plots sensitivity (true positive rate) on the y-axis against 1 minus specificity (false positive rate) on the x-axis, tracing out the trade-off as the classification threshold varies from 1 to 0.

14.6.1. How to Read an ROC Curve

- The **diagonal line** from (0,0) to (1,1) represents a model with no discriminative ability — equivalent to flipping a coin.
- A curve that bows toward the **upper-left corner** indicates good discrimination. The model achieves high sensitivity without sacrificing too much specificity.
- The **Area Under the ROC Curve (AUC or C-statistic)** summarises the curve in a single number. It equals the probability that, given one randomly chosen person with the disease and one without, the model assigns a higher predicted risk to the diseased person.

Table 14.2.: Rough guide to AUC interpretation. These benchmarks are context-dependent.

AUC Range	Interpretation
0.90–1.00	Excellent discrimination
0.80–0.90	Good discrimination
0.70–0.80	Acceptable
0.60–0.70	Poor
0.50–0.60	Fail (near chance)

AUC Is Not Enough

A model can have a high AUC but still produce poorly calibrated probabilities. Two models with the same AUC can have very different clinical utility. AUC tells you about ranking (discrimination), not about the accuracy of the predicted probabilities themselves. Always supplement AUC with calibration assessment (see Chapter 18).

14.6.2. Plotting ROC Curves

14.6.2.1. R

```
library(pROC)

# Using the simulated data from above
set.seed(42)
n <- 1000
true_outcome <- rbinom(n, 1, 0.15)
pred_prob <- plogis(rnorm(n, mean = -1 + 2 * true_outcome, sd = 1.2))

roc_obj <- roc(true_outcome, pred_prob)

# Plot the ROC curve
plot(roc_obj,
     main = paste("ROC Curve (AUC =", round(auc(roc_obj), 3), ")"),
     col = "steelblue", lwd = 2,
     print.auc = TRUE, print.auc.y = 0.4,
     legacy.axes = TRUE)  # Use 1-Specificity on x-axis

# Add confidence interval for AUC
ci_auc <- ci.auc(roc_obj)
cat("95% CI for AUC:", round(ci_auc[1], 3), "-", round(ci_auc[3], 3), "\n")

# Find the optimal threshold (Youden's J)
coords_best <- coords(roc_obj, "best", ret = c("threshold", "sensitivity", "s
cat("Optimal threshold (Youden):", round(coords_best$threshold, 3), "\n")
cat("Sensitivity at optimal:", round(coords_best$sensitivity, 3), "\n")
cat("Specificity at optimal:", round(coords_best$specificity, 3), "\n")
```

14.6.2.2. Python

```python
from sklearn.metrics import roc_curve, roc_auc_score
import matplotlib.pyplot as plt
import numpy as np
from scipy.special import expit

np.random.seed(42)
n = 1000
true_outcome = np.random.binomial(1, 0.15, n)
pred_prob = expit(np.random.normal(-1 + 2 * true_outcome, 1.2))

fpr, tpr, thresholds = roc_curve(true_outcome, pred_prob)
auc_score = roc_auc_score(true_outcome, pred_prob)

plt.figure(figsize=(7, 7))
plt.plot(fpr, tpr, color="steelblue", lw=2,
         label=f"Model (AUC = {auc_score:.3f})")
plt.plot([0, 1], [0, 1], color="grey", linestyle="--", label="Chance")
plt.xlabel("False Positive Rate (1 - Specificity)")
plt.ylabel("True Positive Rate (Sensitivity)")
plt.title("ROC Curve")
plt.legend(loc="lower right")
plt.tight_layout()
plt.show()

# Youden's J statistic to find optimal threshold
j_scores = tpr - fpr
best_idx = np.argmax(j_scores)
print(f"Optimal threshold (Youden): {thresholds[best_idx]:.3f}")
print(f"Sensitivity: {tpr[best_idx]:.3f}")
print(f"Specificity: {1 - fpr[best_idx]:.3f}")
```

What the code is showing. This draws the ROC curve and pulls two numbers off it. The plot traces sensitivity against 1 − specificity as the threshold slides from high to low; a curve hugging the top-left corner is good, the diagonal is a coin flip. The reported AUC (with its 95% confidence interval) summarises the whole curve in one figure — recall it is the probability the model scores a randomly chosen diseased patient higher than a non-diseased one, so the confidence interval matters because a wide one means the estimate could be much better or much worse than the point value suggests. The code also finds the threshold that maximises Youden's J (sensitivity + specificity − 1), reporting the sensitivity and specificity there. Treat that "optimal" threshold as a *statistical* suggestion only: as the later section warns, it assumes a false negative and a false positive cost the same, which is rarely true in medicine.

14.7. Precision-Recall Curves

When outcomes are highly imbalanced — as they often are in clinical prediction — ROC curves can paint an overly optimistic picture. Because specificity is calculated over the large number of true negatives, even a substantial number of false positives barely moves the false positive rate. The **precision-recall (PR) curve** addresses this by focusing entirely on the positive class.

- **Precision** (= PPV): of all predicted positives, how many are truly positive?
- **Recall** (= sensitivity): of all true positives, how many were detected?

The PR curve plots precision on the y-axis against recall on the x-axis. A good model maintains high precision even as recall increases. The baseline for a PR curve is a horizontal line at the prevalence level (not the diagonal, as for ROC).

The **Area Under the Precision-Recall Curve (AUPRC)** is a useful summary, particularly for rare outcomes. While a model may boast an AUC-ROC of 0.95, its AUPRC might reveal that achieving high recall requires accepting very low precision.

14.7.1. Exercise: Comparing ROC and PR Curves

14.7.1.1. R

```
library(PRROC)

# Simulate an imbalanced dataset (2% prevalence)
set.seed(123)
n <- 5000
true_outcome <- rbinom(n, 1, 0.02)
pred_prob <- plogis(rnorm(n, mean = -2 + 3 * true_outcome, sd = 1.5))

# PR curve
pr <- pr.curve(scores.class0 = pred_prob[true_outcome == 1],
               scores.class1 = pred_prob[true_outcome == 0],
               curve = TRUE)
plot(pr, main = paste("Precision-Recall Curve\nAUPRC =", round(pr$auc.integral, 3)))

# For comparison, the ROC curve
roc <- roc.curve(scores.class0 = pred_prob[true_outcome == 1],
                 scores.class1 = pred_prob[true_outcome == 0],
                 curve = TRUE)
plot(roc, main = paste("ROC Curve\nAUROC =", round(roc$auc, 3)))

cat("Notice how the AUROC looks excellent but the AUPRC reveals\n")
cat("the difficulty of achieving high precision with rare outcomes.\n")
```

14.7.1.2. Python

```
from sklearn.metrics import (precision_recall_curve, average_precision_score
                              roc_curve, roc_auc_score)
import matplotlib.pyplot as plt
import numpy as np
from scipy.special import expit

np.random.seed(123)
n = 5000
true_outcome = np.random.binomial(1, 0.02, n)  # 2% prevalence
pred_prob = expit(np.random.normal(-2 + 3 * true_outcome, 1.5))

fig, axes = plt.subplots(1, 2, figsize=(14, 6))

# ROC curve
fpr, tpr, _ = roc_curve(true_outcome, pred_prob)
auroc = roc_auc_score(true_outcome, pred_prob)
axes[0].plot(fpr, tpr, color="steelblue", lw=2)
axes[0].plot([0, 1], [0, 1], "--", color="grey")
axes[0].set_title(f"ROC Curve (AUROC = {auroc:.3f})")
axes[0].set_xlabel("False Positive Rate")
axes[0].set_ylabel("True Positive Rate")

# PR curve
precision, recall, _ = precision_recall_curve(true_outcome, pred_prob)
auprc = average_precision_score(true_outcome, pred_prob)
prevalence = true_outcome.mean()
axes[1].plot(recall, precision, color="darkorange", lw=2)
axes[1].axhline(y=prevalence, linestyle="--", color="grey", label=f"Baseline
axes[1].set_title(f"Precision-Recall Curve (AUPRC = {auprc:.3f})")
axes[1].set_xlabel("Recall (Sensitivity)")
axes[1].set_ylabel("Precision (PPV)")
```

```
axes[1].legend()

plt.tight_layout()
plt.show()

print("The AUROC looks impressive, but AUPRC reveals the real challenge.")
```

What the code is showing. This is a deliberate side-by-side demonstration of how ROC curves can flatter a model on rare outcomes. We simulate a realistically imbalanced dataset — only 2% of 5,000 patients have the event — and plot the ROC and precision-recall curves for the *same* predictions. The ROC curve and its AUROC look excellent, because specificity is computed against the huge pool of true negatives, so even hundreds of false positives barely nudge the false-positive rate. The PR curve tells the uncomfortable truth: its baseline is the prevalence (the dashed line at 0.02), and the curve shows that to catch most true cases (high recall) you must wade through many false positives (precision collapses). The much lower AUPRC is the honest summary for a rare-disease screening problem. The lesson for a health researcher: when your outcome is rare, report the precision-recall curve, not just the ROC.

14.8. Choosing the Threshold: A Clinical Decision

A prediction model typically outputs a probability. To make a binary decision (treat or don't treat, refer or don't refer), you must choose a **threshold**. This is fundamentally a **clinical** decision, not a statistical one.

Consider a model predicting 30-day mortality after surgery:

- If the intervention for high-risk patients is simply closer monitoring (low cost, minimal harm), you might choose a **low threshold** (e.g.,

5%), accepting many false positives to catch as many at-risk patients as possible.
- If the intervention is a risky re-operation, you might choose a **high threshold** (e.g., 30%), requiring strong evidence before acting.

The optimal threshold depends on the **relative costs** of false positives and false negatives, which are determined by the clinical context, not by the data.

> **Youden's Index Is Not Always Optimal**
>
> Youden's J statistic (sensitivity + specificity - 1) finds the threshold that maximises the sum of sensitivity and specificity. This implicitly assumes equal costs for false positives and false negatives — an assumption that is rarely true in medicine. Use Youden's index as a starting point, but always adjust based on clinical reasoning.

14.8.1. The Threshold-Performance Trade-off

14.8.1.1. R

```r
library(pROC)

set.seed(42)
n <- 1000
true_outcome <- rbinom(n, 1, 0.15)
pred_prob <- plogis(rnorm(n, mean = -1 + 2 * true_outcome, sd = 1.2))

roc_obj <- roc(true_outcome, pred_prob)

# Extract sensitivity and specificity at various thresholds
thresholds <- seq(0.05, 0.95, by = 0.05)
```

```r
results <- data.frame(
  threshold = thresholds,
  sensitivity = sapply(thresholds, function(t) {
    coords(roc_obj, t, input = "threshold", ret = "sensitivity")
  }),
  specificity = sapply(thresholds, function(t) {
    coords(roc_obj, t, input = "threshold", ret = "specificity")
  })
)

# Plot the trade-off
plot(results$threshold, results$sensitivity, type = "l", lwd = 2, col = "red",
     xlab = "Classification Threshold", ylab = "Metric Value",
     main = "Sensitivity-Specificity Trade-off Across Thresholds",
     ylim = c(0, 1), las = 1)
lines(results$threshold, results$specificity, lwd = 2, col = "blue")
legend("right", legend = c("Sensitivity", "Specificity"),
       col = c("red", "blue"), lwd = 2)
```

14.8.1.2. Python

```python
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import roc_curve
from scipy.special import expit

np.random.seed(42)
n = 1000
true_outcome = np.random.binomial(1, 0.15, n)
pred_prob = expit(np.random.normal(-1 + 2 * true_outcome, 1.2))

fpr, tpr, thresholds = roc_curve(true_outcome, pred_prob)
```

```
specificity = 1 - fpr

plt.figure(figsize=(8, 5))
plt.plot(thresholds, tpr[:-1] if len(tpr) > len(thresholds) else tpr,
         color="red", lw=2, label="Sensitivity")
plt.plot(thresholds, specificity[:-1] if len(specificity) > len(thresholds) e
         color="blue", lw=2, label="Specificity")
plt.xlabel("Classification Threshold")
plt.ylabel("Metric Value")
plt.title("Sensitivity-Specificity Trade-off Across Thresholds")
plt.legend()
plt.xlim(0, 1)
plt.ylim(0, 1)
plt.tight_layout()
plt.show()
```

What the code is showing. This makes the threshold trade-off visible as a single picture. We step the classification threshold across its range and, at each value, record the model's sensitivity and specificity, then plot both as lines on the same axes. The two curves move in opposite directions: as you raise the threshold (demand stronger evidence before calling a patient positive), sensitivity falls while specificity rises; lower it, and the reverse happens. Where the lines cross is the threshold that balances the two, but — as the surrounding text stresses — the *right* operating point is wherever the clinical costs of a missed case and a false alarm balance, which may be far from the crossing point. Reading this plot lets you pick a threshold deliberately rather than defaulting to 0.5.

14.9. Fairness: Does the Model Work for Everyone?

A model that performs well on average may perform very differently across subgroups defined by age, sex, race, ethnicity, or socioeconomic status.

This is not a theoretical concern — it has been documented repeatedly in clinical prediction models.

A well-known example is the pulse oximeter, which systematically overestimates blood oxygen saturation in patients with darker skin pigmentation. Prediction models trained on predominantly White populations may have lower sensitivity for detecting disease in Black or Hispanic patients, potentially widening existing health disparities.

14.9.1. Key Fairness Concepts

- **Equal performance**: Does the model have similar sensitivity and specificity across demographic groups?
- **Calibration fairness**: Are predicted probabilities equally well-calibrated in each group? A model might predict 20% risk for two groups, but if the actual event rate is 20% in one group and 30% in another, the model is miscalibrated for the latter.
- **Demographic parity**: Are positive predictions made at similar rates across groups? (Note: this may conflict with calibration if true prevalence differs.)

14.9.2. Exercise: Evaluating Fairness

14.9.2.1. R

```r
library(pROC)

# Simulate data with two demographic groups
set.seed(42)
n <- 2000

group <- sample(c("A", "B"), n, replace = TRUE)
```

```
# Group B has slightly different disease prevalence and predictor distributio
true_outcome <- ifelse(group == "A",
                       rbinom(n, 1, 0.10),
                       rbinom(n, 1, 0.15))
# Model performs slightly worse for Group B
pred_prob <- ifelse(group == "A",
                    plogis(rnorm(n, -1.5 + 2.5 * true_outcome, 1.0)),
                    plogis(rnorm(n, -1.5 + 1.8 * true_outcome, 1.2)))

# Calculate AUC by group
for (g in c("A", "B")) {
  idx <- group == g
  roc_g <- roc(true_outcome[idx], pred_prob[idx])
  cat(sprintf("Group %s: AUC = %.3f, Prevalence = %.1f%%\n",
              g, auc(roc_g), mean(true_outcome[idx]) * 100))
}

# Compare sensitivity at a fixed threshold
threshold <- 0.3
for (g in c("A", "B")) {
  idx <- group == g
  pred_class <- ifelse(pred_prob[idx] >= threshold, 1, 0)
  sens <- sum(pred_class == 1 & true_outcome[idx] == 1) / sum(true_outcome[id
  spec <- sum(pred_class == 0 & true_outcome[idx] == 0) / sum(true_outcome[id
  cat(sprintf("Group %s at threshold %.1f: Sensitivity = %.3f, Specificity =
              g, threshold, sens, spec))
}
```

14.9.2.2. Python

```
import numpy as np
from sklearn.metrics import roc_auc_score
```

```
from scipy.special import expit

np.random.seed(42)
n = 2000

group = np.random.choice(["A", "B"], n)
true_outcome = np.where(group == "A",
                        np.random.binomial(1, 0.10, n),
                        np.random.binomial(1, 0.15, n))
pred_prob = np.where(group == "A",
                     expit(np.random.normal(-1.5 + 2.5 * true_outcome, 1.0)),
                     expit(np.random.normal(-1.5 + 1.8 * true_outcome, 1.2)))

# AUC by group
for g in ["A", "B"]:
    mask = group == g
    auc = roc_auc_score(true_outcome[mask], pred_prob[mask])
    prev = true_outcome[mask].mean()
    print(f"Group {g}: AUC = {auc:.3f}, Prevalence = {prev*100:.1f}%")

# Sensitivity and specificity at a fixed threshold
threshold = 0.3
for g in ["A", "B"]:
    mask = group == g
    pred_class = (pred_prob[mask] >= threshold).astype(int)
    tp = ((pred_class == 1) & (true_outcome[mask] == 1)).sum()
    fn = ((pred_class == 0) & (true_outcome[mask] == 1)).sum()
    tn = ((pred_class == 0) & (true_outcome[mask] == 0)).sum()
    fp = ((pred_class == 1) & (true_outcome[mask] == 0)).sum()
    sens = tp / (tp + fn) if (tp + fn) > 0 else 0
    spec = tn / (tn + fp) if (tn + fp) > 0 else 0
    print(f"Group {g} at threshold {threshold}: Sensitivity = {sens:.3f}, Specificity = {spe
```

What the code is showing. This is a minimal but realistic fairness audit. We simulate two groups, A and B, where group B has both a different disease prevalence and a model that genuinely discriminates a little worse (we built that gap in deliberately). The first loop reports the AUC and prevalence *separately* for each group — the single overall AUC would have hidden the disparity. The second loop fixes one threshold and reports sensitivity and specificity per group, exposing the more clinically alarming problem: at the same cut-off, the model may catch a smaller fraction of true cases in group B, meaning patients in that group are more likely to be missed. The takeaway for a health researcher is procedural: always disaggregate your performance metrics by the demographic groups you care about, because a model that looks fine on average can quietly underperform for a subgroup and widen existing health inequities.

14.10. Putting It All Together: A Complete Evaluation

When evaluating a clinical prediction model, no single metric tells the whole story. The following checklist summarises what you should report:

1. **Sample description**: How many patients? How many events? What is the prevalence?
2. **Discrimination**: AUC-ROC with confidence interval. Consider AUPRC for rare outcomes.
3. **Calibration**: Are predicted probabilities accurate? (Covered in detail in Chapter 18.)
4. **Confusion matrix**: At a clinically relevant threshold, not just the default 0.5.
5. **Sensitivity, specificity, PPV, NPV**: At the chosen threshold.
6. **Subgroup performance**: Does the model perform equitably across key demographic groups?

7. **Clinical utility**: Would using this model lead to better decisions than the alternatives? (Covered in Chapter 18.)

> **!** Remember
>
> A model is not good or bad in isolation. It is good or bad **for a specific purpose, in a specific population, at a specific threshold**. Always evaluate with the intended clinical use in mind.

14.11. Summary

- **Accuracy** is misleading for imbalanced outcomes; decompose errors into sensitivity and specificity.
- **Sensitivity** matters most for screening; **specificity** matters most for confirmation.
- **PPV and NPV** depend on prevalence, so a test's clinical usefulness changes with the population.
- **ROC curves** display the sensitivity-specificity trade-off across all thresholds; **AUC** summarises discriminative ability.
- **Precision-recall curves** are more informative than ROC for rare outcomes.
- The **classification threshold** should be chosen based on clinical consequences, not statistical optimality.
- **Fairness** requires checking that model performance is consistent across demographic groups.

14.12. References and Further Reading

The foundational concepts of sensitivity, specificity, and predictive values are covered in every epidemiology textbook, but their application to prediction models requires additional nuance. Van Calster and colleagues provide an excellent overview of performance assessment in their 2025 paper in The Lancet Digital Health, which organises evaluation into five domains: discrimination, calibration, overall performance, classification, and clinical utility. This framework is essential reading for anyone developing or evaluating clinical prediction models. The comprehensive textbook by Smits, van Kuijk, and Wynants (2026) devotes several chapters to model evaluation and provides practical guidance on choosing appropriate metrics for different clinical contexts. For the statistical foundations of ROC analysis, the classic text by Pepe (2003), "The Statistical Evaluation of Medical Tests for Classification and Prediction," remains the definitive reference. Saito and Rehmsmeier (2015) explain why precision-recall curves are preferable to ROC curves for imbalanced datasets in their paper "The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets" in PLOS ONE. On the topic of fairness in clinical prediction, Obermeyer and colleagues published a landmark 2019 paper in Science demonstrating racial bias in a widely used healthcare algorithm, which is required reading for anyone working in this space. Vickers and Elkin (2006) introduced decision curve analysis and net benefit as tools for evaluating clinical utility, which we discuss further in the next chapter.

Part V.

Dimensionality Reduction and Unsupervised Learning

15. PCA, t-SNE, and UMAP: Making Sense of High-Dimensional Data

Imagine you have measured 500 gene expression values, or two dozen lab tests, on each of your patients. You cannot plot 500 axes, and your eyes can only make sense of two or three. Dimensionality reduction is the set of techniques that squeeze those many measurements down to a handful of summary axes you *can* plot — ideally without throwing away the patterns that matter. The reward is a picture: a single scatter plot where each dot is a patient, and where patients with similar profiles sit close together, often revealing subgroups, gradients, or outliers that were invisible in the raw spreadsheet. This chapter shows three ways to make that picture and, just as importantly, how to read it without being misled.

> **Note**
>
> These chapters are set to `eval: false`, so the rendered page shows the code but not its live output. As you read, picture the figure each block *would* produce — the surrounding text describes what to expect so you can run the code yourself and recognise a sensible result.

15.1. Introduction

The previous chapters covered supervised learning — methods that predict an outcome from labelled data. In this part, we turn to **unsupervised** methods that explore and summarize data without a specific outcome in mind.

A single patient encounter can generate hundreds of variables: vital signs, lab results, imaging features, medication lists, genomic markers, clinical notes. When the number of features (p) is large relative to the number of observations (n), standard statistical methods break down. Regression models become unstable, distances between data points become uninformative, and visualisation is impossible.

Dimensionality reduction addresses this by projecting high-dimensional data into a lower-dimensional space while preserving as much meaningful structure as possible. This chapter covers three complementary methods: **PCA** (principal component analysis), **t-SNE** (t-distributed stochastic neighbour embedding), and **UMAP** (uniform manifold approximation and projection). Each serves a different purpose, and understanding when to use which is a core skill in modern data analysis.

15.2. The Curse of Dimensionality

The "curse of dimensionality" refers to a collection of phenomena that arise when data live in high-dimensional spaces. The most important for clinical data analysis:

Distances become meaningless. In high dimensions, the distance between the nearest and farthest data points converges. If all points are approximately equidistant, distance-based methods (k-nearest neighbours, clustering) lose their ability to discriminate.

Sparsity. The volume of a high-dimensional space grows exponentially with the number of dimensions. Data points become increasingly sparse — you need exponentially more observations to maintain the same data density. A study with 200 patients and 500 gene expression features has data that are extraordinarily sparse.

Overfitting. With many features, models can find spurious patterns in the noise. A logistic regression with 200 predictors and 100 observations is almost guaranteed to overfit, even with regularisation.

Visualisation is impossible. Humans perceive at most three spatial dimensions. To explore the structure of a 50-dimensional dataset, we must reduce it to 2 or 3 dimensions.

15.3. PCA: Principal Component Analysis

PCA is the oldest and most widely used dimensionality reduction technique. It finds the directions (linear combinations of the original variables) along which the data vary the most.

15.3.1. The Core Idea

Given a data matrix $\mathbf{X}$ with n observations and p variables:

1. **Centre** the data: subtract the mean of each variable.
2. **Compute the covariance matrix** $\mathbf{C} = \frac{1}{n-1}\mathbf{X}^T\mathbf{X}$.
3. **Find the eigenvectors** of $\mathbf{C}$. These are the **principal components** — the directions of maximum variance.
4. **The eigenvalues** tell you how much variance each component captures.

The first principal component (PC1) captures the most variance, PC2 captures the second most (orthogonal to PC1), and so on. By keeping only the first k components, you reduce the data from p dimensions to k dimensions.

In plain terms, a **principal component** is just a recipe: a new variable made by mixing the original measurements in fixed proportions (the weights in that recipe are called the **loadings**). PC1 is the single mixture that spreads patients out as much as possible — the direction along which they differ most. If glucose, HbA1c, and triglycerides all rise and fall together, PCA notices and bundles them into one axis, so a patient's position on PC1 might summarise their whole "metabolic" profile in a single number. "Variance explained" is then simply the share of the data's total spread that each component accounts for: PC1 might capture 40% of all the variation, meaning that one axis alone tells you a large chunk of what distinguishes your patients.

15.3.2. Key Properties

- PCA is a **linear** method: each component is a weighted sum of the original variables.
- It is **deterministic**: running PCA twice on the same data gives the same result (up to sign flips).
- It preserves **global structure** (distances and variance) well, but may miss non-linear patterns.

15.3.3. Choosing the Number of Components

Scree plot. Plot the eigenvalues (or proportion of variance explained) against the component number. Look for an "elbow" — a point where the curve flattens, suggesting that additional components contribute little.

Cumulative variance threshold. Keep enough components to explain a target percentage of total variance (commonly 80–90%).

Kaiser criterion. Keep components with eigenvalues > 1 (only meaningful when data are standardised).

15.3.4. Clinical Example: Metabolic Panel PCA

A common application: reducing a panel of correlated lab values to a few interpretable dimensions. Consider a dataset with glucose, HbA1c, triglycerides, LDL, HDL, creatinine, ALT, and albumin.

15.3.5. R

```
set.seed(42)
n <- 300

# Simulate correlated metabolic data
# Create correlation structure: glucose/HbA1c/triglycerides cluster;
# creatinine/albumin cluster
library(MASS)
Sigma <- matrix(c(
  1.0, 0.7, 0.5, 0.3,-0.2, 0.1, 0.2,-0.1,
  0.7, 1.0, 0.4, 0.2,-0.2, 0.1, 0.1,-0.1,
  0.5, 0.4, 1.0, 0.4,-0.3, 0.1, 0.3,-0.1,
  0.3, 0.2, 0.4, 1.0,-0.5, 0.1, 0.2, 0.0,
 -0.2,-0.2,-0.3,-0.5, 1.0,-0.1,-0.1, 0.2,
  0.1, 0.1, 0.1, 0.1,-0.1, 1.0, 0.1,-0.3,
  0.2, 0.1, 0.3, 0.2,-0.1, 0.1, 1.0,-0.2,
 -0.1,-0.1,-0.1, 0.0, 0.2,-0.3,-0.2, 1.0
), nrow = 8)
```

```
z <- mvrnorm(n, mu = rep(0, 8), Sigma = Sigma)
metabolic <- tibble(
  glucose      = round(z[,1] * 30 + 100),
  hba1c        = round(z[,2] * 1.0 + 5.8, 1),
  triglycerides = round(z[,3] * 50 + 150),
  ldl          = round(z[,4] * 30 + 120),
  hdl          = round(z[,5] * 12 + 55),
  creatinine   = round(z[,6] * 0.3 + 1.0, 2),
  alt          = round(z[,7] * 15 + 30),
  albumin      = round(z[,8] * 0.4 + 4.0, 1)
)

# PCA on scaled data
pca_result <- prcomp(metabolic, scale. = TRUE)

# Scree plot and biplot side by side
par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))

# Scree plot
var_explained <- pca_result$sdev^2 / sum(pca_result$sdev^2)
barplot(var_explained, names.arg = paste0("PC", 1:8),
        col = "steelblue", main = "Scree Plot",
        ylab = "Proportion of Variance", xlab = "Component",
        ylim = c(0, 0.4))
abline(h = 1/8, lty = 2, col = "firebrick")  # Kaiser criterion line
text(9, 1/8 + 0.015, "Kaiser threshold", col = "firebrick", cex = 0.8)

# Biplot
biplot(pca_result, scale = 0, cex = 0.6,
       col = c("grey70", "firebrick"),
       main = "PCA Biplot: Metabolic Panel")
```

15.3.6. Python

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

np.random.seed(42)
n = 300

Sigma = np.array([
    [1.0, 0.7, 0.5, 0.3,-0.2, 0.1, 0.2,-0.1],
    [0.7, 1.0, 0.4, 0.2,-0.2, 0.1, 0.1,-0.1],
    [0.5, 0.4, 1.0, 0.4,-0.3, 0.1, 0.3,-0.1],
    [0.3, 0.2, 0.4, 1.0,-0.5, 0.1, 0.2, 0.0],
    [-0.2,-0.2,-0.3,-0.5, 1.0,-0.1,-0.1, 0.2],
    [0.1, 0.1, 0.1, 0.1,-0.1, 1.0, 0.1,-0.3],
    [0.2, 0.1, 0.3, 0.2,-0.1, 0.1, 1.0,-0.2],
    [-0.1,-0.1,-0.1, 0.0, 0.2,-0.3,-0.2, 1.0]
])

z = np.random.multivariate_normal(np.zeros(8), Sigma, n)
labels = ["glucose", "hba1c", "triglycerides", "ldl",
          "hdl", "creatinine", "alt", "albumin"]
metabolic = pd.DataFrame(z, columns=labels)

# Scale and fit PCA
scaler = StandardScaler()
X_scaled = scaler.fit_transform(metabolic)
pca = PCA()
scores = pca.fit_transform(X_scaled)
```

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Scree plot
axes[0].bar(range(1, 9), pca.explained_variance_ratio_,
            color="steelblue", edgecolor="white")
axes[0].axhline(y=1/8, color="firebrick", linestyle="--", label="Kaiser thres
axes[0].set_xlabel("Component")
axes[0].set_ylabel("Proportion of Variance")
axes[0].set_title("Scree Plot")
axes[0].set_xticks(range(1, 9))
axes[0].legend()

# Biplot
loadings = pca.components_[:2].T  # first 2 PCs
scale_factor = 3
axes[1].scatter(scores[:, 0], scores[:, 1], alpha=0.3, s=10, color="grey")
for i, lab in enumerate(labels):
    axes[1].annotate("", xy=(loadings[i, 0]*scale_factor,
                              loadings[i, 1]*scale_factor),
                     xytext=(0, 0),
                     arrowprops=dict(arrowstyle="->", color="firebrick", lw=1
    axes[1].text(loadings[i, 0]*scale_factor*1.15,
                 loadings[i, 1]*scale_factor*1.15,
                 lab, fontsize=9, color="firebrick", ha="center")
axes[1].set_xlabel(f"PC1 ({pca.explained_variance_ratio_[0]:.1%} var)")
axes[1].set_ylabel(f"PC2 ({pca.explained_variance_ratio_[1]:.1%} var)")
axes[1].set_title("PCA Biplot: Metabolic Panel")
axes[1].axhline(0, color="grey", linewidth=0.5)
axes[1].axvline(0, color="grey", linewidth=0.5)

plt.tight_layout()
plt.show()
```

What the code is doing. We simulate 300 patients with eight correlated lab values, standardise them (so glucose in the hundreds does not drown out HbA1c around 6), and run PCA. The code then draws two figures. The **scree plot** (left) is a bar chart of how much variance each of the eight components captures, tallest on the left; the dashed "Kaiser threshold" line marks the level a component would reach if it explained no more than its fair share — bars above it are worth keeping. You would look for the bars to drop off sharply after the first two or three, telling you most of the signal lives in just a few dimensions. The **biplot** (right) places every patient on the PC1–PC2 plane and overlays arrows for the original labs; how to read those arrows is explained next.

15.3.7. Interpreting the Biplot

The biplot overlays two things:

- **Points** (grey): individual patients projected onto the first two principal components.
- **Arrows** (red): the original variables. The direction and length of each arrow show how strongly and in which direction that variable contributes to each PC.

In our metabolic panel:

- **PC1** is dominated by glucose, HbA1c, and triglycerides pointing in one direction, with HDL pointing in the opposite direction. This is a "metabolic syndrome" axis: patients to the right tend to have higher glucose and triglycerides and lower HDL.
- **PC2** separates creatinine and albumin, capturing a "renal/hepatic" dimension.

This kind of interpretation is exactly what makes PCA useful in clinical research: it reveals which variables "travel together" and suggests underlying biological dimensions.

15.4. t-SNE: Non-Linear Dimensionality Reduction

15.4.1. The Core Idea

PCA preserves global, linear structure. But many datasets contain non-linear patterns — clusters, manifolds, or curves — that PCA cannot capture well. **t-SNE** (t-distributed stochastic neighbour embedding) was designed specifically for **visualisation** of high-dimensional data in 2 or 3 dimensions.

t-SNE works in two steps:

1. In the high-dimensional space, compute the probability that any two points are "neighbours" (using a Gaussian kernel). Nearby points have high probability; distant points have low probability.
2. In the low-dimensional (2D) embedding, try to arrange points so that the same neighbourhood probabilities are preserved, using a **t-distribution** kernel (heavier tails than a Gaussian, which helps prevent crowding).

The algorithm minimises the KL divergence between the high-dimensional and low-dimensional probability distributions using gradient descent.

The intuition is worth holding onto even if the maths is not: t-SNE asks "who are each patient's closest neighbours?" in the full high-dimensional space, then arranges the 2D plot so that those same neighbourly relationships are preserved. The payoff for a health researcher is a map where patients who are genuinely alike clump together, making subtypes pop out visually — which is exactly why t-SNE became popular for single-cell and other omics data where distinct cell populations or patient phenotypes hide in hundreds of dimensions.

15.4.2. The Perplexity Parameter

Perplexity (typically 5–50) controls how many neighbours each point considers. It roughly corresponds to the effective number of local neighbours.

- **Low perplexity** (5–10): focuses on very local structure. May produce many small, tight clusters.
- **High perplexity** (30–50): considers broader neighbourhoods. Produces smoother, more global layouts.

There is no single correct value; try several and see how the structure changes.

15.4.3. Critical Pitfalls and Misinterpretations

> ⚠ t-SNE is for exploration, not inference
>
> t-SNE is a visualisation tool. It is not a statistical test. Do not draw conclusions about statistical significance from a t-SNE plot.

Distances between clusters are meaningless. Two clusters that appear far apart may not actually be more different than two clusters that appear close. t-SNE distorts global distances to preserve local neighbourhoods.

Cluster sizes are meaningless. A tight cluster in t-SNE space does not mean those points are more similar than a spread-out cluster.

Different runs give different layouts. t-SNE uses stochastic optimisation. Running it twice with different seeds gives different-looking plots (though the same clusters should emerge).

It does not scale well. Standard t-SNE is $O(n^2)$, making it slow for datasets with more than ~10,000 observations. Approximations (Barnes-Hut, FIt-SNE) help.

15.4.4. Clinical Example: Visualising Patient Phenotypes

15.4.5. R

```
library(Rtsne)

set.seed(42)

# Simulate 3 patient subtypes with 20 features
n_per_group <- 100
p <- 20

group1 <- matrix(rnorm(n_per_group * p, mean = 0, sd = 1), ncol = p)
group2 <- matrix(rnorm(n_per_group * p, mean = 2, sd = 1.2), ncol = p)
group3 <- matrix(rnorm(n_per_group * p, mean = -1, sd = 0.8), ncol = p)
# Make groups differ in specific feature subsets
group2[, 1:5] <- group2[, 1:5] + 3
group3[, 10:15] <- group3[, 10:15] - 4

X <- rbind(group1, group2, group3)
labels <- factor(rep(c("Type A", "Type B", "Type C"), each = n_per_group))

# Run t-SNE with two perplexity values
tsne_low  <- Rtsne(X, perplexity = 10, dims = 2, verbose = FALSE, max_iter =
tsne_high <- Rtsne(X, perplexity = 40, dims = 2, verbose = FALSE, max_iter =

par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
cols <- c("steelblue", "firebrick", "forestgreen")

plot(tsne_low$Y, col = cols[labels], pch = 16, cex = 0.8,
     main = "t-SNE (perplexity = 10)", xlab = "t-SNE 1", ylab = "t-SNE 2")
legend("topright", levels(labels), col = cols, pch = 16, cex = 0.8)
```

```
plot(tsne_high$Y, col = cols[labels], pch = 16, cex = 0.8,
     main = "t-SNE (perplexity = 40)", xlab = "t-SNE 1", ylab = "t-SNE 2")
legend("topright", levels(labels), col = cols, pch = 16, cex = 0.8)
```

15.4.6. Python

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.manifold import TSNE

np.random.seed(42)
n_per = 100
p = 20

g1 = np.random.normal(0, 1, (n_per, p))
g2 = np.random.normal(2, 1.2, (n_per, p))
g3 = np.random.normal(-1, 0.8, (n_per, p))
g2[:, :5] += 3
g3[:, 10:15] -= 4

X = np.vstack([g1, g2, g3])
labels = np.repeat(["Type A", "Type B", "Type C"], n_per)
colors = {"Type A": "steelblue", "Type B": "firebrick", "Type C": "forestgreen"}

fig, axes = plt.subplots(1, 2, figsize=(12, 5))
for ax, perp in zip(axes, [10, 40]):
    tsne = TSNE(n_components=2, perplexity=perp, random_state=42, max_iter=1000)
    emb = tsne.fit_transform(X)
    for lab in ["Type A", "Type B", "Type C"]:
        mask = labels == lab
        ax.scatter(emb[mask, 0], emb[mask, 1], c=colors[lab],
                   s=15, alpha=0.7, label=lab)
```

```
    ax.set_title(f"t-SNE (perplexity = {perp})")
    ax.set_xlabel("t-SNE 1")
    ax.set_ylabel("t-SNE 2")
    ax.legend(fontsize=9)

plt.tight_layout()
plt.show()
```

What the code is showing. We build a fake dataset of 300 patients across three "disease subtypes" (each subtype differs on a different handful of the 20 features), then run t-SNE twice — once with perplexity 10, once with 40 — and plot the two resulting maps side by side, colouring each dot by its true subtype. Each dot is a patient; the axes ("t-SNE 1/2") have no units and no inherent meaning, so do not try to read values off them. What you look for is *separation*: three coloured blobs that do not overlap means the subtypes really do have distinct profiles.

Notice how perplexity changes the visual appearance but the three groups are consistently separated. This is reassuring — the cluster structure is real, not an artefact of a particular perplexity setting. (The perplexity setting, recall, is roughly the number of near neighbours each point pays attention to: a small value zooms in on tight local clumps, a large value smooths things into broader groupings.)

15.5. UMAP: Uniform Manifold Approximation and Projection

15.5.1. How UMAP Differs from t-SNE

UMAP (McInnes, Healy, and Melville, 2018) is a newer method that has largely replaced t-SNE in many applications. Key advantages:

Table 15.1.: Comparison of t-SNE and UMAP

Feature	t-SNE	UMAP
Speed	Slow ($O(n^2)$ or $O(n \log n)$ approximate)	Fast (typically 3–10x faster)
Global structure	Poorly preserved	Better preserved
Reproducibility	Stochastic, varies across runs	More stable (with fixed seed)
Scalability	Struggles above ~50k points	Handles millions of points
Embedding dimensions	Typically 2–3	2 to arbitrary d
Use beyond visualisation	Rarely	Commonly used as a preprocessing step

UMAP works by constructing a topological representation (a weighted graph) of the data in high dimensions, then optimising a low-dimensional layout that preserves the graph structure.

15.5.2. Key Parameters

n_neighbors (analogous to perplexity in t-SNE): controls the balance between local and global structure. Small values (5–15) emphasise local clusters; large values (50–200) emphasise global connectivity.

min_dist: controls how tightly points are packed in the embedding. Small values (0.0–0.1) produce tight, well-separated clusters; large values (0.5–1.0) produce a more uniform spread.

15.5.3. When to Use Which Method

- **PCA**: use for initial exploration, as a preprocessing step (reduce to 20–50 PCs before t-SNE/UMAP), or when you need interpretable components. Use when linear relationships are sufficient.
- **t-SNE**: use for detailed visualisation of moderate-sized datasets (< 10,000 points) when you want to emphasise local cluster structure. Avoid for quantitative downstream analysis.
- **UMAP**: use as your default non-linear method. Faster, better global structure, and can serve as a preprocessing step for clustering.

15.5.4. R

```
library(Rtsne)
library(uwot)

set.seed(42)

pca_scores <- prcomp(X, scale. = TRUE)$x[, 1:2]
tsne_emb   <- Rtsne(X, perplexity = 30, dims = 2, verbose = FALSE)$Y
umap_emb   <- umap(X, n_neighbors = 15, min_dist = 0.1, verbose = FALSE)

par(mfrow = c(1, 3), mar = c(4, 4, 3, 1))
cols <- c("steelblue", "firebrick", "forestgreen")[as.numeric(labels)]

plot(pca_scores, col = cols, pch = 16, cex = 0.8,
     main = "PCA", xlab = "PC1", ylab = "PC2")

plot(tsne_emb, col = cols, pch = 16, cex = 0.8,
     main = "t-SNE (perplexity = 30)", xlab = "t-SNE 1", ylab = "t-SNE 2")
```

```
plot(umap_emb, col = cols, pch = 16, cex = 0.8,
     main = "UMAP (n_neighbors = 15)", xlab = "UMAP 1", ylab = "UMAP 2")
```

15.5.5. Python

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE
from umap import UMAP

pca_2d = PCA(n_components=2).fit_transform(X)
tsne_2d = TSNE(n_components=2, perplexity=30, random_state=42).fit_transform(X)
umap_2d = UMAP(n_components=2, n_neighbors=15, min_dist=0.1,
                random_state=42).fit_transform(X)

fig, axes = plt.subplots(1, 3, figsize=(16, 5))
titles = ["PCA", "t-SNE (perplexity=30)", "UMAP (n_neighbors=15)"]
embeddings = [pca_2d, tsne_2d, umap_2d]

for ax, emb, title in zip(axes, embeddings, titles):
    for lab, col in colors.items():
        mask = labels == lab
        ax.scatter(emb[mask, 0], emb[mask, 1], c=col, s=15, alpha=0.7, label=lab)
    ax.set_title(title)
    ax.set_xlabel("Dim 1")
    ax.set_ylabel("Dim 2")
    ax.legend(fontsize=8)

plt.tight_layout()
plt.show()
```

What the code is showing. We take the *same* three-subtype dataset and run all three methods on it — PCA, t-SNE, and UMAP — then plot their 2D outputs in one row so you can compare them like for like, with patients coloured by true subtype. The point of the figure is not any single panel but the contrast between them: it lets you see, on real-looking data, how a linear method (PCA) and two non-linear methods behave on the same problem.

All three methods recover the three groups, but with different visual characteristics. PCA shows the groups with some overlap (because the separation is partly non-linear). t-SNE and UMAP separate them more clearly, with UMAP preserving more of the relative distances between clusters.

15.6. Practical Workflow: Scale, Reduce, Visualise, Interpret

A standard workflow for high-dimensional clinical data:

1. **Scale** the data. PCA, t-SNE, and UMAP are all sensitive to the scale of variables. Always standardise (mean = 0, SD = 1) before applying any dimensionality reduction.
2. **Reduce with PCA first.** If the original data have hundreds or thousands of features, first reduce to 20–50 principal components. This removes noise and speeds up t-SNE/UMAP dramatically.
3. **Visualise with UMAP (or t-SNE).** Apply the non-linear method to the PCA-reduced data. Colour points by known labels (diagnosis, treatment group) or by continuous variables (lab values, scores).
4. **Interpret cautiously.** Look for clusters, gradients, or outliers. Generate hypotheses — but remember that these are visualisations,

not tests. Validate any apparent structure with formal statistical methods.

15.6.1. R

```
library(uwot)

set.seed(42)

# Simulate high-dimensional data: 500 patients, 100 features
n <- 500; p <- 100
group <- sample(c("Healthy", "Mild", "Severe"), n, replace = TRUE,
                prob = c(0.4, 0.35, 0.25))

X_hd <- matrix(rnorm(n * p), ncol = p)
# Add signal in first few features
X_hd[group == "Mild", 1:10]   <- X_hd[group == "Mild", 1:10] + 1.5
X_hd[group == "Severe", 1:10] <- X_hd[group == "Severe", 1:10] + 3
X_hd[group == "Severe", 11:20] <- X_hd[group == "Severe", 11:20] - 2

# Step 1: Scale
X_scaled <- scale(X_hd)

# Step 2: PCA to 20 components
pca_20 <- prcomp(X_scaled)$x[, 1:20]

# Step 3: UMAP on PCA-reduced data
umap_result <- umap(pca_20, n_neighbors = 15, min_dist = 0.1, verbose = FALSE)

# Step 4: Visualise
plot_df <- tibble(
  UMAP1 = umap_result[, 1],
```

```
  UMAP2 = umap_result[, 2],
  Group = factor(group, levels = c("Healthy", "Mild", "Severe"))
)

ggplot(plot_df, aes(x = UMAP1, y = UMAP2, colour = Group)) +
  geom_point(alpha = 0.6, size = 1.5) +
  scale_colour_manual(values = c("Healthy" = "steelblue",
                                 "Mild" = "goldenrod",
                                 "Severe" = "firebrick")) +
  labs(title = "UMAP After PCA Reduction",
       subtitle = "500 patients, 100 features → 20 PCs → 2D UMAP") +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom")
```

15.6.2. Python

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from umap import UMAP

np.random.seed(42)
n, p = 500, 100
group = np.random.choice(["Healthy", "Mild", "Severe"], n, p=[0.4, 0.35, 0.25

X_hd = np.random.normal(0, 1, (n, p))
X_hd[group == "Mild", :10] += 1.5
X_hd[group == "Severe", :10] += 3
X_hd[group == "Severe", 10:20] -= 2

# Workflow
```

```
X_scaled = StandardScaler().fit_transform(X_hd)
pca_20 = PCA(n_components=20).fit_transform(X_scaled)
umap_2d = UMAP(n_neighbors=15, min_dist=0.1, random_state=42).fit_transform(pca_20)

fig, ax = plt.subplots(figsize=(8, 6))
color_map = {"Healthy": "steelblue", "Mild": "goldenrod", "Severe": "firebrick"}
for g in ["Healthy", "Mild", "Severe"]:
    mask = group == g
    ax.scatter(umap_2d[mask, 0], umap_2d[mask, 1],
               c=color_map[g], s=12, alpha=0.6, label=g)
ax.set_xlabel("UMAP 1")
ax.set_ylabel("UMAP 2")
ax.set_title("UMAP After PCA Reduction\n500 patients, 100 features → 20 PCs → 2D")
ax.legend()
plt.tight_layout()
plt.show()
```

What the code is showing. This puts the four-step recipe into practice on a larger, more realistic problem: 500 patients, 100 features, and three severity groups (Healthy, Mild, Severe) whose differences are buried in only the first 20 features. We scale, compress the 100 features down to 20 principal components, then hand those to UMAP to draw a single 2D map, colouring each patient by severity. You would expect to see the three colours form a gradient or three loose clusters — Healthy at one end, Severe at the other — which would confirm that the severity signal survived the squeeze from 100 dimensions to 2. Doing PCA first is not just for speed: it strips out noise so UMAP focuses on the real structure.

15.7. Exercises

15.7.1. Exercise 1: PCA on Clinical Lab Data

Using the simulated metabolic panel data from this chapter (or a real dataset if available):

a. Perform PCA on the standardised data.
b. Create a scree plot and determine how many components to retain using the cumulative variance threshold (80%) and the Kaiser criterion.
c. Examine the loadings of the first two PCs. Which variables load most strongly on each? Propose a clinical interpretation.
d. Create a biplot coloured by a simulated diabetes status variable. Do diabetic patients separate along PC1?

15.7.2. R

```
library(tidyverse)  # %>% pipe, select()

# Add diabetes status (correlated with glucose/HbA1c)
set.seed(99)
metabolic$diabetes <- factor(ifelse(metabolic$glucose > 110 & metabolic$hba1c
                                     "Diabetic", "Non-diabetic"))

pca_fit <- prcomp(metabolic %>% select(-diabetes), scale. = TRUE)

# (a) & (b) Scree plot with cumulative variance
var_prop <- pca_fit$sdev^2 / sum(pca_fit$sdev^2)
cum_var  <- cumsum(var_prop)

par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
```

```r
barplot(var_prop, names.arg = paste0("PC", 1:8), col = "steelblue",
        main = "Variance Explained", ylab = "Proportion")
abline(h = 1/8, col = "firebrick", lty = 2)

plot(1:8, cum_var, type = "b", pch = 16, col = "steelblue",
     main = "Cumulative Variance", xlab = "Components",
     ylab = "Cumulative proportion", ylim = c(0, 1))
abline(h = 0.80, col = "firebrick", lty = 2)
text(6, 0.82, "80% threshold", col = "firebrick", cex = 0.8)

cat("Components for 80% variance:", which(cum_var >= 0.80)[1], "\n")

# (c) Loadings
cat("\nPC1 loadings:\n")
print(round(sort(pca_fit$rotation[, 1], decreasing = TRUE), 3))
cat("\nPC2 loadings:\n")
print(round(sort(pca_fit$rotation[, 2], decreasing = TRUE), 3))
```

15.7.3. Python

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

# Recreate metabolic data (using earlier arrays)
diabetes = ["Diabetic" if metabolic.iloc[i]['glucose'] > 110 and
            metabolic.iloc[i]['hba1c'] > 6.2 else "Non-diabetic"
            for i in range(len(metabolic))]

X = metabolic[['glucose','hba1c','triglycerides','ldl','hdl',
```

```
               'creatinine','alt','albumin']].values
X_s = StandardScaler().fit_transform(X)
pca = PCA().fit(X_s)

cum_var = np.cumsum(pca.explained_variance_ratio_)
n_80 = np.argmax(cum_var >= 0.80) + 1
print(f"Components for 80% variance: {n_80}")

# Loadings
feat_names = ['glucose','hba1c','triglycerides','ldl','hdl',
              'creatinine','alt','albumin']
for pc in [0, 1]:
    loadings = pca.components_[pc]
    order = np.argsort(np.abs(loadings))[::-1]
    print(f"\nPC{pc+1} loadings:")
    for idx in order:
        print(f"  {feat_names[idx]:>15s}: {loadings[idx]:+.3f}")
```

What the solution shows. The code labels patients as diabetic or not (based on glucose and HbA1c), re-runs PCA, and prints two things: how many components are needed to reach 80% of the variance, and the loadings of PC1 and PC2 sorted by size. Reading the printed loadings is the key step — the labs with the largest positive or negative numbers are the ones that define each axis. If glucose, HbA1c, and triglycerides top the PC1 list, then PC1 is effectively a "metabolic" score, and you would expect diabetic patients to sit toward one end of that axis.

15.7.4. Exercise 2: t-SNE Sensitivity to Perplexity

Using the three-group simulated dataset:

a. Run t-SNE with perplexity values of 5, 15, 30, and 50.

b. Create a 2x2 panel of the resulting embeddings.
c. At which perplexity value do the three groups first become clearly separated?
d. Are the distances between clusters consistent across perplexity values? What does this tell you about interpreting t-SNE?

15.7.5. Exercise 3: UMAP Parameter Exploration

Using the same dataset:

a. Run UMAP with `n_neighbors` in {5, 15, 50, 100} while holding `min_dist = 0.1`.
b. Run UMAP with `min_dist` in {0.0, 0.1, 0.5, 1.0} while holding `n_neighbors = 15`.
c. Create a panel of plots for each parameter sweep.
d. Describe how each parameter affects the visual appearance. Which combination would you recommend for this dataset?

15.7.6. Exercise 4: Full Workflow on Simulated Genomic Data

Simulate a dataset of 1,000 patients with 500 gene expression features and 4 cancer subtypes.

a. Scale the data and perform PCA. How many PCs are needed for 80% variance?
b. Apply UMAP to the first 30 PCs. Colour by cancer subtype. Are the subtypes visually separable?
c. Repeat with t-SNE. Compare the two visualisations.
d. One of the four subtypes is rare (5% of patients). Can you still identify it in the UMAP plot?

15.8. Summary

Dimensionality reduction transforms high-dimensional clinical data into interpretable, lower-dimensional representations. PCA provides linear, interpretable components that are ideal for initial exploration and preprocessing. t-SNE and UMAP offer non-linear embeddings that reveal complex structure invisible to PCA, though their outputs must be interpreted with care. The standard workflow — scale, reduce with PCA, embed with UMAP, interpret cautiously — is a reliable starting point for any high-dimensional dataset in clinical research.

Key Takeaways

1. Always scale your data before dimensionality reduction.
2. PCA is linear, deterministic, and interpretable. Use it first.
3. t-SNE is for visualisation only. Distances between clusters are meaningless; cluster sizes are meaningless.
4. UMAP is faster, preserves more global structure, and can serve as a preprocessing step for downstream analysis.
5. Non-linear methods are exploratory. Validate any apparent structure with formal statistical methods or domain expertise.

15.9. References and Further Reading

The foundational reference for UMAP is McInnes, Healy, and Melville's 2018 paper "UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction," available on arXiv (1802.03426). For t-SNE, van der Maaten and Hinton's 2008 paper in the *Journal of Machine Learning Research* introduced the method, and Wattenberg, Viegas, and Johnson's 2016 interactive article "How to Use t-SNE Effectively" on Distill.pub remains the best practical guide to understanding its behaviour and pitfalls.

For PCA in a clinical context, Jolliffe and Cadima's 2016 tutorial "Principal Component Analysis: A Review and Recent Developments" in the *Philosophical Transactions of the Royal Society A* provides a modern overview. Becht et al.'s 2019 paper "Dimensionality Reduction for Visualizing Single-Cell Data Using UMAP" in *Nature Biotechnology* demonstrates the advantages of UMAP over t-SNE in a biological context that generalises well to other high-dimensional clinical data.

A comprehensive 2024 review by Xiang et al. in *BMC Bioinformatics* compares PCA, t-SNE, UMAP, and newer methods (PHATE, TriMap) across multiple benchmarks, providing guidance on method selection for different data types. For the mathematical foundations, Murphy's *Probabilistic Machine Learning: An Introduction* (2022) covers PCA in depth, while the UMAP documentation (umap-learn.readthedocs.io) includes excellent tutorials with worked examples. James, Witten, Hastie, and Tibshirani's *An Introduction to Statistical Learning* (2nd edition, 2021) provides an accessible introduction to PCA that is well suited for students with a biostatistics background.

16. Clustering: Discovering Patient Subgroups

Clustering answers a question every clinician has asked: "are all my patients really the same, or are there hidden subgroups in here?" Given a table of measurements and no labels, a clustering algorithm proposes a grouping — this patient belongs with those patients — so that you can then ask whether the groups differ in outcome, biology, or treatment response. That is how the now-famous sepsis and ARDS subtypes were discovered. This chapter walks through three popular algorithms and, crucially, how to tell a real subgroup from a mirage, because clustering will hand you groups whether or not any exist.

> **Note**
>
> These chapters use `eval: false`, so the rendered page shows code but no live figures or printouts. Each code block is followed by a description of what its output *would* look like, so you can run it yourself and know what counts as a sensible result.

16.1. Introduction

Every classification method we have studied so far is **supervised**: you train a model on labelled data and use it to predict labels for new observations. Clustering is fundamentally different. It is **unsupervised** — there are no labels. The goal is to discover natural groupings in the data that you did not know existed.

In clinical research, clustering is used to identify **patient phenotypes** — subgroups of patients who share similar clinical profiles but may not correspond to any established diagnostic category. These phenotypes can reveal heterogeneity within a disease, suggest new treatment targets, or identify patients who respond differently to therapy. Sepsis, ARDS, heart failure, and asthma have all been re-examined through clustering analyses that revealed clinically meaningful subtypes hidden within traditional diagnostic labels.

This chapter covers three major clustering algorithms (K-means, hierarchical clustering, and DBSCAN), methods for choosing the number of clusters and validating results, and the integration of clustering with the dimensionality reduction methods from the previous chapter.

> **!** Clusters can be artefacts
>
> Clustering algorithms will always produce clusters, even in completely random data. The existence of clusters in your output does not mean they are real. Validation — both statistical and clinical — is essential.

16.2. K-Means Clustering

16.2.1. The Algorithm

K-means is the simplest and most widely used clustering algorithm. Given k (the number of clusters), it works as follows:

1. **Initialise**: randomly place k cluster centroids in the feature space.
2. **Assign**: assign each data point to the nearest centroid (using Euclidean distance).

3. **Update**: recompute each centroid as the mean of all points assigned to it.
4. **Repeat** steps 2–3 until assignments stop changing (convergence).

K-means minimises the **within-cluster sum of squares (WCSS)**: the total squared distance from each point to its assigned centroid.

$$\text{WCSS} = \sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - \mu_k\|^2$$

Read this formula as "for every patient, measure how far they sit from their own cluster's centre (μ_k is that centre, or **centroid**), square it, and add it all up." Tight, well-defined clusters give a small WCSS; loose, scattered ones give a large WCSS. K-means simply shuffles the group assignments until this total is as small as it can make it.

16.2.2. Choosing K: Elbow Method, Silhouette Scores, Gap Statistic

The most important decision in K-means is the number of clusters k. There is no single correct answer, but several heuristics help.

Elbow method. Plot WCSS against k. As k increases, WCSS always decreases (more clusters = points closer to centroids). Look for an "elbow" where the rate of decrease sharply changes. The elbow suggests that adding more clusters yields diminishing returns.

Silhouette scores. For each point, the silhouette score measures how much closer it is to its own cluster than to the nearest other cluster. Values range from –1 (misclassified) to +1 (well clustered). The average silhouette score across all points summarises clustering quality. Higher is better; choose k that maximises the average silhouette score.

Gap statistic. Compares the WCSS of your data to the WCSS of data drawn from a reference null distribution (uniform). The gap is largest at the k where your data's structure most exceeds what would be expected by chance.

16.2.3. Limitations of K-Means

- **Assumes spherical, equally sized clusters.** K-means uses Euclidean distance, so it finds globular clusters. Elongated, ring-shaped, or irregularly shaped clusters will be poorly recovered.
- **Sensitive to initialisation.** Different random starting points can give different results. Modern implementations (K-means++) use smart initialisation to mitigate this, and running the algorithm multiple times with different seeds is standard practice.
- **Sensitive to outliers.** A single extreme point can pull a centroid substantially.
- **Requires scaled data.** Features on different scales will dominate the distance calculations. Always standardise before running K-means.

16.2.4. Clinical Example: Patient Phenotyping from Lab Data

16.2.5. R

```
library(cluster)

set.seed(42)
n <- 400

# Simulate 3 patient phenotypes with distinct lab profiles
pheno1 <- tibble(  # Metabolic syndrome phenotype
  glucose = rnorm(n * 0.35, 140, 20),
  crp     = rnorm(n * 0.35, 5, 2),
```

```
  albumin = rnorm(n * 0.35, 3.5, 0.3),
  wbc     = rnorm(n * 0.35, 8, 2)
)
pheno2 <- tibble(  # Inflammatory phenotype
  glucose = rnorm(n * 0.35, 100, 15),
  crp     = rnorm(n * 0.35, 15, 4),
  albumin = rnorm(n * 0.35, 3.0, 0.4),
  wbc     = rnorm(n * 0.35, 14, 3)
)
pheno3 <- tibble(  # Healthy-ish phenotype
  glucose = rnorm(n * 0.30, 90, 10),
  crp     = rnorm(n * 0.30, 2, 1),
  albumin = rnorm(n * 0.30, 4.2, 0.2),
  wbc     = rnorm(n * 0.30, 7, 1.5)
)

lab_data <- bind_rows(pheno1, pheno2, pheno3)
lab_scaled <- scale(lab_data)

# Elbow and silhouette plots
wcss <- numeric(10)
sil_avg <- numeric(10)
for (k in 2:10) {
  km <- kmeans(lab_scaled, centers = k, nstart = 25)
  wcss[k] <- km$tot.withinss
  sil_avg[k] <- mean(silhouette(km$cluster, dist(lab_scaled))[, 3])
}

par(mfrow = c(1, 3), mar = c(4, 4, 3, 1))

# Elbow plot
plot(2:10, wcss[2:10], type = "b", pch = 16, col = "steelblue",
     xlab = "Number of clusters (k)", ylab = "Within-cluster SS",
```

```
     main = "Elbow Method")

# Silhouette plot
plot(2:10, sil_avg[2:10], type = "b", pch = 16, col = "firebrick",
     xlab = "Number of clusters (k)", ylab = "Average silhouette",
     main = "Silhouette Scores")

# K-means with k=3, visualise on first 2 PCs
km3 <- kmeans(lab_scaled, centers = 3, nstart = 25)
pca2 <- prcomp(lab_scaled)$x[, 1:2]
plot(pca2, col = c("steelblue", "firebrick", "forestgreen")[km3$cluster],
     pch = 16, cex = 0.7,
     main = "K-means (k=3) on PCA", xlab = "PC1", ylab = "PC2")
```

16.2.6. Python

```python
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.decomposition import PCA

np.random.seed(42)

# Simulate phenotypes
g1 = np.column_stack([np.random.normal(140, 20, 140),
                       np.random.normal(5, 2, 140),
                       np.random.normal(3.5, 0.3, 140),
                       np.random.normal(8, 2, 140)])
g2 = np.column_stack([np.random.normal(100, 15, 140),
                       np.random.normal(15, 4, 140),
```

```
                        np.random.normal(3.0, 0.4, 140),
                        np.random.normal(14, 3, 140)])
g3 = np.column_stack([np.random.normal(90, 10, 120),
                        np.random.normal(2, 1, 120),
                        np.random.normal(4.2, 0.2, 120),
                        np.random.normal(7, 1.5, 120)])

X = np.vstack([g1, g2, g3])
X_scaled = StandardScaler().fit_transform(X)

ks = range(2, 11)
wcss = []
sils = []
for k in ks:
    km = KMeans(n_clusters=k, n_init=25, random_state=42)
    km.fit(X_scaled)
    wcss.append(km.inertia_)
    sils.append(silhouette_score(X_scaled, km.labels_))

fig, axes = plt.subplots(1, 3, figsize=(16, 5))

axes[0].plot(list(ks), wcss, "o-", color="steelblue")
axes[0].set_xlabel("Number of clusters (k)")
axes[0].set_ylabel("Within-cluster SS")
axes[0].set_title("Elbow Method")

axes[1].plot(list(ks), sils, "o-", color="firebrick")
axes[1].set_xlabel("Number of clusters (k)")
axes[1].set_ylabel("Average silhouette")
axes[1].set_title("Silhouette Scores")

km3 = KMeans(n_clusters=3, n_init=25, random_state=42).fit(X_scaled)
pca2 = PCA(n_components=2).fit_transform(X_scaled)
```

```
colors = ["steelblue", "firebrick", "forestgreen"]
for c in range(3):
    mask = km3.labels_ == c
    axes[2].scatter(pca2[mask, 0], pca2[mask, 1], c=colors[c],
                    s=12, alpha=0.6, label=f"Cluster {c+1}")
axes[2].set_xlabel("PC1")
axes[2].set_ylabel("PC2")
axes[2].set_title("K-means (k=3) on PCA")
axes[2].legend()

plt.tight_layout()
plt.show()
```

What the code is showing. We simulate 400 patients drawn from three hidden lab-profile phenotypes (a metabolic, an inflammatory, and a healthier group), scale the labs, and then ask K-means to try every value of k from 2 to 10, recording two diagnostics each time. The figure has three panels. The **elbow plot** (left) shows within-cluster spread falling as k rises — you hunt for the "elbow" where it stops dropping steeply, the point of diminishing returns. The **silhouette plot** (middle) shows average cluster quality; here you want the *peak*. The **PCA scatter** (right) shows the actual $k = 3$ clustering, with each patient coloured by assigned cluster and placed on the first two principal components (a 2D summary, since we cannot plot four lab axes at once) — well-separated colours mean the clusters occupy distinct regions.

Both the elbow and silhouette plots suggest $k = 3$ — consistent with our simulation of three phenotypes. In practice, the signal is rarely this clean, and multiple values of k may appear reasonable. Domain knowledge should guide the final choice.

16.3. Hierarchical Clustering

16.3.1. Agglomerative (Bottom-Up) Clustering

Hierarchical clustering does not require specifying k in advance. The **agglomerative** approach starts with each observation as its own cluster and iteratively merges the closest pair of clusters until all observations belong to a single cluster.

The result is a **dendrogram** — a tree-like diagram that shows the merging history. You choose the number of clusters by “cutting” the dendrogram at a height that produces a meaningful number of groups.

16.3.2. Linkage Methods

The key decision is how to define the distance between two clusters (each containing multiple points):

Single linkage. Distance = minimum distance between any two points in the two clusters. Tends to produce long, chain-like clusters. Sensitive to outliers and noise.

Complete linkage. Distance = maximum distance between any two points. Produces compact, roughly equal-sized clusters but can be distorted by outliers.

Average linkage. Distance = average of all pairwise distances. A compromise between single and complete.

Ward’s method. Merges the pair of clusters that causes the smallest increase in total WCSS. Tends to produce compact, spherical clusters of similar size. **This is usually the best default choice.**

16.3.3. R

```r
# Use a smaller subset for readable dendrogram
set.seed(42)
idx <- sample(nrow(lab_scaled), 80)
hc <- hclust(dist(lab_scaled[idx, ]), method = "ward.D2")

# Plot dendrogram
plot(hc, labels = FALSE, main = "Hierarchical Clustering (Ward's Method)",
     xlab = "", sub = "", hang = -1, cex = 0.6)
rect.hclust(hc, k = 3, border = c("steelblue", "firebrick", "forestgreen"))
```

```r
par(mfrow = c(1, 4), mar = c(2, 2, 3, 1))
methods <- c("single", "complete", "average", "ward.D2")
titles  <- c("Single", "Complete", "Average", "Ward's")

for (i in seq_along(methods)) {
  hc_i <- hclust(dist(lab_scaled[idx, ]), method = methods[i])
  plot(hc_i, labels = FALSE, main = titles[i], xlab = "", sub = "",
       hang = -1, cex = 0.5)
  rect.hclust(hc_i, k = 3, border = "firebrick")
}
```

16.3.4. Python

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.cluster.hierarchy import dendrogram, linkage, fcluster

np.random.seed(42)
idx = np.random.choice(len(X_scaled), 80, replace=False)
```

```
X_sub = X_scaled[idx]

Z = linkage(X_sub, method='ward')

fig, ax = plt.subplots(figsize=(12, 5))
dendrogram(Z, ax=ax, no_labels=True, color_threshold=Z[-3, 2])
ax.set_title("Hierarchical Clustering (Ward's Method)")
ax.set_xlabel("Patients")
ax.set_ylabel("Distance")
ax.axhline(y=Z[-3, 2], color='firebrick', linestyle='--', linewidth=1)
ax.text(5, Z[-3, 2] + 0.5, "Cut for k=3", color="firebrick", fontsize=10)
plt.tight_layout()
plt.show()
```

```
fig, axes = plt.subplots(1, 4, figsize=(18, 4))
methods = ['single', 'complete', 'average', 'ward']
titles = ['Single', 'Complete', 'Average', "Ward's"]

for ax, method, title in zip(axes, methods, titles):
    Z_i = linkage(X_sub, method=method)
    dendrogram(Z_i, ax=ax, no_labels=True, color_threshold=0)
    ax.set_title(title)
    ax.set_xlabel("")

plt.tight_layout()
plt.show()
```

What the code is showing. The first block builds a **dendrogram** — the upside-down tree that records, branch by branch, which patients (then which groups of patients) got merged and at what distance. The coloured rectangles show where we "cut" the tree to get three clusters. The second block redraws the dendrogram four times, once per linkage rule (single, complete, average, Ward's), so you can see how the choice of linkage reshapes the tree. The practical lesson from that comparison: single linkage often produces one lopsided "chain" with stragglers, while Ward's gives balanced, compact groups — which is why Ward's is the recommended default for clinical data.

16.3.5. Reading a Dendrogram

The dendrogram encodes the full merging history:

- The **height** of each horizontal line represents the distance at which two clusters were merged.
- **Long vertical lines** before a merge indicate well-separated clusters.
- **Short vertical lines** indicate merges of similar clusters — less clear separation.

Cutting the dendrogram at different heights gives different numbers of clusters. A large "gap" between merging heights (a long vertical line) suggests a natural number of clusters.

16.4. DBSCAN: Density-Based Clustering

16.4.1. When K-Means Fails

K-means assumes clusters are spherical and roughly equal in size. In clinical data, clusters may be irregularly shaped, overlap, or contain noise (outlier patients who do not belong to any group). **DBSCAN** (Density-Based Spatial Clustering of Applications with Noise) addresses these limitations.

16.4.2. How It Works

DBSCAN defines clusters as dense regions separated by sparse regions. Two parameters control its behaviour:

- **eps** (ε): the radius of the neighbourhood around each point.
- **min_samples**: the minimum number of points required within radius ε to form a dense region.

The algorithm classifies each point as:

1. **Core point**: has at least `min_samples` neighbours within radius ε.
2. **Border point**: within ε of a core point but does not have enough neighbours to be core itself.
3. **Noise point**: neither core nor border. These are **outliers** — DBSCAN explicitly identifies them.

Clusters are formed by connecting core points that are within ε of each other.

16.4.3. Advantages and Limitations

Advantages:

- Does not require specifying k in advance.
- Can find arbitrarily shaped clusters.
- Identifies outliers explicitly.

Limitations:

- Sensitive to the choice of ε and `min_samples`.
- Struggles when clusters have very different densities.
- Not ideal for very high-dimensional data (use after dimensionality reduction).

16.4.4. R

```
library(dbscan)

set.seed(42)

# Create data with two crescent-shaped clusters + noise
n_each <- 150
theta1 <- seq(0, pi, length.out = n_each)
theta2 <- seq(0, pi, length.out = n_each)

x1 <- cos(theta1) + rnorm(n_each, 0, 0.08)
y1 <- sin(theta1) + rnorm(n_each, 0, 0.08)
x2 <- 1 - cos(theta2) + rnorm(n_each, 0, 0.08)
y2 <- 1 - sin(theta2) - 0.5 + rnorm(n_each, 0, 0.08)

# Add noise
x_noise <- runif(30, -0.5, 2)
y_noise <- runif(30, -1, 1.5)

crescent_data <- rbind(
  cbind(x1, y1),
  cbind(x2, y2),
  cbind(x_noise, y_noise)
)

# Compare K-means and DBSCAN
km_crescents <- kmeans(crescent_data, centers = 2, nstart = 25)
db_crescents <- dbscan(crescent_data, eps = 0.15, minPts = 5)

par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
cols_km <- c("steelblue", "firebrick")[km_crescents$cluster]
plot(crescent_data, col = cols_km, pch = 16, cex = 0.7,
```

```
     main = "K-means (k=2)", xlab = "x", ylab = "y")

# DBSCAN: cluster 0 = noise (grey), others = colours
db_cols <- c("grey50", "steelblue", "firebrick", "forestgreen")
plot(crescent_data,
     col = db_cols[db_crescents$cluster + 1],
     pch = ifelse(db_crescents$cluster == 0, 4, 16),
     cex = 0.7,
     main = "DBSCAN (eps=0.15)", xlab = "x", ylab = "y")
legend("topright", c("Noise", "Cluster 1", "Cluster 2"),
       col = db_cols[1:3], pch = c(4, 16, 16), cex = 0.8)
```

16.4.5. Python

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans, DBSCAN

np.random.seed(42)
n_each = 150

theta1 = np.linspace(0, np.pi, n_each)
theta2 = np.linspace(0, np.pi, n_each)

x1 = np.cos(theta1) + np.random.normal(0, 0.08, n_each)
y1 = np.sin(theta1) + np.random.normal(0, 0.08, n_each)
x2 = 1 - np.cos(theta2) + np.random.normal(0, 0.08, n_each)
y2 = 1 - np.sin(theta2) - 0.5 + np.random.normal(0, 0.08, n_each)

x_noise = np.random.uniform(-0.5, 2, 30)
y_noise = np.random.uniform(-1, 1.5, 30)
```

```
X_crescent = np.vstack([
    np.column_stack([x1, y1]),
    np.column_stack([x2, y2]),
    np.column_stack([x_noise, y_noise])
])

km = KMeans(n_clusters=2, n_init=25, random_state=42).fit(X_crescent)
db = DBSCAN(eps=0.15, min_samples=5).fit(X_crescent)

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

colors_km = ["steelblue" if c == 0 else "firebrick" for c in km.labels_]
axes[0].scatter(X_crescent[:, 0], X_crescent[:, 1], c=colors_km, s=10)
axes[0].set_title("K-means (k=2)")

color_map = {-1: "grey", 0: "steelblue", 1: "firebrick"}
colors_db = [color_map.get(c, "forestgreen") for c in db.labels_]
markers = ["x" if c == -1 else "o" for c in db.labels_]
for c_val in set(db.labels_):
    mask = db.labels_ == c_val
    m = "x" if c_val == -1 else "o"
    label = "Noise" if c_val == -1 else f"Cluster {c_val+1}"
    axes[1].scatter(X_crescent[mask, 0], X_crescent[mask, 1],
                    c=color_map.get(c_val, "forestgreen"),
                    marker=m, s=15, label=label)
axes[1].set_title("DBSCAN (eps=0.15)")
axes[1].legend(fontsize=9)

plt.tight_layout()
plt.show()
```

What the code is showing. We deliberately build a hard case: two interlocking crescent (banana) shapes plus a scatter of random "noise" patients who belong to neither. We then cluster it two ways and plot them side by side. The left panel is K-means asked for two clusters; because it can only carve out round blobs, it slices each crescent in half — the wrong answer. The right panel is DBSCAN, which grows clusters by following dense trails of points, so it traces each crescent correctly and marks the scattered outliers as "noise" (the crosses). Here `eps = 0.15` sets how close two patients must be to count as neighbours, and `minPts = 5` sets how many neighbours a point needs to anchor a dense region.

The figure shows a classic example: K-means splits the crescent-shaped clusters incorrectly, while DBSCAN identifies them perfectly and correctly labels the noise points.

16.5. Cluster Validation

16.5.1. Internal Metrics

Internal metrics evaluate clustering quality using only the data itself (no external labels).

Silhouette score. For each point i, define $a(i)$ as the mean distance to other points in the same cluster and $b(i)$ as the mean distance to points in the nearest other cluster. The silhouette score is:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

Values range from –1 to +1. A high average silhouette score indicates well-separated, compact clusters. The intuition: for each patient, compare "how far am I from my own group" (a) with "how far am I from the nearest rival group" (b). If the rival is much farther away ($b \gg a$), the score is near +1 and the patient is comfortably in the right cluster; if the patient is actually closer to a rival group, the score goes negative, flagging a likely misassignment.

Calinski-Harabasz index. The ratio of between-cluster variance to within-cluster variance (higher is better). It favours compact, well-separated clusters.

16.5.2. Stability-Based Validation

Internal metrics can be misleading — they can be high even for meaningless clusters. **Bootstrap stability** provides a stronger test:

1. Resample the data with replacement many times.
2. Re-run the clustering on each bootstrap sample.
3. Measure how often the same points end up in the same cluster.

If clusters are stable across resamples, they likely reflect real structure. If membership changes dramatically, the clusters may be artefacts.

16.5.3. R

```r
library(cluster)

km3 <- kmeans(lab_scaled, centers = 3, nstart = 25)
sil <- silhouette(km3$cluster, dist(lab_scaled))

plot(sil, col = c("steelblue", "firebrick", "forestgreen"),
     main = "Silhouette Plot (k = 3)",
     border = NA)
cat("Average silhouette width:", round(mean(sil[, 3]), 3), "\n")
```

16.5.4. Python

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import silhouette_score, silhouette_samples

km3 = KMeans(n_clusters=3, n_init=25, random_state=42).fit(X_scaled)
sil_vals = silhouette_samples(X_scaled, km3.labels_)
avg_sil = silhouette_score(X_scaled, km3.labels_)

fig, ax = plt.subplots(figsize=(8, 5))
y_lower = 0
colors = ["steelblue", "firebrick", "forestgreen"]
for c in range(3):
    c_sil = np.sort(sil_vals[km3.labels_ == c])
    y_upper = y_lower + len(c_sil)
    ax.barh(range(y_lower, y_upper), c_sil, height=1.0,
            color=colors[c], edgecolor="none")
    y_lower = y_upper + 5

ax.axvline(avg_sil, color="black", linestyle="--",
           label=f"Average: {avg_sil:.3f}")
ax.set_xlabel("Silhouette Coefficient")
ax.set_ylabel("Patients (sorted within cluster)")
ax.set_title("Silhouette Plot (k=3)")
ax.legend()
plt.tight_layout()
plt.show()
```

What the code is showing. This draws a **silhouette plot** for the $k = 3$ lab-data solution: one horizontal bar per patient, grouped and coloured by cluster, with each bar's length equal to that patient's silhouette score. A healthy cluster looks like a wide, mostly positive block of bars; bars that are short or dip below zero are patients sitting near (or on the wrong side of) a cluster boundary. The dashed line marks the overall average, the single number you would quote to summarise cluster quality. If one cluster's bars are systematically shorter than the others, that cluster is poorly defined and worth scrutinising.

16.6. Clinical Applications

16.6.1. Patient Phenotyping

The most impactful use of clustering in clinical research is identifying disease subtypes that were previously unrecognised. Notable examples:

- **ARDS phenotypes.** Calfee et al. (2014) applied latent class analysis to two ARDS clinical trial datasets and identified two phenotypes — "hyperinflammatory" and "hypoinflammatory" — with different mortality rates and different responses to treatment. The hyperinflammatory phenotype had higher IL-6, higher vasopressor use, and worse outcomes.
- **Sepsis endotypes.** Seymour et al. (2019) used K-means clustering on EHR data from over 20,000 sepsis patients and identified four phenotypes (alpha, beta, gamma, delta) with distinct organ dysfunction patterns and 28-day mortality rates ranging from 5% to 40%.
- **Heart failure subtypes.** Shah et al. (2015) applied hierarchical clustering to echocardiographic and clinical data from HFpEF patients and identified three phenotypes with different pathophysiology and prognosis.

16.6.2. Disease Subtyping from Lab Panels

A practical workflow for phenotyping:

1. Select clinically relevant variables (labs, vitals, demographics).
2. Handle missing data (imputation or exclusion).
3. Standardise all variables.
4. Reduce dimensionality (PCA to 10–20 components if many variables).
5. Apply multiple clustering algorithms (K-means, hierarchical) and compare.
6. Validate: silhouette scores, stability analysis, and — most importantly — clinical interpretation.
7. Compare clusters on outcomes (mortality, length of stay, treatment response).

16.7. Integrating Clustering with PCA/UMAP

The standard workflow for high-dimensional clinical data combines dimensionality reduction and clustering:

16.7.1. R

```r
library(uwot)
library(cluster)

set.seed(42)

# High-dimensional data: 500 patients, 50 features, 4 subtypes
n <- 500; p <- 50
true_subtype <- sample(1:4, n, replace = TRUE, prob = c(0.3, 0.3, 0.25, 0.15))
X_hd <- matrix(rnorm(n * p), ncol = p)

# Add subtype-specific signals
for (s in 1:4) {
  feature_start <- (s - 1) * 8 + 1
```

```
  feature_end   <- min(s * 8, p)
  X_hd[true_subtype == s, feature_start:feature_end] <-
    X_hd[true_subtype == s, feature_start:feature_end] + 2.5
}

# Step 1: Scale
X_s <- scale(X_hd)

# Step 2: PCA to 15 components
pca_15 <- prcomp(X_s)$x[, 1:15]

# Step 3: K-means on PCA scores
km4 <- kmeans(pca_15, centers = 4, nstart = 50)

# Step 4: UMAP for visualisation
umap_2d <- umap(pca_15, n_neighbors = 15, min_dist = 0.1, verbose = FALSE)

plot_df <- tibble(
  UMAP1 = umap_2d[, 1],
  UMAP2 = umap_2d[, 2],
  Cluster = factor(km4$cluster),
  True_subtype = factor(true_subtype)
)

p1 <- ggplot(plot_df, aes(x = UMAP1, y = UMAP2, colour = Cluster)) +
  geom_point(alpha = 0.6, size = 1.2) +
  scale_colour_manual(values = c("steelblue", "firebrick",
                                "forestgreen", "goldenrod")) +
  labs(title = "K-means Clusters on UMAP") +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom")

p2 <- ggplot(plot_df, aes(x = UMAP1, y = UMAP2, colour = True_subtype)) +
```

```
  geom_point(alpha = 0.6, size = 1.2) +
  scale_colour_manual(values = c("steelblue", "firebrick",
                                 "forestgreen", "goldenrod")) +
  labs(title = "True Subtypes on UMAP", colour = "Subtype") +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom")

library(patchwork)
p1 + p2
```

16.7.2. Python

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from umap import UMAP

np.random.seed(42)
n, p = 500, 50
true_sub = np.random.choice(4, n, p=[0.3, 0.3, 0.25, 0.15])
X_hd = np.random.normal(0, 1, (n, p))

for s in range(4):
    start = s * 8
    end = min((s + 1) * 8, p)
    X_hd[true_sub == s, start:end] += 2.5

X_s = StandardScaler().fit_transform(X_hd)
pca_15 = PCA(n_components=15).fit_transform(X_s)
km4 = KMeans(n_clusters=4, n_init=50, random_state=42).fit(pca_15)
```

```
umap_2d = UMAP(n_neighbors=15, min_dist=0.1, random_state=42).fit_transform(p

fig, axes = plt.subplots(1, 2, figsize=(14, 5))
colors_4 = ["steelblue", "firebrick", "forestgreen", "goldenrod"]

for c in range(4):
    mask = km4.labels_ == c
    axes[0].scatter(umap_2d[mask, 0], umap_2d[mask, 1],
                    c=colors_4[c], s=10, alpha=0.6, label=f"Cluster {c+1}")
axes[0].set_title("K-means Clusters on UMAP")
axes[0].legend(fontsize=8)

for s in range(4):
    mask = true_sub == s
    axes[1].scatter(umap_2d[mask, 0], umap_2d[mask, 1],
                    c=colors_4[s], s=10, alpha=0.6, label=f"Subtype {s+1}")
axes[1].set_title("True Subtypes on UMAP")
axes[1].legend(fontsize=8)

plt.tight_layout()
plt.show()
```

What the code is showing. This is the full pipeline end to end: 500 patients with 50 features and four built-in subtypes, which we scale, compress to 15 principal components, cluster with K-means, and finally lay out on a 2D UMAP map for viewing. The figure has two panels showing the *same* UMAP coordinates twice — on the left, dots are coloured by the cluster K-means assigned; on the right, by the patients' true (simulated) subtype. Because the layout is identical, you read them by comparing colours position-for-position: if the two panels colour the same blobs the same way, the algorithm recovered the real structure.

Comparing the two panels lets you assess how well the clustering algorithm recovered the true subtypes. In practice you do not know the true subtypes — the comparison would instead be against clinical outcomes, treatment response, or other external validation.

16.8. Cautionary Notes

> **⚠ Clusters are hypotheses, not facts**
>
> Finding three clusters in your ICU dataset does not mean there are three types of ICU patients. It means that your data, with your chosen features, using your chosen algorithm, with your chosen parameters, can be partitioned into three groups. Whether those groups are biologically meaningful requires external validation.

Always validate clinically. Clusters should differ on outcomes, biomarkers, or treatment response — not just on the features used to create them. Clustering on lab values and then showing that the clusters differ on lab values is circular reasoning.

Try multiple algorithms. If K-means, hierarchical clustering, and DBSCAN all find the same groups, you have more confidence that the structure is real. If they disagree, the structure may be fragile.

Report negative results. Many clustering analyses find no meaningful structure. This is a legitimate finding — it means the disease population is more homogeneous than expected.

Beware of overfitting to noise. With enough features, clustering will always find patterns. Use stability analysis and, if possible, validate clusters in an independent dataset.

16.9. Exercises

16.9.1. Exercise 1: K-Means on Simulated Patient Data

Simulate a dataset of 600 patients with 6 clinical variables (heart rate, respiratory rate, temperature, systolic BP, creatinine, lactate) drawn from 3 underlying phenotypes.

a. Scale the data and apply K-means for $k = 2, 3, 4, 5, 6$.
b. Create elbow and silhouette plots. Which k appears optimal?
c. Visualise the $k = 3$ solution using PCA.
d. Profile the clusters: compute the mean of each variable within each cluster. Do the profiles make clinical sense?

16.9.2. R

```r
library(tidyverse)  # tibble(), bind_rows(), group_by/summarise/across

set.seed(42)
n <- 600

# Phenotype 1: Septic shock (high HR, RR, lactate, low SBP)
p1 <- tibble(hr = rnorm(200, 115, 12), rr = rnorm(200, 28, 5),
             temp = rnorm(200, 38.5, 0.8), sbp = rnorm(200, 80, 12),
             creat = rnorm(200, 2.5, 0.8), lactate = rnorm(200, 5, 2))
# Phenotype 2: Stable critical (moderate vitals)
p2 <- tibble(hr = rnorm(200, 90, 10), rr = rnorm(200, 20, 3),
             temp = rnorm(200, 37.2, 0.5), sbp = rnorm(200, 110, 15),
             creat = rnorm(200, 1.2, 0.3), lactate = rnorm(200, 1.5, 0.5))
# Phenotype 3: Febrile, preserved hemodynamics
p3 <- tibble(hr = rnorm(200, 100, 10), rr = rnorm(200, 22, 4),
             temp = rnorm(200, 39.2, 0.7), sbp = rnorm(200, 120, 10),
```

```
              creat = rnorm(200, 1.0, 0.2), lactate = rnorm(200, 2.0, 0.8))

dat <- bind_rows(p1, p2, p3)
dat_scaled <- scale(dat)

# Elbow + silhouette
library(cluster)
wcss <- sil <- numeric(6)
for (k in 2:6) {
  km <- kmeans(dat_scaled, k, nstart = 25)
  wcss[k] <- km$tot.withinss
  sil[k] <- mean(silhouette(km$cluster, dist(dat_scaled))[, 3])
}

par(mfrow = c(1, 2))
plot(2:6, wcss[2:6], type = "b", pch = 16, col = "steelblue",
     xlab = "k", ylab = "WCSS", main = "Elbow Method")
plot(2:6, sil[2:6], type = "b", pch = 16, col = "firebrick",
     xlab = "k", ylab = "Avg Silhouette", main = "Silhouette")

# Profile clusters
km3 <- kmeans(dat_scaled, 3, nstart = 25)
dat$cluster <- km3$cluster
cat("\nCluster profiles (original scale):\n")
dat %>%
  group_by(cluster) %>%
  summarise(across(hr:lactate, mean), .groups = "drop") %>%
  print()
```

16.9.3. Python

```
import numpy as np
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

np.random.seed(42)

p1 = np.column_stack([np.random.normal(115,12,200), np.random.normal(28,5,200
                       np.random.normal(38.5,0.8,200), np.random.normal(80,12
                       np.random.normal(2.5,0.8,200), np.random.normal(5,2,20
p2 = np.column_stack([np.random.normal(90,10,200), np.random.normal(20,3,200)
                       np.random.normal(37.2,0.5,200), np.random.normal(110,1
                       np.random.normal(1.2,0.3,200), np.random.normal(1.5,0
p3 = np.column_stack([np.random.normal(100,10,200), np.random.normal(22,4,200
                       np.random.normal(39.2,0.7,200), np.random.normal(120,1
                       np.random.normal(1.0,0.2,200), np.random.normal(2.0,0

X = np.vstack([p1, p2, p3])
cols = ['hr', 'rr', 'temp', 'sbp', 'creat', 'lactate']
df = pd.DataFrame(X, columns=cols)
X_s = StandardScaler().fit_transform(X)

for k in range(2, 7):
    km = KMeans(n_clusters=k, n_init=25, random_state=42).fit(X_s)
    sil = silhouette_score(X_s, km.labels_)
    print(f"k={k}: WCSS={km.inertia_:.0f}, Silhouette={sil:.3f}")

km3 = KMeans(n_clusters=3, n_init=25, random_state=42).fit(X_s)
df['cluster'] = km3.labels_
print("\nCluster profiles:")
```

```
print(df.groupby('cluster')[cols].mean().round(2))
```

What the solution shows. The code simulates 600 patients from three vital-sign phenotypes (a septic-shock pattern, a stable pattern, and a febrile pattern), then runs the elbow and silhouette diagnostics and prints a **cluster profile table** — the mean of each variable within each cluster, on the original scale. That profile table is the payoff: it is where you turn anonymous "Cluster 1, 2, 3" labels into clinical stories by reading off, say, that one cluster has high heart rate, high lactate, and low blood pressure and recognising it as the septic-shock group. A clustering you cannot interpret this way is rarely useful.

16.9.4. Exercise 2: Hierarchical Clustering with Different Linkage Methods

Using the same dataset from Exercise 1:

a. Apply hierarchical clustering with single, complete, average, and Ward's linkage.
b. Cut each dendrogram at $k = 3$ and compare the resulting cluster assignments.
c. Which linkage method produces clusters most similar to the K-means solution? Use the adjusted Rand index (ARI) to quantify agreement.
d. Which linkage method would you recommend for clinical data and why?

16.9.5. Exercise 3: DBSCAN for Outlier Detection

Add 30 "outlier" patients to the Exercise 1 dataset: patients with extreme values across multiple variables (e.g., HR = 180, lactate = 15, creatinine = 8).

a. Run K-means with $k = 3$. Where do the outliers end up?
b. Run DBSCAN with appropriate eps and min_samples. How many outliers does it identify?
c. Compare the cluster assignments for non-outlier patients between K-means and DBSCAN.
d. In a clinical phenotyping study, which approach would you prefer for handling outliers?

16.9.6. Exercise 4: Full Pipeline — PCA + Clustering + UMAP Visualisation

Simulate a dataset of 800 patients with 30 clinical variables and 4 underlying subtypes.

a. Standardise the data and apply PCA. Use a scree plot to choose the number of components.
b. Run K-means ($k = 2, 3, 4, 5$) on the PCA scores. Use silhouette scores to choose k.
c. Visualise the chosen clustering on a UMAP embedding.
d. Profile the clusters: compute mean values for each variable and present in a table.
e. Discuss: how would you validate these clusters in a real clinical study?

16.10. Summary

Clustering is a powerful tool for discovering patient subgroups, but it requires careful application and sceptical interpretation. K-means is simple and effective for spherical clusters but requires specifying k and is sensitive to outliers. Hierarchical clustering provides a dendrogram that reveals the full structure of merging and does not require pre-specifying k. DBSCAN handles irregular cluster shapes and identifies outliers. No single method is best for all situations; using multiple methods and checking for consistency is good practice. Above all, clusters are statistical constructs — they must be validated against clinical outcomes and replicated in independent datasets before they can inform patient care.

Key Takeaways

1. Always scale your data before clustering.
2. K-means is fast and effective but assumes spherical clusters and requires choosing k.
3. Hierarchical clustering (Ward's method) is a solid default that does not require pre-specifying k.
4. DBSCAN excels when clusters are irregularly shaped or when outlier detection matters.
5. Validate clusters with internal metrics (silhouette), stability analysis, and — most importantly — clinical outcomes.
6. Clustering will always find groups, even in random data. The burden of proof is on you to show the groups are meaningful.

16.11. References and Further Reading

The most clinically impactful clustering study in recent critical care literature is Seymour et al.'s 2019 paper "Derivation, Validation, and Potential Treatment Implications of Novel Clinical Phenotypes for Sepsis" in *JAMA*, which identified four sepsis phenotypes using K-means clustering on electronic health record data. Calfee et al.'s 2014 paper "Subphenotypes in Acute Respiratory Distress Syndrome: Latent Class Analysis of Data From Two Randomised Controlled Trials" in *The Lancet Respiratory Medicine* applied a related method (latent class analysis) to identify ARDS subtypes with differential treatment response. Shah et al.'s 2015 paper in *Circulation* demonstrated the use of hierarchical clustering for heart failure with preserved ejection fraction phenotyping.

For the statistical foundations, Hastie, Tibshirani, and Friedman's *The Elements of Statistical Learning* (2nd edition, 2009) provides comprehensive coverage of K-means, hierarchical clustering, and spectral methods. James, Witten, Hastie, and Tibshirani's *An Introduction to Statistical Learning* (2nd edition, 2021) offers a more accessible treatment suitable for students. The original DBSCAN paper by Ester et al. (1996) is surprisingly readable and remains the standard reference for the algorithm.

For practical guidance, a 2024 review by Rodriguez et al. in *Critical Care Medicine* synthesises the rapidly growing literature on clustering-based phenotyping in critical illness, covering methodological best practices and the gap between statistical clusters and actionable clinical subtypes. Hennig et al.'s *Handbook of Cluster Analysis* (2016) is the most comprehensive reference, covering both methodology and applications. For validation, the clValid R package and its accompanying paper by Brock et al. (2008) in the *Journal of Statistical Software* provide tools for comparing clustering algorithms using internal, stability-based, and biological metrics.

Part VI.

Clinical Prediction Models

17. Developing Clinical Prediction Models: A Complete Workflow

17.1. Introduction

Building a clinical prediction model is not simply a matter of fitting a regression or running a machine learning algorithm. It is a structured process that begins with a clearly defined clinical question and proceeds through study design, data preparation, model specification, and evaluation. At every step, decisions must be guided by clinical knowledge, not just statistical convenience.

This chapter walks you through the complete workflow of developing a clinical prediction model, following the framework described by Smits, van Kuijk, and Wynants (2026) and the foundational principles laid out by Steyerberg (2019). We will use the Framingham Heart Study data as a running example to illustrate each step, building a model to predict 10-year cardiovascular disease (CVD) risk.

17.2. Step 1: Define the Clinical Question

Before touching any data, you must answer four questions:

1. **Who** is the target population? (e.g., adults aged 40–70 without prior CVD)
2. **What** is the outcome? (e.g., first CVD event within 10 years)

3. **When** is the prediction made? (e.g., at a routine primary care visit)
4. **Why** is the prediction needed? (e.g., to guide statin therapy decisions)

The intended use determines everything that follows. A model designed to triage emergency department patients needs different predictors, a different time horizon, and different performance requirements than a model for long-term cardiovascular risk stratification.

> The PICOTS Framework
>
> You can adapt the familiar PICOTS framework from clinical research: **P**opulation, **I**ntended use (Index model), **C**omparators (existing tools), **O**utcome, **T**iming, **S**etting.

17.2.1. Common Pitfalls in Problem Definition

- **Vague outcomes**: "clinical deterioration" is not a well-defined outcome. Define exactly what events count, how they are ascertained, and over what time window.
- **Leaky predictors**: including information that would not be available at the moment of prediction. If you are predicting ICU admission at the time of hospital arrival, you cannot use ICU vital signs as predictors.
- **Wrong population**: developing a model in hospitalised patients and hoping to apply it in primary care. The case-mix will be entirely different.

17.3. Step 2: Study Design and Sample Size

17.3.1. Study Design

Clinical prediction models are typically developed from **cohort studies** (prospective or retrospective) or **existing databases** (electronic health records, registries). Randomised trials can also be used, though their selected populations may limit generalisability.

Key design requirements:

- **Inception cohort**: all patients should be enrolled at a similar, well-defined point in their clinical trajectory (the "moment of prediction").
- **Complete follow-up**: outcomes must be ascertained for as many patients as possible. Differential loss to follow-up can bias estimates.
- **Outcome rate**: there must be a sufficient number of events to support the number of candidate predictors.

17.3.2. Sample Size: The Events Per Variable Rule and Beyond

The classical rule of thumb requires at least **10 events per variable (EPV)** in the model. For a logistic regression with 8 predictors, you would need at least 80 events. With 10% prevalence, that means at least 800 patients.

However, this rule is a rough guide. Riley and colleagues (2020) proposed a more rigorous sample size framework based on four criteria:

1. **Small optimism**: the expected shrinkage factor should be close to 1 (e.g., > 0.9).
2. **Small absolute difference** in the apparent and adjusted Nagelkerke R-squared.
3. **Precise estimation** of the overall outcome proportion (intercept).
4. **Precise estimation** of the overall model performance (C-statistic).

The `pmsampsize` package in R implements these calculations.

17.3.2.1. R

```
library(pmsampsize)

# Sample size for a logistic regression prediction model
# 10 candidate predictors, anticipated outcome prevalence of 15%
# Expected C-statistic of 0.75, Cox-Snell R-squared of 0.1
ss <- pmsampsize(type = "b",        # binary outcome
                 rsquared = 0.10,    # anticipated Cox-Snell R-squared
                 parameters = 10,    # number of candidate predictors
                 prevalence = 0.15)  # anticipated prevalence

print(ss)
cat("\nMinimum sample size:", ss$sample_size, "\n")
cat("Minimum number of events:", ss$events, "\n")
```

17.3.2.2. Python

```
# Riley et al. sample size criteria (simplified implementation)
# For full implementation, see the pmsampsize R package
import numpy as np

def pmsampsize_binary(rsquared, parameters, prevalence, shrinkage=0.9):
    """
    Simplified sample size calculation for binary outcome prediction models.
    Based on Riley et al. (2020) criteria.
    """
    # Criterion 1: Target shrinkage
    # n >= p / ((s - 1) * ln(1 - R^2_cs / s))
```

```
    # where s = target shrinkage, p = parameters, R^2_cs = Cox-Snell R^2
    n1 = parameters / ((shrinkage - 1) * np.log(1 - rsquared / shrinkage))

    # Criterion 2: Small absolute difference in R^2
    n2 = parameters / (0.05 * np.log(1 - rsquared))  # delta = 0.05

    # Criterion 3: Precise intercept (SE of log odds of prevalence)
    n3 = (1.96 / 0.05)**2 * (1 / (prevalence * (1 - prevalence)))

    n_min = max(n1, n2, n3)
    events = int(np.ceil(n_min * prevalence))

    print(f"Criterion 1 (shrinkage): n = {int(np.ceil(n1))}")
    print(f"Criterion 2 (R-squared): n = {int(np.ceil(n2))}")
    print(f"Criterion 3 (intercept): n = {int(np.ceil(n3))}")
    print(f"\nMinimum sample size: {int(np.ceil(n_min))}")
    print(f"Minimum events: {events}")

    return int(np.ceil(n_min))

pmsampsize_binary(rsquared=0.10, parameters=10, prevalence=0.15)
```

What the code shows. This runs a sample-size calculation *before* any data are collected. You supply the expected outcome prevalence (here 15%), the number of candidate predictors (10), and an anticipated model performance (expressed through the Cox-Snell R-squared). The function returns the minimum number of patients and the minimum number of events you would need, along with the implied events-per-variable (EPV). The point is feasibility: an underpowered study produces a model that looks impressive in development but falls apart in validation. In R, `pmsampsize` reports the maximum across all four Riley criteria; the simplified Python version prints each criterion separately so you can see which one is driving the requirement. If the required sample size is far larger than what you can realistically recruit, that is a signal to reduce the number of candidate predictors or reconsider the study.

> **i** What is EPV?
>
> **Events per variable (EPV)** is the number of outcome events divided by the number of parameters you are estimating. If you have 80 events and 8 predictors, you have an EPV of 10. Low EPV is the single most common cause of overfitting in clinical models: with too few events, the model "memorises" the development sample rather than learning generalisable patterns.

17.4. Step 3: Handle Missing Data

Missing data is the norm in clinical research, not the exception. Laboratory values are missing because they were not ordered. Follow-up data is missing because patients moved. Lifestyle variables are missing because patients did not answer the questionnaire.

17.4.1. Missing Data Mechanisms

Understanding **why** data are missing is essential for choosing the right approach:

- **MCAR (Missing Completely At Random)**: the probability of missingness is unrelated to any observed or unobserved data. Example: a laboratory sample is accidentally dropped. This is rare in practice.
- **MAR (Missing At Random)**: the probability of missingness depends on **observed** data but not on the missing value itself. Example: HbA1c is more likely to be measured in patients already known to

have diabetes. Given diabetes status (observed), the missingness is random.
- **MNAR (Missing Not At Random)**: the probability of missingness depends on the **unobserved** value itself. Example: patients with severe depression are less likely to attend follow-up appointments, so their depression scores are missing precisely because they are severe.

17.4.2. Why Complete Case Analysis Fails

Deleting rows with any missing data (complete case analysis) is the default in most software, and it is almost always wrong:

- It **wastes data**: if 10 variables each have 5% missing (independently), only 60% of rows are complete.
- It **biases estimates**: if missingness is related to the outcome or predictors (MAR or MNAR), the complete cases are a non-random subset.
- It **reduces power**: smaller effective sample sizes lead to wider confidence intervals and less stable models.

17.4.3. Multiple Imputation

Multiple imputation (MI) is the recommended approach for handling missing data under the MAR assumption. The procedure has three steps:

1. **Impute**: create m complete datasets by filling in missing values with plausible values drawn from the predictive distribution of the missing data, given the observed data.
2. **Analyse**: fit the prediction model separately in each of the m imputed datasets.
3. **Pool**: combine the m sets of results using Rubin's rules, which account for both within-imputation and between-imputation variability.

Typically, $m = 20$ or more imputations are recommended.

17.4.3.1. R

```
library(mice)
library(dplyr)

# Simulate clinical data with missing values
set.seed(42)
n <- 500
data <- data.frame(
  age = rnorm(n, 55, 10),
  sbp = rnorm(n, 130, 20),
  cholesterol = rnorm(n, 220, 40),
  smoking = rbinom(n, 1, 0.25),
  bmi = rnorm(n, 27, 5)
)
data$cvd <- rbinom(n, 1, plogis(-4 + 0.03*data$age + 0.01*data$sbp +
                                   0.005*data$cholesterol + 0.5*data$smoking)

# Introduce MAR missingness
# Cholesterol more likely missing in younger patients
data$cholesterol[data$age < 50 & runif(n) < 0.3] <- NA
# BMI more likely missing in non-smokers
data$bmi[data$smoking == 0 & runif(n) < 0.2] <- NA

cat("Missing data pattern:\n")
md.pattern(data, rotate.names = TRUE)

# Perform multiple imputation
imp <- mice(data, m = 20, method = "pmm", seed = 42, printFlag = FALSE)
```

```r
# Fit model in each imputed dataset and pool
fit_imp <- with(imp, glm(cvd ~ age + sbp + cholesterol + smoking + bmi,
                          family = binomial))
pooled <- pool(fit_imp)
summary(pooled)
```

17.4.3.2. Python

```python
import numpy as np
import pandas as pd
from sklearn.experimental import enable_iterative_imputer
from sklearn.impute import IterativeImputer
from sklearn.linear_model import LogisticRegression
from scipy.special import expit

np.random.seed(42)
n = 500

data = pd.DataFrame({
    'age': np.random.normal(55, 10, n),
    'sbp': np.random.normal(130, 20, n),
    'cholesterol': np.random.normal(220, 40, n),
    'smoking': np.random.binomial(1, 0.25, n),
    'bmi': np.random.normal(27, 5, n)
})
lp = -4 + 0.03*data['age'] + 0.01*data['sbp'] + 0.005*data['cholesterol'] + 0.5*data['smokin
data['cvd'] = np.random.binomial(1, expit(lp))

# Introduce MAR missingness
mask_chol = (data['age'] < 50) & (np.random.uniform(size=n) < 0.3)
data.loc[mask_chol, 'cholesterol'] = np.nan
mask_bmi = (data['smoking'] == 0) & (np.random.uniform(size=n) < 0.2)
```

```
data.loc[mask_bmi, 'bmi'] = np.nan

print("Missing values per column:")
print(data.isnull().sum())

# Simple multiple imputation using IterativeImputer
# Note: for proper MI with pooling, consider the miceforest package
predictors = ['age', 'sbp', 'cholesterol', 'smoking', 'bmi']
X = data[predictors].values
y = data['cvd'].values

# Create multiple imputed datasets and pool coefficients
m = 20
coefs = []
for i in range(m):
    imputer = IterativeImputer(random_state=i, max_iter=10, sample_posterior=
    X_imp = imputer.fit_transform(X)
    model = LogisticRegression(max_iter=1000, penalty=None)
    model.fit(X_imp, y)
    coefs.append(np.concatenate([model.intercept_, model.coef_[0]]))

# Pool results (simplified Rubin's rules)
coefs = np.array(coefs)
pooled_mean = coefs.mean(axis=0)
within_var = coefs.var(axis=0)  # simplified
names = ['intercept'] + predictors
print("\nPooled coefficients:")
for name, coef in zip(names, pooled_mean):
    print(f"  {name}: {coef:.4f}")
```

What the code shows. This demonstrates multiple imputation end-to-end on a simulated cohort where cholesterol and BMI are made deliberately missing in a way that depends on observed variables (MAR). In R, `md.pattern()` prints a table showing which combinations of variables are missing together — useful for spotting structure in the missingness. `mice()` then creates 20 filled-in datasets, the model is fitted in each, and `pool()` combines them with Rubin's rules so the reported standard errors honestly reflect the extra uncertainty from imputing. The pooled coefficients are what you would report. The key takeaway for a health researcher: you do not get a single "best guess" for each missing value — you get several plausible values, and the spread between them is folded into your confidence intervals. The Python version mimics this with `IterativeImputer`; note the comment that for fully correct pooling in Python you would reach for a dedicated MI package such as `miceforest`.

17.5. Step 4: Variable Selection

17.5.1. Choosing Candidate Predictors

The best predictor selection strategy is **expert knowledge**. Decades of cardiovascular research tell us that age, sex, blood pressure, cholesterol, smoking, and diabetes are important predictors of CVD. Starting with established clinical knowledge prevents the discovery of spurious associations and produces more generalisable models.

When expert knowledge is insufficient or the domain is novel, data-driven selection can supplement clinical reasoning. However, the approach matters enormously.

17.5.2. The Dangers of Stepwise Selection

Stepwise selection (forward, backward, or both) is among the most commonly used and most harmful variable selection strategies:

- It inflates the apparent significance of selected variables (multiple testing without correction).
- It produces unstable models: small changes in the data can lead to entirely different variable selections.
- It biases regression coefficients away from zero (selected variables appear stronger than they are).
- It yields overly optimistic performance estimates.

> **Avoid Univariable Screening**
>
> A particularly damaging practice is pre-selecting variables based on univariable p-values (e.g., keeping only variables with $p < 0.20$). This ignores confounding, suppression, and the fact that weak individual predictors can be strong in combination. If you must reduce the candidate set, use clinical knowledge, not univariable screening.

17.5.3. Better Approaches

- **Full model with all pre-specified predictors**: if sample size permits, include all clinically plausible predictors and apply shrinkage (see below). This is the recommended approach by Steyerberg (2019).
- **Penalised regression**: LASSO (L1 penalty) performs automatic variable selection by shrinking some coefficients to zero. Elastic net combines L1 and L2 penalties. These are covered in detail in Chapter 7.
- **Background knowledge with data-driven fine-tuning**: start with a core set of established predictors and consider a small number of additional candidates.

17.6. Step 5: Understand and Combat Overfitting

17.6.1. What Is Overfitting?

Overfitting occurs when a model captures not only the true underlying patterns in the data but also the **noise** — the random variation specific to the development sample. An overfitted model performs well on the data it was trained on but poorly on new data.

Clinical data is especially prone to overfitting because:

- **Sample sizes are often small** relative to the number of candidate predictors.
- **Events may be rare**, limiting the information available to estimate parameters.
- **Complex interactions** are tempting to include but expensive in terms of degrees of freedom.

17.6.2. Detecting Overfitting: Optimism

The **optimism** of a model is the difference between its apparent performance (on the training data) and its expected performance on new data. You can estimate optimism using internal validation techniques such as bootstrapping (covered in Chapter 18).

If a model shows high optimism, it is overfitted. The solution is **shrinkage**.

17.6.3. Shrinkage Methods

Uniform shrinkage multiplies all regression coefficients by a factor s (where $0 < s \leq 1$) estimated from the data, typically via bootstrapping. The intercept is then re-estimated to preserve the overall calibration. This pulls extreme predictions toward the average, reducing the impact of noise.

Penalised methods incorporate shrinkage directly into the estimation process:

- **Ridge regression** (L2 penalty) shrinks all coefficients toward zero but never sets any to exactly zero.
- **LASSO** (L1 penalty) can shrink coefficients to exactly zero, performing variable selection.
- **Elastic net** combines both penalties.

The penalty strength is chosen by cross-validation.

17.6.3.1. R

```
library(glmnet)
library(rms)

# Simulate Framingham-like data
set.seed(42)
n <- 800
age <- rnorm(n, 55, 10)
male <- rbinom(n, 1, 0.5)
sbp <- rnorm(n, 130, 20)
chol <- rnorm(n, 220, 40)
smoking <- rbinom(n, 1, 0.25)
diabetes <- rbinom(n, 1, 0.10)

# True model
lp <- -6 + 0.05*age + 0.3*male + 0.01*sbp + 0.005*chol + 0.4*smoking + 0.5*di
cvd <- rbinom(n, 1, plogis(lp))
```

```r
df <- data.frame(age, male, sbp, chol, smoking, diabetes, cvd)

# Unpenalised logistic regression
fit_full <- glm(cvd ~ ., data = df, family = binomial)
cat("=== Unpenalised coefficients ===\n")
print(round(coef(fit_full), 4))

# Ridge regression (alpha = 0)
X <- model.matrix(cvd ~ ., data = df)[, -1]
y <- df$cvd

fit_ridge <- cv.glmnet(X, y, family = "binomial", alpha = 0, nfolds = 10)
cat("\n=== Ridge coefficients (lambda.min) ===\n")
print(round(as.vector(coef(fit_ridge, s = "lambda.min")), 4))

# LASSO (alpha = 1)
fit_lasso <- cv.glmnet(X, y, family = "binomial", alpha = 1, nfolds = 10)
cat("\n=== LASSO coefficients (lambda.min) ===\n")
print(round(as.vector(coef(fit_lasso, s = "lambda.min")), 4))

# Compare: note how penalised coefficients are shrunk toward zero
par(mfrow = c(1, 2))
plot(fit_ridge, main = "Ridge: CV Error vs log(lambda)")
plot(fit_lasso, main = "LASSO: CV Error vs log(lambda)")
```

17.6.3.2. Python

```python
import numpy as np
import pandas as pd
from sklearn.linear_model import LogisticRegression, LogisticRegressionCV
from scipy.special import expit
import matplotlib.pyplot as plt
```

```
np.random.seed(42)
n = 800
age = np.random.normal(55, 10, n)
male = np.random.binomial(1, 0.5, n)
sbp = np.random.normal(130, 20, n)
chol = np.random.normal(220, 40, n)
smoking = np.random.binomial(1, 0.25, n)
diabetes = np.random.binomial(1, 0.10, n)

lp = -6 + 0.05*age + 0.3*male + 0.01*sbp + 0.005*chol + 0.4*smoking + 0.5*dia
cvd = np.random.binomial(1, expit(lp))

X = np.column_stack([age, male, sbp, chol, smoking, diabetes])
feature_names = ['age', 'male', 'sbp', 'chol', 'smoking', 'diabetes']

# Unpenalised logistic regression
model_full = LogisticRegression(penalty=None, max_iter=1000)
model_full.fit(X, cvd)
print("=== Unpenalised coefficients ===")
for name, coef in zip(feature_names, model_full.coef_[0]):
    print(f"  {name}: {coef:.4f}")

# Ridge regression (L2)
model_ridge = LogisticRegressionCV(penalty='l2', cv=10, max_iter=1000, randon
model_ridge.fit(X, cvd)
print("\n=== Ridge coefficients ===")
for name, coef in zip(feature_names, model_ridge.coef_[0]):
    print(f"  {name}: {coef:.4f}")

# LASSO (L1)
model_lasso = LogisticRegressionCV(penalty='l1', solver='saga', cv=10,
                                   max_iter=5000, random_state=42)
model_lasso.fit(X, cvd)
```

```
print("\n=== LASSO coefficients ===")
for name, coef in zip(feature_names, model_lasso.coef_[0]):
    print(f"  {name}: {coef:.4f}")
```

What the code shows. This fits the same model three ways and prints the coefficients side by side: an ordinary (unpenalised) logistic regression, ridge, and LASSO. The thing to look for is that the ridge and LASSO coefficients are pulled *closer to zero* than the unpenalised ones — that pulling-in is shrinkage in action, and it is deliberate: it trades a little bias for a lot less variance, so the model generalises better to new patients. LASSO goes further and sets some coefficients to *exactly* zero, effectively dropping those predictors. The two R plots show cross-validation error against $\log(\lambda)$, where λ is the penalty strength; the left-hand dashed line marks the λ that minimises error (`lambda.min`). The penalty is chosen by cross-validation rather than by hand so the data, not the analyst, decides how much shrinkage is appropriate.

Shrinkage in plain terms

A **shrinkage factor** is a number between 0 and 1 that you multiply coefficients by to make predictions less extreme. An overfitted model produces predictions that are too confident — very high risks too high, very low risks too low. Shrinkage pulls those predictions back toward the average event rate, which is usually closer to the truth for new patients. A shrinkage factor of 0.85 means the development model was about 15% too optimistic and its effects should be scaled back accordingly.

17.7. Step 6: Build the Model — A Framingham Example

Let us now walk through a complete model development process using simulated data based on the Framingham Heart Study. We will predict 10-year CVD risk using established risk factors.

17.7.1. Data Preparation

17.7.1.1. R

```
library(rms)

# Simulate Framingham-like cohort
set.seed(2024)
n <- 2000

framingham <- data.frame(
  age = round(runif(n, 30, 74)),
  male = rbinom(n, 1, 0.48),
  sbp = round(rnorm(n, 130, 18)),
  total_chol = round(rnorm(n, 210, 38)),
  hdl_chol = round(rnorm(n, 52, 15)),
  smoking = rbinom(n, 1, 0.22),
  diabetes = rbinom(n, 1, 0.08),
  bp_treatment = rbinom(n, 1, 0.15)
)

# Generate outcome based on known risk factors
lp <- with(framingham,
  -7.5 + 0.06 * age + 0.4 * male + 0.012 * sbp +
  0.005 * total_chol - 0.02 * hdl_chol +
```

```r
  0.5 * smoking + 0.7 * diabetes + 0.3 * bp_treatment
)
framingham$cvd_10yr <- rbinom(n, 1, plogis(lp))

cat("Event rate:", round(mean(framingham$cvd_10yr) * 100, 1), "%\n")
cat("Number of events:", sum(framingham$cvd_10yr), "\n")
cat("Events per variable:", round(sum(framingham$cvd_10yr) / 8, 1), "\n")

# Set up the rms environment for enhanced model building
dd <- datadist(framingham)
options(datadist = "dd")

# Fit the model using lrm (logistic regression model from rms)
# Allow nonlinearity for continuous variables using restricted cubic splines
fit <- lrm(cvd_10yr ~ rcs(age, 4) + male + rcs(sbp, 3) +
             total_chol + hdl_chol + smoking + diabetes + bp_treatment,
           data = framingham, x = TRUE, y = TRUE)

print(fit)

# Check for nonlinearity
anova(fit)

# Visualise partial effects
plot(Predict(fit, age), main = "Effect of Age on CVD Risk")
plot(Predict(fit, sbp), main = "Effect of SBP on CVD Risk")
```

17.7.1.2. Python

```python
import numpy as np
import pandas as pd
import statsmodels.api as sm
```

```
from scipy.special import expit

np.random.seed(2024)
n = 2000

framingham = pd.DataFrame({
    'age': np.random.randint(30, 75, n),
    'male': np.random.binomial(1, 0.48, n),
    'sbp': np.round(np.random.normal(130, 18, n)),
    'total_chol': np.round(np.random.normal(210, 38, n)),
    'hdl_chol': np.round(np.random.normal(52, 15, n)),
    'smoking': np.random.binomial(1, 0.22, n),
    'diabetes': np.random.binomial(1, 0.08, n),
    'bp_treatment': np.random.binomial(1, 0.15, n)
})

lp = (-7.5 + 0.06 * framingham['age'] + 0.4 * framingham['male'] +
      0.012 * framingham['sbp'] + 0.005 * framingham['total_chol'] -
      0.02 * framingham['hdl_chol'] + 0.5 * framingham['smoking'] +
      0.7 * framingham['diabetes'] + 0.3 * framingham['bp_treatment'])

framingham['cvd_10yr'] = np.random.binomial(1, expit(lp))

print(f"Event rate: {framingham['cvd_10yr'].mean()*100:.1f}%")
print(f"Number of events: {framingham['cvd_10yr'].sum()}")
print(f"Events per variable: {framingham['cvd_10yr'].sum() / 8:.1f}")

# Fit logistic regression
predictors = ['age', 'male', 'sbp', 'total_chol', 'hdl_chol',
              'smoking', 'diabetes', 'bp_treatment']
X = sm.add_constant(framingham[predictors])
y = framingham['cvd_10yr']
```

```
model = sm.Logit(y, X)
result = model.fit(disp=0)
print("\n", result.summary())

# Calculate predicted probabilities
framingham['pred_prob'] = result.predict(X)

# Summary of predicted risk distribution
print("\nPredicted risk distribution:")
print(framingham['pred_prob'].describe())
```

What the code shows. This builds the actual prediction model on a simulated Framingham-style cohort. The first printed lines report the event rate, the number of events, and the EPV — a sanity check that we have enough events for the number of predictors before trusting anything else. In R, `lrm()` fits the logistic model and `print(fit)` reports the coefficients plus the apparent C-statistic; `rcs(age, 4)` lets age have a flexible (non-straight-line) relationship with risk using restricted cubic splines with 4 knots. The `anova(fit)` output tests whether that flexibility is actually needed (a significant nonlinear term means a straight line would misfit), and `plot(Predict(...))` draws the fitted risk curve for a single predictor while holding the others fixed. The Python version fits the equivalent model with `statsmodels` and summarises the distribution of predicted risks, so you can see whether the model produces a useful spread of probabilities or bunches everyone near the average.

17.7.2. Model Specification Decisions

In the Framingham example above, several important decisions were made:

1. **Pre-specified predictors**: we included the established Framingham risk factors, not variables discovered through data-dredging.

2. **Nonlinearity**: for continuous variables like age and blood pressure, we allowed flexible functional forms using restricted cubic splines (in R's `rms` package). The relationship between age and CVD risk is not linear on the log-odds scale.
3. **No interactions without prior evidence**: we did not fish for interactions. If clinical knowledge suggests an interaction (e.g., the effect of cholesterol might differ by sex), it should be pre-specified.
4. **Sample size check**: with approximately 250 events and 8 predictors (plus a few extra degrees of freedom for splines), we have approximately 25 EPV, which is comfortable.

17.8. Step 7: Assess the Final Model

After fitting, we need to examine the model critically:

17.8.0.1. R

```
library(rms)

# Assuming fit and framingham from above
# 1. Check the discrimination (C-statistic)
cat("C-statistic (AUC):", round(fit$stats["C"], 3), "\n")

# 2. Predicted risk distribution
pred_probs <- predict(fit, type = "fitted")
hist(pred_probs, breaks = 50, col = "steelblue",
     main = "Distribution of Predicted 10-Year CVD Risk",
     xlab = "Predicted Probability")

# 3. Predicted risks in events vs non-events
boxplot(pred_probs ~ framingham$cvd_10yr,
```

```
        names = c("No CVD", "CVD"),
        main = "Predicted Risk by Outcome",
        ylab = "Predicted Probability",
        col = c("lightblue", "salmon"))

# 4. Quick calibration check
val <- val.prob(pred_probs, framingham$cvd_10yr)

# 5. Internal validation via bootstrapping (optimism-corrected)
set.seed(123)
val_boot <- validate(fit, B = 200)
print(val_boot)

cat("\nOptimism-corrected C-statistic:",
    round(0.5 * (val_boot["Dxy", "index.corrected"] + 1), 3), "\n")
```

17.8.0.2. Python

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import roc_auc_score
from sklearn.calibration import calibration_curve

# Assuming framingham and result from above
pred_probs = framingham['pred_prob'].values
y_true = framingham['cvd_10yr'].values

# 1. Discrimination
auc = roc_auc_score(y_true, pred_probs)
print(f"C-statistic (AUC): {auc:.3f}")

# 2. Predicted risk distribution
```

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

axes[0].hist(pred_probs, bins=50, color="steelblue", edgecolor="white")
axes[0].set_xlabel("Predicted Probability")
axes[0].set_ylabel("Frequency")
axes[0].set_title("Distribution of Predicted 10-Year CVD Risk")

# 3. Risk by outcome group
axes[1].boxplot([pred_probs[y_true == 0], pred_probs[y_true == 1]],
                labels=["No CVD", "CVD"])
axes[1].set_ylabel("Predicted Probability")
axes[1].set_title("Predicted Risk by Outcome")

plt.tight_layout()
plt.show()

# 4. Calibration plot
prob_true, prob_pred = calibration_curve(y_true, pred_probs, n_bins=10, strat

plt.figure(figsize=(7, 7))
plt.plot(prob_pred, prob_true, 's-', color="steelblue", label="Model")
plt.plot([0, 1], [0, 1], '--', color='grey', label="Perfect calibration")
plt.xlabel("Mean Predicted Probability")
plt.ylabel("Observed Proportion")
plt.title("Calibration Plot")
plt.legend()
plt.tight_layout()
plt.show()
```

What the code shows. This is a first critical look at the finished model. The C-statistic summarises discrimination (can the model rank a patient who has the event above one who does not?). The histogram of predicted risks shows whether the model meaningfully separates patients or squashes everyone into a narrow band — a model that predicts roughly the same risk for everyone is not clinically useful even if its C-statistic looks acceptable. The boxplot contrasts predicted risk in those who did and did not have CVD: good discrimination shows up as the "CVD" box sitting clearly higher. In R, `val.prob()` produces a calibration check, and `validate(fit, B = 200)` performs bootstrap internal validation — it reruns the whole modelling process on 200 resamples to estimate how much the apparent performance is inflated by overfitting (the *optimism*), then reports an optimism-corrected C-statistic that is a more honest guide to how the model will behave on new patients. Internal validation is covered in full in the next chapter.

> **i** Optimism
>
> **Optimism** is the gap between how well a model appears to perform on the data it was built from and how well it actually performs on new data. Because the model has partly fitted the noise in its own development sample, the apparent C-statistic is always a little too rosy. Bootstrap validation estimates this gap so you can subtract it off and report a realistic number.

17.9. The Complete Workflow: A Summary

The complete prediction model development workflow can be summarised in the following steps:

1. **Define** the clinical question (population, outcome, timing, intended use).
2. **Design** the study with adequate sample size (Riley criteria).
3. **Prepare** the data: handle missing values with multiple imputation.
4. **Select** predictors based on clinical knowledge; avoid stepwise procedures.

5. **Specify** the model: consider nonlinear terms for continuous predictors.
6. **Fit** the model, applying shrinkage to combat overfitting.
7. **Evaluate** performance: discrimination, calibration, clinical utility.
8. **Validate** internally using bootstrapping or cross-validation.
9. **Report** transparently following TRIPOD+AI guidelines (see Chapter 20).

> **! The Modelling Pipeline Is Not Linear**
>
> In practice, you will iterate. Poor calibration might send you back to model specification. Inadequate discrimination might prompt reconsidering the candidate predictors. Internal validation might reveal severe overfitting, requiring stronger shrinkage. The important thing is that each decision is documented and justified.

17.10. Common Mistakes to Avoid

Table 17.1.: Common mistakes in clinical prediction model development.

Mistake	Why It Is Harmful	Better Approach
Stepwise variable selection	Inflates significance, unstable	Pre-specify predictors; use penalisation
Univariable screening	Ignores confounding and suppression	Include all clinically plausible variables
Complete case analysis	Biases estimates, wastes data	Multiple imputation
Evaluating only on training data	Overestimates performance	Internal validation (bootstrap)

Mistake	Why It Is Harmful	Better Approach
Dichotomising continuous predictors	Loses information, creates arbitrary groups	Model continuous variables continuously
Ignoring nonlinearity	Poor fit, biased predictions	Use splines or fractional polynomials
Reporting only accuracy or AUC	Incomplete picture	Report calibration, discrimination, clinical utility

17.11. Exercises

1. **Sample size calculation**: You are developing a model to predict pre-eclampsia (expected prevalence 4%) using 12 candidate predictors. Use `pmsampsize` to determine the minimum required sample size. How many pregnancies would you need to observe?

2. **Missing data simulation**: Take the simulated Framingham dataset above and compare the regression coefficients obtained from (a) complete case analysis, (b) single mean imputation, and (c) multiple imputation. Which approach yields coefficients closest to the true values?

3. **Variable selection**: Fit the Framingham model using (a) all pre-specified predictors, (b) forward stepwise selection, and (c) LASSO. Compare the selected variables and the stability of coefficients across 100 bootstrap samples.

17.12. Summary

- A prediction model starts with a **clearly defined clinical question**, not with data analysis.
- **Sample size** should be determined prospectively using the Riley et al. criteria, not just the EPV rule of thumb.
- **Missing data** should be handled with multiple imputation, not deletion.
- **Variable selection** should be driven by clinical knowledge. Stepwise methods are harmful.
- **Overfitting** is the central challenge; combat it with adequate sample size and shrinkage.
- **Penalised regression** (ridge, LASSO, elastic net) embeds shrinkage into the estimation process.
- The final model should be **internally validated** before any claims about performance.

17.13. References and Further Reading

The most comprehensive modern reference for clinical prediction model development is the textbook by Smits, van Kuijk, and Wynants (2026), which provides practical guidance grounded in decades of methodological research. Steyerberg's "Clinical Prediction Models" (2019, second edition, Springer) remains the foundational text, covering study design, missing data, variable selection, and model evaluation with exceptional rigour. For sample size calculations, Riley and colleagues published a series of papers in Statistics in Medicine (2020) establishing formal criteria that go well beyond the traditional events-per-variable rule; the `pmsampsize` R package implements these methods. Van Buuren's "Flexible Imputation of Missing Data" (2018, second edition, Chapman and Hall) is the definitive resource on multiple imputation, while the `mice` package in R provides the practical tools. Harrell's "Regression Modeling Strategies" (2015, second edition, Springer) is essential reading on variable selection, nonlinearity, and shrinkage in regression modelling. For penalised regression, Hastie, Tibshirani, and Friedman's "The Elements of Statistical Learning" (2009, second edition, Springer) provides the theoretical foundations, while the `glmnet` R package by Friedman, Hastie, and Tibshirani is the standard implementation.

18. Performance Assessment and Model Validation

18.1. Introduction

You have developed a clinical prediction model. Its coefficients look reasonable, the predictors make clinical sense, and the apparent performance seems promising. But how good is this model, really? And will it work on patients beyond your development dataset?

This chapter addresses these questions by covering the five domains of model performance — discrimination, calibration, overall performance, classification, and clinical utility — and the methods for internal and external validation. The structure follows the comprehensive framework proposed by Van Calster and colleagues (2025) in The Lancet Digital Health, which we strongly recommend as a companion reading.

The core message is this: **no single metric tells the full story.** A model with excellent discrimination (high AUC) can still produce dangerously miscalibrated risk predictions. A well-calibrated model might not discriminate well enough to be useful. Clinical utility must be assessed separately from statistical performance. You need the full picture.

18.2. The Five Performance Domains

Van Calster et al. (2025) organise model performance assessment into five complementary domains:

Domain	Question	Key Metrics
Discrimination	Can the model rank patients by risk?	C-statistic, AUC
Calibration	Are predicted probabilities accurate?	Calibration plot, O:E ratio, calibration slope
Overall performance	How close are predictions to observed outcomes?	Brier score, Nagelkerke R-squared
Classification	How well does the model classify at a threshold?	Sensitivity, specificity, PPV, NPV
Clinical utility	Does using the model improve clinical decisions?	Net benefit, decision curve analysis

We will examine each in detail.

18.3. Domain 1: Discrimination

18.3.1. The C-Statistic / AUC

Discrimination measures how well a model separates patients who experience the outcome from those who do not. The most common measure is the **concordance statistic (C-statistic)**, which for binary outcomes is equivalent to the area under the ROC curve (AUC) discussed in Chapter 14.

Recall the interpretation: the C-statistic is the probability that a randomly selected patient with the outcome has a higher predicted probability than a randomly selected patient without the outcome.

18.3.2. R

```
library(rms)

# Simulate clinical data: predicting 30-day mortality after stroke
set.seed(2024)
n <- 1500

stroke_data <- data.frame(
  age = round(rnorm(n, 72, 12)),
  nihss = round(pmax(0, rnorm(n, 8, 6))),       # NIH Stroke Scale
  glucose = round(rnorm(n, 140, 50)),
  afib = rbinom(n, 1, 0.25),                     # Atrial fibrillation
  thrombolysis = rbinom(n, 1, 0.30)
)

# True model for 30-day mortality
lp <- -5 + 0.04 * stroke_data$age +
  0.12 * stroke_data$nihss +
  0.003 * stroke_data$glucose +
  0.3 * stroke_data$afib -
  0.5 * stroke_data$thrombolysis

stroke_data$death_30d <- rbinom(n, 1, plogis(lp))
cat("30-day mortality rate:", mean(stroke_data$death_30d), "\n")
```

```r
# Fit prediction model
dd <- datadist(stroke_data)
options(datadist = "dd")

fit <- lrm(death_30d ~ age + nihss + glucose + afib + thrombolysis,
           data = stroke_data, x = TRUE, y = TRUE)

cat("C-statistic (apparent):", fit$stats["C"], "\n")
```

18.3.3. Python

```python
import numpy as np
import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_auc_score
from scipy.special import expit

np.random.seed(2024)
n = 1500

stroke_data = pd.DataFrame({
    "age": np.round(np.random.normal(72, 12, n)),
    "nihss": np.round(np.maximum(0, np.random.normal(8, 6, n))),
    "glucose": np.round(np.random.normal(140, 50, n)),
    "afib": np.random.binomial(1, 0.25, n),
    "thrombolysis": np.random.binomial(1, 0.30, n)
})

lp = (-5 + 0.04 * stroke_data["age"] +
      0.12 * stroke_data["nihss"] +
      0.003 * stroke_data["glucose"] +
      0.3 * stroke_data["afib"] -
```

```
      0.5 * stroke_data["thrombolysis"])

stroke_data["death_30d"] = np.random.binomial(1, expit(lp))
print(f"30-day mortality rate: {stroke_data['death_30d'].mean():.3f}")

predictors = ["age", "nihss", "glucose", "afib", "thrombolysis"]
X = stroke_data[predictors].values
y = stroke_data["death_30d"].values

model = LogisticRegression(max_iter=5000, random_state=42)
model.fit(X, y)

pred_prob = model.predict_proba(X)[:, 1]
auc = roc_auc_score(y, pred_prob)
print(f"C-statistic (apparent): {auc:.3f}")
```

What the code shows. This simulates a stroke cohort, fits a model for 30-day mortality, and prints the C-statistic. The number to read is that final C-statistic: 0.5 means the model is no better than a coin toss at ranking patients, while 1.0 means it perfectly ranks every patient who died above every patient who survived. In clinical practice values around 0.7–0.8 are common and often useful. This is the *apparent* C-statistic — measured on the same data the model was built from — so treat it as an optimistic upper bound until it has been validated. Note that discrimination only asks whether the *ranking* is right; it says nothing about whether the predicted probabilities themselves are accurate, which is what calibration (next section) checks.

18.3.4. Limitations of the C-Statistic

The C-statistic has important limitations that Van Calster et al. (2025) highlight:

1. **Insensitive to calibration:** A model can perfectly rank patients (C = 1.0) but assign completely wrong probabilities.
2. **Context-blind:** It weights all parts of the ROC curve equally, even clinically irrelevant regions.
3. **Hard to improve:** In many clinical domains, the achievable C-statistic is inherently limited. Adding a new biomarker might genuinely improve predictions but barely move the C-statistic.
4. **Misleading comparisons:** A C-statistic of 0.75 in one population is not comparable to 0.75 in another if the case-mix differs.

18.4. Domain 2: Calibration

18.4.1. Why Calibration Matters More Than You Think

Calibration assesses whether predicted probabilities match observed outcomes. If your model says a patient has a 20% risk, do approximately 20 out of 100 such patients actually experience the outcome?

This matters enormously in clinical practice. When a clinician tells a patient "your model-predicted 10-year cardiovascular risk is 15%," the patient and clinician rely on that number being accurate — not just the patient's relative ranking. **Risk-based treatment decisions require calibrated predictions.**

18.4.2. Calibration-in-the-Large: The O:E Ratio

The simplest calibration check is the **observed-to-expected (O:E) ratio:**

$$\text{O:E ratio} = \frac{\text{Observed event rate}}{\text{Mean predicted probability}}$$

An O:E ratio of 1 indicates perfect calibration-in-the-large. An O:E ratio of 1.5 means the model underestimates risk by 50%.

18.4.3. Calibration Slope

The **calibration slope** is obtained by regressing the outcome on the linear predictor (log-odds of predictions). A calibration slope of 1 indicates perfect calibration. A slope less than 1 indicates overfitting (predictions are too extreme — high risks are too high, low risks are too low). A slope greater than 1 suggests the opposite.

18.4.4. Calibration Plots

The most informative way to assess calibration is visually, through a **calibration plot.** This plots predicted probabilities (x-axis) against observed proportions (y-axis). A perfectly calibrated model follows the 45-degree diagonal.

Two types are common:

- **Grouped calibration plot:** Patients are divided into groups (e.g., deciles) by predicted probability, and the observed proportion within each group is plotted.
- **Smoothed calibration plot:** A smooth curve (e.g., loess or spline) is fit to the predicted vs. observed data.

18.4.5. R

```
library(rms)

# Use the stroke model from above
pred_prob <- predict(fit, type = "fitted")
```

```r
# Grouped calibration plot
cal <- val.prob(pred_prob, stroke_data$death_30d,
                m = 100, cex = 0.5,
                main = "Calibration Plot: 30-Day Stroke Mortality Model")
```

```r
# Calculate calibration metrics manually
obs_rate <- mean(stroke_data$death_30d)
mean_pred <- mean(pred_prob)

cat("Observed event rate:", round(obs_rate, 3), "\n")
cat("Mean predicted probability:", round(mean_pred, 3), "\n")
cat("O:E ratio:", round(obs_rate / mean_pred, 3), "\n")

# Calibration slope
cal_model <- glm(stroke_data$death_30d ~ qlogis(pred_prob),
                 family = binomial)
cat("Calibration slope:", round(coef(cal_model)[2], 3), "\n")
cat("Calibration intercept:", round(coef(cal_model)[1], 3), "\n")

# Brier score
brier <- mean((pred_prob - stroke_data$death_30d)^2)
cat("Brier score:", round(brier, 4), "\n")
```

18.4.6. Python

```python
import numpy as np
import matplotlib.pyplot as plt
from sklearn.calibration import calibration_curve
from sklearn.metrics import brier_score_loss

# Use the stroke model predictions from above
```

```
pred_prob = model.predict_proba(X)[:, 1]
y_true = stroke_data["death_30d"].values

# Grouped calibration curve
prob_true, prob_pred = calibration_curve(y_true, pred_prob,
                                         n_bins=10, strategy="quantile")

fig, axes = plt.subplots(1, 2, figsize=(14, 6))

# Left panel: calibration plot
axes[0].plot(prob_pred, prob_true, "o-", color="steelblue", lw=2,
             markersize=8, label="Model")
axes[0].plot([0, 1], [0, 1], "--", color="grey", label="Perfect calibration")
axes[0].set_xlabel("Mean Predicted Probability")
axes[0].set_ylabel("Observed Proportion")
axes[0].set_title("Calibration Plot (Decile Groups)")
axes[0].legend()
axes[0].set_xlim(0, max(prob_pred) * 1.1)
axes[0].set_ylim(0, max(prob_true) * 1.1)

# Right panel: distribution of predicted probabilities
axes[1].hist(pred_prob[y_true == 0], bins=30, alpha=0.5, color="steelblue",
             label="No event", density=True)
axes[1].hist(pred_prob[y_true == 1], bins=30, alpha=0.5, color="coral",
             label="Event", density=True)
axes[1].set_xlabel("Predicted Probability")
axes[1].set_ylabel("Density")
axes[1].set_title("Distribution of Predicted Probabilities")
axes[1].legend()

plt.tight_layout()
plt.show()
```

```
# Calibration metrics
obs_rate = y_true.mean()
mean_pred = pred_prob.mean()
oe_ratio = obs_rate / mean_pred
brier = brier_score_loss(y_true, pred_prob)

print(f"\nCalibration metrics:")
print(f"  Observed event rate: {obs_rate:.3f}")
print(f"  Mean predicted probability: {mean_pred:.3f}")
print(f"  O:E ratio: {oe_ratio:.3f}")
print(f"  Brier score: {brier:.4f}")
```

What the code shows. This produces and quantifies a calibration plot, which answers the question: when the model says "20% risk," do about 20 in 100 such patients actually have the event? In R, `val.prob()` draws the plot and reports calibration statistics in one step. In Python the left panel plots observed proportion against mean predicted probability for groups of patients (here deciles), and the right panel shows how predicted risks are distributed among those with and without the event. Read the plot by comparing the model's curve to the 45-degree diagonal: points on the diagonal mean perfect calibration, points above mean the model underestimates risk, points below mean it overestimates. The printed metrics summarise the same thing numerically — the O:E ratio for overall calibration-in-the-large, the calibration slope for whether predictions are too extreme, and the Brier score for overall accuracy.

Calibration slope, in one line

The **calibration slope** comes from regressing the outcome on the model's own log-odds predictions. A slope of 1 is ideal. A slope below 1 (the usual sign of overfitting) means the predictions are too spread out — the high risks are too high and the low risks too low — and is exactly the kind of problem that shrinkage in the previous chapter is designed to prevent.

18.4.7. Interpreting Miscalibration

Common patterns and their clinical meaning:

- **O:E > 1 (underestimation):** The model systematically underestimates risk. Patients may not receive treatments they need.
- **O:E < 1 (overestimation):** The model systematically overestimates risk. Patients may receive unnecessary treatments.
- **Calibration slope < 1:** Predictions are too extreme. Patients at high predicted risk actually have lower risk than predicted, and patients at low predicted risk have higher risk. This is the hallmark of overfitting.
- **Calibration slope > 1:** Predictions are too moderate. This is less common and may occur after excessive shrinkage.

18.5. Domain 3: Overall Performance

The **Brier score** combines discrimination and calibration into a single measure:

$$\text{Brier score} = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2$$

where p_i is the predicted probability and y_i is the observed outcome (0 or 1). The Brier score ranges from 0 (perfect) to 1 (worst). For a reference, a model that always predicts the overall event rate achieves a Brier score equal to the prevalence times (1 minus the prevalence).

The **scaled Brier score** (or Brier skill score) expresses improvement over a non-informative model:

$$\text{Scaled Brier} = 1 - \frac{\text{Brier score}}{\text{Brier}_{\text{max}}}$$

where $\text{Brier}_{\text{max}} = \bar{y}(1 - \bar{y})$.

18.6. Domain 4: Classification

Classification metrics (sensitivity, specificity, PPV, NPV) were covered in Chapter 14. The key point for model validation is that classification performance depends entirely on the chosen threshold. When reporting classification metrics, **always specify the threshold and justify its choice based on the clinical context.**

18.7. Domain 5: Clinical Utility

18.7.1. Net Benefit and Decision Curve Analysis

A model might have good discrimination and calibration, yet using it may not actually improve clinical decisions. **Decision curve analysis (DCA)**, proposed by Vickers and Elkin (2006), addresses this by quantifying the **net benefit** of using the model compared to the default strategies of treating all patients or treating no patients.

The net benefit at a given threshold probability p_t is:

$$\text{Net benefit} = \frac{TP}{N} - \frac{FP}{N} \times \frac{p_t}{1 - p_t}$$

The term $\frac{p_t}{1-p_t}$ represents the exchange rate between false positives and true positives, determined by the threshold probability. If the threshold is 10%, you are willing to accept up to 9 false positives for every true positive identified.

18.7.2. R

```
library(dcurves)

# Add predicted probabilities to the dataset
stroke_data$pred_risk <- predict(fit, type = "fitted")

# Decision curve analysis
dca_result <- dca(death_30d ~ pred_risk,
                  data = stroke_data,
                  thresholds = seq(0, 0.5, by = 0.01))

plot(dca_result,
     smooth = TRUE,
     show_ggplot_code = FALSE) +
  ggplot2::ggtitle("Decision Curve Analysis:\n30-Day Stroke Mortality Model")
```

18.7.3. Python

```
import numpy as np
import matplotlib.pyplot as plt

def net_benefit(y_true, y_pred, threshold):
    """Calculate net benefit at a given threshold."""
    n = len(y_true)
    pred_pos = (y_pred >= threshold).astype(int)
```

```
    tp = np.sum((pred_pos == 1) & (y_true == 1))
    fp = np.sum((pred_pos == 1) & (y_true == 0))
    nb = tp / n - fp / n * (threshold / (1 - threshold))
    return nb

thresholds = np.arange(0.01, 0.51, 0.01)
pred_prob_model = model.predict_proba(X)[:, 1]
y_true = stroke_data["death_30d"].values
n_total = len(y_true)

# Net benefit for model
nb_model = [net_benefit(y_true, pred_prob_model, t) for t in thresholds]

# Net benefit for "treat all" strategy
nb_all = [(y_true.mean() - (1 - y_true.mean()) * t / (1 - t))
          for t in thresholds]

# Net benefit for "treat none" is always 0

plt.figure(figsize=(9, 6))
plt.plot(thresholds, nb_model, color="steelblue", lw=2,
         label="Prediction Model")
plt.plot(thresholds, nb_all, color="coral", lw=2, ls="--",
         label="Treat All")
plt.axhline(y=0, color="black", lw=1, label="Treat None")
plt.xlabel("Threshold Probability")
plt.ylabel("Net Benefit")
plt.title("Decision Curve Analysis: 30-Day Stroke Mortality Model")
plt.legend()
plt.ylim(-0.05, max(max(nb_model), max(nb_all)) * 1.1)
plt.tight_layout()
plt.show()
```

What the code shows. This computes net benefit across a range of decision thresholds and plots three curves: using the model, treating everyone, and treating no-one. In R, `dca()` from the `dcurves` package does the calculation; the Python version writes out the net-benefit formula explicitly in the `net_benefit()` function so you can see the arithmetic. The result you care about is whether the model's curve sits *above* both the "treat all" and "treat none" lines over the thresholds clinicians would actually use — if it does, the model improves decisions there; if it dips below either default, the model adds nothing at that threshold and you may as well follow the simpler default strategy.

> **i** Net benefit in clinician's terms
>
> **Net benefit** puts true positives and false positives on a common scale so they can be added up. The threshold probability you pick reflects how you trade them off: a 10% threshold means you would accept treating 9 patients unnecessarily to catch 1 who truly needs it. Net benefit then counts the true positives the model finds and subtracts the false positives, weighted by that exchange rate. Unlike the C-statistic, it directly reflects the clinical consequences of acting on the model.

18.7.4. Interpreting the Decision Curve

The decision curve shows net benefit (y-axis) across a range of threshold probabilities (x-axis). Three strategies are compared:

- **Treat none** (horizontal line at 0): no patients receive the intervention.
- **Treat all** (declining curve): all patients receive the intervention regardless of risk.
- **Model-guided** (the model's curve): only patients above the threshold receive the intervention.

The model is clinically useful at thresholds where its net benefit exceeds both the "treat all" and "treat none" lines. If the model curve is always below one of the default strategies, the model adds no value.

18.8. Internal Validation

18.8.1. Why Internal Validation Is Necessary

As discussed in Chapter 17, apparent performance (measured on the training data) is optimistically biased. Internal validation estimates and corrects for this optimism. It does not replace external validation, but it provides a more honest assessment of likely performance.

18.8.2. Bootstrap Optimism Correction

The recommended approach (Steyerberg 2019, Van Calster et al. 2025) is **bootstrap optimism correction:**

1. Draw a bootstrap sample (with replacement) from the original data.
2. Develop the model in the bootstrap sample (including all modelling decisions).
3. Measure performance in the bootstrap sample (apparent bootstrap performance).
4. Apply the bootstrap model to the original data (test performance).
5. Optimism = apparent bootstrap performance minus test performance.
6. Repeat steps 1–5 many times (e.g., 200+).
7. Average optimism across repetitions.
8. Corrected performance = original apparent performance minus average optimism.

18.8.3. R

```r
library(rms)

# Using the stroke model
fit <- lrm(death_30d ~ age + nihss + glucose + afib + thrombolysis,
           data = stroke_data, x = TRUE, y = TRUE)

# Bootstrap validation with 200 resamples
set.seed(42)
val <- validate(fit, B = 200)

cat("Bootstrap Validation Results:\n")
print(val)

# Extract optimism-corrected C-statistic
dxy_corrected <- val["Dxy", "index.corrected"]
c_corrected <- (dxy_corrected + 1) / 2

cat("\nApparent C-statistic:", fit$stats["C"], "\n")
cat("Optimism:", val["Dxy", "optimism"] / 2, "\n")
cat("Optimism-corrected C-statistic:", c_corrected, "\n")

# Calibration slope from bootstrap
cat("\nCalibration slope (optimism-corrected):",
    val["Slope", "index.corrected"], "\n")
```

18.8.4. Python

```python
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_auc_score, brier_score_loss
```

```
np.random.seed(42)

predictors = ["age", "nihss", "glucose", "afib", "thrombolysis"]
X = stroke_data[predictors].values
y = stroke_data["death_30d"].values
n = len(y)

# Apparent performance
full_model = LogisticRegression(max_iter=5000, random_state=42)
full_model.fit(X, y)
pred_full = full_model.predict_proba(X)[:, 1]
apparent_auc = roc_auc_score(y, pred_full)
apparent_brier = brier_score_loss(y, pred_full)

# Bootstrap optimism correction
n_boot = 200
optimism_auc = []
optimism_brier = []

for b in range(n_boot):
    boot_idx = np.random.choice(n, n, replace=True)
    X_boot, y_boot = X[boot_idx], y[boot_idx]

    boot_model = LogisticRegression(max_iter=5000, random_state=42)
    boot_model.fit(X_boot, y_boot)

    # Apparent performance on bootstrap sample
    pred_boot = boot_model.predict_proba(X_boot)[:, 1]
    auc_boot = roc_auc_score(y_boot, pred_boot)
    brier_boot = brier_score_loss(y_boot, pred_boot)

    # Test performance on original data
    pred_orig = boot_model.predict_proba(X)[:, 1]
```

```
    auc_orig = roc_auc_score(y, pred_orig)
    brier_orig = brier_score_loss(y, pred_orig)

    optimism_auc.append(auc_boot - auc_orig)
    optimism_brier.append(brier_boot - brier_orig)

mean_opt_auc = np.mean(optimism_auc)
mean_opt_brier = np.mean(optimism_brier)
corrected_auc = apparent_auc - mean_opt_auc
corrected_brier = apparent_brier - mean_opt_brier

print(f"Apparent AUC:                  {apparent_auc:.3f}")
print(f"Mean optimism (AUC):           {mean_opt_auc:.4f}")
print(f"Corrected AUC:                 {corrected_auc:.3f}")
print(f"\nApparent Brier:                {apparent_brier:.4f}")
print(f"Mean optimism (Brier):         {mean_opt_brier:.4f}")
print(f"Corrected Brier:               {corrected_brier:.4f}")
```

What the code shows. This implements bootstrap optimism correction, the recommended way to get an honest internal estimate of performance. In R, `validate(fit, B = 200)` does everything: it reports the apparent statistics, the average optimism, and the optimism-corrected versions. (Note that `rms` works in Somers' Dxy, which relates to the C-statistic by C = (Dxy + 1) / 2 — that is why the code converts it.) The Python version spells out the loop: for each of 200 bootstrap resamples it refits the model, measures performance on the resample, then measures the *same* model on the original data; the gap between the two is the optimism. The numbers to compare are the apparent versus corrected AUC. A large drop (say from 0.85 to 0.72) signals serious overfitting and tells you the apparent figure should never have been reported on its own; a small drop means the model is reasonably stable.

18.8.5. Cross-Validation

Cross-validation (typically 10-fold) provides an alternative internal validation approach. The data is split into k folds; each fold serves once as the validation set while the remaining folds are used for model development. The average performance across folds estimates the model's performance on new data.

Cross-validation is computationally simpler but has limitations for prediction model development: it does not directly yield a single final model or a clear calibration slope correction. Bootstrap optimism correction is generally preferred for clinical prediction models.

18.9. External Validation

18.9.1. Types of External Validation

Internal validation tells you how optimistic your apparent performance is. **External validation** tells you whether the model works on genuinely new data — the real test. Van Calster et al. (2025) and Smits et al. (2026) distinguish:

- **Temporal validation:** Same setting, different time period. Example: model developed on 2015–2019 data, validated on 2020–2022 data. This captures changes in clinical practice and patient populations over time.
- **Geographical validation:** Different hospital or region, same time period. This tests transportability across settings.
- **Domain validation:** Different clinical context entirely (e.g., model developed in a tertiary care centre, validated in primary care). This is the most stringent test.

18.9.2. What to Report in External Validation

Following Van Calster et al. (2025), an external validation study should report at minimum:

1. **C-statistic** with 95% confidence interval
2. **Calibration plot** (both grouped and smoothed)
3. **O:E ratio** with 95% confidence interval
4. **Calibration slope**
5. **Net benefit** (decision curve) at clinically relevant thresholds
6. **Distribution of predicted probabilities** (to understand the range and spread)
7. **Performance by subgroups** (age, sex, comorbidities, etc.)

18.10. Model Updating

18.10.1. When Performance Degrades

External validation often reveals that a model does not perform as well in new settings as it did internally. This does not necessarily mean the model is useless. **Model updating** can adapt an existing model to a new population without developing a completely new model.

18.10.2. Updating Strategies

From least to most complex:

1. **Recalibration-in-the-large:** Adjust only the intercept. Corrects for differences in baseline risk (e.g., higher mortality rate in the new population).

2. **Logistic recalibration:** Adjust the intercept and apply a correction to the calibration slope. This corrects for both level and spread of predictions.

3. **Model revision:** Re-estimate some or all coefficients, or add new predictors. This requires larger sample sizes in the new population.

18.10.3. R

```
# Simulate external validation data (different setting)
set.seed(99)
n_ext <- 800

# New population: older, sicker
ext_data <- data.frame(
  age = round(rnorm(n_ext, 78, 10)),   # Older
  nihss = round(pmax(0, rnorm(n_ext, 11, 7))),  # Higher severity
  glucose = round(rnorm(n_ext, 155, 55)),
  afib = rbinom(n_ext, 1, 0.35),
  thrombolysis = rbinom(n_ext, 1, 0.20)  # Less thrombolysis
)

lp_ext <- -5 + 0.04 * ext_data$age +
  0.12 * ext_data$nihss +
  0.003 * ext_data$glucose +
  0.3 * ext_data$afib -
  0.5 * ext_data$thrombolysis

ext_data$death_30d <- rbinom(n_ext, 1, plogis(lp_ext))

# Apply original model to external data
ext_data$pred_original <- predict(fit, newdata = ext_data, type = "fitted")
```

```
cat("External validation (before updating):\n")
cat("  Observed mortality:", mean(ext_data$death_30d), "\n")
cat("  Mean predicted:", mean(ext_data$pred_original), "\n")
cat("  O:E ratio:", round(mean(ext_data$death_30d) / mean(ext_data$pred_original), 3), "\n")

# Calibration slope in external data
lp_orig <- qlogis(ext_data$pred_original)
cal_ext <- glm(death_30d ~ lp_orig, data = ext_data, family = binomial)
cat("  Calibration slope:", round(coef(cal_ext)[2], 3), "\n")
cat("  Calibration intercept:", round(coef(cal_ext)[1], 3), "\n")

# Recalibration: update intercept and slope
ext_data$pred_recalibrated <- predict(cal_ext, type = "response")

cat("\nAfter logistic recalibration:\n")
cat("  Mean predicted:", round(mean(ext_data$pred_recalibrated), 3), "\n")
cat("  O:E ratio:", round(mean(ext_data$death_30d) / mean(ext_data$pred_recalibrated), 3), '
```

18.10.4. Python

```
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_auc_score

np.random.seed(99)
n_ext = 800

ext_data = {
    "age": np.round(np.random.normal(78, 10, n_ext)),
    "nihss": np.round(np.maximum(0, np.random.normal(11, 7, n_ext))),
    "glucose": np.round(np.random.normal(155, 55, n_ext)),
    "afib": np.random.binomial(1, 0.35, n_ext),
```

```
    "thrombolysis": np.random.binomial(1, 0.20, n_ext)
}

lp_ext = (-5 + 0.04 * ext_data["age"] +
          0.12 * ext_data["nihss"] +
          0.003 * ext_data["glucose"] +
          0.3 * ext_data["afib"] -
          0.5 * ext_data["thrombolysis"])

y_ext = np.random.binomial(1, expit(lp_ext))

# Apply original model
X_ext = np.column_stack([ext_data[k] for k in predictors])
pred_original = full_model.predict_proba(X_ext)[:, 1]

obs_rate = y_ext.mean()
mean_pred = pred_original.mean()
print(f"External validation (before updating):")
print(f"  Observed mortality: {obs_rate:.3f}")
print(f"  Mean predicted: {mean_pred:.3f}")
print(f"  O:E ratio: {obs_rate / mean_pred:.3f}")
print(f"  AUC: {roc_auc_score(y_ext, pred_original):.3f}")

# Logistic recalibration: regress outcome on logit(predictions)
from scipy.special import logit
lp_original = logit(np.clip(pred_original, 1e-6, 1 - 1e-6))

recal_model = LogisticRegression(max_iter=5000)
recal_model.fit(lp_original.reshape(-1, 1), y_ext)

pred_recalibrated = recal_model.predict_proba(lp_original.reshape(-1, 1))[:,

print(f"\nAfter logistic recalibration:")
```

```
print(f"  Mean predicted: {pred_recalibrated.mean():.3f}")
print(f"  O:E ratio: {y_ext.mean() / pred_recalibrated.mean():.3f}")
print(f"  Calibration slope: {recal_model.coef_[0][0]:.3f}")
print(f"  Calibration intercept: {recal_model.intercept_[0]:.3f}")
```

What the code shows. This simulates applying the stroke model to a *new, sicker* population and then repairs it. The first block of output is the external check before any updating: compare the observed mortality to the mean predicted risk, look at the O:E ratio (far from 1 means miscalibration), and note the calibration slope. Because the new population is older and more severe, the original model typically underestimates risk, so the O:E ratio comes out above 1. The code then performs logistic recalibration — regressing the outcome on the original model's predictions to re-estimate an intercept and slope — and reprints the metrics, which should now sit much closer to perfect calibration. The clinical message: a model that miscalibrates in a new setting is often not broken, just mis-tuned, and a light-touch update can rescue it without collecting enough data to build a model from scratch.

18.11. Recommended Reporting: The Minimum Set

Based on Van Calster et al. (2025) and Smits et al. (2026), every prediction model evaluation should include at minimum:

1. **AUROC (C-statistic)** with confidence interval — for discrimination
2. **Calibration plot** (smoothed and/or grouped) — for calibration
3. **Decision curve** — for clinical utility
4. **Risk distribution plot** — to show the spread of predicted probabilities

These four visualisations together give a comprehensive picture. A model may look excellent on one plot and problematic on another — which is exactly why you need all four.

18.11.1. R

```
library(pROC)

pred_prob <- predict(fit, type = "fitted")
y_obs <- stroke_data$death_30d

par(mfrow = c(2, 2))

# Panel 1: ROC curve
roc_obj <- roc(y_obs, pred_prob, quiet = TRUE)
plot(roc_obj, col = "steelblue", lwd = 2,
     main = "A. Discrimination (ROC Curve)",
     print.auc = TRUE, print.auc.y = 0.4)

# Panel 2: Calibration plot
val.prob(pred_prob, y_obs, m = 100, cex = 0.5,
         main = "B. Calibration Plot")

# Panel 3: Risk distribution
hist(pred_prob[y_obs == 0], breaks = 30, col = rgb(0.4, 0.6, 0.8, 0.5),
     main = "C. Risk Distribution",
     xlab = "Predicted Probability", freq = FALSE, ylim = c(0, 12))
hist(pred_prob[y_obs == 1], breaks = 30, col = rgb(0.9, 0.4, 0.4, 0.5),
     add = TRUE, freq = FALSE)
legend("topright", legend = c("No event", "Event"),
       fill = c(rgb(0.4, 0.6, 0.8, 0.5), rgb(0.9, 0.4, 0.4, 0.5)))
```

```r
# Panel 4: Simplified decision curve
thresholds <- seq(0.01, 0.5, by = 0.01)
nb_model <- nb_all <- numeric(length(thresholds))
n_total <- length(y_obs)

for (i in seq_along(thresholds)) {
  t <- thresholds[i]
  pred_pos <- as.numeric(pred_prob >= t)
  tp <- sum(pred_pos == 1 & y_obs == 1)
  fp <- sum(pred_pos == 1 & y_obs == 0)
  nb_model[i] <- tp / n_total - fp / n_total * (t / (1 - t))
  nb_all[i] <- mean(y_obs) - (1 - mean(y_obs)) * (t / (1 - t))
}

plot(thresholds, nb_model, type = "l", lwd = 2, col = "steelblue",
     main = "D. Decision Curve Analysis",
     xlab = "Threshold Probability", ylab = "Net Benefit",
     ylim = c(-0.05, max(nb_model, nb_all) * 1.1))
lines(thresholds, nb_all, lwd = 2, col = "coral", lty = 2)
abline(h = 0, col = "black")
legend("topright", legend = c("Model", "Treat All", "Treat None"),
       col = c("steelblue", "coral", "black"),
       lty = c(1, 2, 1), lwd = 2, cex = 0.8)

par(mfrow = c(1, 1))
```

18.11.2. Python

```python
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import roc_curve, auc
from sklearn.calibration import calibration_curve
```

```
pred_prob = full_model.predict_proba(X)[:, 1]
y_obs = stroke_data["death_30d"].values

fig, axes = plt.subplots(2, 2, figsize=(12, 10))

# Panel A: ROC curve
fpr, tpr, _ = roc_curve(y_obs, pred_prob)
roc_auc = auc(fpr, tpr)
axes[0, 0].plot(fpr, tpr, color="steelblue", lw=2,
                label=f"AUC = {roc_auc:.3f}")
axes[0, 0].plot([0, 1], [0, 1], "--", color="grey")
axes[0, 0].set_xlabel("1 - Specificity")
axes[0, 0].set_ylabel("Sensitivity")
axes[0, 0].set_title("A. Discrimination (ROC Curve)")
axes[0, 0].legend()

# Panel B: Calibration plot
prob_true, prob_pred = calibration_curve(y_obs, pred_prob, n_bins=10,
                                          strategy="quantile")
axes[0, 1].plot(prob_pred, prob_true, "o-", color="steelblue", lw=2)
axes[0, 1].plot([0, 1], [0, 1], "--", color="grey")
axes[0, 1].set_xlabel("Mean Predicted Probability")
axes[0, 1].set_ylabel("Observed Proportion")
axes[0, 1].set_title("B. Calibration Plot")

# Panel C: Risk distribution
axes[1, 0].hist(pred_prob[y_obs == 0], bins=30, alpha=0.5,
                color="steelblue", label="No event", density=True)
axes[1, 0].hist(pred_prob[y_obs == 1], bins=30, alpha=0.5,
                color="coral", label="Event", density=True)
axes[1, 0].set_xlabel("Predicted Probability")
axes[1, 0].set_ylabel("Density")
axes[1, 0].set_title("C. Risk Distribution")
```

```
axes[1, 0].legend()

# Panel D: Decision curve
thresholds = np.arange(0.01, 0.51, 0.01)
nb_model = []
nb_all = []
n_total = len(y_obs)

for t in thresholds:
    pred_pos = (pred_prob >= t).astype(int)
    tp = np.sum((pred_pos == 1) & (y_obs == 1))
    fp = np.sum((pred_pos == 1) & (y_obs == 0))
    nb_model.append(tp / n_total - fp / n_total * (t / (1 - t)))
    nb_all.append(y_obs.mean() - (1 - y_obs.mean()) * (t / (1 - t)))

axes[1, 1].plot(thresholds, nb_model, color="steelblue", lw=2,
                label="Model")
axes[1, 1].plot(thresholds, nb_all, color="coral", lw=2, ls="--",
                label="Treat All")
axes[1, 1].axhline(y=0, color="black", lw=1, label="Treat None")
axes[1, 1].set_xlabel("Threshold Probability")
axes[1, 1].set_ylabel("Net Benefit")
axes[1, 1].set_title("D. Decision Curve Analysis")
axes[1, 1].legend(fontsize=9)

plt.tight_layout()
plt.show()
```

What the code shows. This assembles the four plots that, together, give the complete picture of a model's performance and that you should report as a set. Panel A (ROC curve) shows discrimination; B (calibration plot) shows whether predicted risks are accurate; C (risk distribution) shows whether the model produces a useful spread of predictions across patients with and without the event; and D (decision curve) shows clinical utility. The reason to view all four at once is that a model can look excellent on one and alarming on another — strong discrimination with poor calibration, for instance — and only the combined display reveals that. When you read someone else's model paper, check whether they reported all four; a paper that shows only the AUC has told you only a quarter of the story.

18.12. Exercises

18.12.1. Exercise 1: Calibration Assessment

Using the stroke mortality model developed in this chapter:

a. Create a calibration plot using deciles of predicted risk. Is the model well-calibrated?
b. Calculate the O:E ratio. What does it tell you?
c. Calculate the calibration slope. Is there evidence of overfitting?
d. Apply bootstrap optimism correction. How much does the C-statistic decrease? How much does the calibration slope decrease?

18.12.2. Exercise 2: External Validation Simulation

18.12.3. R

```
# Create three external validation populations that differ from development:
# Population A: Same demographics, 3 years later (temporal)
# Population B: Different hospital, younger patients (geographical)
# Population C: Primary care setting with lower severity (domain)

# For each population:
# a. Calculate C-statistic, O:E ratio, calibration slope
```

```
# b. Create calibration plots
# c. Determine which population shows worst calibration and explain why
# d. Perform logistic recalibration and show the improvement
```

18.12.4. Python

```
# Same exercise as R tab - create three external populations
# and assess model performance in each
```

18.12.5. Exercise 3: Decision Curve Interpretation

Consider a model for predicting preeclampsia in pregnant women. The model has an AUC of 0.82 and good calibration. You plot the decision curve.

a. At what range of threshold probabilities is the model useful?
b. A colleague argues that the model should not be used because the AUC is "only" 0.82. Using the decision curve, construct a counter-argument.
c. How would the decision curve change if the follow-up action (closer monitoring) were very low-cost versus very high-cost?

18.12.6. Exercise 4: Complete Evaluation

Choose a clinical prediction model from the literature (e.g., QRISK3, Wells score, APACHE II). Using published validation data:

a. Report the C-statistic with confidence interval.
b. Describe the calibration (if a calibration plot is available).
c. Is there a decision curve analysis? If not, what threshold range would be clinically relevant?

d. Has the model been externally validated? In what populations?
e. Based on your assessment, would you recommend implementing this model in your setting?

18.13. References and Further Reading

The primary framework for this chapter is drawn from Van Calster, McLernon, van Smeden, Worster,"; Steyerberg, and colleagues (2025) in The Lancet Digital Health, whose guide to assessing the performance of prediction models is the most comprehensive and up-to-date reference available on this topic, covering all five performance domains and providing clear recommendations for what to report. The textbook by Smits, van Kuijk, and Wynants (2026) complements this with a highly accessible treatment of internal and external validation, model updating, and practical guidance for applied researchers. Decision curve analysis was introduced by Vickers and Elkin (2006) in Medical Decision Making and is now considered an essential component of model evaluation; the `dcurves` R package and accompanying tutorials make it straightforward to implement. For calibration assessment specifically, the work of Van Calster, Nieboer, Vergouwe, De Cock, Pencina, and Steyerberg (2016) in the Journal of Clinical Epidemiology provides an excellent treatment of different calibration measures and their interpretation. Bootstrap validation methodology is thoroughly described in Steyerberg's"Clinical Prediction Models" (2019, Springer) and in Harrell's "Regression Modeling Strategies" (2015, Springer). The Brier score and its decomposition were formalised by Brier (1950) in Monthly Weather Review, and its application to clinical prediction models is discussed in Steyerberg and colleagues (2010) in Epidemiology.

19. Reporting Standards, TRIPOD+AI, and Clinical Impact

20. Reporting Standards, TRIPOD+AI, and Clinical Impact

20.1. Introduction

Thousands of clinical prediction models are published every year. Yet only a small fraction are ever validated externally, and fewer still make it into clinical practice. One of the major reasons is poor reporting: papers that omit critical methodological details, overstate performance, or fail to describe the model in enough detail for anyone to reproduce or implement it. In 2024, the TRIPOD+AI statement was published to address this problem head-on.

This chapter walks you through the current reporting standards for clinical prediction models, explains why they matter, and gives you the tools to produce work that meets the expectations of top medical journals.

20.2. Why reporting standards matter

Consider trying to use a published prediction model in your own clinical setting. You would need to know:

- What predictors are included, and exactly how they were measured
- What outcome was predicted, and at what time horizon

- How missing data were handled
- How the model was validated, and on what population
- The model's calibration, not just its discrimination
- Whether the model would be feasible to use in your context

Remarkably, many published prediction model studies omit several of these details. A systematic review by Collins et al. found that reporting quality was poor across most published prediction model studies, motivating the development of formal reporting guidelines.

20.3. The TRIPOD+AI statement

The **Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis** (TRIPOD) guideline was first published in 2015. In 2024, it was updated as **TRIPOD+AI** to address machine learning and AI-based models.

In plain terms, TRIPOD+AI is a checklist of things every prediction-model paper should report so that a reader can judge whether the model is trustworthy and could actually be used. Think of it the same way you think of CONSORT for randomised trials or STROBE for observational studies: not a hurdle, but a shared expectation that makes your work credible and reproducible. Journals increasingly require it, and reviewers will look for it.

TRIPOD+AI applies to studies that develop, validate, or update a prediction model, whether based on regression, machine learning, or deep learning. The key additions in the +AI version include:

- **Fairness evaluation**: assessing whether the model performs equitably across demographic groups
- **Open science**: sharing code, data (where possible), and the model itself

- **Handling of AI-specific issues**: hyperparameter tuning, computational requirements, software dependencies

20.3.1. Key TRIPOD+AI items

The full checklist contains items across title, abstract, introduction, methods, results, and discussion. Here are some of the most commonly missed items:

Study design and participants:

- Describe the study design (e.g., cohort, case-control, registry)
- Specify the eligibility criteria clearly
- Report dates of recruitment and follow-up

Predictors and outcome:

- List all candidate predictors, how they were defined, and when they were measured
- Define the outcome precisely, including the time horizon for prognostic models

Missing data:

- Report the amount of missing data for each variable
- Describe the method used to handle missing data (complete case analysis, imputation)
- If multiple imputation was used, report the number of imputations and imputation model

Model development:

- Describe the modelling approach (logistic regression, random forest, etc.)

- Report how continuous predictors were handled (linear terms, splines, categorised)
- For ML models: report hyperparameter tuning strategy, software, and random seeds

Model performance:

- Report discrimination (e.g., C-statistic) with confidence intervals
- Report calibration (calibration plot, calibration slope and intercept)
- Report clinical utility (decision curve analysis or net benefit) where relevant

Validation:

- Describe the validation approach (internal, external, temporal)
- Report performance in the validation data, not just the development data

If some of these terms are still fuzzy, here is the quick clinician's translation. *Discrimination* (the C-statistic) is whether the model ranks a patient who has the outcome above one who does not — like a diagnostic test's ability to separate cases from non-cases. *Calibration* is whether the predicted risks are numerically honest: when the model says "20% risk", do roughly 20 in 100 such patients actually have the event? A model can discriminate well yet be badly calibrated (consistently over- or under-stating risk), which is dangerous if clinicians act on the actual numbers. *Clinical utility* (decision curve analysis / net benefit) goes one step further and asks whether using the model to guide decisions does more good than harm at a realistic risk threshold. A complete paper reports all three, in the validation data, not just the development data where performance is always flattering.

Exercise 12.1: TRIPOD+AI audit

Find a recently published clinical prediction model paper (try searching PubMed for "prediction model" in your area of interest). Using the

TRIPOD+AI checklist, evaluate how many items the paper reports adequately.
Focus on:

1. Is the outcome clearly defined with a time horizon?
2. Is calibration reported, or only discrimination?
3. How was missing data handled?
4. Were continuous predictors appropriately modelled (splines) or just categorised?
5. Is the model available for others to use (coefficients, code, web calculator)?

Write a brief (one paragraph) critical appraisal.

20.4. From model to bedside: the implementation gap

Even well-developed, well-validated models face an uphill battle to clinical implementation. Smits, van Kuijk & Wynants (2026) dedicate five chapters of their book to this topic, covering the journey from model selection through innovation development, impact evaluation, and implementation.

20.4.1. Key barriers to implementation

1. **The model doesn't address a real clinical need.** Models built for academic interest rather than to solve a genuine decision problem rarely get implemented.

2. **The model isn't embedded in clinical workflow.** A model that requires a clinician to manually enter 15 variables into a spreadsheet will not be used, regardless of how accurate it is.

3. **Performance hasn't been demonstrated in the target population.** External validation in the specific setting where the model will be used is essential.

4. **Clinicians don't trust the model.** Transparency, explainability, and evidence of clinical utility (not just statistical performance) build trust.

5. **There is no plan for maintenance.** Models can degrade over time as populations and practices change. A plan for monitoring and updating is essential.

20.4.2. The prediction model-based innovation (PMBI) framework

Smits et al. (2026) introduce the concept of a PMBI: the complete clinical tool that wraps around a prediction model, including:

- The user interface (how predictions are displayed)
- Decision support (what actions are recommended at different risk levels)
- Integration with electronic health records
- Training materials for end users
- A monitoring and updating plan

> **Note**
>
> **Key insight:** Building a good prediction model is necessary but not sufficient for clinical impact. The model must be embedded in a workable clinical tool, validated in the target setting, and demonstrated to improve patient outcomes.

20.5. Risk communication

Presenting predicted probabilities to patients and clinicians is not straightforward. Research in health literacy shows that:

- **Frequencies are easier to understand than probabilities.** "3 out of 100 patients like you" is clearer than "3% probability."
- **Visual aids help.** Icon arrays (showing 100 faces with 3 highlighted) are effective.
- **Framing matters.** "97% chance of survival" feels different from "3% chance of death" — both are true.
- **Uncertainty should be communicated.** Presenting a point estimate without a confidence interval overstates precision.

20.5.1. Presenting results to different audiences

Audience	Recommended format
Patients	Frequencies, icon arrays, plain language
Clinicians	Risk categories with action thresholds, decision curves
Researchers	Full performance metrics, calibration plots, code
Policymakers	Population-level impact, cost-effectiveness

20.6. Ethical considerations

20.6.1. Algorithmic bias

Prediction models can perpetuate or amplify existing health disparities if:

- Training data underrepresents certain populations

- Predictors serve as proxies for protected characteristics (e.g., postcode as proxy for race/ethnicity)
- Performance is not evaluated across subgroups

TRIPOD+AI specifically requires fairness evaluation: reporting model performance stratified by relevant demographic groups.

20.6.2. Informed consent and transparency

Patients have a right to know when a prediction model is being used in their care, what data it uses, and how its output influences clinical decisions. This is particularly important for models that inform treatment decisions or resource allocation.

20.6.3. The EU AI Act and regulatory considerations

The European Union's AI Act (2024) classifies medical AI applications as "high risk" and imposes requirements for transparency, human oversight, and documentation. Clinical prediction models that qualify as medical devices may need regulatory approval (e.g., CE marking in the EU, FDA clearance in the US).

20.7. Writing a statistical methods section

A well-written methods section for a prediction model study should include:

1. **Study design and setting** (one paragraph)
2. **Participants** — eligibility criteria, dates, sample size (one paragraph)
3. **Predictors** — list, definitions, measurement timing (one paragraph)

4. **Outcome** — definition, ascertainment, time horizon (one paragraph)
5. **Missing data** — amount, mechanism assumed, handling method (one paragraph)
6. **Model development** — method, how continuous variables were handled, variable selection approach (one paragraph)
7. **Model performance** — discrimination, calibration, clinical utility measures (one paragraph)
8. **Validation** — internal and/or external validation approach (one paragraph)
9. **Software** — R/Python version, key packages, random seed (one sentence)

Exercise 12.2: Write a methods section

Using the prediction model you developed in earlier chapters (or a hypothetical one), write a complete statistical methods section following the structure above. Aim for approximately 500 words. Check your methods section against the TRIPOD+AI checklist. Are all key items covered?

Example methods section

Study design and setting. We conducted a retrospective cohort study using data from the Framingham Heart Study (original cohort, examinations 1–10). Participants were adults aged 30–74 years free of cardiovascular disease at baseline.

Outcome. The primary outcome was incident coronary heart disease (CHD) within 10 years of baseline examination, defined as myocardial infarction, coronary insufficiency, or CHD death.

Predictors. Candidate predictors were age, sex, systolic blood pressure (modelled with restricted cubic splines, 4 knots), total cholesterol (restricted cubic splines, 4 knots), HDL cholesterol, current smoking status, diabetes status, and use of antihypertensive medication.

Missing data. Missing data ranged from 0% (age, sex) to 8% (HDL cholesterol). We used multiple imputation by chained equations (MICE, 20 imputations) assuming data were missing at random.

Model development. We fitted a logistic regression model. Continuous predictors were modelled with restricted cubic splines to allow for non-linear associations. No automated variable selection was performed; all pre-specified predictors were retained. The model was fitted on the full imputed dataset using Rubin's rules.

Model performance. Discrimination was quantified using the C-statistic with 95% confidence intervals. Calibration was assessed using calibration plots (loess smoother) and the calibration slope and intercept. Clinical utility was evaluated using decision curve analysis, reporting net benefit across decision thresholds from 5% to 40%.

Validation. Internal validation was performed using bootstrap resampling (500 samples) to estimate optimism-corrected performance.

Software. All analyses were performed in R 4.4.1 using the rms, mice, and dcurves packages.

20.8. The future of clinical prediction

Several trends are shaping the next decade of clinical prediction modelling:

- **Dynamic prediction models** that update as new patient data become available (e.g., during a hospital stay)
- **Federated learning** that trains models across institutions without sharing patient data
- **Foundation models** adapted for clinical tasks (large language models, multimodal models)
- **Continuous model monitoring** with automated detection of performance degradation
- **Patient-facing prediction tools** integrated into health apps and patient portals

Regardless of the technology, the fundamentals covered in this course — proper validation, calibration assessment, clinical utility evaluation, and transparent reporting — will remain essential.

Exercise 12.3: Critical appraisal

Find a prediction model paper from 2024 or 2025 in a journal relevant to your field. Answer the following:

1. What clinical question does the model address?
2. Was the model developed with regression, ML, or both?
3. Was calibration reported? If so, was it assessed using a calibration plot?
4. Was clinical utility (decision curve analysis) reported?
5. Could you implement this model in your own setting with the information provided?
6. Does the paper comply with TRIPOD+AI?

Discuss your findings with a colleague or in the course discussion forum.

20.9. References and further reading

- Collins GS, Moons KGM, Dhiman P, et al. **TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods.** *BMJ* 2024;385:e078378. The definitive reporting guideline for prediction models.
- Smits LJM, van Kuijk SMJ, Wynants L. **Improving Health Care with Clinical Prediction Models: From Idea to Impact.** Maastricht University Press, 2026. Chapters 10–15 cover model selection for impact, innovation development, research question formulation, impact evaluation (decision modelling and empirical), and implementation.
- Van Calster B, Collins GS, Vickers AJ, et al. **Evaluation of performance measures in predictive artificial intelligence models to support medical decisions: overview and guidance.** *Lancet Digital Health* 2025;7:e100916. Comprehensive guide to choosing appropriate performance measures.
- Steyerberg EW. **Clinical Prediction Models: A Practical Approach to Development, Validation, and Updating.** Springer, 2019. Chapter 23 covers implementation.
- Vickers AJ, van Calster B, Steyerberg EW. **A simple, step-by-step guide to interpreting decision curve analysis.** *Diagnostic and Prognostic Research* 2019;3:18.
- **EU AI Act** (Regulation 2024/1689). Official text available at eur-lex.europa.eu.
- Gigerenzer G, Edwards A. **Simple tools for understanding risks: from innumeracy to insight.** *BMJ* 2003;327:741–744. Classic paper on risk communication.

Part VII.

Advanced Research Toolkit

21. Causal Inference for Observational Health Data

21.1. Introduction

Most clinical research relies on observational data — electronic health records, claims databases, registries, and cohort studies. Yet the questions we care about most are causal: *Does* this drug reduce mortality? *Would* this surgery improve outcomes compared with conservative management? Answering causal questions with observational data requires careful methodology, because the fundamental problem is that patients who receive a treatment are systematically different from those who do not.

This chapter equips you with the conceptual frameworks and practical tools to move from "Drug X is associated with lower mortality" to a defensible causal claim — or, equally important, to recognise when the data cannot support one.

> **!** The stakes are real
>
> Confounding by indication — where sicker patients receive more aggressive treatment — has led to published studies concluding that treatments *cause harm* when they actually help, and vice versa. The Women's Health Initiative overturned decades of observational evidence on hormone replacement therapy. Causal inference methods

do not guarantee correct answers, but they make your assumptions explicit and testable.

21.2. Correlation vs Causation in Medicine

21.2.1. Why simple regression is not enough

Consider a hospital database showing that patients who receive mechanical ventilation have higher mortality than those who do not. A naive analyst might conclude ventilation is harmful. But the confounding is obvious: ventilation is given to the sickest patients.

This is **confounding by indication** — the indication for treatment is itself a risk factor for the outcome. Standard regression can adjust for *measured* confounders, but:

1. You must know which variables to include (and which to exclude).
2. You must model the relationships correctly (linearity, interactions).
3. You cannot adjust for *unmeasured* confounders.

Causal inference methods address problems 1 and 2 more systematically, and provide tools to assess sensitivity to problem 3.

21.2.2. The potential outcomes framework

The modern causal inference framework, formalised by Rubin (1974) and extended by Hernan and Robins, rests on **potential outcomes**:

- $Y^{a=1}$: the outcome a patient *would have* under treatment
- $Y^{a=0}$: the outcome a patient *would have* under control

The **individual causal effect** is $Y^{a=1} - Y^{a=0}$, but we can never observe both for the same person. This is the **fundamental problem of causal inference**.

We can, however, estimate the **average treatment effect (ATE)**:

$$\text{ATE} = E[Y^{a=1}] - E[Y^{a=0}]$$

or the **average treatment effect on the treated (ATT)**:

$$\text{ATT} = E[Y^{a=1} - Y^{a=0} \mid A = 1]$$

The key assumption that allows us to estimate these from observational data is **exchangeability** (no unmeasured confounding): conditional on measured covariates L, treatment assignment is independent of potential outcomes.

$$Y^a \perp\!\!\!\perp A \mid L$$

21.3. Directed Acyclic Graphs (DAGs)

21.3.1. What is a DAG?

A **directed acyclic graph** is a visual representation of your causal assumptions. Each node represents a variable. Each arrow represents a direct causal effect. "Acyclic" means no variable can cause itself through a chain of arrows.

Figure 21.1.: A simple DAG showing confounding. Age affects both statin use and cardiovascular death.

21.3.2. Key structures in DAGs

There are three fundamental structures you must recognise:

1. Confounders (common causes)

A variable that causes both treatment and outcome. You MUST adjust for confounders to remove bias.

```
L --> A --> Y
L ---------> Y
```

If `L` is diabetes severity, `A` is insulin therapy, and `Y` is HbA1c, then diabetes severity confounds the insulin-HbA1c relationship.

2. Mediators (intermediate variables)

A variable on the causal pathway from treatment to outcome. You should NOT adjust for mediators if you want the total effect.

```
A --> M --> Y
```

If `A` is exercise, `M` is weight loss, and `Y` is blood pressure, then weight loss mediates the exercise-BP relationship. Adjusting for weight loss would block part of the effect you want to estimate.

3. Colliders (common effects)

A variable caused by both treatment and outcome (or by variables on each path). You should NOT adjust for colliders — doing so *creates* a spurious association (collider bias).

```
A --> C <-- Y
```

If `A` is obesity, `Y` is genetic predisposition to diabetes, and `C` is being in a diabetes clinic, then conditioning on clinic attendance (a collider) creates a spurious negative association between obesity and genetic risk.

The "d-separation" rule of thumb

A path between treatment and outcome is **blocked** if:

- It passes through a variable you condition on (confounder or mediator), OR
- It passes through a collider you do NOT condition on.

A path is **open** if it is not blocked. Bias exists when there are open non-causal paths (backdoor paths) between treatment and outcome.

21.3.3. Drawing your own DAG

Before any causal analysis, draw a DAG. Steps:

1. List all variables relevant to treatment assignment and the outcome.
2. Use clinical knowledge to draw arrows (not data — DAGs encode assumptions).
3. Identify all backdoor paths from treatment to outcome.
4. Determine the minimal adjustment set that blocks all backdoor paths without opening new ones.

Tools like DAGitty automate step 4 once you have drawn the DAG.

21.4. Propensity Score Methods

21.4.1. The propensity score

The **propensity score** is the probability of receiving treatment given observed covariates:

$$e(L) = P(A = 1 \mid L)$$

Rosenbaum and Rubin (1983) showed that, if exchangeability holds conditional on L, it also holds conditional on the scalar $e(L)$. This reduces a high-dimensional adjustment problem to a single dimension.

21.4.2. Estimating the propensity score

We typically estimate $e(L)$ using logistic regression (or more flexible methods like gradient boosting):

```
# Simulate clinical data
set.seed(42)
n <- 2000

dat <- tibble(
  age = rnorm(n, 65, 10),
  diabetes = rbinom(n, 1, 0.3),
  egfr = rnorm(n, 60, 15),
  smoking = rbinom(n, 1, 0.25),
  # Treatment assignment depends on covariates (confounding)
  ps_true = plogis(-2 + 0.03 * age + 0.5 * diabetes - 0.02 * egfr + 0.4 * smo
  treatment = rbinom(n, 1, ps_true),
  # Outcome depends on treatment AND covariates
  mortality = rbinom(n, 1, plogis(-3 + 0.04 * age + 0.6 * diabetes -
```

```
                                                    0.03 * egfr + 0.3 * smoking -
                                                    0.5 * treatment))
)

# Estimate propensity score via logistic regression
ps_model <- glm(treatment ~ age + diabetes + egfr + smoking,
                data = dat, family = binomial)
dat$ps <- predict(ps_model, type = "response")

# Visualise overlap
ggplot(dat, aes(x = ps, fill = factor(treatment))) +
  geom_density(alpha = 0.5) +
  labs(x = "Propensity Score", fill = "Treatment",
       title = "Propensity Score Distribution by Treatment Group") +
  theme_minimal()
```

⚠ Positivity assumption

Every patient must have a non-zero probability of receiving *either* treatment. If some patients have propensity scores near 0 or 1, the overlap assumption is violated and estimates become unstable. Always check the propensity score distributions visually.

21.4.3. Propensity score matching

Matching creates a pseudo-randomised subset by pairing treated and control patients with similar propensity scores.

```
# Using MatchIt for nearest-neighbour matching with caliper
m_out <- matchit(treatment ~ age + diabetes + egfr + smoking,
                 data = dat,
                 method = "nearest",
```

```
                  distance = "glm",        # logistic regression PS
                  caliper = 0.2,           # 0.2 SD of logit PS
                  ratio = 1)               # 1:1 matching

summary(m_out)

# Check covariate balance using cobalt
love.plot(m_out,
          thresholds = c(m = 0.1),    # SMD threshold of 0.1
          binary = "std",
          title = "Covariate Balance: Before and After Matching")

# Extract matched data and estimate treatment effect
m_data <- match.data(m_out)

# Outcome model in matched sample
outcome_model <- glm(mortality ~ treatment,
                     data = m_data,
                     family = binomial,
                     weights = weights)

tidy(outcome_model, conf.int = TRUE, exponentiate = TRUE)
```

Checking balance is the most important step after matching. A standardised mean difference (SMD) below 0.1 for all covariates is the conventional threshold for acceptable balance.

21.4.4. Inverse probability of treatment weighting (IPTW)

Instead of discarding unmatched patients, **IPTW** reweights the entire sample so that the distribution of covariates is balanced between groups:

$$w_i = \frac{A_i}{e(L_i)} + \frac{1 - A_i}{1 - e(L_i)}$$

This estimates the ATE. For the ATT, the weights are:

$$w_i = A_i + \frac{(1 - A_i) \cdot e(L_i)}{1 - e(L_i)}$$

```
# Using WeightIt
w_out <- weightit(treatment ~ age + diabetes + egfr + smoking,
                  data = dat,
                  method = "ps",
                  estimand = "ATE")

# Check balance
bal.tab(w_out, thresholds = c(m = 0.1))

# Stabilised weights (recommended to reduce variance)
summary(w_out)

# Estimate treatment effect with IPTW
library(survey)
d_weighted <- svydesign(ids = ~1, weights = w_out$weights, data = dat)
svyglm(mortality ~ treatment, design = d_weighted, family = quasibinomial) |>
  tidy(conf.int = TRUE, exponentiate = TRUE)
```

Stabilised weights

Extreme weights can inflate variance. **Stabilised weights** replace the numerator 1 with $P(A = 1)$ for treated and $P(A = 0)$ for untreated. The `WeightIt` package does this by default. Always check that weights are not excessively large (e.g., max weight > 20 warrants investigation).

21.4.5. Doubly robust estimation

Doubly robust methods combine a propensity score model and an outcome model. The estimate is consistent if *either* model is correctly specified (but not necessarily both). This provides an extra layer of protection against model misspecification.

The augmented inverse probability weighted (AIPW) estimator is:

$$\hat{\tau}_{DR} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{A_i Y_i}{e(L_i)} - \frac{A_i - e(L_i)}{e(L_i)}\hat{\mu}_1(L_i)\right] - \frac{1}{n}\sum_{i=1}^{n}\left[\frac{(1-A_i)Y_i}{1-e(L_i)} + \frac{A_i - e(L_i)}{1-e(L_i)}\hat{\mu}_0(L_i)\right]$$

where $\hat{\mu}_a(L)$ is the predicted outcome under treatment a.

In practice, packages like `AIPW` in R or `DoWhy` in Python handle this computation.

21.5. Target Trial Emulation

21.5.1. The framework

Target trial emulation, developed by Hernan and Robins, asks: "What randomised trial would we *like* to conduct?" and then designs the observational analysis to mimic that trial as closely as possible.

The key components to specify are:

Protocol component	Target trial	Observational emulation
Eligibility criteria	Defined inclusion/exclusion	Same criteria applied to database

Protocol component	Target trial	Observational emulation
Treatment strategies	e.g., start drug A vs drug B	Same, defined at “time zero”
Treatment assignment	Random	Adjusted via PS methods or g-estimation
Start of follow-up	Randomisation date	Date eligibility + treatment criteria met
Outcome	e.g., 5-year mortality	Same
Causal contrast	Intention-to-treat or per-protocol	Same (with appropriate methods)
Analysis plan	Intention-to-treat analysis	Clone-censor-weight or similar

21.5.2. Why this reduces bias

Target trial emulation directly addresses several common biases:

Immortal time bias: In a naive observational study comparing drug users to non-users, the time between cohort entry and first prescription is “immortal” — the patient must survive to receive the drug. This artificially inflates survival in the treated group. Target trial emulation avoids this by aligning “time zero” (start of follow-up) with treatment assignment.

Selection bias from prevalent user designs: Studying patients already on a drug (prevalent users) conflates treatment initiation effects with survival conditioning. Target trial emulation uses a **new-user (incident user) design**.

Vague eligibility and time zero: When eligibility criteria and the start of follow-up are not clearly defined, analyses can be plagued by look-ahead bias. The target trial framework forces explicit specification.

> The TARGET reporting guideline (2025)
>
> The TARGET (TrAnsparent ReportinG of observational studies Emulating a Target trial) guideline provides a structured checklist for reporting target trial emulation studies. It was published in the BMJ in 2025 and should be followed for any observational study making causal claims.

21.5.3. Example: Statin initiation and cardiovascular events

Suppose we want to estimate the effect of initiating statins within 6 months of a type 2 diabetes diagnosis on 5-year cardiovascular events.

Target trial specification:

- **Eligibility**: Adults aged 40–75 with new T2DM diagnosis, no prior CVD, no prior statin use
- **Treatment strategies**: (1) Initiate statin within 6 months, (2) Do not initiate statin within 6 months
- **Time zero**: Date of T2DM diagnosis
- **Outcome**: First major adverse cardiovascular event (MACE) within 5 years
- **Analysis**: Intention-to-treat using IPTW for confounding, clone-censor-weight for treatment strategy adherence

21.6. Regression Discontinuity Design

Regression discontinuity (RD) exploits arbitrary thresholds in clinical decision-making. When treatment is assigned based on whether a continuous variable exceeds a cutoff, patients just above and just below the cutoff are essentially randomly assigned.

Clinical example: Many guidelines recommend antihypertensive therapy when systolic blood pressure exceeds 140 mmHg. Patients with SBP of 141 are very similar to patients with SBP of 139, but the former are much more likely to receive treatment. Comparing outcomes near the cutoff provides a quasi-experimental estimate.

Key requirements for a valid RD design:

1. **Treatment probability changes sharply at the cutoff** (first stage)
2. **Other covariates do not jump at the cutoff** (continuity assumption)
3. **Patients cannot precisely manipulate their score** (no bunching)

RD designs are less common in clinical research than propensity score methods, but when applicable, they provide strong evidence because the assumptions are more testable.

21.7. Implementation in Python

Python's causal inference ecosystem is maturing. Here we demonstrate propensity score matching and IPTW using `dowhy` and `causalinference`.

```
import numpy as np
import pandas as pd
from sklearn.linear_model import LogisticRegression
from scipy import stats
import matplotlib.pyplot as plt

# Simulate clinical data
np.random.seed(42)
n = 2000
```

```
age = np.random.normal(65, 10, n)
diabetes = np.random.binomial(1, 0.3, n)
egfr = np.random.normal(60, 15, n)
smoking = np.random.binomial(1, 0.25, n)

# Confounded treatment assignment
ps_true = 1 / (1 + np.exp(-(-2 + 0.03*age + 0.5*diabetes - 0.02*egfr + 0.4*sm
treatment = np.random.binomial(1, ps_true)

# Outcome
mortality_prob = 1 / (1 + np.exp(-(-3 + 0.04*age + 0.6*diabetes -
                                    0.03*egfr + 0.3*smoking -
                                    0.5*treatment)))
mortality = np.random.binomial(1, mortality_prob)

df = pd.DataFrame({
    'age': age, 'diabetes': diabetes, 'egfr': egfr,
    'smoking': smoking, 'treatment': treatment, 'mortality': mortality
})

# --- Propensity Score Estimation ---
covariates = ['age', 'diabetes', 'egfr', 'smoking']
ps_model = LogisticRegression(max_iter=1000)
ps_model.fit(df[covariates], df['treatment'])
df['ps'] = ps_model.predict_proba(df[covariates])[:, 1]

# --- IPTW ---
# Compute ATE weights
df['iptw'] = np.where(
    df['treatment'] == 1,
    1 / df['ps'],
    1 / (1 - df['ps'])
)
```

```
# Stabilised weights
p_treat = df['treatment'].mean()
df['sw'] = np.where(
    df['treatment'] == 1,
    p_treat / df['ps'],
    (1 - p_treat) / (1 - df['ps'])
)

# Check balance: weighted standardised mean differences
def weighted_smd(data, var, treatment_col, weight_col):
    treated = data[data[treatment_col] == 1]
    control = data[data[treatment_col] == 0]
    mean_t = np.average(treated[var], weights=treated[weight_col])
    mean_c = np.average(control[var], weights=control[weight_col])
    sd_pooled = np.sqrt((treated[var].var() + control[var].var()) / 2)
    return (mean_t - mean_c) / sd_pooled

print("Covariate balance (weighted SMD):")
for var in covariates:
    smd = weighted_smd(df, var, 'treatment', 'sw')
    print(f"  {var}: {smd:.4f}")

# --- Weighted outcome estimation ---
from statsmodels.api import WLS, add_constant

X = add_constant(df['treatment'])
wls_model = WLS(df['mortality'], X, weights=df['sw']).fit()
print("\nIPTW Treatment Effect Estimate:")
print(f"  Coefficient: {wls_model.params[1]:.4f}")
print(f"  95% CI: ({wls_model.conf_int()[1][0]:.4f}, {wls_model.conf_int()[1][1]:.4f})")
```

21.7.1. Using DoWhy for causal analysis

```
# DoWhy provides a structured causal inference workflow
# Install: pip install dowhy

import dowhy
from dowhy import CausalModel

# Step 1: Model - encode causal assumptions
model = CausalModel(
    data=df,
    treatment='treatment',
    outcome='mortality',
    common_causes=['age', 'diabetes', 'egfr', 'smoking']
)

# Step 2: Identify - find valid adjustment strategy
identified_estimand = model.identify_effect()
print(identified_estimand)

# Step 3: Estimate - using propensity score weighting
estimate = model.estimate_effect(
    identified_estimand,
    method_name="backdoor.propensity_score_weighting",
    method_params={"weighting_scheme": "ips_stabilized_weight"}
)
print(f"Causal Estimate: {estimate.value:.4f}")

# Step 4: Refute - sensitivity analysis
refutation = model.refute_estimate(
    identified_estimand, estimate,
    method_name="random_common_cause"
```

```
)
print(refutation)
```

21.8. Sensitivity Analysis for Unmeasured Confounding

No observational study can prove the absence of unmeasured confounding. **Sensitivity analysis** quantifies how strong unmeasured confounding would need to be to explain away the observed effect.

The **E-value** (VanderWeele and Ding, 2017) provides a simple summary: it is the minimum strength of association (on the risk ratio scale) that an unmeasured confounder would need to have with both the treatment and the outcome to fully explain away the observed treatment-outcome association.

$$\text{E-value} = RR + \sqrt{RR \times (RR - 1)}$$

where RR is the observed risk ratio. The E-value for the confidence interval bound closest to the null is also reported.

```
# E-value calculation
library(EValue)

# Suppose our IPTW analysis found RR = 0.65 (95% CI: 0.50, 0.85)
# Convert protective RR to above-null: 1/0.65 = 1.538
evalues.RR(est = 1/0.65, lo = 1/0.85, hi = 1/0.50)
```

21.9. Practical Guidance: Choosing Your Method

Scenario	Recommended approach
Binary treatment, many controls available	Propensity score matching
Binary treatment, need full sample	IPTW
Concern about model misspecification	Doubly robust estimation
Time-varying treatment	Marginal structural models (IPTW over time)
Natural threshold for treatment	Regression discontinuity
Emulating a specific trial design	Target trial emulation framework

21.10. Exercises

21.10.1. Exercise 1: DAG construction (Conceptual)

You are studying the relationship between **ACE inhibitor use** and **acute kidney injury (AKI)** in hospitalised patients.

a) List at least five variables that might be relevant to this relationship.
b) Draw a DAG (on paper or using DAGitty) encoding your causal assumptions.
c) Identify the minimal sufficient adjustment set for estimating the effect of ACE inhibitors on AKI.
d) Identify at least one potential collider. What would happen if you adjusted for it?

21.10.2. Exercise 2: Propensity score matching in R

Using the simulated dataset below, perform propensity score matching and estimate the treatment effect.

```r
set.seed(123)
n <- 1500
exercise_dat <- tibble(
  age = rnorm(n, 70, 8),
  creatinine = rnorm(n, 1.2, 0.4),
  heart_failure = rbinom(n, 1, 0.35),
  prior_mi = rbinom(n, 1, 0.20),
  # Confounded treatment (beta-blocker use)
  treatment = rbinom(n, 1, plogis(-1 + 0.02*age + 0.3*heart_failure +
                                  0.5*prior_mi - 0.8*creatinine)),
  # Outcome: 1-year mortality
  death_1yr = rbinom(n, 1, plogis(-2 + 0.05*age + 0.4*heart_failure +
                                  0.6*prior_mi + 1.0*creatinine -
                                  0.7*treatment))
)
```

a) Estimate propensity scores using logistic regression.
b) Perform 1:1 nearest-neighbour matching with a caliper of 0.2 SD.
c) Create a Love plot to assess balance before and after matching.
d) Estimate the ATT for beta-blocker use on 1-year mortality.
e) Calculate the E-value for your estimate. Interpret it.

21.10.3. Exercise 3: IPTW in Python

Using the same data-generating process as Exercise 2 (replicate it in Python):

```python
import numpy as np
import pandas as pd

np.random.seed(123)
n = 1500
age = np.random.normal(70, 8, n)
creatinine = np.random.normal(1.2, 0.4, n)
heart_failure = np.random.binomial(1, 0.35, n)
prior_mi = np.random.binomial(1, 0.20, n)

ps_true = 1 / (1 + np.exp(-(-1 + 0.02*age + 0.3*heart_failure +
                            0.5*prior_mi - 0.8*creatinine)))
treatment = np.random.binomial(1, ps_true)

mort_prob = 1 / (1 + np.exp(-(-2 + 0.05*age + 0.4*heart_failure +
                              0.6*prior_mi + 1.0*creatinine -
                              0.7*treatment)))
death_1yr = np.random.binomial(1, mort_prob)

df = pd.DataFrame({
    'age': age, 'creatinine': creatinine, 'heart_failure': heart_failure,
    'prior_mi': prior_mi, 'treatment': treatment, 'death_1yr': death_1yr
})
```

a) Fit a propensity score model and compute stabilised IPTW weights.
b) Assess covariate balance using weighted standardised mean differences.
c) Estimate the ATE using weighted regression.
d) Perform a sensitivity analysis: add a simulated unmeasured confounder and show how the estimate changes.

21.10.4. Exercise 4: Target trial emulation (Conceptual)

You have access to a large electronic health records database and want to estimate the effect of **early metformin initiation** (within 3 months of type 2 diabetes diagnosis) vs **delayed initiation** (3–12 months after diagnosis) on 5-year cardiovascular events.

a) Write a complete target trial protocol table (eligibility, treatment strategies, assignment, start of follow-up, outcome, causal contrast, analysis plan).
b) Identify potential sources of immortal time bias in a naive analysis.
c) Describe how the new-user, active comparator design addresses confounding by indication.
d) What unmeasured confounders might remain? How would you conduct a sensitivity analysis?

21.11. Summary

Concept	Key point
Confounding by indication	Sicker patients get more treatment — naive comparisons are biased
DAGs	Encode causal assumptions visually; identify what to adjust for
Propensity score	$P(\text{treatment} \mid \text{covariates})$; reduces dimensionality
Matching	Creates balanced pseudo-randomised groups; check balance
IPTW	Reweights full sample; preserves sample size
Doubly robust	Combines PS + outcome model; consistent if either is correct

Concept	Key point
Target trial emulation	Design observational analysis to mimic an ideal RCT
E-value	Quantifies robustness to unmeasured confounding

21.12. References and Further Reading

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4. **TARGET Guideline.** TrAnsparent ReportinG of observational studies Emulating a Target trial. *BMJ.* 2025. Reporting checklist for target trial emulation studies.

5. **VanderWeele TJ, Ding P.** Sensitivity analysis in observational research: introducing the E-value. *Annals of Internal Medicine.* 2017;167(4):268–274.

6. **Textor J, van der Zander B, Gilthorpe MS, et al.** Robust causal inference using directed acyclic graphs: the R package 'dagitty'. *International Journal of Epidemiology.* 2016;45(6):1887–1894.

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8. **Greifer N.** WeightIt: weighting for covariate balance in observational studies. R package. https://ngreifer.github.io/WeightIt/.

9. **Sharma A, Kiciman E.** DoWhy: An end-to-end library for causal inference. arXiv:2011.04216, 2020.

10. **Lodi S, Phillips A, Lundgren J, et al.** Effect estimates in randomized trials and observational studies: comparing apples with apples. *American Journal of Epidemiology.* 2019;188(8):1569–1577.

22. Meta-Analysis Methods for Evidence Synthesis

22.1. Introduction

A single study, no matter how well designed, provides only one piece of evidence. **Meta-analysis** is the statistical method for combining results from multiple independent studies addressing the same question, producing a more precise and generalisable estimate of an effect.

Meta-analysis sits at the top of the evidence hierarchy in evidence-based medicine. When done well, it can resolve conflicting findings, increase statistical power for rare outcomes, and identify sources of variation across studies. When done poorly, it can amplify biases present in the original studies — "garbage in, garbage out."

This chapter covers the statistical foundations of meta-analysis, practical implementation in R and Python, and critical appraisal of heterogeneity and publication bias.

> **i** Systematic review vs meta-analysis
>
> A **systematic review** is the process of comprehensively searching for, selecting, and appraising studies. A **meta-analysis** is the *statistical* component — combining the numbers. You can have a systematic review without a meta-analysis (if studies are too heterogeneous to combine), but you should never have a meta-analysis without a

systematic review.

22.2. What Does a Meta-Analysis Combine?

Each study contributes an **effect estimate** and a measure of its **precision** (standard error or confidence interval). The meta-analysis computes a **weighted average**, where studies with greater precision (typically larger studies) receive more weight.

The general formula for a weighted average effect:

$$\hat{\theta}_{\text{pooled}} = \frac{\sum_{i=1}^{k} w_i \hat{\theta}_i}{\sum_{i=1}^{k} w_i}$$

where $\hat{\theta}_i$ is the effect estimate from study i and w_i is its weight.

22.3. Effect Measures

The choice of effect measure depends on the type of outcome:

22.3.1. Binary outcomes

Measure	Formula	When to use
Odds ratio (OR)	$\frac{a/c}{b/d} = \frac{ad}{bc}$	Case-control studies; logistic regression
Risk ratio (RR)	$\frac{a/(a+b)}{c/(c+d)}$	Cohort studies; more interpretable than OR

Measure	Formula	When to use
Risk difference (RD)	$\frac{a}{a+b} - \frac{c}{c+d}$	When absolute risk is clinically relevant

Where a, b, c, d are the cells of a 2x2 table (treatment events, treatment non-events, control events, control non-events).

ORs and RRs are typically meta-analysed on the **log scale** (because their sampling distributions are approximately normal on that scale) and then back-transformed for presentation.

22.3.2. Continuous outcomes

Measure	Formula	When to use
Mean difference (MD)	$\bar{X}_T - \bar{X}_C$	All studies use the same measurement scale
Standardised mean difference (SMD)	$\frac{\bar{X}_T - \bar{X}_C}{S_{\text{pooled}}}$	Studies use different scales measuring the same construct (e.g., different depression questionnaires)

The SMD is also known as **Hedges' g** (with a small-sample correction) or **Cohen's d**.

22.4. Fixed-Effect vs Random-Effects Models

22.4.1. Fixed-effect model

Assumes all studies estimate the **same true effect** θ. Differences between study results are due solely to sampling variation.

Weights are the inverse of the within-study variance:

$$w_i^{FE} = \frac{1}{\hat{\sigma}_i^2}$$

This model is appropriate when studies are clinically and methodologically very similar (e.g., exact replications).

22.4.2. Random-effects model

Assumes each study estimates its **own true effect** θ_i, drawn from a distribution of true effects: $\theta_i \sim N(\mu, \tau^2)$.

The between-study variance τ^2 is estimated from the data (commonly using the DerSimonian-Laird, REML, or Paule-Mandel methods). Weights incorporate both within-study and between-study variance:

$$w_i^{RE} = \frac{1}{\hat{\sigma}_i^2 + \hat{\tau}^2}$$

The random-effects model is almost always more appropriate in clinical research, because studies differ in populations, interventions, settings, and outcome definitions.

! The random-effects model gives more weight to small studies

Because $\hat{\tau}^2$ is added to every study's variance, the relative weights become more equal. This means small (potentially biased or low-quality) studies have more influence in a random-effects analysis. This is a feature when heterogeneity is real, but a bug when small studies are biased (e.g., publication bias favoring small positive studies).

22.4.3. Prediction intervals

The confidence interval for the pooled effect tells you about the *average* true effect. The **prediction interval** tells you the range within which the true effect of a *future study* is likely to fall:

$$\hat{\mu} \pm t_{k-2,0.975} \times \sqrt{\text{SE}(\hat{\mu})^2 + \hat{\tau}^2}$$

Prediction intervals are often much wider than confidence intervals and provide a more honest picture of the uncertainty. A pooled effect may be statistically significant while the prediction interval includes the null — meaning that some settings may see no benefit or even harm.

22.5. Heterogeneity

22.5.1. Measuring heterogeneity

Cochran's Q test: Tests H_0: all studies share the same true effect.

$$Q = \sum_{i=1}^{k} w_i^{FE}(\hat{\theta}_i - \hat{\theta}_{FE})^2$$

Under H_0, $Q \sim \chi^2_{k-1}$. The test has low power when k is small.

I^2 **statistic**: The percentage of total variability due to between-study heterogeneity rather than chance.

$$I^2 = \max\left(0, \frac{Q-(k-1)}{Q}\right) \times 100\%$$

Rules of thumb (Higgins et al., 2003):

- $I^2 \approx 25\%$: low heterogeneity
- $I^2 \approx 50\%$: moderate heterogeneity
- $I^2 \approx 75\%$: high heterogeneity

τ^2: The estimated between-study variance on the effect scale. Unlike I^2, it is not influenced by study precision and is more interpretable for clinical decision-making.

22.5.2. Investigating heterogeneity

When heterogeneity is substantial:

1. **Subgroup analysis**: Split studies by a pre-specified characteristic (e.g., drug dose, patient age group, study quality) and compare pooled effects.
2. **Meta-regression**: Model the effect as a function of study-level covariates.
3. **Sensitivity analysis**: Exclude outlier studies or studies at high risk of bias.

> ⚠ Ecological fallacy in meta-regression
>
> Study-level associations do not imply individual-level associations. If trials with older populations show larger treatment effects, this

does NOT prove that older individuals benefit more — the trials may also differ in other ways. This is the **ecological fallacy**, and it is a fundamental limitation of aggregate data meta-analysis.

22.6. Forest Plots

The **forest plot** is the standard visualisation for meta-analysis. Each study is shown as a square (size proportional to weight) with a horizontal line (95% CI). The pooled estimate is shown as a diamond.

```
# Example data: RCTs of statins for secondary CV prevention
statin_data <- data.frame(
  study = c("4S (1994)", "CARE (1996)", "LIPID (1998)", "HPS (2002)",
            "PROSPER (2002)", "PROVE-IT (2004)", "TNT (2005)",
            "JUPITER (2008)", "HOPE-3 (2016)"),
  events_treat = c(111, 212, 287, 1328, 127, 147, 334, 142, 235),
  n_treat      = c(2221, 2081, 4512, 10269, 2891, 2099, 5006, 8901, 6361),
  events_ctrl  = c(189, 274, 373, 1507, 143, 172, 418, 251, 260),
  n_ctrl       = c(2223, 2078, 4502, 10267, 2913, 2063, 5006, 8901, 6344)
)

# Run random-effects meta-analysis
m1 <- metabin(
  event.e = events_treat, n.e = n_treat,
  event.c = events_ctrl,  n.c = n_ctrl,
  studlab = study,
  data = statin_data,
  sm = "RR",                      # Risk ratio
  method.tau = "REML",            # Restricted maximum likelihood for tau^2
  prediction = TRUE               # Include prediction interval
)
```

```
# Forest plot
forest(m1,
       sortvar = TE,
       label.left = "Favours statins",
       label.right = "Favours control",
       col.diamond = "steelblue",
       col.square = "darkblue",
       print.tau2 = TRUE,
       print.I2 = TRUE,
       print.pval.Q = TRUE)
```

22.6.1. Reading a forest plot

Key elements to examine:

1. **Individual study estimates**: Are they all on the same side of the null?
2. **Confidence intervals**: Do they overlap?
3. **Pooled estimate (diamond)**: Where does it fall? Does the CI cross the null?
4. **Prediction interval**: If shown, does it cross the null?
5. **Heterogeneity statistics**: High I^2? Significant Q test?
6. **Study weights**: Are results driven by one dominant study?

22.7. Funnel Plots and Publication Bias

22.7.1. The problem

Studies with statistically significant results are more likely to be published. This **publication bias** means the available evidence may overestimate the true effect.

22.7.2. Funnel plots

A **funnel plot** displays each study's effect estimate (x-axis) against its precision, typically the standard error (y-axis, inverted). In the absence of bias, the plot should resemble a symmetric inverted funnel.

```
funnel(m1,
       xlab = "Risk Ratio (log scale)",
       studlab = TRUE,
       col = "steelblue",
       pch = 16)
```

Asymmetry (typically an excess of small studies with large positive effects) suggests publication bias — but also other causes like genuine heterogeneity, methodological differences in small studies, or chance.

22.7.3. Statistical tests for funnel plot asymmetry

Egger's test: A weighted regression of the effect estimate on its standard error. A significant intercept suggests asymmetry.

```
metabias(m1, method.bias = "Egger")
```

Peters' test: Preferred for binary outcomes (Egger's test can be biased with odds ratios).

22.7.4. Trim-and-fill method

The **trim-and-fill** method estimates the number of "missing" studies, imputes them, and recalculates the pooled effect. It provides a sensitivity analysis rather than a definitive correction.

```r
tf <- trimfill(m1)
summary(tf)
funnel(tf)
```

> **Publication bias is not the only explanation for asymmetry**
>
> Small-study effects can also arise from:
>
> - Genuine heterogeneity (treatment works better in specific populations studied in smaller trials)
> - Methodological flaws in smaller studies (less rigorous protocols)
> - Chance (especially with fewer than 10 studies)
>
> Egger's test should only be applied when there are at least 10 studies.

22.8. Advanced Meta-Analysis using metafor

The `metafor` package provides the most comprehensive and flexible meta-analysis framework in R.

```r
library(metafor)

# Compute log risk ratios and sampling variances
statin_es <- escalc(
  measure = "RR",
  ai = events_treat, n1i = n_treat,
  ci = events_ctrl,  n2i = n_ctrl,
  data = statin_data
)

# Random-effects model with REML
```

```r
res <- rma(yi, vi, data = statin_es, method = "REML")
summary(res)

# Prediction interval
predict(res)

# Meta-regression: effect of year of publication
statin_es$year <- c(1994, 1996, 1998, 2002, 2002, 2004, 2005, 2008, 2016)
res_mr <- rma(yi, vi, mods = ~ year, data = statin_es, method = "REML")
summary(res_mr)

# Leave-one-out sensitivity analysis
leave1out(res)

# Influence diagnostics
inf <- influence(res)
plot(inf)
```

22.8.1. Subgroup analysis

```r
# Classify as primary vs secondary prevention
statin_es$prevention <- c("secondary", "secondary", "secondary", "secondary",
                          "primary", "secondary", "secondary",
                          "primary", "primary")

# Subgroup analysis
res_sub <- rma(yi, vi, mods = ~ prevention, data = statin_es, method = "REML")
summary(res_sub)

# Forest plot by subgroup using meta package
```

```
update(m1, subgroup = statin_es$prevention, print.subgroup.name = TRUE) |>
  forest(sortvar = TE)
```

22.9. Network Meta-Analysis (Brief Overview)

Standard pairwise meta-analysis can only compare treatments that have been directly compared in head-to-head trials. **Network meta-analysis (NMA)**, also called mixed-treatment comparison, simultaneously compares multiple treatments using both **direct evidence** (from head-to-head trials) and **indirect evidence** (inferred through a common comparator).

22.9.1. When is NMA useful?

Consider three antihypertensive drugs: A, B, and C. If trials have compared A vs B and B vs C, but never A vs C, NMA can provide an indirect estimate for A vs C:

$$\hat{d}_{AC} = \hat{d}_{AB} + \hat{d}_{BC}$$

This relies on the **transitivity assumption**: the studies comparing A vs B and those comparing B vs C are similar enough that indirect comparison is valid.

22.9.2. Key concepts

- **Network geometry**: Nodes represent treatments, edges represent direct comparisons. The network should be well connected.
- **Consistency**: Direct and indirect evidence agree. Inconsistency suggests the transitivity assumption is violated.

- **Ranking**: NMA can rank treatments using the surface under the cumulative ranking curve (SUCRA) or P-scores, though rankings should be interpreted cautiously.

```r
library(netmeta)

# Example: network meta-analysis of antidepressants
# (using built-in dataset)
data(Senn2013)

# Run NMA
net <- netmeta(TE, seTE, treat1, treat2, studlab,
               data = Senn2013,
               sm = "MD",
               random = TRUE,
               reference.group = "plac")

summary(net)

# Network graph
netgraph(net, plastic = FALSE, thickness = "w.random",
         col = "steelblue")

# Forest plot of all comparisons vs reference
forest(net, reference.group = "plac", sortvar = TE)

# League table
netleague(net)
```

22.10. Individual Participant Data (IPD) Meta-Analysis

In standard meta-analysis, you combine **aggregate data** (summary statistics from each study). In an **IPD meta-analysis**, you obtain the raw, individual-level data from each study and analyse it directly.

22.10.1. Advantages of IPD meta-analysis

1. **Patient-level subgroup analyses** without ecological fallacy
2. **Standardised outcome definitions** and follow-up times
3. **Updated analyses** with extended follow-up
4. **Prediction model development** and validation across multiple settings

22.10.2. Two-stage vs one-stage approaches

- **Two-stage**: Analyse each study separately, then combine study-level estimates using standard meta-analysis. Familiar and transparent.
- **One-stage**: Fit a single model to all individual data, accounting for study clustering. More flexible, especially with sparse data.

```
# Two-stage IPD meta-analysis example
# Suppose we have a stacked dataset with individual data from 5 studies

# Stage 1: Fit model in each study
library(lme4)

# One-stage approach (mixed model with random study effects)
ipd_model <- glmer(
  event ~ treatment + age + sex + (1 + treatment | study_id),
```

```
  data = ipd_data,
  family = binomial
)

summary(ipd_model)

# The fixed effect for 'treatment' is the pooled effect
# The random slope for 'treatment' captures between-study heterogeneity
```

22.10.3. IPD meta-analysis of prediction models

Riley et al. (2010, 2021) describe frameworks for developing and validating clinical prediction models across multiple studies using IPD:

1. Develop the model using one-stage IPD-MA (internal-external cross-validation)
2. Assess calibration and discrimination in each study
3. Examine heterogeneity in model performance across settings
4. Update model intercept or recalibrate for new settings

22.11. Implementation in Python

Python's meta-analysis ecosystem is less mature than R's, but basic analyses are feasible.

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy import stats

# Study data: log risk ratios and standard errors
```

```python
studies = pd.DataFrame({
    'study': ['4S (1994)', 'CARE (1996)', 'LIPID (1998)', 'HPS (2002)',
              'PROSPER (2002)', 'PROVE-IT (2004)', 'TNT (2005)',
              'JUPITER (2008)', 'HOPE-3 (2016)'],
    'events_t': [111, 212, 287, 1328, 127, 147, 334, 142, 235],
    'n_t': [2221, 2081, 4512, 10269, 2891, 2099, 5006, 8901, 6361],
    'events_c': [189, 274, 373, 1507, 143, 172, 418, 251, 260],
    'n_c': [2223, 2078, 4502, 10267, 2913, 2063, 5006, 8901, 6344]
})

# Compute log risk ratios and variances
studies['rr'] = (studies['events_t'] / studies['n_t']) / \
                (studies['events_c'] / studies['n_c'])
studies['log_rr'] = np.log(studies['rr'])

# Variance of log RR
studies['var_log_rr'] = (
    1/studies['events_t'] - 1/studies['n_t'] +
    1/studies['events_c'] - 1/studies['n_c']
)
studies['se_log_rr'] = np.sqrt(studies['var_log_rr'])

# --- Fixed-effect meta-analysis (inverse variance) ---
w_fe = 1 / studies['var_log_rr']
pooled_fe = np.sum(w_fe * studies['log_rr']) / np.sum(w_fe)
se_fe = np.sqrt(1 / np.sum(w_fe))
ci_fe = (pooled_fe - 1.96*se_fe, pooled_fe + 1.96*se_fe)

print(f"Fixed-effect pooled log RR: {pooled_fe:.4f}")
print(f"Fixed-effect pooled RR: {np.exp(pooled_fe):.4f} "
      f"(95% CI: {np.exp(ci_fe[0]):.4f} - {np.exp(ci_fe[1]):.4f})")

# --- Random-effects: DerSimonian-Laird ---
```

```
k = len(studies)
Q = np.sum(w_fe * (studies['log_rr'] - pooled_fe)**2)
C = np.sum(w_fe) - np.sum(w_fe**2) / np.sum(w_fe)
tau2 = max(0, (Q - (k - 1)) / C)

w_re = 1 / (studies['var_log_rr'] + tau2)
pooled_re = np.sum(w_re * studies['log_rr']) / np.sum(w_re)
se_re = np.sqrt(1 / np.sum(w_re))
ci_re = (pooled_re - 1.96*se_re, pooled_re + 1.96*se_re)

I2 = max(0, (Q - (k-1)) / Q) * 100

print(f"\nRandom-effects pooled log RR: {pooled_re:.4f}")
print(f"Random-effects pooled RR: {np.exp(pooled_re):.4f} "
      f"(95% CI: {np.exp(ci_re[0]):.4f} - {np.exp(ci_re[1]):.4f})")
print(f"tau^2: {tau2:.4f}, I^2: {I2:.1f}%")
print(f"Q statistic: {Q:.2f}, p = {1 - stats.chi2.cdf(Q, k-1):.4f}")
```

22.11.1. Forest plot in Python

```
def forest_plot(studies, pooled_est, pooled_ci, weights, title="Forest Plot"):
    """Create a basic forest plot."""
    fig, ax = plt.subplots(figsize=(10, 8))

    k = len(studies)
    y_pos = list(range(k, 0, -1))

    # Individual studies
    for i, (_, row) in enumerate(studies.iterrows()):
        ci_lo = row['log_rr'] - 1.96 * row['se_log_rr']
        ci_hi = row['log_rr'] + 1.96 * row['se_log_rr']
```

```
        ax.plot([np.exp(ci_lo), np.exp(ci_hi)], [y_pos[i], y_pos[i]],
                'k-', linewidth=1)
        size = weights[i] / weights.max() * 200
        ax.scatter(np.exp(row['log_rr']), y_pos[i],
                   s=size, c='steelblue', zorder=5, edgecolors='darkblue')

    # Pooled estimate (diamond)
    ax.axvline(x=np.exp(pooled_est), color='steelblue',
               linestyle='--', alpha=0.5)
    ax.axvline(x=1, color='black', linestyle='-', linewidth=0.5)

    # Pooled diamond
    diamond_y = 0
    diamond_x = [np.exp(pooled_ci[0]), np.exp(pooled_est),
                 np.exp(pooled_ci[1]), np.exp(pooled_est)]
    diamond_yy = [diamond_y, diamond_y + 0.3, diamond_y, diamond_y - 0.3]
    ax.fill(diamond_x, diamond_yy, color='steelblue', alpha=0.7)

    # Labels
    ax.set_yticks(y_pos + [0])
    ax.set_yticklabels(list(studies['study']) + ['Pooled'])
    ax.set_xlabel('Risk Ratio')
    ax.set_title(title)
    ax.set_xscale('log')
    plt.tight_layout()
    plt.show()

forest_plot(studies, pooled_re, ci_re, w_re,
            title="Statins for CV Prevention: Random-Effects Meta-Analysis")
```

22.11.2. Using PythonMeta or PyMeta

```
# For more complete meta-analysis in Python, consider the 'meta-analysis' package
# Install: pip install meta-analysis

# Alternatively, statsmodels has some capabilities
import statsmodels.api as sm

# Egger's test equivalent: weighted regression of effect on SE
X = sm.add_constant(studies['se_log_rr'])
model = sm.WLS(studies['log_rr'], X, weights=w_fe).fit()
print("Egger's test (intercept):")
print(f"  Intercept: {model.params[0]:.4f}, p = {model.pvalues[0]:.4f}")
```

22.12. Reporting a Meta-Analysis: PRISMA 2020

The **PRISMA 2020** statement (Page et al., 2021) provides an updated checklist for reporting systematic reviews and meta-analyses. Key statistical reporting requirements:

1. **Describe the effect measure** and its rationale
2. **Specify the synthesis model** (fixed/random) and estimation method
3. **Present heterogeneity statistics** (τ^2, I^2, prediction interval)
4. **Report assessments of bias** (risk of bias, publication bias)
5. **Include a forest plot** for the primary outcome
6. **Describe sensitivity analyses** and their results
7. **Register the protocol** (PROSPERO) and follow it

22.13. Exercises

22.13.1. Exercise 1: Basic meta-analysis in R

The following data come from randomised trials of a hypothetical new anticoagulant vs warfarin for stroke prevention in atrial fibrillation.

```r
af_trials <- data.frame(
  study = c("TRAIL-1", "GUARD-AF", "SHIELD", "ORBIT-AF",
            "VENTURE", "COMPASS-AF", "PIONEER-2", "ATLAS-AF"),
  events_new = c(28, 45, 112, 67, 33, 89, 52, 41),
  n_new      = c(1200, 2500, 5400, 3100, 1800, 4200, 2800, 2100),
  events_warf = c(42, 58, 148, 84, 29, 102, 61, 53),
  n_warf      = c(1200, 2500, 5400, 3100, 1800, 4200, 2800, 2100)
)
```

a) Compute the risk ratio and 95% CI for each trial by hand (or using `escalc`).
b) Perform a random-effects meta-analysis using the `meta` or `metafor` package.
c) Create a forest plot. Does the pooled effect favour the new anticoagulant?
d) Calculate and interpret I^2 and the prediction interval.
e) Create a funnel plot and perform Egger's test. Is there evidence of publication bias?
f) Perform a leave-one-out sensitivity analysis. Is the result robust?

22.13.2. Exercise 2: Meta-analysis from scratch in Python

Using the same trial data as Exercise 1:

```python
import pandas as pd
import numpy as np

af_trials = pd.DataFrame({
    'study': ['TRAIL-1', 'GUARD-AF', 'SHIELD', 'ORBIT-AF',
              'VENTURE', 'COMPASS-AF', 'PIONEER-2', 'ATLAS-AF'],
    'events_new':  [28, 45, 112, 67, 33, 89, 52, 41],
    'n_new':       [1200, 2500, 5400, 3100, 1800, 4200, 2800, 2100],
    'events_warf': [42, 58, 148, 84, 29, 102, 61, 53],
    'n_warf':      [1200, 2500, 5400, 3100, 1800, 4200, 2800, 2100]
})
```

a) Compute log risk ratios and their variances for each study.
b) Implement the fixed-effect inverse-variance method.
c) Implement the DerSimonian-Laird random-effects method.
d) Compute Q, I^2, and τ^2.
e) Create a forest plot using matplotlib.
f) Create a funnel plot and implement Egger's regression test.

22.13.3. Exercise 3: Subgroup analysis and meta-regression in R

Suppose the trials in Exercise 1 were conducted in different settings:

```r
af_trials$region <- c("Europe", "North America", "Europe", "Asia",
                       "North America", "Europe", "Asia", "North America")
af_trials$mean_age <- c(72, 68, 74, 65, 70, 71, 63, 69)
af_trials$pct_female <- c(38, 42, 35, 48, 40, 37, 52, 44)
```

a) Perform a subgroup analysis by region. Do treatment effects differ by region?
b) Perform meta-regression with mean age as a moderator. Is there a relationship between mean age and treatment effect?

c) Perform meta-regression with percentage female. Interpret the result, noting the ecological fallacy.
d) Create a bubble plot showing the meta-regression of effect size on mean age.

22.13.4. Exercise 4: Critical appraisal (Conceptual)

You are reviewing a published meta-analysis of 12 trials comparing a new surgical technique to standard care for knee osteoarthritis. The reported results are:

- Pooled standardised mean difference for pain: -0.62 (95% CI: -0.89 to -0.35), $p < 0.001$
- $I^2 = 78\%$, $\tau^2 = 0.15$, Q test $p < 0.001$
- Prediction interval: -1.42 to 0.18
- Egger's test: $p = 0.03$
- 8 of 12 trials were single-centre with fewer than 100 participants

a) Interpret the pooled effect and its clinical significance.
b) What does the prediction interval tell you that the confidence interval does not?
c) What are the implications of $I^2 = 78\%$?
d) Given the Egger's test and the predominance of small trials, what concerns do you have?
e) What additional analyses would you want to see?
f) Would you change clinical practice based on this meta-analysis? Why or why not?

22.14. Summary

Concept	Key point
Fixed-effect model	Assumes one true effect; weights = $1/\sigma_i^2$
Random-effects model	Allows varying true effects; weights = $1/(\sigma_i^2 + \tau^2)$
I^2	Proportion of variability due to heterogeneity (25/50/75% thresholds)
Prediction interval	Range for the true effect in a future setting — often wider than CI
Forest plot	The primary visualisation; always include one
Funnel plot	Checks for small-study effects / publication bias
Network meta-analysis	Combines direct and indirect evidence across multiple treatments
IPD meta-analysis	Uses individual data; avoids ecological fallacy
PRISMA 2020	The reporting guideline for systematic reviews

22.15. References and Further Reading

1. **Higgins JPT, Thomas J, Chandler J, et al.** (eds). *Cochrane Handbook for Systematic Reviews of Interventions.* Version 6.4, 2024. Available at https://training.cochrane.org/handbook. The authoritative guide to systematic review methodology.

2. **Riley RD, Debray TPA, Collins GS, et al.** Individual participant data meta-analysis to examine interactions between treatment effect and participant-level covariates. *Statistical Methods in Medical Research.* 2020;29(12):3531–3556.

3. **Riley RD, Moons KGM, Snell KIE, et al.** A guide to systematic review and meta-analysis of prognostic factor studies. *BMJ.* 2019;364:k4597.

4. **Page MJ, McKenzie JE, Bossuyt PM, et al.** The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ.* 2021;372:n71.

5. **Viechtbauer W.** Conducting meta-analyses in R with the metafor package. *Journal of Statistical Software.* 2010;36(3):1–48.

6. **Schwarzer G, Carpenter JR, Rucker G.** *Meta-Analysis with R.* Springer, 2015. Comprehensive practical guide using the `meta` and `metafor` packages.

7. **DerSimonian R, Laird N.** Meta-analysis in clinical trials. *Controlled Clinical Trials.* 1986;7(3):177–188.

8. **Higgins JPT, Thompson SG, Deeks JJ, Altman DG.** Measuring inconsistency in meta-analyses. *BMJ.* 2003;327(7414):557–560.

9. **Egger M, Davey Smith G, Schneider M, Minder C.** Bias in meta-analysis detected by a simple, graphical test. *BMJ.* 1997;315(7109):629–634.

10. **Salanti G.** Indirect and mixed-treatment comparison, network, or multiple-treatments meta-analysis: many names, many benefits, many concerns for the next generation evidence synthesis tool. *Research Synthesis Methods.* 2012;3(2):80–97.

11. **IntHout J, Ioannidis JPA, Rovers MM, Goeman JJ.** Plea for routinely presenting prediction intervals in meta-analysis. *BMJ Open.* 2016;6(7):e010247.

23. Producing Journal-Ready Statistical Analyses

23.1. Introduction

You have learned how to fit models, validate predictions, and draw causal inferences. The final — and often most undervalued — skill is **communicating your results** in a form that meets the standards of peer-reviewed medical journals. Reviewers and editors routinely reject papers not because the statistics are wrong, but because the presentation is unclear, tables are poorly formatted, figures are low resolution, or the statistical methods section is incomplete.

This chapter walks you through the complete workflow: from creating publication-quality tables and figures to writing a statistical methods section, meeting reporting guideline requirements, and building a reproducible research pipeline. We conclude with three capstone project descriptions that bring together everything in the course.

23.2. What Do Top Journals Expect?

23.2.1. General standards

Major medical journals (The Lancet, BMJ, JAMA, NEJM, Lancet Digital Health) share common expectations:

1. **Transparent reporting**: Follow the relevant reporting guideline (CONSORT for RCTs, STROBE for observational studies, TRIPOD+AI for prediction models, PRISMA for systematic reviews).
2. **Pre-specified analysis plan**: Distinguish confirmatory from exploratory analyses.
3. **Effect estimates with uncertainty**: Report point estimates, confidence intervals, and (where appropriate) p-values. Never report p-values alone.
4. **Reproducibility**: Many journals now require data availability statements and code sharing.
5. **Figure quality**: Minimum 300 DPI, specific file formats (TIFF, EPS, or high-resolution PDF), accessible colour palettes.

23.2.2. Journal-specific requirements

Journal	Key statistical requirements
Lancet / Lancet Digital Health	TRIPOD+AI for prediction models; structured statistical methods; data sharing encouraged
BMJ	EQUATOR reporting guidelines mandatory; patient involvement statement; open access to data
JAMA	Statistical review by dedicated biostatistician; eTable/eFigure supplements expected
NEJM	Extreme rigour; independent statistical review; full protocol as supplement

! The statistical review

All top medical journals employ dedicated statistical reviewers. Common reasons for rejection on statistical grounds include: inappropriate handling of missing data, failure to account for multiple comparisons, presenting associations as causal claims without proper methodology, and inadequate model validation.

23.3. Table 1: Baseline Characteristics

"Table 1" is the standard first table in any clinical paper, describing the study population by group (treatment arms, exposure groups, or clusters).

23.3.1. What belongs in Table 1

- Demographics: age, sex, race/ethnicity
- Clinical characteristics: comorbidities, disease severity, vital signs
- Laboratory values: relevant biomarkers
- Medications and prior treatments
- Summary statistics: mean (SD) or median (IQR) for continuous variables; n (%) for categorical variables

23.3.2. Creating Table 1 in R with gtsummary

```
library(gtsummary)

# Using built-in trial dataset (modify for your data)
trial |>
```

```
  select(trt, age, grade, stage, marker, response) |>
  tbl_summary(
    by = trt,
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = list(
      all_continuous() ~ 1,
      all_categorical() ~ c(0, 1)
    ),
    label = list(
      age ~ "Age, years",
      grade ~ "Tumour grade",
      stage ~ "Cancer stage",
      marker ~ "Biomarker level, ng/mL",
      response ~ "Tumour response"
    ),
    missing_text = "Missing"
  ) |>
  add_overall() |>
  add_p(
    test = list(
      all_continuous() ~ "t.test",
      all_categorical() ~ "chisq.test"
    )
  ) |>
  add_n() |>
  modify_header(label = "**Characteristic**") |>
  modify_spanning_header(c("stat_1", "stat_2") ~ "**Treatment Group**") |>
  bold_labels() |>
  as_gt() |>
  gt::tab_header(
```

```
    title = "Table 1. Baseline Characteristics of the Study Population"
)
```

Should Table 1 include p-values?

In a randomised trial, baseline differences are due to chance, so p-values for baseline comparisons are widely considered inappropriate (they test a hypothesis you know to be true). In observational studies, standardised mean differences (SMDs) are preferred over p-values because SMDs are not influenced by sample size.

23.3.3. Creating Table 1 in Python with tableone

```
import pandas as pd
from tableone import TableOne
import numpy as np

# Simulate clinical trial data
np.random.seed(42)
n = 500

data = pd.DataFrame({
    'Treatment': np.random.choice(['Drug A', 'Drug B'], n),
    'Age': np.random.normal(65, 10, n).round(1),
    'Sex': np.random.choice(['Male', 'Female'], n, p=[0.55, 0.45]),
    'BMI': np.random.normal(28, 5, n).round(1),
    'Diabetes': np.random.choice(['Yes', 'No'], n, p=[0.3, 0.7]),
    'eGFR': np.random.normal(62, 18, n).round(0),
    'Systolic_BP': np.random.normal(140, 20, n).round(0),
    'Smoking': np.random.choice(['Never', 'Former', 'Current'], n, p=[0.4, 0.35, 0.25])
})
```

```
# Define variable types
columns = ['Age', 'Sex', 'BMI', 'Diabetes', 'eGFR', 'Systolic_BP', 'Smoking']
categorical = ['Sex', 'Diabetes', 'Smoking']
nonnormal = ['eGFR']  # Report as median [IQR]

# Create Table 1
table1 = TableOne(
    data,
    columns=columns,
    categorical=categorical,
    groupby='Treatment',
    nonnormal=nonnormal,
    pval=True,
    smd=True  # Standardised mean differences
)

print(table1.tabulate(tablefmt="github"))

# Export to various formats
# table1.to_excel('table1.xlsx')
# table1.to_latex('table1.tex')
# table1.to_html('table1.html')
```

23.4. Regression Tables

23.4.1. Presenting regression results

A well-formatted regression table should include:

- Variable names (not raw column names)
- Effect estimates (OR, HR, RR, or coefficients) with 95% CIs

- P-values (to appropriate decimal places)
- Sample size and number of events
- Model fit statistics (R-squared, C-statistic, AIC)

23.4.2. Regression tables in R

```r
library(gtsummary)
library(survival)

# Fit a Cox model
cox_model <- coxph(Surv(time, status) ~ age + sex + ph.ecog + ph.karno + wt.loss,
                   data = lung)

# Publication-quality table
tbl_regression(
  cox_model,
  exponentiate = TRUE,
  label = list(
    age ~ "Age, per year",
    sex ~ "Sex (female vs male)",
    ph.ecog ~ "ECOG performance status",
    ph.karno ~ "Karnofsky performance score",
    wt.loss ~ "Weight loss, kg"
  ),
  pvalue_fun = ~ style_pvalue(.x, digits = 3)
) |>
  add_global_p() |>
  bold_p(t = 0.05) |>
  modify_header(label = "**Variable**") |>
  modify_footnote(
    estimate ~ "HR = hazard ratio; CI = confidence interval",
  ) |>
```

```
  as_gt() |>
  gt::tab_header(
    title = "Table 2. Multivariable Cox Regression for Overall Survival"
  )
```

23.4.3. Regression tables in Python

```
import pandas as pd
import numpy as np
from lifelines import CoxPHFitter
from lifelines.datasets import load_lung

# Load and prepare data
lung = load_lung()
lung = lung[['T', 'status', 'age', 'sex', 'ph.ecog', 'ph.karno', 'wt.loss']]
lung['status'] = lung['status'] - 1  # Convert to 0/1

# Fit Cox model
cph = CoxPHFitter()
cph.fit(lung, duration_col='T', event_col='status')

# Extract results as a formatted table
summary_df = cph.summary[['exp(coef)', 'exp(coef) lower 95%',
                           'exp(coef) upper 95%', 'p']].copy()
summary_df.columns = ['HR', 'Lower 95% CI', 'Upper 95% CI', 'p-value']
summary_df = summary_df.round({'HR': 2, 'Lower 95% CI': 2,
                                'Upper 95% CI': 2, 'p-value': 4})

# Create formatted CI column
summary_df['HR (95% CI)'] = summary_df.apply(
    lambda r: f"{r['HR']:.2f} ({r['Lower 95% CI']:.2f}-{r['Upper 95% CI']:.2
```

```
    axis=1
)

# Rename index for publication
name_map = {
    'age': 'Age, per year',
    'sex': 'Sex (female vs male)',
    'ph.ecog': 'ECOG performance status',
    'ph.karno': 'Karnofsky score, per 10 points',
    'wt.loss': 'Weight loss, per kg'
}
summary_df.index = summary_df.index.map(lambda x: name_map.get(x, x))

print(summary_df[['HR (95% CI)', 'p-value']].to_string())
```

23.5. Publication-Quality Figures

23.5.1. Journal specifications

Most journals require:

- **Resolution**: minimum 300 DPI (600 DPI for line art)
- **Format**: TIFF, EPS, or high-resolution PDF
- **Size**: single column (89 mm) or double column (183 mm)
- **Fonts**: Arial, Helvetica, or Times New Roman, minimum 8pt
- **Colours**: Accessible to colour-blind readers (avoid red-green combinations)

23.5.2. Creating journal-quality figures in R

```
library(ggplot2)
library(survival)
library(survminer)
library(RColorBrewer)

# Define a publication theme
theme_publication <- function(base_size = 11) {
  theme_minimal(base_size = base_size) %+replace%
    theme(
      text = element_text(family = "Arial"),
      plot.title = element_text(size = base_size + 2, face = "bold",
                                hjust = 0, margin = margin(b = 10)),
      axis.title = element_text(size = base_size, face = "bold"),
      axis.text = element_text(size = base_size - 1, colour = "black"),
      legend.title = element_text(size = base_size, face = "bold"),
      legend.text = element_text(size = base_size - 1),
      legend.position = "bottom",
      panel.grid.minor = element_blank(),
      panel.grid.major = element_line(colour = "grey90"),
      strip.text = element_text(size = base_size, face = "bold"),
      plot.margin = margin(10, 10, 10, 10)
    )
}

# Colour-blind friendly palette
cb_palette <- c("#0072B2", "#D55E00", "#009E73", "#CC79A7",
                "#F0E442", "#56B4E9", "#E69F00", "#000000")

# --- Kaplan-Meier survival curve ---
fit <- survfit(Surv(time, status) ~ sex, data = lung)
```

```
p_surv <- ggsurvplot(
  fit,
  data = lung,
  palette = cb_palette[1:2],
  risk.table = TRUE,
  risk.table.col = "strata",
  pval = TRUE,
  conf.int = TRUE,
  xlab = "Time (days)",
  ylab = "Survival probability",
  legend.labs = c("Male", "Female"),
  legend.title = "Sex",
  font.main = c(14, "bold"),
  font.x = c(12, "bold"),
  font.y = c(12, "bold"),
  font.tickslab = 11,
  risk.table.fontsize = 3.5,
  ggtheme = theme_publication()
)

print(p_surv)

# --- Saving at journal quality ---
# For TIFF (most journals)
ggsave("figure1_survival.tiff",
       plot = p_surv$plot,
       width = 183, height = 120,
       units = "mm", dpi = 300,
       compression = "lzw")

# For PDF (vector format, scalable)
ggsave("figure1_survival.pdf",
       plot = p_surv$plot,
```

```
       width = 183, height = 120,
       units = "mm", device = cairo_pdf)
```

23.5.3. Publication figures in Python

```
import matplotlib.pyplot as plt
import matplotlib as mpl
import numpy as np
from lifelines import KaplanMeierFitter
from lifelines.datasets import load_lung

# Set publication defaults
plt.rcParams.update({
    'font.family': 'Arial',
    'font.size': 11,
    'axes.labelsize': 12,
    'axes.titlesize': 13,
    'axes.labelweight': 'bold',
    'figure.dpi': 300,
    'savefig.dpi': 300,
    'figure.figsize': (7.2, 4.7),  # ~183mm x 120mm
    'axes.spines.top': False,
    'axes.spines.right': False,
})

# Colour-blind friendly palette
cb_palette = ['#0072B2', '#D55E00', '#009E73', '#CC79A7']

# Prepare data
lung = load_lung()
lung['status'] = lung['status'] - 1
```

```
fig, ax = plt.subplots()

kmf = KaplanMeierFitter()

for i, (sex_val, label) in enumerate([(1, 'Male'), (2, 'Female')]):
    mask = lung['sex'] == sex_val
    kmf.fit(lung.loc[mask, 'T'], lung.loc[mask, 'status'], label=label)
    kmf.plot_survival_function(ax=ax, color=cb_palette[i], ci_show=True)

ax.set_xlabel('Time (days)')
ax.set_ylabel('Survival Probability')
ax.set_title('Kaplan-Meier Survival by Sex')
ax.legend(loc='lower left', frameon=False)
ax.set_ylim(0, 1.05)

plt.tight_layout()
plt.savefig('figure1_survival.tiff', dpi=300, bbox_inches='tight')
plt.savefig('figure1_survival.pdf', bbox_inches='tight')
plt.show()
```

23.5.4. Multi-panel figures

Journals often prefer combined multi-panel figures (Figure 1A, 1B, etc.) rather than separate images.

```
library(patchwork)

p1 <- ggplot(lung, aes(x = factor(sex), y = age)) +
  geom_boxplot(fill = cb_palette[1:2], alpha = 0.7) +
  labs(x = "Sex", y = "Age (years)", title = "A") +
  scale_x_discrete(labels = c("Male", "Female")) +
  theme_publication()
```

```
p2 <- ggplot(lung, aes(x = age, y = ph.karno)) +
  geom_point(alpha = 0.5, colour = cb_palette[1]) +
  geom_smooth(method = "lm", colour = cb_palette[2], se = TRUE) +
  labs(x = "Age (years)", y = "Karnofsky Score", title = "B") +
  theme_publication()

combined <- p1 + p2 +
  plot_layout(ncol = 2, widths = c(1, 1))

ggsave("figure2_panels.tiff",
       plot = combined,
       width = 183, height = 90,
       units = "mm", dpi = 300)
```

23.6. Writing a Statistical Methods Section

23.6.1. Structure

A complete statistical methods section should address:

1. **Study design and population**: Brief recap (detailed in Methods)
2. **Primary and secondary outcomes**: How each was defined and measured
3. **Sample size / power calculation**: How you determined the required sample size
4. **Descriptive statistics**: How continuous and categorical variables are summarised
5. **Primary analysis**: The statistical model, its assumptions, and how they were checked
6. **Missing data**: How much was missing, the assumed mechanism (MCAR/MAR/MNAR), and the handling method

7. **Sensitivity analyses**: Pre-specified alternative approaches
8. **Multiple comparisons**: Whether and how adjusted (Bonferroni, FDR, etc.)
9. **Software**: Packages, versions, and random seeds

23.6.2. Example statistical methods paragraph

Baseline characteristics were summarised as means (SD) for normally distributed continuous variables, medians (IQR) for skewed continuous variables, and frequencies (percentages) for categorical variables. The primary outcome (time to first major adverse cardiovascular event) was analysed using multivariable Cox proportional hazards regression, adjusting for age, sex, diabetes, eGFR, systolic blood pressure, and smoking status. The proportional hazards assumption was tested using Schoenfeld residuals. Missing covariate data (3.2% overall, assumed missing at random) were handled using multiple imputation by chained equations (50 imputed datasets, 20 iterations). Rubin's rules were used to pool effect estimates. Model discrimination was assessed using Harrell's C-statistic with 95% confidence intervals estimated via 500 bootstrap resamples. Calibration was assessed graphically and using the Greenwood-Nam-D'Agostino test. Internal-external cross-validation was performed across the three recruitment centres. All analyses were pre-specified and conducted using R version 4.4.1 (R Foundation for Statistical Computing) with the survival (v3.7), mice (v3.16), and rms (v6.8) packages. Two-sided p-values < 0.05 were considered statistically significant.

> **Common mistakes in statistical methods sections**
>
> 1. Not specifying the handling of missing data
> 2. Claiming "data were normally distributed" without stating how this was assessed
> 3. Using stepwise variable selection without acknowledging its limitations
> 4. Reporting only p-values without effect estimates and confidence intervals
> 5. Not specifying the software and package versions
> 6. Not describing how the proportional hazards or linearity assumptions were checked
> 7. Claiming causation from an observational study without using causal inference methods

23.7. TRIPOD+AI Compliance

The **TRIPOD+AI** statement (Collins et al., 2024) extends the original TRIPOD guideline for prediction models to include AI-specific items. If your study develops, validates, or updates a clinical prediction model — whether based on logistic regression or deep learning — you should follow TRIPOD+AI.

23.7.1. Key TRIPOD+AI items

Item	Requirement	How to address
Title	State it is a prediction model study and the type (development, validation, both)	"Development and external validation of a machine learning model for…"

Item	Requirement	How to address
Source of data	Describe the data source, setting, dates	EHR data from Hospital X, 2015-2023
Participants	Eligibility criteria, sampling, flow diagram	Include a CONSORT-style flow diagram
Outcome	Clear definition, timing, blinding	"30-day all-cause mortality ascertained from hospital records"
Sample size	Justify based on Riley et al. criteria	Use `pmsampsize` package
Missing data	Report extent and handling	Multiple imputation; report complete case sensitivity analysis
Model development	Full specification including hyperparameters	Report all hyperparameters; include code supplement
Model performance	Discrimination (C-statistic, AUROC) AND calibration (calibration plot, slope, intercept)	Never report AUROC alone — calibration is equally important
Model fairness	Assess performance across demographic subgroups	Report C-statistic and calibration by sex, age group, ethnicity
Code and data	Availability statement	Share code via GitHub; data via controlled access repository

23.7.2. Step-by-step TRIPOD+AI walkthrough

```
# Step 1: Sample size calculation for a prediction model
library(pmsampsize)

# Binary outcome model
# Assume: 15 candidate predictors, outcome prevalence 10%,
# target C-statistic 0.80, max shrinkage 10%
ss <- pmsampsize(type = "b",
                 rsquared = 0.15,  # Cox-Snell R-squared
                 parameters = 15,
                 prevalence = 0.10)
ss

# Step 2: Develop model with full pre-specification
library(rms)

# Set up data distribution summaries
dd <- datadist(your_data)
options(datadist = "dd")

# Fit model
full_model <- lrm(outcome ~ age + sex + rcs(sbp, 4) + diabetes + egfr +
                    smoking + rcs(bmi, 3),
                  data = your_data, x = TRUE, y = TRUE)

# Step 3: Internal validation with bootstrap
val <- validate(full_model, B = 200)
print(val)

# Optimism-corrected C-statistic
c_stat_apparent <- full_model$stats["C"]
c_stat_corrected <- c_stat_apparent - (val["Dxy", "index.orig"] -
```

```r
                                        val["Dxy", "index.corrected"]) / 2

# Step 4: Calibration plot
cal <- calibrate(full_model, B = 200)
plot(cal, xlab = "Predicted probability", ylab = "Observed proportion",
     main = "Calibration Plot (Bootstrap-Corrected)")

# Step 5: Fairness assessment
# Evaluate performance in subgroups
for (subgroup in c("Male", "Female")) {
  sub_data <- your_data |> filter(sex == subgroup)
  pred <- predict(full_model, newdata = sub_data, type = "fitted")
  auc <- pROC::auc(sub_data$outcome, pred)
  cat(subgroup, ": C-statistic =", round(auc, 3), "\n")
}
```

23.8. Reproducible Research Workflow

23.8.1. Project structure

A reproducible research project should follow a consistent directory structure:

```
project/
  |-- README.md                # Project overview
  |-- renv.lock                 # R package versions (or environment.yml for conda)
  |-- _targets.R                # Pipeline definition (or Makefile)
  |-- data/
  |     |-- raw/                # Never modify raw data
  |     |-- processed/          # Cleaned datasets
  |     |-- metadata/           # Data dictionaries
```

```
|-- R/                          # R functions
|-- python/                     # Python scripts
|-- analysis/
|      |-- 01_data_cleaning.qmd
|      |-- 02_descriptive.qmd
|      |-- 03_primary_analysis.qmd
|      |-- 04_sensitivity.qmd
|-- output/
|      |-- figures/
|      |-- tables/
|-- manuscript/
|      |-- manuscript.qmd     # The paper itself
|      |-- references.bib
|-- Dockerfile                  # For full computational reproducibility
```

23.8.2. Package management

R (renv):

```r
# Initialise renv in your project
renv::init()

# After installing packages, take a snapshot
renv::snapshot()

# Collaborator restores exact versions
renv::restore()
```

Python (conda/pip):

```python
# Create environment
# conda create -n my_project python=3.11
# conda activate my_project
```

```
# pip install pandas numpy scikit-learn lifelines tableone

# Export environment
# pip freeze > requirements.txt
# OR
# conda env export > environment.yml

# Collaborator recreates environment
# pip install -r requirements.txt
# OR
# conda env create -f environment.yml
```

23.8.3. Version control with Git

Essential git practices for research:

1. **Commit analysis scripts, not data** (unless data are small and non-sensitive)
2. **Use `.gitignore`** to exclude data files, outputs, and sensitive information
3. **Write meaningful commit messages**: "Add multiple imputation for missing eGFR values" not "update script"
4. **Tag milestones**: `git tag -a v1.0-submission -m "Initial journal submission"`
5. **Use branches** for exploratory analyses that may not make it into the paper

23.8.4. Docker for full reproducibility (brief)

A Dockerfile captures the complete computational environment:

```
FROM rocker/verse:4.4.1

# Install system dependencies
RUN apt-get update && apt-get install -y libglpk-dev

# Install R packages
RUN R -e "install.packages(c('survival', 'rms', 'mice', 'gtsummary', 'pmsamps

# Copy project files
COPY . /home/rstudio/project

WORKDIR /home/rstudio/project
```

This guarantees that anyone, anywhere, at any time can reproduce your exact results.

23.9. Case Study: Replicating a Lancet Digital Health Analytical Approach

To illustrate the complete workflow, we replicate the analytical structure of a 2025 Lancet Digital Health prediction model study.

Study design: Development and external validation of a machine learning model to predict 30-day hospital readmission in heart failure patients.

```
# --- STEP 1: Data preparation ---
library(tidyverse)
library(mice)

# Load data (using a public dataset as proxy)
# In practice, this would be your EHR extract
```

```
data("heart", package = "survival")  # Stanford heart transplant data

# Document missingness
missing_summary <- data.frame(
  variable = names(heart),
  n_missing = sapply(heart, function(x) sum(is.na(x))),
  pct_missing = sapply(heart, function(x) mean(is.na(x)) * 100)
)
print(missing_summary)

# Multiple imputation
imp <- mice(heart, m = 50, maxit = 20, method = "pmm", seed = 42,
            printFlag = FALSE)

# --- STEP 2: Table 1 ---
library(gtsummary)

heart |>
  tbl_summary(
    by = transplant,
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} ({p}%)"
    )
  ) |>
  add_overall() |>
  add_p()

# --- STEP 3: Model development (in imputed data) ---
library(rms)
library(pROC)

# Pool results across imputations using Rubin's rules
```

```r
# (Simplified example with complete cases for illustration)
dd <- datadist(heart)
options(datadist = "dd")

# Pre-specify model based on clinical knowledge
model <- lrm(event ~ age + year + surgery + rcs(futime, 4),
             data = heart, x = TRUE, y = TRUE)

print(model)

# --- STEP 4: Internal validation ---
set.seed(42)
val <- validate(model, B = 500)
print(val)

# Optimism-corrected C-statistic
cat("Apparent C-statistic:", model$stats["C"], "\n")
cat("Optimism:", (val["Dxy", "index.orig"] - val["Dxy", "index.corrected"])/2

# --- STEP 5: Calibration plot ---
cal <- calibrate(model, B = 200)
plot(cal,
     xlab = "Predicted Probability",
     ylab = "Observed Proportion",
     main = "Calibration Plot (Internal Bootstrap Validation)")
abline(0, 1, lty = 2, col = "grey50")

# --- STEP 6: Decision curve analysis ---
library(dcurves)

dca_result <- dca(event ~ predicted_prob,
                  data = heart_with_preds,
                  thresholds = seq(0, 0.5, by = 0.01))
```

```r
plot(dca_result)

# --- STEP 7: Export all results ---
# Tables to Word/HTML
tbl_summary_result |> as_gt() |> gt::gtsave("output/tables/table1.docx")

# Figures at publication quality
ggsave("output/figures/figure1_calibration.tiff",
       width = 89, height = 89, units = "mm", dpi = 300)
```

23.10. Capstone Projects

Choose one of the following three capstone projects. Each requires you to produce a complete, journal-ready analysis with Table 1, regression/model results, publication-quality figures, and a statistical methods section.

23.10.1. Capstone 1: Cardiovascular Risk Prediction Model

Objective: Develop a cardiovascular risk prediction model using Framingham Heart Study methodology and validate it in NHANES data.

Dataset: Framingham Heart Study (public teaching dataset from `riskCommunicator` R package) for development; NHANES 2017-2020 (via `nhanesA` package) for external validation.

Requirements:

1. **Table 1**: Baseline characteristics of the development cohort, stratified by 10-year CVD event status.
2. **Model development**: Logistic regression predicting 10-year CVD risk using age, sex, total cholesterol, HDL cholesterol, systolic blood pressure, blood pressure treatment, smoking, and diabetes.

3. **Internal validation**: Bootstrap-corrected C-statistic and calibration slope.
4. **External validation in NHANES**: Apply the model, report discrimination and calibration in the external population.
5. **Fairness assessment**: Report model performance stratified by sex and race/ethnicity.
6. **Figures**: (A) Calibration plot (development, bootstrap-corrected), (B) Calibration plot (NHANES external validation), (C) Decision curve analysis.
7. **Statistical methods section**: Following TRIPOD+AI structure.
8. **Code**: Fully reproducible R or Python scripts with renv/conda environment.

23.10.2. Capstone 2: Metabolic Phenotype Clustering

Objective: Identify metabolic phenotypes in the US adult population using unsupervised learning and assess their association with mortality.

Dataset: NHANES 2017-2020 (demographics, laboratory, examination, and mortality linkage files).

Requirements:

1. **Data preparation**: Select metabolic variables (glucose, HbA1c, insulin, triglycerides, HDL, LDL, waist circumference, BMI, blood pressure, CRP). Handle missing data with multiple imputation.
2. **Dimensionality reduction**: PCA to identify major axes of metabolic variation. Present scree plot and loadings table.
3. **Clustering**: Apply k-means and Gaussian mixture models to PC scores. Use the gap statistic, silhouette width, and BIC to select the number of clusters. Visualise clusters in PC space.
4. **Table 1**: Baseline characteristics stratified by cluster assignment.
5. **Cluster characterisation**: Describe the clinical profile of each cluster (the "metabolic phenotypes").

6. **Survival analysis**: Kaplan-Meier curves and Cox regression for all-cause mortality by cluster, adjusting for age, sex, race/ethnicity, smoking, and education.
7. **Figures**: (A) Scree plot, (B) Cluster visualisation in PC1-PC2 space, (C) Kaplan-Meier curves by cluster, (D) Forest plot of hazard ratios.
8. **Statistical methods section**: Including justification for unsupervised approach and cluster validation.

23.10.3. Capstone 3: Bayesian Survival Analysis

Objective: Fit a Bayesian hierarchical survival model to the primary biliary cholangitis (PBC) dataset, comparing Bayesian and frequentist approaches.

Dataset: `pbc` dataset from the `survival` R package (Mayo Clinic PBC trial, 418 patients).

Requirements:

1. **Table 1**: Baseline characteristics stratified by treatment arm (D-penicillamine vs placebo).
2. **Frequentist Cox model**: Standard Cox PH regression with selected predictors. Check proportional hazards assumption.
3. **Bayesian Cox model**: Fit using `brms` (R) or `PyMC` (Python) with:
   - Weakly informative priors on log-hazard ratios: $\beta \sim N(0, 1)$
   - Compare with informative priors based on prior literature
   - Report posterior medians, 95% credible intervals, and posterior probability of benefit
4. **Hierarchical extension**: If the data include centre/site information, fit a model with random intercepts. Otherwise, use histological stage as a grouping variable.
5. **Model comparison**: Compare DIC/WAIC/LOO-CV between Bayesian models.

6. **Posterior predictive checks**: Simulate survival curves from the posterior and overlay on observed Kaplan-Meier.
7. **Figures**: (A) Prior vs posterior distributions for key parameters, (B) Posterior predictive survival curves, (C) Forest plot comparing Bayesian vs frequentist estimates.
8. **Statistical methods section**: Justify the Bayesian approach, describe priors and sensitivity to prior choice.

Capstone assessment criteria

Each capstone will be assessed on:

- **Statistical correctness** (40%): Appropriate methods, valid assumptions, correct interpretation
- **Reproducibility** (20%): Code runs from start to finish; environment documented
- **Presentation quality** (20%): Journal-ready tables, figures, and methods section
- **Critical thinking** (20%): Discussion of limitations, sensitivity analyses, clinical implications

23.11. Exercises

23.11.1. Exercise 1: Table 1 in both R and Python

Using the `lung` dataset (R) or an equivalent simulated dataset (Python):

a) Create a Table 1 stratified by sex, with appropriate summary statistics for age, ECOG score, Karnofsky score, meal calories, and weight loss.
b) Include standardised mean differences instead of p-values.
c) Export the table to a format suitable for journal submission (Word or LaTeX).

R:

```
data(lung, package = "survival")
# Your code here using gtsummary
```

Python:

```
from lifelines.datasets import load_lung
lung = load_lung()
# Your code here using tableone
```

23.11.2. Exercise 2: Publication-quality figure

Create a multi-panel figure (at least 3 panels) summarising a survival analysis of the `lung` dataset:

- Panel A: Kaplan-Meier curve by sex
- Panel B: Forest plot of hazard ratios from a multivariable Cox model
- Panel C: Calibration plot for a 1-year survival prediction

Requirements: colour-blind accessible palette, Arial font, 300 DPI, single figure file suitable for a double-column journal layout (183 mm wide).

23.11.3. Exercise 3: Write a statistical methods section

Write a complete statistical methods section (250-400 words) for the following study:

> A retrospective cohort study of 5,000 patients with type 2 diabetes from a hospital EHR database (2015-2023). The primary outcome is time to first major adverse cardiovascular event (MACE). The exposure is SGLT2 inhibitor use (vs DPP-4

> inhibitor use). Covariates include age, sex, BMI, HbA1c, eGFR, prior CVD, hypertension, and smoking. Missing data range from 2% (age) to 15% (BMI). The analysis uses propensity score matching followed by Cox regression in the matched cohort.

Your methods section should cover all items listed in the "Writing a statistical methods section" subsection above.

23.11.4. Exercise 4: TRIPOD+AI checklist

Download the TRIPOD+AI checklist (from the EQUATOR Network website). For one of the three capstone projects above, complete the checklist, indicating:

- The page/section number where each item is addressed
- Items that are not applicable and why
- Items that you cannot fully address and what you would need

23.12. Summary

Component	Tool/Package	Key considerations
Table 1	gtsummary (R), tableone (Python)	SMDs over p-values in observational studies
Regression tables	gtsummary, broom, gt (R)	Always include CIs; exponentiate where appropriate
Figures	ggplot2 + patchwork (R), matplotlib (Python)	300+ DPI, colour-blind safe, journal size specs

Component	Tool/Package	Key considerations
Methods section	-	Missing data, assumptions, software versions
Reporting guideline	TRIPOD+AI, STROBE, CONSORT	Choose based on study design
Reproducibility	renv (R), conda (Python), Git, Docker	Capture full computational environment
Calibration	rms, probably (R), scikit-learn (Python)	Never report discrimination without calibration

23.13. References and Further Reading

1. **Collins GS, Moons KGM, Dhiman P, et al.** TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ.* 2024;385:e078378.
2. **von Elm E, Altman DG, Egger M, et al.** The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement. *Lancet.* 2007;370(9596):1453–1457.
3. **Sjoberg DD, Whiting K, Curry M, Lavery JA, Larmarange J.** Reproducible summary tables with the gtsummary package. *The R Journal.* 2021;13(1):570–580.
4. **Riley RD, Ensor J, Snell KIE, et al.** Calculating the sample size required for developing a clinical prediction model. *BMJ.* 2020;368:m441.
5. **Steyerberg EW.** *Clinical Prediction Models: A Practical Approach to Development, Validation, and Updating.* 2nd ed. Springer, 2019.

6. **Van Calster B, McLernon DJ, van Smeden M, et al.** Calibration: the Achilles heel of predictive analytics. *BMC Medicine.* 2019;17:230.

7. **Vickers AJ, Elkin EB.** Decision curve analysis: a novel method for evaluating prediction models. *Medical Decision Making.* 2006;26(6):565–574.

8. **Hernan MA, Robins JM.** *Causal Inference: What If.* Chapman & Hall/CRC, 2024.

9. **Harrell FE.** *Regression Modeling Strategies.* 2nd ed. Springer, 2015.

10. **The Lancet Digital Health.** Author guidelines. Available at https://www.thelancet.com/digital-health.

11. **EQUATOR Network.** Reporting guidelines for health research. Available at https://www.equator-network.org.

12. **Wickham H.** *ggplot2: Elegant Graphics for Data Analysis.* 3rd ed. Springer, 2024.

13. **Wilson G, Bryan J, Cranston K, et al.** Good enough practices in scientific computing. *PLoS Computational Biology.* 2017;13(6):e1005510.

A. Dataset Codebook

Dataset Codebook

This appendix describes all datasets used throughout the course. Each dataset is stored as a CSV file in the `data/` directory.

A.1. Framingham Heart Study (Teaching Dataset)

File: `data/framingham.csv` **Source:** NHLBI Biologic Specimen and Data Repository (BioLINCC) teaching dataset. Also available via the R package `riskCommunicator`. **Description:** Prospective cohort study data from Framingham, Massachusetts. This is the classic cardiovascular epidemiology dataset used to develop the Framingham Risk Score.

Variable	Type	Description
`age`	Continuous	Age at baseline examination (years)
`sex`	Binary	0 = Female, 1 = Male
`bmi`	Continuous	Body mass index (kg/m^2)
`sbp`	Continuous	Systolic blood pressure (mmHg)
`dbp`	Continuous	Diastolic blood pressure (mmHg)
`totchol`	Continuous	Total cholesterol (mg/dL)
`hdl`	Continuous	HDL cholesterol (mg/dL)
`glucose`	Continuous	Fasting blood glucose (mg/dL)
`smoking`	Binary	0 = Non-smoker, 1 = Current smoker
`diabetes`	Binary	0 = No, 1 = Yes
`bp_meds`	Binary	0 = Not on BP meds, 1 = On BP meds
`chd_10yr`	Binary	10-year incident CHD (0 = No, 1 = Yes)

Variable	Type	Description
time_chd	Continuous	Time to CHD event or censoring (days)

Used in: Chapters 3, 4, 5, 10, 11, 19 (Capstone 1)

A.2. Wisconsin Diagnostic Breast Cancer (WDBC)

File: `data/wdbc.csv` **Source:** UCI Machine Learning Repository. Also available via `sklearn.datasets.load_breast_cancer()` in Python and `mlbench::BreastCancer` in R. **Description:** Features computed from digitised images of fine needle aspirates (FNA) of breast masses. Each row represents one tumour sample.

Variable	Type	Description
id	Integer	Patient ID
diagnosis	Binary	M = Malignant, B = Benign
radius_mean	Continuous	Mean of distances from centre to perimeter
texture_mean	Continuous	Standard deviation of grey-scale values
perimeter_mean	Continuous	Mean tumour perimeter
area_mean	Continuous	Mean tumour area
smoothness_mean	Continuous	Local variation in radius lengths
compactness_mean	Continuous	Perimeter^2 / area - 1.0
concavity_mean	Continuous	Severity of concave portions
concave_points_mean	Continuous	Number of concave portions
symmetry_mean	Continuous	Tumour symmetry

Variable	Type	Description
`fractal_dimension_mean`	Continuous	"Coastline approximation" - 1
Plus `_se` and `_worst` variants of each		Standard error and worst (largest) value

Total variables: 32 (ID + diagnosis + 30 features) **Observations:** 569 (357 benign, 212 malignant) **Used in:** Chapters 5, 8, 9, 15

A.3. Primary Biliary Cholangitis (PBC)

File: `data/pbc.csv` **Source:** Mayo Clinic trial in primary biliary cholangitis (formerly primary biliary cirrhosis). Available via `survival::pbc` in R. **Description:** Data from 418 patients enrolled in a randomised trial of D-penicillamine vs placebo for PBC, a chronic liver disease.

Variable	Type	Description
`id`	Integer	Patient ID
`time`	Continuous	Days from registration to death, transplant, or censoring
`status`	Integer	0 = Censored, 1 = Transplant, 2 = Dead
`trt`	Binary	1 = D-penicillamine, 2 = Placebo
`age`	Continuous	Age (years)
`sex`	Binary	0 = Male, 1 = Female
`ascites`	Binary	Presence of ascites
`hepato`	Binary	Presence of hepatomegaly
`spiders`	Binary	Presence of spider angiomata

Variable	Type	Description
`edema`	Ordinal	0 = None, 0.5 = Untreated/resolved, 1 = Despite treatment
`bili`	Continuous	Serum bilirubin (mg/dL)
`chol`	Continuous	Serum cholesterol (mg/dL)
`albumin`	Continuous	Serum albumin (g/dL)
`copper`	Continuous	Urine copper (μg/day)
`alk_phos`	Continuous	Alkaline phosphatase (U/L)
`ast`	Continuous	Aspartate aminotransferase (U/L)
`trig`	Continuous	Triglycerides (mg/dL)
`platelet`	Continuous	Platelet count
`protime`	Continuous	Prothrombin time (seconds)
`stage`	Ordinal	Histologic stage (1–4)

Used in: Chapters 4, 6, 14 (Capstone 3)

A.4. NHANES Subset

File: `data/nhanes_subset.csv` **Source:** National Health and Nutrition Examination Survey (CDC). Extracted using the **`nhanesA`** R package from NHANES 2017–2020 cycle. **Description:** A curated subset of NHANES data with demographics, lab results, and health indicators for dimensionality reduction and clustering exercises.

Variable	Type	Description
`seqn`	Integer	Respondent sequence number

Variable	Type	Description
`age`	Continuous	Age (years)
`sex`	Binary	1 = Male, 2 = Female
`race_ethnicity`	Categorical	Race/ethnicity category
`education`	Ordinal	Education level
`bmi`	Continuous	Body mass index (kg/m^2)
`sbp`	Continuous	Systolic blood pressure (mmHg)
`dbp`	Continuous	Diastolic blood pressure (mmHg)
`hba1c`	Continuous	Glycated haemoglobin (%)
`totchol`	Continuous	Total cholesterol (mg/dL)
`hdl`	Continuous	HDL cholesterol (mg/dL)
`ldl`	Continuous	LDL cholesterol (mg/dL)
`triglycerides`	Continuous	Triglycerides (mg/dL)
`creatinine`	Continuous	Serum creatinine (mg/dL)
`egfr`	Continuous	Estimated GFR (mL/min/1.73m^2)
`albumin`	Continuous	Serum albumin (g/dL)
`diabetes`	Binary	Self-reported diabetes
`hypertension`	Binary	Hypertension (SBP 140 or DBP 90 or on meds)
`smoking`	Categorical	Never / Former / Current
`physical_activity`	Continuous	Minutes of moderate-vigorous activity per week

Used in: Chapters 2, 10, 15, 16, 19 (Capstone 2)

A.5. Simulated Meningitis Data

File: `data/meningitis_sim.csv` **Source:** Simulated dataset inspired by the case study in Lopez-Ayala et al. (*BMJ* 2025). Variable distributions are based on published summary statistics from the Duke University Medical Center meningitis cohort. **Description:** Simulated data on acute meningitis patients for demonstrating spline modelling of non-linear associations.

Variable	Type	Description
`id`	Integer	Patient ID
`age`	Continuous	Age (years)
`sex`	Binary	0 = Female, 1 = Male
`csf_glucose`	Continuous	CSF glucose (mg/dL)
`csf_leuk`	Continuous	CSF leucocyte count (cells/mm^3)
`csf_protein`	Continuous	CSF protein (mg/dL)
`blood_glucose`	Continuous	Blood glucose (mg/dL)
`bacterial`	Binary	1 = Acute bacterial meningitis, 0 = Viral

Observations: 501 **Used in:** Chapter 4 (primary), Chapter 3

A.6. Downloading the data

All datasets are included in the course repository. If you have cloned the repo, they are in the `data/` directory.

To load them:

A.7. R

```r
library(readr)
framingham <- read_csv("data/framingham.csv")
wdbc <- read_csv("data/wdbc.csv")
pbc <- read_csv("data/pbc.csv")
nhanes <- read_csv("data/nhanes_subset.csv")
meningitis <- read_csv("data/meningitis_sim.csv")
```

A.8. Python

```python
import pandas as pd
framingham = pd.read_csv("data/framingham.csv")
wdbc = pd.read_csv("data/wdbc.csv")
pbc = pd.read_csv("data/pbc.csv")
nhanes = pd.read_csv("data/nhanes_subset.csv")
meningitis = pd.read_csv("data/meningitis_sim.csv")
```

B. Mathematical Notation Reference

Mathematical Notation Reference

This appendix provides a quick reference for mathematical notation used throughout the course. You do not need to memorise these — use this page as a lookup when you encounter unfamiliar symbols.

B.1. Basic notation

Symbol	Meaning	Example
x	A variable (lowercase)	Blood pressure measurement
X	A random variable (uppercase)	Blood pressure in the population
$\bar{x}$	Sample mean	Mean blood pressure in your study
μ	Population mean	True mean blood pressure
s or $\hat{\sigma}$	Sample standard deviation	SD of blood pressures in your data
σ	Population standard deviation	True SD of blood pressures
n	Sample size	Number of patients
p	Probability or proportion	Prevalence of a disease
$\hat{p}$	Estimated proportion	Observed prevalence in your sample

B.2. Probability

Symbol	Meaning
$P(A)$	Probability of event A
$P(A \mid B)$	Probability of A given B (conditional probability)
$P(A \cap B)$	Probability of both A and B
$P(A \cup B)$	Probability of A or B (or both)

B.3. Distributions

Notation	Distribution	Parameters
$X \sim N(\mu, \sigma^2)$	Normal	Mean μ, variance σ^2
$X \sim \text{Bin}(n, p)$	Binomial	Trials n, success probability p
$X \sim \text{Pois}(\lambda)$	Poisson	Rate λ
$X \sim \text{Beta}(\alpha, \beta)$	Beta	Shape parameters α, β

B.4. Regression

Symbol	Meaning
y_i	Outcome for patient i
x_i	Predictor value for patient i
β_0	Intercept
$\beta_1, \beta_2, \ldots$	Regression coefficients
$\hat{y}_i$	Predicted value for patient i
ϵ_i	Residual (error) for patient i
$\hat{\beta}$	Estimated coefficient

Symbol	Meaning

Linear regression: $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \epsilon_i$

Logistic regression: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots$

B.5. Penalised regression

Symbol	Meaning
λ	Penalty strength (regularisation parameter)
α	Elastic net mixing parameter (0 = ridge, 1 = LASSO)
$\lVert\beta\rVert_1 = \sum \lvert\beta_j\rvert$	L1 norm (sum of absolute values)
$\lVert\beta\rVert_2^2 = \sum \beta_j^2$	L2 norm squared (sum of squares)

B.6. Survival analysis

Symbol	Meaning
T	Survival time (random variable)
t	A specific time point
$S(t) = P(T > t)$	Survival function
$h(t)$	Hazard function (instantaneous risk)
$H(t)$	Cumulative hazard
HR	Hazard ratio (from Cox model)

Cox model: $h(t \mid \mathbf{x}) = h_0(t)\exp(\beta_1 x_1 + \beta_2 x_2 + \cdots)$

B.7. Bayesian statistics

Symbol	Meaning
θ	Parameter of interest
$P(\theta)$ or $\pi(\theta)$	Prior distribution
$P(D \mid \theta)$ or $L(\theta \mid D)$	Likelihood
$P(\theta \mid D)$	Posterior distribution
$P(D)$	Marginal likelihood (evidence)

Bayes' theorem: $P(\theta \mid D) = \frac{P(D|\theta)\,P(\theta)}{P(D)}$

Or in words: Posterior $\propto$ Likelihood $\times$ Prior

B.8. Model performance

Symbol	Meaning
TP, FP, TN, FN	True/false positives/negatives
Se = TP / (TP + FN)	Sensitivity (recall)
Sp = TN / (TN + FP)	Specificity
PPV = TP / (TP + FP)	Positive predictive value (precision)
NPV = TN / (TN + FN)	Negative predictive value
AUC	Area under the ROC curve
NB	Net benefit (from decision curve analysis)

B.9. Subscripts and superscripts

Notation	Meaning
x_i	Value for individual i
x_{ij}	Value of variable j for individual i
$\hat{\theta}$	Estimated value of θ
θ^*	Bootstrap replicate of θ
x^2	x squared
$\sqrt{x}$	Square root of x

B.10. Greek letters commonly used in statistics

Letter	Name	Common use
α	Alpha	Significance level, elastic net mixing
β	Beta	Regression coefficients, Type II error rate
γ	Gamma	Effect modifier, shape parameter
δ	Delta	Difference, effect size
ϵ	Epsilon	Error term, small quantity
λ	Lambda	Rate parameter, penalty parameter
μ	Mu	Population mean
π	Pi	Proportion, prior
σ	Sigma	Standard deviation
τ	Tau	Between-study variance (meta-analysis)
χ^2	Chi-squared	Chi-squared test statistic

C. Further Reading and Resources

Further Reading and Resources

This annotated bibliography organises key references by topic. It emphasises recent (2024–2026) publications from top medical journals, along with foundational textbooks. Resources are grouped by course part for easy navigation.

C.1. Textbooks

- **Smits LJM, van Kuijk SMJ, Wynants L.** *Improving Health Care with Clinical Prediction Models: From Idea to Impact.* Maastricht University Press, 2026. Open access (CC BY 4.0). Covers the complete prediction model journey from development to implementation. Ideal companion to this course.
- **Steyerberg EW.** *Clinical Prediction Models: A Practical Approach to Development, Validation, and Updating.* 2nd ed. Springer, 2019. The definitive reference on clinical prediction model methodology. Supplementary materials with R code at clinicalpredictionmodels.org.
- **Harrell FE.** *Regression Modeling Strategies.* 2nd ed. Springer, 2015. Comprehensive treatment of regression modelling with emphasis on correct handling of continuous variables, splines, and validation. Free course materials at hbiostat.org.
- **McElreath R.** *Statistical Rethinking: A Bayesian Course with Examples in R and Stan.* 2nd ed. CRC Press, 2020. The most

accessible introduction to Bayesian statistics. Lecture videos freely available online.

- **Gelman A, Carlin JB, Stern HS, et al.** *Bayesian Data Analysis.* 3rd ed. CRC Press, 2013. The comprehensive Bayesian reference. PDF freely available from the authors.

- **Hernán MA, Robins JM.** *Causal Inference: What If.* Chapman & Hall/CRC, 2024. The standard text on causal inference from observational data. Freely available at miguelhernan.org.

- **Boehmke B, Greenwell B.** *Hands-On Machine Learning with R.* CRC Press, 2019. Practical ML in R. Free online at bradley-boehmke.github.io/HOML.

- **Vittinghoff E, Glidden DV, Shiboski SC, McCulloch CE.** *Regression Methods in Biostatistics.* 2nd ed. Springer, 2012. Clear, practical regression modelling for biomedical researchers.

C.2. Reporting Guidelines

- **Collins GS, et al.** TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* 2024;385:e078378. The current standard for reporting prediction models.

- **CONSORT 2025 statement.** Updated guideline for reporting randomised trials. *BMJ* 2025. The latest CONSORT update.

- **TARGET reporting guideline.** For target trial emulation studies. *JAMA* 2025. 21-item checklist analogous to CONSORT.

C.3. Statistical Methods in Medicine (Key Papers 2024–2026)

C.3.1. Splines and Continuous Variables

- **Lopez-Ayala P, Riley RD, Collins GS, Zimmermann T.** Dealing with continuous variables and modelling non-linear associations in healthcare data: practical guide. *BMJ* 2025;390:e082440. Essential reading on splines and fractional polynomials.
- **Austin PC.** Graphical methods to illustrate the nature of the relation between a continuous variable and the outcome when using restricted cubic splines with a Cox proportional hazards model. *Statistical Methods in Medical Research* 2025.

C.3.2. Prediction Model Performance

- **Van Calster B, et al.** Evaluation of performance measures in predictive AI models to support medical decisions: overview and guidance. *Lancet Digital Health* 2025;7:e100916. Reviews 32 performance measures across 5 domains. Essential for understanding discrimination, calibration, and clinical utility.
- **Riley RD, et al.** Evaluation of clinical prediction models (parts 1–3). *BMJ* 2024. Three-part series on development, external validation, and sample size.

C.3.3. Bayesian Methods

- **FDA.** Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products: Draft Guidance for Industry. January 2026. Signals regulatory acceptance of Bayesian methods.

- **Bürkner PC.** brms: An R Package for Bayesian Multilevel Models Using Stan. *Journal of Statistical Software* 2017;80(1). The definitive reference for the brms package.
- **Gelman A, Vehtari A, McElreath R.** Statistical Workflow. December 2025.

C.3.4. Missing Data

- **Wijesuriya R, et al.** Multiple Imputation for Longitudinal Data: A Tutorial. *Statistics in Medicine* 2025. Practical MI tutorial with R and Stata code.

C.3.5. Causal Inference

- **Hernán MA.** Target trial emulation: methods and applications. *NEJM* 2024 (Perspective). Key framework paper.
- **TARGET Reporting Guideline.** *JAMA* 2025. 21-item checklist for target trial emulation.
- **The Causal Roadmap.** Improving the rigor and transparency of causal analyses. *Epidemiology* 2024.

C.3.6. Survival Analysis

- **Austin PC.** Restricted cubic splines with Cox proportional hazards models. *Statistical Methods in Medical Research* 2025. Graphical methods for non-linear survival associations.

C.3.7. Decision Curve Analysis

- **Vickers AJ, et al.** A simple, step-by-step guide to interpreting decision curve analysis. *Diagnostic and Prognostic Research* 2019. Practical tutorial on clinical utility assessment.
- **Vickers AJ, Elkin EB.** Decision curve analysis: a novel method for evaluating prediction models. *Medical Decision Making* 2006;26(6):565–574.

C.3.8. Meta-Analysis

- **Cochrane Handbook for Systematic Reviews of Interventions.** Version 6.5, 2024. The standard reference. Available at training.cochrane.org/handbook.
- **Riley RD, et al.** Individual participant data meta-analysis of prediction model studies. Various publications covering development, validation, and updating.

C.3.9. Machine Learning in Medicine

- **Boehmke B, Greenwell B.** *Hands-On Machine Learning with R.* Free online. Practical R-based ML with good coverage of tree-based methods.
- **Lgatto.** Introduction to Machine Learning with R. Free online at lgatto.github.io/IntroMachineLearningWithR. Concise, exercise-driven introduction.

C.3.10. Deep Learning

- **Goodfellow IJ, Bengio Y, Courville A.** *Deep Learning.* MIT Press, 2016. Freely available at deeplearningbook.org. The definitive theoretical reference for neural networks, CNNs, and RNNs.

- **Zhang A, Lipton ZC, Li M, Smola AJ.** *Dive into Deep Learning.* Cambridge University Press, 2023. Freely available at d2l.ai. Interactive textbook with executable code. Adopted at 500+ universities.
- **Howard J, Gugger S.** *Deep Learning for Coders with fastai and PyTorch.* O'Reilly, 2020. Companion to fast.ai. Coding-first approach ideal for applied researchers.
- **Grinsztajn L, Oyallon E, Varoquaux G.** Why do tree-based models still outperform deep learning on typical tabular data? *NeurIPS* 2022. Benchmark demonstrating that XGBoost beats neural networks on tabular data.
- **Ma J, et al.** Segment anything in medical images. *Nature Communications* 2024;15:654. MedSAM foundation model for universal medical image segmentation.
- **Bedi S, et al.** MedHELM: Holistic Evaluation of Large Language Models for Medical Tasks. *Nature Medicine* 2025. Most comprehensive evaluation of LLMs for clinical tasks.
- **Wiegrebe S, et al.** Deep learning for survival analysis: a review. *Artificial Intelligence Review*, Springer, 2024. Comprehensive taxonomy of deep survival methods.

C.4. R Packages (Key Packages, Recently Updated)

Package	Purpose	Last Updated
`tidyverse`	Data wrangling and visualisation	2024
`rms`	Regression modelling strategies (Harrell)	2024
`glmnet`	Penalised regression (LASSO, ridge, elastic net)	2024
`survival`	Survival analysis	2024
`survminer` / `ggsurvfit`	Survival plot visualisation	2024
`tidymodels`	Unified ML framework	2024
`ranger`	Fast random forests	2024
`xgboost`	Gradient boosting	2025

Package	Purpose	Last Updated
`brms`	Bayesian regression with Stan	2024
`rstanarm`	Bayesian regression (simpler interface)	2024
`mice`	Multiple imputation by chained equations	2024
`dcurves`	Decision curve analysis	2024
`CalibrationCurves`	Calibration plots	2024
`pROC`	ROC curves and AUC	2023
`gtsummary`	Publication-ready summary tables	2024
`gt`	Publication-quality tables	2024
`MatchIt`	Propensity score matching	2024
`WeightIt`	Propensity score weighting	2024
`cobalt`	Covariate balance assessment	2024
`meta` / `metafor`	Meta-analysis	2024
`uwot`	UMAP in R	2024
`Rtsne`	t-SNE in R	2023
`cluster`	Clustering algorithms	2023
`factoextra`	Clustering visualisation	2023
`keras3`	Deep learning (TensorFlow/JAX backend)	2025
`torch`	Deep learning (PyTorch backend)	2025
`naniar`	Missing data visualisation	2024
`finalfit`	Regression modelling and reporting	2024
`marginaleffects`	Marginal effects, predictions, contrasts	2025
`performance`	Model performance and diagnostics	2025
`parameters`	Model parameter extraction	2025
`easystats`	Meta-package for modern statistics	2025

C.5. Python Packages (Key Packages, Recently Updated)

Package	Purpose	Last Updated
`pandas`	Data manipulation	2025
`numpy`	Numerical computing	2025
`scikit-learn`	Machine learning	2025
`statsmodels`	Statistical modelling	2024
`lifelines`	Survival analysis	2024
`xgboost`	Gradient boosting	2025
`lightgbm`	Gradient boosting (fast)	2025
`pymc`	Bayesian modelling	2025
`bambi`	Bayesian regression (brms-like)	2025
`arviz`	Bayesian visualisation and diagnostics	2025
`umap-learn`	UMAP	2024
`matplotlib`	Visualisation	2025
`seaborn`	Statistical visualisation	2024
`plotnine`	ggplot2 for Python	2024
`miceforest`	Multiple imputation (random forest)	2024
`tableone`	Baseline characteristics tables	2024
`great_tables`	Publication-quality tables	2025
`shap`	ML model interpretation	2024
`dowhy`	Causal inference	2024
`tensorflow` / `keras`	Deep learning	2025
`torch`	Deep learning (PyTorch)	2025
`pycox`	Deep learning survival analysis	2024
`torchsurv`	Deep survival analysis (Novartis/FDA)	2024
`plotly`	Interactive visualisation	2025
`patsy`	Design matrices (spline bases)	2023
`pygam`	Generalised additive models	2024

C.6. JAMA Guide to Statistics and Methods

The JAMA Guide to Statistics and Methods is an ongoing essay series explaining statistical techniques in plain language for clinicians. Available at jamanetwork.com. Key recent topics include:

- Bayesian hierarchical models (2024)
- Instrumental variables and heterogeneous treatment effects (2025)
- Target trial emulation (2025)
- Nonparametric statistical analysis (2025)

C.7. BMJ Statistics Notes

The BMJ Statistics Notes series, edited by Doug Altman and Martin Bland, provides short, accessible guides on statistical topics for medical researchers. The series has been running since 1994. Full listing at bmj.com/specialties/statistics-notes.

C.8. Online Courses and Tutorials

- **Frank Harrell's Biostatistics for Biomedical Research** — hbiostat.org/bbr. Free, comprehensive course with R code. Excellent coverage of regression, prediction, and Bayesian methods.
- **Harrell's Regression Modeling Strategies course** — hbiostat.org/course. 4-day intensive course materials.
- **Statistics in Medicine tutorials** — The journal publishes tutorial papers on a wide range of topics. Accessible from onlinelibrary.wiley.com.

D. Exercise Solutions

Exercise Solutions

This appendix collects solutions for all exercises in the course. Each solution is provided in both R and Python. Use them to check your work after attempting the exercises yourself.

> **Try First**
>
> These solutions are most valuable after you have made a genuine attempt at each exercise. Resist the temptation to read the solution before trying.

The solution files are also available in the course repository under `solutions/R/` and `solutions/python/`.

D.1. Chapter 04: Continuous Variables and Splines

D.1.1. Exercise 1

D.1.1.1. R

```
# ==============================================================================
# Chapter 4, Exercise 1: Recognising the Problem
# Categorisation of systolic blood pressure and stroke risk
# ==============================================================================
# This exercise is primarily conceptual. Answers are provided as comments,
```

```
# with supporting code to illustrate the points.

library(ggplot2)

# --- Question (a) ---
# What assumptions does categorising SBP into Normal (<120), Elevated (120-12
# Stage 1 HTN (130-139), Stage 2 HTN (>=140) impose?
#
# ANSWER:
# 1. Within each category, the relationship between SBP and stroke risk is FL
#    A patient with SBP=121 is treated identically to SBP=129.
# 2. At category boundaries, there is a SUDDEN JUMP in risk.
#    SBP=129 and SBP=130 are treated as fundamentally different despite only
#    1 mmHg difference.
# 3. The cut-points (120, 130, 140) are assumed to be biologically meaningful
#    boundaries, which may not reflect the true continuous biology.

# --- Question (b) ---
# A patient with SBP 131 is classified the same as SBP 139. Why is this probl
#
# ANSWER:
# The difference of 8 mmHg (131 vs 139) is clinically substantial. In reality
# stroke risk increases with higher SBP. By lumping these patients together,
# we lose the ability to distinguish their different risk levels. The patient
# at 139 mmHg (close to Stage 2) likely has meaningfully higher stroke risk
# than the patient at 131 mmHg, but the categorised model assigns them
# identical predicted risk. This information loss reduces statistical power
# and can bias effect estimates.

# --- Question (c) ---
# Suggest a better modelling approach.
#
# ANSWER:
```

```r
# Use a restricted cubic spline (RCS) with 3-5 knots to model SBP as a
# continuous predictor. This:
#   - Preserves the full information in SBP
#   - Allows the relationship to be non-linear (capturing any steepening
#     at higher pressures)
#   - Constrains the curve to be linear in the tails (sensible extrapolation)
#   - Uses only k-1 degrees of freedom for k knots (parsimonious)

# --- Supporting demonstration ---
# Simulate data to visually compare categorised vs spline approaches

library(rms)

set.seed(2025)
n <- 1000
sbp <- rnorm(n, mean = 130, sd = 15)
sbp <- pmax(sbp, 90)
sbp <- pmin(sbp, 200)

# True relationship: risk accelerates at higher SBP
logit_p <- -6 + 0.02 * sbp + 0.0002 * (sbp - 130)^2
y <- rbinom(n, 1, plogis(logit_p))

sim_data <- data.frame(sbp = sbp, stroke = y)

# Categorise as described in the exercise
sim_data$sbp_cat <- cut(sim_data$sbp,
                         breaks = c(-Inf, 120, 130, 140, Inf),
                         labels = c("Normal", "Elevated", "Stage1", "Stage2"))

# Set up datadist for rms
dd <- datadist(sim_data)
options(datadist = "dd")
```

```r
# Fit three models
fit_cat <- lrm(stroke ~ sbp_cat, data = sim_data)
fit_linear <- lrm(stroke ~ sbp, data = sim_data)
fit_rcs <- lrm(stroke ~ rcs(sbp, 4), data = sim_data)

# Compare AIC
cat("AIC Comparison:\n")
cat("  Categorised:", AIC(fit_cat), "\n")
cat("  Linear:     ", AIC(fit_linear), "\n")
cat("  RCS (4 knots):", AIC(fit_rcs), "\n")
cat("\nLower AIC = better fit. The RCS model captures the true non-linear\n")
cat("relationship without imposing arbitrary cut-points.\n")

# Plot comparison
sbp_seq <- seq(min(sim_data$sbp), max(sim_data$sbp), length.out = 200)

pred_rcs <- Predict(fit_rcs, sbp = sbp_seq)
pred_linear_vals <- predict(fit_linear,
                            newdata = data.frame(sbp = sbp_seq),
                            type = "fitted")

# True probability
true_logit <- -6 + 0.02 * sbp_seq + 0.0002 * (sbp_seq - 130)^2
true_prob <- plogis(true_logit)

plot_df <- data.frame(
  sbp = rep(sbp_seq, 3),
  probability = c(true_prob, pred_linear_vals,
                  plogis(as.data.frame(pred_rcs)$yhat)),
  Model = rep(c("True relationship", "Linear", "RCS (4 knots)"),
              each = length(sbp_seq))
)
```

```r
p <- ggplot(plot_df, aes(x = sbp, y = probability, colour = Model)) +
  geom_line(linewidth = 1.1) +
  scale_colour_manual(values = c("True relationship" = "black",
                                 "Linear" = "#3498db",
                                 "RCS (4 knots)" = "#e74c3c")) +
  labs(x = "Systolic Blood Pressure (mmHg)",
       y = "Predicted Probability of Stroke",
       title = "Comparing modelling approaches for SBP-stroke relationship") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "bottom")

print(p)
```

D.1.1.2. Python

```python
# ============================================================================
# Chapter 4, Exercise 1: Recognising the Problem
# Categorisation of systolic blood pressure and stroke risk
# ============================================================================
# This exercise is primarily conceptual. Answers are provided as comments,
# with supporting code to illustrate the points.

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.api as sm
from patsy import dmatrix
from scipy.special import expit

# --- Question (a) ---
# What assumptions does categorising SBP into Normal (<120), Elevated (120-129),
# Stage 1 HTN (130-139), Stage 2 HTN (>=140) impose?
```

```
#
# ANSWER:
# 1. Within each category, the relationship between SBP and stroke risk is FL
#    A patient with SBP=121 is treated identically to SBP=129.
# 2. At category boundaries, there is a SUDDEN JUMP in risk.
#    SBP=129 and SBP=130 are treated as fundamentally different despite only
#    1 mmHg difference.
# 3. The cut-points (120, 130, 140) are assumed to be biologically meaningful
#    boundaries, which may not reflect the true continuous biology.

# --- Question (b) ---
# A patient with SBP 131 is classified the same as SBP 139. Why is this probl
#
# ANSWER:
# The difference of 8 mmHg (131 vs 139) is clinically substantial. In reality
# stroke risk increases with higher SBP. By lumping these patients together,
# we lose the ability to distinguish their different risk levels. The patient
# at 139 mmHg (close to Stage 2) likely has meaningfully higher stroke risk
# than the patient at 131 mmHg, but the categorised model assigns them
# identical predicted risk. This information loss reduces statistical power
# and can bias effect estimates.

# --- Question (c) ---
# Suggest a better modelling approach.
#
# ANSWER:
# Use a restricted cubic spline (RCS) with 3-5 knots to model SBP as a
# continuous predictor. This:
#   - Preserves the full information in SBP
#   - Allows the relationship to be non-linear (capturing any steepening
#     at higher pressures)
#   - Constrains the curve to be linear in the tails (sensible extrapolation)
#   - Uses only k-1 degrees of freedom for k knots (parsimonious)
```

```python
# --- Supporting demonstration ---
# Simulate data to visually compare categorised vs spline approaches

np.random.seed(2025)
n = 1000
sbp = np.random.normal(130, 15, size=n)
sbp = np.clip(sbp, 90, 200)

# True relationship: risk accelerates at higher SBP
logit_p = -6 + 0.02 * sbp + 0.0002 * (sbp - 130)**2
y = np.random.binomial(1, expit(logit_p))

df = pd.DataFrame({'sbp': sbp, 'stroke': y})

# --- Model 1: Categorised ---
df['sbp_cat'] = pd.cut(df['sbp'],
                        bins=[-np.inf, 120, 130, 140, np.inf],
                        labels=['Normal', 'Elevated', 'Stage1', 'Stage2'])
X_cat = pd.get_dummies(df['sbp_cat'], drop_first=True).astype(float)
X_cat = sm.add_constant(X_cat)
fit_cat = sm.GLM(df['stroke'], X_cat, family=sm.families.Binomial()).fit()

# --- Model 2: Linear ---
X_lin = sm.add_constant(df[['sbp']])
fit_lin = sm.GLM(df['stroke'], X_lin, family=sm.families.Binomial()).fit()

# --- Model 3: RCS (natural cubic spline with df=3, i.e., 4 knots) ---
X_rcs = dmatrix("cr(sbp, df=3)", data=df, return_type='dataframe')
fit_rcs = sm.GLM(df['stroke'], X_rcs, family=sm.families.Binomial()).fit()

# Compare AIC
print("AIC Comparison:")
print(f"  Categorised: {fit_cat.aic:.1f}")
```

```
print(f"  Linear:      {fit_lin.aic:.1f}")
print(f"  RCS (4 knots): {fit_rcs.aic:.1f}")
print("\nLower AIC = better fit. The RCS model captures the true non-linear"
print("relationship without imposing arbitrary cut-points.")

# --- Plot comparison ---
sbp_range = np.linspace(df['sbp'].min(), df['sbp'].max(), 200)

# True probability
true_logit = -6 + 0.02 * sbp_range + 0.0002 * (sbp_range - 130)**2
true_prob = expit(true_logit)

# Linear predictions
X_lin_pred = sm.add_constant(pd.DataFrame({'sbp': sbp_range}))
lin_prob = fit_lin.predict(X_lin_pred)

# RCS predictions
X_rcs_pred = dmatrix("cr(sbp, df=3)",
                      data=pd.DataFrame({'sbp': sbp_range}),
                      return_type='dataframe')
rcs_prob = fit_rcs.predict(X_rcs_pred)

plt.figure(figsize=(10, 6))
plt.plot(sbp_range, true_prob, 'k--', linewidth=2, label='True relationship'
plt.plot(sbp_range, lin_prob, color='#3498db', linewidth=1.5, label='Linear'
plt.plot(sbp_range, rcs_prob, color='#e74c3c', linewidth=1.5, label='RCS (4 
plt.xlabel('Systolic Blood Pressure (mmHg)')
plt.ylabel('Predicted Probability of Stroke')
plt.title('Comparing modelling approaches for SBP-stroke relationship')
plt.legend()
plt.tight_layout()
plt.show()
```

D.1.2. Exercise 2

D.1.2.1. R

```r
# ==============================================================================
# Chapter 4, Exercise 2: Fitting and Interpreting Spline Models
# PBC dataset: bilirubin and transplant status
# ==============================================================================

library(rms)
library(ggplot2)

# --- Load the PBC data ---
data(pbc, package = "survival")
pbc <- pbc[!is.na(pbc$trt), ]  # Remove incomplete cases

# Create a binary outcome: transplanted (status == 1) vs not
pbc$transplant <- as.numeric(pbc$status == 1)

# Set up datadist (required by rms)
dd <- datadist(pbc)
options(datadist = "dd")

# --- Task 1: Fit logistic regression with bilirubin as a linear term ---
fit_linear <- lrm(transplant ~ bili, data = pbc)
cat("=== Linear Model ===\n")
print(fit_linear)

# --- Task 2: Fit logistic regression with bilirubin using RCS (4 knots) ---
fit_rcs <- lrm(transplant ~ rcs(bili, 4), data = pbc)
cat("\n=== RCS Model (4 knots) ===\n")
print(fit_rcs)
```

```
# --- Task 3: Test for non-linearity ---
cat("\n=== ANOVA for RCS model (test of non-linearity) ===\n")
print(anova(fit_rcs))
# The ANOVA output shows:
#   - Total effect of bili (tests whether bili has ANY association)
#   - Nonlinear component (tests whether the relationship departs from linear
# If the nonlinear p-value is significant, a linear term is insufficient.

# --- Task 4: Compare AIC ---
cat("\n=== AIC Comparison ===\n")
cat("Linear model AIC:", AIC(fit_linear), "\n")
cat("RCS model AIC:   ", AIC(fit_rcs), "\n")
cat("Lower AIC = better fit (penalised for complexity)\n")

# --- Task 5: Plot the relationship ---
p <- Predict(fit_rcs, bili)
pred_df <- as.data.frame(p)

plot_rcs <- ggplot(pred_df, aes(x = bili, y = yhat)) +
  geom_ribbon(aes(ymin = lower, ymax = upper), alpha = 0.2, fill = "#3498db")
  geom_line(linewidth = 1.2, colour = "#2c3e50") +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
  labs(x = "Serum Bilirubin (mg/dL)",
       y = "Log-Odds of Transplant",
       title = "Non-linear effect of bilirubin on transplant (RCS, 4 knots)")
  theme_minimal(base_size = 14)

print(plot_rcs)

# --- Task 6: Interpretation ---
cat("\n=== Interpretation ===\n")
cat("The plot shows the estimated log-odds of transplant as a function of\n")
cat("serum bilirubin. If the curve is not a straight line, the relationship\r
```

```
cat("is non-linear, and a linear term would be insufficient.\n")
cat("\nThe ANOVA non-linearity test (above) formally tests this. If the\n")
cat("non-linear p-value is < 0.05, there is evidence against linearity.\n")
cat("\nCompare the AIC values: a lower AIC for the RCS model confirms\n")
cat("that the additional flexibility improves fit beyond what a linear\n")
cat("term provides.\n")
```

D.1.2.2. Python

```
# ============================================================================
# Chapter 4, Exercise 2: Fitting and Interpreting Spline Models
# PBC dataset: bilirubin and transplant status
# ============================================================================

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.api as sm
from patsy import dmatrix

# --- Simulate data similar to PBC bilirubin-transplant relationship ---
# (Following the chapter's Python approach since PBC is an R-native dataset)
np.random.seed(42)
n = 300
bili = np.random.exponential(scale=2, size=n)
bili = np.clip(bili, 0.3, 25)
logit_p = -3 + 0.1 * bili + 0.005 * bili**2  # Non-linear true relationship
prob = 1 / (1 + np.exp(-logit_p))
transplant = np.random.binomial(1, prob)

df = pd.DataFrame({'bili': bili, 'transplant': transplant})
```

```
# --- Task 1: Fit logistic regression with linear bilirubin ---
X_linear = sm.add_constant(df[['bili']])
fit_linear = sm.GLM(df['transplant'], X_linear,
                     family=sm.families.Binomial()).fit()
print("=== Linear Model ===")
print(fit_linear.summary())

# --- Task 2: Create spline basis and fit logistic regression with splines --
# cr() creates a natural cubic spline (restricted cubic spline)
# df=3 means 4 knots (df = number of knots - 1)
spline_basis = dmatrix("cr(bili, df=3)", data=df, return_type='dataframe')
fit_spline = sm.GLM(df['transplant'], spline_basis,
                      family=sm.families.Binomial()).fit()
print("\n=== Spline Model (RCS, 4 knots) ===")
print(fit_spline.summary())

# --- Task 3: Compare AIC ---
print(f"\n=== AIC Comparison ===")
print(f"Linear model AIC: {fit_linear.aic:.1f}")
print(f"Spline model AIC: {fit_spline.aic:.1f}")
print("Lower AIC = better fit (penalised for complexity)")

# --- Task 4: Test for non-linearity ---
# Compare the linear model to the spline model using a likelihood ratio test
# The difference in deviance follows a chi-squared distribution with
# df = (spline params - linear params) degrees of freedom.
from scipy import stats

lr_stat = fit_linear.deviance - fit_spline.deviance
df_diff = fit_linear.df_model - fit_spline.df_model  # difference in paramete
# Note: df_model counts differently in statsmodels; compute manually
n_params_linear = len(fit_linear.params)
n_params_spline = len(fit_spline.params)
```

```
df_diff = n_params_spline - n_params_linear

p_value = 1 - stats.chi2.cdf(lr_stat, df_diff)
print(f"\n=== Likelihood Ratio Test for Non-linearity ===")
print(f"LR statistic: {lr_stat:.3f}")
print(f"Degrees of freedom: {df_diff}")
print(f"p-value: {p_value:.4f}")
if p_value < 0.05:
    print("Result: Significant non-linearity detected. The spline model fits better.")
else:
    print("Result: No significant non-linearity. A linear term may suffice.")

# --- Task 5: Plot the predicted probabilities across bilirubin range ---
bili_range = np.linspace(df['bili'].min(), df['bili'].max(), 200)

# Linear predictions
X_lin_pred = sm.add_constant(pd.DataFrame({'bili': bili_range}))
lin_prob = fit_linear.predict(X_lin_pred)

# Spline predictions
spline_pred = dmatrix("cr(bili, df=3)",
                       data=pd.DataFrame({'bili': bili_range}),
                       return_type='dataframe')
rcs_prob = fit_spline.predict(spline_pred)

plt.figure(figsize=(10, 6))
plt.plot(bili_range, lin_prob, color='#3498db', linewidth=1.5,
         label='Linear model')
plt.plot(bili_range, rcs_prob, color='#2c3e50', linewidth=2,
         label='RCS model (4 knots)')
plt.scatter(df['bili'], df['transplant'], alpha=0.15, color='grey', s=20)
plt.xlabel('Serum Bilirubin (mg/dL)')
plt.ylabel('Predicted Probability of Transplant')
```

```
plt.title('Non-linear effect of bilirubin on transplant (RCS)')
plt.legend()
plt.tight_layout()
plt.show()

# --- Task 6: Interpretation ---
print("\n=== Interpretation ===")
print("The plot shows the predicted probability of transplant as a function o
print("serum bilirubin, comparing the linear and RCS models.")
print("If the RCS curve departs notably from a straight line, the relationshi
print("is non-linear and the linear model is insufficient.")
print("The AIC comparison and likelihood ratio test formally assess this.")
print("The RCS model captures the accelerating risk at higher bilirubin level
print("that the linear model cannot represent.")
```

D.1.3. Exercise 3

D.1.3.1. R

```
# ==========================================================================
# Chapter 4, Exercise 3: Categorisation vs Splines Head-to-Head
# Simulate U-shaped relationship and compare approaches
# ==========================================================================

library(rms)
library(ggplot2)

set.seed(2025)
n <- 1000
x <- rnorm(n, mean = 50, sd = 10)

# True U-shaped relationship
```

```
logit_p <- -5 + 0.004 * (x - 50)^2
y <- rbinom(n, 1, plogis(logit_p))

sim_data <- data.frame(x = x, y = y)
dd <- datadist(sim_data)
options(datadist = "dd")

# --- Model 1: Categorise into quartiles ---
sim_data$x_cat <- cut(sim_data$x,
                      breaks = quantile(sim_data$x, c(0, 0.25, 0.5, 0.75, 1)),
                      include.lowest = TRUE)
fit_cat <- lrm(y ~ x_cat, data = sim_data)

# --- Model 2: Linear term ---
fit_lin <- lrm(y ~ x, data = sim_data)

# --- Model 3: RCS with 5 knots ---
fit_rcs <- lrm(y ~ rcs(x, 5), data = sim_data)

# --- Question (a): Compare AIC ---
cat("=== Question (a): AIC Comparison ===\n")
cat("Categorised (quartiles):", AIC(fit_cat), "\n")
cat("Linear:                 ", AIC(fit_lin), "\n")
cat("RCS (5 knots):          ", AIC(fit_rcs), "\n")
cat("\nThe RCS model should have the lowest AIC, indicating best fit.\n")
cat("The linear model has the worst fit because it cannot capture\n")
cat("the U-shape at all.\n")

# --- Question (b): Plot all three fitted curves with the true function ---
x_seq <- seq(min(sim_data$x), max(sim_data$x), length.out = 200)

# True curve
true_logit <- -5 + 0.004 * (x_seq - 50)^2
```

```r
true_prob <- plogis(true_logit)

# RCS predictions
pred_rcs <- Predict(fit_rcs, x = x_seq)
rcs_prob <- plogis(as.data.frame(pred_rcs)$yhat)

# Linear predictions
pred_linear <- predict(fit_linear <- lrm(y ~ x, data = sim_data),
                        newdata = data.frame(x = x_seq), type = "fitted")

# Categorised predictions (need to assign each x_seq value to a category)
x_seq_cat <- cut(x_seq,
                  breaks = quantile(sim_data$x, c(0, 0.25, 0.5, 0.75, 1)),
                  include.lowest = TRUE)
pred_cat <- predict(fit_cat,
                     newdata = data.frame(x_cat = x_seq_cat),
                     type = "fitted")

plot_df <- data.frame(
  x = rep(x_seq, 4),
  probability = c(true_prob, pred_linear, pred_cat, rcs_prob),
  Model = rep(c("True relationship", "Linear", "Categorised (quartiles)",
                "RCS (5 knots)"), each = length(x_seq))
)

p <- ggplot(plot_df, aes(x = x, y = probability, colour = Model,
                          linetype = Model)) +
  geom_line(linewidth = 1.1) +
  scale_colour_manual(values = c("True relationship" = "black",
                                  "Linear" = "#3498db",
                                  "Categorised (quartiles)" = "#e74c3c",
                                  "RCS (5 knots)" = "#27ae60")) +
  scale_linetype_manual(values = c("True relationship" = "dashed",
```

```
                                    "Linear" = "solid",
                                    "Categorised (quartiles)" = "solid",
                                    "RCS (5 knots)" = "solid")) +
  labs(x = "Predictor (x)", y = "Predicted Probability",
       title = "Comparing modelling strategies for a U-shaped relationship") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "bottom")

print(p)

cat("\n=== Question (b): Interpretation ===\n")
cat("The RCS model best recovers the true U-shaped relationship.\n")
cat("The linear model completely misses the U-shape (it fits a flat or\n")
cat("slightly sloped line). The categorised model captures some of the\n")
cat("pattern but in crude steps, losing precision.\n")

# --- Question (c): Change quartiles to tertiles ---
cat("\n=== Question (c): Tertile categorisation ===\n")
sim_data$x_tert <- cut(sim_data$x,
                       breaks = quantile(sim_data$x, c(0, 1/3, 2/3, 1)),
                       include.lowest = TRUE)
fit_tert <- lrm(y ~ x_tert, data = sim_data)
cat("AIC with tertiles: ", AIC(fit_tert), "\n")
cat("AIC with quartiles:", AIC(fit_cat), "\n")
cat("AIC with RCS:      ", AIC(fit_rcs), "\n")

cat("\nChanging from quartiles to tertiles changes the categorised model's\n")
cat("estimates. The AIC may differ, and the estimated effects will change\n")
cat("because different patients are grouped together. This demonstrates\n")
cat("that categorisation results are ARBITRARY and depend on cut-point\n")
cat("choice. The RCS model does not have this problem because it models\n")
cat("the continuous relationship directly.\n")
```

```
# Test for non-linearity in the RCS model
cat("\n=== Non-linearity test for RCS model ===\n")
print(anova(fit_rcs))
```

D.1.3.2. Python

```
# ============================================================================
# Chapter 4, Exercise 3: Categorisation vs Splines Head-to-Head
# Simulate U-shaped relationship and compare approaches
# ============================================================================

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.api as sm
from patsy import dmatrix
from scipy.special import expit

np.random.seed(2025)
n = 1000
x = np.random.normal(50, 10, size=n)

# True U-shaped relationship
logit_p = -5 + 0.004 * (x - 50)**2
y = np.random.binomial(1, expit(logit_p))

df = pd.DataFrame({'x': x, 'y': y})

# --- Model 1: Categorise into quartiles ---
df['x_cat'] = pd.qcut(df['x'], q=4, labels=['Q1', 'Q2', 'Q3', 'Q4'])
X_cat = pd.get_dummies(df['x_cat'], drop_first=True).astype(float)
X_cat = sm.add_constant(X_cat)
```

```
fit_cat = sm.GLM(df['y'], X_cat, family=sm.families.Binomial()).fit()

# --- Model 2: Linear ---
X_lin = sm.add_constant(df[['x']])
fit_lin = sm.GLM(df['y'], X_lin, family=sm.families.Binomial()).fit()

# --- Model 3: RCS with df=4 (5 knots) ---
X_rcs = dmatrix("cr(x, df=4)", data=df, return_type='dataframe')
fit_rcs = sm.GLM(df['y'], X_rcs, family=sm.families.Binomial()).fit()

# --- Question (a): Compare AIC ---
print("=== Question (a): AIC Comparison ===")
print(f"Categorised (quartiles): {fit_cat.aic:.1f}")
print(f"Linear:                  {fit_lin.aic:.1f}")
print(f"RCS (5 knots):           {fit_rcs.aic:.1f}")
print("\nThe RCS model should have the lowest AIC, indicating best fit.")
print("The linear model has the worst fit because it cannot capture")
print("the U-shape at all.\n")

# --- Question (b): Plot all three fitted curves with the true function ---
x_range = np.linspace(df['x'].min(), df['x'].max(), 200)

# True curve
true_logit = -5 + 0.004 * (x_range - 50)**2
true_prob = expit(true_logit)

# RCS predictions
X_rcs_pred = dmatrix("cr(x, df=4)",
                     data=pd.DataFrame({'x': x_range}),
                     return_type='dataframe')
rcs_prob = fit_rcs.predict(X_rcs_pred)

# Linear predictions
```

```python
X_lin_pred = sm.add_constant(pd.DataFrame({'x': x_range}))
lin_prob = fit_lin.predict(X_lin_pred)

# Categorised predictions
# Assign each x_range value to its quartile, then predict
quartile_edges = df['x'].quantile([0, 0.25, 0.5, 0.75, 1.0]).values
quartile_edges[0] = -np.inf
quartile_edges[-1] = np.inf
x_cat_pred = pd.cut(x_range, bins=quartile_edges,
                    labels=['Q1', 'Q2', 'Q3', 'Q4'])
X_cat_pred = pd.get_dummies(x_cat_pred, drop_first=True).astype(float)
X_cat_pred = sm.add_constant(X_cat_pred)
# Ensure columns match
for col in X_cat.columns:
    if col not in X_cat_pred.columns:
        X_cat_pred[col] = 0.0
X_cat_pred = X_cat_pred[X_cat.columns]
cat_prob = fit_cat.predict(X_cat_pred)

plt.figure(figsize=(10, 6))
plt.plot(x_range, true_prob, 'k--', linewidth=2, label='True relationship')
plt.plot(x_range, rcs_prob, color='#27ae60', linewidth=2,
         label='RCS (5 knots)')
plt.plot(x_range, cat_prob, color='#e74c3c', linewidth=1.5,
         label='Categorised (quartiles)')
plt.plot(x_range, lin_prob, color='#3498db', linewidth=1.5, label='Linear')
plt.xlabel('Predictor (x)')
plt.ylabel('Predicted Probability')
plt.title('Comparing modelling strategies for a U-shaped relationship')
plt.legend()
plt.tight_layout()
plt.show()
```

```python
print("=== Question (b): Interpretation ===")
print("The RCS model best recovers the true U-shaped relationship.")
print("The linear model completely misses the U-shape.")
print("The categorised model captures some pattern but in crude steps.\n")

# --- Question (c): Change quartiles to tertiles ---
print("=== Question (c): Tertile categorisation ===")
df['x_tert'] = pd.qcut(df['x'], q=3, labels=['T1', 'T2', 'T3'])
X_tert = pd.get_dummies(df['x_tert'], drop_first=True).astype(float)
X_tert = sm.add_constant(X_tert)
fit_tert = sm.GLM(df['y'], X_tert, family=sm.families.Binomial()).fit()

print(f"AIC with tertiles:  {fit_tert.aic:.1f}")
print(f"AIC with quartiles: {fit_cat.aic:.1f}")
print(f"AIC with RCS:       {fit_rcs.aic:.1f}")
print("\nChanging from quartiles to tertiles changes the categorised model's")
print("estimates. This demonstrates that categorisation results are ARBITRARY")
print("and depend on cut-point choice. The RCS model does not have this")
print("problem because it models the continuous relationship directly.")

# --- Likelihood ratio test for non-linearity ---
from scipy import stats

lr_stat = fit_lin.deviance - fit_rcs.deviance
df_diff = len(fit_rcs.params) - len(fit_lin.params)
p_value = 1 - stats.chi2.cdf(lr_stat, df_diff)
print(f"\n=== Non-linearity test ===")
print(f"LR statistic: {lr_stat:.3f}, df: {df_diff}, p-value: {p_value:.6f}")
if p_value < 0.05:
    print("Significant non-linearity: the U-shaped relationship is confirmed.")
```

D.2. Chapter 05: Penalised Regression

D.2.1. Exercise 1

D.2.1.1. R

```
# ===========================================================================
# Chapter 5, Exercise 1: Conceptual Questions
# Penalised Regression: Ridge, LASSO, Elastic Net
# ===========================================================================
# This exercise is conceptual. Answers are provided as comments.

# --- Question (a) ---
# A colleague says: "I used LASSO and it removed age from my model predicting
# mortality. This means age is not a risk factor for death."
# Explain why this interpretation is incorrect.
#
# ANSWER:
# The LASSO removing age does NOT mean age is unimportant. It means that,
# given the OTHER predictors in the model, age does not provide enough
# ADDITIONAL predictive information to justify its inclusion under the
# penalty. Possible reasons why LASSO dropped age:
#
# 1. Age may be highly CORRELATED with another retained variable (e.g.,
#    number of comorbidities, renal function). The LASSO tends to pick
#    one of a group of correlated predictors and drop the rest arbitrarily.
#
# 2. The sample size may be too small to detect age's effect given the
#    penalty strength.
#
# 3. The LASSO performs variable selection based on PREDICTION, not
#    CAUSAL importance. A variable can be a genuine risk factor but
#    redundant for prediction when other variables are present.
```

```
#
# The correct interpretation is: "Age was not selected by the LASSO model,
# possibly because its predictive information overlaps with other retained
# variables. This does not imply age is not a risk factor for death."

# --- Question (b) ---
# Ridge CV-RMSE = 2.1 days. Standard linear regression RMSE = 1.8 days
# (on the same training data). Colleague says standard model is better.
# What is wrong?
#
# ANSWER:
# The comparison is unfair because:
#
# 1. The Ridge RMSE of 2.1 is CROSS-VALIDATED -- it estimates performance
#    on UNSEEN data by training and testing on different folds.
#
# 2. The standard regression RMSE of 1.8 is likely computed on the SAME
#    training data used to fit the model (apparent performance). This is
#    OVERLY OPTIMISTIC because the model has memorised the training data's
#    noise.
#
# 3. If you computed a cross-validated RMSE for the standard regression,
#    it would likely be HIGHER than 2.1 (worse than Ridge), especially
#    with 20 predictors.
#
# The correct comparison requires evaluating both models using the SAME
# methodology -- either both cross-validated, or both on a held-out test
# set. Never compare a cross-validated metric to a training metric.

# --- Question (c) ---
# Explain in one sentence why the LASSO produces exact zeros but Ridge does not.
#
# ANSWER:
```

```
# The L1 penalty (LASSO) constrains coefficients to a diamond-shaped region
# whose corners lie on the axes, so the loss function contours are much more
# likely to first touch the constraint region at a corner (where one or more
# coefficients are exactly zero) than the smooth circular boundary of the L2
# penalty (Ridge), which has no corners.

# --- Supporting code to illustrate the geometry ---
library(ggplot2)

theta <- seq(0, 2 * pi, length.out = 200)

# L2 constraint (circle)
l2_x <- cos(theta)
l2_y <- sin(theta)

# L1 constraint (diamond)
l1_x <- c(1, 0, -1, 0, 1)
l1_y <- c(0, 1, 0, -1, 0)

p <- ggplot() +
  # L2 constraint region
  geom_polygon(data = data.frame(x = l2_x, y = l2_y),
               aes(x, y), fill = "#3498db", alpha = 0.15) +
  geom_path(data = data.frame(x = l2_x, y = l2_y),
            aes(x, y), colour = "#3498db", linewidth = 1.2) +
  # L1 constraint region
  geom_polygon(data = data.frame(x = l1_x, y = l1_y),
               aes(x, y), fill = "#e74c3c", alpha = 0.15) +
  geom_path(data = data.frame(x = l1_x, y = l1_y),
            aes(x, y), colour = "#e74c3c", linewidth = 1.2) +
  # Labels
  annotate("text", x = 0.7, y = 0.7, label = "L2 (Ridge)",
           colour = "#3498db", size = 4) +
```

```
  annotate("text", x = -0.7, y = -0.3, label = "L1 (LASSO)",
           colour = "#e74c3c", size = 4) +
  annotate("text", x = 0, y = -1.3,
           label = "The diamond has corners on the axes\nwhere coefficients = 0",
           size = 3.5) +
  coord_fixed() +
  labs(x = expression(beta[1]), y = expression(beta[2]),
       title = "L1 vs L2 constraint geometry") +
  theme_minimal(base_size = 14) +
  xlim(-1.5, 1.5) + ylim(-1.5, 1.0)

print(p)

cat("\nThe L1 diamond has corners at the axes, making it more likely that\n")
cat("the optimal solution lies at a corner (coefficient = 0), producing\n")
cat("exact sparsity. The L2 circle has no corners, so coefficients are\n")
cat("shrunk toward zero but never reach it exactly.\n")
```

D.2.1.2. Python

```
# ==============================================================================
# Chapter 5, Exercise 1: Conceptual Questions
# Penalised Regression: Ridge, LASSO, Elastic Net
# ==============================================================================
# This exercise is conceptual. Answers are provided as comments.

import numpy as np
import matplotlib.pyplot as plt

# --- Question (a) ---
# A colleague says: "I used LASSO and it removed age from my model predicting
# mortality. This means age is not a risk factor for death."
```

```
# Explain why this interpretation is incorrect.
#
# ANSWER:
# The LASSO removing age does NOT mean age is unimportant. It means that,
# given the OTHER predictors in the model, age does not provide enough
# ADDITIONAL predictive information to justify its inclusion under the
# penalty. Possible reasons why LASSO dropped age:
#
# 1. Age may be highly CORRELATED with another retained variable (e.g.,
#    number of comorbidities, renal function). The LASSO tends to pick
#    one of a group of correlated predictors and drop the rest arbitrarily.
#
# 2. The sample size may be too small to detect age's effect given the
#    penalty strength.
#
# 3. The LASSO performs variable selection based on PREDICTION, not
#    CAUSAL importance. A variable can be a genuine risk factor but
#    redundant for prediction when other variables are present.
#
# The correct interpretation is: "Age was not selected by the LASSO model,
# possibly because its predictive information overlaps with other retained
# variables. This does not imply age is not a risk factor for death."

# --- Question (b) ---
# Ridge CV-RMSE = 2.1 days. Standard linear regression RMSE = 1.8 days
# (on the same training data). Colleague says standard model is better.
# What is wrong?
#
# ANSWER:
# The comparison is unfair because:
#
# 1. The Ridge RMSE of 2.1 is CROSS-VALIDATED -- it estimates performance
#    on UNSEEN data by training and testing on different folds.
```

```
#
# 2. The standard regression RMSE of 1.8 is likely computed on the SAME
#    training data used to fit the model (apparent performance). This is
#    OVERLY OPTIMISTIC because the model has memorised the training data's
#    noise.
#
# 3. If you computed a cross-validated RMSE for the standard regression,
#    it would likely be HIGHER than 2.1 (worse than Ridge), especially
#    with 20 predictors.
#
# The correct comparison requires evaluating both models using the SAME
# methodology -- either both cross-validated, or both on a held-out test
# set. Never compare a cross-validated metric to a training metric.

# --- Question (c) ---
# Explain in one sentence why the LASSO produces exact zeros but Ridge does not.
#
# ANSWER:
# The L1 penalty (LASSO) constrains coefficients to a diamond-shaped region
# whose corners lie on the axes, so the loss function contours are much more
# likely to first touch the constraint region at a corner (where one or more
# coefficients are exactly zero) than the smooth circular boundary of the L2
# penalty (Ridge), which has no corners.

# --- Supporting code to illustrate the geometry ---

theta = np.linspace(0, 2 * np.pi, 200)

# L2 constraint (circle)
l2_x = np.cos(theta)
l2_y = np.sin(theta)

# L1 constraint (diamond)
```

```
l1_x = [1, 0, -1, 0, 1]
l1_y = [0, 1, 0, -1, 0]

fig, ax = plt.subplots(figsize=(7, 7))

# L2 region
ax.fill(l2_x, l2_y, alpha=0.15, color='#3498db')
ax.plot(l2_x, l2_y, color='#3498db', linewidth=1.5, label='L2 (Ridge)')

# L1 region
ax.fill(l1_x, l1_y, alpha=0.15, color='#e74c3c')
ax.plot(l1_x, l1_y, color='#e74c3c', linewidth=1.5, label='L1 (LASSO)')

# Mark the corners
ax.plot([1, -1, 0, 0], [0, 0, 1, -1], 'ro', markersize=6)
ax.annotate('Corner (exact zero for $\\beta_2$)', xy=(1, 0), fontsize=9,
            xytext=(1.1, 0.3), arrowprops=dict(arrowstyle='->', color='red'))

ax.set_xlabel(r'$\beta_1$', fontsize=14)
ax.set_ylabel(r'$\beta_2$', fontsize=14)
ax.set_title('L1 vs L2 constraint geometry')
ax.set_aspect('equal')
ax.legend(fontsize=11)
ax.set_xlim(-1.5, 1.8)
ax.set_ylim(-1.5, 1.5)
ax.axhline(0, color='grey', linewidth=0.5)
ax.axvline(0, color='grey', linewidth=0.5)
plt.tight_layout()
plt.show()

print("The L1 diamond has corners at the axes, making it more likely that")
print("the optimal solution lies at a corner (coefficient = 0), producing")
print("exact sparsity. The L2 circle has no corners, so coefficients are")
```

```
print("shrunk toward zero but never reach it exactly.")
```

D.2.2. Exercise 2

D.2.2.1. R

```
# ==============================================================================
# Chapter 5, Exercise 2: Predicting Diabetes Onset
# Pima Indians Diabetes dataset
# ==============================================================================

library(glmnet)
library(mlbench)
library(ggplot2)

# --- Load data ---
data(PimaIndiansDiabetes2, package = "mlbench")
pima <- na.omit(PimaIndiansDiabetes2)
cat("Complete cases:", nrow(pima), "\n\n")

# Prepare data
X <- as.matrix(pima[, 1:8])
y <- ifelse(pima$diabetes == "pos", 1, 0)

# --- Task 1: Fit a LASSO model with 10-fold cross-validation ---
set.seed(123)
cv_lasso <- cv.glmnet(X, y, family = "binomial", alpha = 1, nfolds = 10)

cat("=== LASSO Cross-Validation ===\n")
cat("Lambda.min:", round(cv_lasso$lambda.min, 4), "\n")
cat("Lambda.1se:", round(cv_lasso$lambda.1se, 4), "\n\n")
```

```
# --- Task 2: Plot the cross-validation curve ---
plot(cv_lasso, main = "LASSO Cross-Validation Curve")

# --- Task 3: Which variables are selected at lambda.1se? ---
cat("=== Variables selected at lambda.1se ===\n")
coef_1se <- coef(cv_lasso, s = "lambda.1se")
print(coef_1se)

# Count non-zero coefficients (excluding intercept)
n_selected_lasso <- sum(coef_1se[-1, ] != 0)
cat("\nNumber of variables selected by LASSO at lambda.1se:", n_selected_lass

# --- Task 4: Fit Ridge and Elastic Net (alpha = 0.5) ---
set.seed(123)
cv_ridge <- cv.glmnet(X, y, family = "binomial", alpha = 0, nfolds = 10)
set.seed(123)
cv_enet <- cv.glmnet(X, y, family = "binomial", alpha = 0.5, nfolds = 10)

# --- Task 5: Compare the three methods ---

# (a) How many variables does each select at lambda.1se?
coef_ridge_1se <- coef(cv_ridge, s = "lambda.1se")
coef_enet_1se <- coef(cv_enet, s = "lambda.1se")

n_selected_ridge <- sum(abs(coef_ridge_1se[-1, ]) > 1e-6)
n_selected_enet <- sum(abs(coef_enet_1se[-1, ]) > 1e-6)

cat("\n=== Number of variables selected (lambda.1se) ===\n")
cat("LASSO:       ", n_selected_lasso, "of 8\n")
cat("Elastic Net: ", n_selected_enet, "of 8\n")
cat("Ridge:       ", n_selected_ridge, "of 8 (Ridge never sets to exactly 0)

# (b) Cross-validated deviance for each
```

```
cat("\n=== Cross-validated deviance (at lambda.min) ===\n")
cat("LASSO:       ", round(min(cv_lasso$cvm), 4), "\n")
cat("Elastic Net: ", round(min(cv_enet$cvm), 4), "\n")
cat("Ridge:       ", round(min(cv_ridge$cvm), 4), "\n")

# (c) Plot the regularisation path for the LASSO model
fit_lasso <- glmnet(X, y, family = "binomial", alpha = 1)
plot(fit_lasso, xvar = "lambda", label = TRUE,
     main = "LASSO Regularisation Path")
abline(v = log(cv_lasso$lambda.min), lty = 2, col = "blue")
abline(v = log(cv_lasso$lambda.1se), lty = 2, col = "red")
legend("topright", legend = c("lambda.min", "lambda.1se"),
       col = c("blue", "red"), lty = 2, cex = 0.8)

# --- Task 6: Recommendation ---
cat("\n=== Recommendation ===\n")
cat("For the Pima diabetes dataset (8 predictors, ~390 complete cases):\n")
cat("- The number of predictors is small relative to sample size,\n")
cat("  so penalisation is not strictly necessary.\n")
cat("- LASSO is useful here for identifying the most predictive variables.\n")
cat("- All three methods produce similar cross-validated deviance,\n")
cat("  confirming that the choice of penalty type matters less than\n")
cat("  using penalisation vs not.\n")
cat("- For variable selection and a parsimonious model, LASSO at\n")
cat("  lambda.1se is a good choice.\n")
cat("- For best prediction, any of the three at lambda.min would work.\n")

# --- Bonus: Compare coefficients side by side ---
coef_comparison <- data.frame(
  Variable = colnames(X),
  LASSO = as.vector(coef(cv_lasso, s = "lambda.1se"))[-1],
  Ridge = as.vector(coef(cv_ridge, s = "lambda.1se"))[-1],
  ElasticNet = as.vector(coef(cv_enet, s = "lambda.1se"))[-1]
```

```
)

cat("\n=== Coefficient comparison (lambda.1se) ===\n")
print(round(coef_comparison, 4))
```

D.2.2.2. Python

```
# ===========================================================================
# Chapter 5, Exercise 2: Predicting Diabetes Onset
# Pima Indians Diabetes dataset
# ===========================================================================

import numpy as np
import pandas as pd
from sklearn.linear_model import LogisticRegression, LogisticRegressionCV
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import cross_val_score
import matplotlib.pyplot as plt

# --- Load the Pima Indians Diabetes dataset ---
url = ("https://raw.githubusercontent.com/jbrownlee/Datasets/master/"
       "pima-indians-diabetes.data.csv")
columns = ['pregnancies', 'glucose', 'blood_pressure', 'skin_thickness',
           'insulin', 'bmi', 'diabetes_pedigree', 'age', 'outcome']
pima = pd.read_csv(url, names=columns)

# Remove rows with zero values where zero is not meaningful
cols_no_zero = ['glucose', 'blood_pressure', 'skin_thickness',
                'insulin', 'bmi']
pima[cols_no_zero] = pima[cols_no_zero].replace(0, np.nan)
pima = pima.dropna()
```

```
print(f"Complete cases: {len(pima)}\n")

X = pima[columns[:-1]].values
y = pima['outcome'].values

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# --- Task 1: Fit LASSO with cross-validation ---
lasso_cv = LogisticRegressionCV(
    penalty='l1', solver='saga', Cs=50, cv=10,
    scoring='neg_log_loss', max_iter=10000, random_state=42
)
lasso_cv.fit(X_scaled, y)

print("=== LASSO Cross-Validation ===")
print(f"Best C (= 1/lambda): {lasso_cv.C_[0]:.4f}")
print(f"Best lambda: {1/lasso_cv.C_[0]:.4f}\n")

print("LASSO selected features:")
n_selected_lasso = 0
for name, coef in zip(columns[:-1], lasso_cv.coef_[0]):
    if abs(coef) > 1e-6:
        print(f"  {name}: {coef:.4f}")
        n_selected_lasso += 1
print(f"\nNumber selected: {n_selected_lasso} of 8\n")

# --- Task 2: Fit Ridge with cross-validation ---
ridge_cv = LogisticRegressionCV(
    penalty='l2', solver='lbfgs', Cs=50, cv=10,
    scoring='neg_log_loss', max_iter=10000, random_state=42
)
ridge_cv.fit(X_scaled, y)
```

```
# --- Task 3: Fit Elastic Net with cross-validation ---
enet_cv = LogisticRegressionCV(
    penalty='elasticnet', solver='saga', Cs=50, cv=10,
    l1_ratios=[0.25, 0.5, 0.75], scoring='neg_log_loss',
    max_iter=10000, random_state=42
)
enet_cv.fit(X_scaled, y)

print(f"Elastic Net best l1_ratio (alpha): {enet_cv.l1_ratio_[0]:.2f}")
print(f"Elastic Net best C (1/lambda): {enet_cv.C_[0]:.4f}\n")

# --- Task 4: Compare cross-validated scores ---
print("=== Cross-validated log-loss (lower = better) ===")
for name, model in [('LASSO', lasso_cv), ('Ridge', ridge_cv),
                     ('Elastic Net', enet_cv)]:
    scores = cross_val_score(model, X_scaled, y, cv=10,
                             scoring='neg_log_loss')
    print(f"  {name}: {-scores.mean():.4f} (+/- {scores.std():.4f})")

# --- Task 5: Plot regularisation path for LASSO ---
Cs = np.logspace(-2, 2, 80)
coefs = []
for C in Cs:
    model = LogisticRegression(penalty='l1', C=C, solver='saga',
                               max_iter=10000, random_state=42)
    model.fit(X_scaled, y)
    coefs.append(model.coef_[0])

coefs = np.array(coefs)
lambdas = 1.0 / Cs

plt.figure(figsize=(10, 6))
for j, name in enumerate(columns[:-1]):
```

```
    plt.plot(np.log10(lambdas), coefs[:, j], label=name)
plt.axvline(x=np.log10(1/lasso_cv.C_[0]), color='k', linestyle='--',
            label='CV optimal lambda')
plt.xlabel('log10(lambda)')
plt.ylabel('Coefficient')
plt.title('LASSO Regularisation Path - Pima Diabetes')
plt.legend(loc='best', fontsize=8)
plt.tight_layout()
plt.show()

# --- Task 6: Recommendation ---
print("\n=== Recommendation ===")
print("For the Pima diabetes dataset (8 predictors, ~390 complete cases):")
print("- The number of predictors is small relative to sample size,")
print("  so penalisation is not strictly necessary.")
print("- LASSO is useful for identifying the most predictive variables.")
print("- All three methods produce similar cross-validated performance.")
print("- For variable selection, LASSO provides a parsimonious model.")
print("- For best prediction, any method works well here.")

# --- Bonus: Compare coefficients side by side ---
print("\n=== Coefficient comparison ===")
coef_df = pd.DataFrame({
    'Variable': columns[:-1],
    'LASSO': lasso_cv.coef_[0],
    'Ridge': ridge_cv.coef_[0],
    'Elastic Net': enet_cv.coef_[0]
})
print(coef_df.round(4).to_string(index=False))
```

D.2.3. Exercise 3

D.2.3.1. R

```r
# ==============================================================================
# Chapter 5, Exercise 3: The Effect of Correlation
# How LASSO and Ridge handle correlated predictors differently
# ==============================================================================

library(glmnet)
library(MASS)

set.seed(2025)
n <- 200

# --- Create 4 correlated predictors (correlation ~ 0.9) ---
Sigma <- matrix(0.9, 4, 4)
diag(Sigma) <- 1
X_corr <- mvrnorm(n, mu = rep(0, 4), Sigma = Sigma)

# Add 6 independent predictors (noise)
X_indep <- matrix(rnorm(n * 6), n, 6)
X <- cbind(X_corr, X_indep)
colnames(X) <- c(paste0("Corr_", 1:4), paste0("Noise_", 1:6))

# True model: all 4 correlated predictors have effect = 0.5
true_beta <- c(rep(0.5, 4), rep(0, 6))
y <- rbinom(n, 1, plogis(X %*% true_beta))

# --- Task 1: Fit LASSO and examine which variables are selected ---
set.seed(42)
cv_lasso <- cv.glmnet(X, y, family = "binomial", alpha = 1)
cat("=== Task 1: LASSO coefficients (lambda.1se) ===\n")
```

```
print(round(as.matrix(coef(cv_lasso, s = "lambda.1se")), 3))

# --- Task 2: Fit Ridge and examine coefficients ---
set.seed(42)
cv_ridge <- cv.glmnet(X, y, family = "binomial", alpha = 0)
cat("\n=== Task 2: Ridge coefficients (lambda.1se) ===\n")
print(round(as.matrix(coef(cv_ridge, s = "lambda.1se")), 3))

# --- Task 3: Fit Elastic Net (alpha = 0.5) ---
set.seed(42)
cv_enet <- cv.glmnet(X, y, family = "binomial", alpha = 0.5)
cat("\n=== Task 3: Elastic Net coefficients (lambda.1se) ===\n")
print(round(as.matrix(coef(cv_enet, s = "lambda.1se")), 3))

# --- Question (a) ---
cat("\n=== Question (a): Does LASSO select all 4 correlated predictors? ===\n")
lasso_coefs <- as.vector(coef(cv_lasso, s = "lambda.1se"))[-1]
names(lasso_coefs) <- colnames(X)
selected_lasso <- names(lasso_coefs[abs(lasso_coefs) > 1e-6])
cat("LASSO selected:", paste(selected_lasso, collapse = ", "), "\n")
cat("LASSO typically selects only 1-2 of the 4 correlated predictors.\n")
cat("This is because the LASSO arbitrarily picks one predictor from a\n")
cat("correlated group and drops the rest. With correlation ~0.9, any\n")
cat("single correlated predictor carries almost the same information\n")
cat("as all four combined, so the LASSO keeps just one (or two) and\n")
cat("sets the others to zero.\n")

# --- Question (b) ---
cat("\n=== Question (b): How does Ridge handle correlated predictors? ===\n")
ridge_coefs <- as.vector(coef(cv_ridge, s = "lambda.1se"))[-1]
names(ridge_coefs) <- colnames(X)
cat("Ridge coefficients for correlated predictors:\n")
print(round(ridge_coefs[1:4], 4))
```

```
cat("\nRidge distributes the effect EVENLY across all correlated predictors.\
cat("All four correlated predictors retain non-zero (and similar) coefficient
cat("This is the 'grouping effect' -- Ridge shrinks correlated predictors\n")
cat("toward each other rather than picking one arbitrarily.\n")

# --- Question (c) ---
cat("\n=== Question (c): Does Elastic Net improve on LASSO? ===\n")
enet_coefs <- as.vector(coef(cv_enet, s = "lambda.1se"))[-1]
names(enet_coefs) <- colnames(X)
selected_enet <- names(enet_coefs[abs(enet_coefs) > 1e-6])
cat("Elastic Net selected:", paste(selected_enet, collapse = ", "), "\n")
cat("The Elastic Net typically selects MORE of the correlated predictors\n")
cat("than LASSO, thanks to the L2 component (grouping effect).\n")
cat("It provides a middle ground: some sparsity (like LASSO) but\n")
cat("better handling of correlated groups (like Ridge).\n")

# --- Question (d): Stability analysis ---
cat("\n=== Question (d): Stability analysis (10 different seeds) ===\n")
cat("LASSO variable selection across 10 bootstrap samples:\n")
for (seed in 1:10) {
  set.seed(seed)
  idx <- sample(n, n, replace = TRUE)
  cv_boot <- cv.glmnet(X[idx, ], y[idx], family = "binomial", alpha = 1,
                       nfolds = 5)
  boot_coefs <- as.vector(coef(cv_boot, s = "lambda.1se"))[-1]
  selected <- colnames(X)[abs(boot_coefs) > 1e-6]
  cat(sprintf("  Seed %2d: %s\n", seed, paste(selected, collapse = ", ")))
}

cat("\nRidge variable selection (all retained) across 10 bootstrap samples:\n
for (seed in 1:10) {
  set.seed(seed)
  idx <- sample(n, n, replace = TRUE)
```

```
  cv_boot <- cv.glmnet(X[idx, ], y[idx], family = "binomial", alpha = 0,
                       nfolds = 5)
  boot_coefs <- as.vector(coef(cv_boot, s = "lambda.1se"))[-1]
  selected <- colnames(X)[abs(boot_coefs) > 0.01]
  cat(sprintf("  Seed %2d: %s\n", seed, paste(selected, collapse = ", ")))
}

cat("\nConclusion: LASSO variable selection is UNSTABLE for correlated\n")
cat("predictors -- the selected predictor(s) change across samples.\n")
cat("Ridge is stable because it always retains all predictors.\n")
cat("Elastic Net offers a compromise with better stability than LASSO.\n")
```

D.2.3.2. Python

```
# ============================================================================
# Chapter 5, Exercise 3: The Effect of Correlation
# How LASSO and Ridge handle correlated predictors differently
# ============================================================================

import numpy as np
import pandas as pd
from sklearn.linear_model import LogisticRegressionCV
from sklearn.preprocessing import StandardScaler
from scipy.special import expit

np.random.seed(2025)
n = 200

# --- Create 4 correlated predictors (correlation ~ 0.9) ---
mean = np.zeros(4)
cov = np.full((4, 4), 0.9)
np.fill_diagonal(cov, 1.0)
```

```
X_corr = np.random.multivariate_normal(mean, cov, size=n)

# Add 6 independent noise predictors
X_indep = np.random.randn(n, 6)
X = np.hstack([X_corr, X_indep])

feature_names = ([f"Corr_{i+1}" for i in range(4)] +
                 [f"Noise_{i+1}" for i in range(6)])

# True model: all 4 correlated predictors contribute
true_beta = np.array([0.5]*4 + [0.0]*6)
y = np.random.binomial(1, expit(X @ true_beta))

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# --- Task 1: LASSO ---
lasso = LogisticRegressionCV(penalty='l1', solver='saga', Cs=50,
                              cv=10, max_iter=10000, random_state=42)
lasso.fit(X_scaled, y)

print("=== Task 1: LASSO coefficients ===")
for name, coef in zip(feature_names, lasso.coef_[0]):
    marker = " *" if abs(coef) > 1e-6 else ""
    print(f"  {name:10s}: {coef:8.4f}{marker}")

# --- Task 2: Ridge ---
ridge = LogisticRegressionCV(penalty='l2', solver='lbfgs', Cs=50,
                              cv=10, max_iter=10000, random_state=42)
ridge.fit(X_scaled, y)

print("\n=== Task 2: Ridge coefficients ===")
for name, coef in zip(feature_names, ridge.coef_[0]):
```

```
    print(f"  {name:10s}: {coef:8.4f}")

# --- Task 3: Elastic Net ---
enet = LogisticRegressionCV(penalty='elasticnet', solver='saga', Cs=50,
                             cv=10, l1_ratios=[0.5], max_iter=10000,
                             random_state=42)
enet.fit(X_scaled, y)

print("\n=== Task 3: Elastic Net coefficients ===")
for name, coef in zip(feature_names, enet.coef_[0]):
    marker = " *" if abs(coef) > 1e-6 else ""
    print(f"  {name:10s}: {coef:8.4f}{marker}")

# --- Question (a): Does LASSO select all 4 correlated predictors? ---
print("\n=== Question (a) ===")
selected_lasso = [name for name, coef in zip(feature_names, lasso.coef_[0])
                  if abs(coef) > 1e-6]
print(f"LASSO selected: {selected_lasso}")
print("LASSO typically selects only 1-2 of the 4 correlated predictors.")
print("This is because the LASSO arbitrarily picks one predictor from a")
print("correlated group and drops the rest. With correlation ~0.9, any")
print("single predictor carries almost the same information as all four.")

# --- Question (b): How does Ridge handle correlated predictors? ---
print("\n=== Question (b) ===")
print("Ridge coefficients for correlated predictors:")
for i in range(4):
    print(f"  {feature_names[i]}: {ridge.coef_[0][i]:.4f}")
print("Ridge distributes the effect EVENLY across all correlated predictors.")
print("All four retain non-zero (and similar) coefficients.")

# --- Question (c): Does Elastic Net improve on LASSO? ---
print("\n=== Question (c) ===")
```

```
selected_enet = [name for name, coef in zip(feature_names, enet.coef_[0])
                 if abs(coef) > 1e-6]
print(f"Elastic Net selected: {selected_enet}")
print("The Elastic Net typically selects MORE of the correlated predictors")
print("than LASSO, thanks to the L2 component (grouping effect).")

# --- Question (d): Stability analysis ---
print("\n=== Question (d): Stability analysis (10 bootstrap samples) ===")
print("\nLASSO variable selection:")
for seed in range(10):
    rng = np.random.RandomState(seed)
    idx = rng.choice(n, n, replace=True)
    lasso_boot = LogisticRegressionCV(penalty='l1', solver='saga',
                                      Cs=20, cv=5, max_iter=10000,
                                      random_state=42)
    lasso_boot.fit(X_scaled[idx], y[idx])
    selected = [feature_names[j] for j in range(10)
                if abs(lasso_boot.coef_[0][j]) > 1e-6]
    print(f"  Seed {seed}: {selected}")

print("\nRidge - all predictors retained (non-zero coefficients):")
for seed in range(10):
    rng = np.random.RandomState(seed)
    idx = rng.choice(n, n, replace=True)
    ridge_boot = LogisticRegressionCV(penalty='l2', solver='lbfgs',
                                      Cs=20, cv=5, max_iter=10000,
                                      random_state=42)
    ridge_boot.fit(X_scaled[idx], y[idx])
    # Ridge always keeps all predictors; show which have coef > 0.01
    selected = [feature_names[j] for j in range(10)
                if abs(ridge_boot.coef_[0][j]) > 0.01]
    print(f"  Seed {seed}: {selected}")
```

```
print("\nConclusion: LASSO variable selection is UNSTABLE for correlated")
print("predictors -- the selected predictor(s) change across samples.")
print("Ridge is stable because it always retains all predictors.")
print("Elastic Net offers a compromise with better stability than LASSO.")
```

D.3. Chapter 06: Survival Analysis

D.3.1. Exercise 1

D.3.1.1. R

```
# ===============================================================================
# Chapter 6, Exercise 1: Kaplan-Meier Curves by Disease Stage
# PBC dataset: KM curves stratified by stage, log-rank test, median survival
# ===============================================================================

library(survival)
library(survminer)
library(tidyverse)

# --- Load and prepare the PBC dataset ---
data(pbc, package = "survival")

pbc_clean <- pbc %>%
  filter(!is.na(trt)) %>%
  mutate(
    status_binary = ifelse(status == 2, 1, 0),
    time_years = time / 365.25,
    trt_label = factor(trt, labels = c("D-penicillamine", "Placebo"))
  ) %>%
  filter(!is.na(stage))  # Remove missing stage values
```

```
cat("N =", nrow(pbc_clean), "\n")
cat("Events (deaths):", sum(pbc_clean$status_binary), "\n")
cat("Stages:", paste(sort(unique(pbc_clean$stage)), collapse = ", "), "\n\n")

# --- Fit Kaplan-Meier curves stratified by disease stage ---
km_stage <- survfit(Surv(time_years, status_binary) ~ stage, data = pbc_clean
print(km_stage)

# --- Report median survival time for each stage ---
cat("\n=== Median Survival Time by Stage ===\n")
median_surv <- summary(km_stage)$table
print(median_surv)

# Extract and display more clearly
cat("\nMedian survival (years) by stage:\n")
for (s in sort(unique(pbc_clean$stage))) {
  km_s <- survfit(Surv(time_years, status_binary) ~ 1,
                  data = pbc_clean[pbc_clean$stage == s, ])
  med <- surv_median(km_s)
  cat(sprintf("  Stage %d: %.2f years (95%% CI: %.2f - %.2f)\n",
              s, med$median, med$lower, med$upper))
}

# --- Perform the log-rank test ---
cat("\n=== Log-Rank Test ===\n")
logrank <- survdiff(Surv(time_years, status_binary) ~ stage, data = pbc_clean
print(logrank)

# Extract p-value
pvalue <- 1 - pchisq(logrank$chisq, length(logrank$n) - 1)
cat(sprintf("\nLog-rank test p-value: %.6f\n", pvalue))

if (pvalue < 0.05) {
```

```
  cat("Result: Significant difference in survival across disease stages.\n")
} else {
  cat("Result: No significant difference detected.\n")
}

# --- Plot the Kaplan-Meier curves ---
p <- ggsurvplot(
  km_stage,
  data = pbc_clean,
  pval = TRUE,                # Show log-rank p-value
  conf.int = TRUE,             # Show confidence intervals
  risk.table = TRUE,           # Show number at risk table
  xlab = "Time (years)",
  ylab = "Survival probability",
  title = "Kaplan-Meier Curves by Disease Stage (PBC)",
  legend.title = "Stage",
  palette = c("#2ecc71", "#3498db", "#e67e22", "#e74c3c"),
  ggtheme = theme_minimal()
)

print(p)

# --- Interpretation ---
cat("\n=== Interpretation ===\n")
cat("The KM curves should show a clear separation by stage, with higher\n")
cat("stages having worse survival. The log-rank test formally confirms\n")
cat("whether these differences are statistically significant.\n")
cat("As expected in PBC, patients with Stage 4 disease have the worst\n")
cat("prognosis, while Stage 1-2 patients have substantially better survival.\n")
```

D.3.1.2. Python

```
# ================================================================================
# Chapter 6, Exercise 1: Kaplan-Meier Curves by Disease Stage
# PBC dataset (simulated): KM curves by stage, log-rank test, median surviva
# ================================================================================

import numpy as np
import pandas as pd
from lifelines import KaplanMeierFitter
from lifelines.statistics import logrank_test, multivariate_logrank_test
import matplotlib.pyplot as plt

# --- Simulate PBC-like data with stage variable ---
# (Since PBC is an R-native dataset, we simulate its structure as done
#  in the chapter's Python code)
np.random.seed(42)
n = 312

# Simulate stage (1-4) with realistic proportions
stage = np.random.choice([1, 2, 3, 4], size=n, p=[0.05, 0.25, 0.40, 0.30])

# Simulate survival times that depend on stage (higher stage = worse surviva
# Base scale varies by stage
scale_by_stage = {1: 14, 2: 10, 3: 7, 4: 4}
time_years = np.array([np.random.exponential(scale=scale_by_stage[s]) for s i
time_years = np.clip(time_years, 0.1, 15)  # Clip to realistic range

# Event probability increases with stage
event_prob = {1: 0.20, 2: 0.35, 3: 0.50, 4: 0.65}
event = np.array([np.random.binomial(1, event_prob[s]) for s in stage])

pbc = pd.DataFrame({
```

```
    'time_years': time_years,
    'event': event,
    'stage': stage
})

print(f"N = {len(pbc)}")
print(f"Events (deaths): {pbc['event'].sum()}")
print(f"Stages: {sorted(pbc['stage'].unique())}\n")

# --- Fit KM for each stage and report median survival ---
print("=== Median Survival Time by Stage ===")
kmf_dict = {}
for s in sorted(pbc['stage'].unique()):
    mask = pbc['stage'] == s
    kmf = KaplanMeierFitter()
    kmf.fit(pbc.loc[mask, 'time_years'],
            event_observed=pbc.loc[mask, 'event'],
            label=f'Stage {s}')
    kmf_dict[s] = kmf

    median_surv = kmf.median_survival_time_
    ci = kmf.confidence_interval_survival_function_
    print(f"  Stage {s}: Median survival = {median_surv:.2f} years "
          f"(n={mask.sum()}, events={pbc.loc[mask, 'event'].sum()})")

# --- Perform the log-rank test (overall comparison across all stages) ---
print("\n=== Log-Rank Test (overall) ===")
result = multivariate_logrank_test(
    pbc['time_years'], pbc['stage'], pbc['event']
)
print(f"Test statistic: {result.test_statistic:.3f}")
print(f"p-value: {result.p_value:.6f}")
```

```
if result.p_value < 0.05:
    print("Result: Significant difference in survival across disease stages.")
else:
    print("Result: No significant difference detected.")

# --- Pairwise log-rank tests (Stage 1 vs 4, etc.) ---
print("\n=== Pairwise Log-Rank Tests (selected) ===")
for s1, s2 in [(1, 4), (2, 3), (3, 4)]:
    mask1 = pbc['stage'] == s1
    mask2 = pbc['stage'] == s2
    if mask1.sum() > 0 and mask2.sum() > 0:
        lr = logrank_test(
            pbc.loc[mask1, 'time_years'], pbc.loc[mask2, 'time_years'],
            pbc.loc[mask1, 'event'], pbc.loc[mask2, 'event']
        )
        print(f"  Stage {s1} vs {s2}: p = {lr.p_value:.4f}")

# --- Plot KM curves for all stages ---
fig, ax = plt.subplots(figsize=(10, 6))

colors = {1: '#2ecc71', 2: '#3498db', 3: '#e67e22', 4: '#e74c3c'}
for s, kmf in kmf_dict.items():
    kmf.plot_survival_function(ax=ax, ci_show=True, color=colors[s])

ax.set_xlabel("Time (years)", fontsize=12)
ax.set_ylabel("Survival probability", fontsize=12)
ax.set_title("Kaplan-Meier Curves by Disease Stage (PBC)", fontsize=14)
ax.legend(fontsize=10)

# Add p-value annotation
ax.text(0.7, 0.95, f'Log-rank p = {result.p_value:.4f}',
        transform=ax.transAxes, fontsize=11,
        verticalalignment='top',
```

```
        bbox=dict(boxstyle='round', facecolor='white', alpha=0.8))

plt.tight_layout()
plt.show()

# --- Interpretation ---
print("\n=== Interpretation ===")
print("The KM curves show a clear separation by stage, with higher stages")
print("having worse survival. The log-rank test formally confirms whether")
print("these differences are statistically significant.")
print("Patients with Stage 4 disease have the worst prognosis, while")
print("Stage 1-2 patients have substantially better survival.")
```

D.3.2. Exercise 2

D.3.2.1. R

```
# ============================================================================
# Chapter 6, Exercise 2: Build and Evaluate a Cox Model
# PBC dataset: Cox PH with age, log(bili), albumin, protime, edema
# ============================================================================

library(survival)
library(survminer)
library(tidyverse)

# --- Load and prepare data ---
data(pbc, package = "survival")

pbc_clean <- pbc %>%
  filter(!is.na(trt)) %>%
  mutate(
```

```
    status_binary = ifelse(status == 2, 1, 0),
    time_years = time / 365.25,
    trt_label = factor(trt, labels = c("D-penicillamine", "Placebo")),
    log_bili = log(bili)
  ) %>%
  select(time_years, status_binary, age, log_bili, albumin, protime, edema) %
  drop_na()

cat("Complete cases:", nrow(pbc_clean), "\n")
cat("Events:", sum(pbc_clean$status_binary), "\n\n")

# --- Task 1: Fit the Cox model and report hazard ratios ---
cox_model <- coxph(
  Surv(time_years, status_binary) ~ age + log_bili + albumin + protime + eden
  data = pbc_clean
)

cat("=== Task 1: Cox Model Summary ===\n")
print(summary(cox_model))

# Extract and display HRs with 95% CIs
cat("\n=== Hazard Ratios and 95% CIs ===\n")
hr_table <- data.frame(
  Variable = names(coef(cox_model)),
  HR = exp(coef(cox_model)),
  Lower_95 = exp(confint(cox_model))[, 1],
  Upper_95 = exp(confint(cox_model))[, 2],
  p_value = summary(cox_model)$coefficients[, 5]
)
print(round(hr_table, 4))

# Forest plot
ggforest(cox_model, data = pbc_clean)
```

```
# --- Task 2: Check the proportional hazards assumption ---
cat("\n=== Task 2: Proportional Hazards Assumption ===\n")
ph_test <- cox.zph(cox_model)
print(ph_test)

cat("\nInterpretation of PH test:\n")
cat("- A significant p-value (< 0.05) indicates violation of the PH assumption.\n")
cat("- Look at the GLOBAL test and individual covariate tests.\n")

# Which covariates violate PH?
ph_df <- data.frame(
  Variable = rownames(ph_test$table)[-nrow(ph_test$table)],
  p_value = ph_test$table[-nrow(ph_test$table), "p"]
)
violators <- ph_df$Variable[ph_df$p_value < 0.05]
if (length(violators) > 0) {
  cat("Covariates violating PH assumption (p < 0.05):",
      paste(violators, collapse = ", "), "\n")
} else {
  cat("No covariates significantly violate the PH assumption.\n")
}

# Plot Schoenfeld residuals
par(mfrow = c(2, 3))
plot(ph_test)
par(mfrow = c(1, 1))

# --- Task 3: Concordance index ---
cat("\n=== Task 3: Concordance Index ===\n")
c_index <- summary(cox_model)$concordance[1]
c_se <- summary(cox_model)$concordance[2]
cat(sprintf("C-index: %.3f (SE: %.3f)\n", c_index, c_se))
cat(sprintf("95%% CI: %.3f - %.3f\n", c_index - 1.96 * c_se, c_index + 1.96 * c_se))
```

```r
cat("\nInterpretation:\n")
cat("- C-index = 0.5: model has no discriminative ability (random)\n")
cat("- C-index = 1.0: perfect discrimination\n")
cat("- C-index > 0.7: generally considered acceptable\n")
cat("- C-index > 0.8: considered strong discrimination\n")
cat(sprintf("This model's C-index of %.3f indicates %s discrimination.\n",
            c_index,
            ifelse(c_index > 0.8, "strong",
                   ifelse(c_index > 0.7, "acceptable", "modest"))))

# --- Task 4: Predict 5-year survival for a specific patient ---
cat("\n=== Task 4: 5-Year Survival Prediction ===\n")
new_patient <- data.frame(
  age = 55,
  log_bili = log(3),       # bilirubin = 3 mg/dL
  albumin = 3.2,
  protime = 11,
  edema = 0                # no edema
)

cat("Patient profile:\n")
cat("  Age: 55 years\n")
cat("  Bilirubin: 3 mg/dL (log = ", round(log(3), 2), ")\n")
cat("  Albumin: 3.2 g/dL\n")
cat("  Prothrombin time: 11 seconds\n")
cat("  Edema: none\n\n")

# Get predicted survival function
pred_surv <- survfit(cox_model, newdata = new_patient)

# Extract 5-year survival probability
surv_summary <- summary(pred_surv, times = 5)
cat(sprintf("Predicted 5-year survival probability: %.3f (%.1f%%)\n",
```

```
            surv_summary$surv, surv_summary$surv * 100))
cat(sprintf("95%% CI: %.3f - %.3f\n",
            surv_summary$lower, surv_summary$upper))

# Plot the predicted survival curve
plot(pred_surv,
     xlab = "Time (years)",
     ylab = "Survival probability",
     main = "Predicted Survival for Patient Profile",
     lwd = 2, col = "blue")
abline(h = 0.5, lty = 2, col = "grey")
abline(v = 5, lty = 2, col = "red")
legend("topright", c("Predicted survival", "5-year mark"),
       col = c("blue", "red"), lty = c(1, 2), lwd = c(2, 1))
```

D.3.2.2. Python

```
# ============================================================================
# Chapter 6, Exercise 2: Build and Evaluate a Cox Model
# PBC-like dataset: Cox PH with age, log(bili), albumin, protime, edema
# ============================================================================

import numpy as np
import pandas as pd
from lifelines import CoxPHFitter
import matplotlib.pyplot as plt

# --- Simulate PBC-like clinical dataset ---
# (Following the chapter's Python approach with simulated data)
np.random.seed(42)
n = 276
```

```
pbc = pd.DataFrame({
    'time_years': np.abs(np.random.exponential(7, n) +
                         np.random.normal(0, 1, n)),
    'event': np.random.binomial(1, 0.45, n),
    'age': np.random.normal(50, 10, n),
    'log_bili': np.random.normal(0.5, 1.0, n),     # log(bilirubin)
    'albumin': np.random.normal(3.5, 0.4, n),
    'protime': np.random.normal(11, 1, n),
    'edema': np.random.choice([0, 0.5, 1], n, p=[0.75, 0.15, 0.10])
})

# Make survival time depend on covariates to create realistic associations
hazard_linear = (0.04 * pbc['age'] + 0.3 * pbc['log_bili'] -
                 0.5 * pbc['albumin'] + 0.2 * pbc['protime'] +
                 0.8 * pbc['edema'])
# Adjust times based on hazard
pbc['time_years'] = pbc['time_years'] * np.exp(-0.1 * hazard_linear)
pbc['time_years'] = np.clip(pbc['time_years'], 0.05, 15)

print(f"N = {len(pbc)}")
print(f"Events: {pbc['event'].sum()}")
print(f"Median follow-up: {pbc['time_years'].median():.2f} years\n")

# --- Task 1: Fit Cox model and report hazard ratios ---
cph = CoxPHFitter()
cph.fit(pbc, duration_col='time_years', event_col='event',
        formula='age + log_bili + albumin + protime + edema')

print("=== Task 1: Cox Model Summary ===")
cph.print_summary()

# Extract HR table
print("\n=== Hazard Ratios and 95% CIs ===")
```

```
hr_table = pd.DataFrame({
    'HR': np.exp(cph.params_),
    'Lower_95': np.exp(cph.confidence_intervals_.iloc[:, 0]),
    'Upper_95': np.exp(cph.confidence_intervals_.iloc[:, 1]),
    'p_value': cph.summary['p']
})
print(hr_table.round(4))

# Forest plot
fig, ax = plt.subplots(figsize=(8, 5))
cph.plot(ax=ax)
ax.set_title("Cox PH Model - Hazard Ratios")
plt.tight_layout()
plt.show()

# --- Task 2: Check proportional hazards assumption ---
print("\n=== Task 2: Proportional Hazards Assumption ===")
try:
    cph.check_assumptions(pbc, p_value_threshold=0.05, show_plots=True)
except Exception as e:
    print(f"PH assumption check result: {e}")
    print("If no warnings are raised, the PH assumption is not violated.")

# Interpretation
print("\nInterpretation:")
print("- A significant p-value (< 0.05) indicates violation of PH.")
print("- Look at the Schoenfeld residual plots for trends over time.")
print("- If a covariate violates PH, consider stratifying on it or")
print("  adding a time interaction.")

# --- Task 3: Concordance index ---
print(f"\n=== Task 3: Concordance Index ===")
c_index = cph.concordance_index_
```

```
print(f"C-index: {c_index:.3f}")
print(f"\nInterpretation:")
print(f"- C-index = 0.5: no discriminative ability (random)")
print(f"- C-index = 1.0: perfect discrimination")
print(f"- C-index > 0.7: generally considered acceptable")
print(f"- C-index > 0.8: considered strong discrimination")
quality = "strong" if c_index > 0.8 else ("acceptable" if c_index > 0.7 else
print(f"This model's C-index of {c_index:.3f} indicates {quality} discriminat

# --- Task 4: 5-year survival for a specific patient ---
print(f"\n=== Task 4: 5-Year Survival Prediction ===")
print("Patient profile:")
print("  Age: 55 years")
print(f"  Bilirubin: 3 mg/dL (log = {np.log(3):.2f})")
print("  Albumin: 3.2 g/dL")
print("  Prothrombin time: 11 seconds")
print("  Edema: none (0)")

new_patient = pd.DataFrame({
    'age': [55],
    'log_bili': [np.log(3)],    # bilirubin = 3 mg/dL
    'albumin': [3.2],
    'protime': [11],
    'edema': [0]
})

# Get predicted survival function
surv_func = cph.predict_survival_function(new_patient)

# Find 5-year survival
# Get the closest time point to 5 years
times = surv_func.index
closest_5yr = times[np.argmin(np.abs(times - 5))]
```

```
surv_5yr = surv_func.loc[closest_5yr].values[0]

print(f"\nPredicted 5-year survival probability: {surv_5yr:.3f} ({surv_5yr*100:.1f}%)")

# Plot predicted survival curve
fig, ax = plt.subplots(figsize=(8, 5))
surv_func.plot(ax=ax, color='blue', linewidth=2, label='Predicted survival')
ax.axhline(y=0.5, color='grey', linestyle='--', alpha=0.7, label='50% survival')
ax.axvline(x=5, color='red', linestyle='--', alpha=0.7, label='5-year mark')
ax.set_xlabel("Time (years)", fontsize=12)
ax.set_ylabel("Survival probability", fontsize=12)
ax.set_title("Predicted Survival for Patient Profile", fontsize=14)
ax.legend()
plt.tight_layout()
plt.show()
```

D.3.3. Exercise 3

D.3.3.1. R

```
# ==============================================================================
# Chapter 6, Exercise 3: Competing Risks Thinking
# Kidney transplant rejection with death as a competing risk
# ==============================================================================
# This exercise is primarily conceptual (parts 1-2) with a bonus coding
# component (part 3).

# --- Question 1 ---
# Explain why treating death as censoring would bias the results.
# What direction would the bias go?
#
# ANSWER:
```

```
# Treating death as simple censoring VIOLATES the assumption of
# NON-INFORMATIVE censoring. Non-informative censoring means that
# censored patients have the same future risk of the event as those
# who remain under observation. But patients who die are NOT like
# patients who remain alive -- they have ZERO future risk of rejection
# (because dead patients cannot experience rejection).
#
# The bias direction: treating death as censoring INFLATES (overestimates)
# the estimated probability of rejection. This happens because the
# Kaplan-Meier estimator assumes that censored patients would eventually
# experience the event at the same rate as those still at risk. But dead
# patients will NEVER experience rejection, so acting as if they could
# leads to an overestimate of the cumulative incidence of rejection.
#
# In the KM framework, when a patient dies and is "censored," they are
# removed from the risk set, effectively assuming they would have had
# the same rejection rate as surviving patients. Since they actually
# have zero rejection risk, this inflates the estimated rate.

# --- Question 2 ---
# Which approach for estimating probability of rejection within 2 years:
# cause-specific hazards or Fine-Gray?
#
# ANSWER:
# For estimating the PROBABILITY of rejection within 2 years, use the
# FINE-GRAY subdistribution hazard model. Here is why:
#
# - Cause-specific hazards model the RATE of rejection among those
#   currently alive and rejection-free. This answers: "Among patients
#   still alive, what is the instantaneous risk of rejection?" This is
#   useful for understanding ETIOLOGY (what causes rejection).
#
# - The Fine-Gray model directly models the CUMULATIVE INCIDENCE
```

```r
#   FUNCTION, which gives the probability of rejection by time t,
#   accounting for the fact that some patients will die first. This
#   is the right quantity for PREDICTION and CLINICAL COMMUNICATION.
#
# When your goal is prediction ("What is the probability that this
# patient's transplant will be rejected within 2 years?"), you need
# the cumulative incidence, which properly accounts for the competing
# risk of death. The Fine-Gray model provides this directly.

# --- Question 3 (Bonus): Fine-Gray model in R ---

library(survival)
library(tidycmprsk)
library(ggplot2)

# Simulate kidney transplant data with competing risks
set.seed(2025)
n <- 500

# Covariates
age <- rnorm(n, mean = 50, sd = 12)
donor_type <- rbinom(n, 1, 0.4)  # 0 = living, 1 = deceased donor
hla_mismatch <- rpois(n, lambda = 2)

# Simulate competing event times
# Time to rejection (cause 1)
lambda_reject <- exp(-4 + 0.02 * age + 0.5 * donor_type + 0.2 * hla_mismatch)
time_reject <- rexp(n, rate = lambda_reject)

# Time to death without rejection (cause 2)
lambda_death <- exp(-5 + 0.03 * age + 0.3 * donor_type)
time_death <- rexp(n, rate = lambda_death)
```

```
# Administrative censoring at 10 years
time_censor <- runif(n, 5, 10)

# Determine observed time and event type
time_obs <- pmin(time_reject, time_death, time_censor)
event <- ifelse(time_obs == time_censor, 0,
                ifelse(time_obs == time_reject, 1, 2))
# 0 = censored, 1 = rejection, 2 = death

df <- data.frame(
  time = time_obs,
  event = event,
  age = age,
  donor_type = factor(donor_type, labels = c("Living", "Deceased")),
  hla_mismatch = hla_mismatch
)

cat("=== Dataset summary ===\n")
cat("N:", n, "\n")
cat("Rejections (event=1):", sum(event == 1), "\n")
cat("Deaths (event=2):", sum(event == 2), "\n")
cat("Censored (event=0):", sum(event == 0), "\n\n")

# --- Cumulative Incidence Function (CIF) ---
# Using tidycmprsk for a tidy interface
cuminc_fit <- cuminc(Surv(time, event) ~ donor_type, data = df)
print(cuminc_fit)

# Plot cumulative incidence by donor type
p <- ggcuminc(cuminc_fit, outcome = "1") +  # outcome 1 = rejection
  labs(x = "Time (years)",
       y = "Cumulative Incidence of Rejection",
       title = "Cumulative Incidence of Transplant Rejection by Donor Type")
```

```
  theme_minimal(base_size = 14)
print(p)

# --- Fine-Gray subdistribution hazard model ---
cat("\n=== Fine-Gray Model ===\n")
# Using tidycmprsk::crr for Fine-Gray regression
fg_model <- crr(Surv(time, event) ~ age + donor_type + hla_mismatch,
                data = df, failcode = 1)  # failcode 1 = rejection
print(summary(fg_model))

cat("\nInterpretation:\n")
cat("- The Fine-Gray model estimates subdistribution hazard ratios.\n")
cat("- These quantify how covariates affect the cumulative incidence\n")
cat("  of rejection, accounting for the competing risk of death.\n")
cat("- HR > 1 means higher cumulative incidence of rejection.\n")

# --- Compare with cause-specific Cox model ---
cat("\n=== Cause-Specific Cox Model (for comparison) ===\n")
# Censor deaths for cause-specific analysis of rejection
df$event_cs <- ifelse(df$event == 1, 1, 0)  # only rejection is event
cox_cs <- coxph(Surv(time, event_cs) ~ age + donor_type + hla_mismatch,
                data = df)
print(summary(cox_cs))

cat("\n=== Comparison Summary ===\n")
cat("The cause-specific and Fine-Gray HRs can differ because they\n")
cat("answer different questions:\n")
cat("- Cause-specific: 'What affects the rate of rejection among\n")
cat("  those still alive?'\n")
cat("- Fine-Gray: 'What affects the probability of rejection by\n")
cat("  time t, accounting for death?'\n")
cat("Both are valid; the choice depends on whether your goal is\n")
cat("etiological understanding or clinical prediction.\n")
```

D.3.3.2. Python

```
# ==============================================================================
# Chapter 6, Exercise 3: Competing Risks Thinking
# Kidney transplant rejection with death as a competing risk
# ==============================================================================
# This exercise is primarily conceptual (parts 1-2) with a bonus coding
# component (part 3).

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# --- Question 1 ---
# Explain why treating death as censoring would bias the results.
# What direction would the bias go?
#
# ANSWER:
# Treating death as simple censoring VIOLATES the assumption of
# NON-INFORMATIVE censoring. Non-informative censoring means that
# censored patients have the same future risk of the event as those
# who remain under observation. But patients who die are NOT like
# patients who remain alive -- they have ZERO future risk of rejection
# (because dead patients cannot experience rejection).
#
# The bias direction: treating death as censoring INFLATES (overestimates)
# the estimated probability of rejection. This happens because the
# Kaplan-Meier estimator assumes that censored patients would eventually
# experience the event at the same rate as those still at risk. But dead
# patients will NEVER experience rejection, so acting as if they could
# leads to an overestimate of the cumulative incidence of rejection.
#
# In the KM framework, when a patient dies and is "censored," they are
```

```python
# removed from the risk set, effectively assuming they would have had
# the same rejection rate as surviving patients. Since they actually
# have zero rejection risk, this inflates the estimated rate.

# --- Question 2 ---
# Which approach for estimating probability of rejection within 2 years:
# cause-specific hazards or Fine-Gray?
#
# ANSWER:
# For estimating the PROBABILITY of rejection within 2 years, use the
# FINE-GRAY subdistribution hazard model. Here is why:
#
# - Cause-specific hazards model the RATE of rejection among those
#   currently alive and rejection-free. This answers: "Among patients
#   still alive, what is the instantaneous risk of rejection?" This is
#   useful for understanding ETIOLOGY (what causes rejection).
#
# - The Fine-Gray model directly models the CUMULATIVE INCIDENCE
#   FUNCTION, which gives the probability of rejection by time t,
#   accounting for the fact that some patients will die first. This
#   is the right quantity for PREDICTION and CLINICAL COMMUNICATION.
#
# When your goal is prediction ("What is the probability that this
# patient's transplant will be rejected within 2 years?"), you need
# the cumulative incidence, which properly accounts for the competing
# risk of death. The Fine-Gray model provides this directly.

# --- Question 3 (Bonus): Competing risks analysis in Python ---

from lifelines import KaplanMeierFitter, CoxPHFitter

# Simulate kidney transplant data with competing risks
np.random.seed(2025)
```

```
n = 500

# Covariates
age = np.random.normal(50, 12, size=n)
donor_type = np.random.binomial(1, 0.4, size=n)  # 0=living, 1=deceased
hla_mismatch = np.random.poisson(2, size=n)

# Simulate competing event times
# Time to rejection (cause 1)
lambda_reject = np.exp(-4 + 0.02 * age + 0.5 * donor_type + 0.2 * hla_mismatc
time_reject = np.random.exponential(1 / lambda_reject)

# Time to death without rejection (cause 2)
lambda_death = np.exp(-5 + 0.03 * age + 0.3 * donor_type)
time_death = np.random.exponential(1 / lambda_death)

# Administrative censoring at 5-10 years
time_censor = np.random.uniform(5, 10, size=n)

# Determine observed time and event type
time_obs = np.minimum(np.minimum(time_reject, time_death), time_censor)
event = np.where(time_obs == time_censor, 0,
                 np.where(time_obs == time_reject, 1, 2))
# 0 = censored, 1 = rejection, 2 = death

df = pd.DataFrame({
    'time': time_obs,
    'event': event,
    'age': age,
    'donor_type': donor_type,
    'hla_mismatch': hla_mismatch
})
```

```
print("=== Dataset summary ===")
print(f"N: {n}")
print(f"Rejections (event=1): {(event == 1).sum()}")
print(f"Deaths (event=2): {(event == 2).sum()}")
print(f"Censored (event=0): {(event == 0).sum()}\n")

# --- Approach 1: Naive KM (treating death as censoring) ---
# This is WRONG but illustrative
kmf_naive = KaplanMeierFitter()
event_naive = (event == 1).astype(int)  # only rejection = event; death = censored
kmf_naive.fit(time_obs, event_observed=event_naive, label="Naive KM (death = censored)")

# --- Approach 2: Cause-specific Cox model ---
# For rejection: censor deaths and administrative censoring
print("=== Cause-Specific Cox Model (rejection) ===")
df_cs = df.copy()
df_cs['event_rejection'] = (df_cs['event'] == 1).astype(int)

cph_cs = CoxPHFitter()
cph_cs.fit(df_cs, duration_col='time', event_col='event_rejection',
           formula='age + donor_type + hla_mismatch')
cph_cs.print_summary()

# --- Approach 3: Cumulative Incidence Function (Aalen-Johansen estimator) ---
# Compute the CIF manually using the Aalen-Johansen approach
# This properly accounts for competing risks

def compute_cif(times, events, cause=1):
    """
    Compute the cumulative incidence function for a specific cause
    using the Aalen-Johansen estimator.
    """
    # Sort by time
```

```
    order = np.argsort(times)
    t_sorted = times[order]
    e_sorted = events[order]

    unique_times = np.unique(t_sorted[e_sorted > 0])
    n_risk = len(times)

    cif = []
    surv = 1.0  # Overall survival (from all causes)
    cif_times = []

    for t in unique_times:
        at_risk = np.sum(t_sorted >= t)
        d_cause = np.sum((t_sorted == t) & (e_sorted == cause))
        d_all = np.sum((t_sorted == t) & (e_sorted > 0))

        if at_risk > 0:
            # CIF increment: S(t-) * d_cause/n_risk
            cif_increment = surv * (d_cause / at_risk)
            # Update overall survival
            surv *= (1 - d_all / at_risk)

            cif.append(cif_increment)
            cif_times.append(t)

    # Cumulative sum
    cif_cumulative = np.cumsum(cif)
    return np.array(cif_times), cif_cumulative

# Compute CIF for rejection (cause 1)
cif_times, cif_values = compute_cif(time_obs, event, cause=1)

# Compute CIF for death (cause 2)
```

```
cif_times_d, cif_values_d = compute_cif(time_obs, event, cause=2)

# --- Plot: Naive KM vs CIF ---
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Left panel: Naive KM (1 - S(t)) vs CIF for rejection
ax1 = axes[0]
# Naive 1 - KM
naive_times = kmf_naive.survival_function_.index
naive_1_minus_s = 1 - kmf_naive.survival_function_.values.flatten()
ax1.step(naive_times, naive_1_minus_s, color='red', linewidth=2,
         label='Naive 1-KM (BIASED)', where='post')
# Proper CIF
ax1.step(cif_times, cif_values, color='blue', linewidth=2,
         label='Cumulative Incidence (correct)', where='post')
ax1.set_xlabel("Time (years)", fontsize=12)
ax1.set_ylabel("Probability of Rejection", fontsize=12)
ax1.set_title("Naive KM vs Cumulative Incidence", fontsize=13)
ax1.legend(fontsize=10)
ax1.set_xlim(0, 10)

# Right panel: Stacked CIF (rejection + death)
ax2 = axes[1]
# Interpolate to common time grid
from scipy import interpolate

t_grid = np.linspace(0.01, min(cif_times.max(), cif_times_d.max()), 200)
f_reject = interpolate.interp1d(cif_times, cif_values, kind='previous',
                                 bounds_error=False, fill_value=(0, cif_values[-1]))
f_death = interpolate.interp1d(cif_times_d, cif_values_d, kind='previous',
                                bounds_error=False, fill_value=(0, cif_values_d[-1]))

cif_r = f_reject(t_grid)
```

```
cif_d = f_death(t_grid)

ax2.fill_between(t_grid, 0, cif_r, alpha=0.4, color='#e74c3c', label='Rejecti
ax2.fill_between(t_grid, cif_r, cif_r + cif_d, alpha=0.4,
                 color='#3498db', label='Death (competing)')
ax2.plot(t_grid, cif_r + cif_d, color='black', linewidth=1)
ax2.set_xlabel("Time (years)", fontsize=12)
ax2.set_ylabel("Cumulative Incidence", fontsize=12)
ax2.set_title("Stacked Cumulative Incidence Functions", fontsize=13)
ax2.legend(fontsize=10)
ax2.set_xlim(0, 10)
ax2.set_ylim(0, 1)

plt.tight_layout()
plt.show()

# --- 2-year cumulative incidence of rejection ---
# Find closest time to 2 years
idx_2yr = np.argmin(np.abs(cif_times - 2))
ci_2yr = cif_values[idx_2yr]
print(f"\n=== 2-Year Cumulative Incidence of Rejection ===")
print(f"Proper CIF estimate: {ci_2yr:.3f} ({ci_2yr*100:.1f}%)")

# Compare with naive KM
naive_2yr = 1 - kmf_naive.predict(2)
print(f"Naive 1-KM estimate: {naive_2yr:.3f} ({naive_2yr*100:.1f}%)")
print(f"\nThe naive estimate is HIGHER (biased upward) because it treats")
print(f"deaths as censoring, overestimating the probability of rejection.")

# --- Summary ---
print("\n=== Summary ===")
print("1. Treating death as censoring overestimates rejection probability.")
print("2. For prediction, use the Fine-Gray model / cumulative incidence.")
```

```
print("3. For etiology, use cause-specific Cox models.")
print("4. Always report cumulative incidence functions, not 1-KM,")
print("   when competing risks are present.")
```

D.4. Chapter 07: ML Foundations

D.4.1. Exercise 1

D.4.1.1. R

```
# ============================================================================
# Chapter 7, Exercise 1: Cross-Validation Experiment
# Compare logistic regression and SVM (RBF kernel) using 10-fold stratified CV.
# Report AUC for both models.
# ============================================================================

library(tidyverse)
library(tidymodels)
library(kernlab)

# --- Simulate the clinical dataset ---
set.seed(123)
n <- 600
ex_data <- tibble(
  age = rnorm(n, 65, 10),
  creatinine = rlnorm(n, 0, 0.5),
  hemoglobin = rnorm(n, 12, 2),
  platelets = rnorm(n, 250, 70),
  wbc = rlnorm(n, 2, 0.4),
  icu = factor(rbinom(n, 1, plogis(-4 + 0.03 * rnorm(n, 65, 10) +
                                    0.5 * rlnorm(n, 0, 0.5))),
```

```
               labels = c("No", "Yes"))
)

cat("ICU admission rate:", mean(ex_data$icu == "Yes"), "\n")

# --- 10-fold stratified cross-validation ---
set.seed(42)
folds <- vfold_cv(ex_data, v = 10, strata = icu)

# --- Logistic Regression workflow ---
lr_spec <- logistic_reg() %>%
  set_engine("glm")

lr_recipe <- recipe(icu ~ ., data = ex_data) %>%
  step_normalize(all_numeric_predictors())

lr_wf <- workflow() %>%
  add_model(lr_spec) %>%
  add_recipe(lr_recipe)

lr_results <- fit_resamples(lr_wf, resamples = folds,
                            metrics = metric_set(roc_auc))

# --- SVM (RBF kernel) workflow ---
svm_spec <- svm_rbf(cost = 1, rbf_sigma = 0.5) %>%
  set_engine("kernlab") %>%
  set_mode("classification")

svm_recipe <- recipe(icu ~ ., data = ex_data) %>%
  step_normalize(all_numeric_predictors())

svm_wf <- workflow() %>%
  add_model(svm_spec) %>%
```

```
  add_recipe(svm_recipe)

svm_results <- fit_resamples(svm_wf, resamples = folds,
                             metrics = metric_set(roc_auc))

# --- Collect and compare results ---
lr_metrics  <- collect_metrics(lr_results) %>% mutate(model = "Logistic Regression")
svm_metrics <- collect_metrics(svm_results) %>% mutate(model = "SVM (RBF)")

comparison <- bind_rows(lr_metrics, svm_metrics) %>%
  select(model, .metric, mean, std_err)

print(comparison)

# --- Interpretation ---
# Both models are expected to show similar AUC because the simulated data
# has simple, roughly linear relationships between predictors and outcome.
# Logistic regression handles linear relationships well, and an RBF SVM
# adds flexibility that is not needed here.
# With real clinical data that has non-linear or interactive effects,
# SVM may outperform logistic regression.
```

D.4.1.2. Python

```
# ============================================================================
# Chapter 7, Exercise 1: Cross-Validation Experiment
# Compare logistic regression and SVM (RBF kernel) using 10-fold stratified CV.
# Report AUC for both models.
# ============================================================================

import numpy as np
import pandas as pd
```

```
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.model_selection import cross_val_score, StratifiedKFold
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline

# --- Simulate the clinical dataset ---
np.random.seed(123)
n = 600

X = pd.DataFrame({
    'age': np.random.normal(65, 10, n),
    'creatinine': np.random.lognormal(0, 0.5, n),
    'hemoglobin': np.random.normal(12, 2, n),
    'platelets': np.random.normal(250, 70, n),
    'wbc': np.random.lognormal(2, 0.4, n)
})

# Simulate ICU outcome (binary)
prob = 1 / (1 + np.exp(-(-4 + 0.03 * np.random.normal(65, 10, n) +
                          0.5 * np.random.lognormal(0, 0.5, n))))
y = np.random.binomial(1, prob)
print(f"ICU admission rate: {y.mean():.3f}")

# --- 10-fold stratified CV ---
cv = StratifiedKFold(n_splits=10, shuffle=True, random_state=42)

# --- Logistic Regression ---
lr_pipe = make_pipeline(StandardScaler(), LogisticRegression(max_iter=1000))
lr_scores = cross_val_score(lr_pipe, X, y, cv=cv, scoring='roc_auc')
print(f"\nLogistic Regression AUC: {lr_scores.mean():.3f} (+/- {lr_scores.sto

# --- SVM (RBF kernel) ---
```

```
svm_pipe = make_pipeline(
    StandardScaler(),
    SVC(kernel='rbf', C=1.0, gamma=0.5, probability=True)
)
svm_scores = cross_val_score(svm_pipe, X, y, cv=cv, scoring='roc_auc')
print(f"SVM (RBF) AUC:           {svm_scores.mean():.3f} (+/- {svm_scores.std():.3f})")

# --- Comparison ---
results = pd.DataFrame({
    'Model': ['Logistic Regression', 'SVM (RBF)'],
    'Mean AUC': [lr_scores.mean(), svm_scores.mean()],
    'Std AUC': [lr_scores.std(), svm_scores.std()]
})
print("\n", results.to_string(index=False))

# --- Interpretation ---
# Both models are expected to show similar AUC because the simulated data
# has simple, roughly linear relationships. Logistic regression is well-suited
# for such data. The RBF SVM adds flexibility that is not needed here.
# With real clinical data containing non-linear effects, SVM may outperform
# logistic regression.
```

D.4.2. Exercise 2

D.4.2.1. R

```
# ============================================================================
# Chapter 7, Exercise 2: Feature Engineering Challenge
# Engineer clinically meaningful features from raw variables for a diabetes
# prediction model. Implement using tidymodels recipes.
# ============================================================================
```

```r
library(tidyverse)
library(tidymodels)

# --- Simulate raw clinical data ---
set.seed(42)
n <- 500

raw_data <- tibble(
  height_cm = rnorm(n, 170, 10),
  weight_kg = rnorm(n, 80, 15),
  sbp = rnorm(n, 130, 18),
  dbp = rnorm(n, 82, 12),
  fasting_glucose = rnorm(n, 105, 25),
  hba1c = rnorm(n, 5.8, 0.8),
  age = rnorm(n, 55, 12),
  sex = sample(c("Male", "Female"), n, replace = TRUE),
  waist_circumference = rnorm(n, 95, 12),
  hip_circumference = rnorm(n, 100, 10),
  total_cholesterol = rnorm(n, 200, 40),
  hdl = rnorm(n, 50, 15),
  ldl = rnorm(n, 120, 35),
  triglycerides = rnorm(n, 150, 60),
  diabetes = factor(rbinom(n, 1, 0.3), labels = c("No", "Yes"))
)

# --- Part 1: Clinically meaningful engineered features ---
# 1. BMI = weight_kg / (height_m)^2
#    WHY: Standard obesity measure; strong risk factor for Type 2 diabetes.
#
# 2. Pulse Pressure = sbp - dbp
#    WHY: Reflects arterial stiffness; associated with cardiovascular risk
#         and metabolic syndrome.
#
```

```
# 3. Mean Arterial Pressure (MAP) = dbp + (sbp - dbp) / 3
#    WHY: Measures average perfusion pressure; linked to vascular health.
#
# 4. Waist-to-Hip Ratio (WHR) = waist_circumference / hip_circumference
#    WHY: Central adiposity is a stronger predictor of insulin resistance
#         than BMI alone.
#
# 5. Non-HDL Cholesterol = total_cholesterol - hdl
#    WHY: Captures all atherogenic lipoproteins; recommended by guidelines
#         as a secondary target in diabetes management.
#
# 6. Triglyceride-to-HDL Ratio = triglycerides / hdl
#    WHY: A proxy for insulin resistance; high TG/HDL ratio is associated
#         with increased diabetes risk.
#
# 7. LDL/HDL Ratio = ldl / hdl
#    WHY: Captures atherogenic dyslipidemia profile common in diabetes.

# --- Part 2: Implement feature engineering with recipes ---
diabetes_recipe <- recipe(diabetes ~ ., data = raw_data) %>%
  # BMI: weight / height_m^2
  step_mutate(
    height_m = height_cm / 100,
    bmi = weight_kg / height_m^2,
    # Pulse pressure
    pulse_pressure = sbp - dbp,
    # Mean arterial pressure
    map = dbp + (sbp - dbp) / 3,
    # Waist-to-hip ratio
    whr = waist_circumference / hip_circumference,
    # Non-HDL cholesterol
    non_hdl = total_cholesterol - hdl,
    # Triglyceride-to-HDL ratio
```

```
    tg_hdl_ratio = triglycerides / hdl,
    # LDL/HDL ratio
    ldl_hdl_ratio = ldl / hdl
  ) %>%
  # Remove intermediate and redundant variables
  step_rm(height_m) %>%
  step_normalize(all_numeric_predictors()) %>%
  step_dummy(all_nominal_predictors())

# Prepare (bake) the recipe to see the result
prepped <- prep(diabetes_recipe)
engineered_data <- bake(prepped, new_data = NULL)

cat("Original variables:", ncol(raw_data) - 1, "\n")
cat("Engineered dataset columns:", ncol(engineered_data) - 1, "\n")
cat("\nColumn names:\n")
print(names(engineered_data))

# --- Part 3: Discuss redundant features ---
# After engineering:
# - height_cm and weight_kg may become redundant once BMI is computed
#   (though height or weight alone may still carry predictive signal).
# - sbp and dbp are partially captured by pulse_pressure and MAP,
#   though keeping them may still be useful for tree-based models.
# - waist_circumference and hip_circumference are largely captured by WHR.
# - total_cholesterol is partly captured by non_hdl (since non_hdl = TC - HDI
# - Individual lipids (hdl, ldl, triglycerides) overlap with the ratios,
#   but a LASSO or tree model can sort out which representation is most usefu
#
# In practice, include both raw and engineered features and let a
# regularised model (LASSO, elastic net) or tree-based model perform
# implicit feature selection.
```

D.4.2.2. Python

```
# ==============================================================================
# Chapter 7, Exercise 2: Feature Engineering Challenge
# Engineer clinically meaningful features from raw variables for a diabetes
# prediction model. Implement using pandas and sklearn.
# ==============================================================================

import numpy as np
import pandas as pd
from sklearn.preprocessing import StandardScaler

# --- Simulate raw clinical data ---
np.random.seed(42)
n = 500

raw_data = pd.DataFrame({
    'height_cm': np.random.normal(170, 10, n),
    'weight_kg': np.random.normal(80, 15, n),
    'sbp': np.random.normal(130, 18, n),
    'dbp': np.random.normal(82, 12, n),
    'fasting_glucose': np.random.normal(105, 25, n),
    'hba1c': np.random.normal(5.8, 0.8, n),
    'age': np.random.normal(55, 12, n),
    'sex': np.random.choice(['Male', 'Female'], n),
    'waist_circumference': np.random.normal(95, 12, n),
    'hip_circumference': np.random.normal(100, 10, n),
    'total_cholesterol': np.random.normal(200, 40, n),
    'hdl': np.random.normal(50, 15, n),
    'ldl': np.random.normal(120, 35, n),
    'triglycerides': np.random.normal(150, 60, n),
})
y = np.random.binomial(1, 0.3, n)
```

```
# --- Part 1: Clinically meaningful engineered features ---
# 1. BMI = weight_kg / (height_m)^2
#    WHY: Standard obesity measure; strong risk factor for Type 2 diabetes.
#
# 2. Pulse Pressure = sbp - dbp
#    WHY: Reflects arterial stiffness; associated with cardiovascular risk
#         and metabolic syndrome.
#
# 3. Mean Arterial Pressure (MAP) = dbp + (sbp - dbp) / 3
#    WHY: Measures average perfusion pressure; linked to vascular health.
#
# 4. Waist-to-Hip Ratio (WHR) = waist / hip
#    WHY: Central adiposity is a stronger predictor of insulin resistance
#         than BMI alone.
#
# 5. Non-HDL Cholesterol = total_cholesterol - hdl
#    WHY: Captures all atherogenic lipoproteins; recommended by guidelines
#         as a secondary target in diabetes management.
#
# 6. Triglyceride-to-HDL Ratio = triglycerides / hdl
#    WHY: A proxy for insulin resistance; high TG/HDL ratio is associated
#         with increased diabetes risk.
#
# 7. LDL/HDL Ratio = ldl / hdl
#    WHY: Captures atherogenic dyslipidemia profile common in diabetes.

# --- Part 2: Implement feature engineering ---
df = raw_data.copy()

# Compute engineered features
df['bmi'] = df['weight_kg'] / (df['height_cm'] / 100) ** 2
df['pulse_pressure'] = df['sbp'] - df['dbp']
df['map'] = df['dbp'] + (df['sbp'] - df['dbp']) / 3
```

```python
df['whr'] = df['waist_circumference'] / df['hip_circumference']
df['non_hdl'] = df['total_cholesterol'] - df['hdl']
df['tg_hdl_ratio'] = df['triglycerides'] / df['hdl']
df['ldl_hdl_ratio'] = df['ldl'] / df['hdl']

# Encode sex as binary
df['sex_male'] = (df['sex'] == 'Male').astype(int)
df = df.drop(columns=['sex'])

print(f"Original features: {raw_data.shape[1]}")
print(f"Engineered features: {df.shape[1]}")
print(f"\nColumn names:\n{list(df.columns)}")

# Scale numeric features
scaler = StandardScaler()
numeric_cols = df.select_dtypes(include=[np.number]).columns
df_scaled = df.copy()
df_scaled[numeric_cols] = scaler.fit_transform(df[numeric_cols])

print(f"\nFirst few rows of engineered data:")
print(df_scaled.head())

# --- Part 3: Discuss redundant features ---
# After engineering:
# - height_cm and weight_kg may become redundant once BMI is computed
#   (though they may still carry independent predictive signal).
# - sbp and dbp are partially captured by pulse_pressure and MAP,
#   though keeping them may still be useful for tree-based models.
# - waist_circumference and hip_circumference are largely captured by WHR.
# - total_cholesterol is partly captured by non_hdl (since non_hdl = TC - HDL).
# - Individual lipids (hdl, ldl, triglycerides) overlap with the ratios,
#   but a LASSO or tree model can sort out which representation is most useful.
#
```

```
# In practice, include both raw and engineered features and let a
# regularised model (LASSO, elastic net) or tree-based model perform
# implicit feature selection.
```

D.4.3. Exercise 3

D.4.3.1. R

```
# ===========================================================================
# Chapter 7, Exercise 3: The Bias-Variance Tradeoff in Practice
# Fit polynomials of degree 1, 3, 5, 10, and 20 to a training set.
# Plot fitted curves, compute training/test error, identify optimal degree.
# ===========================================================================

library(tidyverse)

# --- Generate data ---
set.seed(42)
x_train <- sort(runif(50, 0, 10))
y_train <- sin(x_train) + rnorm(50, 0, 0.3)
x_test <- sort(runif(200, 0, 10))
y_test <- sin(x_test) + rnorm(200, 0, 0.3)

train_df <- tibble(x = x_train, y = y_train)
test_df  <- tibble(x = x_test,  y = y_test)

# --- Fit polynomials and compute RMSE ---
degrees <- c(1, 3, 5, 10, 20)
results <- tibble(degree = integer(), train_rmse = double(), test_rmse = doub

# Also store predictions for plotting
plot_data <- tibble()
```

```
for (d in degrees) {
  # Fit polynomial of degree d
  fit <- lm(y ~ poly(x, degree = d, raw = TRUE), data = train_df)

  # Predictions on train and test
  pred_train <- predict(fit, newdata = train_df)
  pred_test  <- predict(fit, newdata = test_df)

  # Compute RMSE
  rmse_train <- sqrt(mean((y_train - pred_train)^2))
  rmse_test  <- sqrt(mean((y_test - pred_test)^2))

  results <- bind_rows(results,
                       tibble(degree = d, train_rmse = rmse_train, test_rmse = rmse_test))

  # Smooth curve for plotting
  x_grid <- seq(0, 10, length.out = 300)
  pred_grid <- predict(fit, newdata = tibble(x = x_grid))
  # Clip extreme predictions for high-degree polynomials
  pred_grid <- pmin(pmax(pred_grid, -3), 3)
  plot_data <- bind_rows(plot_data,
                         tibble(x = x_grid, y_pred = pred_grid,
                                degree = paste("Degree", d)))
}

# --- Print RMSE results ---
cat("Polynomial Regression: Training vs Test RMSE\n")
cat("============================================\n")
print(results)

# --- Part 2: Plot fitted curves ---
p1 <- ggplot() +
  geom_point(data = train_df, aes(x, y), alpha = 0.5, size = 2) +
```

```
  geom_line(data = plot_data, aes(x, y_pred, color = degree), linewidth = 1)
  geom_line(data = tibble(x = seq(0, 10, 0.01), y = sin(seq(0, 10, 0.01))),
            aes(x, y), linetype = "dashed", color = "black", linewidth = 0.8)
  labs(title = "Polynomial Fits of Varying Complexity",
       subtitle = "Dashed line = true function sin(x)",
       x = "x", y = "y", color = "Polynomial") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top")

print(p1)

# --- Part 3: Plot training vs test error ---
results_long <- results %>%
  pivot_longer(cols = c(train_rmse, test_rmse),
               names_to = "set", values_to = "rmse") %>%
  mutate(set = ifelse(set == "train_rmse", "Training", "Test"))

p2 <- ggplot(results_long, aes(x = degree, y = rmse, color = set)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 3) +
  labs(title = "Training vs Test RMSE by Polynomial Degree",
       x = "Polynomial Degree", y = "RMSE", color = "Dataset") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top")

print(p2)

# --- Part 4: Interpretation ---
best_degree <- results$degree[which.min(results$test_rmse)]
cat("\nBest polynomial degree (lowest test RMSE):", best_degree, "\n")
cat("\nInterpretation (Bias-Variance Tradeoff):\n")
cat("- Degree 1 (linear): High bias -- too simple to capture the sine curve.\
cat("  Underfits both training and test data.\n")
```

```
cat("- Degree 3-5: Good balance. Captures the main curvature of sin(x)\n")
cat("  without fitting noise. Test error is minimized here.\n")
cat("- Degree 10-20: High variance -- the polynomial wiggles to fit\n")
cat("  training noise. Training error drops but test error increases.\n")
cat("  This is classic overfitting.\n")
cat("\nThe optimal degree (~3-5) sits at the sweet spot of the bias-variance\n")
cat("tradeoff, where the model is flexible enough to capture the true\n")
cat("pattern but not so flexible that it memorizes noise.\n")
```

D.4.3.2. Python

```
# ============================================================================
# Chapter 7, Exercise 3: The Bias-Variance Tradeoff in Practice
# Fit polynomials of degree 1, 3, 5, 10, and 20 to a training set.
# Plot fitted curves, compute training/test error, identify optimal degree.
# ============================================================================

import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

# --- Generate data ---
np.random.seed(42)
x_train = np.sort(np.random.uniform(0, 10, 50))
y_train = np.sin(x_train) + np.random.normal(0, 0.3, 50)
x_test = np.sort(np.random.uniform(0, 10, 200))
y_test = np.sin(x_test) + np.random.normal(0, 0.3, 200)

# --- Fit polynomials and compute RMSE ---
degrees = [1, 3, 5, 10, 20]
```

```
results = []
predictions = {}

for d in degrees:
    # Create polynomial features and fit
    poly = PolynomialFeatures(degree=d, include_bias=False)
    X_train_poly = poly.fit_transform(x_train.reshape(-1, 1))
    X_test_poly = poly.transform(x_test.reshape(-1, 1))

    model = LinearRegression()
    model.fit(X_train_poly, y_train)

    # Predictions
    pred_train = model.predict(X_train_poly)
    pred_test = model.predict(X_test_poly)

    # RMSE
    rmse_train = np.sqrt(mean_squared_error(y_train, pred_train))
    rmse_test = np.sqrt(mean_squared_error(y_test, pred_test))

    results.append({'degree': d, 'train_rmse': rmse_train, 'test_rmse': rmse_

    # Smooth curve for plotting
    x_grid = np.linspace(0, 10, 300).reshape(-1, 1)
    X_grid_poly = poly.transform(x_grid)
    pred_grid = model.predict(X_grid_poly)
    # Clip extreme predictions for high-degree polynomials
    pred_grid = np.clip(pred_grid, -3, 3)
    predictions[d] = (x_grid.ravel(), pred_grid)

# --- Print RMSE results ---
print("Polynomial Regression: Training vs Test RMSE")
print("=" * 50)
```

```
print(f"{'Degree':>8s}  {'Train RMSE':>12s}  {'Test RMSE':>12s}")
print("-" * 50)
for r in results:
    print(f"{r['degree']:>8d}  {r['train_rmse']:>12.4f}  {r['test_rmse']:>12.4f}")

# --- Part 2: Plot fitted curves ---
fig, ax = plt.subplots(figsize=(10, 6))
ax.scatter(x_train, y_train, alpha=0.5, s=30, label='Training data', zorder=5)
x_true = np.linspace(0, 10, 300)
ax.plot(x_true, np.sin(x_true), 'k--', linewidth=1.5, label='True function sin(x)')

colors = ['#E69F00', '#56B4E9', '#009E73', '#D55E00', '#CC79A7']
for (d, color) in zip(degrees, colors):
    x_g, pred_g = predictions[d]
    ax.plot(x_g, pred_g, linewidth=1.5, color=color, label=f'Degree {d}')

ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_title('Polynomial Fits of Varying Complexity')
ax.legend(loc='upper right', fontsize=9)
ax.set_ylim(-3, 3)
plt.tight_layout()
plt.show()

# --- Part 3: Plot training vs test error ---
degs = [r['degree'] for r in results]
train_rmses = [r['train_rmse'] for r in results]
test_rmses = [r['test_rmse'] for r in results]

fig, ax = plt.subplots(figsize=(8, 5))
ax.plot(degs, train_rmses, 'o-', linewidth=2, label='Training RMSE', color='steelblue')
ax.plot(degs, test_rmses, 'o-', linewidth=2, label='Test RMSE', color='darkorange')
ax.set_xlabel('Polynomial Degree')
```

```
ax.set_ylabel('RMSE')
ax.set_title('Training vs Test RMSE by Polynomial Degree')
ax.legend()
plt.tight_layout()
plt.show()

# --- Part 4: Interpretation ---
best_idx = np.argmin(test_rmses)
best_degree = degs[best_idx]
print(f"\nBest polynomial degree (lowest test RMSE): {best_degree}")
print("\nInterpretation (Bias-Variance Tradeoff):")
print("- Degree 1 (linear): High bias -- too simple to capture the sine curve
print("  Underfits both training and test data.")
print("- Degree 3-5: Good balance. Captures the main curvature of sin(x)")
print("  without fitting noise. Test error is minimized here.")
print("- Degree 10-20: High variance -- the polynomial wiggles to fit")
print("  training noise. Training error drops but test error increases.")
print("  This is classic overfitting.")
print("\nThe optimal degree (~3-5) sits at the sweet spot of the bias-varianc
print("tradeoff, where the model is flexible enough to capture the true")
print("pattern but not so flexible that it memorizes noise.")
```

D.5. Chapter 08: Trees and Ensembles

D.5.1. Exercise 1

D.5.1.1. R

```
# ==============================================================================
# Chapter 8, Exercise 1: Build and Prune a Classification Tree
# 1. Fit a full, unpruned tree and count terminal nodes.
```

```
# 2. Use cross-validation to find the optimal cp.
# 3. Prune the tree and plot it.
# 4. Identify the top 3 splitting variables.
# ============================================================================

library(tidyverse)
library(rpart)
library(rpart.plot)

# --- Simulate the readmission dataset (same as chapter) ---
set.seed(42)
n <- 1000

readmit_data <- tibble(
  age = rnorm(n, 68, 12),
  length_of_stay = rpois(n, 5) + 1,
  num_comorbidities = rpois(n, 3),
  prior_admissions = rpois(n, 1),
  discharge_hemoglobin = rnorm(n, 11, 2),
  discharge_creatinine = rlnorm(n, 0.2, 0.5),
  has_diabetes = rbinom(n, 1, 0.35),
  has_chf = rbinom(n, 1, 0.25),
  readmitted = factor(
    rbinom(n, 1, plogis(-3 + 0.02 * (rnorm(n, 68, 12) - 68) +
                          0.15 * rpois(n, 1) +
                          0.1 * rpois(n, 3) +
                          0.3 * rbinom(n, 1, 0.25) -
                          0.1 * rnorm(n, 11, 2))),
    labels = c("No", "Yes")
  )
)

cat("Readmission rate:", mean(readmit_data$readmitted == "Yes"), "\n")
```

```r
# --- Part 1: Fit a full, unpruned tree ---
full_tree <- rpart(readmitted ~ ., data = readmit_data, method = "class",
                   control = rpart.control(cp = 0.001, minsplit = 2, minbucke

n_terminal_full <- sum(full_tree$frame$var == "<leaf>")
cat("\nFull tree terminal nodes:", n_terminal_full, "\n")

# --- Part 2: Cross-validation to find optimal cp ---
cat("\nCP Table:\n")
printcp(full_tree)

# Plot CV error vs cp
plotcp(full_tree)

# Find optimal cp (minimum xerror)
cp_table <- full_tree$cptable
optimal_cp <- cp_table[which.min(cp_table[, "xerror"]), "CP"]
cat("\nOptimal cp:", optimal_cp, "\n")

# Alternative: 1-SE rule (smallest tree within 1 SE of the minimum)
min_xerror <- min(cp_table[, "xerror"])
min_se <- cp_table[which.min(cp_table[, "xerror"]), "xstd"]
cp_1se <- cp_table[cp_table[, "xerror"] <= min_xerror + min_se, "CP"]
optimal_cp_1se <- max(cp_1se)  # largest cp (smallest tree) within 1 SE
cat("Optimal cp (1-SE rule):", optimal_cp_1se, "\n")

# --- Part 3: Prune and plot ---
pruned_tree <- prune(full_tree, cp = optimal_cp)
n_terminal_pruned <- sum(pruned_tree$frame$var == "<leaf>")
cat("\nPruned tree terminal nodes:", n_terminal_pruned, "\n")

rpart.plot(pruned_tree, type = 4, extra = 106, under = TRUE,
           box.palette = "RdYlGn", roundint = FALSE,
```

```r
          main = "Pruned Decision Tree: 30-Day Readmission")

# --- Part 4: Top 3 splitting variables ---
cat("\nVariable Importance:\n")
vi <- sort(full_tree$variable.importance, decreasing = TRUE)
print(vi)

top3 <- names(vi)[1:min(3, length(vi))]
cat("\nTop 3 splitting variables:", paste(top3, collapse = ", "), "\n")
cat("\nClinical interpretation:\n")
cat("- These variables capture patient acuity and complexity.\n")
cat("- Discharge lab values (hemoglobin, creatinine) reflect the patient's\n")
cat("  clinical status at the time of discharge.\n")
cat("- Age, comorbidity count, and prior admissions reflect overall\n")
cat("  disease burden and frailty.\n")
cat("- These are well-established risk factors for 30-day readmission\n")
cat("  in the clinical literature.\n")
```

D.5.1.2. Python

```python
# ============================================================================
# Chapter 8, Exercise 1: Build and Prune a Classification Tree
# 1. Fit trees with different max_depth values and count leaves.
# 2. Cross-validate each to find optimal max_depth.
# 3. Plot the optimal tree.
# 4. Examine feature importances.
# ============================================================================

import numpy as np
import pandas as pd
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.model_selection import cross_val_score, StratifiedKFold
```

```python
import matplotlib.pyplot as plt

# --- Simulate the readmission dataset (same as chapter) ---
np.random.seed(42)
n = 1000

df = pd.DataFrame({
    'age': np.random.normal(68, 12, n),
    'length_of_stay': np.random.poisson(5, n) + 1,
    'num_comorbidities': np.random.poisson(3, n),
    'prior_admissions': np.random.poisson(1, n),
    'discharge_hemoglobin': np.random.normal(11, 2, n),
    'discharge_creatinine': np.random.lognormal(0.2, 0.5, n),
    'has_diabetes': np.random.binomial(1, 0.35, n),
    'has_chf': np.random.binomial(1, 0.25, n)
})
y = np.random.binomial(1, 0.18, n)

feature_names = list(df.columns)
print(f"Readmission rate: {y.mean():.3f}")

# --- Part 1: Fit a full, unpruned tree ---
full_tree = DecisionTreeClassifier(random_state=42)
full_tree.fit(df, y)
n_leaves_full = full_tree.get_n_leaves()
print(f"\nFull (unpruned) tree: {n_leaves_full} terminal nodes")
print(f"Full tree depth: {full_tree.get_depth()}")

# --- Part 2: Cross-validation to find optimal max_depth ---
depths = [2, 3, 4, 5, 7, 10, 15, 20, None]
cv = StratifiedKFold(n_splits=10, shuffle=True, random_state=42)

print("\nCross-validation results by max_depth:")
```

```
print(f"{'max_depth':>10s}  {'Mean AUC':>10s}  {'Std AUC':>10s}")
print("-" * 35)

cv_results = []
for d in depths:
    tree = DecisionTreeClassifier(max_depth=d, random_state=42)
    scores = cross_val_score(tree, df, y, cv=cv, scoring='roc_auc')
    label = str(d) if d is not None else "None"
    cv_results.append({'max_depth': d, 'label': label,
                       'mean_auc': scores.mean(), 'std_auc': scores.std()})
    print(f"{label:>10s}  {scores.mean():>10.3f}  {scores.std():>10.3f}")

# Find the best depth
best = max(cv_results, key=lambda x: x['mean_auc'])
print(f"\nBest max_depth: {best['label']} (AUC = {best['mean_auc']:.3f})")

# --- Part 3: Fit and plot the optimal tree ---
best_depth = best['max_depth']
optimal_tree = DecisionTreeClassifier(max_depth=best_depth, random_state=42)
optimal_tree.fit(df, y)

print(f"\nOptimal tree: {optimal_tree.get_n_leaves()} terminal nodes")

fig, ax = plt.subplots(figsize=(20, 10))
plot_tree(optimal_tree, feature_names=feature_names, class_names=['No', 'Yes'],
          filled=True, rounded=True, ax=ax, fontsize=9,
          max_depth=4)  # show up to depth 4 for readability
ax.set_title(f"Pruned Decision Tree (max_depth={best['label']})")
plt.tight_layout()
plt.show()

# --- Part 4: Top 3 splitting variables ---
importances = pd.Series(optimal_tree.feature_importances_, index=feature_names)
```

```
importances = importances.sort_values(ascending=False)

print("\nFeature Importances:")
print(importances.to_string())

top3 = importances.head(3).index.tolist()
print(f"\nTop 3 splitting variables: {top3}")

# Plot feature importances
fig, ax = plt.subplots(figsize=(8, 5))
importances.sort_values(ascending=True).plot(kind='barh', ax=ax, color='#2E8
ax.set_xlabel("Feature Importance (Gini)")
ax.set_title("Decision Tree Feature Importance")
plt.tight_layout()
plt.show()

print("\nClinical interpretation:")
print("- These variables capture patient acuity and complexity.")
print("- Discharge lab values reflect the patient's clinical status at discha
print("- Age, comorbidity count, and prior admissions reflect disease burden
print("- These are well-established readmission risk factors in the literatur
```

D.5.2. Exercise 2

D.5.2.1. R

```
# ==========================================================================
# Chapter 8, Exercise 2: Random Forest vs XGBoost Tuning Challenge
# 1. Split data 80/20. 2. Tune RF and XGBoost with 5-fold CV.
# 3. Select best hyperparameters. 4. Evaluate on test set.
# 5. Create variable importance plots.
# ==========================================================================
```

```r
library(tidyverse)
library(tidymodels)
library(vip)

# --- Simulate the readmission dataset (same as chapter) ---
set.seed(42)
n <- 1000

readmit_data <- tibble(
  age = rnorm(n, 68, 12),
  length_of_stay = rpois(n, 5) + 1,
  num_comorbidities = rpois(n, 3),
  prior_admissions = rpois(n, 1),
  discharge_hemoglobin = rnorm(n, 11, 2),
  discharge_creatinine = rlnorm(n, 0.2, 0.5),
  has_diabetes = rbinom(n, 1, 0.35),
  has_chf = rbinom(n, 1, 0.25),
  readmitted = factor(
    rbinom(n, 1, plogis(-3 + 0.02 * (rnorm(n, 68, 12) - 68) +
                          0.15 * rpois(n, 1) +
                          0.1 * rpois(n, 3) +
                          0.3 * rbinom(n, 1, 0.25) -
                          0.1 * rnorm(n, 11, 2))),
    labels = c("No", "Yes")
  )
)

# --- Part 1: Train/test split ---
set.seed(42)
split <- initial_split(readmit_data, prop = 0.8, strata = readmitted)
train <- training(split)
test  <- testing(split)
```

```
cat("Training set size:", nrow(train), "\n")
cat("Test set size:", nrow(test), "\n")

# 5-fold CV on training set
folds <- vfold_cv(train, v = 5, strata = readmitted)

# Recipe (shared)
base_recipe <- recipe(readmitted ~ ., data = train)

# --- Part 2a: Tune Random Forest ---
rf_spec <- rand_forest(
  trees = 500,
  mtry = tune(),
  min_n = tune()
) %>%
  set_engine("ranger", importance = "impurity") %>%
  set_mode("classification")

rf_wf <- workflow() %>%
  add_model(rf_spec) %>%
  add_recipe(base_recipe)

rf_grid <- grid_regular(
  mtry(range = c(2, 7)),
  min_n(range = c(5, 30)),
  levels = 5
)

set.seed(42)
rf_tune <- tune_grid(rf_wf, resamples = folds, grid = rf_grid,
                     metrics = metric_set(roc_auc))

cat("\n--- Random Forest Tuning Results (Top 5) ---\n")
```

```r
print(show_best(rf_tune, metric = "roc_auc", n = 5))

best_rf <- select_best(rf_tune, metric = "roc_auc")
cat("\nBest RF params - mtry:", best_rf$mtry, "min_n:", best_rf$min_n, "\n")

# --- Part 2b: Tune XGBoost ---
xgb_spec <- boost_tree(
  trees = 500,
  tree_depth = tune(),
  learn_rate = tune(),
  min_n = tune()
) %>%
  set_engine("xgboost") %>%
  set_mode("classification")

xgb_wf <- workflow() %>%
  add_model(xgb_spec) %>%
  add_recipe(base_recipe)

xgb_grid <- grid_regular(
  tree_depth(range = c(2, 6)),
  learn_rate(range = c(-3, -1)),   # log10 scale: 0.001 to 0.1
  min_n(range = c(5, 20)),
  levels = 4
)

set.seed(42)
xgb_tune <- tune_grid(xgb_wf, resamples = folds, grid = xgb_grid,
                      metrics = metric_set(roc_auc))

cat("\n--- XGBoost Tuning Results (Top 5) ---\n")
print(show_best(xgb_tune, metric = "roc_auc", n = 5))
```

```r
best_xgb <- select_best(xgb_tune, metric = "roc_auc")
cat("\nBest XGB params - depth:", best_xgb$tree_depth,
    "learn_rate:", best_xgb$learn_rate,
    "min_n:", best_xgb$min_n, "\n")

# --- Part 3: Finalize and fit on full training set ---
final_rf_wf <- finalize_workflow(rf_wf, best_rf)
final_xgb_wf <- finalize_workflow(xgb_wf, best_xgb)

rf_final_fit <- fit(final_rf_wf, data = train)
xgb_final_fit <- fit(final_xgb_wf, data = train)

# --- Part 4: Evaluate on test set ---
rf_test_pred  <- predict(rf_final_fit, test, type = "prob") %>%
  bind_cols(test %>% select(readmitted))
xgb_test_pred <- predict(xgb_final_fit, test, type = "prob") %>%
  bind_cols(test %>% select(readmitted))

rf_auc  <- roc_auc(rf_test_pred, truth = readmitted, .pred_Yes)
xgb_auc <- roc_auc(xgb_test_pred, truth = readmitted, .pred_Yes)

cat("\n=== Test Set Performance ===\n")
cat("Random Forest AUC:", rf_auc$.estimate, "\n")
cat("XGBoost AUC:      ", xgb_auc$.estimate, "\n")

if (xgb_auc$.estimate > rf_auc$.estimate) {
  cat("\nXGBoost performs better on the test set.\n")
} else {
  cat("\nRandom Forest performs better on the test set.\n")
}

# --- Part 5: Variable importance plots ---
# Random Forest
```

```r
rf_vi <- rf_final_fit %>%
  extract_fit_parsnip() %>%
  vip(num_features = 8) +
  ggtitle("Random Forest Variable Importance")
print(rf_vi)

# XGBoost
xgb_vi <- xgb_final_fit %>%
  extract_fit_parsnip() %>%
  vip(num_features = 8) +
  ggtitle("XGBoost Variable Importance")
print(xgb_vi)

cat("\nBoth models should generally agree on the most important features,\n")
cat("though rankings may differ. Continuous variables with more possible\n")
cat("split points (e.g., age, creatinine) often rank higher in tree-based\n")
cat("importance measures than binary variables (e.g., has_chf).\n")
```

D.5.2.2. Python

```python
# =============================================================================
# Chapter 8, Exercise 2: Random Forest vs XGBoost Tuning Challenge
# 1. Split data 80/20. 2. Tune RF and XGBoost with 5-fold CV.
# 3. Select best hyperparameters. 4. Evaluate on test set.
# 5. Create variable importance plots.
# =============================================================================

import numpy as np
import pandas as pd
from sklearn.model_selection import (train_test_split, RandomizedSearchCV,
                                      StratifiedKFold)
from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.metrics import roc_auc_score
import xgboost as xgb
import matplotlib.pyplot as plt

# --- Simulate the readmission dataset (same as chapter) ---
np.random.seed(42)
n = 1000

df = pd.DataFrame({
    'age': np.random.normal(68, 12, n),
    'length_of_stay': np.random.poisson(5, n) + 1,
    'num_comorbidities': np.random.poisson(3, n),
    'prior_admissions': np.random.poisson(1, n),
    'discharge_hemoglobin': np.random.normal(11, 2, n),
    'discharge_creatinine': np.random.lognormal(0.2, 0.5, n),
    'has_diabetes': np.random.binomial(1, 0.35, n),
    'has_chf': np.random.binomial(1, 0.25, n)
})
y = np.random.binomial(1, 0.18, n)

feature_names = list(df.columns)

# --- Part 1: Train/test split ---
X_train, X_test, y_train, y_test = train_test_split(
    df, y, test_size=0.2, stratify=y, random_state=42
)
print(f"Training set: {X_train.shape[0]} | Test set: {X_test.shape[0]}")

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

# --- Part 2a: Tune Random Forest ---
rf_param_dist = {
    'n_estimators': [200, 500],
```

```python
    'max_features': ['sqrt', 'log2', 3, 5, 7],
    'min_samples_leaf': [5, 10, 15, 20, 30],
    'max_depth': [5, 10, 15, 20, None]
}

rf = RandomForestClassifier(random_state=42, n_jobs=-1)

rf_search = RandomizedSearchCV(
    rf, rf_param_dist, n_iter=30, cv=cv, scoring='roc_auc',
    random_state=42, n_jobs=-1, verbose=0
)
rf_search.fit(X_train, y_train)

print(f"\n--- Random Forest Tuning ---")
print(f"Best CV AUC: {rf_search.best_score_:.3f}")
print(f"Best params: {rf_search.best_params_}")

# --- Part 2b: Tune XGBoost ---
xgb_param_dist = {
    'n_estimators': [200, 500, 1000],
    'learning_rate': [0.01, 0.05, 0.1],
    'max_depth': [2, 3, 4, 5, 6],
    'subsample': [0.7, 0.8, 0.9, 1.0],
    'colsample_bytree': [0.7, 0.8, 0.9, 1.0],
    'min_child_weight': [1, 5, 10]
}

xgb_model = xgb.XGBClassifier(
    random_state=42, use_label_encoder=False, eval_metric='logloss'
)

xgb_search = RandomizedSearchCV(
    xgb_model, xgb_param_dist, n_iter=30, cv=cv, scoring='roc_auc',
```

```
    random_state=42, n_jobs=-1, verbose=0
)
xgb_search.fit(X_train, y_train)

print(f"\n--- XGBoost Tuning ---")
print(f"Best CV AUC: {xgb_search.best_score_:.3f}")
print(f"Best params: {xgb_search.best_params_}")

# --- Part 4: Evaluate on test set ---
rf_best = rf_search.best_estimator_
xgb_best = xgb_search.best_estimator_

rf_test_probs = rf_best.predict_proba(X_test)[:, 1]
xgb_test_probs = xgb_best.predict_proba(X_test)[:, 1]

rf_test_auc = roc_auc_score(y_test, rf_test_probs)
xgb_test_auc = roc_auc_score(y_test, xgb_test_probs)

print(f"\n=== Test Set Performance ===")
print(f"Random Forest AUC: {rf_test_auc:.3f}")
print(f"XGBoost AUC:       {xgb_test_auc:.3f}")

if xgb_test_auc > rf_test_auc:
    print("\nXGBoost performs better on the test set.")
else:
    print("\nRandom Forest performs better on the test set.")

# --- Part 5: Variable importance plots ---
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Random Forest importance
rf_imp = pd.Series(rf_best.feature_importances_, index=feature_names)
rf_imp = rf_imp.sort_values(ascending=True)
```

```
rf_imp.plot(kind='barh', ax=axes[0], color='#2E86AB')
axes[0].set_xlabel("Feature Importance (Gini)")
axes[0].set_title("Random Forest Variable Importance")

# XGBoost importance
xgb_imp = pd.Series(xgb_best.feature_importances_, index=feature_names)
xgb_imp = xgb_imp.sort_values(ascending=True)
xgb_imp.plot(kind='barh', ax=axes[1], color='#D55E00')
axes[1].set_xlabel("Feature Importance (Gain)")
axes[1].set_title("XGBoost Variable Importance")

plt.tight_layout()
plt.show()

# Compare top features
print("\nRF top 3 features:", rf_imp.sort_values(ascending=False).head(3).index.tolist())
print("XGB top 3 features:", xgb_imp.sort_values(ascending=False).head(3).index.tolist())
print("\nBoth models should generally agree on the most important features,")
print("though rankings may differ due to how each algorithm uses features.")
```

D.5.3. Exercise 3

D.5.3.1. R

```
# =============================================================================
# Chapter 8, Exercise 3: Interpreting a Clinical Prediction Model
# Conceptual exercise: write answers as comments.
# =============================================================================

# =============================================================================
# Part 1: Non-technical summary for a hospital quality committee
# =============================================================================
```

```
#
# "We developed a computer-based prediction model that estimates each patient
# risk of being readmitted to the hospital within 30 days of discharge. The
# model uses information routinely collected during hospitalisation -- such a
# lab values, age, prior admissions, and existing medical conditions -- to
# assign each patient a risk score between 0% and 100%. In cross-validated
# testing, the model correctly distinguished between patients who were and
# were not readmitted approximately [AUC]% of the time. This tool could help
# care teams focus discharge planning resources on the patients at highest
# risk, potentially reducing readmission rates and associated penalties."

# ==============================================================================
# Part 2: Explaining variable importance to a clinical audience
# ==============================================================================
#
# "The model identified 'number of prior admissions' and 'discharge creatinin
# as the two variables that contribute most to the model's predictions. This
# means that knowing these values provides the most useful information for
# distinguishing patients who will be readmitted from those who will not.
#
# WHAT THIS MEANS:
# - Patients with more prior admissions tend to have higher predicted risk.
# - Patients with elevated discharge creatinine (indicating impaired kidney
#   function) also tend to have higher predicted risk.
# - These findings align with clinical intuition: patients with a history of
#   recurrent hospitalisations and those with renal impairment are known to
#   be at elevated risk.
#
# WHAT THIS DOES NOT MEAN:
# - Variable importance does NOT imply causation. We cannot say that reducing
#   creatinine at discharge will reduce readmission risk. The model identifie
#   associations, not causes.
# - A variable with high importance may be a proxy for something else. For
```

```
#   example, 'prior admissions' may reflect underlying disease severity,
#   social determinants, or healthcare access patterns rather than being
#   a direct cause of readmission.
# - We should not intervene on these variables based on importance alone.
#   Clinical trials or causal inference methods would be needed to establish
#   whether modifying these factors actually changes outcomes."

# =============================================================================
# Part 3: Validation strategy before clinical deployment
# =============================================================================
#
# PROPOSED VALIDATION PLAN:
#
# 1. Temporal validation: Test the model on data from a time period AFTER the
#    training data (e.g., train on 2020-2022, validate on 2023-2024). This
#    tests whether the model's performance holds over time, as clinical
#    practices and patient populations may shift.
#
# 2. External validation: Apply the model to data from a different hospital
#    system. A model developed at one institution may not generalise to
#    another due to differences in patient demographics, coding practices,
#    discharge protocols, and local disease patterns.
#
# 3. Subgroup analysis: Evaluate performance across key demographic groups
#    (age, sex, race/ethnicity, insurance status). A model that performs
#    well overall but poorly for specific populations could worsen existing
#    health disparities.
#
# 4. Calibration assessment: Verify that predicted probabilities match
#    observed readmission rates. A model that says "30% risk" should be
#    right about 30% of the time across all risk levels.
#
# 5. Prospective pilot: Before full deployment, run the model in parallel
```

```
#    alongside current practice (silent mode) to monitor performance in
#    real-time without affecting clinical decisions.
#
# RISKS OF SKIPPING EXTERNAL VALIDATION:
# - The model may be overfit to idiosyncrasies of the development data
#   (specific EMR system, local coding conventions, patient mix).
# - Performance reported from internal validation (even cross-validation)
#   tends to be optimistically biased.
# - Deploying an unvalidated model could misallocate resources, either
#   missing high-risk patients (false negatives) or overwhelming care
#   teams with false alarms (false positives).
# - Regulatory and ethical risks: deploying a model without adequate
#   validation may violate institutional policies and could cause patient
#   harm.

cat("This exercise is conceptual. See the comments in this file for the\n")
cat("complete answers to all three parts.\n")
```

D.5.3.2. Python

```
# ============================================================================
# Chapter 8, Exercise 3: Interpreting a Clinical Prediction Model
# Conceptual exercise: write answers as comments.
# ============================================================================

# ============================================================================
# Part 1: Non-technical summary for a hospital quality committee
# ============================================================================
#
# "We developed a computer-based prediction model that estimates each patient
# risk of being readmitted to the hospital within 30 days of discharge. The
# model uses information routinely collected during hospitalisation -- such a
```

```
# lab values, age, prior admissions, and existing medical conditions -- to
# assign each patient a risk score between 0% and 100%. In cross-validated
# testing, the model correctly distinguished between patients who were and
# were not readmitted approximately [AUC]% of the time. This tool could help
# care teams focus discharge planning resources on the patients at highest
# risk, potentially reducing readmission rates and associated penalties."

# ===============================================================================
# Part 2: Explaining variable importance to a clinical audience
# ===============================================================================
#
# "The model identified 'number of prior admissions' and 'discharge creatinine'
# as the two variables that contribute most to the model's predictions. This
# means that knowing these values provides the most useful information for
# distinguishing patients who will be readmitted from those who will not.
#
# WHAT THIS MEANS:
# - Patients with more prior admissions tend to have higher predicted risk.
# - Patients with elevated discharge creatinine (indicating impaired kidney
#   function) also tend to have higher predicted risk.
# - These findings align with clinical intuition: patients with a history of
#   recurrent hospitalisations and those with renal impairment are known to
#   be at elevated risk.
#
# WHAT THIS DOES NOT MEAN:
# - Variable importance does NOT imply causation. We cannot say that reducing
#   creatinine at discharge will reduce readmission risk. The model identifies
#   associations, not causes.
# - A variable with high importance may be a proxy for something else. For
#   example, 'prior admissions' may reflect underlying disease severity,
#   social determinants, or healthcare access patterns rather than being
#   a direct cause of readmission.
# - We should not intervene on these variables based on importance alone.
```

```
#   Clinical trials or causal inference methods would be needed to establish
#   whether modifying these factors actually changes outcomes."

# ============================================================================
# Part 3: Validation strategy before clinical deployment
# ============================================================================
#
# PROPOSED VALIDATION PLAN:
#
# 1. Temporal validation: Test the model on data from a time period AFTER the
#    training data (e.g., train on 2020-2022, validate on 2023-2024). This
#    tests whether the model's performance holds over time.
#
# 2. External validation: Apply the model to data from a different hospital
#    system. A model developed at one institution may not generalise to
#    another due to differences in patient demographics, coding practices,
#    discharge protocols, and local disease patterns.
#
# 3. Subgroup analysis: Evaluate performance across key demographic groups
#    (age, sex, race/ethnicity, insurance status). A model that performs
#    well overall but poorly for specific populations could worsen existing
#    health disparities.
#
# 4. Calibration assessment: Verify that predicted probabilities match
#    observed readmission rates. A model that says "30% risk" should be
#    right about 30% of the time across all risk levels.
#
# 5. Prospective pilot: Before full deployment, run the model in parallel
#    alongside current practice (silent mode) to monitor performance in
#    real-time without affecting clinical decisions.
#
# RISKS OF SKIPPING EXTERNAL VALIDATION:
# - The model may be overfit to idiosyncrasies of the development data.
```

```
# - Performance from internal validation tends to be optimistically biased.
# - Deploying an unvalidated model could misallocate resources, either
#   missing high-risk patients or overwhelming care teams with false alarms.
# - Regulatory and ethical risks: deploying without adequate validation may
#   violate institutional policies and could cause patient harm.

print("This exercise is conceptual. See the comments in this file for the")
print("complete answers to all three parts.")
```

D.6. Chapter 08b: Introduction to Deep Learning

D.6.1. Exercise 1

D.6.1.1. R

```
# ============================================================================
# Chapter 8b, Exercise 1: Neural Network vs XGBoost on Tabular Data
# Fit both a neural network and XGBoost on the readmission dataset.
# Compare 5-fold cross-validated AUC.
# ============================================================================

library(tidyverse)
library(tidymodels)
library(keras3)

# --- Simulate readmission data (same as chapter) ---
set.seed(42)
n <- 1000

readmit_data <- tibble(
  age = rnorm(n, 68, 12),
```

```r
  length_of_stay = rpois(n, 5) + 1,
  num_comorbidities = rpois(n, 3),
  prior_admissions = rpois(n, 1),
  discharge_hgb = rnorm(n, 11, 2),
  discharge_creatinine = rlnorm(n, 0.2, 0.5),
  has_diabetes = rbinom(n, 1, 0.35),
  has_chf = rbinom(n, 1, 0.25)
)

readmit_prob <- plogis(-3 + 0.02 * (readmit_data$age - 68) +
                         0.15 * readmit_data$prior_admissions +
                         0.1 * readmit_data$num_comorbidities +
                         0.3 * readmit_data$has_chf -
                         0.1 * readmit_data$discharge_hgb)
readmit_data$readmitted <- factor(rbinom(n, 1, readmit_prob),
                                   labels = c("No", "Yes"))

cat("Readmission rate:", mean(readmit_data$readmitted == "Yes"), "\n")

# --- XGBoost with 5-fold CV (using tidymodels) ---
set.seed(42)
folds <- vfold_cv(readmit_data, v = 5, strata = readmitted)

xgb_spec <- boost_tree(trees = 500, tree_depth = 4, learn_rate = 0.05,
                        min_n = 10) %>%
  set_engine("xgboost") %>%
  set_mode("classification")

xgb_wf <- workflow() %>%
  add_model(xgb_spec) %>%
  add_recipe(recipe(readmitted ~ ., data = readmit_data))

xgb_res <- fit_resamples(xgb_wf, resamples = folds,
```

```
                         metrics = metric_set(roc_auc))

xgb_metrics <- collect_metrics(xgb_res)
cat("\nXGBoost CV AUC:", xgb_metrics$mean, "+/-", xgb_metrics$std_err, "\n")

# --- Neural Network with 5-fold CV (manual loop) ---
x_all <- readmit_data %>% select(-readmitted) %>% as.matrix()
y_all <- as.numeric(readmit_data$readmitted == "Yes")

# Standardize features
x_mean <- apply(x_all, 2, mean)
x_sd   <- apply(x_all, 2, sd)
x_scaled <- scale(x_all, center = x_mean, scale = x_sd)

set.seed(42)
fold_ids <- vfold_cv(readmit_data, v = 5, strata = readmitted)
nn_aucs <- numeric(5)

for (i in seq_len(5)) {
  # Get train/validation indices
  train_idx <- fold_ids$splits[[i]] %>% analysis() %>% rownames() %>% as.integer()
  val_idx   <- fold_ids$splits[[i]] %>% assessment() %>% rownames() %>% as.integer()

  x_train <- x_scaled[train_idx, ]
  y_train <- y_all[train_idx]
  x_val   <- x_scaled[val_idx, ]
  y_val   <- y_all[val_idx]

  # Build neural network
  model <- keras_model_sequential(input_shape = ncol(x_train)) %>%
    layer_dense(units = 32, activation = "relu") %>%
    layer_dropout(rate = 0.3) %>%
    layer_dense(units = 16, activation = "relu") %>%
```

```
    layer_dropout(rate = 0.3) %>%
    layer_dense(units = 1, activation = "sigmoid")

  model %>% compile(
    optimizer = optimizer_adam(learning_rate = 0.001),
    loss = "binary_crossentropy",
    metrics = "AUC"
  )

  # Train with early stopping
  history <- model %>% fit(
    x_train, y_train,
    epochs = 50,
    batch_size = 32,
    validation_data = list(x_val, y_val),
    callbacks = list(
      callback_early_stopping(patience = 5, restore_best_weights = TRUE)
    ),
    verbose = 0
  )

  # Evaluate
  results <- model %>% evaluate(x_val, y_val, verbose = 0)
  nn_aucs[i] <- results[[2]]  # AUC metric
  cat(sprintf("  Fold %d: NN AUC = %.3f\n", i, nn_aucs[i]))
}

cat("\nNeural Network CV AUC:", mean(nn_aucs), "+/-", sd(nn_aucs) / sqrt(5),

# --- Comparison ---
cat("\n=== Comparison ===\n")
cat("XGBoost CV AUC:        ", round(xgb_metrics$mean, 3), "\n")
cat("Neural Network CV AUC: ", round(mean(nn_aucs), 3), "\n")
```

```
cat("\nInterpretation:\n")
cat("XGBoost typically matches or outperforms neural networks on tabular\n")
cat("clinical data. This is expected -- the Grinsztajn et al. (2022)\n")
cat("NeurIPS benchmark showed that tree-based models consistently\n")
cat("outperform neural networks on typical tabular datasets. Deep learning\n")
cat("excels on images, text, and sequences, not spreadsheets.\n")
```

D.6.1.2. Python

```
# ============================================================================
# Chapter 8b, Exercise 1: Neural Network vs XGBoost on Tabular Data
# Fit both a neural network and XGBoost on the readmission dataset.
# Compare 5-fold cross-validated AUC.
# ============================================================================

import numpy as np
import pandas as pd
from sklearn.model_selection import StratifiedKFold, cross_val_score
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import roc_auc_score
import xgboost as xgb
import tensorflow as tf
from tensorflow.keras import layers, models, callbacks

# --- Simulate readmission data (same as chapter) ---
np.random.seed(42)
n = 1000

age = np.random.normal(68, 12, n)
length_of_stay = np.random.poisson(5, n) + 1
num_comorbidities = np.random.poisson(3, n)
```

```
prior_admissions = np.random.poisson(1, n)
discharge_hgb = np.random.normal(11, 2, n)
discharge_creatinine = np.random.lognormal(0.2, 0.5, n)
has_diabetes = np.random.binomial(1, 0.35, n)
has_chf = np.random.binomial(1, 0.25, n)

X = np.column_stack([age, length_of_stay, num_comorbidities,
                     prior_admissions, discharge_hgb,
                     discharge_creatinine, has_diabetes, has_chf])

# Generate outcome with known logistic relationship
prob = 1 / (1 + np.exp(-(-3 + 0.02 * (age - 68) +
                          0.15 * prior_admissions +
                          0.1 * num_comorbidities +
                          0.3 * has_chf -
                          0.1 * discharge_hgb)))
y = np.random.binomial(1, prob)
print(f"Readmission rate: {y.mean():.3f}")

# --- XGBoost with 5-fold CV ---
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

xgb_model = xgb.XGBClassifier(
    n_estimators=500, learning_rate=0.05, max_depth=4,
    subsample=0.8, colsample_bytree=0.8,
    random_state=42, use_label_encoder=False, eval_metric='logloss'
)

xgb_scores = cross_val_score(xgb_model, X, y, cv=cv, scoring='roc_auc')
print(f"\nXGBoost CV AUC: {xgb_scores.mean():.3f} (+/- {xgb_scores.std():.3f}

# --- Neural Network with 5-fold CV (manual loop) ---
nn_aucs = []
```

```
fold_idx = 0

for train_idx, val_idx in cv.split(X, y):
    fold_idx += 1

    # Split and scale
    scaler = StandardScaler()
    X_train = scaler.fit_transform(X[train_idx])
    X_val = scaler.transform(X[val_idx])
    y_train = y[train_idx]
    y_val = y[val_idx]

    # Build neural network (same architecture as chapter)
    tf.random.set_seed(42)
    model = models.Sequential([
        layers.Dense(32, activation="relu", input_shape=(X_train.shape[1],)),
        layers.Dropout(0.3),
        layers.Dense(16, activation="relu"),
        layers.Dropout(0.3),
        layers.Dense(1, activation="sigmoid")
    ])

    model.compile(
        optimizer=tf.keras.optimizers.Adam(learning_rate=0.001),
        loss="binary_crossentropy",
        metrics=[tf.keras.metrics.AUC(name="auc")]
    )

    # Train with early stopping
    history = model.fit(
        X_train, y_train,
        epochs=50,
        batch_size=32,
```

```
        validation_data=(X_val, y_val),
        callbacks=[
            callbacks.EarlyStopping(patience=5, restore_best_weights=True)
        ],
        verbose=0
    )

    # Evaluate
    y_pred_prob = model.predict(X_val, verbose=0).ravel()
    fold_auc = roc_auc_score(y_val, y_pred_prob)
    nn_aucs.append(fold_auc)
    print(f"  Fold {fold_idx}: NN AUC = {fold_auc:.3f}")

nn_mean_auc = np.mean(nn_aucs)
nn_std_auc = np.std(nn_aucs)
print(f"\nNeural Network CV AUC: {nn_mean_auc:.3f} (+/- {nn_std_auc:.3f})")

# --- Comparison ---
print(f"\n=== Comparison ===")
print(f"XGBoost CV AUC:        {xgb_scores.mean():.3f}")
print(f"Neural Network CV AUC: {nn_mean_auc:.3f}")

print("\nInterpretation:")
print("XGBoost typically matches or outperforms neural networks on tabular")
print("clinical data. This is expected -- the Grinsztajn et al. (2022)")
print("NeurIPS benchmark showed that tree-based models consistently")
print("outperform neural networks on typical tabular datasets. Deep learning"
print("excels on images, text, and sequences, not spreadsheets.")
```

D.6.2. Exercise 2

D.6.2.1. R

```
# ==============================================================================
# Chapter 8b, Exercise 2: Architecture Matching
# For each clinical task, identify the best DL architecture, whether DL
# is likely to outperform gradient-boosted trees, and the reporting guideline.
# ==============================================================================

# ==============================================================================
# Task 1: Predicting 30-day mortality from 15 structured EHR variables
# ==============================================================================
#
# (a) Architecture: Feedforward neural network (multi-layer perceptron) or,
#     better yet, gradient-boosted trees (XGBoost/LightGBM). With only 15
#     structured variables, a simple architecture suffices.
#
# (b) DL likely to outperform GBT? NO. This is classic tabular data with a
#     modest number of features. The Grinsztajn et al. (2022) NeurIPS
#     benchmark and subsequent clinical benchmarks consistently show that
#     tree-based models match or outperform neural networks on tabular data.
#     Logistic regression or XGBoost is the appropriate starting point.
#
# (c) Reporting guideline: TRIPOD+AI (Collins et al., BMJ 2024). This is a
#     standard clinical prediction model using structured data.

# ==============================================================================
# Task 2: Classifying skin lesions as benign/malignant from dermoscopy images
# ==============================================================================
#
# (a) Architecture: Convolutional Neural Network (CNN), specifically a
#     pretrained model like ResNet, EfficientNet, or Inception fine-tuned
```

```
#     on the dermoscopy images. A domain-specific foundation model could
#     also be used if available.
#
# (b) DL likely to outperform GBT? YES. Image data contains spatial
#     structure (edges, textures, shapes) that CNNs are specifically designed
#     to exploit. GBTs cannot process raw images and would require manual
#     feature extraction, which is inferior to learned CNN features. Esteva
#     et al. (Nature 2017) demonstrated dermatologist-level performance.
#
# (c) Reporting guideline: CLAIM (Checklist for AI in Medical Imaging,
#     Mongan et al., Radiology: AI 2020), supplemented by TRIPOD+AI.

# ============================================================================
# Task 3: Detecting atrial fibrillation from 12-lead ECG tracings
# ============================================================================
#
# (a) Architecture: Transformer-based model or temporal CNN. The chapter
#     notes that transformers are the current preferred architecture for ECG
#     data. Medformer (NeurIPS 2024) is specifically designed for medical
#     time series classification.
#
# (b) DL likely to outperform GBT? YES. ECG data is sequential with complex
#     temporal patterns. GBTs would require extensive manual feature
#     engineering (interval measurements, morphology features), whereas DL
#     can learn directly from the raw waveform. Hannun et al. (Nature
#     Medicine 2019) demonstrated cardiologist-level arrhythmia detection.
#
# (c) Reporting guideline: TRIPOD+AI for the prediction model, potentially
#     supplemented by CLAIM if imaging is involved (e.g., ECG images rather
#     than signal data).

# ============================================================================
# Task 4: Extracting medication names from unstructured discharge summaries
```

```
# ==============================================================================
#
# (a) Architecture: Transformer-based language model, such as ClinicalBERT
#     or PubMedBERT for named entity recognition (NER). These pretrained
#     models understand medical vocabulary and can be fine-tuned for NER.
#
# (b) DL likely to outperform GBT? YES, decisively. This is a natural
#     language processing task. GBTs cannot process raw text meaningfully.
#     Transformer-based models understand context, synonyms, and medical
#     abbreviations. Rule-based and dictionary approaches are alternatives,
#     but modern NER with transformers is superior.
#
# (c) Reporting guideline: TRIPOD+AI. No specific imaging guideline applies.
#     MINIMAR (Hernandez-Boussard et al., JAMIA 2020) may also be relevant
#     as a minimum reporting standard.

# ==============================================================================
# Task 5: Predicting length of stay from structured EHR + chest X-ray
# ==============================================================================
#
# (a) Architecture: Multimodal fusion model. A CNN (e.g., pretrained
#     ResNet) processes the chest X-ray to extract image features. These
#     are concatenated with the structured EHR features and fed into a
#     combined prediction head (either a feedforward network or GBT on
#     the fused features).
#
# (b) DL likely to outperform GBT? PARTIALLY. The DL component is necessary
#     for the image. For the structured data alone, GBTs may be equal or
#     better. The optimal approach may be a hybrid: use a CNN to extract
#     image features, then combine those features with structured data in
#     a GBT. Recent work on multimodal fusion suggests this hybrid approach
#     can outperform either modality alone.
#
```

```r
# (c) Reporting guideline: TRIPOD+AI for the overall prediction model,
#     supplemented by CLAIM for the imaging component. Both should be
#     addressed since the model involves medical imaging.

cat("This exercise is conceptual. See the comments in this file for the\n")
cat("complete answers to all five clinical tasks.\n")
```

D.6.2.2. Python

```python
# ===========================================================================
# Chapter 8b, Exercise 2: Architecture Matching
# For each clinical task, identify the best DL architecture, whether DL
# is likely to outperform gradient-boosted trees, and the reporting guideline
# ===========================================================================

# ===========================================================================
# Task 1: Predicting 30-day mortality from 15 structured EHR variables
# ===========================================================================
#
# (a) Architecture: Feedforward neural network (MLP) or, better yet,
#     gradient-boosted trees (XGBoost/LightGBM). With only 15 structured
#     variables, a simple architecture suffices.
#
# (b) DL likely to outperform GBT? NO. This is classic tabular data with a
#     modest number of features. The Grinsztajn et al. (2022) NeurIPS
#     benchmark and subsequent clinical benchmarks consistently show that
#     tree-based models match or outperform neural networks on tabular data.
#
# (c) Reporting guideline: TRIPOD+AI (Collins et al., BMJ 2024). Standard
#     clinical prediction model using structured data.

# ===========================================================================
```

```
# Task 2: Classifying skin lesions as benign/malignant from dermoscopy images
# ==============================================================================
#
# (a) Architecture: CNN (e.g., pretrained ResNet, EfficientNet, or Inception
#     fine-tuned on dermoscopy images). A domain-specific foundation model
#     could also be used if available.
#
# (b) DL likely to outperform GBT? YES. Image data contains spatial
#     structure that CNNs exploit. GBTs cannot process raw images. Esteva
#     et al. (Nature 2017) demonstrated dermatologist-level performance.
#
# (c) Reporting guideline: CLAIM (Mongan et al., Radiology: AI 2020),
#     supplemented by TRIPOD+AI.

# ==============================================================================
# Task 3: Detecting atrial fibrillation from 12-lead ECG tracings
# ==============================================================================
#
# (a) Architecture: Transformer-based model or temporal CNN. Transformers
#     are the current preferred architecture for ECG data. Medformer
#     (NeurIPS 2024) is designed for medical time series classification.
#
# (b) DL likely to outperform GBT? YES. ECG data is sequential with complex
#     temporal patterns. DL can learn directly from raw waveforms, whereas
#     GBTs require extensive manual feature engineering.
#
# (c) Reporting guideline: TRIPOD+AI, potentially supplemented by CLAIM if
#     ECG images rather than signal data are used.

# ==============================================================================
# Task 4: Extracting medication names from unstructured discharge summaries
# ==============================================================================
#
```

```
# (a) Architecture: Transformer-based language model (ClinicalBERT or
#     PubMedBERT) for named entity recognition (NER). These models
#     understand medical vocabulary and context.
#
# (b) DL likely to outperform GBT? YES, decisively. This is an NLP task.
#     GBTs cannot process raw text meaningfully. Transformer-based NER
#     models understand context, synonyms, and medical abbreviations.
#
# (c) Reporting guideline: TRIPOD+AI. MINIMAR (Hernandez-Boussard et al.,
#     JAMIA 2020) also applicable as minimum reporting standard.

# ==============================================================================
# Task 5: Predicting length of stay from structured EHR + chest X-ray
# ==============================================================================
#
# (a) Architecture: Multimodal fusion model. A CNN (pretrained ResNet)
#     extracts image features from the chest X-ray. These are concatenated
#     with structured EHR features and fed into a combined prediction head.
#
# (b) DL likely to outperform GBT? PARTIALLY. DL is necessary for the image
#     component. For structured data alone, GBTs may be equal or better.
#     A hybrid approach (CNN for image features, then GBT on fused features)
#     may be optimal.
#
# (c) Reporting guideline: TRIPOD+AI for the prediction model, supplemented
#     by CLAIM for the imaging component.

print("This exercise is conceptual. See the comments in this file for the")
print("complete answers to all five clinical tasks.")
```

D.6.3. Exercise 3

D.6.3.1. R

```
# ==============================================================================
# Chapter 8b, Exercise 3: Critical Appraisal of a Deep Learning Study
# Evaluate a DL paper against the CLAIM or TRIPOD+AI checklist.
# This is a conceptual/guided exercise -- the template below provides
# the framework for appraising any DL study.
# ==============================================================================

# ==============================================================================
# INSTRUCTIONS:
# Find a recent (2024 or later) paper applying deep learning to a clinical
# task in your area of interest. Use this template to evaluate it.
# ==============================================================================

# ==============================================================================
# Question 1: Was the model externally validated?
# ==============================================================================
#
# Look for:
# - Was the model tested on data from a DIFFERENT institution, time period,
#   or geographic region than the training data?
# - If yes, how did external performance compare to internal (e.g., was there
#   a drop in AUC)?
#
# Example answer:
# "The model was validated on data from Hospital B after training on Hospital A.
#  Internal AUC was 0.92; external AUC dropped to 0.84 -- a 0.08 decrease.
#  This is consistent with the systematic review finding that 81% of DL models
#  show decreased accuracy on external datasets."
#
```

```
# Red flag: If NO external validation was performed, the results should be
# treated as preliminary. Internal CV alone is insufficient for clinical cla

# ==========================================================================
# Question 2: Were subgroup analyses reported?
# ==========================================================================
#
# Look for:
# - Performance broken down by age, sex, race/ethnicity
# - Any mention of fairness or equity analysis
# - Performance in clinically important subgroups (e.g., patients with
#   comorbidities, different disease severity)
#
# Example answer:
# "The paper reported AUC by sex (male: 0.90, female: 0.87) but did not
#  report performance by race/ethnicity or age group. Given the known issue
#  of fairness non-transferability across sites (Nature Medicine 2024),
#  this is a significant omission."

# ==========================================================================
# Question 3: Was the model compared to a simpler baseline?
# ==========================================================================
#
# Look for:
# - Comparison against logistic regression, random forest, or XGBoost
# - If the DL model only marginally outperforms the baseline, the added
#   complexity may not be justified
#
# Example answer:
# "The paper compared DL (AUC 0.91) to logistic regression (AUC 0.85) and
#  random forest (AUC 0.88). The improvement over RF is modest (0.03),
#  raising questions about whether the DL model's added complexity and
#  reduced interpretability are justified."
```

```
# ==============================================================================
# Question 4: Were training data, code, and model weights shared?
# ==============================================================================
#
# Look for:
# - Public dataset or data sharing agreement
# - Code repository (GitHub, GitLab)
# - Pretrained model weights available for download
# - FAIR data principles
#
# Example answer:
# "Code was shared on GitHub. The training data is from a private hospital
#  system and is not publicly available, though the authors describe a
#  data sharing agreement. Model weights were not released."

# ==============================================================================
# Question 5: How close is this model to clinical deployment?
# ==============================================================================
#
# Consider:
# - Has it been externally validated across multiple sites?
# - Has it been tested prospectively (not just retrospectively)?
# - Has a clinical workflow been designed for how it would be used?
# - Has a regulatory pathway been identified (e.g., FDA 510(k), EU MDR)?
# - Has a monitoring plan for post-deployment performance been described?
# - What are the potential failure modes and harms?
#
# Example answer:
# "This model is at an early research stage. While the internal results are
#  promising, the model has been validated at only one external site with a
#  modest sample. Before clinical deployment, I would want to see:
#  (1) multi-site external validation,
#  (2) a prospective pilot study in clinical workflow,
```

```
#  (3) subgroup analysis across demographics,
#  (4) calibration assessment,
#  (5) a monitoring plan for detecting model drift over time."

cat("This exercise is a guided template for critical appraisal.\n")
cat("See the comments for the framework to evaluate any DL paper.\n")
cat("Students should find their own paper and fill in the answers.\n")
```

D.6.3.2. Python

```
# ============================================================================
# Chapter 8b, Exercise 3: Critical Appraisal of a Deep Learning Study
# Evaluate a DL paper against the CLAIM or TRIPOD+AI checklist.
# This is a conceptual/guided exercise -- the template below provides
# the framework for appraising any DL study.
# ============================================================================

# ============================================================================
# INSTRUCTIONS:
# Find a recent (2024 or later) paper applying deep learning to a clinical
# task in your area of interest. Use this template to evaluate it.
# ============================================================================

# ============================================================================
# Question 1: Was the model externally validated?
# ============================================================================
#
# Look for:
# - Was the model tested on data from a DIFFERENT institution, time period,
#   or geographic region than the training data?
# - If yes, how did external performance compare to internal (e.g., was there
#   a drop in AUC)?
```

```
#
# Example answer:
# "The model was validated on data from Hospital B after training on Hospital A.
#  Internal AUC was 0.92; external AUC dropped to 0.84 -- a 0.08 decrease.
#  This is consistent with the systematic review finding that 81% of DL models
#  show decreased accuracy on external datasets."
#
# Red flag: If NO external validation was performed, the results should be
# treated as preliminary.

# ==============================================================================
# Question 2: Were subgroup analyses reported?
# ==============================================================================
#
# Look for:
# - Performance broken down by age, sex, race/ethnicity
# - Any mention of fairness or equity analysis
# - Performance in clinically important subgroups
#
# Example answer:
# "The paper reported AUC by sex (male: 0.90, female: 0.87) but did not
#  report performance by race/ethnicity or age group. Given the known issue
#  of fairness non-transferability across sites, this is a significant
#  omission."

# ==============================================================================
# Question 3: Was the model compared to a simpler baseline?
# ==============================================================================
#
# Look for:
# - Comparison against logistic regression, random forest, or XGBoost
# - If the DL model only marginally outperforms the baseline, the added
#   complexity may not be justified
```

```
#
# Example answer:
# "The paper compared DL (AUC 0.91) to logistic regression (AUC 0.85) and
#  random forest (AUC 0.88). The improvement over RF is modest (0.03)."

# ============================================================================
# Question 4: Were training data, code, and model weights shared?
# ============================================================================
#
# Look for:
# - Public dataset or data sharing agreement
# - Code repository (GitHub, GitLab)
# - Pretrained model weights available
#
# Example answer:
# "Code was shared on GitHub. Training data is from a private hospital
#  system. Model weights were not released."

# ============================================================================
# Question 5: How close is this model to clinical deployment?
# ============================================================================
#
# Consider:
# - Has it been externally validated across multiple sites?
# - Has it been tested prospectively?
# - Has a clinical workflow been designed?
# - Has a regulatory pathway been identified?
# - Has a monitoring plan been described?
#
# Example answer:
# "This model is at an early research stage. Before clinical deployment,
#  I would want to see:
#  (1) multi-site external validation,
```

```
#  (2) a prospective pilot study,
#  (3) subgroup analysis across demographics,
#  (4) calibration assessment,
#  (5) a monitoring plan for model drift."

print("This exercise is a guided template for critical appraisal.")
print("See the comments for the framework to evaluate any DL paper.")
print("Students should find their own paper and fill in the answers.")
```

D.7. Chapter 09: Model Evaluation

D.7.1. Exercise 1

D.7.1.1. R

```
# ============================================================================
# Chapter 9, Exercise 1: Bayes' Theorem in Practice (PPV Calculations)
# Calculate PPV at different prevalence levels and plot the relationship.
# ============================================================================

library(tidyverse)

# --- PPV function using Bayes' theorem ---
# PPV = (Sensitivity * Prevalence) /
#       (Sensitivity * Prevalence + (1 - Specificity) * (1 - Prevalence))
calculate_ppv <- function(sensitivity, specificity, prevalence) {
  numerator <- sensitivity * prevalence
  denominator <- numerator + (1 - specificity) * (1 - prevalence)
  return(numerator / denominator)
}
```

```
# --- Calculate NPV ---
calculate_npv <- function(sensitivity, specificity, prevalence) {
  numerator <- specificity * (1 - prevalence)
  denominator <- numerator + (1 - sensitivity) * prevalence
  return(numerator / denominator)
}

# --- Example: HIV rapid test (sensitivity = 99.7%, specificity = 99.5%) ---

# In a general population (prevalence ~ 0.4%)
ppv_general <- calculate_ppv(0.997, 0.995, 0.004)
npv_general <- calculate_npv(0.997, 0.995, 0.004)
cat("=== HIV Rapid Test (Sens=99.7%, Spec=99.5%) ===\n\n")
cat("General population (prevalence 0.4%):\n")
cat("  PPV:", round(ppv_general * 100, 1), "%\n")
cat("  NPV:", round(npv_general * 100, 1), "%\n")

# In an STI clinic population (prevalence ~ 5%)
ppv_clinic <- calculate_ppv(0.997, 0.995, 0.05)
npv_clinic <- calculate_npv(0.997, 0.995, 0.05)
cat("\nSTI clinic (prevalence 5%):\n")
cat("  PPV:", round(ppv_clinic * 100, 1), "%\n")
cat("  NPV:", round(npv_clinic * 100, 1), "%\n")

# In a population with known exposure (prevalence ~ 30%)
ppv_exposed <- calculate_ppv(0.997, 0.995, 0.30)
npv_exposed <- calculate_npv(0.997, 0.995, 0.30)
cat("\nKnown exposure (prevalence 30%):\n")
cat("  PPV:", round(ppv_exposed * 100, 1), "%\n")
cat("  NPV:", round(npv_exposed * 100, 1), "%\n")

# --- Plot PPV across a range of prevalences ---
prevalences <- seq(0.001, 0.5, by = 0.001)
```

```r
ppvs <- sapply(prevalences, function(p) calculate_ppv(0.997, 0.995, p))
npvs <- sapply(prevalences, function(p) calculate_npv(0.997, 0.995, p))

plot_df <- tibble(
  prevalence = rep(prevalences, 2),
  value = c(ppvs, npvs),
  metric = rep(c("PPV", "NPV"), each = length(prevalences))
)

ggplot(plot_df, aes(x = prevalence * 100, y = value * 100, color = metric)) +
  geom_line(linewidth = 1.5) +
  geom_hline(yintercept = 50, linetype = "dashed", color = "grey50") +
  scale_color_manual(values = c("PPV" = "steelblue", "NPV" = "darkorange")) +
  labs(x = "Prevalence (%)",
       y = "Predictive Value (%)",
       title = "PPV and NPV Depend Heavily on Prevalence",
       subtitle = "HIV rapid test: Sensitivity=99.7%, Specificity=99.5%",
       color = "Metric") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top")

# --- Additional: Compare tests with different sensitivity/specificity ---
cat("\n\n=== Comparing Tests at 1% Prevalence ===\n")

tests <- tibble(
  Test = c("High Sens/Low Spec", "Balanced", "Low Sens/High Spec"),
  Sensitivity = c(0.99, 0.95, 0.80),
  Specificity = c(0.90, 0.95, 0.99)
)

for (i in seq_len(nrow(tests))) {
  ppv <- calculate_ppv(tests$Sensitivity[i], tests$Specificity[i], 0.01)
  npv <- calculate_npv(tests$Sensitivity[i], tests$Specificity[i], 0.01)
```

```
  cat(sprintf("%s (Sens=%.0f%%, Spec=%.0f%%): PPV=%.1f%%, NPV=%.1f%%\n",
              tests$Test[i], tests$Sensitivity[i]*100, tests$Specificity[i]*1
              ppv*100, npv*100))
}

cat("\nKey takeaway: Even excellent tests have low PPV when prevalence is lo
cat("This is why screening should target high-risk populations.\n")
```

D.7.1.2. Python

```
# ==============================================================================
# Chapter 9, Exercise 1: Bayes' Theorem in Practice (PPV Calculations)
# Calculate PPV at different prevalence levels and plot the relationship.
# ==============================================================================

import numpy as np
import matplotlib.pyplot as plt


# --- PPV function using Bayes' theorem ---
def calculate_ppv(sensitivity, specificity, prevalence):
    """PPV = (Sens * Prev) / (Sens * Prev + (1-Spec) * (1-Prev))"""
    numerator = sensitivity * prevalence
    denominator = numerator + (1 - specificity) * (1 - prevalence)
    return numerator / denominator


def calculate_npv(sensitivity, specificity, prevalence):
    """NPV = (Spec * (1-Prev)) / (Spec * (1-Prev) + (1-Sens) * Prev)"""
    numerator = specificity * (1 - prevalence)
    denominator = numerator + (1 - sensitivity) * prevalence
    return numerator / denominator
```

```
# --- Example: HIV rapid test (sensitivity = 99.7%, specificity = 99.5%) ---
print("=== HIV Rapid Test (Sens=99.7%, Spec=99.5%) ===\n")

# General population (prevalence ~ 0.4%)
ppv_general = calculate_ppv(0.997, 0.995, 0.004)
npv_general = calculate_npv(0.997, 0.995, 0.004)
print(f"General population (prevalence 0.4%):")
print(f"  PPV: {ppv_general*100:.1f}%")
print(f"  NPV: {npv_general*100:.1f}%")

# STI clinic (prevalence ~ 5%)
ppv_clinic = calculate_ppv(0.997, 0.995, 0.05)
npv_clinic = calculate_npv(0.997, 0.995, 0.05)
print(f"\nSTI clinic (prevalence 5%):")
print(f"  PPV: {ppv_clinic*100:.1f}%")
print(f"  NPV: {npv_clinic*100:.1f}%")

# Known exposure (prevalence ~ 30%)
ppv_exposed = calculate_ppv(0.997, 0.995, 0.30)
npv_exposed = calculate_npv(0.997, 0.995, 0.30)
print(f"\nKnown exposure (prevalence 30%):")
print(f"  PPV: {ppv_exposed*100:.1f}%")
print(f"  NPV: {npv_exposed*100:.1f}%")

# --- Plot PPV and NPV across a range of prevalences ---
prevalences = np.linspace(0.001, 0.5, 500)
ppvs = [calculate_ppv(0.997, 0.995, p) for p in prevalences]
npvs = [calculate_npv(0.997, 0.995, p) for p in prevalences]

fig, ax = plt.subplots(figsize=(8, 5))
ax.plot(prevalences * 100, np.array(ppvs) * 100, lw=2, color="steelblue",
        label="PPV")
```

```python
ax.plot(prevalences * 100, np.array(npvs) * 100, lw=2, color="darkorange",
        label="NPV")
ax.axhline(y=50, linestyle="--", color="grey", alpha=0.7)
ax.set_xlabel("Prevalence (%)")
ax.set_ylabel("Predictive Value (%)")
ax.set_title("PPV and NPV Depend Heavily on Prevalence\n"
             "HIV rapid test: Sens=99.7%, Spec=99.5%")
ax.legend()
plt.tight_layout()
plt.show()

# --- Additional: Compare tests with different sensitivity/specificity ---
print("\n=== Comparing Tests at 1% Prevalence ===")

tests = [
    ("High Sens/Low Spec", 0.99, 0.90),
    ("Balanced",           0.95, 0.95),
    ("Low Sens/High Spec", 0.80, 0.99),
]

for name, sens, spec in tests:
    ppv = calculate_ppv(sens, spec, 0.01)
    npv = calculate_npv(sens, spec, 0.01)
    print(f"{name} (Sens={sens*100:.0f}%, Spec={spec*100:.0f}%): "
          f"PPV={ppv*100:.1f}%, NPV={npv*100:.1f}%")

print("\nKey takeaway: Even excellent tests have low PPV when prevalence is l
print("This is why screening should target high-risk populations.")
```

D.7.2. Exercise 2

D.7.2.1. R

```
# ==============================================================================
# Chapter 9, Exercise 2: Comparing ROC and Precision-Recall Curves
# Simulate an imbalanced dataset (2% prevalence), plot ROC and PR curves,
# and discuss why PR curves are more informative for rare outcomes.
# ==============================================================================

library(tidyverse)
library(pROC)
library(PRROC)

# --- Simulate an imbalanced dataset (2% prevalence) ---
set.seed(123)
n <- 5000
true_outcome <- rbinom(n, 1, 0.02)
# Simulate a moderately good model
pred_prob <- plogis(rnorm(n, mean = -2 + 3 * true_outcome, sd = 1.5))

cat("Number of observations:", n, "\n")
cat("Number of events:", sum(true_outcome), "\n")
cat("Prevalence:", mean(true_outcome), "\n")

# --- ROC Curve ---
roc_obj <- roc(true_outcome, pred_prob, quiet = TRUE)
auroc <- auc(roc_obj)

par(mfrow = c(1, 2))

# Plot ROC
plot(roc_obj,
```

```
     main = paste("ROC Curve\nAUROC =", round(auroc, 3)),
     col = "steelblue", lwd = 2,
     legacy.axes = TRUE)
abline(0, 1, lty = 2, col = "grey50")

# --- Precision-Recall Curve ---
pr_obj <- pr.curve(
  scores.class0 = pred_prob[true_outcome == 1],
  scores.class1 = pred_prob[true_outcome == 0],
  curve = TRUE
)
auprc <- pr_obj$auc.integral

# Plot PR curve
plot(pr_obj,
     main = paste("Precision-Recall Curve\nAUPRC =", round(auprc, 3)),
     color = "darkorange", lwd = 2)
abline(h = mean(true_outcome), lty = 2, col = "grey50")

par(mfrow = c(1, 1))

# --- Report metrics ---
cat("\n=== Summary ===\n")
cat("AUROC:", round(auroc, 3), "\n")
cat("AUPRC:", round(auprc, 3), "\n")
cat("Baseline AUPRC (prevalence):", round(mean(true_outcome), 3), "\n")

# --- Find optimal threshold (Youden's J) ---
coords_best <- coords(roc_obj, "best", ret = c("threshold", "sensitivity", "s
cat("\nOptimal threshold (Youden's J):", round(coords_best$threshold, 3), "\n
cat("Sensitivity:", round(coords_best$sensitivity, 3), "\n")
cat("Specificity:", round(coords_best$specificity, 3), "\n")
```

```r
# PPV at this threshold
pred_class <- ifelse(pred_prob >= coords_best$threshold, 1, 0)
tp <- sum(pred_class == 1 & true_outcome == 1)
fp <- sum(pred_class == 1 & true_outcome == 0)
ppv_at_optimal <- tp / (tp + fp)
cat("PPV at optimal threshold:", round(ppv_at_optimal, 3), "\n")

# --- Interpretation ---
cat("\n=== Interpretation ===\n")
cat("The AUROC looks excellent (", round(auroc, 3), "), suggesting the model\n")
cat("discriminates well. However, the AUPRC (", round(auprc, 3), ") reveals\n")
cat("the real challenge: achieving high recall while maintaining reasonable\n")
cat("precision is difficult with a 2% prevalence rate.\n\n")
cat("At the Youden-optimal threshold, the PPV is only", round(ppv_at_optimal * 100, 1), "%.\
cat("This means that even at the 'best' threshold, most positive predictions\n")
cat("are false positives.\n\n")
cat("KEY LESSON: For rare outcomes, always examine the PR curve alongside\n")
cat("the ROC curve. AUROC can paint an overly optimistic picture because\n")
cat("specificity is calculated over the large majority class.\n")
```

D.7.2.2. Python

```python
# ============================================================================
# Chapter 9, Exercise 2: Comparing ROC and Precision-Recall Curves
# Simulate an imbalanced dataset (2% prevalence), plot ROC and PR curves,
# and discuss why PR curves are more informative for rare outcomes.
# ============================================================================

import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import (roc_curve, roc_auc_score,
                             precision_recall_curve, average_precision_score)
```

```
from scipy.special import expit

# --- Simulate an imbalanced dataset (2% prevalence) ---
np.random.seed(123)
n = 5000
true_outcome = np.random.binomial(1, 0.02, n)
# Simulate a moderately good model
pred_prob = expit(np.random.normal(-2 + 3 * true_outcome, 1.5))

print(f"Number of observations: {n}")
print(f"Number of events: {true_outcome.sum()}")
print(f"Prevalence: {true_outcome.mean():.3f}")

# --- ROC Curve ---
fpr, tpr, roc_thresholds = roc_curve(true_outcome, pred_prob)
auroc = roc_auc_score(true_outcome, pred_prob)

# --- Precision-Recall Curve ---
precision, recall, pr_thresholds = precision_recall_curve(true_outcome, pred_
auprc = average_precision_score(true_outcome, pred_prob)
prevalence = true_outcome.mean()

# --- Plot side by side ---
fig, axes = plt.subplots(1, 2, figsize=(14, 6))

# ROC curve
axes[0].plot(fpr, tpr, color="steelblue", lw=2)
axes[0].plot([0, 1], [0, 1], "--", color="grey")
axes[0].set_title(f"ROC Curve (AUROC = {auroc:.3f})")
axes[0].set_xlabel("False Positive Rate (1 - Specificity)")
axes[0].set_ylabel("True Positive Rate (Sensitivity)")

# PR curve
```

```
axes[1].plot(recall, precision, color="darkorange", lw=2)
axes[1].axhline(y=prevalence, linestyle="--", color="grey",
                label=f"Baseline (prevalence={prevalence:.3f})")
axes[1].set_title(f"Precision-Recall Curve (AUPRC = {auprc:.3f})")
axes[1].set_xlabel("Recall (Sensitivity)")
axes[1].set_ylabel("Precision (PPV)")
axes[1].legend()

plt.tight_layout()
plt.show()

# --- Report metrics ---
print(f"\n=== Summary ===")
print(f"AUROC: {auroc:.3f}")
print(f"AUPRC: {auprc:.3f}")
print(f"Baseline AUPRC (prevalence): {prevalence:.3f}")

# --- Find optimal threshold (Youden's J) ---
j_scores = tpr - fpr
best_idx = np.argmax(j_scores)
optimal_threshold = roc_thresholds[best_idx]
print(f"\nOptimal threshold (Youden's J): {optimal_threshold:.3f}")
print(f"Sensitivity: {tpr[best_idx]:.3f}")
print(f"Specificity: {1 - fpr[best_idx]:.3f}")

# PPV at this threshold
pred_class = (pred_prob >= optimal_threshold).astype(int)
tp = ((pred_class == 1) & (true_outcome == 1)).sum()
fp = ((pred_class == 1) & (true_outcome == 0)).sum()
ppv_at_optimal = tp / (tp + fp) if (tp + fp) > 0 else 0
print(f"PPV at optimal threshold: {ppv_at_optimal:.3f}")

# --- Interpretation ---
```

```
print(f"\n=== Interpretation ===")
print(f"The AUROC looks excellent ({auroc:.3f}), suggesting the model")
print(f"discriminates well. However, the AUPRC ({auprc:.3f}) reveals")
print(f"the real challenge: achieving high recall while maintaining")
print(f"reasonable precision is difficult with a 2% prevalence rate.")
print(f"\nAt the Youden-optimal threshold, the PPV is only {ppv_at_optimal*1
print(f"This means that even at the 'best' threshold, most positive")
print(f"predictions are false positives.")
print(f"\nKEY LESSON: For rare outcomes, always examine the PR curve")
print(f"alongside the ROC curve. AUROC can paint an overly optimistic")
print(f"picture because specificity is calculated over the large")
print(f"majority class.")
```

D.7.3. Exercise 3

D.7.3.1. R

```
# ==========================================================================
# Chapter 9, Exercise 3: Evaluating Fairness
# Simulate data with two demographic groups, evaluate whether the model
# performs equitably across groups.
# ==========================================================================

library(tidyverse)
library(pROC)

# --- Simulate data with two demographic groups ---
set.seed(42)
n <- 2000

group <- sample(c("A", "B"), n, replace = TRUE)
```

```
# Group B has slightly different disease prevalence and predictor distribution
true_outcome <- ifelse(group == "A",
                       rbinom(n, 1, 0.10),
                       rbinom(n, 1, 0.15))

# Model performs slightly worse for Group B (weaker signal)
pred_prob <- ifelse(group == "A",
                    plogis(rnorm(n, -1.5 + 2.5 * true_outcome, 1.0)),
                    plogis(rnorm(n, -1.5 + 1.8 * true_outcome, 1.2)))

df <- tibble(group = group, true_outcome = true_outcome, pred_prob = pred_prob)

# --- 1. Calculate AUC by group ---
cat("=== Discrimination (AUC) by Group ===\n")
for (g in c("A", "B")) {
  idx <- df$group == g
  roc_g <- roc(df$true_outcome[idx], df$pred_prob[idx], quiet = TRUE)
  ci_g <- ci.auc(roc_g)
  cat(sprintf("Group %s: AUC = %.3f (95%% CI: %.3f - %.3f), Prevalence = %.1f%%\n",
              g, auc(roc_g), ci_g[1], ci_g[3],
              mean(df$true_outcome[idx]) * 100))
}

# --- 2. Sensitivity and specificity at various thresholds ---
cat("\n=== Performance at Different Thresholds ===\n")
thresholds <- c(0.15, 0.20, 0.30, 0.40, 0.50)

for (threshold in thresholds) {
  cat(sprintf("\nThreshold = %.2f:\n", threshold))
  for (g in c("A", "B")) {
    idx <- df$group == g
    pred_class <- ifelse(df$pred_prob[idx] >= threshold, 1, 0)
    actual <- df$true_outcome[idx]
```

```
    tp <- sum(pred_class == 1 & actual == 1)
    fn <- sum(pred_class == 0 & actual == 1)
    tn <- sum(pred_class == 0 & actual == 0)
    fp <- sum(pred_class == 1 & actual == 0)

    sens <- ifelse(tp + fn > 0, tp / (tp + fn), NA)
    spec <- ifelse(tn + fp > 0, tn / (tn + fp), NA)
    ppv  <- ifelse(tp + fp > 0, tp / (tp + fp), NA)

    cat(sprintf("  Group %s: Sens=%.3f  Spec=%.3f  PPV=%.3f  (TP=%d FP=%d FN=
                g, sens, spec, ppv, tp, fp, fn, tn))
  }
}

# --- 3. Positive prediction rate (demographic parity) ---
cat("\n=== Positive Prediction Rate (Demographic Parity) ===\n")
threshold <- 0.30
for (g in c("A", "B")) {
  idx <- df$group == g
  pred_class <- ifelse(df$pred_prob[idx] >= threshold, 1, 0)
  pos_rate <- mean(pred_class)
  cat(sprintf("Group %s: %.1f%% predicted positive at threshold %.2f\n",
              g, pos_rate * 100, threshold))
}

# --- 4. ROC curves overlaid ---
roc_a <- roc(df$true_outcome[df$group == "A"],
             df$pred_prob[df$group == "A"], quiet = TRUE)
roc_b <- roc(df$true_outcome[df$group == "B"],
             df$pred_prob[df$group == "B"], quiet = TRUE)

plot(roc_a, col = "steelblue", lwd = 2, legacy.axes = TRUE,
     main = "ROC Curves by Demographic Group")
```

```r
plot(roc_b, col = "darkorange", lwd = 2, add = TRUE)
legend("bottomright",
       legend = c(paste("Group A (AUC =", round(auc(roc_a), 3), ")"),
                  paste("Group B (AUC =", round(auc(roc_b), 3), ")")),
       col = c("steelblue", "darkorange"), lwd = 2)

# --- Interpretation ---
cat("\n=== Interpretation ===\n")
cat("1. The model shows different AUC values across groups, indicating\n")
cat("   unequal discrimination. Group B (higher prevalence, noisier data)\n")
cat("   has lower AUC than Group A.\n\n")
cat("2. At any fixed threshold, sensitivity and specificity differ between\n")
cat("   groups. This means a single threshold does not provide equitable\n")
cat("   performance. Group-specific thresholds could equalize sensitivity\n")
cat("   but would raise questions about fairness.\n\n")
cat("3. Different positive prediction rates reflect both different prevalence\n")
cat("   and different model performance. Demographic parity (equal positive\n")
cat("   rates) may conflict with calibration if true prevalence differs.\n\n")
cat("4. CLINICAL IMPLICATION: Before deploying a model, always evaluate\n")
cat("   performance across demographic subgroups. A model that works well\n")
cat("   'on average' may perform poorly for specific populations, potentially\n")
cat("   widening health disparities. Report subgroup-specific metrics.\n")
```

D.7.3.2. Python

```python
# ============================================================================
# Chapter 9, Exercise 3: Evaluating Fairness
# Simulate data with two demographic groups, evaluate whether the model
# performs equitably across groups.
# ============================================================================

import numpy as np
```

```
import matplotlib.pyplot as plt
from sklearn.metrics import roc_auc_score, roc_curve
from scipy.special import expit

# --- Simulate data with two demographic groups ---
np.random.seed(42)
n = 2000

group = np.random.choice(["A", "B"], n)

# Group B has slightly different disease prevalence
true_outcome = np.where(group == "A",
                        np.random.binomial(1, 0.10, n),
                        np.random.binomial(1, 0.15, n))

# Model performs slightly worse for Group B (weaker signal, more noise)
pred_prob = np.where(group == "A",
                     expit(np.random.normal(-1.5 + 2.5 * true_outcome, 1.0))
                     expit(np.random.normal(-1.5 + 1.8 * true_outcome, 1.2))

# --- 1. Calculate AUC by group ---
print("=== Discrimination (AUC) by Group ===")
for g in ["A", "B"]:
    mask = group == g
    auc = roc_auc_score(true_outcome[mask], pred_prob[mask])
    prev = true_outcome[mask].mean()
    print(f"Group {g}: AUC = {auc:.3f}, Prevalence = {prev*100:.1f}%")

# --- 2. Sensitivity and specificity at various thresholds ---
print("\n=== Performance at Different Thresholds ===")
thresholds = [0.15, 0.20, 0.30, 0.40, 0.50]

for threshold in thresholds:
```

```
    print(f"\nThreshold = {threshold:.2f}:")
    for g in ["A", "B"]:
        mask = group == g
        pred_class = (pred_prob[mask] >= threshold).astype(int)
        actual = true_outcome[mask]

        tp = ((pred_class == 1) & (actual == 1)).sum()
        fn = ((pred_class == 0) & (actual == 1)).sum()
        tn = ((pred_class == 0) & (actual == 0)).sum()
        fp = ((pred_class == 1) & (actual == 0)).sum()

        sens = tp / (tp + fn) if (tp + fn) > 0 else 0
        spec = tn / (tn + fp) if (tn + fp) > 0 else 0
        ppv = tp / (tp + fp) if (tp + fp) > 0 else 0

        print(f"  Group {g}: Sens={sens:.3f}  Spec={spec:.3f}  "
              f"PPV={ppv:.3f}  (TP={tp} FP={fp} FN={fn} TN={tn})")

# --- 3. Positive prediction rate (demographic parity) ---
print("\n=== Positive Prediction Rate (Demographic Parity) ===")
threshold = 0.30
for g in ["A", "B"]:
    mask = group == g
    pred_class = (pred_prob[mask] >= threshold).astype(int)
    pos_rate = pred_class.mean()
    print(f"Group {g}: {pos_rate*100:.1f}% predicted positive "
          f"at threshold {threshold:.2f}")

# --- 4. ROC curves overlaid ---
fig, ax = plt.subplots(figsize=(7, 7))

for g, color, label_prefix in [("A", "steelblue", "Group A"),
                                ("B", "darkorange", "Group B")]:
```

```
    mask = group == g
    fpr, tpr, _ = roc_curve(true_outcome[mask], pred_prob[mask])
    auc = roc_auc_score(true_outcome[mask], pred_prob[mask])
    ax.plot(fpr, tpr, color=color, lw=2,
            label=f"{label_prefix} (AUC = {auc:.3f})")

ax.plot([0, 1], [0, 1], "--", color="grey")
ax.set_xlabel("False Positive Rate (1 - Specificity)")
ax.set_ylabel("True Positive Rate (Sensitivity)")
ax.set_title("ROC Curves by Demographic Group")
ax.legend(loc="lower right")
plt.tight_layout()
plt.show()

# --- Interpretation ---
print("\n=== Interpretation ===")
print("1. The model shows different AUC values across groups, indicating")
print("   unequal discrimination. Group B (higher prevalence, noisier data)")
print("   has lower AUC than Group A.")
print("\n2. At any fixed threshold, sensitivity and specificity differ betwe
print("   groups. A single threshold does not provide equitable performance.
print("   Group-specific thresholds could equalize sensitivity but raise")
print("   questions about fairness.")
print("\n3. Different positive prediction rates reflect both different preva
print("   and different model performance. Demographic parity (equal positiv
print("   rates) may conflict with calibration if true prevalence differs.")
print("\n4. CLINICAL IMPLICATION: Before deploying a model, always evaluate")
print("   performance across demographic subgroups. A model that works well")
print("   'on average' may perform poorly for specific populations, potentia
print("   widening health disparities. Report subgroup-specific metrics.")
```

D.8. Chapter 11: Performance and Validation

D.8.1. Exercise 1

D.8.1.1. R

```
# Chapter 11, Exercise 1: Calibration Assessment
# Using the stroke mortality model from the chapter

library(rms)

# ---- Simulate the stroke data (same as chapter) ----
set.seed(2024)
n <- 1500

stroke_data <- data.frame(
  age = round(rnorm(n, 72, 12)),
  nihss = round(pmax(0, rnorm(n, 8, 6))),
  glucose = round(rnorm(n, 140, 50)),
  afib = rbinom(n, 1, 0.25),
  thrombolysis = rbinom(n, 1, 0.30)
)

lp <- -5 + 0.04 * stroke_data$age +
  0.12 * stroke_data$nihss +
  0.003 * stroke_data$glucose +
  0.3 * stroke_data$afib -
  0.5 * stroke_data$thrombolysis

stroke_data$death_30d <- rbinom(n, 1, plogis(lp))

# Fit the prediction model
dd <- datadist(stroke_data)
```

```r
options(datadist = "dd")

fit <- lrm(death_30d ~ age + nihss + glucose + afib + thrombolysis,
           data = stroke_data, x = TRUE, y = TRUE)

pred_prob <- predict(fit, type = "fitted")

# ---- (a) Calibration plot using deciles of predicted risk ----
cat("=== Part (a): Calibration Plot ===\n")
val.prob(pred_prob, stroke_data$death_30d, m = 150, cex = 0.5,
         main = "Calibration Plot (Deciles): 30-Day Stroke Mortality")
# The model appears well-calibrated since the points cluster near the diagona
# This is expected because we are evaluating apparent performance on the
# training data.

# ---- (b) O:E ratio ----
cat("\n=== Part (b): O:E Ratio ===\n")
obs_rate <- mean(stroke_data$death_30d)
mean_pred <- mean(pred_prob)
oe_ratio <- obs_rate / mean_pred

cat("Observed event rate:", round(obs_rate, 3), "\n")
cat("Mean predicted probability:", round(mean_pred, 3), "\n")
cat("O:E ratio:", round(oe_ratio, 3), "\n")
# An O:E ratio near 1.0 indicates good calibration-in-the-large.
# The model's average predictions match the observed event rate.

# ---- (c) Calibration slope ----
cat("\n=== Part (c): Calibration Slope ===\n")
cal_model <- glm(stroke_data$death_30d ~ qlogis(pred_prob), family = binomial
cal_slope <- coef(cal_model)[2]
cal_intercept <- coef(cal_model)[1]
```

```r
cat("Calibration slope:", round(cal_slope, 3), "\n")
cat("Calibration intercept:", round(cal_intercept, 3), "\n")
# A calibration slope of 1.0 indicates no overfitting.
# The apparent slope is typically close to 1 on the training data.
# Values < 1 on new data would suggest overfitting (predictions too extreme).

# ---- (d) Bootstrap optimism correction ----
cat("\n=== Part (d): Bootstrap Optimism Correction ===\n")
set.seed(42)
val <- validate(fit, B = 200)

cat("Bootstrap Validation Results:\n")
print(val)

# Extract optimism-corrected C-statistic
dxy_corrected <- val["Dxy", "index.corrected"]
c_apparent <- fit$stats["C"]
c_corrected <- (dxy_corrected + 1) / 2

cat("\nApparent C-statistic:", round(c_apparent, 4), "\n")
cat("Optimism-corrected C-statistic:", round(c_corrected, 4), "\n")
cat("C-statistic decrease:", round(c_apparent - c_corrected, 4), "\n")

# Calibration slope from bootstrap
slope_corrected <- val["Slope", "index.corrected"]
cat("\nApparent calibration slope: 1.000\n")
cat("Optimism-corrected calibration slope:", round(slope_corrected, 4), "\n")
cat("Slope decrease:", round(1 - slope_corrected, 4), "\n")

# The C-statistic decreases slightly after optimism correction, reflecting
# the mild optimism in apparent performance. The calibration slope also
# decreases below 1, indicating some overfitting that inflates apparent
# performance.
```

D.8.1.2. Python

```python
# Chapter 11, Exercise 1: Calibration Assessment
# Using the stroke mortality model from the chapter

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_auc_score, brier_score_loss
from sklearn.calibration import calibration_curve
from scipy.special import expit, logit

# ---- Simulate the stroke data (same as chapter) ----
np.random.seed(2024)
n = 1500

stroke_data = pd.DataFrame({
    "age": np.round(np.random.normal(72, 12, n)),
    "nihss": np.round(np.maximum(0, np.random.normal(8, 6, n))),
    "glucose": np.round(np.random.normal(140, 50, n)),
    "afib": np.random.binomial(1, 0.25, n),
    "thrombolysis": np.random.binomial(1, 0.30, n)
})

lp = (-5 + 0.04 * stroke_data["age"] +
      0.12 * stroke_data["nihss"] +
      0.003 * stroke_data["glucose"] +
      0.3 * stroke_data["afib"] -
      0.5 * stroke_data["thrombolysis"])

stroke_data["death_30d"] = np.random.binomial(1, expit(lp))
```

```
predictors = ["age", "nihss", "glucose", "afib", "thrombolysis"]
X = stroke_data[predictors].values
y = stroke_data["death_30d"].values

model = LogisticRegression(max_iter=5000, random_state=42)
model.fit(X, y)
pred_prob = model.predict_proba(X)[:, 1]

# ---- (a) Calibration plot using deciles ----
print("=== Part (a): Calibration Plot ===")
prob_true, prob_pred = calibration_curve(y, pred_prob, n_bins=10, strategy="quantile")

fig, ax = plt.subplots(figsize=(7, 6))
ax.plot(prob_pred, prob_true, "o-", color="steelblue", lw=2, markersize=8, label="Model")
ax.plot([0, 1], [0, 1], "--", color="grey", label="Perfect calibration")
ax.set_xlabel("Mean Predicted Probability")
ax.set_ylabel("Observed Proportion")
ax.set_title("Calibration Plot (Deciles): 30-Day Stroke Mortality")
ax.legend()
plt.tight_layout()
plt.savefig("ch11_ex1_calibration.png", dpi=150)
plt.show()
# The points cluster near the diagonal, suggesting good apparent calibration.

# ---- (b) O:E ratio ----
print("\n=== Part (b): O:E Ratio ===")
obs_rate = y.mean()
mean_pred = pred_prob.mean()
oe_ratio = obs_rate / mean_pred

print(f"Observed event rate: {obs_rate:.3f}")
print(f"Mean predicted probability: {mean_pred:.3f}")
print(f"O:E ratio: {oe_ratio:.3f}")
```

```
# An O:E ratio near 1.0 indicates good calibration-in-the-large.

# ---- (c) Calibration slope ----
print("\n=== Part (c): Calibration Slope ===")
lp_pred = logit(np.clip(pred_prob, 1e-8, 1 - 1e-8))
cal_model = LogisticRegression(max_iter=5000, penalty=None)
cal_model.fit(lp_pred.reshape(-1, 1), y)

print(f"Calibration slope: {cal_model.coef_[0][0]:.3f}")
print(f"Calibration intercept: {cal_model.intercept_[0]:.3f}")
# A slope near 1.0 on training data is expected. Values < 1 on new data
# indicate overfitting.

# ---- (d) Bootstrap optimism correction ----
print("\n=== Part (d): Bootstrap Optimism Correction ===")
np.random.seed(42)

# Apparent performance
apparent_auc = roc_auc_score(y, pred_prob)
apparent_brier = brier_score_loss(y, pred_prob)

n_boot = 200
optimism_auc = []
optimism_brier = []

for b in range(n_boot):
    boot_idx = np.random.choice(n, n, replace=True)
    X_boot, y_boot = X[boot_idx], y[boot_idx]

    boot_model = LogisticRegression(max_iter=5000, random_state=42)
    boot_model.fit(X_boot, y_boot)

    # Apparent on bootstrap sample
```

```
    pred_boot = boot_model.predict_proba(X_boot)[:, 1]
    auc_boot = roc_auc_score(y_boot, pred_boot)

    # Test on original data
    pred_orig = boot_model.predict_proba(X)[:, 1]
    auc_orig = roc_auc_score(y, pred_orig)

    optimism_auc.append(auc_boot - auc_orig)

mean_opt_auc = np.mean(optimism_auc)
corrected_auc = apparent_auc - mean_opt_auc

print(f"Apparent C-statistic:          {apparent_auc:.4f}")
print(f"Mean optimism (AUC):           {mean_opt_auc:.4f}")
print(f"Optimism-corrected C-statistic:{corrected_auc:.4f}")
print(f"C-statistic decrease:          {mean_opt_auc:.4f}")

# The C-statistic decreases slightly after correction, reflecting mild
# optimism in the apparent performance. With 1500 observations and 5
# predictors, overfitting is modest.
```

D.8.2. Exercise 2

D.8.2.1. R

```
# Chapter 11, Exercise 2: External Validation Simulation
# Create three external validation populations and assess model performance

library(rms)
library(pROC)

# ---- Simulate development data and fit model (same as chapter) ----
```

```r
set.seed(2024)
n <- 1500

stroke_data <- data.frame(
  age = round(rnorm(n, 72, 12)),
  nihss = round(pmax(0, rnorm(n, 8, 6))),
  glucose = round(rnorm(n, 140, 50)),
  afib = rbinom(n, 1, 0.25),
  thrombolysis = rbinom(n, 1, 0.30)
)

lp <- -5 + 0.04 * stroke_data$age +
  0.12 * stroke_data$nihss +
  0.003 * stroke_data$glucose +
  0.3 * stroke_data$afib -
  0.5 * stroke_data$thrombolysis

stroke_data$death_30d <- rbinom(n, 1, plogis(lp))

dd <- datadist(stroke_data)
options(datadist = "dd")

fit <- lrm(death_30d ~ age + nihss + glucose + afib + thrombolysis,
           data = stroke_data, x = TRUE, y = TRUE)

# ---- Helper function: evaluate model performance ----
evaluate_ext <- function(ext_data, fit, label) {
  ext_data$pred <- predict(fit, newdata = ext_data, type = "fitted")
  obs_rate <- mean(ext_data$death_30d)
  mean_pred <- mean(ext_data$pred)
  oe <- obs_rate / mean_pred

  # C-statistic
```

```
  roc_obj <- roc(ext_data$death_30d, ext_data$pred, quiet = TRUE)
  c_stat <- auc(roc_obj)


  # Calibration slope
  lp_ext <- qlogis(ext_data$pred)
  cal_model <- glm(death_30d ~ lp_ext, data = ext_data, family = binomial)
  cal_slope <- coef(cal_model)[2]
  cal_int <- coef(cal_model)[1]

  cat(sprintf("\n=== %s ===\n", label))
  cat(sprintf("  N = %d, Observed mortality: %.3f\n", nrow(ext_data), obs_rate))
  cat(sprintf("  Mean predicted: %.3f\n", mean_pred))
  cat(sprintf("  C-statistic: %.3f\n", c_stat))
  cat(sprintf("  O:E ratio: %.3f\n", oe))
  cat(sprintf("  Calibration slope: %.3f\n", cal_slope))
  cat(sprintf("  Calibration intercept: %.3f\n", cal_int))

  return(ext_data)
}

# ---- Population A: Temporal validation (3 years later) ----
# Same demographics, slightly different practice patterns
set.seed(101)
n_a <- 800
pop_a <- data.frame(
  age = round(rnorm(n_a, 73, 12)),         # Similar age
  nihss = round(pmax(0, rnorm(n_a, 8, 6))),
  glucose = round(rnorm(n_a, 138, 48)),
  afib = rbinom(n_a, 1, 0.27),
  thrombolysis = rbinom(n_a, 1, 0.40)      # More thrombolysis over time
)
```

```
lp_a <- -5 + 0.04 * pop_a$age + 0.12 * pop_a$nihss +
  0.003 * pop_a$glucose + 0.3 * pop_a$afib - 0.5 * pop_a$thrombolysis
pop_a$death_30d <- rbinom(n_a, 1, plogis(lp_a))

pop_a <- evaluate_ext(pop_a, fit, "Population A: Temporal (3 years later)")

# ---- Population B: Geographical (younger patients) ----
set.seed(202)
n_b <- 600
pop_b <- data.frame(
  age = round(rnorm(n_b, 62, 10)),        # Younger
  nihss = round(pmax(0, rnorm(n_b, 6, 5))),  # Lower severity
  glucose = round(rnorm(n_b, 130, 45)),
  afib = rbinom(n_b, 1, 0.15),            # Less AF
  thrombolysis = rbinom(n_b, 1, 0.35)
)

lp_b <- -5 + 0.04 * pop_b$age + 0.12 * pop_b$nihss +
  0.003 * pop_b$glucose + 0.3 * pop_b$afib - 0.5 * pop_b$thrombolysis
pop_b$death_30d <- rbinom(n_b, 1, plogis(lp_b))

pop_b <- evaluate_ext(pop_b, fit, "Population B: Geographical (younger patie

# ---- Population C: Domain (primary care, lower severity) ----
set.seed(303)
n_c <- 500
pop_c <- data.frame(
  age = round(rnorm(n_c, 68, 14)),
  nihss = round(pmax(0, rnorm(n_c, 3, 3))),  # Much lower severity
  glucose = round(rnorm(n_c, 120, 35)),
  afib = rbinom(n_c, 1, 0.20),
  thrombolysis = rbinom(n_c, 1, 0.10)        # Rarely used in primary care
)
```

```r
# Different outcome model: lower baseline risk in primary care
lp_c <- -6 + 0.03 * pop_c$age + 0.10 * pop_c$nihss +
  0.002 * pop_c$glucose + 0.2 * pop_c$afib - 0.3 * pop_c$thrombolysis
pop_c$death_30d <- rbinom(n_c, 1, plogis(lp_c))

pop_c <- evaluate_ext(pop_c, fit, "Population C: Domain (primary care)")

# ---- (b) Calibration plots ----
par(mfrow = c(1, 3), mar = c(4, 4, 3, 1))
val.prob(pop_a$pred, pop_a$death_30d, m = 80, cex = 0.5,
         main = "A: Temporal")
val.prob(pop_b$pred, pop_b$death_30d, m = 60, cex = 0.5,
         main = "B: Geographical")
val.prob(pop_c$pred, pop_c$death_30d, m = 50, cex = 0.5,
         main = "C: Domain")

# ---- (c) Which population shows worst calibration? ----
cat("\n=== Part (c): Worst Calibration ===\n")
cat("Population C (primary care / domain validation) shows the worst calibration.\n")
cat("This is because the outcome model differs from the development setting:\n")
cat("  - Different baseline risk (intercept)\n")
cat("  - Different predictor-outcome relationships (coefficients)\n")
cat("  - Different case-mix (lower severity patients)\n")
cat("Domain validation is the most stringent test of transportability.\n")

# ---- (d) Logistic recalibration ----
cat("\n=== Part (d): Logistic Recalibration ===\n")

recalibrate <- function(ext_data, label) {
  lp_ext <- qlogis(ext_data$pred)
  cal_fit <- glm(death_30d ~ lp_ext, data = ext_data, family = binomial)
  ext_data$pred_recal <- predict(cal_fit, type = "response")
```

```
  oe_before <- mean(ext_data$death_30d) / mean(ext_data$pred)
  oe_after  <- mean(ext_data$death_30d) / mean(ext_data$pred_recal)

  cat(sprintf("\n%s:\n", label))
  cat(sprintf("  O:E before recalibration: %.3f\n", oe_before))
  cat(sprintf("  O:E after recalibration:  %.3f\n", oe_after))
}

recalibrate(pop_a, "Population A")
recalibrate(pop_b, "Population B")
recalibrate(pop_c, "Population C")

cat("\nLogistic recalibration corrects for differences in baseline risk\n")
cat("and prediction spread, bringing O:E ratios closer to 1.0.\n")
```

D.8.2.2. Python

```
# Chapter 11, Exercise 2: External Validation Simulation
# Create three external validation populations and assess model performance

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_auc_score, brier_score_loss
from sklearn.calibration import calibration_curve
from scipy.special import expit, logit

# ---- Simulate development data and fit model (same as chapter) ----
np.random.seed(2024)
n = 1500
```

```
stroke_data = pd.DataFrame({
    "age": np.round(np.random.normal(72, 12, n)),
    "nihss": np.round(np.maximum(0, np.random.normal(8, 6, n))),
    "glucose": np.round(np.random.normal(140, 50, n)),
    "afib": np.random.binomial(1, 0.25, n),
    "thrombolysis": np.random.binomial(1, 0.30, n)
})

lp = (-5 + 0.04 * stroke_data["age"] +
      0.12 * stroke_data["nihss"] +
      0.003 * stroke_data["glucose"] +
      0.3 * stroke_data["afib"] -
      0.5 * stroke_data["thrombolysis"])

stroke_data["death_30d"] = np.random.binomial(1, expit(lp))

predictors = ["age", "nihss", "glucose", "afib", "thrombolysis"]
X = stroke_data[predictors].values
y = stroke_data["death_30d"].values

model = LogisticRegression(max_iter=5000, random_state=42)
model.fit(X, y)


# ---- Helper function ----
def evaluate_ext(ext_df, model, predictors, label):
    """Evaluate model on external data and return predictions."""
    X_ext = ext_df[predictors].values
    y_ext = ext_df["death_30d"].values
    pred = model.predict_proba(X_ext)[:, 1]

    obs_rate = y_ext.mean()
    mean_pred = pred.mean()
```

```
    oe = obs_rate / mean_pred
    auc = roc_auc_score(y_ext, pred)

    # Calibration slope
    lp_ext = logit(np.clip(pred, 1e-8, 1 - 1e-8))
    cal = LogisticRegression(max_iter=5000, penalty=None)
    cal.fit(lp_ext.reshape(-1, 1), y_ext)

    print(f"\n=== {label} ===")
    print(f"  N = {len(y_ext)}, Observed mortality: {obs_rate:.3f}")
    print(f"  Mean predicted: {mean_pred:.3f}")
    print(f"  C-statistic: {auc:.3f}")
    print(f"  O:E ratio: {oe:.3f}")
    print(f"  Calibration slope: {cal.coef_[0][0]:.3f}")
    print(f"  Calibration intercept: {cal.intercept_[0]:.3f}")

    return pred


# ---- Population A: Temporal validation (3 years later) ----
np.random.seed(101)
n_a = 800
pop_a = pd.DataFrame({
    "age": np.round(np.random.normal(73, 12, n_a)),
    "nihss": np.round(np.maximum(0, np.random.normal(8, 6, n_a))),
    "glucose": np.round(np.random.normal(138, 48, n_a)),
    "afib": np.random.binomial(1, 0.27, n_a),
    "thrombolysis": np.random.binomial(1, 0.40, n_a)  # More thrombolysis
})
lp_a = (-5 + 0.04 * pop_a["age"] + 0.12 * pop_a["nihss"] +
        0.003 * pop_a["glucose"] + 0.3 * pop_a["afib"] -
        0.5 * pop_a["thrombolysis"])
pop_a["death_30d"] = np.random.binomial(1, expit(lp_a))
```

```
pred_a = evaluate_ext(pop_a, model, predictors, "Population A: Temporal")

# ---- Population B: Geographical (younger patients) ----
np.random.seed(202)
n_b = 600
pop_b = pd.DataFrame({
    "age": np.round(np.random.normal(62, 10, n_b)),
    "nihss": np.round(np.maximum(0, np.random.normal(6, 5, n_b))),
    "glucose": np.round(np.random.normal(130, 45, n_b)),
    "afib": np.random.binomial(1, 0.15, n_b),
    "thrombolysis": np.random.binomial(1, 0.35, n_b)
})
lp_b = (-5 + 0.04 * pop_b["age"] + 0.12 * pop_b["nihss"] +
        0.003 * pop_b["glucose"] + 0.3 * pop_b["afib"] -
        0.5 * pop_b["thrombolysis"])
pop_b["death_30d"] = np.random.binomial(1, expit(lp_b))
pred_b = evaluate_ext(pop_b, model, predictors, "Population B: Geographical")

# ---- Population C: Domain (primary care, lower severity) ----
np.random.seed(303)
n_c = 500
pop_c = pd.DataFrame({
    "age": np.round(np.random.normal(68, 14, n_c)),
    "nihss": np.round(np.maximum(0, np.random.normal(3, 3, n_c))),
    "glucose": np.round(np.random.normal(120, 35, n_c)),
    "afib": np.random.binomial(1, 0.20, n_c),
    "thrombolysis": np.random.binomial(1, 0.10, n_c)
})
# Different outcome model in primary care
lp_c = (-6 + 0.03 * pop_c["age"] + 0.10 * pop_c["nihss"] +
        0.002 * pop_c["glucose"] + 0.2 * pop_c["afib"] -
        0.3 * pop_c["thrombolysis"])
pop_c["death_30d"] = np.random.binomial(1, expit(lp_c))
```

```
pred_c = evaluate_ext(pop_c, model, predictors, "Population C: Domain")

# ---- (b) Calibration plots ----
fig, axes = plt.subplots(1, 3, figsize=(16, 5))
pops = [(pop_a, pred_a, "A: Temporal"), (pop_b, pred_b, "B: Geographical"),
        (pop_c, pred_c, "C: Domain")]

for ax, (pop, pred, title) in zip(axes, pops):
    y_ext = pop["death_30d"].values
    prob_true, prob_pred = calibration_curve(y_ext, pred, n_bins=10,
                                             strategy="quantile")
    ax.plot(prob_pred, prob_true, "o-", color="steelblue", lw=2)
    ax.plot([0, 1], [0, 1], "--", color="grey")
    ax.set_xlabel("Predicted Probability")
    ax.set_ylabel("Observed Proportion")
    ax.set_title(title)

plt.tight_layout()
plt.savefig("ch11_ex2_calibration_plots.png", dpi=150)
plt.show()

# ---- (c) Worst calibration ----
print("\n=== Part (c): Worst Calibration ===")
print("Population C (primary care / domain validation) shows worst calibratio
print("The outcome model differs from the development setting:")
print("  - Different baseline risk and predictor-outcome relationships")
print("  - Much lower severity patients (NIHSS ~ 3 vs 8)")
print("  - Domain validation is the most stringent test of transportability."

# ---- (d) Logistic recalibration ----
print("\n=== Part (d): Logistic Recalibration ===")

for pop, pred, label in pops:
```

```
    y_ext = pop["death_30d"].values
    lp_ext = logit(np.clip(pred, 1e-8, 1 - 1e-8))
    recal = LogisticRegression(max_iter=5000, penalty=None)
    recal.fit(lp_ext.reshape(-1, 1), y_ext)
    pred_recal = recal.predict_proba(lp_ext.reshape(-1, 1))[:, 1]

    oe_before = y_ext.mean() / pred.mean()
    oe_after = y_ext.mean() / pred_recal.mean()

    print(f"\n{label}:")
    print(f"  O:E before: {oe_before:.3f}")
    print(f"  O:E after:  {oe_after:.3f}")

print("\nLogistic recalibration adjusts intercept and slope, bringing")
print("O:E ratios closer to 1.0 in each external population.")
```

D.8.3. Exercise 3

D.8.3.1. R

```
# Chapter 11, Exercise 3: Decision Curve Interpretation
# Conceptual exercise about a preeclampsia prediction model

# This exercise is primarily conceptual. We simulate a plausible preeclampsia
# model and generate a decision curve to support the discussion.

set.seed(42)
n <- 2000

# Simulate data for preeclampsia prediction
preeclampsia_data <- data.frame(
  age = round(rnorm(n, 30, 5)),
```

```
  bmi = round(rnorm(n, 27, 5), 1),
  nulliparous = rbinom(n, 1, 0.4),
  history_pe = rbinom(n, 1, 0.05),
  map = round(rnorm(n, 85, 10))  # mean arterial pressure
)

lp <- -5.5 + 0.03 * preeclampsia_data$age +
  0.04 * preeclampsia_data$bmi +
  0.5 * preeclampsia_data$nulliparous +
  1.5 * preeclampsia_data$history_pe +
  0.02 * preeclampsia_data$map

preeclampsia_data$pe <- rbinom(n, 1, plogis(lp))
cat("Preeclampsia rate:", mean(preeclampsia_data$pe), "\n")

# Fit model
model <- glm(pe ~ age + bmi + nulliparous + history_pe + map,
             data = preeclampsia_data, family = binomial)

pred_prob <- predict(model, type = "response")
y_obs <- preeclampsia_data$pe

# ---- Generate decision curve ----
thresholds <- seq(0.01, 0.50, by = 0.01)
n_total <- length(y_obs)
nb_model <- nb_all <- numeric(length(thresholds))

for (i in seq_along(thresholds)) {
  t <- thresholds[i]
  pred_pos <- as.numeric(pred_prob >= t)
  tp <- sum(pred_pos == 1 & y_obs == 1)
  fp <- sum(pred_pos == 1 & y_obs == 0)
  nb_model[i] <- tp / n_total - fp / n_total * (t / (1 - t))
```

```r
  nb_all[i] <- mean(y_obs) - (1 - mean(y_obs)) * (t / (1 - t))
}

plot(thresholds, nb_model, type = "l", lwd = 2, col = "steelblue",
     main = "Decision Curve: Preeclampsia Prediction Model\n(AUC ~ 0.82)",
     xlab = "Threshold Probability", ylab = "Net Benefit",
     ylim = c(-0.05, max(c(nb_model, nb_all)) * 1.1))
lines(thresholds, nb_all, lwd = 2, col = "coral", lty = 2)
abline(h = 0, col = "black")
legend("topright", legend = c("Model", "Treat All", "Treat None"),
       col = c("steelblue", "coral", "black"),
       lty = c(1, 2, 1), lwd = 2)

# ---- Part (a): Range of useful thresholds ----
cat("\n=== Part (a): Range of Useful Thresholds ===\n")
cat("The model is useful at thresholds where its net benefit curve\n")
cat("exceeds BOTH the 'treat all' and 'treat none' lines.\n")
cat("From the decision curve, the model appears useful approximately\n")
cat("in the range of threshold probabilities from about 2% to 30-40%.\n")
cat("Below ~2%, 'treat all' has similar or higher net benefit.\n")
cat("Above ~30-40%, the model's net benefit approaches zero.\n")

# ---- Part (b): Counter-argument to 'AUC is only 0.82' ----
cat("\n=== Part (b): Counter-argument ===\n")
cat("An AUC of 0.82 is not 'only' -- it is strong discrimination.\n")
cat("More importantly, the AUC does not directly tell you whether\n")
cat("using the model improves clinical decisions.\n\n")
cat("The decision curve shows that across the clinically relevant\n")
cat("threshold range (e.g., 5-20%), the model provides positive\n")
cat("net benefit above both default strategies.\n\n")
cat("This means that using the model to guide closer monitoring\n")
cat("decisions would identify more true preeclampsia cases per\n")
cat("'unnecessary' monitoring than either monitoring everyone or\n")
```

```
cat("monitoring no one. Clinical utility is what matters for\n")
cat("patient care -- not the AUC alone.\n")

# ---- Part (c): Effect of action cost on the decision curve ----
cat("\n=== Part (c): Effect of Action Cost ===\n")
cat("The threshold probability reflects the implicit cost-benefit\n")
cat("ratio of the action (closer monitoring).\n\n")
cat("LOW-COST action (e.g., closer monitoring, extra visits):\n")
cat("  - Clinicians would use a LOW threshold (e.g., 3-5%)\n")
cat("  - They accept many false positives to avoid missing cases\n")
cat("  - The relevant region of the decision curve shifts LEFT\n")
cat("  - 'Treat all' remains competitive at low thresholds\n\n")
cat("HIGH-COST action (e.g., prophylactic medication with side effects):\n")
cat("  - Clinicians would use a HIGHER threshold (e.g., 15-25%)\n")
cat("  - They want more certainty before acting\n")
cat("  - The relevant region shifts RIGHT\n")
cat("  - The model has more value here because it avoids unnecessary\n")
cat("    treatment of low-risk patients\n\n")
cat("In summary: the decision curve itself does not change, but the\n")
cat("clinically relevant REGION changes depending on the cost of action.\n")
```

D.8.3.2. Python

```
# Chapter 11, Exercise 3: Decision Curve Interpretation
# Conceptual exercise about a preeclampsia prediction model

import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_auc_score
from scipy.special import expit
```

```python
# Simulate a plausible preeclampsia model to generate a decision curve
np.random.seed(42)
n = 2000

age = np.round(np.random.normal(30, 5, n))
bmi = np.round(np.random.normal(27, 5, n), 1)
nulliparous = np.random.binomial(1, 0.4, n)
history_pe = np.random.binomial(1, 0.05, n)
map_val = np.round(np.random.normal(85, 10, n))

lp = (-5.5 + 0.03 * age + 0.04 * bmi + 0.5 * nulliparous +
      1.5 * history_pe + 0.02 * map_val)
pe = np.random.binomial(1, expit(lp))
print(f"Preeclampsia rate: {pe.mean():.3f}")

X = np.column_stack([age, bmi, nulliparous, history_pe, map_val])
model = LogisticRegression(max_iter=5000, random_state=42)
model.fit(X, pe)
pred_prob = model.predict_proba(X)[:, 1]
print(f"AUC: {roc_auc_score(pe, pred_prob):.3f}")

# ---- Generate decision curve ----
thresholds = np.arange(0.01, 0.51, 0.01)
n_total = len(pe)


def net_benefit(y_true, y_pred, threshold):
    pred_pos = (y_pred >= threshold).astype(int)
    tp = np.sum((pred_pos == 1) & (y_true == 1))
    fp = np.sum((pred_pos == 1) & (y_true == 0))
    return tp / n_total - fp / n_total * (threshold / (1 - threshold))
```

```
nb_model = [net_benefit(pe, pred_prob, t) for t in thresholds]
nb_all = [pe.mean() - (1 - pe.mean()) * t / (1 - t) for t in thresholds]

plt.figure(figsize=(9, 6))
plt.plot(thresholds, nb_model, color="steelblue", lw=2, label="Prediction Moc
plt.plot(thresholds, nb_all, color="coral", lw=2, ls="--", label="Treat All")
plt.axhline(y=0, color="black", lw=1, label="Treat None")
plt.xlabel("Threshold Probability")
plt.ylabel("Net Benefit")
plt.title("Decision Curve: Preeclampsia Prediction Model (AUC ~ 0.82)")
plt.legend()
plt.ylim(-0.05, max(max(nb_model), max(nb_all)) * 1.1)
plt.tight_layout()
plt.savefig("ch11_ex3_decision_curve.png", dpi=150)
plt.show()

# ---- Part (a): Range of useful thresholds ----
print("\n=== Part (a): Range of Useful Thresholds ===")
print("The model is useful at thresholds where its net benefit curve")
print("exceeds BOTH 'treat all' and 'treat none'.")
print("From the decision curve, the model is useful approximately")
print("from about 2% to 30-40% threshold probability.")
print("Below ~2%, 'treat all' has similar or higher net benefit.")
print("Above ~30-40%, the model's net benefit approaches zero.")

# ---- Part (b): Counter-argument to 'AUC is only 0.82' ----
print("\n=== Part (b): Counter-argument ===")
print("An AUC of 0.82 is strong discrimination, not 'only'.")
print("More importantly, AUC does not directly measure clinical utility.")
print("")
print("The decision curve shows the model provides positive net benefit")
print("above both default strategies across the clinically relevant")
print("threshold range (e.g., 5-20%). This means using the model to")
```

```
print("guide monitoring decisions identifies more true preeclampsia")
print("cases per 'unnecessary' monitoring episode than either monitoring")
print("everyone or no one. Clinical utility -- not AUC -- determines")
print("whether a model should be used in practice.")

# ---- Part (c): Effect of action cost ----
print("\n=== Part (c): Effect of Action Cost ===")
print("The threshold probability reflects the cost-benefit ratio of acting.")
print("")
print("LOW-COST action (closer monitoring, extra visits):")
print("  - Clinicians use a LOW threshold (e.g., 3-5%)")
print("  - Many false positives are tolerable")
print("  - The relevant region of the decision curve is on the LEFT")
print("  - 'Treat all' remains competitive at low thresholds")
print("")
print("HIGH-COST action (prophylactic medication with side effects):")
print("  - Clinicians use a HIGHER threshold (e.g., 15-25%)")
print("  - More certainty is needed before acting")
print("  - The relevant region shifts RIGHT")
print("  - The model adds more value by avoiding unnecessary treatment")
print("")
print("The curve itself does not change, but the clinically relevant")
print("region changes depending on the cost of the follow-up action.")
```

D.8.4. Exercise 4

D.8.4.1. R

```
# Chapter 11, Exercise 4: Complete Evaluation of a Published Model
# Example: QRISK3 Cardiovascular Risk Prediction Model
#
# This exercise is primarily a literature review exercise. Below we provide
```

```
# a structured answer based on published validation data for QRISK3.

# ---- Part (a): C-statistic with confidence interval ----
cat("=== Part (a): C-statistic ===\n")
cat("QRISK3 (Hippisley-Cox et al., 2017, BMJ) was developed to predict\n")
cat("10-year cardiovascular disease risk.\n\n")
cat("Development cohort:\n")
cat("  C-statistic (women): 0.880 (95% CI: 0.878-0.882)\n")
cat("  C-statistic (men):   0.858 (95% CI: 0.856-0.860)\n\n")
cat("Validation cohort (separate 25% held out):\n")
cat("  C-statistic (women): 0.879 (95% CI: 0.876-0.882)\n")
cat("  C-statistic (men):   0.858 (95% CI: 0.855-0.861)\n")
cat("The discrimination is excellent and stable between development\n")
cat("and validation sets.\n")

# ---- Part (b): Calibration ----
cat("\n=== Part (b): Calibration ===\n")
cat("QRISK3 provides calibration plots in the original publication.\n")
cat("In the validation cohort:\n")
cat("  - The calibration was generally good, with predicted risks\n")
cat("    closely matching observed event rates across deciles.\n")
cat("  - Some overestimation of risk was observed at higher risk\n")
cat("    levels, particularly in older age groups.\n")
cat("  - External validations in different populations (e.g., outside\n")
cat("    the UK) have shown variable calibration, often with\n")
cat("    overestimation in lower-risk populations.\n")

# ---- Part (c): Decision curve analysis ----
cat("\n=== Part (c): Decision Curve Analysis ===\n")
cat("The original QRISK3 paper does not include decision curve analysis.\n\n
cat("Clinically relevant threshold range for statin initiation:\n")
cat("  - UK NICE guidelines: 10% 10-year CVD risk threshold\n")
cat("  - US ACC/AHA guidelines: 7.5% threshold\n")
```

```
cat("  - A decision curve would be most informative in the 5-20%\n")
cat("    threshold range, where the decision to start statins is\n")
cat("    most uncertain.\n")
cat("  - Below 5%, most clinicians would not start statins\n")
cat("  - Above 20%, most clinicians would start statins regardless\n")

# ---- Part (d): External validation ----
cat("\n=== Part (d): External Validation ===\n")
cat("QRISK3 has been validated in multiple populations:\n")
cat("  - Internal-external: UK CPRD data (separate time period)\n")
cat("  - Geographic: Various European populations\n")
cat("  - Ethnic subgroups: South Asian, Black, Chinese populations\n")
cat("  - Temporal: Validated across different calendar periods\n\n")
cat("Key findings from external validations:\n")
cat("  - Discrimination generally remains good (C > 0.80)\n")
cat("  - Calibration can deteriorate in non-UK populations\n")
cat("  - May overestimate risk in populations with lower baseline\n")
cat("    cardiovascular event rates\n")

# ---- Part (e): Recommendation for implementation ----
cat("\n=== Part (e): Recommendation ===\n")
cat("Recommendation depends on the clinical setting:\n\n")
cat("FOR a UK primary care setting:\n")
cat("  - YES, recommend implementation. QRISK3 was developed and\n")
cat("    validated in UK primary care, has excellent discrimination,\n")
cat("    good calibration, and is already integrated into UK\n")
cat("    clinical guidelines.\n\n")
cat("FOR a non-UK setting:\n")
cat("  - CONDITIONAL recommendation. Would first require:\n")
cat("    1. External validation in the local population\n")
cat("    2. Assessment of calibration (likely needs recalibration)\n")
cat("    3. Decision curve analysis at locally relevant thresholds\n")
cat("    4. Comparison with locally developed risk scores\n")
```

```
cat("  - If calibration is poor, logistic recalibration should be\n")
cat("    considered before implementation.\n")
```

D.8.4.2. Python

```
# Chapter 11, Exercise 4: Complete Evaluation of a Published Model
# Example: QRISK3 Cardiovascular Risk Prediction Model
#
# This exercise is primarily a literature review exercise. Below we provide
# a structured answer based on published validation data for QRISK3.

# ---- Part (a): C-statistic with confidence interval ----
print("=== Part (a): C-statistic ===")
print("QRISK3 (Hippisley-Cox et al., 2017, BMJ) was developed to predict")
print("10-year cardiovascular disease risk.")
print()
print("Development cohort:")
print("  C-statistic (women): 0.880 (95% CI: 0.878-0.882)")
print("  C-statistic (men):   0.858 (95% CI: 0.856-0.860)")
print()
print("Validation cohort (separate 25% held out):")
print("  C-statistic (women): 0.879 (95% CI: 0.876-0.882)")
print("  C-statistic (men):   0.858 (95% CI: 0.855-0.861)")
print("The discrimination is excellent and stable between development")
print("and validation sets.")

# ---- Part (b): Calibration ----
print("\n=== Part (b): Calibration ===")
print("QRISK3 provides calibration plots in the original publication.")
print("In the validation cohort:")
print("  - Calibration was generally good, with predicted risks closely")
print("    matching observed event rates across deciles.")
```

```
print("  - Some overestimation at higher risk levels, particularly in")
print("    older age groups.")
print("  - External validations outside the UK have shown variable")
print("    calibration, often with overestimation in lower-risk populations.")

# ---- Part (c): Decision curve analysis ----
print("\n=== Part (c): Decision Curve Analysis ===")
print("The original QRISK3 paper does not include decision curve analysis.")
print()
print("Clinically relevant threshold range for statin initiation:")
print("  - UK NICE guidelines: 10% 10-year CVD risk threshold")
print("  - US ACC/AHA guidelines: 7.5% threshold")
print("  - A decision curve would be most informative in the 5-20%")
print("    threshold range, where the statin decision is most uncertain.")
print("  - Below 5%, most clinicians would not start statins")
print("  - Above 20%, most clinicians would start statins regardless")

# ---- Part (d): External validation ----
print("\n=== Part (d): External Validation ===")
print("QRISK3 has been validated in multiple populations:")
print("  - Internal-external: UK CPRD data (separate time period)")
print("  - Geographic: Various European populations")
print("  - Ethnic subgroups: South Asian, Black, Chinese populations")
print("  - Temporal: Validated across different calendar periods")
print()
print("Key findings from external validations:")
print("  - Discrimination generally remains good (C > 0.80)")
print("  - Calibration can deteriorate in non-UK populations")
print("  - May overestimate risk in populations with lower baseline")
print("    cardiovascular event rates")

# ---- Part (e): Recommendation ----
print("\n=== Part (e): Recommendation ===")
```

```
print("Recommendation depends on the clinical setting:")
print()
print("FOR a UK primary care setting:")
print("  YES, recommend implementation. QRISK3 was developed and")
print("  validated in UK primary care, has excellent discrimination,")
print("  good calibration, and is integrated into UK clinical guidelines.")
print()
print("FOR a non-UK setting:")
print("  CONDITIONAL recommendation. Would first require:")
print("    1. External validation in the local population")
print("    2. Assessment of calibration (likely needs recalibration)")
print("    3. Decision curve analysis at locally relevant thresholds")
print("    4. Comparison with locally developed risk scores")
print("  If calibration is poor, logistic recalibration should be")
print("  considered before implementation.")
```

D.9. Chapter 13: Bayesian Inference

D.9.1. Exercise 1

D.9.1.1. R

```
# Chapter 13, Exercise 1: Diagnostic Test Updating
# Sequential Bayesian updating with mammogram and ultrasound

# ---- Part (a): Posterior probability after positive mammogram ----
cat("=== Part (a): Positive Mammogram ===\n")

sens_mammo <- 0.90   # Sensitivity of mammogram
spec_mammo <- 0.92   # Specificity of mammogram
prevalence <- 0.02   # Prior probability (prevalence in women aged 50-59)
```

```
# Apply Bayes' theorem:
# P(cancer | +) = P(+ | cancer) * P(cancer) / P(+)
# P(+) = P(+ | cancer)*P(cancer) + P(+ | no cancer)*P(no cancer)

p_pos <- sens_mammo * prevalence + (1 - spec_mammo) * (1 - prevalence)
post_mammo <- (sens_mammo * prevalence) / p_pos

cat("Prior (prevalence): ", prevalence, "\n")
cat("P(positive test):   ", round(p_pos, 4), "\n")
cat("P(cancer | positive mammogram):", round(post_mammo, 4), "\n")
cat("\nDespite 90% sensitivity and 92% specificity, the posterior\n")
cat("probability is only about", round(post_mammo * 100, 1), "% because\n")
cat("the prior probability (prevalence) is very low at 2%.\n")

# ---- Part (b): Sequential update with positive ultrasound ----
cat("\n=== Part (b): Sequential Update with Positive Ultrasound ===\n")

sens_us <- 0.95   # Sensitivity of ultrasound
spec_us <- 0.85   # Specificity of ultrasound

# The posterior from (a) becomes the new prior
prior_us <- post_mammo

p_pos_us <- sens_us * prior_us + (1 - spec_us) * (1 - prior_us)
post_us <- (sens_us * prior_us) / p_pos_us

cat("Prior (posterior from mammogram):", round(prior_us, 4), "\n")
cat("P(cancer | positive mammogram AND positive ultrasound):",
    round(post_us, 4), "\n")
cat("\nAfter two positive tests, the posterior probability rises to about",
    round(post_us * 100, 1), "%.\n")

# ---- Part (c): What does this illustrate? ----
```

```
cat("\n=== Part (c): What Sequential Updating Illustrates ===\n")
cat("1. NATURAL SEQUENTIAL UPDATING: In the Bayesian framework,\n")
cat("   today's posterior becomes tomorrow's prior. Each new piece\n")
cat("   of evidence updates our belief incrementally.\n\n")
cat("2. THE PRIOR MATTERS: Starting from a low prevalence (2%),\n")
cat("   even a sensitive test only raises the probability to ~19%.\n")
cat("   A second positive test raises it further to ~", round(post_us * 100)
cat("3. CLINICAL REASONING IS BAYESIAN: Clinicians intuitively do\n")
cat("   this -- they order confirmatory tests precisely because a\n")
cat("   single screening test in a low-prevalence population is\n")
cat("   insufficient. The Bayesian framework formalises this logic.\n\n")
cat("4. COHERENCE: The Bayesian approach naturally handles sequential\n")
cat("   evidence without the multiple-testing corrections that\n")
cat("   frequentist methods would require.\n")

# ---- Visualise the updating process ----
stages <- c("Prior\n(prevalence)", "After\nmammogram (+)", "After\nultrasound
probs <- c(prevalence, post_mammo, post_us)

barplot(probs, names.arg = stages, col = c("steelblue", "goldenrod", "firebri
        main = "Sequential Bayesian Updating\nBreast Cancer Diagnosis",
        ylab = "P(Cancer)", ylim = c(0, max(probs) * 1.2),
        border = NA)
text(x = c(0.7, 1.9, 3.1), y = probs + 0.02,
     labels = paste0(round(probs * 100, 1), "%"), cex = 0.9)
```

D.9.1.2. Python

```
# Chapter 13, Exercise 1: Diagnostic Test Updating
# Sequential Bayesian updating with mammogram and ultrasound

import matplotlib.pyplot as plt
```

```
# ---- Part (a): Posterior probability after positive mammogram ----
print("=== Part (a): Positive Mammogram ===")

sens_mammo = 0.90   # Sensitivity of mammogram
spec_mammo = 0.92   # Specificity of mammogram
prevalence = 0.02   # Prior probability (prevalence in women aged 50-59)

# Apply Bayes' theorem:
# P(cancer | +) = P(+ | cancer) * P(cancer) / P(+)
p_pos = sens_mammo * prevalence + (1 - spec_mammo) * (1 - prevalence)
post_mammo = (sens_mammo * prevalence) / p_pos

print(f"Prior (prevalence):  {prevalence}")
print(f"P(positive test):    {p_pos:.4f}")
print(f"P(cancer | positive mammogram): {post_mammo:.4f}")
print(f"\nDespite 90% sensitivity and 92% specificity, the posterior")
print(f"probability is only about {post_mammo*100:.1f}% because the")
print(f"prior probability (prevalence) is very low at 2%.")

# ---- Part (b): Sequential update with positive ultrasound ----
print("\n=== Part (b): Sequential Update with Positive Ultrasound ===")

sens_us = 0.95   # Sensitivity of ultrasound
spec_us = 0.85   # Specificity of ultrasound

# The posterior from (a) becomes the new prior
prior_us = post_mammo
p_pos_us = sens_us * prior_us + (1 - spec_us) * (1 - prior_us)
post_us = (sens_us * prior_us) / p_pos_us

print(f"Prior (posterior from mammogram): {prior_us:.4f}")
print(f"P(cancer | pos mammogram AND pos ultrasound): {post_us:.4f}")
print(f"\nAfter two positive tests, the posterior probability rises to about {post_us*100:.1
```

```
# ---- Part (c): What sequential updating illustrates ----
print("\n=== Part (c): What Sequential Updating Illustrates ===")
print("1. NATURAL SEQUENTIAL UPDATING: In the Bayesian framework,")
print("   today's posterior becomes tomorrow's prior. Each new piece")
print("   of evidence updates our belief incrementally.")
print()
print("2. THE PRIOR MATTERS: Starting from a low prevalence (2%),")
print(f"   even a sensitive test only raises the probability to ~{post_mammo
print(f"   A second positive test raises it further to ~{post_us*100:.0f}%."
print()
print("3. CLINICAL REASONING IS BAYESIAN: Clinicians intuitively do")
print("   this -- they order confirmatory tests precisely because a")
print("   single screening test in a low-prevalence population is")
print("   insufficient. The Bayesian framework formalises this logic.")
print()
print("4. COHERENCE: The Bayesian approach naturally handles sequential")
print("   evidence without the multiple-testing corrections that")
print("   frequentist methods would require.")

# ---- Visualise the updating process ----
stages = ["Prior\n(prevalence)", "After\nmammogram (+)", "After\nultrasound
probs = [prevalence, post_mammo, post_us]
colors = ["steelblue", "goldenrod", "firebrick"]

fig, ax = plt.subplots(figsize=(7, 5))
bars = ax.bar(stages, probs, color=colors, edgecolor="none", width=0.6)
for bar, p in zip(bars, probs):
    ax.text(bar.get_x() + bar.get_width() / 2, bar.get_height() + 0.01,
            f"{p*100:.1f}%", ha="center", fontsize=11)
ax.set_ylabel("P(Cancer)")
ax.set_title("Sequential Bayesian Updating\nBreast Cancer Diagnosis")
ax.set_ylim(0, max(probs) * 1.3)
plt.tight_layout()
```

```
plt.savefig("ch13_ex1_updating.png", dpi=150)
plt.show()
```

D.9.2. Exercise 2

D.9.2.1. R

```
# Chapter 13, Exercise 2: Prior Sensitivity Analysis
# Phase II antibiotic trial: 21/30 patients achieve clinical cure

library(tidyverse)

y <- 21  # successes
n <- 30  # total patients

# Define three priors
priors <- list(
  list(name = "Beta(1,1) - Uninformative", a = 1, b = 1),
  list(name = "Beta(5,5) - Weakly informative", a = 5, b = 5),
  list(name = "Beta(20,10) - Informative", a = 20, b = 10)
)

theta <- seq(0, 1, length.out = 500)

# ---- Part (a): Posterior mean and 95% credible interval ----
cat("=== Part (a): Posterior Summaries ===\n\n")

plot_df <- tibble()

for (p in priors) {
  a_post <- p$a + y
  b_post <- p$b + n - y
```

```
  post_mean <- a_post / (a_post + b_post)
  ci <- qbeta(c(0.025, 0.975), a_post, b_post)

  cat(sprintf("Prior: %s\n", p$name))
  cat(sprintf("  Prior mean: %.3f\n", p$a / (p$a + p$b)))
  cat(sprintf("  Posterior: Beta(%d, %d)\n", a_post, b_post))
  cat(sprintf("  Posterior mean: %.3f\n", post_mean))
  cat(sprintf("  95%% credible interval: [%.3f, %.3f]\n\n", ci[1], ci[2]))

  plot_df <- bind_rows(plot_df,
    tibble(
      theta = theta,
      Density = dbeta(theta, a_post, b_post),
      Prior = p$name
    )
  )
}

# ---- Part (b): Plot all three posteriors ----
cat("=== Part (b): Posterior Plot ===\n")

ggplot(plot_df, aes(x = theta, y = Density, colour = Prior)) +
  geom_line(linewidth = 1.2) +
  geom_vline(xintercept = y / n, linetype = "dashed", colour = "grey40") +
  annotate("text", x = y / n + 0.02, y = max(plot_df$Density) * 0.9,
           label = paste0("MLE = ", round(y / n, 3)),
           hjust = 0, size = 3.5) +
  labs(x = expression(theta ~ "(cure rate)"),
       y = "Density",
       title = "Posterior Distributions Under Different Priors",
       subtitle = "21/30 patients cured in Phase II trial") +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom")
```

```r
# ---- Part (c): Which prior has most influence? ----
cat("=== Part (c): Which Prior Has Most Influence? ===\n\n")
cat("The Beta(5,5) prior has the MOST influence on the posterior.\n\n")
cat("Why? The effective sample size of a Beta(a,b) prior is a + b.\n")
cat("  - Beta(1,1):   effective n = 2  (trivial influence)\n")
cat("  - Beta(5,5):   effective n = 10 (modest influence)\n")
cat("  - Beta(20,10): effective n = 30 (substantial influence)\n\n")
cat("However, while Beta(20,10) has the largest effective sample size,\n")
cat("its prior mean (0.667) is close to the data (21/30 = 0.700),\n")
cat("so the prior and data 'agree', and the posterior is not pulled\n")
cat("far from the MLE.\n\n")
cat("Beta(5,5) has a prior mean of 0.500, which is DIFFERENT from\n")
cat("the data. It pulls the posterior mean toward 0.5, producing the\n")
cat("most visible shift from the MLE. With n=30 observed data, even\n")
cat("an effective sample size of 10 produces noticeable pull when\n")
cat("the prior and data disagree.\n\n")
cat("Key insight: prior influence depends on BOTH the effective sample\n")
cat("size AND how much the prior disagrees with the data.\n")
```

D.9.2.2. Python

```python
# Chapter 13, Exercise 2: Prior Sensitivity Analysis
# Phase II antibiotic trial: 21/30 patients achieve clinical cure

import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import beta

y, n = 21, 30  # successes and total
theta = np.linspace(0, 1, 500)
```

```
# Define three priors
priors = [
    ("Beta(1,1) - Uninformative", 1, 1),
    ("Beta(5,5) - Weakly informative", 5, 5),
    ("Beta(20,10) - Informative", 20, 10),
]

# ---- Part (a): Posterior mean and 95% credible interval ----
print("=== Part (a): Posterior Summaries ===\n")

fig, ax = plt.subplots(figsize=(8, 5))

for name, a0, b0 in priors:
    a_post = a0 + y
    b_post = b0 + n - y
    post_mean = a_post / (a_post + b_post)
    ci = beta.ppf([0.025, 0.975], a_post, b_post)

    print(f"Prior: {name}")
    print(f"  Prior mean: {a0 / (a0 + b0):.3f}")
    print(f"  Posterior: Beta({a_post}, {b_post})")
    print(f"  Posterior mean: {post_mean:.3f}")
    print(f"  95% credible interval: [{ci[0]:.3f}, {ci[1]:.3f}]")
    print()

    # ---- Part (b): Plot all three posteriors ----
    ax.plot(theta, beta.pdf(theta, a_post, b_post), label=name, linewidth=1.5

ax.axvline(y / n, linestyle="--", color="grey", linewidth=1)
ax.text(y / n + 0.02, ax.get_ylim()[1] * 0.1,
        f"MLE = {y/n:.3f}", fontsize=9, color="grey")
ax.set_xlabel(r"$\theta$ (cure rate)")
ax.set_ylabel("Density")
```

```
ax.set_title("Posterior Distributions Under Different Priors\n"
             "21/30 patients cured in Phase II trial")
ax.legend(fontsize=9)
plt.tight_layout()
plt.savefig("ch13_ex2_posteriors.png", dpi=150)
plt.show()

# ---- Part (c): Which prior has the most influence? ----
print("=== Part (c): Which Prior Has Most Influence? ===\n")
print("The Beta(5,5) prior has the MOST influence on the posterior.")
print()
print("Why? The effective sample size of a Beta(a,b) prior is a + b.")
print("  - Beta(1,1):   effective n = 2  (trivial influence)")
print("  - Beta(5,5):   effective n = 10 (modest influence)")
print("  - Beta(20,10): effective n = 30 (substantial influence)")
print()
print("However, while Beta(20,10) has the largest effective sample size,")
print("its prior mean (0.667) is close to the data (21/30 = 0.700),")
print("so the prior and data 'agree', and the posterior is not pulled")
print("far from the MLE.")
print()
print("Beta(5,5) has a prior mean of 0.500, which DISAGREES with the")
print("data. It pulls the posterior mean toward 0.5, producing the most")
print("visible shift from the MLE. With n=30 observed data, even an")
print("effective sample size of 10 produces noticeable pull when the")
print("prior and data disagree.")
print()
print("Key insight: prior influence depends on BOTH the effective sample")
print("size AND how much the prior disagrees with the data.")
```

D.9.3. Exercise 3

D.9.3.1. R

```
# Chapter 13, Exercise 3: Interpreting MCMC Diagnostics
# Conceptual exercise about convergence problems

# Given diagnostics for a Bayesian logistic regression treatment effect:
#   - R-hat: 1.08
#   - Bulk ESS: 150
#   - Tail ESS: 85
#   - Trace plot: two chains exploring different regions

# ---- Part (a): Is there evidence of convergence problems? ----
cat("=== Part (a): Evidence of Convergence Problems ===\n\n")

cat("YES, there is clear evidence of convergence problems.\n")
cat("ALL THREE diagnostics are concerning:\n\n")

cat("1. R-hat = 1.08 (threshold: < 1.01)\n")
cat("   - R-hat compares between-chain to within-chain variance.\n")
cat("   - A value of 1.08 far exceeds the 1.01 threshold.\n")
cat("   - This means chains are NOT sampling from the same distribution.\n")
cat("   - The trace plot confirms this: chains explore different regions.\n\

cat("2. Bulk ESS = 150 (recommended: > 400 per chain, i.e., > 1600 total)\n"
cat("   - Bulk ESS estimates the effective number of independent samples\n")
cat("     for estimating posterior means and medians.\n")
cat("   - 150 is far too low for reliable posterior summaries.\n")
cat("   - High autocorrelation or poor mixing causes low ESS.\n\n")

cat("3. Tail ESS = 85 (recommended: > 400)\n")
cat("   - Tail ESS measures the effective samples for tail quantiles\n")
```

```
cat("     (e.g., 2.5th and 97.5th percentiles for credible intervals).\n")
cat("   - 85 is dangerously low -- credible intervals will be unreliable.\n")
cat("   - Tail ESS is almost always lower than bulk ESS, so if bulk is\n")
cat("     already too low, tail ESS will be even worse.\n\n")

cat("4. Trace plot: chains exploring different regions\n")
cat("   - This is the most visually obvious sign of non-convergence.\n")
cat("   - Well-behaved chains should overlap ('hairy caterpillar').\n")
cat("   - Chains stuck in different regions indicate multimodality or\n")
cat("     insufficient sampling to traverse the parameter space.\n")

# ---- Part (b): Steps to fix these problems ----
cat("\n=== Part (b): Steps to Fix These Problems ===\n\n")

cat("1. INCREASE WARMUP AND ITERATIONS:\n")
cat("   - Current chains may not have had enough warmup to find\n")
cat("     the typical set. Try warmup = 2000-4000, iter = 4000-8000.\n\n")

cat("2. INCREASE adapt_delta (Stan/brms):\n")
cat("   - Set control = list(adapt_delta = 0.95 or 0.99).\n")
cat("   - This makes the sampler take smaller steps, reducing\n")
cat("     divergent transitions at the cost of slower sampling.\n\n")

cat("3. CHECK THE MODEL SPECIFICATION:\n")
cat("   - Priors may be too vague or conflicting with the likelihood.\n")
cat("   - Use prior predictive checks to ensure priors are reasonable.\n")
cat("   - Consider stronger (more informative) priors if justified.\n\n")

cat("4. REPARAMETERISE THE MODEL:\n")
cat("   - Some parameterisations create difficult posterior geometries.\n")
cat("   - For hierarchical models, use non-centred parameterisation.\n")
cat("   - Standardise predictors to improve sampler efficiency.\n\n")
```

```r
cat("5. SIMPLIFY THE MODEL:\n")
cat("   - If the model is too complex for the data, consider\n")
cat("     removing weak predictors or reducing random effects.\n\n")

cat("6. RUN MORE CHAINS:\n")
cat("   - Running 6-8 chains (instead of 4) with different\n")
cat("     starting values helps diagnose multimodality.\n")

# ---- Part (c): Trust the posterior summaries? ----
cat("\n=== Part (c): Can We Trust the Posterior Summaries? ===\n\n")

cat("NO, absolutely not. The posterior summaries should NOT be trusted.\n\n")

cat("Reasons:\n")
cat("1. With R-hat = 1.08, the chains have not converged to the same\n")
cat("   distribution. Any summary (mean, median, CI) is averaging\n")
cat("   across different distributions -- the result is meaningless.\n\n")

cat("2. With bulk ESS = 150, the posterior mean is estimated from\n")
cat("   only ~150 effective independent samples. The Monte Carlo\n")
cat("   error is too large for the estimates to be precise.\n\n")

cat("3. With tail ESS = 85, the 95% credible interval is based on\n")
cat("   only ~85 effective tail samples. The interval bounds could\n")
cat("   shift substantially with additional sampling.\n\n")

cat("4. Reporting results from a non-converged model is a serious\n")
cat("   methodological error. The correct action is to fix the\n")
cat("   convergence issues BEFORE interpreting any results.\n\n")

cat("Rule: If R-hat > 1.01 or ESS < 400, do NOT interpret results.\n")
cat("Fix the model first.\n")
```

D.9.3.2. Python

```
# Chapter 13, Exercise 3: Interpreting MCMC Diagnostics
# Conceptual exercise about convergence problems

# Given diagnostics for a Bayesian logistic regression treatment effect:
#   - R-hat: 1.08
#   - Bulk ESS: 150
#   - Tail ESS: 85
#   - Trace plot: two chains exploring different regions

# ---- Part (a): Is there evidence of convergence problems? ----
print("=== Part (a): Evidence of Convergence Problems ===\n")

print("YES, there is clear evidence of convergence problems.")
print("ALL THREE diagnostics are concerning:\n")

print("1. R-hat = 1.08 (threshold: < 1.01)")
print("   - R-hat compares between-chain to within-chain variance.")
print("   - A value of 1.08 far exceeds the 1.01 threshold.")
print("   - This means chains are NOT sampling from the same distribution.")
print("   - The trace plot confirms this: chains explore different regions.\n")

print("2. Bulk ESS = 150 (recommended: > 400 per chain, i.e., > 1600 total)")
print("   - Bulk ESS estimates the effective number of independent samples")
print("     for estimating posterior means and medians.")
print("   - 150 is far too low for reliable posterior summaries.")
print("   - High autocorrelation or poor mixing causes low ESS.\n")

print("3. Tail ESS = 85 (recommended: > 400)")
print("   - Tail ESS measures effective samples for tail quantiles")
print("     (2.5th and 97.5th percentiles for credible intervals).")
print("   - 85 is dangerously low -- credible intervals will be unreliable.")
```

```
print("   - Tail ESS is almost always lower than bulk ESS.\n")

print("4. Trace plot: chains exploring different regions")
print("   - Well-behaved chains should overlap ('hairy caterpillar').")
print("   - Chains in different regions indicate multimodality or")
print("     insufficient sampling.\n")

# ---- Part (b): Steps to fix ----
print("=== Part (b): Steps to Fix These Problems ===\n")

print("1. INCREASE WARMUP AND ITERATIONS:")
print("   Try warmup=2000-4000, draws=4000-8000.\n")

print("2. INCREASE target_accept (PyMC) / adapt_delta (Stan):")
print("   Set target_accept=0.95 or 0.99 to reduce divergences.\n")

print("3. CHECK MODEL SPECIFICATION:")
print("   - Are priors too vague or conflicting with the likelihood?")
print("   - Run prior predictive checks.")
print("   - Consider stronger priors if justified.\n")

print("4. REPARAMETERISE THE MODEL:")
print("   - Use non-centred parameterisation for hierarchical models.")
print("   - Standardise predictors.\n")

print("5. SIMPLIFY THE MODEL:")
print("   - Remove weak predictors or reduce random effects.\n")

print("6. RUN MORE CHAINS:")
print("   - 6-8 chains with different starting values to diagnose multimodal

# ---- Part (c): Trust the posterior summaries? ----
print("=== Part (c): Can We Trust the Posterior Summaries? ===\n")
```

```
print("NO, absolutely not. The posterior summaries should NOT be trusted.\n")

print("Reasons:")
print("1. With R-hat = 1.08, chains have not converged to the same")
print("   distribution. Any summary averages across different")
print("   distributions -- the result is meaningless.\n")

print("2. With bulk ESS = 150, the posterior mean is estimated from")
print("   only ~150 effective samples. Monte Carlo error is too large.\n")

print("3. With tail ESS = 85, the 95% credible interval is based on")
print("   ~85 effective tail samples. Bounds could shift substantially.\n")

print("4. Reporting results from a non-converged model is a serious")
print("   methodological error.\n")

print("Rule: If R-hat > 1.01 or ESS < 400, do NOT interpret results.")
print("Fix the model first.")
```

D.10. Chapter 14: Applied Bayesian Modelling

D.10.1. Exercise 1

D.10.1.1. R

```
# Chapter 14, Exercise 1: Bayesian Logistic Regression for ICU Mortality
# 500 ICU admissions with age, APACHE II, ventilation status, 28-day mortality

library(brms)
library(tidyverse)
```

```
library(bayesplot)

# ---- (a) Simulate dataset ----
set.seed(101)
n <- 500

icu_data <- tibble(
  age = round(rnorm(n, 62, 15)),
  apache = round(rnorm(n, 18, 7)),
  ventilated = rbinom(n, 1, 0.35)
)

# True model: mortality increases with age, APACHE, and ventilation
lp <- -4.5 + 0.02 * icu_data$age +
  0.12 * icu_data$apache +
  0.8 * icu_data$ventilated

icu_data$mortality <- rbinom(n, 1, plogis(lp))

cat("Dataset summary:\n")
cat("  N:", n, "\n")
cat("  Mortality rate:", mean(icu_data$mortality), "\n")
cat("  Mean age:", round(mean(icu_data$age), 1), "\n")
cat("  Mean APACHE:", round(mean(icu_data$apache), 1), "\n")
cat("  % Ventilated:", round(mean(icu_data$ventilated) * 100, 1), "%\n")

# ---- (b) Fit Bayesian logistic regression ----
fit_icu <- brm(
  mortality ~ age + apache + ventilated,
  data = icu_data,
  family = bernoulli(link = "logit"),
  prior = c(
    prior(normal(0, 2.5), class = "b"),       # weakly informative on log-odo
```

```
    prior(normal(0, 5), class = "Intercept")
  ),
  chains = 4,
  iter = 2000,
  warmup = 1000,
  seed = 42,
  silent = 2,
  refresh = 0
)

cat("\nModel summary:\n")
summary(fit_icu)

# ---- (c) Prior predictive check ----
cat("\n=== Part (c): Prior Predictive Check ===\n")

# Fit with priors only (no data influence)
fit_prior <- brm(
  mortality ~ age + apache + ventilated,
  data = icu_data,
  family = bernoulli(link = "logit"),
  prior = c(
    prior(normal(0, 2.5), class = "b"),
    prior(normal(0, 5), class = "Intercept")
  ),
  sample_prior = "only",
  chains = 4,
  iter = 2000,
  warmup = 1000,
  seed = 42,
  silent = 2,
  refresh = 0
)
```

```
# Simulate from prior predictive
pp_prior <- posterior_predict(fit_prior)
prior_mort_rates <- rowMeans(pp_prior)

cat("Prior predictive mortality rates:\n")
cat("  Mean:", round(mean(prior_mort_rates), 3), "\n")
cat("  SD:", round(sd(prior_mort_rates), 3), "\n")
cat("  Range:", round(min(prior_mort_rates), 3), "to",
    round(max(prior_mort_rates), 3), "\n")
cat("The priors allow mortality rates from near 0 to near 1,\n")
cat("which covers all plausible ICU mortality rates. The priors\n")
cat("are appropriately weakly informative.\n")

# ---- (d) Posterior odds ratios with 95% credible intervals ----
cat("\n=== Part (d): Posterior Odds Ratios ===\n")

post <- as_draws_df(fit_icu)

or_age <- exp(post$b_age)
or_apache <- exp(post$b_apache)
or_vent <- exp(post$b_ventilated)

cat(sprintf("OR Age:        %.3f [%.3f, %.3f]\n",
            mean(or_age), quantile(or_age, 0.025), quantile(or_age, 0.975)))
cat(sprintf("OR APACHE:     %.3f [%.3f, %.3f]\n",
            mean(or_apache), quantile(or_apache, 0.025), quantile(or_apache,
cat(sprintf("OR Ventilated: %.3f [%.3f, %.3f]\n",
            mean(or_vent), quantile(or_vent, 0.025), quantile(or_vent, 0.975

# Plot posterior OR distributions
mcmc_areas(fit_icu, pars = c("b_age", "b_apache", "b_ventilated"),
           prob = 0.95, prob_outer = 0.99,
           transformations = exp) +
```

```
  geom_vline(xintercept = 1, linetype = "dashed", colour = "grey40") +
  labs(title = "Posterior Odds Ratios (95% CrI)",
       x = "Odds Ratio") +
  theme_minimal(base_size = 13)

# ---- (e) P(OR_APACHE > 1.10 | data) ----
cat("\n=== Part (e): P(OR_APACHE > 1.10 | data) ===\n")

prob_or_gt_110 <- mean(or_apache > 1.10)
cat(sprintf("P(OR_APACHE > 1.10 | data) = %.3f\n", prob_or_gt_110))
cat("\nInterpretation: There is a", round(prob_or_gt_110 * 100, 1),
    "% posterior probability\n")
cat("that each unit increase in APACHE score increases the odds of\n")
cat("28-day mortality by more than 10%. This is a direct probability\n")
cat("statement about the parameter -- something only Bayesian inference\n")
cat("can provide.\n")
```

D.10.1.2. Python

```
# Chapter 14, Exercise 1: Bayesian Logistic Regression for ICU Mortality
# 500 ICU admissions with age, APACHE II, ventilation status, 28-day mortality

import numpy as np
import pandas as pd
import pymc as pm
import arviz as az
from scipy.special import expit

# ---- (a) Simulate dataset ----
np.random.seed(101)
n = 500
```

```
icu_data = pd.DataFrame({
    'age': np.round(np.random.normal(62, 15, n)),
    'apache': np.round(np.random.normal(18, 7, n)),
    'ventilated': np.random.binomial(1, 0.35, n)
})

# True model
lp = (-4.5 + 0.02 * icu_data['age'] +
      0.12 * icu_data['apache'] +
      0.8 * icu_data['ventilated'])
icu_data['mortality'] = np.random.binomial(1, expit(lp))

print("Dataset summary:")
print(f"  N: {n}")
print(f"  Mortality rate: {icu_data['mortality'].mean():.3f}")
print(f"  Mean age: {icu_data['age'].mean():.1f}")
print(f"  Mean APACHE: {icu_data['apache'].mean():.1f}")
print(f"  % Ventilated: {icu_data['ventilated'].mean()*100:.1f}%")

# ---- (b) Fit Bayesian logistic regression ----
with pm.Model() as icu_model:
    # Weakly informative priors
    intercept = pm.Normal("Intercept", mu=0, sigma=5)
    b_age = pm.Normal("b_age", mu=0, sigma=2.5)
    b_apache = pm.Normal("b_apache", mu=0, sigma=2.5)
    b_vent = pm.Normal("b_ventilated", mu=0, sigma=2.5)

    # Linear predictor
    logit_p = (intercept +
               b_age * icu_data['age'].values +
               b_apache * icu_data['apache'].values +
               b_vent * icu_data['ventilated'].values)
```

```
    # Likelihood
    y_obs = pm.Bernoulli("mortality", logit_p=logit_p,
                         observed=icu_data['mortality'].values)

    # Sample
    trace = pm.sample(1000, tune=1000, chains=4, random_seed=42,
                      progressbar=True)

print("\nModel summary:")
print(az.summary(trace, var_names=["Intercept", "b_age", "b_apache",
                                     "b_ventilated"]))

# ---- (c) Prior predictive check ----
print("\n=== Part (c): Prior Predictive Check ===")

# Simulate mortality rates from the priors
np.random.seed(42)
n_sim = 2000
intercepts = np.random.normal(0, 5, n_sim)
b_ages = np.random.normal(0, 2.5, n_sim)
b_apaches = np.random.normal(0, 2.5, n_sim)
b_vents = np.random.normal(0, 2.5, n_sim)

# For a "typical" patient: age=62, apache=18, ventilated=0
sim_lp = intercepts + b_ages * 62 + b_apaches * 18 + b_vents * 0
sim_mort = expit(sim_lp)

print(f"Prior predictive mortality rates (typical patient):")
print(f"  Mean: {sim_mort.mean():.3f}")
print(f"  SD: {sim_mort.std():.3f}")
print(f"  Range: {sim_mort.min():.3f} to {sim_mort.max():.3f}")
print("The priors allow mortality rates from near 0 to near 1,")
print("covering all plausible ICU mortality rates. The priors are")
```

```
print("appropriately weakly informative.")

# ---- (d) Posterior odds ratios with 95% credible intervals ----
print("\n=== Part (d): Posterior Odds Ratios ===")

posterior = az.extract(trace)

for var_name, label in [("b_age", "Age"), ("b_apache", "APACHE"),
                         ("b_ventilated", "Ventilated")]:
    or_vals = np.exp(posterior[var_name].values)
    mean_or = or_vals.mean()
    ci_low = np.percentile(or_vals, 2.5)
    ci_high = np.percentile(or_vals, 97.5)
    print(f"OR {label:>12s}: {mean_or:.3f} [{ci_low:.3f}, {ci_high:.3f}]")

# ---- (e) P(OR_APACHE > 1.10 | data) ----
print("\n=== Part (e): P(OR_APACHE > 1.10 | data) ===")

or_apache = np.exp(posterior["b_apache"].values)
prob_gt_110 = (or_apache > 1.10).mean()

print(f"P(OR_APACHE > 1.10 | data) = {prob_gt_110:.3f}")
print(f"\nInterpretation: There is a {prob_gt_110*100:.1f}% posterior probabi
print("that each unit increase in APACHE score increases the odds of")
print("28-day mortality by more than 10%. This is a direct probability")
print("statement about the parameter -- something only Bayesian inference")
print("can provide.")
```

D.10.2. Exercise 2

D.10.2.1. R

```
# Chapter 14, Exercise 2: Hierarchical Model for Multi-Site Drug Trial
# 12 hospitals, LDL cholesterol change, new statin vs standard care

library(brms)
library(tidyverse)

# ---- (a) Simulate data ----
set.seed(789)
n_hospitals <- 12
patients_per_hospital <- c(20, 25, 30, 40, 50, 60, 70, 80, 100, 120, 150, 200)
true_grand_effect <- -25  # mg/dL reduction in LDL
tau_true <- 5             # between-hospital SD

hospital_effects <- rnorm(n_hospitals, 0, tau_true)

trial_data <- map2_dfr(1:n_hospitals, patients_per_hospital, function(j, nj) {
  tibble(
    hospital = factor(j),
    treatment = rep(0:1, length.out = nj),
    ldl_change = (true_grand_effect + hospital_effects[j]) * treatment +
      rnorm(nj, 0, 20)  # residual SD = 20 mg/dL
  )
})

cat("Data summary:\n")
cat("  Total patients:", nrow(trial_data), "\n")
cat("  Hospitals:", n_hospitals, "\n")
cat("  Patients per hospital:", paste(patients_per_hospital, collapse = ", "), "\n")
cat("  True grand effect:", true_grand_effect, "mg/dL\n")
```

```r
cat("  True between-hospital SD:", tau_true, "mg/dL\n")

# ---- (b) Fit Bayesian hierarchical model ----
fit_hier <- brm(
  ldl_change ~ treatment + (treatment | hospital),
  data = trial_data,
  prior = c(
    prior(normal(0, 30), class = "b"),
    prior(normal(0, 30), class = "Intercept"),
    prior(cauchy(0, 5), class = "sd"),
    prior(exponential(0.05), class = "sigma")
  ),
  chains = 4,
  iter = 2000,
  warmup = 1000,
  seed = 42,
  silent = 2,
  refresh = 0,
  control = list(adapt_delta = 0.95)
)

cat("\nHierarchical model summary:\n")
summary(fit_hier)

# Grand mean treatment effect
grand_mean <- fixef(fit_hier)["treatment", "Estimate"]
cat("\nGrand mean treatment effect:", round(grand_mean, 1), "mg/dL\n")

# ---- (c) Shrinkage plot ----
# Extract hospital-specific treatment effects (partial pooling)
ranefs <- ranef(fit_hier)$hospital[, , "treatment"]
partial_pool <- grand_mean + ranefs[, "Estimate"]
```

```
# No-pooling estimates (separate OLS per hospital)
no_pool <- trial_data %>%
  group_by(hospital) %>%
  summarise(
    no_pool_est = coef(lm(ldl_change ~ treatment))[2],
    n = n(),
    .groups = "drop"
  ) %>%
  mutate(
    hospital_num = as.numeric(hospital),
    partial_pool_est = partial_pool
  )

# Plot shrinkage
ggplot(no_pool, aes(y = reorder(hospital, n))) +
  geom_point(aes(x = no_pool_est, colour = "No pooling"), size = 3) +
  geom_point(aes(x = partial_pool_est, colour = "Partial pooling"), size = 3) +
  geom_vline(xintercept = grand_mean, linetype = "dashed", colour = "grey40") +
  geom_segment(aes(x = no_pool_est, xend = partial_pool_est,
                   yend = reorder(hospital, n)),
               arrow = arrow(length = unit(0.15, "cm")),
               colour = "grey60") +
  annotate("text", x = grand_mean + 1, y = 12.5,
           label = paste0("Grand mean = ", round(grand_mean, 1)),
           hjust = 0, size = 3.5) +
  scale_colour_manual(values = c("No pooling" = "steelblue",
                                 "Partial pooling" = "firebrick")) +
  labs(x = "Treatment Effect (mg/dL change in LDL)",
       y = "Hospital (ordered by sample size)",
       colour = "Estimate Type",
       title = "Shrinkage in Hierarchical Model",
       subtitle = "Arrows show partial pooling pulling estimates toward the grand mean") +
  theme_minimal(base_size = 13) +
```

```
  theme(legend.position = "bottom")

# ---- (d) Compare hierarchical to separate OLS regressions ----
cat("\n=== Part (d): Comparison ===\n\n")
cat("Hospital  |  N   | No-Pooling | Partial Pooling | Shrinkage\n")
cat("----------|------|------------|-----------------|----------\n")

for (i in 1:nrow(no_pool)) {
  shrinkage <- abs(no_pool$no_pool_est[i] - no_pool$partial_pool_est[i])
  cat(sprintf("   %2d     | %3d  |   %6.1f   |     %6.1f      |  %5.1f\n",
              no_pool$hospital_num[i], no_pool$n[i],
              no_pool$no_pool_est[i], no_pool$partial_pool_est[i], shrinkage)
}

cat("\nThe hierarchical model is MORE APPROPRIATE because:\n")
cat("1. Small hospitals (n=20-30) have noisy OLS estimates that are\n")
cat("   shrunk toward the grand mean, reducing estimation error.\n")
cat("2. Large hospitals (n=150-200) retain their individual estimates\n")
cat("   since their data are informative enough.\n")
cat("3. The between-hospital SD (tau) is estimated from the data,\n")
cat("   quantifying the degree of heterogeneity across sites.\n")
cat("4. Hospital-by-hospital OLS ignores the shared structure --\n")
cat("   all hospitals are studying the same drug. The hierarchical\n")
cat("   model borrows strength across sites while allowing for\n")
cat("   genuine between-site variation.\n")
```

D.10.2.2. Python

```
# Chapter 14, Exercise 2: Hierarchical Model for Multi-Site Drug Trial
# 12 hospitals, LDL cholesterol change, new statin vs standard care

import numpy as np
```

```python
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# ---- (a) Simulate data ----
np.random.seed(789)
n_hospitals = 12
patients = [20, 25, 30, 40, 50, 60, 70, 80, 100, 120, 150, 200]
grand_effect = -25  # mg/dL
tau = 5             # between-hospital SD

hospital_effects = np.random.normal(0, tau, n_hospitals)

rows = []
for j in range(n_hospitals):
    nj = patients[j]
    trt = np.tile([0, 1], nj // 2 + 1)[:nj]
    ldl_change = ((grand_effect + hospital_effects[j]) * trt +
                  np.random.normal(0, 20, nj))
    for i in range(nj):
        rows.append({
            'hospital': j,
            'treatment': trt[i],
            'ldl_change': ldl_change[i]
        })

trial_data = pd.DataFrame(rows)

print("Data summary:")
print(f"  Total patients: {len(trial_data)}")
print(f"  Hospitals: {n_hospitals}")
print(f"  True grand effect: {grand_effect} mg/dL")
print(f"  True between-hospital SD: {tau} mg/dL")
```

```
# ---- No-pooling estimates (OLS per hospital) ----
no_pool = []
for j in range(n_hospitals):
    df_j = trial_data[trial_data['hospital'] == j]
    trt_mean = df_j[df_j['treatment'] == 1]['ldl_change'].mean()
    ctl_mean = df_j[df_j['treatment'] == 0]['ldl_change'].mean()
    no_pool.append(trt_mean - ctl_mean)

no_pool = np.array(no_pool)

# ---- (b) & (c) Partial pooling approximation ----
# For a full Bayesian fit, use PyMC. Here we approximate shrinkage
# to demonstrate the concept without requiring MCMC sampling.

# Estimate within-hospital variance from data
within_var = []
for j in range(n_hospitals):
    df_j = trial_data[trial_data['hospital'] == j]
    trt_vals = df_j[df_j['treatment'] == 1]['ldl_change'].values
    ctl_vals = df_j[df_j['treatment'] == 0]['ldl_change'].values
    # Variance of the treatment effect estimate
    se_j = np.sqrt(np.var(trt_vals, ddof=1) / len(trt_vals) +
                   np.var(ctl_vals, ddof=1) / len(ctl_vals))
    within_var.append(se_j**2)

within_var = np.array(within_var)

# Estimate between-hospital variance using method of moments
grand_mean_est = np.mean(no_pool)
tau_est_sq = max(0, np.var(no_pool, ddof=1) - np.mean(within_var))
tau_est = np.sqrt(tau_est_sq)

# Shrinkage factor: B_j = within_var_j / (within_var_j + tau^2)
```

```python
shrinkage = within_var / (within_var + tau_est_sq)
partial_pool = grand_mean_est + (1 - shrinkage) * (no_pool - grand_mean_est)

print(f"\nEstimated grand mean effect: {grand_mean_est:.1f} mg/dL")
print(f"Estimated between-hospital SD: {tau_est:.1f} mg/dL")

# ---- (c) Shrinkage plot ----
fig, ax = plt.subplots(figsize=(10, 6))
order = np.argsort(patients)
y_pos = np.arange(n_hospitals)

for i, idx in enumerate(order):
    ax.annotate("", xy=(partial_pool[idx], i), xytext=(no_pool[idx], i),
                arrowprops=dict(arrowstyle="->", color="grey"))

ax.scatter(no_pool[order], y_pos, color="steelblue", s=60, zorder=2,
           label="No pooling (OLS)")
ax.scatter(partial_pool[order], y_pos, color="firebrick", s=60, zorder=2,
           label="Partial pooling")
ax.axvline(grand_mean_est, linestyle="--", color="grey", linewidth=1)
ax.text(grand_mean_est + 0.5, n_hospitals - 0.5,
        f"Grand mean = {grand_mean_est:.1f}", fontsize=9)

ax.set_yticks(y_pos)
ax.set_yticklabels([f"Hospital {order[i]+1}\n(n={patients[order[i]]})"
                     for i in range(n_hospitals)], fontsize=9)
ax.set_xlabel("Treatment Effect (mg/dL change in LDL)")
ax.set_title("Shrinkage in Hierarchical Model\n"
             "Arrows show estimates pulled toward the grand mean")
ax.legend()
plt.tight_layout()
plt.savefig("ch14_ex2_shrinkage.png", dpi=150)
plt.show()
```

```
# ---- (d) Compare approaches ----
print("\n=== Part (d): Comparison ===\n")
print(f"{'Hospital':>8} | {'N':>4} | {'No-Pool':>10} | {'Partial Pool':>12}
print("-" * 55)
for j in range(n_hospitals):
    shrink_amt = abs(no_pool[j] - partial_pool[j])
    print(f"   {j+1:>2}     | {patients[j]:>3}  | {no_pool[j]:>9.1f}  | {part

print("\nThe hierarchical model is MORE APPROPRIATE because:")
print("1. Small hospitals have noisy OLS estimates that are shrunk")
print("   toward the grand mean, reducing estimation error.")
print("2. Large hospitals retain their individual estimates since")
print("   their data are informative enough.")
print("3. Between-hospital SD (tau) is estimated from data,")
print("   quantifying heterogeneity across sites.")
print("4. Hospital-by-hospital OLS ignores shared structure --")
print("   all hospitals study the same drug. The hierarchical model")
print("   borrows strength across sites while allowing genuine")
print("   between-site variation.")

# NOTE: For a full Bayesian hierarchical model, use PyMC:
#
# import pymc as pm
# with pm.Model() as hier_model:
#     mu_trt = pm.Normal("mu_trt", mu=0, sigma=30)
#     tau = pm.HalfCauchy("tau", beta=5)
#     trt_j = pm.Normal("trt_j", mu=mu_trt, sigma=tau, shape=n_hospitals)
#     sigma = pm.Exponential("sigma", lam=0.05)
#     mu = trt_j[hospital_idx] * treatment
#     y = pm.Normal("y", mu=mu, sigma=sigma, observed=ldl_change)
#     trace = pm.sample(1000, tune=1000, chains=4, random_seed=42)
```

D.10.3. Exercise 3

D.10.3.1. R

```
# Chapter 14, Exercise 3: Prior Sensitivity for Rare Events
# New surgical technique: 0 adverse events in 40 patients

# ---- (a) Compute posterior distributions under three priors ----
cat("=== Part (a): Posteriors Under Three Priors ===\n\n")

y <- 0   # adverse events
n <- 40  # total patients

priors <- list(
  list(name = "Beta(1,1) - Uniform", a = 1, b = 1),
  list(name = "Beta(0.5,0.5) - Jeffreys", a = 0.5, b = 0.5),
  list(name = "Beta(1,9) - Informative (~10%)", a = 1, b = 9)
)

theta <- seq(0, 0.3, length.out = 500)

par(mfrow = c(1, 1))
plot(NULL, xlim = c(0, 0.2), ylim = c(0, 50),
     xlab = "Adverse Event Rate", ylab = "Density",
     main = "Posterior Distributions: 0/40 Adverse Events")

colors <- c("steelblue", "firebrick", "forestgreen")

for (i in seq_along(priors)) {
  p <- priors[[i]]
  a_post <- p$a + y
  b_post <- p$b + n - y
```

```
  post_mean <- a_post / (a_post + b_post)
  ci <- qbeta(c(0.025, 0.975), a_post, b_post)
  upper_95 <- qbeta(0.95, a_post, b_post)

  cat(sprintf("Prior: %s\n", p$name))
  cat(sprintf("  Posterior: Beta(%.1f, %.1f)\n", a_post, b_post))
  cat(sprintf("  Posterior mean: %.4f (%.2f%%)\n", post_mean, post_mean * 100
  cat(sprintf("  95%% credible interval: [%.4f, %.4f]\n", ci[1], ci[2]))
  cat(sprintf("  95%% upper credible bound: %.4f (%.2f%%)\n\n",
              upper_95, upper_95 * 100))

  lines(theta, dbeta(theta, a_post, b_post), col = colors[i], lwd = 2)
}

legend("topright", sapply(priors, function(p) p$name),
       col = colors, lwd = 2, cex = 0.8)

# ---- (b) Summary table ----
cat("\n=== Part (b): Summary Table ===\n\n")
cat("Prior              | Post Mean | 95% Upper Bound\n")
cat("-------------------|-----------|----------------\n")

for (p in priors) {
  a_post <- p$a + y
  b_post <- p$b + n - y
  post_mean <- a_post / (a_post + b_post)
  upper_95 <- qbeta(0.95, a_post, b_post)
  cat(sprintf("%-19s| %7.4f   | %7.4f (%.1f%%)\n",
              p$name, post_mean, upper_95, upper_95 * 100))
}

# ---- (c) Why Bayesian estimates are more useful ----
cat("\n=== Part (c): Why Bayesian Estimates Are More Useful ===\n\n")
```

```r
cat("The frequentist point estimate is 0/40 = 0 (0%).\n\n")

cat("This is PROBLEMATIC for regulatory decision-making because:\n\n")

cat("1. ZERO IS NOT CREDIBLE: Just because no adverse events were\n")
cat("   observed in 40 patients does not mean the true rate is zero.\n")
cat("   The 'rule of 3' (frequentist) gives an upper bound of 3/40 = 7.5%,\n")
cat("   but this is ad hoc and does not provide a full distribution.\n\n")

cat("2. BAYESIAN ESTIMATES ARE HONEST: Each prior gives a non-zero\n")
cat("   estimate of the adverse event rate, which is more realistic.\n")
cat("   Even with the Jeffreys prior (minimal information), the\n")
cat("   posterior mean is about 1.2%, acknowledging that rare events\n")
cat("   can occur even if none were observed.\n\n")

cat("3. UPPER CREDIBLE BOUNDS: Regulators need worst-case estimates.\n")
cat("   The 95% upper credible bound provides a direct probability\n")
cat("   statement: 'There is a 95% probability that the true adverse\n")
cat("   event rate is below X%'. This is exactly what is needed for\n")
cat("   risk-benefit assessments.\n\n")

cat("4. PRIOR INFORMATION IS VALUABLE: If similar surgical procedures\n")
cat("   have known complication rates (~10%), the informative prior\n")
cat("   Beta(1,9) incorporates this, giving a more realistic estimate.\n")
cat("   The frequentist approach ignores all prior knowledge.\n\n")

cat("5. DECISION SUPPORT: The full posterior distribution allows\n")
cat("   calculation of quantities like P(rate < 5% | data), which\n")
cat("   directly supports regulatory decisions.\n")

# Compute P(rate < 5% | data) for each prior
cat("\n  P(rate < 5% | data):\n")
for (p in priors) {
```

```
  a_post <- p$a + y
  b_post <- p$b + n - y
  prob_lt_5 <- pbeta(0.05, a_post, b_post)
  cat(sprintf("    %s: %.3f\n", p$name, prob_lt_5))
}
```

D.10.3.2. Python

```python
# Chapter 14, Exercise 3: Prior Sensitivity for Rare Events
# New surgical technique: 0 adverse events in 40 patients

import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import beta

y = 0   # adverse events
n = 40  # total patients

priors = [
    ("Beta(1,1) - Uniform", 1, 1),
    ("Beta(0.5,0.5) - Jeffreys", 0.5, 0.5),
    ("Beta(1,9) - Informative (~10%)", 1, 9),
]

theta = np.linspace(0, 0.25, 500)
colors = ["steelblue", "firebrick", "forestgreen"]

# ---- (a) Compute posterior distributions ----
print("=== Part (a): Posteriors Under Three Priors ===\n")

fig, ax = plt.subplots(figsize=(8, 5))
```

```
for (name, a0, b0), color in zip(priors, colors):
    a_post = a0 + y
    b_post = b0 + n - y
    post_mean = a_post / (a_post + b_post)
    ci = beta.ppf([0.025, 0.975], a_post, b_post)
    upper_95 = beta.ppf(0.95, a_post, b_post)

    print(f"Prior: {name}")
    print(f"  Posterior: Beta({a_post}, {b_post})")
    print(f"  Posterior mean: {post_mean:.4f} ({post_mean*100:.2f}%)")
    print(f"  95% credible interval: [{ci[0]:.4f}, {ci[1]:.4f}]")
    print(f"  95% upper credible bound: {upper_95:.4f} ({upper_95*100:.2f}%)")
    print()

    ax.plot(theta, beta.pdf(theta, a_post, b_post),
            color=color, lw=2, label=name)

ax.set_xlabel("Adverse Event Rate")
ax.set_ylabel("Density")
ax.set_title("Posterior Distributions: 0/40 Adverse Events")
ax.legend(fontsize=9)
ax.set_xlim(0, 0.2)
plt.tight_layout()
plt.savefig("ch14_ex3_rare_events.png", dpi=150)
plt.show()

# ---- (b) Summary table ----
print("=== Part (b): Summary Table ===\n")
print(f"{'Prior':<30s} | {'Post Mean':>9s} | {'95% Upper Bound':>15s}")
print("-" * 60)

for name, a0, b0 in priors:
    a_post = a0 + y
```

```
    b_post = b0 + n - y
    post_mean = a_post / (a_post + b_post)
    upper_95 = beta.ppf(0.95, a_post, b_post)
    print(f"{name:<30s} | {post_mean:>9.4f} | {upper_95:>9.4f} ({upper_95*10

# ---- (c) Why Bayesian estimates are more useful ----
print("\n=== Part (c): Why Bayesian Estimates Are More Useful ===\n")

print("The frequentist point estimate is 0/40 = 0 (0%).\n")

print("This is PROBLEMATIC for regulatory decision-making because:\n")

print("1. ZERO IS NOT CREDIBLE: Just because no adverse events were")
print("   observed in 40 patients does not mean the true rate is zero.")
print("   The 'rule of 3' gives an upper bound of 3/40 = 7.5%, but")
print("   this is ad hoc and does not provide a full distribution.\n")

print("2. BAYESIAN ESTIMATES ARE HONEST: Each prior gives a non-zero")
print("   estimate, which is more realistic. Even the Jeffreys prior")
print("   yields ~1.2%, acknowledging rare events can occur.\n")

print("3. UPPER CREDIBLE BOUNDS: Regulators need worst-case estimates.")
print("   The 95% upper bound provides a direct probability statement:")
print("   'There is 95% probability the true rate is below X%'.\n")

print("4. PRIOR INFORMATION IS VALUABLE: Known complication rates from")
print("   similar procedures can be formally incorporated. The frequentist")
print("   approach ignores all prior knowledge.\n")

print("5. DECISION SUPPORT: The full posterior enables quantities like")
print("   P(rate < 5% | data), directly supporting regulatory decisions.\n")

print("  P(rate < 5% | data):")
```

```
for name, a0, b0 in priors:
    a_post = a0 + y
    b_post = b0 + n - y
    prob_lt_5 = beta.cdf(0.05, a_post, b_post)
    print(f"    {name}: {prob_lt_5:.3f}")
```

D.11. Chapter 15: Dimensionality Reduction

D.11.1. Exercise 1

D.11.1.1. R

```
# Chapter 15, Exercise 1: PCA on Clinical Lab Data
# Using the simulated metabolic panel data from the chapter

library(tidyverse)
library(MASS)

# ---- Simulate metabolic data (same as chapter) ----
set.seed(42)
n <- 300

Sigma <- matrix(c(
  1.0, 0.7, 0.5, 0.3,-0.2, 0.1, 0.2,-0.1,
  0.7, 1.0, 0.4, 0.2,-0.2, 0.1, 0.1,-0.1,
  0.5, 0.4, 1.0, 0.4,-0.3, 0.1, 0.3,-0.1,
  0.3, 0.2, 0.4, 1.0,-0.5, 0.1, 0.2, 0.0,
 -0.2,-0.2,-0.3,-0.5, 1.0,-0.1,-0.1, 0.2,
  0.1, 0.1, 0.1, 0.1,-0.1, 1.0, 0.1,-0.3,
  0.2, 0.1, 0.3, 0.2,-0.1, 0.1, 1.0,-0.2,
 -0.1,-0.1,-0.1, 0.0, 0.2,-0.3,-0.2, 1.0
```

```
), nrow = 8)

z <- mvrnorm(n, mu = rep(0, 8), Sigma = Sigma)
metabolic <- tibble(
  glucose       = round(z[,1] * 30 + 100),
  hba1c         = round(z[,2] * 1.0 + 5.8, 1),
  triglycerides = round(z[,3] * 50 + 150),
  ldl           = round(z[,4] * 30 + 120),
  hdl           = round(z[,5] * 12 + 55),
  creatinine    = round(z[,6] * 0.3 + 1.0, 2),
  alt           = round(z[,7] * 15 + 30),
  albumin       = round(z[,8] * 0.4 + 4.0, 1)
)

# ---- (a) Perform PCA on standardised data ----
cat("=== Part (a): PCA on Standardised Data ===\n")
pca_fit <- prcomp(metabolic, scale. = TRUE)
cat("PCA completed. Summary:\n")
print(summary(pca_fit))

# ---- (b) Scree plot and number of components ----
cat("\n=== Part (b): Scree Plot and Component Selection ===\n")

var_prop <- pca_fit$sdev^2 / sum(pca_fit$sdev^2)
cum_var  <- cumsum(var_prop)

par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))

# Scree plot
barplot(var_prop, names.arg = paste0("PC", 1:8), col = "steelblue",
        main = "Scree Plot", ylab = "Proportion of Variance",
        xlab = "Component", ylim = c(0, 0.4))
abline(h = 1/8, lty = 2, col = "firebrick")
```

```
text(9, 1/8 + 0.015, "Kaiser threshold (1/p)", col = "firebrick", cex = 0.8)

# Cumulative variance
plot(1:8, cum_var, type = "b", pch = 16, col = "steelblue",
     main = "Cumulative Variance", xlab = "Number of Components",
     ylab = "Cumulative Proportion", ylim = c(0, 1))
abline(h = 0.80, col = "firebrick", lty = 2)
text(6, 0.82, "80% threshold", col = "firebrick", cex = 0.8)

# Kaiser criterion: eigenvalues > 1 (which equals variance > 1/p for scaled data)
eigenvalues <- pca_fit$sdev^2
n_kaiser <- sum(eigenvalues > 1)
n_80 <- which(cum_var >= 0.80)[1]

cat("\nKaiser criterion (eigenvalue > 1):", n_kaiser, "components\n")
cat("80% cumulative variance threshold:", n_80, "components\n")
cat("Eigenvalues:", round(eigenvalues, 3), "\n")
cat("Cumulative variance:", round(cum_var, 3), "\n")

# ---- (c) Loadings of first two PCs ----
cat("\n=== Part (c): Loadings and Interpretation ===\n")

cat("\nPC1 loadings (sorted by absolute value):\n")
pc1_loadings <- pca_fit$rotation[, 1]
print(round(sort(abs(pc1_loadings), decreasing = TRUE), 3))
cat("Signed loadings:\n")
print(round(pc1_loadings[order(abs(pc1_loadings), decreasing = TRUE)], 3))

cat("\nPC2 loadings (sorted by absolute value):\n")
pc2_loadings <- pca_fit$rotation[, 2]
print(round(sort(abs(pc2_loadings), decreasing = TRUE), 3))
cat("Signed loadings:\n")
print(round(pc2_loadings[order(abs(pc2_loadings), decreasing = TRUE)], 3))
```

```
cat("\nClinical interpretation:\n")
cat("PC1: Dominated by glucose, HbA1c, triglycerides (positive) and\n")
cat("     HDL (negative). This is a 'metabolic syndrome' axis.\n")
cat("     Patients scoring high on PC1 tend to have elevated glucose,\n")
cat("     HbA1c, triglycerides and lower HDL.\n\n")
cat("PC2: Dominated by creatinine and albumin (with opposite signs).\n")
cat("     This captures a 'renal/hepatic function' dimension.\n")
cat("     Patients with high creatinine and low albumin (suggesting\n")
cat("     renal impairment or liver dysfunction) score high on PC2.\n")

# ---- (d) Biplot coloured by diabetes status ----
cat("\n=== Part (d): Biplot with Diabetes Status ===\n")

set.seed(99)
metabolic$diabetes <- factor(
  ifelse(metabolic$glucose > 110 & metabolic$hba1c > 6.2,
         "Diabetic", "Non-diabetic")
)

cat("Diabetes prevalence:", mean(metabolic$diabetes == "Diabetic"), "\n")

# Extract PC scores
pc_scores <- as.data.frame(pca_fit$x[, 1:2])
pc_scores$diabetes <- metabolic$diabetes

par(mfrow = c(1, 1))

# Biplot with diabetes colouring
cols <- ifelse(metabolic$diabetes == "Diabetic", "firebrick", "steelblue")
plot(pc_scores$PC1, pc_scores$PC2, col = cols, pch = 16, cex = 0.7,
     xlab = paste0("PC1 (", round(var_prop[1] * 100, 1), "% var)"),
     ylab = paste0("PC2 (", round(var_prop[2] * 100, 1), "% var)"),
     main = "PCA Biplot Coloured by Diabetes Status")
```

```r
# Add loading arrows
loadings <- pca_fit$rotation[, 1:2]
scale_factor <- 3
for (i in 1:nrow(loadings)) {
  arrows(0, 0, loadings[i, 1] * scale_factor, loadings[i, 2] * scale_factor,
         col = "grey30", length = 0.1, lwd = 1.5)
  text(loadings[i, 1] * scale_factor * 1.15,
       loadings[i, 2] * scale_factor * 1.15,
       rownames(loadings)[i], cex = 0.7, col = "grey20")
}

legend("topright", c("Diabetic", "Non-diabetic"),
       col = c("firebrick", "steelblue"), pch = 16, cex = 0.8)

cat("\nDiabetic patients tend to cluster toward higher PC1 values,\n")
cat("which aligns with the 'metabolic syndrome' interpretation.\n")
cat("This confirms that PC1 captures the metabolic dimension\n")
cat("along which diabetic patients differ from non-diabetic patients.\n")
```

D.11.1.2. Python

```python
# Chapter 15, Exercise 1: PCA on Clinical Lab Data
# Using the simulated metabolic panel data from the chapter

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

# ---- Simulate metabolic data (same as chapter) ----
```

```
np.random.seed(42)
n = 300

Sigma = np.array([
    [1.0, 0.7, 0.5, 0.3,-0.2, 0.1, 0.2,-0.1],
    [0.7, 1.0, 0.4, 0.2,-0.2, 0.1, 0.1,-0.1],
    [0.5, 0.4, 1.0, 0.4,-0.3, 0.1, 0.3,-0.1],
    [0.3, 0.2, 0.4, 1.0,-0.5, 0.1, 0.2, 0.0],
    [-0.2,-0.2,-0.3,-0.5, 1.0,-0.1,-0.1, 0.2],
    [0.1, 0.1, 0.1, 0.1,-0.1, 1.0, 0.1,-0.3],
    [0.2, 0.1, 0.3, 0.2,-0.1, 0.1, 1.0,-0.2],
    [-0.1,-0.1,-0.1, 0.0, 0.2,-0.3,-0.2, 1.0]
])

z = np.random.multivariate_normal(np.zeros(8), Sigma, n)
labels = ["glucose", "hba1c", "triglycerides", "ldl",
          "hdl", "creatinine", "alt", "albumin"]

metabolic = pd.DataFrame({
    "glucose": np.round(z[:, 0] * 30 + 100),
    "hba1c": np.round(z[:, 1] * 1.0 + 5.8, 1),
    "triglycerides": np.round(z[:, 2] * 50 + 150),
    "ldl": np.round(z[:, 3] * 30 + 120),
    "hdl": np.round(z[:, 4] * 12 + 55),
    "creatinine": np.round(z[:, 5] * 0.3 + 1.0, 2),
    "alt": np.round(z[:, 6] * 15 + 30),
    "albumin": np.round(z[:, 7] * 0.4 + 4.0, 1)
})

# ---- (a) PCA on standardised data ----
print("=== Part (a): PCA on Standardised Data ===")
X = metabolic.values
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)
pca = PCA()
scores = pca.fit_transform(X_scaled)

print("Explained variance ratio:", np.round(pca.explained_variance_ratio_, 3))

# ---- (b) Scree plot and component selection ----
print("\n=== Part (b): Scree Plot and Component Selection ===")

var_prop = pca.explained_variance_ratio_
cum_var = np.cumsum(var_prop)

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# Scree plot
axes[0].bar(range(1, 9), var_prop, color="steelblue", edgecolor="white")
axes[0].axhline(y=1/8, color="firebrick", linestyle="--", label="Kaiser threshold")
axes[0].set_xlabel("Component")
axes[0].set_ylabel("Proportion of Variance")
axes[0].set_title("Scree Plot")
axes[0].set_xticks(range(1, 9))
axes[0].legend()

# Cumulative variance
axes[1].plot(range(1, 9), cum_var, "o-", color="steelblue", lw=2)
axes[1].axhline(y=0.80, color="firebrick", linestyle="--", label="80% threshold")
axes[1].set_xlabel("Number of Components")
axes[1].set_ylabel("Cumulative Proportion")
axes[1].set_title("Cumulative Variance")
axes[1].set_xticks(range(1, 9))
axes[1].legend()

plt.tight_layout()
```

```
plt.savefig("ch15_ex1_scree.png", dpi=150)
plt.show()

# Kaiser criterion: eigenvalues > 1
eigenvalues = pca.explained_variance_
n_kaiser = np.sum(eigenvalues > 1)
n_80 = np.argmax(cum_var >= 0.80) + 1

print(f"\nKaiser criterion (eigenvalue > 1): {n_kaiser} components")
print(f"80% cumulative variance threshold: {n_80} components")
print(f"Eigenvalues: {np.round(eigenvalues, 3)}")
print(f"Cumulative variance: {np.round(cum_var, 3)}")

# ---- (c) Loadings of first two PCs ----
print("\n=== Part (c): Loadings and Interpretation ===")

for pc_idx in [0, 1]:
    loadings = pca.components_[pc_idx]
    order = np.argsort(np.abs(loadings))[::-1]
    print(f"\nPC{pc_idx+1} loadings (sorted by |loading|):")
    for idx in order:
        print(f"  {labels[idx]:>15s}: {loadings[idx]:+.3f}")

print("\nClinical interpretation:")
print("PC1: Dominated by glucose, HbA1c, triglycerides (positive) and")
print("     HDL (negative). This is a 'metabolic syndrome' axis.")
print("PC2: Dominated by creatinine and albumin (opposite signs).")
print("     This captures a 'renal/hepatic function' dimension.")

# ---- (d) Biplot coloured by diabetes status ----
print("\n=== Part (d): Biplot with Diabetes Status ===")

diabetes = np.where(
```

```
    (metabolic["glucose"] > 110) & (metabolic["hba1c"] > 6.2),
    "Diabetic", "Non-diabetic"
)
print(f"Diabetes prevalence: {(diabetes == 'Diabetic').mean():.3f}")

fig, ax = plt.subplots(figsize=(9, 7))

# Plot scores
for label, color in [("Non-diabetic", "steelblue"), ("Diabetic", "firebrick")]:
    mask = diabetes == label
    ax.scatter(scores[mask, 0], scores[mask, 1], c=color, s=20,
               alpha=0.6, label=label)

# Add loading arrows
loadings_2d = pca.components_[:2].T
scale_factor = 3
for i, lab in enumerate(labels):
    ax.annotate("", xy=(loadings_2d[i, 0]*scale_factor,
                        loadings_2d[i, 1]*scale_factor),
                 xytext=(0, 0),
                 arrowprops=dict(arrowstyle="->", color="grey30", lw=1.5))
    ax.text(loadings_2d[i, 0]*scale_factor*1.15,
             loadings_2d[i, 1]*scale_factor*1.15,
             lab, fontsize=8, color="grey20", ha="center")

ax.set_xlabel(f"PC1 ({var_prop[0]:.1%} var)")
ax.set_ylabel(f"PC2 ({var_prop[1]:.1%} var)")
ax.set_title("PCA Biplot Coloured by Diabetes Status")
ax.axhline(0, color="grey", linewidth=0.5)
ax.axvline(0, color="grey", linewidth=0.5)
ax.legend()
plt.tight_layout()
plt.savefig("ch15_ex1_biplot.png", dpi=150)
```

```
plt.show()

print("\nDiabetic patients tend to cluster toward higher PC1 values,")
print("aligning with the 'metabolic syndrome' interpretation.")
print("PC1 captures the metabolic dimension along which diabetic")
print("patients differ from non-diabetic patients.")
```

D.11.2. Exercise 2

D.11.2.1. R

```
# Chapter 15, Exercise 2: t-SNE Sensitivity to Perplexity
# Using the three-group simulated dataset from the chapter

library(Rtsne)

# ---- Simulate data (same as chapter) ----
set.seed(42)
n_per_group <- 100
p <- 20

group1 <- matrix(rnorm(n_per_group * p, mean = 0, sd = 1), ncol = p)
group2 <- matrix(rnorm(n_per_group * p, mean = 2, sd = 1.2), ncol = p)
group3 <- matrix(rnorm(n_per_group * p, mean = -1, sd = 0.8), ncol = p)
group2[, 1:5] <- group2[, 1:5] + 3
group3[, 10:15] <- group3[, 10:15] - 4

X <- rbind(group1, group2, group3)
labels <- factor(rep(c("Type A", "Type B", "Type C"), each = n_per_group))
cols <- c("steelblue", "firebrick", "forestgreen")

# ---- (a) & (b) Run t-SNE with 4 perplexity values ----
```

```r
perplexities <- c(5, 15, 30, 50)

par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))

for (perp in perplexities) {
  set.seed(42)  # Same seed for comparability
  tsne_result <- Rtsne(X, perplexity = perp, dims = 2,
                        verbose = FALSE, max_iter = 1000)

  plot(tsne_result$Y, col = cols[labels], pch = 16, cex = 0.8,
       main = paste("t-SNE (perplexity =", perp, ")"),
       xlab = "t-SNE 1", ylab = "t-SNE 2")

  if (perp == 5) {
    legend("topright", levels(labels), col = cols, pch = 16, cex = 0.7)
  }
}

# ---- (c) When do groups become clearly separated? ----
cat("=== Part (c): When Are Groups Clearly Separated? ===\n\n")
cat("The three groups become clearly separated at perplexity = 15.\n")
cat("At perplexity = 5, the embedding focuses on very local structure,\n")
cat("which can fragment the groups into smaller sub-clusters.\n")
cat("At perplexity = 15 and above, the groups are well-separated\n")
cat("with clear boundaries between them.\n\n")
cat("Higher perplexities (30, 50) also separate the groups clearly\n")
cat("but with slightly different visual arrangements and more\n")
cat("spread-out clusters.\n")

# ---- (d) Are distances between clusters consistent? ----
cat("\n=== Part (d): Consistency of Inter-Cluster Distances ===\n\n")
cat("NO, the distances between clusters are NOT consistent across\n")
cat("perplexity values. Key observations:\n\n")
```

```
cat("1. The relative positions of the three clusters change across\n")
cat("   perplexity settings. Type A might be nearest to Type C at\n")
cat("   one perplexity but nearest to Type B at another.\n\n")
cat("2. The absolute distances between cluster centres vary\n")
cat("   substantially. At low perplexity, clusters may appear close;\n")
cat("   at high perplexity, they may be farther apart (or vice versa).\n\n")
cat("3. Cluster sizes (spread) also change with perplexity.\n\n")
cat("What this tells us about interpreting t-SNE:\n")
cat("- Distances between clusters in t-SNE are MEANINGLESS.\n")
cat("- t-SNE preserves local neighbourhood structure, not global\n")
cat("  distances.\n")
cat("- You should NEVER conclude that two clusters are 'more similar'\n")
cat("  because they appear closer in a t-SNE plot.\n")
cat("- The only reliable information is WHETHER clusters exist,\n")
cat("  not HOW FAR APART they are.\n")
cat("- Always try multiple perplexity values. If the same clusters\n")
cat("  appear consistently, the structure is likely real.\n")
```

D.11.2.2. Python

```
# Chapter 15, Exercise 2: t-SNE Sensitivity to Perplexity
# Using the three-group simulated dataset from the chapter

import numpy as np
import matplotlib.pyplot as plt
from sklearn.manifold import TSNE

# ---- Simulate data (same as chapter) ----
np.random.seed(42)
n_per = 100
p = 20
```

```
g1 = np.random.normal(0, 1, (n_per, p))
g2 = np.random.normal(2, 1.2, (n_per, p))
g3 = np.random.normal(-1, 0.8, (n_per, p))
g2[:, :5] += 3
g3[:, 10:15] -= 4

X = np.vstack([g1, g2, g3])
labels = np.repeat(["Type A", "Type B", "Type C"], n_per)
colors = {"Type A": "steelblue", "Type B": "firebrick", "Type C": "forestgreen"}

# ---- (a) & (b) Run t-SNE with 4 perplexity values ----
perplexities = [5, 15, 30, 50]

fig, axes = plt.subplots(2, 2, figsize=(12, 10))
axes = axes.flatten()

for ax, perp in zip(axes, perplexities):
    tsne = TSNE(n_components=2, perplexity=perp, random_state=42, max_iter=1000)
    emb = tsne.fit_transform(X)

    for lab in ["Type A", "Type B", "Type C"]:
        mask = labels == lab
        ax.scatter(emb[mask, 0], emb[mask, 1], c=colors[lab],
                   s=15, alpha=0.7, label=lab)

    ax.set_title(f"t-SNE (perplexity = {perp})")
    ax.set_xlabel("t-SNE 1")
    ax.set_ylabel("t-SNE 2")
    ax.legend(fontsize=8)

plt.tight_layout()
plt.savefig("ch15_ex2_tsne_perplexity.png", dpi=150)
plt.show()
```

```
# ---- (c) When do groups become clearly separated? ----
print("=== Part (c): When Are Groups Clearly Separated? ===\n")
print("The three groups become clearly separated at perplexity = 15.")
print("At perplexity = 5, the embedding focuses on very local structure,")
print("which can fragment groups into smaller sub-clusters.")
print("At perplexity = 15 and above, groups are well-separated.")
print("Higher perplexities (30, 50) also separate groups clearly but")
print("with different visual arrangements and more spread-out clusters.")

# ---- (d) Are distances consistent? ----
print("\n=== Part (d): Consistency of Inter-Cluster Distances ===\n")
print("NO, the distances between clusters are NOT consistent across")
print("perplexity values. Key observations:\n")
print("1. Relative positions of the three clusters change across")
print("   perplexity settings.\n")
print("2. Absolute distances between cluster centres vary substantially.\n")
print("3. Cluster sizes (spread) also change with perplexity.\n")
print("What this tells us about interpreting t-SNE:")
print("- Distances between clusters are MEANINGLESS.")
print("- t-SNE preserves local neighbourhood structure, not global distances
print("- NEVER conclude two clusters are 'more similar' because they")
print("  appear closer in a t-SNE plot.")
print("- The only reliable information is WHETHER clusters exist,")
print("  not HOW FAR APART they are.")
print("- Always try multiple perplexity values. If clusters appear")
print("  consistently, the structure is likely real.")
```

D.11.3. Exercise 3

D.11.3.1. R

```
# Chapter 15, Exercise 3: UMAP Parameter Exploration
# Using the three-group simulated dataset from the chapter

library(uwot)

# ---- Simulate data (same as chapter) ----
set.seed(42)
n_per_group <- 100
p <- 20

group1 <- matrix(rnorm(n_per_group * p, mean = 0, sd = 1), ncol = p)
group2 <- matrix(rnorm(n_per_group * p, mean = 2, sd = 1.2), ncol = p)
group3 <- matrix(rnorm(n_per_group * p, mean = -1, sd = 0.8), ncol = p)
group2[, 1:5] <- group2[, 1:5] + 3
group3[, 10:15] <- group3[, 10:15] - 4

X <- rbind(group1, group2, group3)
labels <- factor(rep(c("Type A", "Type B", "Type C"), each = n_per_group))
cols <- c("steelblue", "firebrick", "forestgreen")

# ---- (a) UMAP with varying n_neighbors, min_dist = 0.1 ----
n_neighbors_vals <- c(5, 15, 50, 100)

par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))

for (nn in n_neighbors_vals) {
  set.seed(42)
  umap_result <- umap(X, n_neighbors = nn, min_dist = 0.1, verbose = FALSE)
```

```
  plot(umap_result, col = cols[labels], pch = 16, cex = 0.8,
       main = paste("n_neighbors =", nn, ", min_dist = 0.1"),
       xlab = "UMAP 1", ylab = "UMAP 2")

  if (nn == 5) {
    legend("topright", levels(labels), col = cols, pch = 16, cex = 0.7)
  }
}

# ---- (b) UMAP with varying min_dist, n_neighbors = 15 ----
min_dist_vals <- c(0.0, 0.1, 0.5, 1.0)

par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))

for (md in min_dist_vals) {
  set.seed(42)
  umap_result <- umap(X, n_neighbors = 15, min_dist = md, verbose = FALSE)

  plot(umap_result, col = cols[labels], pch = 16, cex = 0.8,
       main = paste("n_neighbors = 15, min_dist =", md),
       xlab = "UMAP 1", ylab = "UMAP 2")

  if (md == 0.0) {
    legend("topright", levels(labels), col = cols, pch = 16, cex = 0.7)
  }
}

# ---- (c) Already done above via the two panels of plots ----

# ---- (d) Description and recommendation ----
cat("=== Part (d): How Each Parameter Affects Visual Appearance ===\n\n")

cat("n_neighbors (with min_dist = 0.1 fixed):\n")
```

```r
cat("  n_neighbors = 5:   Very tight, fragmented clusters. Each point\n")
cat("    considers only 5 neighbors, emphasising micro-structure.\n")
cat("    May split true clusters into sub-groups.\n")
cat("  n_neighbors = 15:  Well-separated, compact clusters. Good\n")
cat("    balance between local and global structure.\n")
cat("  n_neighbors = 50:  Broader clusters, more connected. Groups\n")
cat("    start to merge slightly as the algorithm considers wider\n")
cat("    neighbourhoods.\n")
cat("  n_neighbors = 100: Even more spread out. Global structure is\n")
cat("    emphasised over local detail. Clusters are less tightly packed.\n\n")

cat("min_dist (with n_neighbors = 15 fixed):\n")
cat("  min_dist = 0.0:  Very tight, dense clusters. Points are packed\n")
cat("    as close together as possible. Maximum visual separation.\n")
cat("  min_dist = 0.1:  Slightly looser clusters. Good default.\n")
cat("  min_dist = 0.5:  More spread-out embedding. Internal cluster\n")
cat("    structure becomes more visible but inter-cluster gaps shrink.\n")
cat("  min_dist = 1.0:  Very spread out, almost uniform. Clusters\n")
cat("    overlap, and the embedding loses much of its structure.\n\n")

cat("Recommendation for this dataset:\n")
cat("  n_neighbors = 15, min_dist = 0.1\n")
cat("  This combination provides well-separated, compact clusters\n")
cat("  that clearly reveal the three-group structure. It is also\n")
cat("  the default in most implementations, and for good reason:\n")
cat("  it balances local detail with global structure effectively.\n")
```

D.11.3.2. Python

```python
# Chapter 15, Exercise 3: UMAP Parameter Exploration
# Using the three-group simulated dataset from the chapter
```

```
import numpy as np
import matplotlib.pyplot as plt
from umap import UMAP

# ---- Simulate data (same as chapter) ----
np.random.seed(42)
n_per = 100
p = 20

g1 = np.random.normal(0, 1, (n_per, p))
g2 = np.random.normal(2, 1.2, (n_per, p))
g3 = np.random.normal(-1, 0.8, (n_per, p))
g2[:, :5] += 3
g3[:, 10:15] -= 4

X = np.vstack([g1, g2, g3])
labels = np.repeat(["Type A", "Type B", "Type C"], n_per)
colors = {"Type A": "steelblue", "Type B": "firebrick", "Type C": "forestgre

# ---- (a) UMAP with varying n_neighbors, min_dist = 0.1 ----
n_neighbors_vals = [5, 15, 50, 100]

fig, axes = plt.subplots(2, 2, figsize=(12, 10))
axes = axes.flatten()

for ax, nn in zip(axes, n_neighbors_vals):
    umap_emb = UMAP(n_components=2, n_neighbors=nn, min_dist=0.1,
                    random_state=42).fit_transform(X)
    for lab in ["Type A", "Type B", "Type C"]:
        mask = labels == lab
        ax.scatter(umap_emb[mask, 0], umap_emb[mask, 1], c=colors[lab],
                   s=15, alpha=0.7, label=lab)
    ax.set_title(f"n_neighbors = {nn}, min_dist = 0.1")
```

```
    ax.set_xlabel("UMAP 1")
    ax.set_ylabel("UMAP 2")
    ax.legend(fontsize=8)

plt.suptitle("UMAP: Varying n_neighbors", fontsize=14, y=1.02)
plt.tight_layout()
plt.savefig("ch15_ex3_umap_neighbors.png", dpi=150)
plt.show()

# ---- (b) UMAP with varying min_dist, n_neighbors = 15 ----
min_dist_vals = [0.0, 0.1, 0.5, 1.0]

fig, axes = plt.subplots(2, 2, figsize=(12, 10))
axes = axes.flatten()

for ax, md in zip(axes, min_dist_vals):
    umap_emb = UMAP(n_components=2, n_neighbors=15, min_dist=md,
                      random_state=42).fit_transform(X)
    for lab in ["Type A", "Type B", "Type C"]:
        mask = labels == lab
        ax.scatter(umap_emb[mask, 0], umap_emb[mask, 1], c=colors[lab],
                   s=15, alpha=0.7, label=lab)
    ax.set_title(f"n_neighbors = 15, min_dist = {md}")
    ax.set_xlabel("UMAP 1")
    ax.set_ylabel("UMAP 2")
    ax.legend(fontsize=8)

plt.suptitle("UMAP: Varying min_dist", fontsize=14, y=1.02)
plt.tight_layout()
plt.savefig("ch15_ex3_umap_mindist.png", dpi=150)
plt.show()

# ---- (d) Description and recommendation ----
```

```
print("=== Part (d): How Each Parameter Affects Visual Appearance ===\n")

print("n_neighbors (with min_dist = 0.1 fixed):")
print("  n_neighbors = 5:   Very tight, fragmented clusters. Emphasises")
print("    micro-structure, may split true clusters.")
print("  n_neighbors = 15:  Well-separated, compact clusters. Good balance")
print("    between local and global structure.")
print("  n_neighbors = 50:  Broader clusters, more connected. Groups")
print("    start to merge as wider neighbourhoods are considered.")
print("  n_neighbors = 100: Spread out. Global structure emphasised.\n")

print("min_dist (with n_neighbors = 15 fixed):")
print("  min_dist = 0.0:  Very tight, dense clusters. Maximum separation.")
print("  min_dist = 0.1:  Slightly looser. Good default.")
print("  min_dist = 0.5:  Spread-out embedding. Internal structure visible")
print("    but inter-cluster gaps shrink.")
print("  min_dist = 1.0:  Very spread out, near-uniform. Clusters overlap.\n"

print("Recommendation for this dataset:")
print("  n_neighbors = 15, min_dist = 0.1")
print("  This provides well-separated, compact clusters that clearly")
print("  reveal the three-group structure. It is the default for good")
print("  reason: it balances local detail with global structure.")
```

D.11.4. Exercise 4

D.11.4.1. R

```
# Chapter 15, Exercise 4: Full Workflow on Simulated Genomic Data
# 1000 patients, 500 gene expression features, 4 cancer subtypes

library(tidyverse)
```

```
library(Rtsne)
library(uwot)

# ---- Simulate data ----
set.seed(42)
n <- 1000
p <- 500

# 4 cancer subtypes with different prevalences (one rare at 5%)
subtype_probs <- c(0.35, 0.30, 0.30, 0.05)
subtype <- sample(1:4, n, replace = TRUE, prob = subtype_probs)

cat("Subtype distribution:\n")
print(table(subtype))

# Generate base gene expression
X <- matrix(rnorm(n * p), ncol = p)

# Add subtype-specific signals in different gene subsets
# Subtype 1: upregulated in genes 1-30
X[subtype == 1, 1:30] <- X[subtype == 1, 1:30] + 2.0

# Subtype 2: upregulated in genes 31-60, downregulated in 61-80
X[subtype == 2, 31:60] <- X[subtype == 2, 31:60] + 2.5
X[subtype == 2, 61:80] <- X[subtype == 2, 61:80] - 1.5

# Subtype 3: upregulated in genes 81-120
X[subtype == 3, 81:120] <- X[subtype == 3, 81:120] + 2.0

# Subtype 4 (rare): strong signal in genes 121-160
X[subtype == 4, 121:160] <- X[subtype == 4, 121:160] + 3.5

# ---- (a) Scale and PCA ----
```

```
cat("\n=== Part (a): PCA ===\n")
X_scaled <- scale(X)
pca_result <- prcomp(X_scaled)

var_prop <- pca_result$sdev^2 / sum(pca_result$sdev^2)
cum_var <- cumsum(var_prop)
n_80 <- which(cum_var >= 0.80)[1]

cat("PCs needed for 80% variance:", n_80, "\n")
cat("First 10 cumulative variance:", round(cum_var[1:10], 3), "\n")

# Scree plot (first 30 components)
par(mfrow = c(1, 1))
barplot(var_prop[1:30], names.arg = 1:30, col = "steelblue",
        main = "Scree Plot (first 30 PCs)",
        xlab = "Component", ylab = "Proportion of Variance")
abline(h = 1/p, col = "firebrick", lty = 2)

# ---- (b) UMAP on first 30 PCs ----
cat("\n=== Part (b): UMAP on First 30 PCs ===\n")
pca_30 <- pca_result$x[, 1:30]

set.seed(42)
umap_result <- umap(pca_30, n_neighbors = 15, min_dist = 0.1, verbose = FALSE

subtype_labels <- factor(subtype, labels = c("Subtype 1", "Subtype 2",
                                             "Subtype 3", "Subtype 4 (rare)"
cols <- c("steelblue", "firebrick", "forestgreen", "goldenrod")

plot_df <- tibble(
  UMAP1 = umap_result[, 1],
  UMAP2 = umap_result[, 2],
  Subtype = subtype_labels
```

```
)

# UMAP plot
par(mfrow = c(1, 1))
plot(umap_result, col = cols[subtype], pch = 16, cex = 0.7,
     main = "UMAP of Simulated Genomic Data (4 Cancer Subtypes)",
     xlab = "UMAP 1", ylab = "UMAP 2")
legend("topright", levels(subtype_labels), col = cols, pch = 16, cex = 0.8)

cat("The four subtypes are visually separable in the UMAP embedding.\n")
cat("Each subtype forms a distinct cluster, reflecting the different\n")
cat("gene expression profiles we simulated.\n")

# ---- (c) t-SNE comparison ----
cat("\n=== Part (c): t-SNE Comparison ===\n")

set.seed(42)
tsne_result <- Rtsne(pca_30, perplexity = 30, dims = 2,
                     verbose = FALSE, max_iter = 1000)

par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))

plot(umap_result, col = cols[subtype], pch = 16, cex = 0.7,
     main = "UMAP (n_neighbors=15)", xlab = "UMAP 1", ylab = "UMAP 2")
legend("topright", levels(subtype_labels), col = cols, pch = 16, cex = 0.6)

plot(tsne_result$Y, col = cols[subtype], pch = 16, cex = 0.7,
     main = "t-SNE (perplexity=30)", xlab = "t-SNE 1", ylab = "t-SNE 2")
legend("topright", levels(subtype_labels), col = cols, pch = 16, cex = 0.6)

cat("Both UMAP and t-SNE successfully separate the four subtypes.\n")
cat("UMAP tends to preserve more global structure (relative distances\n")
cat("between clusters are more meaningful), while t-SNE may produce\n")
```

```r
cat("more compact and well-separated clusters but with unreliable\n")
cat("inter-cluster distances.\n")

# ---- (d) Can the rare subtype be identified? ----
cat("\n=== Part (d): Identifying the Rare Subtype ===\n")

n_rare <- sum(subtype == 4)
cat("Rare subtype (Subtype 4) has", n_rare, "patients (5% of total).\n\n")

cat("YES, the rare subtype can be identified in the UMAP plot.\n")
cat("Despite having only ~50 patients, Subtype 4 forms a distinct\n")
cat("cluster (shown in goldenrod/yellow). This is because:\n")
cat("  1. The signal is strong (effect size = 3.5 in 40 genes)\n")
cat("  2. UMAP preserves local structure well, so even small groups\n")
cat("     remain cohesive\n")
cat("  3. The subtype's gene expression pattern is qualitatively\n")
cat("     different from the other subtypes (different genes)\n\n")
cat("In practice, rare subtypes CAN be identified if:\n")
cat("  - Their molecular profile is sufficiently distinct\n")
cat("  - The sample size is not too small (>20-30 patients)\n")
cat("  - Dimensionality reduction preserves the relevant structure\n")
cat("If the signal were weaker, the rare subtype might merge with\n")
cat("another group or be scattered as outliers.\n")
```

D.11.4.2. Python

```python
# Chapter 15, Exercise 4: Full Workflow on Simulated Genomic Data
# 1000 patients, 500 gene expression features, 4 cancer subtypes

import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE
from umap import UMAP

# ---- Simulate data ----
np.random.seed(42)
n = 1000
p = 500

# 4 cancer subtypes (one rare at 5%)
subtype = np.random.choice(4, n, p=[0.35, 0.30, 0.30, 0.05])
print("Subtype distribution:", {i: (subtype == i).sum() for i in range(4)})

# Generate base gene expression
X = np.random.normal(0, 1, (n, p))

# Add subtype-specific signals
X[subtype == 0, :30] += 2.0       # Subtype 1: genes 0-29
X[subtype == 1, 30:60] += 2.5     # Subtype 2: genes 30-59
X[subtype == 1, 60:80] -= 1.5     # Subtype 2: downregulated genes 60-79
X[subtype == 2, 80:120] += 2.0    # Subtype 3: genes 80-119
X[subtype == 3, 120:160] += 3.5   # Subtype 4 (rare): genes 120-159

# ---- (a) Scale and PCA ----
print("\n=== Part (a): PCA ===")
X_scaled = StandardScaler().fit_transform(X)
pca = PCA()
scores = pca.fit_transform(X_scaled)

cum_var = np.cumsum(pca.explained_variance_ratio_)
n_80 = np.argmax(cum_var >= 0.80) + 1
print(f"PCs needed for 80% variance: {n_80}")
print(f"First 10 cumulative variance: {np.round(cum_var[:10], 3)}")
```

```python
# Scree plot
fig, ax = plt.subplots(figsize=(10, 4))
ax.bar(range(1, 31), pca.explained_variance_ratio_[:30],
       color="steelblue", edgecolor="white")
ax.axhline(y=1/p, color="firebrick", linestyle="--", label=f"1/p = {1/p:.4f}"
ax.set_xlabel("Component")
ax.set_ylabel("Proportion of Variance")
ax.set_title("Scree Plot (first 30 PCs)")
ax.legend()
plt.tight_layout()
plt.savefig("ch15_ex4_scree.png", dpi=150)
plt.show()

# ---- (b) UMAP on first 30 PCs ----
print("\n=== Part (b): UMAP on First 30 PCs ===")
pca_30 = scores[:, :30]
umap_2d = UMAP(n_components=2, n_neighbors=15, min_dist=0.1,
                random_state=42).fit_transform(pca_30)

subtype_names = ["Subtype 1", "Subtype 2", "Subtype 3", "Subtype 4 (rare)"]
colors_4 = ["steelblue", "firebrick", "forestgreen", "goldenrod"]

fig, ax = plt.subplots(figsize=(8, 6))
for s in range(4):
    mask = subtype == s
    ax.scatter(umap_2d[mask, 0], umap_2d[mask, 1], c=colors_4[s],
               s=12, alpha=0.6, label=subtype_names[s])
ax.set_xlabel("UMAP 1")
ax.set_ylabel("UMAP 2")
ax.set_title("UMAP of Simulated Genomic Data (4 Cancer Subtypes)")
ax.legend()
plt.tight_layout()
plt.savefig("ch15_ex4_umap.png", dpi=150)
```

```
plt.show()

print("The four subtypes are visually separable in the UMAP embedding.")

# ---- (c) t-SNE comparison ----
print("\n=== Part (c): t-SNE Comparison ===")
tsne_2d = TSNE(n_components=2, perplexity=30, random_state=42,
                max_iter=1000).fit_transform(pca_30)

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

for s in range(4):
    mask = subtype == s
    axes[0].scatter(umap_2d[mask, 0], umap_2d[mask, 1], c=colors_4[s],
                    s=12, alpha=0.6, label=subtype_names[s])
    axes[1].scatter(tsne_2d[mask, 0], tsne_2d[mask, 1], c=colors_4[s],
                    s=12, alpha=0.6, label=subtype_names[s])

axes[0].set_title("UMAP (n_neighbors=15)")
axes[0].set_xlabel("UMAP 1")
axes[0].set_ylabel("UMAP 2")
axes[0].legend(fontsize=8)

axes[1].set_title("t-SNE (perplexity=30)")
axes[1].set_xlabel("t-SNE 1")
axes[1].set_ylabel("t-SNE 2")
axes[1].legend(fontsize=8)

plt.tight_layout()
plt.savefig("ch15_ex4_comparison.png", dpi=150)
plt.show()

print("Both UMAP and t-SNE successfully separate the four subtypes.")
```

```
print("UMAP preserves more global structure; t-SNE may produce more")
print("compact clusters but with unreliable inter-cluster distances.")

# ---- (d) Identifying the rare subtype ----
print("\n=== Part (d): Identifying the Rare Subtype ===")
n_rare = (subtype == 3).sum()
print(f"Rare subtype (Subtype 4) has {n_rare} patients (5% of total).\n")

print("YES, the rare subtype can be identified in the UMAP plot.")
print("Despite having only ~50 patients, Subtype 4 forms a distinct")
print("cluster (shown in goldenrod). This is because:")
print("  1. The signal is strong (effect size = 3.5 in 40 genes)")
print("  2. UMAP preserves local structure, so small groups remain cohesive")
print("  3. The gene expression pattern is qualitatively different\n")
print("In practice, rare subtypes CAN be identified if:")
print("  - Their molecular profile is sufficiently distinct")
print("  - Sample size is not too small (>20-30 patients)")
print("  - Dimensionality reduction preserves the relevant structure")
print("If the signal were weaker, the rare subtype might merge with")
print("another group or scatter as outliers.")
```

D.12. Chapter 16: Clustering

D.12.1. Exercise 1

D.12.1.1. R

```
# Chapter 16, Exercise 1: K-Means on Simulated Patient Data
# 600 patients, 6 clinical variables, 3 underlying phenotypes

library(tidyverse)
```

```r
library(cluster)

# ---- Simulate data ----
set.seed(42)
n <- 600

# Phenotype 1: Septic shock (high HR, RR, lactate, low SBP)
p1 <- tibble(
  hr = rnorm(200, 115, 12), rr = rnorm(200, 28, 5),
  temp = rnorm(200, 38.5, 0.8), sbp = rnorm(200, 80, 12),
  creat = rnorm(200, 2.5, 0.8), lactate = rnorm(200, 5, 2)
)

# Phenotype 2: Stable critical (moderate vitals)
p2 <- tibble(
  hr = rnorm(200, 90, 10), rr = rnorm(200, 20, 3),
  temp = rnorm(200, 37.2, 0.5), sbp = rnorm(200, 110, 15),
  creat = rnorm(200, 1.2, 0.3), lactate = rnorm(200, 1.5, 0.5)
)

# Phenotype 3: Febrile, preserved hemodynamics
p3 <- tibble(
  hr = rnorm(200, 100, 10), rr = rnorm(200, 22, 4),
  temp = rnorm(200, 39.2, 0.7), sbp = rnorm(200, 120, 10),
  creat = rnorm(200, 1.0, 0.2), lactate = rnorm(200, 2.0, 0.8)
)

dat <- bind_rows(p1, p2, p3)
true_labels <- rep(1:3, each = 200)

# ---- (a) Scale and apply K-means for k = 2 to 6 ----
dat_scaled <- scale(dat)
```

```
wcss <- numeric(6)
sil_avg <- numeric(6)

for (k in 2:6) {
  km <- kmeans(dat_scaled, centers = k, nstart = 25)
  wcss[k] <- km$tot.withinss
  sil_avg[k] <- mean(silhouette(km$cluster, dist(dat_scaled))[, 3])
}

# ---- (b) Elbow and silhouette plots ----
par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))

plot(2:6, wcss[2:6], type = "b", pch = 16, col = "steelblue",
     xlab = "Number of clusters (k)", ylab = "Within-cluster SS",
     main = "Elbow Method")

plot(2:6, sil_avg[2:6], type = "b", pch = 16, col = "firebrick",
     xlab = "Number of clusters (k)", ylab = "Average Silhouette",
     main = "Silhouette Scores")

cat("=== Part (b): Optimal k ===\n")
cat("Elbow plot: The elbow appears at k = 3, where WCSS decreases\n")
cat("  sharply and then flattens.\n")
cat("Silhouette: Maximum average silhouette at k =", which.max(sil_avg[2:6])
cat("Both methods suggest k = 3, consistent with the 3 simulated phenotypes.

# ---- (c) Visualise k=3 solution using PCA ----
km3 <- kmeans(dat_scaled, centers = 3, nstart = 25)
pca2 <- prcomp(dat_scaled)$x[, 1:2]

par(mfrow = c(1, 1))
cols <- c("steelblue", "firebrick", "forestgreen")
plot(pca2, col = cols[km3$cluster], pch = 16, cex = 0.7,
```

```
     main = "K-means (k=3) on First 2 PCs",
     xlab = "PC1", ylab = "PC2")
legend("topright", paste("Cluster", 1:3), col = cols, pch = 16, cex = 0.8)

# ---- (d) Profile the clusters ----
cat("\n=== Part (d): Cluster Profiles ===\n\n")

dat$cluster <- km3$cluster
profiles <- dat %>%
  group_by(cluster) %>%
  summarise(across(hr:lactate, mean), n = n(), .groups = "drop")

print(profiles)

cat("\nClinical interpretation of cluster profiles:\n\n")

# Identify which cluster matches which phenotype
for (cl in 1:3) {
  prof <- profiles %>% filter(cluster == cl)
  cat(sprintf("Cluster %d (n = %d):\n", cl, prof$n))
  cat(sprintf("  HR=%.0f, RR=%.0f, Temp=%.1f, SBP=%.0f, Creat=%.1f, Lactate=%.1f\n",
              prof$hr, prof$rr, prof$temp, prof$sbp, prof$creat, prof$lactate))

  if (prof$lactate > 3 && prof$sbp < 90) {
    cat("  -> Matches SEPTIC SHOCK phenotype: high HR, RR, lactate; low SBP\n")
  } else if (prof$temp > 38.5) {
    cat("  -> Matches FEBRILE phenotype: high temp, preserved hemodynamics\n")
  } else {
    cat("  -> Matches STABLE CRITICAL phenotype: moderate vitals\n")
  }
  cat("\n")
}
```

```
cat("The cluster profiles make clinical sense. The algorithm has\n")
cat("successfully recovered the three simulated phenotypes, each\n")
cat("with a distinct clinical profile that a clinician would recognise.\n")
```

D.12.1.2. Python

```
# Chapter 16, Exercise 1: K-Means on Simulated Patient Data
# 600 patients, 6 clinical variables, 3 underlying phenotypes

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.decomposition import PCA

# ---- Simulate data ----
np.random.seed(42)

# Phenotype 1: Septic shock
p1 = np.column_stack([
    np.random.normal(115, 12, 200),   # hr
    np.random.normal(28, 5, 200),     # rr
    np.random.normal(38.5, 0.8, 200), # temp
    np.random.normal(80, 12, 200),    # sbp
    np.random.normal(2.5, 0.8, 200),  # creat
    np.random.normal(5, 2, 200)       # lactate
])

# Phenotype 2: Stable critical
p2 = np.column_stack([
```

```
    np.random.normal(90, 10, 200),
    np.random.normal(20, 3, 200),
    np.random.normal(37.2, 0.5, 200),
    np.random.normal(110, 15, 200),
    np.random.normal(1.2, 0.3, 200),
    np.random.normal(1.5, 0.5, 200)
])

# Phenotype 3: Febrile
p3 = np.column_stack([
    np.random.normal(100, 10, 200),
    np.random.normal(22, 4, 200),
    np.random.normal(39.2, 0.7, 200),
    np.random.normal(120, 10, 200),
    np.random.normal(1.0, 0.2, 200),
    np.random.normal(2.0, 0.8, 200)
])

X = np.vstack([p1, p2, p3])
cols = ['hr', 'rr', 'temp', 'sbp', 'creat', 'lactate']
df = pd.DataFrame(X, columns=cols)

# ---- (a) Scale and K-means for k = 2 to 6 ----
X_scaled = StandardScaler().fit_transform(X)

ks = range(2, 7)
wcss_list = []
sil_list = []

for k in ks:
    km = KMeans(n_clusters=k, n_init=25, random_state=42)
    km.fit(X_scaled)
    wcss_list.append(km.inertia_)
```

```
    sil_list.append(silhouette_score(X_scaled, km.labels_))

# ---- (b) Elbow and silhouette plots ----
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

axes[0].plot(list(ks), wcss_list, "o-", color="steelblue")
axes[0].set_xlabel("Number of clusters (k)")
axes[0].set_ylabel("Within-cluster SS")
axes[0].set_title("Elbow Method")

axes[1].plot(list(ks), sil_list, "o-", color="firebrick")
axes[1].set_xlabel("Number of clusters (k)")
axes[1].set_ylabel("Average Silhouette")
axes[1].set_title("Silhouette Scores")

plt.tight_layout()
plt.savefig("ch16_ex1_elbow_silhouette.png", dpi=150)
plt.show()

best_k = list(ks)[np.argmax(sil_list)]
print(f"=== Part (b): Optimal k ===")
print(f"Elbow: Elbow appears at k = 3")
print(f"Silhouette: Maximum at k = {best_k}")
print(f"Both methods suggest k = 3, matching the 3 simulated phenotypes.")

# ---- (c) Visualise k=3 with PCA ----
km3 = KMeans(n_clusters=3, n_init=25, random_state=42).fit(X_scaled)
pca2 = PCA(n_components=2).fit_transform(X_scaled)

fig, ax = plt.subplots(figsize=(8, 6))
colors_3 = ["steelblue", "firebrick", "forestgreen"]
for c in range(3):
    mask = km3.labels_ == c
```

```python
    ax.scatter(pca2[mask, 0], pca2[mask, 1], c=colors_3[c],
               s=15, alpha=0.6, label=f"Cluster {c+1}")
ax.set_xlabel("PC1")
ax.set_ylabel("PC2")
ax.set_title("K-means (k=3) on First 2 PCs")
ax.legend()
plt.tight_layout()
plt.savefig("ch16_ex1_pca_clusters.png", dpi=150)
plt.show()

# ---- (d) Profile the clusters ----
print("\n=== Part (d): Cluster Profiles ===\n")

df['cluster'] = km3.labels_
profiles = df.groupby('cluster')[cols].mean()
print(profiles.round(1))

print("\nClinical interpretation:")
for cl in range(3):
    prof = profiles.loc[cl]
    n_cl = (km3.labels_ == cl).sum()
    print(f"\nCluster {cl+1} (n = {n_cl}):")
    print(f"  HR={prof['hr']:.0f}, RR={prof['rr']:.0f}, Temp={prof['temp']:.1f}, "
          f"SBP={prof['sbp']:.0f}, Creat={prof['creat']:.1f}, Lactate={prof['lactate']:.1f}'

    if prof['lactate'] > 3 and prof['sbp'] < 90:
        print("  -> SEPTIC SHOCK: high HR, RR, lactate; low SBP")
    elif prof['temp'] > 38.5:
        print("  -> FEBRILE: high temperature, preserved hemodynamics")
    else:
        print("  -> STABLE CRITICAL: moderate vitals")

print("\nThe cluster profiles make clinical sense. The algorithm")
```

```
print("successfully recovered the three simulated phenotypes.")
```

D.12.2. Exercise 2

D.12.2.1. R

```
# Chapter 16, Exercise 2: Hierarchical Clustering with Different Linkage Meth
# Using the same dataset from Exercise 1

library(tidyverse)
library(cluster)
library(mclust)  # for adjustedRandIndex

# ---- Simulate data (same as Exercise 1) ----
set.seed(42)

p1 <- tibble(hr = rnorm(200, 115, 12), rr = rnorm(200, 28, 5),
             temp = rnorm(200, 38.5, 0.8), sbp = rnorm(200, 80, 12),
             creat = rnorm(200, 2.5, 0.8), lactate = rnorm(200, 5, 2))
p2 <- tibble(hr = rnorm(200, 90, 10), rr = rnorm(200, 20, 3),
             temp = rnorm(200, 37.2, 0.5), sbp = rnorm(200, 110, 15),
             creat = rnorm(200, 1.2, 0.3), lactate = rnorm(200, 1.5, 0.5))
p3 <- tibble(hr = rnorm(200, 100, 10), rr = rnorm(200, 22, 4),
             temp = rnorm(200, 39.2, 0.7), sbp = rnorm(200, 120, 10),
             creat = rnorm(200, 1.0, 0.2), lactate = rnorm(200, 2.0, 0.8))

dat <- bind_rows(p1, p2, p3)
dat_scaled <- scale(dat)

# K-means reference solution
km3 <- kmeans(dat_scaled, centers = 3, nstart = 25)
```

```r
# ---- (a) Hierarchical clustering with 4 linkage methods ----
cat("=== Part (a): Hierarchical Clustering ===\n\n")

d <- dist(dat_scaled)
methods <- c("single", "complete", "average", "ward.D2")
titles <- c("Single", "Complete", "Average", "Ward's")

# Use a subset for readable dendrograms
set.seed(42)
idx <- sample(nrow(dat_scaled), 80)

par(mfrow = c(2, 2), mar = c(2, 3, 3, 1))
for (i in seq_along(methods)) {
  hc <- hclust(dist(dat_scaled[idx, ]), method = methods[i])
  plot(hc, labels = FALSE, main = titles[i], xlab = "", sub = "",
       hang = -1, cex = 0.5)
  rect.hclust(hc, k = 3, border = "firebrick")
}

# ---- (b) Cut each dendrogram at k=3, compare assignments ----
cat("=== Part (b): Cluster Assignments ===\n\n")

hc_clusters <- list()
for (i in seq_along(methods)) {
  hc <- hclust(d, method = methods[i])
  hc_clusters[[methods[i]]] <- cutree(hc, k = 3)
}

# Crosstabulation between methods
cat("Crosstab: Ward's vs Complete linkage:\n")
print(table(Ward = hc_clusters[["ward.D2"]],
            Complete = hc_clusters[["complete"]]))
```

```
cat("\nCrosstab: Ward's vs Single linkage:\n")
print(table(Ward = hc_clusters[["ward.D2"]],
            Single = hc_clusters[["single"]]))

# ---- (c) ARI comparison with K-means ----
cat("\n=== Part (c): Adjusted Rand Index vs K-means ===\n\n")

for (i in seq_along(methods)) {
  ari <- adjustedRandIndex(km3$cluster, hc_clusters[[methods[i]]])
  cat(sprintf("ARI(%s vs K-means): %.3f\n", titles[i], ari))
}

cat("\nWard's method produces clusters most similar to K-means (highest ARI)
cat("This is expected because both Ward's method and K-means minimise\n")
cat("within-cluster variance (WCSS), so they have similar objectives.\n")

# ---- (d) Recommendation for clinical data ----
cat("\n=== Part (d): Recommendation ===\n\n")

cat("Ward's method is the recommended default for clinical data because:\n\n
cat("1. OBJECTIVE: Ward's minimises within-cluster variance, which\n")
cat("   aligns with the goal of finding compact, homogeneous clusters.\n\n")
cat("2. CONSISTENCY: It produces results most similar to K-means,\n")
cat("   which is the most widely used method. Using both and checking\n")
cat("   for agreement strengthens confidence in the results.\n\n")
cat("3. ROBUSTNESS: Complete linkage can be distorted by outliers;\n")
cat("   single linkage creates chaining effects (long, straggling\n")
cat("   clusters). Ward's avoids both issues.\n\n")
cat("4. COMPACT CLUSTERS: Clinical phenotypes are typically expected\n")
cat("   to be compact groups of similar patients, which Ward's\n")
cat("   naturally produces.\n\n")
cat("5. INTERPRETABILITY: The dendrogram from Ward's method is\n")
cat("   usually the easiest to interpret, with clear height gaps\n")
```

```
cat("   indicating natural cluster boundaries.\n\n")
cat("CAVEAT: If you suspect non-spherical clusters (e.g., disease\n")
cat("trajectories that form elongated shapes), average linkage\n")
cat("might be more appropriate. But for most clinical phenotyping\n")
cat("applications, Ward's is the safe default.\n")
```

D.12.2.2. Python

```
# Chapter 16, Exercise 2: Hierarchical Clustering with Different Linkage Methods
# Using the same dataset from Exercise 1

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.metrics import adjusted_rand_score
from scipy.cluster.hierarchy import dendrogram, linkage, fcluster

# ---- Simulate data (same as Exercise 1) ----
np.random.seed(42)

p1 = np.column_stack([
    np.random.normal(115, 12, 200), np.random.normal(28, 5, 200),
    np.random.normal(38.5, 0.8, 200), np.random.normal(80, 12, 200),
    np.random.normal(2.5, 0.8, 200), np.random.normal(5, 2, 200)
])
p2 = np.column_stack([
    np.random.normal(90, 10, 200), np.random.normal(20, 3, 200),
    np.random.normal(37.2, 0.5, 200), np.random.normal(110, 15, 200),
    np.random.normal(1.2, 0.3, 200), np.random.normal(1.5, 0.5, 200)
])
```

```
p3 = np.column_stack([
    np.random.normal(100, 10, 200), np.random.normal(22, 4, 200),
    np.random.normal(39.2, 0.7, 200), np.random.normal(120, 10, 200),
    np.random.normal(1.0, 0.2, 200), np.random.normal(2.0, 0.8, 200)
])

X = np.vstack([p1, p2, p3])
X_scaled = StandardScaler().fit_transform(X)

# K-means reference
km3 = KMeans(n_clusters=3, n_init=25, random_state=42).fit(X_scaled)

# ---- (a) Hierarchical clustering with 4 linkage methods ----
print("=== Part (a): Hierarchical Clustering ===\n")

methods = ['single', 'complete', 'average', 'ward']
titles = ['Single', 'Complete', 'Average', "Ward's"]

# Dendrograms (use subset for readability)
np.random.seed(42)
idx = np.random.choice(len(X_scaled), 80, replace=False)
X_sub = X_scaled[idx]

fig, axes = plt.subplots(2, 2, figsize=(16, 10))
axes = axes.flatten()

for ax, method, title in zip(axes, methods, titles):
    Z = linkage(X_sub, method=method)
    dendrogram(Z, ax=ax, no_labels=True, color_threshold=0)
    ax.set_title(title)
    ax.set_xlabel("")

plt.suptitle("Hierarchical Clustering: Linkage Method Comparison", fontsize=
```

```
plt.tight_layout()
plt.savefig("ch16_ex2_dendrograms.png", dpi=150)
plt.show()

# ---- (b) Cut at k=3 and compare ----
print("=== Part (b): Cluster Assignments ===\n")

hc_labels = {}
for method in methods:
    Z = linkage(X_scaled, method=method)
    hc_labels[method] = fcluster(Z, t=3, criterion='maxclust')

# Crosstabs
print("Crosstab: Ward's vs Complete:")
ct = pd.crosstab(pd.Series(hc_labels['ward'], name='Ward'),
                  pd.Series(hc_labels['complete'], name='Complete'))
print(ct)

print("\nCrosstab: Ward's vs Single:")
ct2 = pd.crosstab(pd.Series(hc_labels['ward'], name='Ward'),
                   pd.Series(hc_labels['single'], name='Single'))
print(ct2)

# ---- (c) ARI comparison with K-means ----
print("\n=== Part (c): Adjusted Rand Index vs K-means ===\n")

for method, title in zip(methods, titles):
    ari = adjusted_rand_score(km3.labels_, hc_labels[method])
    print(f"ARI({title} vs K-means): {ari:.3f}")

print("\nWard's method produces clusters most similar to K-means (highest ARI).")
print("Both minimise within-cluster variance, so they share similar objectives.")
```

```
# ---- (d) Recommendation ----
print("\n=== Part (d): Recommendation ===\n")

print("Ward's method is recommended for clinical data because:\n")
print("1. OBJECTIVE: Minimises within-cluster variance, producing")
print("   compact, homogeneous clusters.\n")
print("2. CONSISTENCY: Most similar to K-means, strengthening")
print("   confidence when both methods agree.\n")
print("3. ROBUSTNESS: Avoids chaining (single linkage) and outlier")
print("   sensitivity (complete linkage).\n")
print("4. COMPACT CLUSTERS: Clinical phenotypes are typically compact")
print("   groups, which Ward's naturally produces.\n")
print("5. INTERPRETABILITY: Dendrogram with clear height gaps at")
print("   natural cluster boundaries.\n")
print("CAVEAT: For non-spherical clusters, average linkage may be")
print("more appropriate. But Ward's is the safe default for most")
print("clinical phenotyping applications.")
```

D.12.3. Exercise 3

D.12.3.1. R

```
# Chapter 16, Exercise 3: DBSCAN for Outlier Detection
# Add outlier patients to the Exercise 1 dataset

library(tidyverse)
library(cluster)
library(dbscan)

# ---- Simulate data (same as Exercise 1) ----
set.seed(42)
```

```r
p1 <- tibble(hr = rnorm(200, 115, 12), rr = rnorm(200, 28, 5),
             temp = rnorm(200, 38.5, 0.8), sbp = rnorm(200, 80, 12),
             creat = rnorm(200, 2.5, 0.8), lactate = rnorm(200, 5, 2))
p2 <- tibble(hr = rnorm(200, 90, 10), rr = rnorm(200, 20, 3),
             temp = rnorm(200, 37.2, 0.5), sbp = rnorm(200, 110, 15),
             creat = rnorm(200, 1.2, 0.3), lactate = rnorm(200, 1.5, 0.5))
p3 <- tibble(hr = rnorm(200, 100, 10), rr = rnorm(200, 22, 4),
             temp = rnorm(200, 39.2, 0.7), sbp = rnorm(200, 120, 10),
             creat = rnorm(200, 1.0, 0.2), lactate = rnorm(200, 2.0, 0.8))

dat <- bind_rows(p1, p2, p3)
is_outlier <- rep(FALSE, 600)

# Add 30 outlier patients with extreme values
set.seed(99)
outliers <- tibble(
  hr = rnorm(30, 180, 10),        # Extreme heart rate
  rr = rnorm(30, 40, 5),          # Extreme respiratory rate
  temp = rnorm(30, 40.5, 0.5),    # Very high temperature
  sbp = rnorm(30, 50, 10),        # Very low blood pressure
  creat = rnorm(30, 8, 1.5),      # Very high creatinine
  lactate = rnorm(30, 15, 3)      # Very high lactate
)

dat <- bind_rows(dat, outliers)
is_outlier <- c(is_outlier, rep(TRUE, 30))
dat_scaled <- scale(dat)

cat("Total patients:", nrow(dat), "(600 normal + 30 outliers)\n\n")

# ---- (a) K-means with k=3 ----
cat("=== Part (a): K-means with k=3 ===\n")
```

```r
km3 <- kmeans(dat_scaled, centers = 3, nstart = 25)

# Where do outliers end up?
outlier_clusters <- km3$cluster[is_outlier]
cat("Outlier distribution across K-means clusters:\n")
print(table(outlier_clusters))

cat("\nK-means assigns outliers to one of the existing clusters,\n")
cat("typically the cluster whose centroid is nearest (even if far).\n")
cat("This can distort the centroid and corrupt the cluster profiles.\n")

# Show how outliers affect cluster means
dat$km_cluster <- km3$cluster
dat$is_outlier <- is_outlier

for (cl in 1:3) {
  n_outlier_in_cl <- sum(dat$km_cluster == cl & dat$is_outlier)
  n_total_in_cl <- sum(dat$km_cluster == cl)
  cat(sprintf("Cluster %d: %d patients (%d outliers)\n",
              cl, n_total_in_cl, n_outlier_in_cl))
}

# ---- (b) DBSCAN ----
cat("\n=== Part (b): DBSCAN ===\n")

# Use kNNdistplot to help choose eps
kNNdistplot(dat_scaled, k = 5)
abline(h = 2.5, col = "firebrick", lty = 2)

# Run DBSCAN
db <- dbscan(dat_scaled, eps = 2.5, minPts = 10)

cat("DBSCAN results:\n")
```

```r
cat("  Number of clusters:", max(db$cluster), "\n")
cat("  Noise points (outliers):", sum(db$cluster == 0), "\n")
cat("  Actual outliers detected as noise:",
    sum(db$cluster == 0 & is_outlier), "out of 30\n")

cat("\nDBSCAN cluster sizes:\n")
print(table(db$cluster))

# ---- (c) Compare non-outlier assignments ----
cat("\n=== Part (c): Comparison for Non-Outlier Patients ===\n")

# Get cluster assignments for non-outlier patients only
normal_idx <- !is_outlier
km_normal <- km3$cluster[normal_idx]
db_normal <- db$cluster[normal_idx]

# For DBSCAN, remove any normal patients assigned as noise
db_noise_normal <- sum(db_normal == 0)
cat("Normal patients misclassified as noise by DBSCAN:", db_noise_normal, "\n")

# ARI for non-noise, non-outlier patients
both_assigned <- db_normal > 0
if (sum(both_assigned) > 0) {
  # Use mclust for ARI
  library(mclust)
  ari <- adjustedRandIndex(km_normal[both_assigned], db_normal[both_assigned])
  cat("ARI between K-means and DBSCAN (non-outlier, non-noise):", round(ari, 3), "\n")
}

# ---- (d) Which approach for outlier handling? ----
cat("\n=== Part (d): Preferred Approach for Outlier Handling ===\n\n")

cat("DBSCAN is preferred for clinical phenotyping when outliers are expected.\n\n")
```

```r
cat("Reasons:\n")
cat("1. EXPLICIT OUTLIER IDENTIFICATION: DBSCAN labels outliers as noise,\n")
cat("   making them visible for clinical review. K-means silently absorbs\n")
cat("   them into clusters, distorting the results.\n\n")

cat("2. CLINICAL RELEVANCE: Outlier patients (e.g., with extreme vitals)\n")
cat("   may represent data errors, rare presentations, or patients who\n")
cat("   need individual assessment. Identifying them is valuable.\n\n")

cat("3. CLUSTER INTEGRITY: By excluding outliers, DBSCAN preserves the\n")
cat("   purity of the main clusters. K-means cluster centroids can be\n")
cat("   pulled by outliers, producing misleading profiles.\n\n")

cat("4. PRACTICAL WORKFLOW:\n")
cat("   - Use DBSCAN to identify outliers first.\n")
cat("   - Review outliers clinically (data errors? rare cases?).\n")
cat("   - Apply K-means to the remaining non-outlier patients for\n")
cat("     cleaner phenotyping.\n\n")

cat("5. CAVEAT: DBSCAN requires tuning eps and minPts, which can be\n")
cat("   challenging. The kNN distance plot helps, but the choice\n")
cat("   affects how many points are labelled as noise.\n")

# Visualise comparison
pca2 <- prcomp(dat_scaled)$x[, 1:2]

par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))

# K-means
cols_km <- c("steelblue", "firebrick", "forestgreen")[km3$cluster]
pch_km <- ifelse(is_outlier, 4, 16)
plot(pca2, col = cols_km, pch = pch_km, cex = 0.7,
     main = "K-means (k=3)\n(X = outlier patients)",
```

```
     xlab = "PC1", ylab = "PC2")

# DBSCAN
db_cols <- c("grey50", "steelblue", "firebrick", "forestgreen")
cols_db <- db_cols[db$cluster + 1]
pch_db <- ifelse(db$cluster == 0, 4, 16)
plot(pca2, col = cols_db, pch = pch_db, cex = 0.7,
     main = "DBSCAN\n(X/grey = noise)",
     xlab = "PC1", ylab = "PC2")
legend("topright", c("Noise", "Cluster 1", "Cluster 2", "Cluster 3"),
       col = db_cols, pch = c(4, 16, 16, 16), cex = 0.7)
```

D.12.3.2. Python

```
# Chapter 16, Exercise 3: DBSCAN for Outlier Detection
# Add outlier patients to the Exercise 1 dataset

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans, DBSCAN
from sklearn.metrics import adjusted_rand_score
from sklearn.decomposition import PCA
from sklearn.neighbors import NearestNeighbors

# ---- Simulate data (same as Exercise 1) ----
np.random.seed(42)

p1 = np.column_stack([
    np.random.normal(115, 12, 200), np.random.normal(28, 5, 200),
    np.random.normal(38.5, 0.8, 200), np.random.normal(80, 12, 200),
```

```
    np.random.normal(2.5, 0.8, 200), np.random.normal(5, 2, 200)
])
p2 = np.column_stack([
    np.random.normal(90, 10, 200), np.random.normal(20, 3, 200),
    np.random.normal(37.2, 0.5, 200), np.random.normal(110, 15, 200),
    np.random.normal(1.2, 0.3, 200), np.random.normal(1.5, 0.5, 200)
])
p3 = np.column_stack([
    np.random.normal(100, 10, 200), np.random.normal(22, 4, 200),
    np.random.normal(39.2, 0.7, 200), np.random.normal(120, 10, 200),
    np.random.normal(1.0, 0.2, 200), np.random.normal(2.0, 0.8, 200)
])

X_normal = np.vstack([p1, p2, p3])
is_outlier = np.array([False] * 600)

# Add 30 outlier patients
np.random.seed(99)
outliers = np.column_stack([
    np.random.normal(180, 10, 30),     # Extreme HR
    np.random.normal(40, 5, 30),       # Extreme RR
    np.random.normal(40.5, 0.5, 30),   # Very high temp
    np.random.normal(50, 10, 30),      # Very low SBP
    np.random.normal(8, 1.5, 30),      # Very high creatinine
    np.random.normal(15, 3, 30)        # Very high lactate
])

X = np.vstack([X_normal, outliers])
is_outlier = np.concatenate([is_outlier, np.array([True] * 30)])
X_scaled = StandardScaler().fit_transform(X)

print(f"Total patients: {len(X)} (600 normal + 30 outliers)\n")
```

```python
# ---- (a) K-means with k=3 ----
print("=== Part (a): K-means with k=3 ===")

km3 = KMeans(n_clusters=3, n_init=25, random_state=42).fit(X_scaled)

outlier_clusters = km3.labels_[is_outlier]
print("Outlier distribution across K-means clusters:")
for c in range(3):
    n_out = (outlier_clusters == c).sum()
    n_tot = (km3.labels_ == c).sum()
    print(f"  Cluster {c+1}: {n_tot} patients ({n_out} outliers)")

print("\nK-means assigns outliers to existing clusters, distorting centroids.")

# ---- (b) DBSCAN ----
print("\n=== Part (b): DBSCAN ===")

# kNN distance plot to choose eps
nn = NearestNeighbors(n_neighbors=5).fit(X_scaled)
distances, _ = nn.kneighbors(X_scaled)
knn_dist = np.sort(distances[:, -1])

plt.figure(figsize=(8, 4))
plt.plot(knn_dist, color="steelblue")
plt.axhline(y=2.5, color="firebrick", linestyle="--", label="eps = 2.5")
plt.xlabel("Points (sorted)")
plt.ylabel("5-NN Distance")
plt.title("kNN Distance Plot for eps Selection")
plt.legend()
plt.tight_layout()
plt.savefig("ch16_ex3_knn_dist.png", dpi=150)
plt.show()
```

```
# Run DBSCAN
db = DBSCAN(eps=2.5, min_samples=10).fit(X_scaled)

n_clusters = len(set(db.labels_)) - (1 if -1 in db.labels_ else 0)
n_noise = (db.labels_ == -1).sum()
outliers_as_noise = ((db.labels_ == -1) & is_outlier).sum()

print(f"Number of clusters: {n_clusters}")
print(f"Noise points (outliers): {n_noise}")
print(f"Actual outliers detected as noise: {outliers_as_noise} out of 30")

# ---- (c) Compare non-outlier assignments ----
print("\n=== Part (c): Comparison for Non-Outlier Patients ===")

normal_mask = ~is_outlier
km_normal = km3.labels_[normal_mask]
db_normal = db.labels_[normal_mask]

db_noise_normal = (db_normal == -1).sum()
print(f"Normal patients misclassified as noise: {db_noise_normal}")

# ARI for non-noise, non-outlier patients
both_assigned = db_normal >= 0
if both_assigned.sum() > 0:
    ari = adjusted_rand_score(km_normal[both_assigned], db_normal[both_assig
    print(f"ARI (K-means vs DBSCAN, non-outlier, non-noise): {ari:.3f}")

# ---- Visualise comparison ----
pca2 = PCA(n_components=2).fit_transform(X_scaled)

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# K-means
```

```
colors_km = ["steelblue", "firebrick", "forestgreen"]
for c in range(3):
    mask = (km3.labels_ == c) & ~is_outlier
    axes[0].scatter(pca2[mask, 0], pca2[mask, 1], c=colors_km[c], s=12, alpha=0.6)
    mask_out = (km3.labels_ == c) & is_outlier
    axes[0].scatter(pca2[mask_out, 0], pca2[mask_out, 1], c=colors_km[c],
                    marker="x", s=50, linewidths=2)
axes[0].set_title("K-means (k=3)\n(X = outlier patients)")
axes[0].set_xlabel("PC1")
axes[0].set_ylabel("PC2")

# DBSCAN
color_map = {-1: "grey", 0: "steelblue", 1: "firebrick", 2: "forestgreen"}
for c_val in sorted(set(db.labels_)):
    mask = db.labels_ == c_val
    marker = "x" if c_val == -1 else "o"
    label = "Noise" if c_val == -1 else f"Cluster {c_val+1}"
    axes[1].scatter(pca2[mask, 0], pca2[mask, 1],
                    c=color_map.get(c_val, "purple"),
                    marker=marker, s=15 if c_val >= 0 else 40,
                    alpha=0.6, label=label)
axes[1].set_title("DBSCAN\n(X/grey = noise)")
axes[1].set_xlabel("PC1")
axes[1].set_ylabel("PC2")
axes[1].legend(fontsize=8)

plt.tight_layout()
plt.savefig("ch16_ex3_comparison.png", dpi=150)
plt.show()

# ---- (d) Preferred approach ----
print("\n=== Part (d): Preferred Approach ===\n")
print("DBSCAN is preferred for clinical phenotyping when outliers are expected.\n")
```

```
print("1. EXPLICIT OUTLIER IDENTIFICATION: DBSCAN labels outliers as noise,")
print("   making them visible. K-means silently absorbs them.\n")
print("2. CLINICAL RELEVANCE: Outlier patients may represent data errors,")
print("   rare presentations, or patients needing individual assessment.\n")
print("3. CLUSTER INTEGRITY: DBSCAN preserves cluster purity by excluding")
print("   outliers. K-means centroids can be pulled by outliers.\n")
print("4. PRACTICAL WORKFLOW: Use DBSCAN first to identify outliers,")
print("   review them clinically, then apply K-means to non-outliers.\n")
print("5. CAVEAT: DBSCAN requires tuning eps and min_samples, which")
print("   affects how many points are labelled as noise.")
```

D.12.4. Exercise 4

D.12.4.1. R

```r
# Chapter 16, Exercise 4: Full Pipeline - PCA + Clustering + UMAP Visualisat
# 800 patients, 30 clinical variables, 4 underlying subtypes

library(tidyverse)
library(cluster)
library(uwot)

# ---- Simulate data ----
set.seed(42)
n <- 800
p <- 30

# 4 subtypes with different prevalences
subtype_probs <- c(0.30, 0.30, 0.25, 0.15)
true_subtype <- sample(1:4, n, replace = TRUE, prob = subtype_probs)

cat("True subtype distribution:\n")
```

```
print(table(true_subtype))

# Generate base data
X <- matrix(rnorm(n * p), ncol = p)

# Add subtype-specific signals in different feature subsets
for (s in 1:4) {
  feature_start <- (s - 1) * 6 + 1
  feature_end <- min(s * 6, p)
  X[true_subtype == s, feature_start:feature_end] <-
    X[true_subtype == s, feature_start:feature_end] + 2.5
}

# ---- (a) Standardise and PCA ----
cat("\n=== Part (a): PCA ===\n")

X_scaled <- scale(X)
pca_result <- prcomp(X_scaled)

var_prop <- pca_result$sdev^2 / sum(pca_result$sdev^2)
cum_var <- cumsum(var_prop)

# Scree plot
par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
barplot(var_prop[1:15], names.arg = 1:15, col = "steelblue",
        main = "Scree Plot (first 15 PCs)",
        xlab = "Component", ylab = "Proportion of Variance")
abline(h = 1/p, col = "firebrick", lty = 2)

plot(1:15, cum_var[1:15], type = "b", pch = 16, col = "steelblue",
     main = "Cumulative Variance", xlab = "Components",
     ylab = "Cumulative Proportion", ylim = c(0, 1))
abline(h = 0.80, col = "firebrick", lty = 2)
```

```r
n_80 <- which(cum_var >= 0.80)[1]
cat("PCs needed for 80% variance:", n_80, "\n")

# Use first 10 PCs for clustering
n_pcs <- 10
pca_scores <- pca_result$x[, 1:n_pcs]

# ---- (b) K-means on PCA scores ----
cat("\n=== Part (b): K-means on PCA Scores ===\n")

sil_scores <- numeric(5)
for (k in 2:5) {
  km <- kmeans(pca_scores, centers = k, nstart = 50)
  sil_scores[k] <- mean(silhouette(km$cluster, dist(pca_scores))[, 3])
  cat(sprintf("k = %d: Silhouette = %.3f\n", k, sil_scores[k]))
}

best_k <- which.max(sil_scores[2:5]) + 1
cat("Best k by silhouette:", best_k, "\n")

# Fit final clustering
km_final <- kmeans(pca_scores, centers = best_k, nstart = 50)

# ---- (c) UMAP visualisation ----
cat("\n=== Part (c): UMAP Visualisation ===\n")

set.seed(42)
umap_result <- umap(pca_scores, n_neighbors = 15, min_dist = 0.1,
                     verbose = FALSE)

plot_df <- tibble(
  UMAP1 = umap_result[, 1],
  UMAP2 = umap_result[, 2],
```

```
  Cluster = factor(km_final$cluster),
  True_subtype = factor(true_subtype)
)

cols4 <- c("steelblue", "firebrick", "forestgreen", "goldenrod")

par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))

# Discovered clusters
plot(umap_result, col = cols4[km_final$cluster], pch = 16, cex = 0.6,
     main = "K-means Clusters on UMAP", xlab = "UMAP 1", ylab = "UMAP 2")
legend("topright", paste("Cluster", 1:best_k), col = cols4[1:best_k],
       pch = 16, cex = 0.7)

# True subtypes
plot(umap_result, col = cols4[true_subtype], pch = 16, cex = 0.6,
     main = "True Subtypes on UMAP", xlab = "UMAP 1", ylab = "UMAP 2")
legend("topright", paste("Subtype", 1:4), col = cols4, pch = 16, cex = 0.7)

# ---- (d) Cluster profiles ----
cat("\n=== Part (d): Cluster Profiles ===\n\n")

# Name variables for interpretability
var_names <- paste0("V", 1:p)
dat_df <- as.data.frame(X)
colnames(dat_df) <- var_names
dat_df$cluster <- km_final$cluster

# Compute mean of each variable by cluster
profiles <- dat_df %>%
  group_by(cluster) %>%
  summarise(across(everything(), mean), n = n(), .groups = "drop")
```

```
cat("Cluster sizes:\n")
print(table(km_final$cluster))

cat("\nCluster means for key variables (first 24, showing subtype signals):\n

# Show means for the signal variables
signal_vars <- paste0("V", 1:24)
profile_table <- profiles %>%
  select(cluster, n, all_of(signal_vars))

# Print in a compact format
for (cl in 1:best_k) {
  prof <- profile_table %>% filter(cluster == cl)
  cat(sprintf("Cluster %d (n = %d):\n", cl, prof$n))
  for (s in 1:4) {
    start <- (s - 1) * 6 + 1
    end <- s * 6
    vars <- paste0("V", start:end)
    means <- round(unlist(prof[vars]), 2)
    cat(sprintf("  Subtype %d signal vars (V%d-V%d): mean = %.2f\n",
                s, start, end, mean(means)))
  }
  cat("\n")
}

# ---- (e) Discussion ----
cat("=== Part (e): Validation Discussion ===\n\n")

cat("To validate these clusters in a real clinical study:\n\n")

cat("1. EXTERNAL OUTCOMES: Compare clusters on outcomes NOT used in\n")
cat("   clustering (mortality, length of stay, treatment response).\n")
cat("   If clusters predict outcomes they were not trained on, the\n")
```

```
cat("   structure is likely clinically meaningful.\n\n")

cat("2. STABILITY ANALYSIS: Resample the data (bootstrap) and re-run\n")
cat("   clustering. If the same patients consistently end up in the\n")
cat("   same clusters, the results are robust. If clusters change\n")
cat("   substantially across resamples, they may be artefacts.\n\n")

cat("3. INDEPENDENT REPLICATION: Apply the same pipeline to an\n")
cat("   independent dataset (different hospital, different time period).\n")
cat("   If similar clusters emerge, the findings are generalisable.\n\n")

cat("4. CLINICAL EXPERT REVIEW: Present cluster profiles to clinicians.\n")
cat("   Do the clusters correspond to recognisable patient types?\n")
cat("   Do they suggest different management strategies?\n\n")

cat("5. MULTIPLE ALGORITHMS: Compare results from K-means, hierarchical\n")
cat("   clustering, and DBSCAN. Agreement across methods strengthens\n")
cat("   confidence. Disagreement suggests fragile structure.\n\n")

cat("6. AVOID CIRCULAR REASONING: Do NOT validate clusters by showing\n")
cat("   they differ on the same variables used to create them.\n")
cat("   This is guaranteed by construction and proves nothing.\n")
```

D.12.4.2. Python

```
# Chapter 16, Exercise 4: Full Pipeline - PCA + Clustering + UMAP Visualisation
# 800 patients, 30 clinical variables, 4 underlying subtypes

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from umap import UMAP

# ---- Simulate data ----
np.random.seed(42)
n = 800
p = 30

# 4 subtypes
true_subtype = np.random.choice(4, n, p=[0.30, 0.30, 0.25, 0.15])
print("True subtype distribution:", {i: (true_subtype == i).sum() for i in ra

X = np.random.normal(0, 1, (n, p))

# Add subtype-specific signals
for s in range(4):
    start = s * 6
    end = min((s + 1) * 6, p)
    X[true_subtype == s, start:end] += 2.5

# ---- (a) Standardise and PCA ----
print("\n=== Part (a): PCA ===")

X_scaled = StandardScaler().fit_transform(X)
pca = PCA()
scores = pca.fit_transform(X_scaled)

cum_var = np.cumsum(pca.explained_variance_ratio_)
n_80 = np.argmax(cum_var >= 0.80) + 1
print(f"PCs needed for 80% variance: {n_80}")
```

```
# Scree plot
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

axes[0].bar(range(1, 16), pca.explained_variance_ratio_[:15],
            color="steelblue", edgecolor="white")
axes[0].axhline(y=1/p, color="firebrick", linestyle="--", label="1/p")
axes[0].set_xlabel("Component")
axes[0].set_ylabel("Proportion of Variance")
axes[0].set_title("Scree Plot")
axes[0].legend()

axes[1].plot(range(1, 16), cum_var[:15], "o-", color="steelblue")
axes[1].axhline(y=0.80, color="firebrick", linestyle="--", label="80%")
axes[1].set_xlabel("Components")
axes[1].set_ylabel("Cumulative Proportion")
axes[1].set_title("Cumulative Variance")
axes[1].legend()

plt.tight_layout()
plt.savefig("ch16_ex4_scree.png", dpi=150)
plt.show()

# Use 10 PCs
n_pcs = 10
pca_scores = scores[:, :n_pcs]

# ---- (b) K-means on PCA scores ----
print("\n=== Part (b): K-means on PCA Scores ===")

sil_list = []
for k in range(2, 6):
    km = KMeans(n_clusters=k, n_init=50, random_state=42).fit(pca_scores)
    sil = silhouette_score(pca_scores, km.labels_)
```

```
    sil_list.append(sil)
    print(f"k = {k}: Silhouette = {sil:.3f}")

best_k = list(range(2, 6))[np.argmax(sil_list)]
print(f"Best k by silhouette: {best_k}")

# Fit final clustering
km_final = KMeans(n_clusters=best_k, n_init=50, random_state=42).fit(pca_scor

# ---- (c) UMAP visualisation ----
print("\n=== Part (c): UMAP Visualisation ===")

umap_2d = UMAP(n_components=2, n_neighbors=15, min_dist=0.1,
                random_state=42).fit_transform(pca_scores)

colors_4 = ["steelblue", "firebrick", "forestgreen", "goldenrod"]

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Discovered clusters
for c in range(best_k):
    mask = km_final.labels_ == c
    axes[0].scatter(umap_2d[mask, 0], umap_2d[mask, 1], c=colors_4[c],
                    s=10, alpha=0.6, label=f"Cluster {c+1}")
axes[0].set_title("K-means Clusters on UMAP")
axes[0].set_xlabel("UMAP 1")
axes[0].set_ylabel("UMAP 2")
axes[0].legend(fontsize=8)

# True subtypes
for s in range(4):
    mask = true_subtype == s
    axes[1].scatter(umap_2d[mask, 0], umap_2d[mask, 1], c=colors_4[s],
```

```
                    s=10, alpha=0.6, label=f"Subtype {s+1}")
axes[1].set_title("True Subtypes on UMAP")
axes[1].set_xlabel("UMAP 1")
axes[1].set_ylabel("UMAP 2")
axes[1].legend(fontsize=8)

plt.tight_layout()
plt.savefig("ch16_ex4_umap.png", dpi=150)
plt.show()

# ---- (d) Cluster profiles ----
print("\n=== Part (d): Cluster Profiles ===\n")

var_names = [f"V{i+1}" for i in range(p)]
df = pd.DataFrame(X, columns=var_names)
df['cluster'] = km_final.labels_

print("Cluster sizes:")
print(df['cluster'].value_counts().sort_index())

print("\nCluster means for signal variables:")
for cl in range(best_k):
    mask = df['cluster'] == cl
    n_cl = mask.sum()
    print(f"\nCluster {cl+1} (n = {n_cl}):")
    for s in range(4):
        start = s * 6
        end = (s + 1) * 6
        signal_vars = [f"V{i+1}" for i in range(start, min(end, p))]
        mean_val = df.loc[mask, signal_vars].mean().mean()
        print(f"  Subtype {s+1} signal vars (V{start+1}-V{min(end,p)}): mean = {mean_val:.2f
```

```
# ---- (e) Discussion ----
```

```
print("\n=== Part (e): Validation Discussion ===\n")

print("To validate these clusters in a real clinical study:\n")
print("1. EXTERNAL OUTCOMES: Compare clusters on outcomes NOT used in")
print("   clustering (mortality, LOS, treatment response). Clusters")
print("   that predict external outcomes are clinically meaningful.\n")
print("2. STABILITY ANALYSIS: Bootstrap resampling + re-clustering.")
print("   Consistent membership = robust; variable = artefact.\n")
print("3. INDEPENDENT REPLICATION: Apply pipeline to an independent")
print("   dataset (different hospital/time). Similar clusters = generalisable
print("4. CLINICAL EXPERT REVIEW: Do clusters correspond to")
print("   recognisable patient types? Do they suggest different management?\n
print("5. MULTIPLE ALGORITHMS: Compare K-means, hierarchical, DBSCAN.")
print("   Agreement across methods strengthens confidence.\n")
print("6. AVOID CIRCULAR REASONING: Do NOT validate on the same")
print("   variables used for clustering. That proves nothing.")
```

D.13. Chapter 17: Causal Inference

D.13.1. Exercise 1

D.13.1.1. R

```
# ==========================================================================
# Chapter 17 - Exercise 1: DAG Construction (Conceptual)
# Studying the relationship between ACE inhibitor use and Acute Kidney Injury
# ==========================================================================

# This is a conceptual exercise. The answers are provided as detailed comment
# with optional code to illustrate DAG concepts using the dagitty package.
```

```
# --- Part (a): List at least five relevant variables ---
#
# 1. Baseline kidney function (eGFR / serum creatinine)
#    - Patients with worse kidney function are more likely to receive ACE inhibitors
#      (for renoprotective effects) AND are at higher risk of AKI.
#
# 2. Heart failure severity
#    - Heart failure is a major indication for ACE inhibitors AND independently
#      increases AKI risk (through haemodynamic changes).
#
# 3. Diabetes
#    - Diabetes is an indication for ACE inhibitors (nephroprotection) AND a risk
#      factor for AKI.
#
# 4. Age
#    - Older patients are more likely to be on ACE inhibitors and are at higher
#      risk of AKI.
#
# 5. Hypertension
#    - Primary indication for ACE inhibitors and can contribute to kidney injury.
#
# 6. Concomitant nephrotoxic drugs (e.g., NSAIDs, contrast agents)
#    - May be prescribed alongside or instead of ACE inhibitors and directly
#      cause AKI.
#
# 7. Volume status / dehydration
#    - Dehydration increases AKI risk and may influence whether ACE inhibitors
#      are held or continued.
#
# 8. Proteinuria
#    - An indication for ACE inhibitors and a marker of kidney disease severity
#      (related to AKI risk).
```

```r
# --- Part (b): Draw a DAG ---
# We use the dagitty package to encode the DAG programmatically.

# install.packages("dagitty")
library(dagitty)

dag <- dagitty('dag {
  ACEi [exposure]
  AKI [outcome]
  eGFR [confounder]
  HeartFailure [confounder]
  Diabetes [confounder]
  Age [confounder]
  Hypertension [confounder]
  NephrotoxicDrugs [confounder]
  VolumeStatus [confounder]
  ICUAdmission [collider]

  Age -> ACEi

  Age -> AKI
  Age -> eGFR
  Age -> HeartFailure
  Age -> Diabetes

  eGFR -> ACEi
  eGFR -> AKI

  HeartFailure -> ACEi
  HeartFailure -> AKI
  HeartFailure -> VolumeStatus

  Diabetes -> ACEi
```

```
  Diabetes -> AKI
  Diabetes -> eGFR

  Hypertension -> ACEi
  Hypertension -> AKI

  NephrotoxicDrugs -> AKI

  VolumeStatus -> AKI

  ACEi -> AKI

  # Collider: ICU Admission is caused by both ACEi use patterns
  # and by AKI itself (patients with AKI go to ICU)
  ACEi -> ICUAdmission
  AKI -> ICUAdmission
}')

# Visualise the DAG (if running interactively)
# plot(dag)

# --- Part (c): Minimal sufficient adjustment set ---
# Using dagitty to compute the adjustment set automatically
adj_set <- adjustmentSets(dag, exposure = "ACEi", outcome = "AKI", type = "minimal")
cat("Minimal sufficient adjustment sets:\n")
print(adj_set)

# Explanation:
# The minimal adjustment set includes confounders that block all backdoor paths
# from ACEi to AKI. Based on our DAG, this includes:
# - Age, eGFR, HeartFailure, Diabetes, Hypertension
# These are the common causes of both ACEi use and AKI.
# We do NOT need to adjust for VolumeStatus (it's not a confounder of ACEi->AKI
```

```
# unless there's a direct path from ACEi to VolumeStatus).
# We do NOT need to adjust for NephrotoxicDrugs (it only affects AKI, not ACE

# --- Part (d): Identify a collider ---
#
# ICU Admission is a COLLIDER: it is caused by both ACEi use (patients on
# ACE inhibitors who develop complications may be admitted to ICU) and AKI
# (AKI itself is a reason for ICU admission).
#
# What happens if we adjust for ICU Admission?
# Conditioning on a collider OPENS a spurious path between ACEi and AKI.
# This would create "collider bias" (also known as Berkson's bias in clinical
# settings). Even if ACEi had no effect on AKI, adjusting for ICU admission
# would create an artificial association because:
# - Among ICU patients, if they were NOT admitted for AKI, they were more
#   likely admitted for ACEi-related reasons (and vice versa).
# - This induces a negative correlation between the two causes of the collide
#
# Practical lesson: Do NOT adjust for variables that are consequences of both
# the exposure and the outcome (or that lie on causal paths from both).

cat("\n--- Summary ---\n")
cat("Key confounders to adjust for: Age, eGFR, Heart Failure, Diabetes, Hyper
cat("Collider to AVOID adjusting for: ICU Admission\n")
cat("Mediators to consider carefully: Volume status changes caused by ACEi\n"
```

D.13.1.2. Python

```
# ==========================================================================
# Chapter 17 - Exercise 1: DAG Construction (Conceptual)
# Studying the relationship between ACE inhibitor use and Acute Kidney Injury
# ==========================================================================
```

```
# This is a conceptual exercise. The answers are provided as detailed comments
# with optional code to illustrate DAG concepts.

# --- Part (a): List at least five relevant variables ---
#
# 1. Baseline kidney function (eGFR / serum creatinine)
#    - Patients with worse kidney function are more likely to receive ACE inhibitors
#      (for renoprotective effects) AND are at higher risk of AKI.
#
# 2. Heart failure severity
#    - Heart failure is a major indication for ACE inhibitors AND independently
#      increases AKI risk (through haemodynamic changes).
#
# 3. Diabetes
#    - Diabetes is an indication for ACE inhibitors (nephroprotection) AND a risk
#      factor for AKI.
#
# 4. Age
#    - Older patients are more likely to be on ACE inhibitors and are at higher
#      risk of AKI.
#
# 5. Hypertension
#    - Primary indication for ACE inhibitors and can contribute to kidney injury.
#
# 6. Concomitant nephrotoxic drugs (e.g., NSAIDs, contrast agents)
#    - May be prescribed alongside or instead of ACE inhibitors and directly
#      cause AKI.
#
# 7. Volume status / dehydration
#    - Dehydration increases AKI risk and may influence whether ACE inhibitors
#      are held or continued.
#
# 8. Proteinuria
```

```
#    - An indication for ACE inhibitors and a marker of kidney disease severi
#      (related to AKI risk).

# --- Part (b): Draw a DAG ---
# We represent the DAG as an adjacency list and visualise with networkx.

import networkx as nx
import matplotlib.pyplot as plt

# Define the DAG
dag = nx.DiGraph()

# Add edges representing causal relationships
edges = [
    # Age affects many variables
    ("Age", "ACEi"), ("Age", "AKI"), ("Age", "eGFR"),
    ("Age", "HeartFailure"), ("Age", "Diabetes"),
    # eGFR confounds ACEi-AKI
    ("eGFR", "ACEi"), ("eGFR", "AKI"),
    # Heart failure confounds ACEi-AKI
    ("HeartFailure", "ACEi"), ("HeartFailure", "AKI"),
    ("HeartFailure", "VolumeStatus"),
    # Diabetes confounds ACEi-AKI
    ("Diabetes", "ACEi"), ("Diabetes", "AKI"), ("Diabetes", "eGFR"),
    # Hypertension confounds ACEi-AKI
    ("Hypertension", "ACEi"), ("Hypertension", "AKI"),
    # Other causes of AKI
    ("NephrotoxicDrugs", "AKI"),
    ("VolumeStatus", "AKI"),
    # Treatment -> Outcome (causal effect of interest)
    ("ACEi", "AKI"),
    # Collider: ICU admission caused by both ACEi complications and AKI
    ("ACEi", "ICUAdmission"), ("AKI", "ICUAdmission"),
```

```
]

dag.add_edges_from(edges)

# Visualise the DAG
plt.figure(figsize=(12, 8))
pos = nx.spring_layout(dag, seed=42, k=2)
nx.draw(dag, pos, with_labels=True, node_color='lightblue',
        node_size=2000, font_size=9, font_weight='bold',
        arrows=True, arrowsize=20, edge_color='gray')
plt.title("DAG: ACE Inhibitor Use and Acute Kidney Injury")
plt.tight_layout()
plt.savefig("dag_acei_aki.png", dpi=300)
plt.show()

# --- Part (c): Minimal sufficient adjustment set ---
#
# To estimate the causal effect of ACEi on AKI, we need to block all
# backdoor paths (non-causal paths from ACEi to AKI).
#
# Backdoor paths from ACEi to AKI:
# 1. ACEi <- Age -> AKI
# 2. ACEi <- eGFR -> AKI
# 3. ACEi <- HeartFailure -> AKI
# 4. ACEi <- Diabetes -> AKI
# 5. ACEi <- Diabetes -> eGFR -> AKI
# 6. ACEi <- Hypertension -> AKI
# 7. ACEi <- Age -> eGFR -> AKI
# 8. ACEi <- Age -> HeartFailure -> AKI
# 9. ACEi <- Age -> Diabetes -> AKI
# 10. ACEi <- HeartFailure -> VolumeStatus -> AKI
#
# Minimal sufficient adjustment set:
```

```
# {Age, eGFR, HeartFailure, Diabetes, Hypertension}
#
# This blocks all backdoor paths. Note:
# - NephrotoxicDrugs only affects AKI (not a confounder), so no need to adjus
# - VolumeStatus is blocked by conditioning on HeartFailure (the path goes
#   through HeartFailure).

print("Minimal sufficient adjustment set:")
print("  {Age, eGFR, HeartFailure, Diabetes, Hypertension}")

# --- Part (d): Identify a collider ---
#
# ICU Admission is a COLLIDER in the DAG:
#   ACEi -> ICUAdmission <- AKI
#
# Both ACEi use (through complications or monitoring) and AKI (as a
# critical condition) can cause ICU admission.
#
# What happens if we adjust for ICU Admission?
#
# Conditioning on a collider OPENS a spurious path between its causes.
# This creates "collider bias" (Berkson's bias):
#
# - Among ICU patients, if a patient was NOT admitted because of AKI,
#   they were more likely admitted due to ACEi-related issues (and vice versa
# - This induces an artificial negative association between ACEi and AKI,
#   even if no true causal relationship exists.
#
# Practical lesson: Never adjust for variables that are consequences of both
# the exposure and the outcome. Restricting the analysis to ICU patients
# only would similarly introduce collider bias.

print("\nCollider identified: ICU Admission")
```

```
print("  ACEi -> ICUAdmission <- AKI")
print("  Adjusting for ICU admission would OPEN a spurious path (collider bias)")

print("\n--- Summary ---")
print("Key confounders to adjust for: Age, eGFR, Heart Failure, Diabetes, Hypertension")
print("Collider to AVOID adjusting for: ICU Admission")
print("Mediators to consider: Volume status changes caused by ACEi")
```

D.13.2. Exercise 2

D.13.2.1. R

```
# ============================================================================
# Chapter 17 - Exercise 2: Propensity Score Matching in R
# Beta-blocker use and 1-year mortality
# ============================================================================

library(tidyverse)
library(MatchIt)
library(cobalt)
library(broom)

# --- Simulate the dataset (from the exercise) ---
set.seed(123)
n <- 1500
exercise_dat <- tibble(
  age = rnorm(n, 70, 8),
  creatinine = rnorm(n, 1.2, 0.4),
  heart_failure = rbinom(n, 1, 0.35),
  prior_mi = rbinom(n, 1, 0.20),
  # Confounded treatment (beta-blocker use)
  treatment = rbinom(n, 1, plogis(-1 + 0.02*age + 0.3*heart_failure +
```

```
                                     0.5*prior_mi - 0.8*creatinine)),
  # Outcome: 1-year mortality
  death_1yr = rbinom(n, 1, plogis(-2 + 0.05*age + 0.4*heart_failure +
                                     0.6*prior_mi + 1.0*creatinine -
                                     0.7*treatment))
)

cat("Dataset dimensions:", nrow(exercise_dat), "patients\n")
cat("Treatment prevalence:", mean(exercise_dat$treatment), "\n")
cat("Outcome prevalence:", mean(exercise_dat$death_1yr), "\n\n")

# --- Part (a): Estimate propensity scores using logistic regression ---
ps_model <- glm(treatment ~ age + creatinine + heart_failure + prior_mi,
                data = exercise_dat, family = binomial)

exercise_dat$ps <- predict(ps_model, type = "response")

cat("Propensity score summary:\n")
print(summary(exercise_dat$ps))

# Visualise propensity score overlap
ggplot(exercise_dat, aes(x = ps, fill = factor(treatment))) +
  geom_density(alpha = 0.5) +
  labs(x = "Propensity Score", fill = "Beta-blocker",
       title = "Propensity Score Distribution by Treatment Group") +
  scale_fill_manual(values = c("#0072B2", "#D55E00"),
                    labels = c("No", "Yes")) +
  theme_minimal()

# --- Part (b): 1:1 nearest-neighbour matching with caliper of 0.2 SD ---
m_out <- matchit(treatment ~ age + creatinine + heart_failure + prior_mi,
                 data = exercise_dat,
                 method = "nearest",
```

```
                 distance = "glm",        # logistic regression PS
                 caliper = 0.2,           # 0.2 SD of logit PS
                 ratio = 1)               # 1:1 matching

cat("\nMatching summary:\n")
summary(m_out)

# --- Part (c): Love plot to assess balance ---
love.plot(m_out,
          thresholds = c(m = 0.1),    # SMD threshold of 0.1
          binary = "std",
          var.order = "unadjusted",
          title = "Covariate Balance: Before and After Matching",
          colors = c("#D55E00", "#0072B2"))

# Also check numeric balance
cat("\nBalance table:\n")
print(bal.tab(m_out, thresholds = c(m = 0.1)))

# --- Part (d): Estimate ATT for beta-blocker use on 1-year mortality ---
m_data <- match.data(m_out)

cat("\nMatched sample size:", nrow(m_data), "patients\n")
cat("Treated in matched sample:", sum(m_data$treatment == 1), "\n")
cat("Control in matched sample:", sum(m_data$treatment == 0), "\n")

# Outcome model in matched sample
outcome_model <- glm(death_1yr ~ treatment,
                     data = m_data,
                     family = binomial,
                     weights = weights)

cat("\nATT estimate (logistic regression in matched sample):\n")
```

```r
result <- tidy(outcome_model, conf.int = TRUE, exponentiate = TRUE)
print(result)

# Risk difference (linear probability model)
rd_model <- lm(death_1yr ~ treatment, data = m_data, weights = weights)
cat("\nRisk difference estimate:\n")
print(tidy(rd_model, conf.int = TRUE))

# Mortality rates in matched sample
cat("\nMortality in matched sample:\n")
cat("  Treated:", mean(m_data$death_1yr[m_data$treatment == 1]), "\n")
cat("  Control:", mean(m_data$death_1yr[m_data$treatment == 0]), "\n")

# --- Part (e): E-value calculation ---
# Extract the odds ratio from the matched analysis
or_est <- result$estimate[result$term == "treatment"]
or_lo <- result$conf.low[result$term == "treatment"]
or_hi <- result$conf.high[result$term == "treatment"]

cat("\nOdds Ratio:", round(or_est, 3),
    "(95% CI:", round(or_lo, 3), "-", round(or_hi, 3), ")\n")

# Convert OR to RR approximation (for rare outcomes, OR ~ RR)
# For the E-value, we need RR on the above-null scale
# Since treatment is protective (OR < 1), convert: RR_above_null = 1/OR
rr_est <- 1 / or_est
rr_lo_bound <- 1 / or_hi  # Note: CI bounds flip when inverting

# E-value formula: E = RR + sqrt(RR * (RR - 1))
e_value <- function(rr) {
  if (rr < 1) rr <- 1 / rr  # Convert to above-null
  return(rr + sqrt(rr * (rr - 1)))
}
```

```r
e_val_point <- e_value(rr_est)

# E-value for the CI bound closest to null
# If protective, the upper bound of RR (lower bound of 1/OR) is closest to null
e_val_ci <- e_value(rr_lo_bound)

cat("\nE-value for point estimate:", round(e_val_point, 2), "\n")
cat("E-value for CI bound closest to null:", round(e_val_ci, 2), "\n")

# Interpretation
cat("\n--- E-value Interpretation ---\n")
cat("The E-value of", round(e_val_point, 2), "means that an unmeasured\n")
cat("confounder would need to be associated with BOTH beta-blocker use AND\n")
cat("1-year mortality by a risk ratio of at least", round(e_val_point, 2), "\n")
cat("(above and beyond measured confounders) to fully explain away the\n")
cat("observed protective association.\n")
cat("\nThe E-value for the confidence interval bound is", round(e_val_ci, 2), ",\n")
cat("meaning a confounder of that strength could shift the CI to include the null.\n")
cat("If these values are larger than plausible confounders, the result is robust.\n")

# Optional: use EValue package if available
# library(EValue)
# evalues.OR(or_est, lo = or_lo, hi = or_hi, rare = FALSE)
```

D.13.2.2. Python

```python
# ============================================================================
# Chapter 17 - Exercise 2: Propensity Score Matching in Python
# Beta-blocker use and 1-year mortality
# ============================================================================
```

```
import numpy as np
import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import NearestNeighbors
from scipy import stats
import matplotlib.pyplot as plt

# --- Simulate the dataset (same DGP as the R exercise) ---
np.random.seed(123)
n = 1500

age = np.random.normal(70, 8, n)
creatinine = np.random.normal(1.2, 0.4, n)
heart_failure = np.random.binomial(1, 0.35, n)
prior_mi = np.random.binomial(1, 0.20, n)

# Confounded treatment assignment
ps_true = 1 / (1 + np.exp(-(-1 + 0.02*age + 0.3*heart_failure +
                              0.5*prior_mi - 0.8*creatinine)))
treatment = np.random.binomial(1, ps_true)

# Outcome
mort_prob = 1 / (1 + np.exp(-(-2 + 0.05*age + 0.4*heart_failure +
                                0.6*prior_mi + 1.0*creatinine -
                                0.7*treatment)))
death_1yr = np.random.binomial(1, mort_prob)

df = pd.DataFrame({
    'age': age, 'creatinine': creatinine, 'heart_failure': heart_failure,
    'prior_mi': prior_mi, 'treatment': treatment, 'death_1yr': death_1yr
})

print(f"Dataset: {len(df)} patients")
```

```python
print(f"Treatment prevalence: {df['treatment'].mean():.3f}")
print(f"Outcome prevalence: {df['death_1yr'].mean():.3f}\n")

# --- Part (a): Estimate propensity scores ---
covariates = ['age', 'creatinine', 'heart_failure', 'prior_mi']
ps_model = LogisticRegression(max_iter=1000)
ps_model.fit(df[covariates], df['treatment'])
df['ps'] = ps_model.predict_proba(df[covariates])[:, 1]

# Compute logit of PS for caliper matching
df['logit_ps'] = np.log(df['ps'] / (1 - df['ps']))

print("Propensity score summary:")
print(df['ps'].describe())

# Visualise PS overlap
fig, ax = plt.subplots(figsize=(8, 5))
df[df['treatment'] == 0]['ps'].plot.kde(ax=ax, label='Control', color='#0072B2')
df[df['treatment'] == 1]['ps'].plot.kde(ax=ax, label='Treated', color='#D55E00')
ax.set_xlabel('Propensity Score')
ax.set_ylabel('Density')
ax.set_title('Propensity Score Distribution by Treatment Group')
ax.legend()
plt.tight_layout()
plt.savefig('ps_overlap.png', dpi=300)
plt.show()

# --- Part (b): 1:1 nearest-neighbour matching with caliper ---
treated = df[df['treatment'] == 1].copy()
control = df[df['treatment'] == 0].copy()

# Caliper: 0.2 * SD of logit PS
caliper = 0.2 * df['logit_ps'].std()
```

```
print(f"\nCaliper (0.2 SD of logit PS): {caliper:.4f}")

# Nearest-neighbour matching on logit PS
nn = NearestNeighbors(n_neighbors=1, metric='euclidean')
nn.fit(control[['logit_ps']].values)

distances, indices = nn.kneighbors(treated[['logit_ps']].values)

# Apply caliper: keep only matches within the caliper
matched_treated_idx = []
matched_control_idx = []
used_controls = set()

for i in range(len(treated)):
    dist = distances[i, 0]
    ctrl_idx = indices[i, 0]
    if dist <= caliper and ctrl_idx not in used_controls:
        matched_treated_idx.append(treated.index[i])
        matched_control_idx.append(control.index[ctrl_idx])
        used_controls.add(ctrl_idx)

# Create matched dataset
matched_df = pd.concat([
    df.loc[matched_treated_idx],
    df.loc[matched_control_idx]
])

print(f"Matched sample: {len(matched_df)} patients")
print(f"  Treated: {len(matched_treated_idx)}")
print(f"  Control: {len(matched_control_idx)}")
print(f"  Unmatched treated: {len(treated) - len(matched_treated_idx)}")

# --- Part (c): Assess balance using SMD (Love plot equivalent) ---
```

```python
def compute_smd(data, var, group_col):
    """Compute standardised mean difference."""
    treated = data[data[group_col] == 1][var]
    control = data[data[group_col] == 0][var]
    pooled_sd = np.sqrt((treated.var() + control.var()) / 2)
    if pooled_sd == 0:
        return 0
    return (treated.mean() - control.mean()) / pooled_sd

print("\nCovariate balance (SMD):")
print(f"{'Variable':<20} {'Before':>10} {'After':>10} {'Balanced?':>10}")
print("-" * 52)

smd_before = {}
smd_after = {}
for var in covariates:
    smd_b = compute_smd(df, var, 'treatment')
    smd_a = compute_smd(matched_df, var, 'treatment')
    smd_before[var] = smd_b
    smd_after[var] = smd_a
    balanced = "Yes" if abs(smd_a) < 0.1 else "No"
    print(f"{var:<20} {smd_b:>10.4f} {smd_a:>10.4f} {balanced:>10}")

# Love plot
fig, ax = plt.subplots(figsize=(8, 5))
y_pos = range(len(covariates))

ax.scatter([abs(smd_before[v]) for v in covariates], y_pos,
           color='#D55E00', label='Before matching', s=80, zorder=5)
ax.scatter([abs(smd_after[v]) for v in covariates], y_pos,
           color='#0072B2', label='After matching', s=80, zorder=5)

# Connect before and after
```

```python
for i, var in enumerate(covariates):
    ax.plot([abs(smd_before[var]), abs(smd_after[var])], [i, i],
            'k-', alpha=0.3)

ax.axvline(x=0.1, color='red', linestyle='--', alpha=0.5, label='SMD = 0.1 th
ax.set_yticks(y_pos)
ax.set_yticklabels(covariates)
ax.set_xlabel('Absolute Standardised Mean Difference')
ax.set_title('Love Plot: Covariate Balance Before and After Matching')
ax.legend(loc='lower right')
plt.tight_layout()
plt.savefig('love_plot.png', dpi=300)
plt.show()

# --- Part (d): Estimate ATT ---
# Mortality rates in matched sample
mort_treated = matched_df[matched_df['treatment'] == 1]['death_1yr'].mean()
mort_control = matched_df[matched_df['treatment'] == 0]['death_1yr'].mean()

print(f"\nMortality in matched sample:")
print(f"  Treated (beta-blocker): {mort_treated:.4f}")
print(f"  Control (no beta-blocker): {mort_control:.4f}")
print(f"  Risk difference (ATT): {mort_treated - mort_control:.4f}")

# Odds ratio
from statsmodels.api import Logit, add_constant

X = add_constant(matched_df['treatment'])
logit_model = Logit(matched_df['death_1yr'], X).fit(disp=0)
print(f"\nLogistic regression in matched sample:")
print(logit_model.summary2().tables[1])

or_est = np.exp(logit_model.params['treatment'])
```

```python
ci = np.exp(logit_model.conf_int().loc['treatment'])
print(f"\nOdds Ratio: {or_est:.3f} (95% CI: {ci[0]:.3f} - {ci[1]:.3f})")

# --- Part (e): E-value ---
# Convert protective OR to RR on above-null scale
rr_est = 1 / or_est  # Flip because protective
rr_ci_closest_null = 1 / ci[1]  # Upper CI of OR -> lower bound of 1/OR

def e_value(rr):
    """Compute E-value for a risk ratio."""
    if rr < 1:
        rr = 1 / rr
    return rr + np.sqrt(rr * (rr - 1))

e_val_point = e_value(rr_est)
e_val_ci = e_value(rr_ci_closest_null)

print(f"\n--- E-value Analysis ---")
print(f"E-value for point estimate: {e_val_point:.2f}")
print(f"E-value for CI bound closest to null: {e_val_ci:.2f}")

print(f"\nInterpretation:")
print(f"An unmeasured confounder would need to be associated with BOTH")
print(f"beta-blocker use AND 1-year mortality by a risk ratio of at least")
print(f"{e_val_point:.2f} to fully explain away the observed protective effect.")
print(f"For the confidence interval, a confounder with RR >= {e_val_ci:.2f}")
print(f"could shift the CI to include the null.")
```

D.13.3. Exercise 3

D.13.3.1. R

```
# ==============================================================================
# Chapter 17 - Exercise 3: IPTW in R
# (Exercise specifies Python, but we provide an R version as well)
# Beta-blocker use and 1-year mortality using IPTW
# ==============================================================================

library(tidyverse)
library(WeightIt)
library(cobalt)
library(survey)
library(broom)

# --- Simulate the dataset (same DGP as Exercise 2) ---
set.seed(123)
n <- 1500

exercise_dat <- tibble(
  age = rnorm(n, 70, 8),
  creatinine = rnorm(n, 1.2, 0.4),
  heart_failure = rbinom(n, 1, 0.35),
  prior_mi = rbinom(n, 1, 0.20),
  treatment = rbinom(n, 1, plogis(-1 + 0.02*age + 0.3*heart_failure +
                                  0.5*prior_mi - 0.8*creatinine)),
  death_1yr = rbinom(n, 1, plogis(-2 + 0.05*age + 0.4*heart_failure +
                                  0.6*prior_mi + 1.0*creatinine -
                                  0.7*treatment))
)

# --- Part (a): Fit PS model and compute stabilised IPTW weights ---
```

```r
w_out <- weightit(treatment ~ age + creatinine + heart_failure + prior_mi,
                  data = exercise_dat,
                  method = "ps",
                  estimand = "ATE")

cat("Weight summary:\n")
summary(w_out)

# Check for extreme weights
cat("\nWeight distribution:\n")
cat("  Min:", min(w_out$weights), "\n")
cat("  Max:", max(w_out$weights), "\n")
cat("  Mean:", mean(w_out$weights), "\n")
cat("  SD:", sd(w_out$weights), "\n")

# --- Part (b): Assess covariate balance using weighted SMDs ---
cat("\nBalance table:\n")
bt <- bal.tab(w_out, thresholds = c(m = 0.1))
print(bt)

# Love plot
love.plot(w_out,
          thresholds = c(m = 0.1),
          binary = "std",
          var.order = "unadjusted",
          title = "Covariate Balance: Before and After IPTW",
          colors = c("#D55E00", "#0072B2"))

# --- Part (c): Estimate ATE using weighted regression ---
d_weighted <- svydesign(ids = ~1, weights = w_out$weights, data = exercise_dat)

# Logistic regression (odds ratio)
ate_model <- svyglm(death_1yr ~ treatment, design = d_weighted,
```

```
                    family = quasibinomial)
cat("\nIPTW ATE estimate (logistic):\n")
print(tidy(ate_model, conf.int = TRUE, exponentiate = TRUE))

# Linear probability model (risk difference)
rd_model <- svyglm(death_1yr ~ treatment, design = d_weighted,
                   family = gaussian())
cat("\nIPTW ATE estimate (risk difference):\n")
print(tidy(rd_model, conf.int = TRUE))

# --- Part (d): Sensitivity analysis with simulated unmeasured confounder ---
cat("\n--- Sensitivity Analysis: Unmeasured Confounder ---\n")

# Simulate an unmeasured confounder U that affects both treatment and outcome
set.seed(456)
U <- rnorm(n, 0, 1)

# Re-simulate treatment and outcome with U included
exercise_dat_u <- tibble(
  age = exercise_dat$age,
  creatinine = exercise_dat$creatinine,
  heart_failure = exercise_dat$heart_failure,
  prior_mi = exercise_dat$prior_mi,
  U = U,
  treatment = rbinom(n, 1, plogis(-1 + 0.02*age + 0.3*heart_failure +
                                     0.5*prior_mi - 0.8*creatinine +
                                     0.6*U)),  # U affects treatment
  death_1yr = rbinom(n, 1, plogis(-2 + 0.05*age + 0.4*heart_failure +
                                     0.6*prior_mi + 1.0*creatinine -
                                     0.7*treatment +
                                     0.8*U))   # U affects outcome
)
```

```r
# Analysis WITHOUT adjusting for U (biased)
w_no_u <- weightit(treatment ~ age + creatinine + heart_failure + prior_mi,
                   data = exercise_dat_u,
                   method = "ps", estimand = "ATE")

d_no_u <- svydesign(ids = ~1, weights = w_no_u$weights, data = exercise_dat_u)
model_no_u <- svyglm(death_1yr ~ treatment, design = d_no_u,
                     family = quasibinomial)
res_no_u <- tidy(model_no_u, conf.int = TRUE, exponentiate = TRUE)

cat("\nWithout adjusting for U:\n")
print(res_no_u)

# Analysis WITH adjusting for U (unbiased)
w_with_u <- weightit(treatment ~ age + creatinine + heart_failure + prior_mi + U,
                     data = exercise_dat_u,
                     method = "ps", estimand = "ATE")

d_with_u <- svydesign(ids = ~1, weights = w_with_u$weights, data = exercise_dat_u)
model_with_u <- svyglm(death_1yr ~ treatment, design = d_with_u,
                       family = quasibinomial)
res_with_u <- tidy(model_with_u, conf.int = TRUE, exponentiate = TRUE)

cat("\nWith adjusting for U:\n")
print(res_with_u)

cat("\n--- Interpretation ---\n")
cat("The unmeasured confounder U (with associations of 0.6 with treatment\n")
cat("and 0.8 with outcome on the log-odds scale) shifts the treatment effect\n")
cat("estimate when not adjusted for. This demonstrates:\n")
cat("  1. IPTW cannot correct for unmeasured confounding.\n")
cat("  2. Sensitivity analysis helps quantify how strong a confounder\n")
cat("     would need to be to change conclusions.\n")
```

```
cat("  3. The E-value provides a formal framework for this assessment.\n")
```

D.13.3.2. Python

```
# ==========================================================================
# Chapter 17 - Exercise 3: IPTW in Python
# Beta-blocker use and 1-year mortality using IPTW
# ==========================================================================

import numpy as np
import pandas as pd
from sklearn.linear_model import LogisticRegression
from scipy import stats
import matplotlib.pyplot as plt
import statsmodels.api as sm

# --- Simulate the dataset (same DGP as Exercise 2) ---
np.random.seed(123)
n = 1500

age = np.random.normal(70, 8, n)
creatinine = np.random.normal(1.2, 0.4, n)
heart_failure = np.random.binomial(1, 0.35, n)
prior_mi = np.random.binomial(1, 0.20, n)

ps_true = 1 / (1 + np.exp(-(-1 + 0.02*age + 0.3*heart_failure +
                               0.5*prior_mi - 0.8*creatinine)))
treatment = np.random.binomial(1, ps_true)

mort_prob = 1 / (1 + np.exp(-(-2 + 0.05*age + 0.4*heart_failure +
                                 0.6*prior_mi + 1.0*creatinine -
                                 0.7*treatment)))
```

```
death_1yr = np.random.binomial(1, mort_prob)

df = pd.DataFrame({
    'age': age, 'creatinine': creatinine, 'heart_failure': heart_failure,
    'prior_mi': prior_mi, 'treatment': treatment, 'death_1yr': death_1yr
})

print(f"Dataset: {len(df)} patients")
print(f"Treatment prevalence: {df['treatment'].mean():.3f}")
print(f"Outcome prevalence: {df['death_1yr'].mean():.3f}\n")

# --- Part (a): Fit PS model and compute stabilised IPTW weights ---
covariates = ['age', 'creatinine', 'heart_failure', 'prior_mi']

ps_model = LogisticRegression(max_iter=1000)
ps_model.fit(df[covariates], df['treatment'])
df['ps'] = ps_model.predict_proba(df[covariates])[:, 1]

# Unstabilised ATE weights: w = A/ps + (1-A)/(1-ps)
df['iptw'] = np.where(
    df['treatment'] == 1,
    1 / df['ps'],
    1 / (1 - df['ps'])
)

# Stabilised weights: replace numerator with marginal treatment probability
p_treat = df['treatment'].mean()
df['sw'] = np.where(
    df['treatment'] == 1,
    p_treat / df['ps'],
    (1 - p_treat) / (1 - df['ps'])
)
```

```python
print("Stabilised weight summary:")
print(f"  Min: {df['sw'].min():.4f}")
print(f"  Max: {df['sw'].max():.4f}")
print(f"  Mean: {df['sw'].mean():.4f}")
print(f"  SD: {df['sw'].std():.4f}")

# Check for extreme weights
extreme = (df['sw'] > 10).sum()
print(f"  Extreme weights (>10): {extreme}")

# --- Part (b): Assess covariate balance using weighted SMDs ---
def weighted_smd(data, var, treatment_col, weight_col):
    """Compute weighted standardised mean difference."""
    treated = data[data[treatment_col] == 1]
    control = data[data[treatment_col] == 0]
    mean_t = np.average(treated[var], weights=treated[weight_col])
    mean_c = np.average(control[var], weights=control[weight_col])
    # Use unweighted pooled SD for standardisation
    sd_pooled = np.sqrt((treated[var].var() + control[var].var()) / 2)
    if sd_pooled == 0:
        return 0
    return (mean_t - mean_c) / sd_pooled

def unweighted_smd(data, var, treatment_col):
    """Compute unweighted SMD."""
    treated = data[data[treatment_col] == 1][var]
    control = data[data[treatment_col] == 0][var]
    sd_pooled = np.sqrt((treated.var() + control.var()) / 2)
    if sd_pooled == 0:
        return 0
    return (treated.mean() - control.mean()) / sd_pooled

print("\nCovariate balance (weighted SMD):")
```

```python
print(f"{'Variable':<20} {'Unweighted':>12} {'Weighted':>12} {'Balanced?':>10}")
print("-" * 56)

smd_before = {}
smd_after = {}
for var in covariates:
    smd_b = unweighted_smd(df, var, 'treatment')
    smd_a = weighted_smd(df, var, 'treatment', 'sw')
    smd_before[var] = smd_b
    smd_after[var] = smd_a
    balanced = "Yes" if abs(smd_a) < 0.1 else "No"
    print(f"{var:<20} {smd_b:>12.4f} {smd_a:>12.4f} {balanced:>10}")

# Love plot equivalent
fig, ax = plt.subplots(figsize=(8, 5))
y_pos = range(len(covariates))

ax.scatter([abs(smd_before[v]) for v in covariates], y_pos,
           color='#D55E00', label='Before IPTW', s=80, zorder=5)
ax.scatter([abs(smd_after[v]) for v in covariates], y_pos,
           color='#0072B2', label='After IPTW', s=80, zorder=5)

for i, var in enumerate(covariates):
    ax.plot([abs(smd_before[var]), abs(smd_after[var])], [i, i],
            'k-', alpha=0.3)

ax.axvline(x=0.1, color='red', linestyle='--', alpha=0.5, label='SMD = 0.1')
ax.set_yticks(y_pos)
ax.set_yticklabels(covariates)
ax.set_xlabel('Absolute Standardised Mean Difference')
ax.set_title('Covariate Balance: Before and After IPTW')
ax.legend(loc='lower right')
plt.tight_layout()
```

```python
plt.savefig('iptw_balance.png', dpi=300)
plt.show()

# --- Part (c): Estimate ATE using weighted regression ---
X = sm.add_constant(df['treatment'])

# Weighted least squares (linear probability model for risk difference)
wls_model = sm.WLS(df['death_1yr'], X, weights=df['sw']).fit()
print("\nIPTW ATE Estimate (Risk Difference):")
print(f"  Coefficient: {wls_model.params.iloc[1]:.4f}")
print(f"  95% CI: ({wls_model.conf_int().iloc[1, 0]:.4f}, "
      f"{wls_model.conf_int().iloc[1, 1]:.4f})")
print(f"  p-value: {wls_model.pvalues.iloc[1]:.4f}")

# Weighted logistic regression (for odds ratio)
from statsmodels.genmod.generalized_linear_model import GLM
from statsmodels.genmod.families import Binomial

glm_model = GLM(df['death_1yr'], X,
                family=Binomial(),
                freq_weights=df['sw']).fit()
or_est = np.exp(glm_model.params.iloc[1])
ci = np.exp(glm_model.conf_int().iloc[1])
print(f"\nIPTW ATE Estimate (Odds Ratio):")
print(f"  OR: {or_est:.3f} (95% CI: {ci[0]:.3f} - {ci[1]:.3f})")

# --- Part (d): Sensitivity analysis with unmeasured confounder ---
print("\n--- Sensitivity Analysis: Unmeasured Confounder ---")
print("Varying confounder strength to show impact on ATE estimate:\n")

# We simulate data with an unmeasured confounder U of varying strength
results = []
gamma_values = [0, 0.2, 0.4, 0.6, 0.8, 1.0]
```

```
for gamma in gamma_values:
    np.random.seed(42)  # Same seed for comparability
    U = np.random.normal(0, 1, n)

    # Re-simulate with confounder U
    ps_u = 1 / (1 + np.exp(-(-1 + 0.02*age + 0.3*heart_failure +
                            0.5*prior_mi - 0.8*creatinine + gamma*U)))
    trt_u = np.random.binomial(1, ps_u)
    mort_u = 1 / (1 + np.exp(-(-2 + 0.05*age + 0.4*heart_failure +
                              0.6*prior_mi + 1.0*creatinine -
                              0.7*trt_u + gamma*U)))
    death_u = np.random.binomial(1, mort_u)

    df_u = pd.DataFrame({
        'age': age, 'creatinine': creatinine, 'heart_failure': heart_failure,
        'prior_mi': prior_mi, 'treatment': trt_u, 'death_1yr': death_u
    })

    # IPTW without U
    ps_mod = LogisticRegression(max_iter=1000)
    ps_mod.fit(df_u[covariates], df_u['treatment'])
    ps_est = ps_mod.predict_proba(df_u[covariates])[:, 1]

    p_t = df_u['treatment'].mean()
    sw = np.where(df_u['treatment'] == 1, p_t / ps_est, (1 - p_t) / (1 - ps_est))

    X_u = sm.add_constant(df_u['treatment'])
    wls_u = sm.WLS(df_u['death_1yr'], X_u, weights=sw).fit()

    results.append({
        'gamma': gamma,
        'ate': wls_u.params.iloc[1],
        'ci_lo': wls_u.conf_int().iloc[1, 0],
```

```
        'ci_hi': wls_u.conf_int().iloc[1, 1]
    })

results_df = pd.DataFrame(results)
print(f"{'Gamma (U strength)':<20} {'ATE':>10} {'95% CI':>25}")
print("-" * 57)
for _, row in results_df.iterrows():
    print(f"{row['gamma']:<20.1f} {row['ate']:>10.4f} "
          f"({row['ci_lo']:.4f}, {row['ci_hi']:.4f})")

# Plot sensitivity analysis
fig, ax = plt.subplots(figsize=(8, 5))
ax.errorbar(results_df['gamma'], results_df['ate'],
            yerr=[results_df['ate'] - results_df['ci_lo'],
                  results_df['ci_hi'] - results_df['ate']],
            fmt='o-', color='#0072B2', capsize=5, markersize=8)
ax.axhline(y=0, color='red', linestyle='--', alpha=0.5, label='Null effect')
ax.set_xlabel('Unmeasured Confounder Strength (gamma)')
ax.set_ylabel('ATE Estimate (Risk Difference)')
ax.set_title('Sensitivity Analysis: Impact of Unmeasured Confounding')
ax.legend()
plt.tight_layout()
plt.savefig('sensitivity_analysis.png', dpi=300)
plt.show()

print("\n--- Interpretation ---")
print("As the strength of the unmeasured confounder increases (gamma),")
print("the IPTW estimate of the treatment effect becomes more biased")
print("because IPTW cannot account for unmeasured confounding.")
print("When gamma = 0 (no unmeasured confounder), the estimate is closest")
print("to the true effect. This demonstrates why sensitivity analysis")
print("is essential for observational causal inference.")
```

D.13.4. Exercise 4

D.13.4.1. R

```
# ============================================================================
# Chapter 17 - Exercise 4: Target Trial Emulation (Conceptual)
# Early vs delayed metformin initiation and 5-year cardiovascular events
# ============================================================================

# This is a conceptual exercise. All answers are provided as detailed comments.

# --- Part (a): Complete target trial protocol table ---
#
# +----------------------------+---------------------------------------------+---------------
# | Protocol component         | Target trial                                | Observational
# +----------------------------+---------------------------------------------+---------------
# | Eligibility criteria       | Adults aged 30-80 with new T2DM diagnosis,  | Same criteria
# |                            | no prior CVD, no prior metformin use,        | first T2DM d
# |                            | eGFR >= 30, no contraindications to          | no prior MAC
# |                            | metformin (e.g., severe renal impairment).   | before index
# |                            |                                             | 6 months.
# +----------------------------+---------------------------------------------+---------------
# | Treatment strategies       | Strategy 1: Initiate metformin within       | Same. Defined
# |                            | 3 months of T2DM diagnosis.                 | prescription
# |                            | Strategy 2: Do not initiate metformin       | diagnosis dat
# |                            | within 3 months (delayed, 3-12 months).     |
# +----------------------------+---------------------------------------------+---------------
# | Treatment assignment       | Random assignment at time of T2DM           | Not random. A
# |                            | diagnosis.                                  | propensity sc
# |                            |                                             | on baseline c
# +----------------------------+---------------------------------------------+---------------
# | Start of follow-up         | Date of randomisation (= date of T2DM       | Date of T2DM
# | (time zero)                | diagnosis).                                 | All eligible
```

```
# +-----------------------------+--------------------------------------------
# | Outcome                     | First major adverse cardiovascular event
# |                             | (MACE): composite of MI, stroke, or CV
# |                             | death within 5 years.
# |                             |
# |                             |
# +-----------------------------+--------------------------------------------
# | Causal contrast             | Intention-to-treat: effect of being
# |                             | assigned to early vs delayed initiation.
# |                             | Per-protocol: effect of adhering to the
# |                             | assigned strategy.
# +-----------------------------+--------------------------------------------
# | Analysis plan               | Cox proportional hazards model.
# |                             | ITT analysis is primary.
# |                             | Per-protocol as sensitivity analysis.
# |                             |
# |                             |
# +-----------------------------+--------------------------------------------

# --- Part (b): Sources of immortal time bias in a naive analysis ---
#
# In a naive analysis comparing "early initiators" (metformin within 3 months
# vs "delayed initiators" (3-12 months), several sources of immortal time
# bias can arise:
#
# 1. SURVIVAL REQUIREMENT FOR CLASSIFICATION:
#    To be classified as a "delayed initiator," a patient must survive at lea
#    3 months (to reach the delayed window). The time between T2DM diagnosis
#    and 3 months is "immortal" for the delayed group -- they cannot experien
#    the outcome during this period by definition. This artificially inflates
#    survival in the delayed group.
#
# 2. MISCLASSIFICATION OF PERSON-TIME:
```

```
#    If follow-up starts at T2DM diagnosis for both groups but treatment
#    classification depends on future events (whether and when metformin is
#    started), person-time before treatment initiation is misclassified.
#    A patient who starts metformin at month 2 contributes 2 months of
#    "untreated" time that is credited to the "early" group.
#
# 3. EXCLUSION OF NON-INITIATORS WHO DIE EARLY:
#    Patients who die before 3 months and never start metformin may be excluded
#    entirely if the study requires treatment initiation. This creates
#    survivorship bias.
#
# HOW TARGET TRIAL EMULATION AVOIDS THIS:
# - Time zero is aligned with eligibility (T2DM diagnosis), not treatment start
# - The clone-censor-weight approach assigns each patient to BOTH strategies
#   at time zero, censors them when they deviate from their assigned strategy,
#   and uses inverse probability of censoring weights (IPCW) to correct for
#   the informative censoring

# --- Part (c): How new-user, active comparator design addresses confounding ---
#
# The NEW-USER (INCIDENT USER) DESIGN:
# - Only includes patients at the TIME OF FIRST metformin prescription
#   (or the decision point: T2DM diagnosis)
# - Avoids "prevalent user bias" where including patients already on metformin
#   selectively includes those who tolerated and responded to the drug
#   (survivors of the initial treatment period)
# - Captures early effects (including side effects) that might be missed
#   in prevalent user analyses
#
# The ACTIVE COMPARATOR DESIGN:
# - Compares early metformin vs delayed metformin (not metformin vs no treatment)
# - Patients choosing delayed initiation are more comparable to early initiators
#   than patients who never initiate (who may differ fundamentally in disease
```

```
#   severity, healthcare access, or physician preferences)
# - Reduces confounding by indication because both groups are deemed to need
#   metformin -- the question is only about TIMING
# - Makes the positivity assumption more plausible: most patients have a
#   realistic probability of being in either group
# - Mimics the clinical question: "Should I start metformin now or wait?"

# --- Part (d): Unmeasured confounders and sensitivity analysis ---
#
# POTENTIAL UNMEASURED CONFOUNDERS:
#
# 1. HbA1c trajectory / diabetes severity:
#    Patients with more rapidly rising HbA1c may receive metformin earlier.
#    If severity also affects CVD risk, this confounds the comparison.
#    (May be partially captured in EHR but with measurement timing issues.)
#
# 2. Patient health literacy and adherence behaviour:
#    Patients who seek care promptly and fill prescriptions early may also
#    engage in other health-promoting behaviours (exercise, diet) that
#    independently reduce CVD risk. This is a form of "healthy user bias."
#
# 3. Physician prescribing patterns / quality of care:
#    Physicians who prescribe metformin early may also provide better
#    overall cardiovascular risk management.
#
# 4. Socioeconomic status / insurance:
#    Patients with better insurance or higher SES may fill prescriptions
#    faster and have better access to follow-up care.
#
# 5. Diet, exercise, and lifestyle factors:
#    Rarely captured in EHR data but strongly associated with both
#    treatment initiation patterns and CVD outcomes.
#
```

```
# SENSITIVITY ANALYSIS APPROACHES:
#
# 1. E-VALUE:
#    Calculate the E-value for the primary estimate to quantify how strong
#    an unmeasured confounder would need to be (in terms of associations with
#    both treatment and outcome) to fully explain away the observed effect.
#
# 2. QUANTITATIVE BIAS ANALYSIS:
#    Use external data (e.g., surveys with lifestyle data) to estimate the
#    likely magnitude of confounding by specific unmeasured factors. Apply
#    the bias formula to adjust the estimate.
#
# 3. NEGATIVE CONTROL OUTCOMES:
#    Test outcomes that should NOT be affected by metformin timing (e.g.,
#    accidental injuries). If an association is found, it suggests residual
#    confounding.
#
# 4. NEGATIVE CONTROL EXPOSURES:
#    Test treatments that share the same confounding structure but should
#    not affect CVD (e.g., timing of proton pump inhibitor initiation).
#
# 5. INSTRUMENTAL VARIABLE ANALYSIS:
#    If a valid instrument exists (e.g., physician preference for early
#    prescribing), IV analysis can estimate the causal effect even with
#    unmeasured confounding.

cat("Exercise 4 is a conceptual exercise.\n")
cat("All answers are provided as detailed comments in this script.\n")
cat("Review the comments for:\n")
cat("  (a) Complete target trial protocol table\n")
cat("  (b) Sources of immortal time bias\n")
cat("  (c) How new-user, active comparator design helps\n")
cat("  (d) Unmeasured confounders and sensitivity approaches\n")
```

D.13.4.2. Python

```
# ==============================================================================
# Chapter 17 - Exercise 4: Target Trial Emulation (Conceptual)
# Early vs delayed metformin initiation and 5-year cardiovascular events
# ==============================================================================

# This is a conceptual exercise. All answers are provided as detailed comment

# --- Part (a): Complete target trial protocol table ---
#
# Protocol Component     | Target Trial                          | Observatio
# -----------------------|---------------------------------------|-----------
# Eligibility criteria   | Adults aged 30-80 with new T2DM       | Same crite
#                        | diagnosis, no prior CVD, no prior      | ICD-10 E1
#                        | metformin use, eGFR >= 30, no          | no metfor
#                        | contraindications to metformin.        | eGFR >= 3
#                        |                                       |
# Treatment strategies   | Strategy 1: Initiate metformin        | Same. Defi
#                        | within 3 months of T2DM diagnosis.    | prescripti
#                        | Strategy 2: Do not initiate within     | T2DM diag
#                        | 3 months (delayed: 3-12 months).      |
#                        |                                       |
# Treatment assignment   | Random assignment at T2DM diagnosis.  | Not random
#                        |                                       | PS matchin
#                        |                                       |
# Start of follow-up     | Date of randomisation (= date of      | Date of T2
# (time zero)            | T2DM diagnosis).                      | All eligib
#                        |                                       | date, rega
#                        |                                       |
# Outcome                | First MACE (MI, stroke, or CV death)  | Same. MACE
#                        | within 5 years.                       | Censored a
#                        |                                       | follow-up
```

```
#                         |                                        |
# Causal contrast         | ITT: effect of assignment to early    | ITT: compare groups as as
#                         | vs delayed. Per-protocol: effect of   | Per-protocol: clone-censo
#                         | adhering to assignment.                |
#                         |                                        |
# Analysis plan           | Cox PH model; ITT primary.             | IPTW-weighted Cox PH for
#                         |                                        | Clone-censor-weight for p

# --- Part (b): Sources of immortal time bias ---
#
# 1. SURVIVAL REQUIREMENT FOR CLASSIFICATION:
#    To be classified as a "delayed initiator," a patient must survive at
#    least 3 months without starting metformin. This period is "immortal
#    time" -- the patient cannot die (by definition) during this window.
#    This artificially inflates survival in the delayed group.
#
# 2. MISCLASSIFICATION OF PERSON-TIME:
#    If follow-up starts at T2DM diagnosis but treatment classification
#    depends on future behavior, person-time before treatment initiation
#    is misclassified. The period before a patient's first metformin
#    prescription is attributed to the wrong treatment group.
#
# 3. EXCLUSION OF EARLY DEATHS:
#    Patients who die before initiating metformin may be excluded from
#    the analysis entirely, creating survivorship bias.
#
# HOW TARGET TRIAL EMULATION AVOIDS THIS:
# - Aligns time zero with eligibility (T2DM diagnosis), not treatment
# - Uses clone-censor-weight: clones each patient into both strategies,
#   censors when they deviate, and applies IPCW to correct for the
#   informative censoring introduced by this approach.

# --- Part (c): New-user, active comparator design ---
```

```
#
# NEW-USER (INCIDENT USER) DESIGN:
# - Includes patients only at the point of first treatment decision
# - Avoids prevalent user bias (selective survival of tolerators)
# - Captures early effects including initial side effects
# - Provides a clear "time zero" aligned with treatment decision
#
# ACTIVE COMPARATOR DESIGN:
# - Compares early metformin vs delayed metformin (not vs no treatment)
# - Patients in both groups are deemed to need metformin
# - Reduces confounding by indication: the comparison is about timing,
#   not about treatment vs no treatment
# - Improves positivity: most T2DM patients can plausibly be in either
#   timing group
# - Reduces healthy user bias: both groups seek treatment
# - Mimics the clinically relevant question

# --- Part (d): Unmeasured confounders and sensitivity analysis ---
#
# POTENTIAL UNMEASURED CONFOUNDERS:
# 1. Diabetes severity trajectory (HbA1c trends)
# 2. Health literacy and medication adherence behaviour
# 3. Physician quality / prescribing culture
# 4. Socioeconomic status / insurance coverage
# 5. Diet, exercise, and lifestyle factors
# 6. Patient preferences and shared decision-making
#
# SENSITIVITY ANALYSIS APPROACHES:
# 1. E-value: quantify minimum confounder strength to nullify the result
# 2. Quantitative bias analysis: use external data to model bias
# 3. Negative control outcomes: test outcomes unrelated to metformin
# 4. Negative control exposures: test unrelated treatments
# 5. Instrumental variable analysis: e.g., physician preference
```

```
print("=" * 70)
print("Exercise 4: Target Trial Emulation (Conceptual)")
print("=" * 70)
print()
print("All answers are provided as comments in this script.")
print("Review the comments for:")
print("  (a) Complete target trial protocol table")
print("  (b) Sources of immortal time bias in a naive analysis")
print("  (c) How new-user, active comparator design addresses confounding")
print("  (d) Unmeasured confounders and sensitivity analysis approaches")
print()
print("Key takeaway: Target trial emulation forces explicit specification")
print("of all protocol components, making assumptions transparent and")
print("reducing common biases like immortal time bias and prevalent user bias.")
```

D.14. Chapter 18: Meta-Analysis

D.14.1. Exercise 1

D.14.1.1. R

```
# ============================================================================
# Chapter 18 - Exercise 1: Basic Meta-Analysis in R
# New anticoagulant vs warfarin for stroke prevention in atrial fibrillation
# ============================================================================

library(tidyverse)
library(meta)
library(metafor)
```

```r
# --- Dataset ---
af_trials <- data.frame(
  study = c("TRAIL-1", "GUARD-AF", "SHIELD", "ORBIT-AF",
            "VENTURE", "COMPASS-AF", "PIONEER-2", "ATLAS-AF"),
  events_new = c(28, 45, 112, 67, 33, 89, 52, 41),
  n_new      = c(1200, 2500, 5400, 3100, 1800, 4200, 2800, 2100),
  events_warf = c(42, 58, 148, 84, 29, 102, 61, 53),
  n_warf      = c(1200, 2500, 5400, 3100, 1800, 4200, 2800, 2100)
)

# --- Part (a): Compute RR and 95% CI for each trial ---
# Using metafor's escalc function
es <- escalc(measure = "RR",
             ai = events_new, n1i = n_new,
             ci = events_warf, n2i = n_warf,
             data = af_trials)

# Add human-readable columns
es$RR <- exp(es$yi)
es$ci_lo <- exp(es$yi - 1.96 * sqrt(es$vi))
es$ci_hi <- exp(es$yi + 1.96 * sqrt(es$vi))

cat("Individual trial risk ratios:\n")
print(es[, c("study", "RR", "ci_lo", "ci_hi")], digits = 3)

# By hand verification for first trial
rr_trail1 <- (28/1200) / (42/1200)
se_trail1 <- sqrt(1/28 - 1/1200 + 1/42 - 1/1200)
cat("\nManual check (TRAIL-1):")
cat("\n  RR =", round(rr_trail1, 3))
cat("\n  95% CI: (", round(exp(log(rr_trail1) - 1.96*se_trail1), 3),
    ",", round(exp(log(rr_trail1) + 1.96*se_trail1), 3), ")\n")
```

```
# --- Part (b): Random-effects meta-analysis ---
# Using the meta package
m1 <- metabin(
  event.e = events_new, n.e = n_new,
  event.c = events_warf, n.c = n_warf,
  studlab = study,
  data = af_trials,
  sm = "RR",
  method.tau = "REML",
  prediction = TRUE
)

cat("\n--- Random-effects meta-analysis ---\n")
print(summary(m1))

# Also using metafor for comparison
res <- rma(yi, vi, data = es, method = "REML")
cat("\n--- metafor results ---\n")
print(summary(res))

# --- Part (c): Forest plot ---
forest(m1,
       sortvar = TE,
       label.left = "Favours new anticoagulant",
       label.right = "Favours warfarin",
       col.diamond = "steelblue",
       col.square = "darkblue",
       print.tau2 = TRUE,
       print.I2 = TRUE,
       print.pval.Q = TRUE,
       prediction = TRUE,
       main = "New Anticoagulant vs Warfarin: Stroke Prevention in AF")
```

```
cat("\n--- Does the pooled effect favour the new anticoagulant? ---\n")
pooled_rr <- exp(m1$TE.random)
cat("Pooled RR:", round(pooled_rr, 3), "\n")
cat("95% CI: (", round(exp(m1$lower.random), 3), ",",
    round(exp(m1$upper.random), 3), ")\n")
if (pooled_rr < 1 & exp(m1$upper.random) < 1) {
  cat("Yes, the pooled effect significantly favours the new anticoagulant.\n
} else if (pooled_rr < 1) {
  cat("The point estimate favours the new anticoagulant, but the CI includes
} else {
  cat("The pooled effect does not favour the new anticoagulant.\n")
}

# --- Part (d): I-squared and prediction interval ---
cat("\n--- Heterogeneity ---\n")
cat("I-squared:", round(m1$I2, 1), "%\n")
cat("tau-squared:", round(m1$tau2, 4), "\n")
cat("Q statistic:", round(m1$Q, 2), ", df =", m1$df.Q,
    ", p =", round(m1$pval.Q, 4), "\n")

cat("\n--- Prediction interval ---\n")
cat("Prediction interval for RR: (",
    round(exp(m1$lower.predict), 3), ",",
    round(exp(m1$upper.predict), 3), ")\n")

# Interpretation
if (m1$I2 < 25) {
  cat("I-squared suggests LOW heterogeneity.\n")
} else if (m1$I2 < 50) {
  cat("I-squared suggests MODERATE heterogeneity.\n")
} else if (m1$I2 < 75) {
  cat("I-squared suggests SUBSTANTIAL heterogeneity.\n")
} else {
```

```
  cat("I-squared suggests CONSIDERABLE heterogeneity.\n")
}

cat("\nThe prediction interval tells us the range within which the true\n")
cat("effect in a future study is expected to fall. If it crosses 1,\n")
cat("some settings might see no benefit from the new anticoagulant.\n")

# --- Part (e): Funnel plot and Egger's test ---
funnel(m1,
       xlab = "Risk Ratio (log scale)",
       studlab = TRUE,
       col = "steelblue",
       pch = 16,
       main = "Funnel Plot")

cat("\n--- Egger's test for funnel plot asymmetry ---\n")
egger <- metabias(m1, method.bias = "Egger")
print(egger)

if (egger$p.value < 0.05) {
  cat("Egger's test is significant (p < 0.05), suggesting potential\n")
  cat("publication bias or small-study effects.\n")
} else {
  cat("Egger's test is not significant, no strong evidence of\n")
  cat("publication bias. However, with only", length(af_trials$study),
      "studies,\n")
  cat("the test has limited power (recommended: >= 10 studies).\n")
}

# Trim-and-fill as additional check
tf <- trimfill(m1)
cat("\n--- Trim-and-fill ---\n")
cat("Imputed missing studies:", tf$k0, "\n")
```

```r
cat("Adjusted pooled RR:", round(exp(tf$TE.random), 3), "\n")

# --- Part (f): Leave-one-out sensitivity analysis ---
cat("\n--- Leave-one-out sensitivity analysis ---\n")
l1o <- leave1out(res)
print(l1o)

# Check if removing any single study changes the conclusion
cat("\nIs the result robust to removing any single study?\n")
all_sig <- all(exp(l1o$ci.lb) < 1 | exp(l1o$ci.ub) > 1)
# Check if all leave-one-out CIs exclude 1 (if pooled is significant)
if (exp(res$ci.ub) < 1) {
  robust <- all(exp(l1o$ci.ub) < 1)
} else {
  robust <- TRUE
}

if (robust) {
  cat("Yes - no single study removal changes the direction or significance.\
} else {
  cat("No - removing at least one study changes the conclusion.\n")
  cat("The result may be driven by specific influential studies.\n")
}

# Influence diagnostics
inf <- influence(res)
plot(inf, main = "Influence Diagnostics")
```

D.14.1.2. Python

```python
# ===========================================================================
# Chapter 18 - Exercise 1: Basic Meta-Analysis in Python
```

```python
# (Python version of the R Exercise 1 for completeness)
# New anticoagulant vs warfarin for stroke prevention in AF
# ==============================================================================

import numpy as np
import pandas as pd
from scipy import stats
import matplotlib.pyplot as plt

# --- Dataset ---
af_trials = pd.DataFrame({
    'study': ['TRAIL-1', 'GUARD-AF', 'SHIELD', 'ORBIT-AF',
              'VENTURE', 'COMPASS-AF', 'PIONEER-2', 'ATLAS-AF'],
    'events_new':  [28, 45, 112, 67, 33, 89, 52, 41],
    'n_new':       [1200, 2500, 5400, 3100, 1800, 4200, 2800, 2100],
    'events_warf': [42, 58, 148, 84, 29, 102, 61, 53],
    'n_warf':      [1200, 2500, 5400, 3100, 1800, 4200, 2800, 2100]
})

# --- Part (a): Compute log RR and variances ---
af_trials['rr'] = (af_trials['events_new'] / af_trials['n_new']) / \
                  (af_trials['events_warf'] / af_trials['n_warf'])
af_trials['log_rr'] = np.log(af_trials['rr'])
af_trials['var_log_rr'] = (
    1/af_trials['events_new'] - 1/af_trials['n_new'] +
    1/af_trials['events_warf'] - 1/af_trials['n_warf']
)
af_trials['se_log_rr'] = np.sqrt(af_trials['var_log_rr'])

# 95% CI for each study
af_trials['ci_lo'] = np.exp(af_trials['log_rr'] - 1.96 * af_trials['se_log_rr'])
af_trials['ci_hi'] = np.exp(af_trials['log_rr'] + 1.96 * af_trials['se_log_rr'])
```

```
print("Individual trial risk ratios:")
print(af_trials[['study', 'rr', 'ci_lo', 'ci_hi']].to_string(
    float_format=lambda x: f"{x:.3f}", index=False))

# --- Part (b): Fixed-effect (inverse variance) ---
w_fe = 1 / af_trials['var_log_rr']
pooled_fe = np.sum(w_fe * af_trials['log_rr']) / np.sum(w_fe)
se_fe = np.sqrt(1 / np.sum(w_fe))
ci_fe = (pooled_fe - 1.96*se_fe, pooled_fe + 1.96*se_fe)

print(f"\n--- Fixed-effect meta-analysis ---")
print(f"Pooled log RR: {pooled_fe:.4f}")
print(f"Pooled RR: {np.exp(pooled_fe):.4f} "
      f"(95% CI: {np.exp(ci_fe[0]):.4f} - {np.exp(ci_fe[1]):.4f})")

# --- Part (c): DerSimonian-Laird random-effects ---
k = len(af_trials)
Q = np.sum(w_fe * (af_trials['log_rr'] - pooled_fe)**2)
C = np.sum(w_fe) - np.sum(w_fe**2) / np.sum(w_fe)
tau2 = max(0, (Q - (k - 1)) / C)

w_re = 1 / (af_trials['var_log_rr'] + tau2)
pooled_re = np.sum(w_re * af_trials['log_rr']) / np.sum(w_re)
se_re = np.sqrt(1 / np.sum(w_re))
ci_re = (pooled_re - 1.96*se_re, pooled_re + 1.96*se_re)

# --- Part (d): Q, I-squared, tau-squared ---
I2 = max(0, (Q - (k-1)) / Q) * 100
p_Q = 1 - stats.chi2.cdf(Q, k-1)

print(f"\n--- Random-effects meta-analysis (DerSimonian-Laird) ---")
print(f"Pooled log RR: {pooled_re:.4f}")
print(f"Pooled RR: {np.exp(pooled_re):.4f} "
```

```
      f"(95% CI: {np.exp(ci_re[0]):.4f} - {np.exp(ci_re[1]):.4f})")
print(f"\nHeterogeneity:")
print(f"  tau^2: {tau2:.4f}")
print(f"  I^2: {I2:.1f}%")
print(f"  Q: {Q:.2f}, df = {k-1}, p = {p_Q:.4f}")

# Prediction interval
t_crit = stats.t.ppf(0.975, k - 2)
pi_lo = pooled_re - t_crit * np.sqrt(se_re**2 + tau2)
pi_hi = pooled_re + t_crit * np.sqrt(se_re**2 + tau2)
print(f"\nPrediction interval for RR: ({np.exp(pi_lo):.4f}, {np.exp(pi_hi):.4f})")

if np.exp(pi_lo) < 1 and np.exp(pi_hi) > 1:
    print("The prediction interval crosses 1, meaning some future settings")
    print("might not see a benefit from the new anticoagulant.")
else:
    print("The prediction interval does not cross 1.")

# --- Part (e): Forest plot ---
fig, ax = plt.subplots(figsize=(10, 8))

y_pos = list(range(k, 0, -1))
weights = w_re / w_re.max()

# Individual studies
for i, (_, row) in enumerate(af_trials.iterrows()):
    ci_lo_i = row['ci_lo']
    ci_hi_i = row['ci_hi']
    ax.plot([ci_lo_i, ci_hi_i], [y_pos[i], y_pos[i]], 'k-', linewidth=1)
    size = weights.iloc[i] * 200
    ax.scatter(row['rr'], y_pos[i], s=size, c='steelblue',
               zorder=5, edgecolors='darkblue')
```

```
# Null line
ax.axvline(x=1, color='black', linestyle='-', linewidth=0.5)

# Pooled estimate line
ax.axvline(x=np.exp(pooled_re), color='steelblue', linestyle='--', alpha=0.5

# Pooled diamond
diamond_y = 0
dx = [np.exp(ci_re[0]), np.exp(pooled_re), np.exp(ci_re[1]), np.exp(pooled_re
dy = [diamond_y, diamond_y + 0.3, diamond_y, diamond_y - 0.3]
ax.fill(dx, dy, color='steelblue', alpha=0.7)

ax.set_yticks(y_pos + [0])
ax.set_yticklabels(list(af_trials['study']) + ['Pooled RE'])
ax.set_xlabel('Risk Ratio')
ax.set_title('Forest Plot: New Anticoagulant vs Warfarin for AF')
ax.set_xscale('log')
plt.tight_layout()
plt.savefig('forest_plot_af.png', dpi=300)
plt.show()

# --- Part (f): Funnel plot and Egger's test ---
fig, ax = plt.subplots(figsize=(8, 6))

ax.scatter(af_trials['log_rr'], af_trials['se_log_rr'],
           c='steelblue', s=80, edgecolors='darkblue', zorder=5)
ax.axvline(x=pooled_re, color='grey', linestyle='--', alpha=0.7)

# Pseudo-confidence region
se_range = np.linspace(0, af_trials['se_log_rr'].max() * 1.1, 100)
ax.plot(pooled_re - 1.96*se_range, se_range, 'k--', alpha=0.3)
ax.plot(pooled_re + 1.96*se_range, se_range, 'k--', alpha=0.3)
```

```python
# Add study labels
for _, row in af_trials.iterrows():
    ax.annotate(row['study'], (row['log_rr'], row['se_log_rr']),
                textcoords="offset points", xytext=(5, 5), fontsize=7)

ax.invert_yaxis()
ax.set_xlabel('Log Risk Ratio')
ax.set_ylabel('Standard Error')
ax.set_title('Funnel Plot')
plt.tight_layout()
plt.savefig('funnel_plot_af.png', dpi=300)
plt.show()

# Egger's test: weighted regression of effect on SE
import statsmodels.api as sm

X_egger = sm.add_constant(af_trials['se_log_rr'])
egger_model = sm.WLS(af_trials['log_rr'], X_egger, weights=w_fe).fit()

print(f"\n--- Egger's test ---")
print(f"Intercept: {egger_model.params.iloc[0]:.4f}")
print(f"p-value: {egger_model.pvalues.iloc[0]:.4f}")

if egger_model.pvalues.iloc[0] < 0.05:
    print("Egger's test is significant, suggesting potential publication bias.")
else:
    print("Egger's test is not significant.")
    print(f"Note: With only {k} studies, the test has limited power.")

# --- Leave-one-out sensitivity analysis ---
print(f"\n--- Leave-one-out sensitivity analysis ---")
print(f"{'Excluded study':<15} {'Pooled RR':>10} {'95% CI':>25} {'I2':>8}")
print("-" * 60)
```

```
for i in range(k):
    # Exclude study i
    mask = af_trials.index != i
    w_i = 1 / af_trials.loc[mask, 'var_log_rr']

    # Fixed-effect pooled
    pooled_i_fe = np.sum(w_i * af_trials.loc[mask, 'log_rr']) / np.sum(w_i)

    # Q and tau2
    Q_i = np.sum(w_i * (af_trials.loc[mask, 'log_rr'] - pooled_i_fe)**2)
    C_i = np.sum(w_i) - np.sum(w_i**2) / np.sum(w_i)
    tau2_i = max(0, (Q_i - (k - 2)) / C_i)

    # Random-effects
    w_re_i = 1 / (af_trials.loc[mask, 'var_log_rr'] + tau2_i)
    pooled_i = np.sum(w_re_i * af_trials.loc[mask, 'log_rr']) / np.sum(w_re_i
    se_i = np.sqrt(1 / np.sum(w_re_i))
    ci_i = (pooled_i - 1.96*se_i, pooled_i + 1.96*se_i)
    I2_i = max(0, (Q_i - (k-2)) / Q_i) * 100

    print(f"{af_trials.loc[i, 'study']:<15} {np.exp(pooled_i):>10.4f} "
          f"({np.exp(ci_i[0]):.4f}, {np.exp(ci_i[1]):.4f})  {I2_i:>6.1f}%")

print("\nIf removing any single study changes the conclusion (CI crosses 1),"
print("the result is sensitive to that study.")
```

D.14.2. Exercise 2

D.14.2.1. R

```r
# ==============================================================================
# Chapter 18 - Exercise 2: Meta-Analysis from Scratch in R
# (Exercise 2 is Python-focused, but we provide an R version as well)
# Manual implementation of meta-analysis computations
# ==============================================================================

library(tidyverse)

# --- Dataset (same as Exercise 1) ---
af_trials <- data.frame(
  study = c("TRAIL-1", "GUARD-AF", "SHIELD", "ORBIT-AF",
            "VENTURE", "COMPASS-AF", "PIONEER-2", "ATLAS-AF"),
  events_new  = c(28, 45, 112, 67, 33, 89, 52, 41),
  n_new       = c(1200, 2500, 5400, 3100, 1800, 4200, 2800, 2100),
  events_warf = c(42, 58, 148, 84, 29, 102, 61, 53),
  n_warf      = c(1200, 2500, 5400, 3100, 1800, 4200, 2800, 2100)
)

# --- Part (a): Compute log RR and variances by hand ---
af_trials$rr <- (af_trials$events_new / af_trials$n_new) /
                (af_trials$events_warf / af_trials$n_warf)
af_trials$log_rr <- log(af_trials$rr)
af_trials$var_log_rr <- (1/af_trials$events_new - 1/af_trials$n_new +
                         1/af_trials$events_warf - 1/af_trials$n_warf)
af_trials$se_log_rr <- sqrt(af_trials$var_log_rr)

cat("Log risk ratios and SEs:\n")
print(af_trials[, c("study", "log_rr", "se_log_rr", "rr")])

# --- Part (b): Fixed-effect inverse-variance method ---
w_fe <- 1 / af_trials$var_log_rr
pooled_fe <- sum(w_fe * af_trials$log_rr) / sum(w_fe)
se_fe <- sqrt(1 / sum(w_fe))
```

```
ci_fe_lo <- pooled_fe - 1.96 * se_fe
ci_fe_hi <- pooled_fe + 1.96 * se_fe
z_fe <- pooled_fe / se_fe
p_fe <- 2 * (1 - pnorm(abs(z_fe)))

cat("\n--- Fixed-effect meta-analysis (by hand) ---\n")
cat("Pooled log RR:", round(pooled_fe, 4), "\n")
cat("Pooled RR:", round(exp(pooled_fe), 4),
    "(95% CI:", round(exp(ci_fe_lo), 4), "-", round(exp(ci_fe_hi), 4), ")\n")
cat("z =", round(z_fe, 3), ", p =", round(p_fe, 4), "\n")

# --- Part (c): DerSimonian-Laird random-effects ---
k <- nrow(af_trials)

# Cochran's Q
Q <- sum(w_fe * (af_trials$log_rr - pooled_fe)^2)

# C constant
C_val <- sum(w_fe) - sum(w_fe^2) / sum(w_fe)

# Between-study variance
tau2 <- max(0, (Q - (k - 1)) / C_val)

# Random-effects weights
w_re <- 1 / (af_trials$var_log_rr + tau2)
pooled_re <- sum(w_re * af_trials$log_rr) / sum(w_re)
se_re <- sqrt(1 / sum(w_re))
ci_re_lo <- pooled_re - 1.96 * se_re
ci_re_hi <- pooled_re + 1.96 * se_re
z_re <- pooled_re / se_re
p_re <- 2 * (1 - pnorm(abs(z_re)))

cat("\n--- Random-effects meta-analysis (DerSimonian-Laird, by hand) ---\n")
```

```r
cat("Pooled log RR:", round(pooled_re, 4), "\n")
cat("Pooled RR:", round(exp(pooled_re), 4),
    "(95% CI:", round(exp(ci_re_lo), 4), "-", round(exp(ci_re_hi), 4), ")\n")
cat("z =", round(z_re, 3), ", p =", round(p_re, 4), "\n")

# --- Part (d): Q, I-squared, tau-squared ---
p_Q <- 1 - pchisq(Q, df = k - 1)
I2 <- max(0, (Q - (k - 1)) / Q) * 100

cat("\n--- Heterogeneity statistics ---\n")
cat("Q =", round(Q, 2), ", df =", k - 1, ", p =", round(p_Q, 4), "\n")
cat("tau^2 =", round(tau2, 4), "\n")
cat("I^2 =", round(I2, 1), "%\n")

# Prediction interval
t_crit <- qt(0.975, df = k - 2)
pi_lo <- pooled_re - t_crit * sqrt(se_re^2 + tau2)
pi_hi <- pooled_re + t_crit * sqrt(se_re^2 + tau2)
cat("Prediction interval for RR: (", round(exp(pi_lo), 4),
    ",", round(exp(pi_hi), 4), ")\n")

# --- Part (e): Forest plot (base R, no packages) ---
af_trials$w_re <- w_re
af_trials$ci_lo <- exp(af_trials$log_rr - 1.96 * af_trials$se_log_rr)
af_trials$ci_hi <- exp(af_trials$log_rr + 1.96 * af_trials$se_log_rr)

par(mar = c(5, 10, 4, 2))
y_pos <- k:1

plot(af_trials$rr, y_pos,
     xlim = c(0.3, 2.0), ylim = c(-0.5, k + 0.5),
     pch = 15, cex = af_trials$w_re / max(af_trials$w_re) * 2,
     xlab = "Risk Ratio", ylab = "", yaxt = "n",
```

```
     main = "Forest Plot (Built from Scratch)",
     log = "x", col = "steelblue")

# Study CIs
segments(af_trials$ci_lo, y_pos, af_trials$ci_hi, y_pos, lwd = 1.5)

# Study labels
axis(2, at = y_pos, labels = af_trials$study, las = 1)

# Null line
abline(v = 1, lty = 2, col = "grey50")

# Pooled estimate diamond
diamond_x <- c(exp(ci_re_lo), exp(pooled_re), exp(ci_re_hi), exp(pooled_re))
diamond_y <- c(0, 0.3, 0, -0.3)
polygon(diamond_x, diamond_y, col = "steelblue", border = "darkblue")

# Pooled reference line
abline(v = exp(pooled_re), col = "steelblue", lty = 3, lwd = 1)

# --- Part (f): Funnel plot and Egger's test (by hand) ---
par(mar = c(5, 5, 4, 2))
plot(af_trials$log_rr, af_trials$se_log_rr,
     pch = 16, col = "steelblue", cex = 1.5,
     xlim = c(min(af_trials$log_rr) - 0.3, max(af_trials$log_rr) + 0.3),
     ylim = rev(c(0, max(af_trials$se_log_rr) * 1.1)),
     xlab = "Log Risk Ratio", ylab = "Standard Error",
     main = "Funnel Plot (Built from Scratch)")

abline(v = pooled_re, col = "grey50", lty = 2)

# Pseudo-CI region
se_seq <- seq(0, max(af_trials$se_log_rr) * 1.1, length.out = 100)
```

```r
lines(pooled_re - 1.96 * se_seq, se_seq, lty = 3, col = "grey70")
lines(pooled_re + 1.96 * se_seq, se_seq, lty = 3, col = "grey70")

# Egger's test: weighted regression of standardised effect on precision
# Regression of yi/sei on 1/sei, weighted by 1/vi
# Equivalent: regress yi on sei, weighted by 1/vi
egger_fit <- lm(log_rr ~ se_log_rr, data = af_trials, weights = w_fe)
cat("\n--- Egger's test (by hand) ---\n")
cat("Intercept:", round(coef(egger_fit)[1], 4), "\n")
cat("SE:", round(summary(egger_fit)$coefficients[1, 2], 4), "\n")
cat("p-value:", round(summary(egger_fit)$coefficients[1, 4], 4), "\n")

if (summary(egger_fit)$coefficients[1, 4] < 0.05) {
  cat("Significant asymmetry detected.\n")
} else {
  cat("No significant asymmetry (limited power with", k, "studies).\n")
}
```

D.14.2.2. Python

```python
# ==============================================================================
# Chapter 18 - Exercise 2: Meta-Analysis from Scratch in Python
# New anticoagulant vs warfarin for stroke prevention in AF
# ==============================================================================

import numpy as np
import pandas as pd
from scipy import stats
import statsmodels.api as sm
import matplotlib.pyplot as plt

# --- Dataset ---
```

```
af_trials = pd.DataFrame({
    'study': ['TRAIL-1', 'GUARD-AF', 'SHIELD', 'ORBIT-AF',
              'VENTURE', 'COMPASS-AF', 'PIONEER-2', 'ATLAS-AF'],
    'events_new':  [28, 45, 112, 67, 33, 89, 52, 41],
    'n_new':       [1200, 2500, 5400, 3100, 1800, 4200, 2800, 2100],
    'events_warf': [42, 58, 148, 84, 29, 102, 61, 53],
    'n_warf':      [1200, 2500, 5400, 3100, 1800, 4200, 2800, 2100]
})

# --- Part (a): Compute log risk ratios and variances ---
af_trials['rr'] = (af_trials['events_new'] / af_trials['n_new']) / \
                  (af_trials['events_warf'] / af_trials['n_warf'])
af_trials['log_rr'] = np.log(af_trials['rr'])

# Variance of log RR: 1/a - 1/n1 + 1/c - 1/n2
af_trials['var_log_rr'] = (
    1/af_trials['events_new'] - 1/af_trials['n_new'] +
    1/af_trials['events_warf'] - 1/af_trials['n_warf']
)
af_trials['se_log_rr'] = np.sqrt(af_trials['var_log_rr'])

print("Part (a): Log risk ratios and standard errors")
print(af_trials[['study', 'log_rr', 'se_log_rr', 'rr']].to_string(
    float_format=lambda x: f"{x:.4f}", index=False))

# --- Part (b): Fixed-effect inverse-variance method ---
w_fe = 1 / af_trials['var_log_rr']
pooled_fe = np.sum(w_fe * af_trials['log_rr']) / np.sum(w_fe)
se_fe = np.sqrt(1 / np.sum(w_fe))
ci_fe = (pooled_fe - 1.96 * se_fe, pooled_fe + 1.96 * se_fe)
z_fe = pooled_fe / se_fe
p_fe = 2 * (1 - stats.norm.cdf(abs(z_fe)))
```

```
print(f"\n--- Part (b): Fixed-effect meta-analysis ---")
print(f"Pooled log RR: {pooled_fe:.4f}")
print(f"Pooled RR: {np.exp(pooled_fe):.4f} "
      f"(95% CI: {np.exp(ci_fe[0]):.4f} - {np.exp(ci_fe[1]):.4f})")
print(f"z = {z_fe:.3f}, p = {p_fe:.4f}")

# Study weights (as percentages)
af_trials['weight_fe_pct'] = (w_fe / w_fe.sum() * 100).round(1)
print(f"\nStudy weights (fixed-effect):")
print(af_trials[['study', 'weight_fe_pct']].to_string(index=False))

# --- Part (c): DerSimonian-Laird random-effects ---
k = len(af_trials)

# Step 1: Cochran's Q statistic
Q = np.sum(w_fe * (af_trials['log_rr'] - pooled_fe)**2)

# Step 2: C constant
C = np.sum(w_fe) - np.sum(w_fe**2) / np.sum(w_fe)

# Step 3: Between-study variance tau^2
tau2 = max(0, (Q - (k - 1)) / C)

# Step 4: Random-effects weights
w_re = 1 / (af_trials['var_log_rr'] + tau2)
pooled_re = np.sum(w_re * af_trials['log_rr']) / np.sum(w_re)
se_re = np.sqrt(1 / np.sum(w_re))
ci_re = (pooled_re - 1.96 * se_re, pooled_re + 1.96 * se_re)
z_re = pooled_re / se_re
p_re = 2 * (1 - stats.norm.cdf(abs(z_re)))

print(f"\n--- Part (c): Random-effects meta-analysis (DerSimonian-Laird) ---")
print(f"Pooled log RR: {pooled_re:.4f}")
```

```
print(f"Pooled RR: {np.exp(pooled_re):.4f} "
      f"(95% CI: {np.exp(ci_re[0]):.4f} - {np.exp(ci_re[1]):.4f})")
print(f"z = {z_re:.3f}, p = {p_re:.4f}")

af_trials['weight_re_pct'] = (w_re / w_re.sum() * 100).round(1)
print(f"\nStudy weights (random-effects):")
print(af_trials[['study', 'weight_fe_pct', 'weight_re_pct']].to_string(index=

# --- Part (d): Q, I-squared, tau-squared ---
p_Q = 1 - stats.chi2.cdf(Q, k - 1)
I2 = max(0, (Q - (k - 1)) / Q) * 100

print(f"\n--- Part (d): Heterogeneity statistics ---")
print(f"Q = {Q:.2f}, df = {k-1}, p = {p_Q:.4f}")
print(f"tau^2 = {tau2:.4f}")
print(f"tau = {np.sqrt(tau2):.4f}")
print(f"I^2 = {I2:.1f}%")

if I2 < 25:
    print("Interpretation: LOW heterogeneity")
elif I2 < 50:
    print("Interpretation: MODERATE heterogeneity")
elif I2 < 75:
    print("Interpretation: SUBSTANTIAL heterogeneity")
else:
    print("Interpretation: CONSIDERABLE heterogeneity")

# Prediction interval
t_crit = stats.t.ppf(0.975, k - 2)
pi_lo = pooled_re - t_crit * np.sqrt(se_re**2 + tau2)
pi_hi = pooled_re + t_crit * np.sqrt(se_re**2 + tau2)
print(f"\nPrediction interval for RR: ({np.exp(pi_lo):.4f}, {np.exp(pi_hi):.4
```

```
# --- Part (e): Forest plot ---
fig, ax = plt.subplots(figsize=(10, 8))

y_pos = list(range(k, 0, -1))
weights_norm = w_re / w_re.max()

for i, (_, row) in enumerate(af_trials.iterrows()):
    ci_lo_i = np.exp(row['log_rr'] - 1.96 * row['se_log_rr'])
    ci_hi_i = np.exp(row['log_rr'] + 1.96 * row['se_log_rr'])

    # CI line
    ax.plot([ci_lo_i, ci_hi_i], [y_pos[i], y_pos[i]], 'k-', linewidth=1)

    # Study point (size proportional to weight)
    size = weights_norm.iloc[i] * 200
    ax.scatter(row['rr'], y_pos[i], s=size, c='steelblue',
               zorder=5, edgecolors='darkblue')

    # Weight annotation
    ax.text(2.0, y_pos[i], f"{row['weight_re_pct']:.1f}%",
            ha='left', va='center', fontsize=9)

# Null line
ax.axvline(x=1, color='black', linestyle='-', linewidth=0.5)

# Pooled estimate line
ax.axvline(x=np.exp(pooled_re), color='steelblue', linestyle='--', alpha=0.5)

# Pooled diamond
diamond_x = [np.exp(ci_re[0]), np.exp(pooled_re),
             np.exp(ci_re[1]), np.exp(pooled_re)]
diamond_y = [0, 0.3, 0, -0.3]
ax.fill(diamond_x, diamond_y, color='steelblue', alpha=0.7)
```

```
# Labels
ax.set_yticks(y_pos + [0])
ax.set_yticklabels(list(af_trials['study']) + ['Pooled RE'])
ax.set_xlabel('Risk Ratio')
ax.set_title('Forest Plot: New Anticoagulant vs Warfarin for AF\n'
             f'RE Model: RR = {np.exp(pooled_re):.3f} '
             f'(95% CI: {np.exp(ci_re[0]):.3f}-{np.exp(ci_re[1]):.3f}), '
             f'I\u00B2 = {I2:.1f}%')
ax.set_xscale('log')

# Add "Favours" labels
ax.text(0.5, -1.2, 'Favours new anticoagulant', ha='center',
        fontsize=9, style='italic')
ax.text(1.8, -1.2, 'Favours warfarin', ha='center',
        fontsize=9, style='italic')

plt.tight_layout()
plt.savefig('forest_plot_af.png', dpi=300)
plt.show()

# --- Part (f): Funnel plot and Egger's test ---
fig, ax = plt.subplots(figsize=(8, 6))

ax.scatter(af_trials['log_rr'], af_trials['se_log_rr'],
           c='steelblue', s=80, edgecolors='darkblue', zorder=5)
ax.axvline(x=pooled_re, color='grey', linestyle='--', alpha=0.7,
           label='Pooled effect')

# Pseudo-confidence regions
se_range = np.linspace(0, af_trials['se_log_rr'].max() * 1.1, 100)
ax.plot(pooled_re - 1.96*se_range, se_range, 'k--', alpha=0.3)
ax.plot(pooled_re + 1.96*se_range, se_range, 'k--', alpha=0.3)
```

```python
# Add study labels
for _, row in af_trials.iterrows():
    ax.annotate(row['study'], (row['log_rr'], row['se_log_rr']),
                textcoords="offset points", xytext=(5, 5), fontsize=7)

ax.invert_yaxis()
ax.set_xlabel('Log Risk Ratio')
ax.set_ylabel('Standard Error')
ax.set_title('Funnel Plot: Checking for Publication Bias')
ax.legend()
plt.tight_layout()
plt.savefig('funnel_plot_af.png', dpi=300)
plt.show()

# Egger's regression test
# Weighted regression of effect estimate on standard error
X_egger = sm.add_constant(af_trials['se_log_rr'])
egger_model = sm.WLS(af_trials['log_rr'], X_egger,
                     weights=1/af_trials['var_log_rr']).fit()

print(f"\n--- Part (f): Egger's test ---")
print(f"Intercept: {egger_model.params.iloc[0]:.4f}")
print(f"SE of intercept: {egger_model.bse.iloc[0]:.4f}")
print(f"t-statistic: {egger_model.tvalues.iloc[0]:.3f}")
print(f"p-value: {egger_model.pvalues.iloc[0]:.4f}")

if egger_model.pvalues.iloc[0] < 0.05:
    print("Egger's test is significant (p < 0.05).")
    print("This suggests potential publication bias or small-study effects.")
else:
    print("Egger's test is not significant.")
    print(f"Note: With only {k} studies, the test has limited power.")
    print("At least 10 studies are recommended for reliable Egger's test.")
```

D.14.3. Exercise 3

D.14.3.1. R

```
# ============================================================================
# Chapter 18 - Exercise 3: Subgroup Analysis and Meta-Regression in R
# ============================================================================

library(tidyverse)
library(meta)
library(metafor)

# --- Dataset ---
af_trials <- data.frame(
  study = c("TRAIL-1", "GUARD-AF", "SHIELD", "ORBIT-AF",
            "VENTURE", "COMPASS-AF", "PIONEER-2", "ATLAS-AF"),
  events_new  = c(28, 45, 112, 67, 33, 89, 52, 41),
  n_new       = c(1200, 2500, 5400, 3100, 1800, 4200, 2800, 2100),
  events_warf = c(42, 58, 148, 84, 29, 102, 61, 53),
  n_warf      = c(1200, 2500, 5400, 3100, 1800, 4200, 2800, 2100),
  region      = c("Europe", "North America", "Europe", "Asia",
                  "North America", "Europe", "Asia", "North America"),
  mean_age    = c(72, 68, 74, 65, 70, 71, 63, 69),
  pct_female  = c(38, 42, 35, 48, 40, 37, 52, 44)
)

# Compute effect sizes using metafor
es <- escalc(measure = "RR",
             ai = events_new, n1i = n_new,
             ci = events_warf, n2i = n_warf,
             data = af_trials)

# Base meta-analysis
```

```r
m1 <- metabin(
  event.e = events_new, n.e = n_new,
  event.c = events_warf, n.c = n_warf,
  studlab = study, data = af_trials,
  sm = "RR", method.tau = "REML"
)

# --- Part (a): Subgroup analysis by region ---
cat("--- Part (a): Subgroup analysis by region ---\n\n")

# Using meta package
m_sub <- update(m1, subgroup = af_trials$region)
cat("Subgroup analysis results:\n")
print(summary(m_sub))

# Forest plot by subgroup
forest(m_sub,
       sortvar = TE,
       label.left = "Favours new anticoagulant",
       label.right = "Favours warfarin",
       col.diamond = "steelblue",
       print.tau2 = TRUE,
       print.I2 = TRUE,
       print.subgroup.name = TRUE,
       main = "Subgroup Analysis by Region")

# Using metafor for the test of subgroup differences
res_sub <- rma(yi, vi, mods = ~ region, data = es, method = "REML")
cat("\nTest for subgroup differences (metafor):\n")
print(summary(res_sub))

cat("\nDo treatment effects differ by region?\n")
cat("Test for moderation: QM =", round(res_sub$QM, 2),
```

```
    ", df =", res_sub$m, ", p =", round(res_sub$QMp, 4), "\n")
if (res_sub$QMp < 0.05) {
  cat("Yes, there is a statistically significant difference between regions.\
} else {
  cat("No, there is no statistically significant difference between regions.\
  cat("However, with only", nrow(es), "studies split across",
      length(unique(es$region)), "regions,\n")
  cat("statistical power for detecting subgroup differences is limited.\n")
}

# --- Part (b): Meta-regression with mean age ---
cat("\n--- Part (b): Meta-regression with mean age ---\n\n")

res_age <- rma(yi, vi, mods = ~ mean_age, data = es, method = "REML")
print(summary(res_age))

cat("\nInterpretation:\n")
cat("Coefficient for mean_age:", round(coef(res_age)["mean_age"], 4), "\n")
cat("p-value:", round(res_age$pval[2], 4), "\n")
cat("R^2 (proportion of heterogeneity explained):",
    round(max(0, res_age$R2), 1), "%\n")

if (res_age$pval[2] < 0.05) {
  cat("There IS a statistically significant relationship between mean age\n")
  cat("and treatment effect. Each 1-year increase in mean age is associated\n
  cat("with a", round(coef(res_age)["mean_age"], 4),
      "change in log RR.\n")
} else {
  cat("There is NO statistically significant relationship between mean age\n
  cat("and treatment effect (p =", round(res_age$pval[2], 3), ").\n")
}

# --- Part (c): Meta-regression with percentage female ---
```

```r
cat("\n--- Part (c): Meta-regression with percentage female ---\n\n")

res_fem <- rma(yi, vi, mods = ~ pct_female, data = es, method = "REML")
print(summary(res_fem))

cat("\nInterpretation:\n")
cat("Coefficient for pct_female:", round(coef(res_fem)["pct_female"], 4), "\n")
cat("p-value:", round(res_fem$pval[2], 4), "\n")
cat("R^2:", round(max(0, res_fem$R2), 1), "%\n")

cat("\n*** ECOLOGICAL FALLACY WARNING ***\n")
cat("Even if there is an association between trial-level percentage female\n")
cat("and the treatment effect, this does NOT prove that individual women\n")
cat("respond differently to the treatment than individual men.\n")
cat("Trials with higher percentage female may differ in other ways:\n")
cat("  - Geographic region (cultural differences in enrolment)\n")
cat("  - Mean age (women often present with AF at older ages)\n")
cat("  - Comorbidity profiles\n")
cat("  - Trial methodology\n")
cat("Only an individual participant data (IPD) meta-analysis with a\n")
cat("treatment-by-sex interaction term can properly assess whether\n")
cat("sex modifies the treatment effect.\n")

# --- Part (d): Bubble plot of meta-regression on mean age ---
cat("\n--- Part (d): Bubble plot ---\n")

# Method 1: Using metafor's built-in bubble plot
regplot(res_age,
        xlab = "Mean Age (years)",
        ylab = "Log Risk Ratio",
        main = "Meta-Regression: Treatment Effect vs Mean Age",
        ci = TRUE,
        pi = TRUE,  # prediction interval
```

```
        col = "steelblue",
        bg = "lightblue",
        las = 1)

# Method 2: Using ggplot2 for more control
es$w_re <- 1 / (es$vi + res_age$tau2)
es$w_norm <- es$w_re / max(es$w_re)

# Predicted line from meta-regression
age_seq <- seq(min(es$mean_age) - 2, max(es$mean_age) + 2, length.out = 100)
pred <- predict(res_age, newmods = age_seq)

pred_df <- data.frame(
  mean_age = age_seq,
  pred = pred$pred,
  ci_lo = pred$ci.lb,
  ci_hi = pred$ci.ub,
  pi_lo = pred$pi.lb,
  pi_hi = pred$pi.ub
)

p_bubble <- ggplot() +
  # Prediction interval
  geom_ribbon(data = pred_df, aes(x = mean_age, ymin = pi_lo, ymax = pi_hi),
              fill = "grey90", alpha = 0.5) +
  # Confidence interval for regression line
  geom_ribbon(data = pred_df, aes(x = mean_age, ymin = ci_lo, ymax = ci_hi),
              fill = "steelblue", alpha = 0.2) +
  # Regression line
  geom_line(data = pred_df, aes(x = mean_age, y = pred),
            colour = "steelblue", linewidth = 1) +
  # Study points (bubbles)
  geom_point(data = es, aes(x = mean_age, y = yi, size = w_norm),
```

```
              colour = "steelblue", alpha = 0.7) +
  # Study labels
  geom_text(data = es, aes(x = mean_age, y = yi, label = study),
            size = 2.5, vjust = -1.5) +
  # Reference line at null
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
  # Aesthetics
  scale_size_continuous(range = c(3, 10), guide = "none") +
  labs(x = "Mean Age (years)",
       y = "Log Risk Ratio",
       title = "Bubble Plot: Meta-Regression of Treatment Effect on Mean Age",
       subtitle = paste0("Slope = ", round(coef(res_age)["mean_age"], 4),
                         ", p = ", round(res_age$pval[2], 3),
                         "; bubble size proportional to study weight")) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 12),
    axis.title = element_text(face = "bold")
  )

print(p_bubble)

ggsave("bubble_plot_age.png", p_bubble, width = 8, height = 6, dpi = 300)

cat("\nBubble plot saved. Larger bubbles indicate more precise studies.\n")
cat("The regression line shows the relationship between mean age\n")
cat("and treatment effect, with shaded confidence and prediction intervals.\n")
```

D.14.3.2. Python

```
# ==============================================================================
# Chapter 18 - Exercise 3: Subgroup Analysis and Meta-Regression in Python
```

```
# ===========================================================================

import numpy as np
import pandas as pd
from scipy import stats
import statsmodels.api as sm
import matplotlib.pyplot as plt

# --- Dataset ---
af_trials = pd.DataFrame({
    'study': ['TRAIL-1', 'GUARD-AF', 'SHIELD', 'ORBIT-AF',
              'VENTURE', 'COMPASS-AF', 'PIONEER-2', 'ATLAS-AF'],
    'events_new':  [28, 45, 112, 67, 33, 89, 52, 41],
    'n_new':       [1200, 2500, 5400, 3100, 1800, 4200, 2800, 2100],
    'events_warf': [42, 58, 148, 84, 29, 102, 61, 53],
    'n_warf':      [1200, 2500, 5400, 3100, 1800, 4200, 2800, 2100],
    'region':      ['Europe', 'North America', 'Europe', 'Asia',
                    'North America', 'Europe', 'Asia', 'North America'],
    'mean_age':    [72, 68, 74, 65, 70, 71, 63, 69],
    'pct_female':  [38, 42, 35, 48, 40, 37, 52, 44]
})

# Compute effect sizes
af_trials['rr'] = (af_trials['events_new'] / af_trials['n_new']) / \
                  (af_trials['events_warf'] / af_trials['n_warf'])
af_trials['log_rr'] = np.log(af_trials['rr'])
af_trials['var_log_rr'] = (
    1/af_trials['events_new'] - 1/af_trials['n_new'] +
    1/af_trials['events_warf'] - 1/af_trials['n_warf']
)
af_trials['se_log_rr'] = np.sqrt(af_trials['var_log_rr'])
```

```
def dl_random_effects(log_rr, var):
    """DerSimonian-Laird random-effects meta-analysis."""
    k = len(log_rr)
    w_fe = 1 / var
    pooled_fe = np.sum(w_fe * log_rr) / np.sum(w_fe)
    Q = np.sum(w_fe * (log_rr - pooled_fe)**2)
    C = np.sum(w_fe) - np.sum(w_fe**2) / np.sum(w_fe)
    tau2 = max(0, (Q - (k - 1)) / C)
    w_re = 1 / (var + tau2)
    pooled_re = np.sum(w_re * log_rr) / np.sum(w_re)
    se_re = np.sqrt(1 / np.sum(w_re))
    I2 = max(0, (Q - (k-1)) / Q) * 100 if Q > 0 else 0
    return pooled_re, se_re, tau2, Q, I2, w_re


# Overall random-effects analysis
pooled_re, se_re, tau2, Q, I2, w_re = dl_random_effects(
    af_trials['log_rr'].values, af_trials['var_log_rr'].values
)

# --- Part (a): Subgroup analysis by region ---
print("=" * 60)
print("Part (a): Subgroup Analysis by Region")
print("=" * 60)

regions = af_trials['region'].unique()
subgroup_results = []

for region in regions:
    mask = af_trials['region'] == region
    sub = af_trials[mask]

    if len(sub) >= 2:
```

```
        p_re, s_re, t2, q, i2, w = dl_random_effects(
            sub['log_rr'].values, sub['var_log_rr'].values
        )
        ci = (p_re - 1.96*s_re, p_re + 1.96*s_re)
    else:
        # Single study: use study estimate directly
        p_re = sub['log_rr'].values[0]
        s_re = sub['se_log_rr'].values[0]
        t2, q, i2 = 0, 0, 0
        ci = (p_re - 1.96*s_re, p_re + 1.96*s_re)

    subgroup_results.append({
        'region': region,
        'k': len(sub),
        'pooled_rr': np.exp(p_re),
        'ci_lo': np.exp(ci[0]),
        'ci_hi': np.exp(ci[1]),
        'I2': i2,
        'tau2': t2
    })

    print(f"\n{region} (k={len(sub)}):")
    print(f"  Pooled RR: {np.exp(p_re):.4f} "
          f"(95% CI: {np.exp(ci[0]):.4f} - {np.exp(ci[1]):.4f})")
    print(f"  I^2: {i2:.1f}%")

# Test for subgroup differences using meta-regression with region dummies
region_dummies = pd.get_dummies(af_trials['region'], drop_first=True).astype
X_sub = sm.add_constant(region_dummies)
wls_sub = sm.WLS(af_trials['log_rr'], X_sub,
                 weights=1/af_trials['var_log_rr']).fit()

# Wald test for region coefficients
```

```
from scipy.stats import chi2
f_stat = wls_sub.f_test(np.eye(len(wls_sub.params))[1:])
print(f"\nTest for subgroup differences:")
print(f"  F-statistic: {f_stat.fvalue[0][0]:.3f}")
print(f"  p-value: {f_stat.pvalue:.4f}")

if f_stat.pvalue < 0.05:
    print("  Significant difference between regions.")
else:
    print("  No significant difference between regions.")
    print(f"  (Limited power with only {len(af_trials)} studies)")

# --- Part (b): Meta-regression with mean age ---
print(f"\n{'=' * 60}")
print("Part (b): Meta-Regression with Mean Age")
print("=" * 60)

X_age = sm.add_constant(af_trials['mean_age'])
wls_age = sm.WLS(af_trials['log_rr'], X_age,
                 weights=1/af_trials['var_log_rr']).fit()

print(f"\nMeta-regression: log_rr ~ mean_age")
print(f"  Intercept: {wls_age.params.iloc[0]:.4f} (p = {wls_age.pvalues.iloc[0]:.4f})")
print(f"  Mean age slope: {wls_age.params.iloc[1]:.4f} (p = {wls_age.pvalues.iloc[1]:.4f})")
print(f"  R-squared: {wls_age.rsquared:.3f}")

if wls_age.pvalues.iloc[1] < 0.05:
    print("\n  There IS a significant relationship between mean age and effect size.")
    if wls_age.params.iloc[1] < 0:
        print("  Trials with older populations show larger treatment benefits.")
    else:
        print("  Trials with older populations show smaller treatment benefits.")
else:
```

```
    print(f"\n  No significant relationship (p = {wls_age.pvalues.iloc[1]:.3f

# --- Part (c): Meta-regression with percentage female ---
print(f"\n{'=' * 60}")
print("Part (c): Meta-Regression with Percentage Female")
print("=" * 60)

X_fem = sm.add_constant(af_trials['pct_female'])
wls_fem = sm.WLS(af_trials['log_rr'], X_fem,
                 weights=1/af_trials['var_log_rr']).fit()

print(f"\nMeta-regression: log_rr ~ pct_female")
print(f"  Intercept: {wls_fem.params.iloc[0]:.4f} (p = {wls_fem.pvalues.iloc
print(f"  Pct female slope: {wls_fem.params.iloc[1]:.4f} (p = {wls_fem.pvalue
print(f"  R-squared: {wls_fem.rsquared:.3f}")

print("\n  *** ECOLOGICAL FALLACY WARNING ***")
print("  This analysis examines TRIAL-LEVEL associations between percentage")
print("  female and treatment effect. Even if significant, this does NOT")
print("  prove that individual women respond differently to treatment.")
print("  Trials with more women may differ in other characteristics:")
print("    - Geographic region and healthcare systems")
print("    - Mean age and comorbidity profiles")
print("    - Trial methodology and inclusion criteria")
print("  Only individual participant data (IPD) meta-analysis with a")
print("  treatment-by-sex interaction can properly test effect modification.

# --- Part (d): Bubble plot ---
print(f"\n{'=' * 60}")
print("Part (d): Bubble Plot")
print("=" * 60)

fig, ax = plt.subplots(figsize=(10, 7))
```

```
# Study weights for bubble sizes
w_plot = 1 / af_trials['var_log_rr']
sizes = (w_plot / w_plot.max()) * 500

# Scatter: bubbles
ax.scatter(af_trials['mean_age'], af_trials['log_rr'],
           s=sizes, alpha=0.6, c='steelblue', edgecolors='darkblue',
           linewidth=1, zorder=5)

# Add study labels
for _, row in af_trials.iterrows():
    ax.annotate(row['study'],
                (row['mean_age'], row['log_rr']),
                textcoords="offset points", xytext=(8, 8),
                fontsize=8, color='grey30')

# Meta-regression line
age_range = np.linspace(af_trials['mean_age'].min() - 2,
                        af_trials['mean_age'].max() + 2, 100)
pred = wls_age.params.iloc[0] + wls_age.params.iloc[1] * age_range
ax.plot(age_range, pred, 'steelblue', linewidth=2,
        label=f'Meta-regression line (slope={wls_age.params.iloc[1]:.4f})')

# Confidence band (approximate)
X_pred = sm.add_constant(age_range)
pred_full = wls_age.get_prediction(X_pred)
pred_ci = pred_full.conf_int(alpha=0.05)
ax.fill_between(age_range, pred_ci[:, 0], pred_ci[:, 1],
                alpha=0.15, color='steelblue', label='95% CI')

# Reference line
ax.axhline(y=0, color='grey', linestyle='--', linewidth=0.8,
           label='Null (log RR = 0)')
```

```
ax.set_xlabel('Mean Age (years)', fontweight='bold', fontsize=12)
ax.set_ylabel('Log Risk Ratio', fontweight='bold', fontsize=12)
ax.set_title('Bubble Plot: Meta-Regression of Treatment Effect on Mean Age\n
             f'(Slope p = {wls_age.pvalues.iloc[1]:.3f}; '
             f'bubble size = study weight)',
             fontweight='bold', fontsize=13)
ax.legend(loc='best', frameon=True, framealpha=0.9)

plt.tight_layout()
plt.savefig('bubble_plot_age.png', dpi=300)
plt.show()

print("\nBubble plot saved as 'bubble_plot_age.png'")
print("Larger bubbles represent more precise (higher-weight) studies.")
```

D.14.4. Exercise 4

D.14.4.1. R

```
# ==============================================================================
# Chapter 18 - Exercise 4: Critical Appraisal of a Meta-Analysis (Conceptual)
# 12 trials of new surgical technique vs standard care for knee OA
# ==============================================================================

# This is a conceptual exercise. Answers are provided as detailed comments.

# Given information:
# - Pooled SMD for pain: -0.62 (95% CI: -0.89 to -0.35), p < 0.001
# - I^2 = 78%, tau^2 = 0.15, Q test p < 0.001
# - Prediction interval: -1.42 to 0.18
# - Egger's test: p = 0.03
```

```
# - 8 of 12 trials were single-centre with fewer than 100 participants

# --- Part (a): Interpret the pooled effect and clinical significance ---
#
# The pooled SMD of -0.62 favours the new surgical technique (negative =
# lower pain scores = better outcome).
#
# Using Cohen's conventions for interpreting SMD:
#   - Small effect: |d| = 0.2
#   - Medium effect: |d| = 0.5
#   - Large effect: |d| = 0.8
#
# An SMD of -0.62 is a MEDIUM-TO-LARGE effect size. The 95% CI (-0.89 to
# -0.35) excludes zero, and the p-value is < 0.001, indicating statistical
# significance.
#
# HOWEVER, statistical significance does not imply clinical significance.
# To assess clinical importance, we should consider:
# 1. The MINIMUM CLINICALLY IMPORTANT DIFFERENCE (MCID) for the pain
#    outcome. For common knee OA pain scales, MCID is typically SMD ~0.3-0.5.
#    The point estimate exceeds this, but the CI includes values near 0.35.
# 2. The ABSOLUTE DIFFERENCE on the original scale would be more
#    interpretable (e.g., "5 points on a 0-100 VAS scale").
# 3. Whether the pain reduction justifies the RISKS AND COSTS of a new
#    surgical procedure (risk-benefit analysis).

# --- Part (b): Prediction interval vs confidence interval ---
#
# The CONFIDENCE INTERVAL (-0.89 to -0.35) tells us about the AVERAGE
# true effect across all settings. It says: "We are 95% confident that
# the mean true effect lies between -0.89 and -0.35."
#
# The PREDICTION INTERVAL (-1.42 to 0.18) tells us where the true effect
```

```
# of a FUTURE STUDY (in a new setting) is likely to fall. It incorporates
# BOTH sampling uncertainty AND between-study heterogeneity.
#
# Key insight: The prediction interval CROSSES ZERO (includes 0.18).
# This means:
# - While the average effect is beneficial, in some settings the new
#   technique might provide NO BENEFIT or even be slightly harmful.
# - A new centre implementing this technique cannot be confident of
#   seeing a benefit.
# - The prediction interval is the more honest representation of
#   uncertainty for clinical decision-making.
#
# This discrepancy (significant CI but prediction interval crossing null)
# is common when I^2 is high and highlights why reporting only the pooled
# CI can be misleading.

# --- Part (c): Implications of I^2 = 78% ---
#
# I^2 = 78% means that 78% of the observed variability in effect sizes
# is due to TRUE BETWEEN-STUDY HETEROGENEITY rather than chance.
# This is classified as CONSIDERABLE heterogeneity (>75%).
#
# Implications:
# 1. The assumption of a single common effect is clearly violated.
# 2. The RANDOM-EFFECTS model is appropriate (and was used), but the
#    pooled estimate should be interpreted as an AVERAGE across heterogeneous
#    true effects, not as a single definitive answer.
# 3. The heterogeneity is statistically significant (Q test p < 0.001),
#    confirming this is not due to sampling variation.
# 4. INVESTIGATING SOURCES of heterogeneity is essential:
#    - Subgroup analysis by study characteristics (e.g., surgical
#      technique variation, patient severity, follow-up duration)
#    - Meta-regression to model effect modifiers
```

```
#    - Sensitivity analysis excluding outlier or high-risk-of-bias studies
# 5. The random-effects model gives MORE WEIGHT TO SMALL STUDIES,
#    which in this case (8 of 12 are small) means the pooled estimate
#    is heavily influenced by potentially biased small studies.

# --- Part (d): Concerns about Egger's test and small trials ---
#
# Egger's test is significant (p = 0.03), suggesting FUNNEL PLOT ASYMMETRY.
# Combined with the fact that 8 of 12 trials are small single-centre
# studies, this raises several concerns:
#
# 1. PUBLICATION BIAS: Small trials showing no benefit may not have been
#    published. The available evidence may overestimate the true effect.
#    This is the most common interpretation of funnel plot asymmetry.
#
# 2. SMALL-STUDY EFFECTS: Small studies may have different effect sizes
#    for legitimate reasons:
#    - More selected patient populations (more severe cases)
#    - More enthusiastic, expert surgeons (performance bias)
#    - Less rigorous outcome assessment (detection bias)
#    - Higher risk of bias in general
#
# 3. SINGLE-CENTRE BIAS: Single-centre trials often report larger effects
#    than multicentre trials because of:
#    - Surgeon expertise (learning curve effects)
#    - Patient selection
#    - Lack of external validity
#    - Potential unblinding or outcome assessment bias
#
# 4. The combination of significant Egger's test + predominantly small
#    single-centre trials is a RED FLAG. The pooled effect may be
#    substantially overestimated.
```

```
# --- Part (e): Additional analyses wanted ---
#
# 1. TRIM-AND-FILL ANALYSIS: To estimate the adjusted pooled effect after
#    accounting for potentially missing studies.
#
# 2. RISK OF BIAS ASSESSMENT: Using Cochrane Risk of Bias tool (RoB 2) for
#    each trial. Present a risk of bias summary plot.
#
# 3. SENSITIVITY ANALYSIS RESTRICTED TO LARGER TRIALS: Re-run the meta-
#    analysis using only the 4 larger/multicentre trials to see if the
#    effect persists.
#
# 4. SENSITIVITY ANALYSIS BY RISK OF BIAS: Exclude high-risk-of-bias
#    studies and re-estimate.
#
# 5. LEAVE-ONE-OUT ANALYSIS: Check if any single study is driving the
#    result.
#
# 6. SUBGROUP ANALYSIS by:
#    - Study size (small vs large)
#    - Single-centre vs multicentre
#    - Blinding status
#    - Follow-up duration
#    - OA severity at baseline
#
# 7. META-REGRESSION: Examine whether study-level characteristics
#    (sample size, year, risk of bias score) moderate the effect.
#
# 8. FUNCTIONAL OUTCOMES: Pain is subjective and susceptible to placebo
#    effects (especially in surgical trials). Objective outcomes (e.g.,
#    range of motion, need for total knee replacement) would be more
#    convincing.
#
```

```
# 9. SHAM SURGERY COMPARISON: Were any trials sham-controlled? Without
#    sham surgery, the "placebo effect of surgery" cannot be distinguished
#    from a true treatment effect.

# --- Part (f): Would you change clinical practice? ---
#
# NO, I would NOT change clinical practice based on this meta-analysis
# alone. The reasons are:
#
# 1. HIGH HETEROGENEITY (I^2 = 78%): The treatment effect is highly
#    variable across settings. Some settings may see no benefit.
#
# 2. PREDICTION INTERVAL CROSSES NULL: A new centre cannot be confident
#    of achieving benefit.
#
# 3. EVIDENCE OF PUBLICATION BIAS: The significant Egger's test and
#    predominance of small positive studies suggest the true effect may
#    be substantially smaller than the pooled estimate.
#
# 4. MOSTLY SMALL SINGLE-CENTRE TRIALS: The evidence base lacks large,
#    multicentre, adequately blinded trials that demonstrate external
#    validity and generalisability.
#
# 5. SURGICAL INTERVENTION: Given the invasiveness, costs, and risks
#    of surgery, a higher evidence bar is appropriate. One would want
#    at least one large, multicentre, preferably sham-controlled RCT
#    showing clinically meaningful benefit.
#
# 6. NO SHAM CONTROL: Surgical procedures are known to have strong
#    placebo effects (cf. Moseley et al., 2002 on arthroscopic surgery
#    for knee OA). Without sham-controlled trials, the observed benefit
#    could be largely or entirely due to placebo.
#
```

```
# RECOMMENDED NEXT STEPS:
# - Commission a large, multicentre, sham-controlled RCT
# - Include objective outcomes alongside patient-reported pain
# - Longer follow-up (surgical effects may wane over time)
# - Standardise the surgical technique across centres
# - Register the protocol prospectively

cat("Exercise 4 is a conceptual exercise.\n")
cat("All answers are provided as detailed comments in this script.\n")
cat("Review the comments for:\n")
cat("  (a) Interpretation of pooled SMD and clinical significance\n")
cat("  (b) Prediction interval vs confidence interval\n")
cat("  (c) Implications of I^2 = 78%\n")
cat("  (d) Concerns about Egger's test and small trials\n")
cat("  (e) Additional analyses wanted\n")
cat("  (f) Whether to change clinical practice\n")
```

D.14.4.2. Python

```
# ==============================================================================
# Chapter 18 - Exercise 4: Critical Appraisal of a Meta-Analysis (Conceptual)
# 12 trials of new surgical technique vs standard care for knee OA
# ==============================================================================

# This is a conceptual exercise. Answers are provided as detailed comments.

# Given information:
# - Pooled SMD for pain: -0.62 (95% CI: -0.89 to -0.35), p < 0.001
# - I^2 = 78%, tau^2 = 0.15, Q test p < 0.001
# - Prediction interval: -1.42 to 0.18
# - Egger's test: p = 0.03
# - 8 of 12 trials were single-centre with fewer than 100 participants
```

```
# --- Part (a): Interpret the pooled effect and clinical significance ---
#
# The pooled SMD of -0.62 favours the new surgical technique (lower pain).
# By Cohen's conventions: |d|=0.2 is small, 0.5 is medium, 0.8 is large.
# So -0.62 represents a MEDIUM-TO-LARGE effect.
#
# The 95% CI (-0.89 to -0.35) excludes zero, and p < 0.001.
#
# HOWEVER, clinical significance requires considering:
# 1. The Minimum Clinically Important Difference (MCID) -- typically
#    SMD ~0.3-0.5 for knee OA pain scales. The CI includes values
#    near MCID threshold.
# 2. The absolute difference on the original pain scale.
# 3. Risk-benefit ratio for a surgical procedure.
# 4. Patient values and preferences.

# --- Part (b): Prediction interval vs confidence interval ---
#
# CI (-0.89 to -0.35): Describes uncertainty about the AVERAGE effect.
# PI (-1.42 to 0.18): Describes where the NEXT STUDY'S true effect
#                      is likely to fall.
#
# Critical difference: The PI CROSSES ZERO (upper bound = 0.18).
# This means in some settings, the technique might provide NO BENEFIT.
# A new centre implementing this cannot be confident of benefit.
# The PI is more relevant for individual clinical decisions.

# --- Part (c): Implications of I^2 = 78% ---
#
# I^2 = 78% = CONSIDERABLE heterogeneity (>75% threshold).
# 78% of variability is due to true between-study differences.
#
# Implications:
```

```
# 1. No single common effect exists -- the pooled estimate is an average.
# 2. Sources of heterogeneity must be investigated (subgroup analysis,
#    meta-regression, sensitivity analysis).
# 3. Random-effects model gives more weight to small studies, which
#    here (8/12 small) means heavy influence from potentially biased work.
# 4. Reporting the pooled estimate without the PI is misleading.

# --- Part (d): Concerns about Egger's test + small trials ---
#
# Egger's test p = 0.03 indicates significant funnel plot asymmetry.
# Combined with 8/12 small single-centre trials, concerns include:
#
# 1. PUBLICATION BIAS: Negative small trials may not have been published.
# 2. SMALL-STUDY EFFECTS: Small trials often show larger effects due to
#    selected populations, enthusiastic surgeons, or less rigorous methods.
# 3. SINGLE-CENTRE BIAS: Lack of external validity, expertise effects.
# 4. The combination is a RED FLAG: the pooled effect is likely
#    OVERESTIMATED.

# --- Part (e): Additional analyses wanted ---
#
# 1. Trim-and-fill to estimate adjusted effect
# 2. Risk of bias assessment (Cochrane RoB 2) for each trial
# 3. Sensitivity analysis restricted to larger/multicentre trials only
# 4. Sensitivity analysis excluding high-risk-of-bias studies
# 5. Leave-one-out analysis
# 6. Subgroup analyses: by study size, single vs multicentre, blinding
# 7. Meta-regression on study-level characteristics
# 8. Objective functional outcomes (not just subjective pain)
# 9. Check for sham-controlled trials (surgical placebo effect)

# --- Part (f): Would you change clinical practice? ---
#
```

```
# NO. Reasons:
# 1. High heterogeneity (I^2 = 78%) -- variable effects across settings
# 2. Prediction interval crosses null -- some settings may see no benefit
# 3. Evidence of publication bias (Egger's p = 0.03)
# 4. Evidence base dominated by small single-centre trials
# 5. Surgery requires higher evidence bar due to invasiveness/costs/risks
# 6. No sham-controlled trials mentioned (surgical placebo effect)
#
# Recommended: Commission a large multicentre sham-controlled RCT with
# objective outcomes and long follow-up before changing practice.

# Print summary for verification
print("=" * 70)
print("Exercise 4: Critical Appraisal of a Published Meta-Analysis")
print("=" * 70)
print()

print("Given:")
print("  Pooled SMD: -0.62 (95% CI: -0.89 to -0.35)")
print("  I^2 = 78%, tau^2 = 0.15, Q test p < 0.001")
print("  Prediction interval: -1.42 to 0.18")
print("  Egger's test: p = 0.03")
print("  8/12 trials: small, single-centre (<100 participants)")
print()

print("(a) POOLED EFFECT: Medium-to-large (SMD = -0.62), statistically")
print("    significant but clinical significance depends on MCID and")
print("    risk-benefit analysis for a surgical intervention.")
print()
print("(b) PREDICTION INTERVAL crosses zero (upper bound 0.18), meaning")
print("    some settings may see no benefit. The CI alone is misleading.")
print()
print("(c) I^2 = 78% indicates considerable heterogeneity. The pooled")
```

```
print("    estimate is an average of very different true effects.")
print()
print("(d) Egger's p = 0.03 + mostly small trials = RED FLAG for")
print("    publication bias and small-study effects. The true effect")
print("    is likely overestimated.")
print()
print("(e) Need: trim-and-fill, risk of bias assessment, sensitivity")
print("    analyses (excluding small/high-bias trials), sham control check.")
print()
print("(f) Do NOT change practice. Evidence is insufficient due to")
print("    heterogeneity, publication bias concerns, and reliance on")
print("    small single-centre trials. Need large multicentre sham-RCT.")
```

D.15. Chapter 19: Journal-Ready Analysis

D.15.1. Exercise 1

D.15.1.1. R

```
# ============================================================================
# Chapter 19 - Exercise 1: Table 1 in R
# Using the lung dataset, create a publication-quality Table 1
# ============================================================================

library(tidyverse)
library(gtsummary)
library(gt)
library(survival)

# --- Load data ---
data(lung, package = "survival")
```

```r
# Prepare data: recode sex for clarity
lung_clean <- lung %>%
  mutate(
    sex = factor(sex, levels = c(1, 2), labels = c("Male", "Female")),
    ph.ecog = factor(ph.ecog, levels = 0:4,
                     labels = c("Fully active", "Restricted",
                                "Ambulatory", "Limited self-care",
                                "Completely disabled")),
    status = factor(status, levels = c(1, 2), labels = c("Censored", "Dead"))
  )

# --- Part (a): Create Table 1 stratified by sex ---
table1 <- lung_clean %>%
  select(sex, age, ph.ecog, ph.karno, meal.cal, wt.loss) %>%
  tbl_summary(
    by = sex,
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = list(
      all_continuous() ~ 1,
      all_categorical() ~ c(0, 1)
    ),
    label = list(
      age ~ "Age, years",
      ph.ecog ~ "ECOG performance status",
      ph.karno ~ "Karnofsky performance score",
      meal.cal ~ "Meal calories, kcal/day",
      wt.loss ~ "Weight loss, kg (last 6 months)"
    ),
    missing_text = "Missing"
  ) %>%
```

```
  add_overall() %>%
  add_n()

# --- Part (b): Add SMDs instead of p-values ---
# gtsummary supports SMD via add_difference()
# For Table 1 with SMDs, we use a different approach
table1_smd <- lung_clean %>%
  select(sex, age, ph.ecog, ph.karno, meal.cal, wt.loss) %>%
  tbl_summary(
    by = sex,
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = list(
      all_continuous() ~ 1,
      all_categorical() ~ c(0, 1)
    ),
    label = list(
      age ~ "Age, years",
      ph.ecog ~ "ECOG performance status",
      ph.karno ~ "Karnofsky performance score",
      meal.cal ~ "Meal calories, kcal/day",
      wt.loss ~ "Weight loss, kg (last 6 months)"
    ),
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_n() %>%
  add_difference(
    estimate_fun = list(all_continuous() ~ function(x) style_sigfig(x, digits
    pvalue_fun = ~ style_pvalue(.x, digits = 3)
  ) %>%
```

```r
  modify_header(label = "**Characteristic**") %>%
  modify_spanning_header(c("stat_1", "stat_2") ~ "**Sex**") %>%
  bold_labels()

# Compute SMDs manually for continuous variables
compute_smd <- function(x, group) {
  g1 <- x[group == "Male"]
  g2 <- x[group == "Female"]
  g1 <- g1[!is.na(g1)]
  g2 <- g2[!is.na(g2)]
  pooled_sd <- sqrt((var(g1) + var(g2)) / 2)
  if (pooled_sd == 0) return(0)
  (mean(g1) - mean(g2)) / pooled_sd
}

cat("Standardised Mean Differences (Male vs Female):\n")
cat("  Age:", round(compute_smd(lung_clean$age, lung_clean$sex), 3), "\n")
cat("  Karnofsky:", round(compute_smd(lung_clean$ph.karno, lung_clean$sex), 3), "\n")
cat("  Meal calories:", round(compute_smd(lung_clean$meal.cal, lung_clean$sex), 3), "\n")
cat("  Weight loss:", round(compute_smd(lung_clean$wt.loss, lung_clean$sex), 3), "\n")

# Create the final publication table with gt formatting
final_table <- table1_smd %>%
  as_gt() %>%
  gt::tab_header(
    title = "Table 1. Baseline Characteristics of the Study Population",
    subtitle = "NCCTG Lung Cancer Dataset, Stratified by Sex"
  ) %>%
  gt::tab_footnote(
    footnote = "Values are mean (SD) for continuous variables and n (%) for categorical vari
    locations = gt::cells_title()
  )
```

```
print(final_table)

# --- Part (c): Export to Word and LaTeX ---
# Export to Word (.docx)
# gt::gtsave(final_table, "table1_lung.docx")
cat("\nTo export to Word: gt::gtsave(final_table, 'table1_lung.docx')\n")

# Export to LaTeX
# gt::as_latex(final_table) %>% writeLines("table1_lung.tex")
cat("To export to LaTeX: gt::as_latex(final_table)\n")

# Export to HTML
# gt::gtsave(final_table, "table1_lung.html")
cat("To export to HTML: gt::gtsave(final_table, 'table1_lung.html')\n")

cat("\nNote: SMDs < 0.1 indicate good balance. In an RCT, p-values for\n")
cat("baseline comparisons are inappropriate because any differences are\n")
cat("due to chance. In observational studies, SMDs are preferred over\n")
cat("p-values because they are independent of sample size.\n")
```

D.15.1.2. Python

```
# ============================================================================
# Chapter 19 - Exercise 1: Table 1 in Python
# Using the lung dataset, create a publication-quality Table 1
# ============================================================================

import pandas as pd
import numpy as np

# --- Load data ---
# Use lifelines' lung dataset
```

```python
from lifelines.datasets import load_lung
lung = load_lung()

# Prepare data
lung['sex_label'] = lung['sex'].map({1: 'Male', 2: 'Female'})
lung['ph.ecog_label'] = lung['ph.ecog'].map({
    0: 'Fully active', 1: 'Restricted', 2: 'Ambulatory',
    3: 'Limited self-care', 4: 'Completely disabled'
})

# --- Part (a): Create Table 1 stratified by sex with SMDs ---
try:
    from tableone import TableOne

    # Define variable types
    columns = ['age', 'ph.ecog', 'ph.karno', 'meal.cal', 'wt.loss']
    categorical = ['ph.ecog']
    nonnormal = []  # all continuous reported as mean (SD)

    # Create Table 1 with SMDs instead of p-values
    table1 = TableOne(
        lung,
        columns=columns,
        categorical=categorical,
        groupby='sex_label',
        nonnormal=nonnormal,
        pval=False,  # No p-values
        smd=True,    # Include SMDs
        rename={
            'age': 'Age, years',
            'ph.ecog': 'ECOG performance status',
            'ph.karno': 'Karnofsky performance score',
            'meal.cal': 'Meal calories, kcal/day',
```

```
            'wt.loss': 'Weight loss, kg (last 6 months)'
        },
        missing=True,
        overall=True
    )

    print("=" * 70)
    print("Table 1. Baseline Characteristics of the Study Population")
    print("NCCTG Lung Cancer Dataset, Stratified by Sex")
    print("=" * 70)
    print(table1.tabulate(tablefmt="github"))

    # --- Part (c): Export ---
    # table1.to_excel('table1_lung.xlsx')
    # table1.to_latex('table1_lung.tex')
    # table1.to_html('table1_lung.html')
    print("\nExport options:")
    print("  table1.to_excel('table1_lung.xlsx')")
    print("  table1.to_latex('table1_lung.tex')")
    print("  table1.to_html('table1_lung.html')")

except ImportError:
    print("tableone package not installed. Building Table 1 manually.\n")

    # --- Manual Table 1 construction ---
    def compute_smd_cont(data, var, group_col):
        """Compute SMD for a continuous variable."""
        g1 = data[data[group_col] == 'Male'][var].dropna()
        g2 = data[data[group_col] == 'Female'][var].dropna()
        pooled_sd = np.sqrt((g1.var() + g2.var()) / 2)
        if pooled_sd == 0:
            return 0
        return (g1.mean() - g2.mean()) / pooled_sd
```

```
def summarise_continuous(data, var, group_col):
    """Summarise continuous variable by group."""
    results = {}
    for group in ['Overall'] + list(data[group_col].unique()):
        if group == 'Overall':
            subset = data[var].dropna()
        else:
            subset = data[data[group_col] == group][var].dropna()
        results[group] = f"{subset.mean():.1f} ({subset.std():.1f})"
    return results

def summarise_categorical(data, var, group_col):
    """Summarise categorical variable by group."""
    results = {}
    categories = sorted(data[var].dropna().unique())
    for group in ['Overall'] + list(data[group_col].unique()):
        if group == 'Overall':
            subset = data[var].dropna()
        else:
            subset = data[data[group_col] == group][var].dropna()
        counts = subset.value_counts()
        total = len(subset)
        group_results = {}
        for cat in categories:
            n = counts.get(cat, 0)
            pct = n / total * 100
            group_results[cat] = f"{n} ({pct:.1f}%)"
        results[group] = group_results
    return results

# Build table
cont_vars = {
    'age': 'Age, years',
```

```
        'ph.karno': 'Karnofsky performance score',
        'meal.cal': 'Meal calories, kcal/day',
        'wt.loss': 'Weight loss, kg (last 6 months)'
    }

    print(f"{'Variable':<40} {'Overall':>18} {'Male':>18} {'Female':>18} {'SM
    print("-" * 104)

    for var, label in cont_vars.items():
        summ = summarise_continuous(lung, var, 'sex_label')
        smd = compute_smd_cont(lung, var, 'sex_label')
        n_miss = lung[var].isna().sum()
        miss_str = f" [missing: {n_miss}]" if n_miss > 0 else ""
        print(f"{label + miss_str:<40} {summ['Overall']:>18} "
              f"{summ.get('Male', 'N/A'):>18} {summ.get('Female', 'N/A'):>18}
              f"{abs(smd):>7.3f}")

    # ECOG as categorical
    print(f"\n{'ECOG performance status':<40}")
    ecog_summ = summarise_categorical(lung, 'ph.ecog', 'sex_label')
    for cat in sorted(lung['ph.ecog'].dropna().unique()):
        label_map = {0: '  Fully active', 1: '  Restricted',
                     2: '  Ambulatory', 3: '  Limited self-care',
                     4: '  Completely disabled'}
        label = label_map.get(cat, f'  {cat}')
        overall = ecog_summ['Overall'].get(cat, '0 (0.0%)')
        male = ecog_summ.get('Male', {}).get(cat, '0 (0.0%)')
        female = ecog_summ.get('Female', {}).get(cat, '0 (0.0%)')
        print(f"{label:<40} {overall:>18} {male:>18} {female:>18}")

# --- Part (b): SMD interpretation ---
print("\n" + "=" * 70)
print("Standardised Mean Differences (SMD)")
```

```python
print("=" * 70)

def compute_smd(data, var, group_col):
    g1 = data[data[group_col] == 'Male'][var].dropna()
    g2 = data[data[group_col] == 'Female'][var].dropna()
    pooled_sd = np.sqrt((g1.var() + g2.var()) / 2)
    if pooled_sd == 0:
        return 0
    return (g1.mean() - g2.mean()) / pooled_sd

for var, label in [('age', 'Age'), ('ph.karno', 'Karnofsky score'),
                   ('meal.cal', 'Meal calories'), ('wt.loss', 'Weight loss')]:
    smd = compute_smd(lung, var, 'sex_label')
    balanced = "Balanced" if abs(smd) < 0.1 else "Imbalanced"
    print(f"  {label:<20}: SMD = {smd:>7.3f}  [{balanced}]")

print("\nNote: SMD < 0.1 indicates good balance between groups.")
print("In observational studies, SMDs are preferred over p-values because")
print("they are independent of sample size, whereas p-values from large")
print("samples can be significant for clinically trivial differences.")
```

D.15.2. Exercise 2

D.15.2.1. R

```r
# ==============================================================================
# Chapter 19 - Exercise 2: Publication-Quality Multi-Panel Figure in R
# Survival analysis of the lung dataset
# ==============================================================================

library(tidyverse)
library(survival)
```

```r
library(survminer)
library(broom)
library(patchwork)

# --- Publication theme and colour-blind palette ---
theme_publication <- function(base_size = 11) {
  theme_minimal(base_size = base_size) %+replace%
    theme(
      text = element_text(family = "Arial"),
      plot.title = element_text(size = base_size + 2, face = "bold",
                                hjust = 0, margin = margin(b = 10)),
      axis.title = element_text(size = base_size, face = "bold"),
      axis.text = element_text(size = base_size - 1, colour = "black"),
      legend.title = element_text(size = base_size, face = "bold"),
      legend.text = element_text(size = base_size - 1),
      legend.position = "bottom",
      panel.grid.minor = element_blank(),
      panel.grid.major = element_line(colour = "grey90"),
      strip.text = element_text(size = base_size, face = "bold"),
      plot.margin = margin(10, 10, 10, 10)
    )
}

cb_palette <- c("#0072B2", "#D55E00", "#009E73", "#CC79A7",
                "#F0E442", "#56B4E9", "#E69F00", "#000000")

# --- Load and prepare data ---
data(lung, package = "survival")
lung <- lung %>%
  mutate(
    sex_label = factor(sex, levels = c(1, 2), labels = c("Male", "Female")),
    status_event = status - 1  # Convert to 0/1
  ) %>%
```

```
  filter(!is.na(ph.ecog) & !is.na(ph.karno) & !is.na(wt.loss))

# ==============================================================================
# Panel A: Kaplan-Meier Curve by Sex
# ==============================================================================
fit_km <- survfit(Surv(time, status_event) ~ sex_label, data = lung)

# Using ggsurvplot for KM curve
p_km <- ggsurvplot(
  fit_km,
  data = lung,
  palette = cb_palette[1:2],
  risk.table = FALSE,
  pval = TRUE,
  pval.coord = c(700, 0.9),
  conf.int = TRUE,
  conf.int.alpha = 0.15,
  xlab = "Time (days)",
  ylab = "Survival probability",
  legend.labs = c("Male", "Female"),
  legend.title = "Sex",
  ggtheme = theme_publication(),
  title = "A"
)

panel_a <- p_km$plot +
  theme(legend.position = c(0.8, 0.85),
        legend.background = element_rect(fill = "white", colour = "grey80"),
        plot.title = element_text(size = 14, face = "bold"))

# ==============================================================================
# Panel B: Forest Plot of Hazard Ratios from Multivariable Cox Model
# ==============================================================================
```

```
cox_model <- coxph(Surv(time, status_event) ~ age + sex_label + ph.ecog +
                     ph.karno + wt.loss,
                   data = lung)

# Extract coefficients
cox_tidy <- tidy(cox_model, conf.int = TRUE, exponentiate = TRUE)
cox_tidy <- cox_tidy %>%
  mutate(
    label = case_when(
      term == "age" ~ "Age (per year)",
      term == "sex_labelFemale" ~ "Female vs Male",
      term == "ph.ecog" ~ "ECOG PS (per level)",
      term == "ph.karno" ~ "Karnofsky (per 10 pts)",
      term == "wt.loss" ~ "Weight loss (per kg)"
    ),
    # Rescale Karnofsky to per-10-point increase for interpretability
    estimate = ifelse(term == "ph.karno", estimate^10, estimate),
    conf.low = ifelse(term == "ph.karno", conf.low^10, conf.low),
    conf.high = ifelse(term == "ph.karno", conf.high^10, conf.high)
  )

panel_b <- ggplot(cox_tidy, aes(x = estimate, y = reorder(label, estimate)))
  geom_vline(xintercept = 1, linetype = "dashed", colour = "grey50") +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high),
                 height = 0.2, colour = cb_palette[1], linewidth = 0.8) +
  geom_point(size = 3, colour = cb_palette[1]) +
  geom_text(aes(label = sprintf("%.2f (%.2f-%.2f)", estimate, conf.low, conf
            hjust = -0.1, size = 3, nudge_y = 0.25) +
  scale_x_log10() +
  labs(x = "Hazard Ratio (95% CI)",
       y = "",
       title = "B") +
  theme_publication() +
```

```
  theme(
    legend.position = "none",
    plot.title = element_text(size = 14, face = "bold")
  )

# ==============================================================================
# Panel C: Calibration Plot for 1-Year Survival Prediction
# ==============================================================================

# Predict 1-year (365 day) survival probabilities
# Use Cox model baseline hazard
surv_pred <- summary(survfit(cox_model), times = 365)

# For calibration, split into deciles of predicted risk
lung$pred_1yr <- 1 - predict(cox_model, type = "expected",
                             newdata = lung) / max(lung$time) * 365
# Alternative: use survfit-based prediction
baseline_surv <- basehaz(cox_model, centered = FALSE)
# Get linear predictor
lung$lp <- predict(cox_model, type = "lp")

# Simple calibration: bin by predicted risk deciles
lung$pred_surv <- exp(-predict(cox_model, type = "expected"))

# Create risk groups based on linear predictor
lung$risk_group <- cut(lung$lp, breaks = quantile(lung$lp, probs = seq(0, 1, 0.1)),
                       include.lowest = TRUE, labels = FALSE)

# Calculate observed 1-year survival in each group
cal_data <- lung %>%
  group_by(risk_group) %>%
  summarise(
    n = n(),
```

```
    events = sum(status_event),
    # KM estimate at 1 year
    obs_surv = {
      fit <- survfit(Surv(time, status_event) ~ 1, data = cur_data())
      s <- summary(fit, times = 365)
      if (length(s$surv) > 0) s$surv else NA
    },
    mean_lp = mean(lp),
    .groups = "drop"
  ) %>%
  filter(!is.na(obs_surv))

# Get predicted survival at each mean LP
h0_365 <- approx(baseline_surv$time, baseline_surv$hazard, xout = 365)$y
if (is.na(h0_365)) h0_365 <- max(baseline_surv$hazard)
cal_data$pred_surv <- exp(-h0_365 * exp(cal_data$mean_lp))

panel_c <- ggplot(cal_data, aes(x = 1 - pred_surv, y = 1 - obs_surv)) +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed", colour = "grey5(
  geom_point(size = 3, colour = cb_palette[1]) +
  geom_smooth(method = "loess", se = TRUE, colour = cb_palette[2],
              fill = cb_palette[2], alpha = 0.2) +
  coord_equal(xlim = c(0, 1), ylim = c(0, 1)) +
  labs(x = "Predicted 1-year mortality risk",
       y = "Observed 1-year mortality",
       title = "C") +
  theme_publication() +
  theme(plot.title = element_text(size = 14, face = "bold"))

# ==========================================================================
# Combine panels using patchwork
# ==========================================================================
# Panel A takes the top row, B and C share the bottom row
```

```r
combined <- panel_a / (panel_b | panel_c) +
  plot_layout(heights = c(1, 1))

# Save at publication quality
# Double-column layout: 183 mm wide
ggsave("figure_survival_analysis.tiff",
       plot = combined,
       width = 183, height = 200,
       units = "mm", dpi = 300,
       compression = "lzw")

ggsave("figure_survival_analysis.pdf",
       plot = combined,
       width = 183, height = 200,
       units = "mm", device = cairo_pdf)

print(combined)

cat("\nMulti-panel figure saved as:\n")
cat("  figure_survival_analysis.tiff (300 DPI, TIFF)\n")
cat("  figure_survival_analysis.pdf (vector PDF)\n")
cat("\nSpecifications:\n")
cat("  Width: 183 mm (double-column)\n")
cat("  Resolution: 300 DPI\n")
cat("  Colour palette: colour-blind accessible\n")
cat("  Font: Arial\n")
```

D.15.2.2. Python

```python
# ===============================================================================
# Chapter 19 - Exercise 2: Publication-Quality Multi-Panel Figure in Python
# Survival analysis of the lung dataset
```

```
# ===========================================================================

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
from lifelines import KaplanMeierFitter, CoxPHFitter
from lifelines.datasets import load_lung
from lifelines.statistics import logrank_test

# --- Publication settings ---
plt.rcParams.update({
    'font.family': 'Arial',
    'font.size': 10,
    'axes.labelsize': 11,
    'axes.titlesize': 12,
    'axes.labelweight': 'bold',
    'figure.dpi': 300,
    'savefig.dpi': 300,
    'axes.spines.top': False,
    'axes.spines.right': False,
})

# Colour-blind friendly palette
cb_palette = ['#0072B2', '#D55E00', '#009E73', '#CC79A7']

# --- Load and prepare data ---
lung = load_lung()
lung['status'] = lung['status'] - 1  # Convert to 0/1
lung = lung.dropna(subset=['ph.ecog', 'ph.karno', 'wt.loss', 'age', 'sex'])

# ===========================================================================
# Create the multi-panel figure
```

```python
# ============================================================================
fig = plt.figure(figsize=(7.2, 9.0))  # ~183mm x ~230mm
gs = gridspec.GridSpec(2, 2, height_ratios=[1, 1], hspace=0.35, wspace=0.35)

# ============================================================================
# Panel A: Kaplan-Meier Curve by Sex
# ============================================================================
ax_a = fig.add_subplot(gs[0, :])  # Top row, full width

kmf = KaplanMeierFitter()

for sex_val, label, color in [(1, 'Male', cb_palette[0]),
                               (2, 'Female', cb_palette[1])]:
    mask = lung['sex'] == sex_val
    kmf.fit(lung.loc[mask, 'T'], lung.loc[mask, 'status'], label=label)
    kmf.plot_survival_function(ax=ax_a, color=color, ci_show=True, ci_alpha=0.15)

# Log-rank test
lr = logrank_test(
    lung.loc[lung['sex'] == 1, 'T'], lung.loc[lung['sex'] == 2, 'T'],
    lung.loc[lung['sex'] == 1, 'status'], lung.loc[lung['sex'] == 2, 'status']
)
ax_a.text(0.7, 0.85, f'Log-rank p = {lr.p_value:.3f}',
          transform=ax_a.transAxes, fontsize=10,
          bbox=dict(boxstyle='round', facecolor='white', edgecolor='grey'))

ax_a.set_xlabel('Time (days)')
ax_a.set_ylabel('Survival Probability')
ax_a.set_title('A', loc='left', fontweight='bold', fontsize=14)
ax_a.set_ylim(0, 1.05)
ax_a.legend(loc='lower left', frameon=True, framealpha=0.9)

# Number at risk
```

```
n_at_risk_times = [0, 200, 400, 600, 800, 1000]
risk_text = "At risk:\n"
for sex_val, label in [(1, 'Male'), (2, 'Female')]:
    mask = lung['sex'] == sex_val
    times = lung.loc[mask, 'T']
    events = lung.loc[mask, 'status']
    counts = []
    for t in n_at_risk_times:
        counts.append(str(sum(times >= t)))
    risk_text += f"  {label}: " + "  ".join(counts) + "\n"

# ==============================================================================
# Panel B: Forest Plot of Hazard Ratios
# ==============================================================================
ax_b = fig.add_subplot(gs[1, 0])

# Fit Cox model
cph = CoxPHFitter()
cox_data = lung[['T', 'status', 'age', 'sex', 'ph.ecog', 'ph.karno', 'wt.loss
cph.fit(cox_data, duration_col='T', event_col='status')

# Extract results
summary_df = cph.summary[['exp(coef)', 'exp(coef) lower 95%', 'exp(coef) uppe
summary_df.columns = ['HR', 'CI_lo', 'CI_hi']

# Rename for display
name_map = {
    'age': 'Age (per year)',
    'sex': 'Sex (Female vs Male)',
    'ph.ecog': 'ECOG PS (per level)',
    'ph.karno': 'Karnofsky (per point)',
    'wt.loss': 'Weight loss (per kg)'
}
```

```python
summary_df.index = summary_df.index.map(lambda x: name_map.get(x, x))

# Plot forest
y_pos = range(len(summary_df) - 1, -1, -1)
labels = list(summary_df.index)

ax_b.axvline(x=1, color='grey', linestyle='--', linewidth=0.8)

for i, (label, row) in enumerate(summary_df.iterrows()):
    y = list(y_pos)[i]
    ax_b.errorbarh(y, row['HR'], xerr=[[row['HR'] - row['CI_lo']],
                                       [row['CI_hi'] - row['HR']]],
                   fmt='o', color=cb_palette[0], capsize=4, markersize=6,
                   linewidth=1.5)
    # Annotate with HR (CI)
    ax_b.text(max(summary_df['CI_hi']) * 1.15, y,
              f"{row['HR']:.2f} ({row['CI_lo']:.2f}-{row['CI_hi']:.2f})",
              va='center', fontsize=7.5)

ax_b.set_yticks(list(y_pos))
ax_b.set_yticklabels(labels, fontsize=9)
ax_b.set_xlabel('Hazard Ratio (95% CI)')
ax_b.set_xscale('log')
ax_b.set_title('B', loc='left', fontweight='bold', fontsize=14)

# ============================================================================
# Panel C: Calibration Plot for 1-Year Survival
# ============================================================================
ax_c = fig.add_subplot(gs[1, 1])

# Predict survival at 1 year (365 days) using the Cox model
lung_pred = cox_data.copy()
lung_pred['lp'] = cph.predict_partial_hazard(lung_pred)
```

```
# Create decile groups based on linear predictor
lung_pred['risk_decile'] = pd.qcut(lung_pred['lp'], q=10, labels=False,
                                    duplicates='drop')

# Predicted and observed 1-year mortality for each decile
cal_results = []
for decile in sorted(lung_pred['risk_decile'].unique()):
    subset = lung_pred[lung_pred['risk_decile'] == decile]

    # Predicted: use Cox model survival function
    predicted_surv = cph.predict_survival_function(subset, times=[365])
    pred_mort = 1 - predicted_surv.mean(axis=1).values[0]

    # Observed: Kaplan-Meier estimate at 365 days
    kmf_cal = KaplanMeierFitter()
    kmf_cal.fit(subset['T'], subset['status'])
    try:
        obs_surv = kmf_cal.predict(365)
        obs_mort = 1 - obs_surv
    except Exception:
        obs_mort = np.nan

    cal_results.append({
        'decile': decile,
        'predicted': pred_mort,
        'observed': obs_mort,
        'n': len(subset)
    })

cal_df = pd.DataFrame(cal_results).dropna()

# Plot calibration
ax_c.plot([0, 1], [0, 1], 'k--', alpha=0.5, linewidth=0.8, label='Perfect cal
```

```
ax_c.scatter(cal_df['predicted'], cal_df['observed'],
             s=cal_df['n'] * 2, c=cb_palette[0], alpha=0.7,
             edgecolors='darkblue', linewidth=0.5)

# Fit LOESS-like smoothing
from numpy.polynomial import polynomial as P
if len(cal_df) > 3:
    z = np.polyfit(cal_df['predicted'], cal_df['observed'], 2)
    p = np.poly1d(z)
    x_smooth = np.linspace(cal_df['predicted'].min(), cal_df['predicted'].max(), 100)
    ax_c.plot(x_smooth, p(x_smooth), color=cb_palette[1], linewidth=1.5,
              label='Observed (smoothed)')

ax_c.set_xlabel('Predicted 1-year mortality')
ax_c.set_ylabel('Observed 1-year mortality')
ax_c.set_title('C', loc='left', fontweight='bold', fontsize=14)
ax_c.set_xlim(0, 1)
ax_c.set_ylim(0, 1)
ax_c.set_aspect('equal')
ax_c.legend(loc='lower right', fontsize=8, frameon=True)

# ============================================================================
# Save figure
# ============================================================================
plt.tight_layout()
plt.savefig('figure_survival_analysis.tiff', dpi=300, bbox_inches='tight')
plt.savefig('figure_survival_analysis.pdf', bbox_inches='tight')
plt.show()

print("Multi-panel figure saved as:")
print("  figure_survival_analysis.tiff (300 DPI, raster)")
print("  figure_survival_analysis.pdf (vector)")
print()
```

```
print("Specifications:")
print("  Width: ~183 mm (double-column)")
print("  Resolution: 300 DPI")
print("  Colour palette: colour-blind accessible")
print("  Font: Arial")
print()
print("Panel A: Kaplan-Meier survival curves by sex with log-rank p-value")
print("Panel B: Forest plot of hazard ratios from multivariable Cox model")
print("Panel C: Calibration plot for 1-year mortality prediction")
```

D.15.3. Exercise 3

D.15.3.1. R

```
# ==============================================================================
# Chapter 19 - Exercise 3: Write a Statistical Methods Section
# Retrospective cohort study: SGLT2i vs DPP4i and MACE in T2DM
# ==============================================================================

# This is a conceptual exercise. The statistical methods section is provided
# as a multi-line character string that could be included in a manuscript.

methods_section <- '
STATISTICAL METHODS

Study Design and Population
This was a retrospective cohort study of 5,000 adults with type 2 diabetes
mellitus identified from the hospital electronic health records (EHR) databas
between 1 January 2015 and 31 December 2023. Patients were eligible if they
had a new prescription for either an SGLT2 inhibitor or a DPP-4 inhibitor, wi
no prior use of the comparator drug class. Patients with a history of major
adverse cardiovascular events (MACE) prior to the index date were excluded.
```

```
The index date (time zero) was defined as the date of first prescription of
the study drug, consistent with a new-user, active comparator design to
minimise immortal time bias and confounding by indication.

Primary and Secondary Outcomes
The primary outcome was time to first MACE, defined as a composite of
myocardial infarction (ICD-10: I21), ischaemic stroke (ICD-10: I63), or
cardiovascular death (underlying cause of death codes I00-I99). Patients were
followed from the index date until the first MACE event, death from non-
cardiovascular causes, loss to follow-up, end of the study period
(31 December 2023), or 5 years after the index date, whichever occurred first.

Sample Size
With 5,000 patients and an anticipated event rate of 8% over 5 years in the
DPP-4 inhibitor group, the study had approximately 80% power to detect a
hazard ratio of 0.70 or smaller at a two-sided alpha of 0.05, assuming a
1:1 treatment group ratio and accounting for 10% loss to follow-up.

Descriptive Statistics
Baseline characteristics were summarised as means (SD) for normally
distributed continuous variables, medians (IQR) for skewed continuous
variables, and frequencies (percentages) for categorical variables.
Standardised mean differences (SMDs) were used to compare baseline
characteristics between treatment groups, with an absolute SMD < 0.1
indicating adequate balance; p-values were not used for baseline comparisons
in accordance with current recommendations.

Propensity Score Estimation and Matching
The propensity score -- the probability of receiving an SGLT2 inhibitor
versus a DPP-4 inhibitor -- was estimated using multivariable logistic
regression. Covariates included age, sex, body mass index (BMI), glycated
haemoglobin (HbA1c), estimated glomerular filtration rate (eGFR), history
of cardiovascular disease, hypertension, and smoking status. These covariates
```

```
were selected a priori based on clinical knowledge and a directed acyclic
graph (DAG) encoding assumed causal relationships. Propensity scores were
used for 1:1 nearest-neighbour matching without replacement, using a caliper
of 0.2 standard deviations of the logit of the propensity score. Covariate
balance after matching was assessed using SMDs, with all covariates required
to achieve an absolute SMD < 0.1. Propensity score overlap was assessed
visually using density plots.

Primary Analysis
The primary analysis estimated the average treatment effect on the treated
(ATT) using a Cox proportional hazards regression model fitted to the matched
cohort, with SGLT2 inhibitor use as the sole covariate. The proportional
hazards assumption was tested using scaled Schoenfeld residuals and
log-log survival plots. If the assumption was violated, a time-varying
coefficient or restricted mean survival time (RMST) analysis was planned
as an alternative. Hazard ratios (HRs) with 95% confidence intervals (CIs)
were reported.

Missing Data
Missing covariate data ranged from 2% (age) to 15% (BMI). Missingness was
assumed to be missing at random (MAR) conditional on observed variables.
Multiple imputation by chained equations (MICE) was performed with 50
imputed datasets and 20 iterations per dataset. The imputation model
included all analysis variables (covariates, exposure, outcome indicator,
and the Nelson-Aalen cumulative hazard estimate) to ensure compatibility
with the substantive analysis model. Propensity score estimation and
matching were performed within each imputed dataset, and results were
pooled using Rubin rules. A complete-case analysis was performed as a
sensitivity analysis.

Sensitivity Analyses
Five pre-specified sensitivity analyses were conducted: (1) complete-case
analysis; (2) inverse probability of treatment weighting (IPTW) with
```

```
stabilised weights as an alternative to matching; (3) inclusion of
additional covariates (income quintile, number of medications) in the
propensity score model; (4) restriction to patients with at least
12 months of follow-up; and (5) E-value calculation to quantify the
minimum strength of association an unmeasured confounder would need with
both the treatment and the outcome to explain away the observed association.

Multiple Comparisons
As this study had a single pre-specified primary outcome, no adjustment
for multiple comparisons was applied to the primary analysis. Secondary
outcomes were interpreted with appropriate caution as hypothesis-generating.

Software
All analyses were conducted using R version 4.4.1 (R Foundation for
Statistical Computing, Vienna, Austria) with the following packages:
MatchIt (v4.5.5) for propensity score matching, cobalt (v4.5.1) for
balance assessment, survival (v3.7-0) for Cox regression, mice (v3.16.0)
for multiple imputation, and survey (v4.4-2) for IPTW analyses. The
random seed was set to 42 for reproducibility. Analysis code is available
at [repository URL]. Two-sided p-values < 0.05 were considered
statistically significant.
'

# Print the methods section
cat(methods_section)

# --- Checklist verification ---
cat("\n\n=== CHECKLIST: Key Items Addressed ===\n")
cat("1. Study design and population:           YES\n")
cat("2. Primary and secondary outcomes:        YES\n")
cat("3. Sample size / power calculation:        YES\n")
cat("4. Descriptive statistics approach:        YES\n")
cat("5. Primary analysis model:                YES\n")
```

```
cat("6. Assumptions and how checked:          YES (PH assumption)\n")
cat("7. Missing data (extent & handling):      YES (MICE, 50 datasets)\n")
cat("8. Sensitivity analyses:                  YES (5 pre-specified)\n")
cat("9. Multiple comparisons:                  YES (single primary)\n")
cat("10. Software and versions:                YES\n")
cat("11. Active comparator, new-user design:   YES\n")
cat("12. DAG for covariate selection:          YES\n")
cat("13. Balance assessment (SMDs):            YES\n")
cat("14. E-value for unmeasured confounding:   YES\n")
```

D.15.3.2. Python

```
# ==============================================================================
# Chapter 19 - Exercise 3: Write a Statistical Methods Section
# Retrospective cohort study: SGLT2i vs DPP4i and MACE in T2DM
# ==============================================================================

# This is a conceptual exercise. The statistical methods section is provided
# as a detailed text output.

methods_section = """
STATISTICAL METHODS

Study Design and Population
This was a retrospective cohort study of 5,000 adults with type 2 diabetes
mellitus identified from the hospital electronic health records (EHR) databas
between 1 January 2015 and 31 December 2023. Patients were eligible if they
had a new prescription for either an SGLT2 inhibitor or a DPP-4 inhibitor,
with no prior use of the comparator drug class. Patients with a history of
major adverse cardiovascular events (MACE) prior to the index date were
excluded. The index date (time zero) was defined as the date of first
prescription of the study drug, consistent with a new-user, active comparator
```

```
design to minimise immortal time bias and confounding by indication.

Primary and Secondary Outcomes
The primary outcome was time to first MACE, defined as a composite of
myocardial infarction (ICD-10: I21), ischaemic stroke (ICD-10: I63), or
cardiovascular death (underlying cause of death codes I00-I99). Patients were
followed from the index date until the first MACE event, death from non-
cardiovascular causes, loss to follow-up, end of the study period
(31 December 2023), or 5 years after the index date, whichever occurred first.

Sample Size
With 5,000 patients and an anticipated event rate of 8% over 5 years in the
DPP-4 inhibitor group, the study had approximately 80% power to detect a
hazard ratio of 0.70 or smaller at a two-sided alpha of 0.05, assuming a
1:1 treatment group ratio and accounting for 10% loss to follow-up.

Descriptive Statistics
Baseline characteristics were summarised as means (SD) for normally
distributed continuous variables, medians (IQR) for skewed continuous
variables, and frequencies (percentages) for categorical variables.
Standardised mean differences (SMDs) were used to compare baseline
characteristics between treatment groups, with an absolute SMD < 0.1
indicating adequate balance; p-values were not used for baseline comparisons
in accordance with current recommendations.

Propensity Score Estimation and Matching
The propensity score -- the probability of receiving an SGLT2 inhibitor
versus a DPP-4 inhibitor -- was estimated using multivariable logistic
regression. Covariates included age, sex, body mass index (BMI), glycated
haemoglobin (HbA1c), estimated glomerular filtration rate (eGFR), history
of cardiovascular disease, hypertension, and smoking status. These covariates
were selected a priori based on clinical knowledge and a directed acyclic
graph (DAG) encoding assumed causal relationships. Propensity scores were
```

```
used for 1:1 nearest-neighbour matching without replacement, using a caliper
of 0.2 standard deviations of the logit of the propensity score. Covariate
balance after matching was assessed using SMDs, with all covariates required
to achieve an absolute SMD < 0.1. Propensity score overlap was assessed
visually using density plots.

Primary Analysis
The primary analysis estimated the average treatment effect on the treated
(ATT) using a Cox proportional hazards regression model fitted to the matched
cohort, with SGLT2 inhibitor use as the sole covariate. The proportional
hazards assumption was tested using scaled Schoenfeld residuals and
log-log survival plots. If the assumption was violated, a time-varying
coefficient or restricted mean survival time (RMST) analysis was planned
as an alternative. Hazard ratios (HRs) with 95% confidence intervals (CIs)
were reported.

Missing Data
Missing covariate data ranged from 2% (age) to 15% (BMI). Missingness was
assumed to be missing at random (MAR) conditional on observed variables.
Multiple imputation by chained equations (MICE) was performed with 50
imputed datasets and 20 iterations per dataset. The imputation model
included all analysis variables (covariates, exposure, outcome indicator,
and the Nelson-Aalen cumulative hazard estimate) to ensure compatibility
with the substantive analysis model. Propensity score estimation and
matching were performed within each imputed dataset, and results were
pooled using Rubin's rules. A complete-case analysis was performed as a
sensitivity analysis.

Sensitivity Analyses
Five pre-specified sensitivity analyses were conducted: (1) complete-case
analysis; (2) inverse probability of treatment weighting (IPTW) with
stabilised weights as an alternative to matching; (3) inclusion of
additional covariates (income quintile, number of medications) in the
```

```
propensity score model; (4) restriction to patients with at least
12 months of follow-up; and (5) E-value calculation to quantify the
minimum strength of association an unmeasured confounder would need with
both the treatment and the outcome to explain away the observed association.

Multiple Comparisons
As this study had a single pre-specified primary outcome, no adjustment
for multiple comparisons was applied to the primary analysis. Secondary
outcomes were interpreted with appropriate caution as hypothesis-generating.

Software
All analyses were conducted using Python version 3.11 with the following
packages: scikit-learn (v1.4.0) for propensity score estimation, lifelines
(v0.29.0) for Cox proportional hazards regression, statsmodels (v0.14.1)
for weighted regression analyses, and tableone (v0.9.1) for baseline
characteristics tables. The random seed was set to 42 for reproducibility.
Analysis code is available at [repository URL]. Two-sided p-values < 0.05
were considered statistically significant.
"""

print(methods_section)

# --- Checklist verification ---
print("=" * 60)
print("CHECKLIST: Key Items Addressed in Methods Section")
print("=" * 60)

checklist = [
    ("Study design and population", True),
    ("Primary and secondary outcomes (defined)", True),
    ("Sample size / power calculation", True),
    ("Descriptive statistics approach", True),
    ("Primary analysis model (Cox PH)", True),
```

```
    ("Assumptions and how checked (PH assumption)", True),
    ("Missing data (extent, mechanism, handling)", True),
    ("Sensitivity analyses (5 pre-specified)", True),
    ("Multiple comparisons addressed", True),
    ("Software and package versions", True),
    ("New-user, active comparator design", True),
    ("DAG for covariate selection", True),
    ("Balance assessment method (SMDs)", True),
    ("E-value for unmeasured confounding", True),
    ("Caliper specification for matching", True),
    ("Number of imputations and iterations", True),
]

for item, addressed in checklist:
    status = "YES" if addressed else "NO"
    print(f"  [{status:>3}] {item}")

print("\nWord count: approximately 550 words")
print("(The exercise requests 250-400 words; this is comprehensive")
print("to cover all required items. Can be condensed for journals")
print("with strict word limits by moving details to a supplement.)")
```

D.15.4. Exercise 4

D.15.4.1. R

```
# ==============================================================================
# Chapter 19 - Exercise 4: TRIPOD+AI Checklist (Conceptual)
# Completed for Capstone 1: Cardiovascular Risk Prediction Model
# ==============================================================================

# This exercise asks you to complete the TRIPOD+AI checklist for one of the
```

```r
# capstone projects. Below is a completed checklist for Capstone 1:
# "Development and external validation of a cardiovascular risk prediction
# model using Framingham methodology, validated in NHANES data."

checklist <- data.frame(
  Item = c(
    "1. Title",
    "2. Abstract",
    "3a. Background/rationale",
    "3b. Objectives",
    "4a. Source of data",
    "4b. Dates of study",
    "5a. Key eligibility criteria",
    "5b. Treatments received",
    "6a. Outcome definition",
    "6b. Outcome timing",
    "6c. Blinding of outcome",
    "7. Predictors",
    "8. Sample size",
    "9. Missing data",
    "10a. Statistical analysis: model development",
    "10b. Model specification",
    "10c. Predictor selection",
    "10d. Model performance measures",
    "11. Risk groups",
    "12a. Internal validation",
    "12b. External validation",
    "13. Fairness assessment (AI-specific)",
    "14. Results: participants",
    "15. Results: model development",
    "16. Results: model performance",
    "17. Results: model updating",
    "18. Discussion: interpretation",
```

```
    "19. Discussion: limitations",
    "20. Discussion: implications",
    "21. Supplementary: code and data",
    "AI-1. Data preprocessing",
    "AI-2. Hyperparameter tuning",
    "AI-3. Model explainability",
    "AI-4. Software and hardware"
  ),
  Section = c(
    "Title page",
    "Abstract",
    "Introduction",
    "Introduction",
    "Methods",
    "Methods",
    "Methods",
    "Methods",
    "Methods",
    "Methods",
    "Methods",
    "Methods",
    "Methods",
    "Methods",
    "Methods",
    "Methods",
    "Methods",
    "Methods",
    "Methods",
    "Methods",
    "Methods",
    "Methods",
    "Results",
    "Results",
```

```
    "Results",
    "Results",
    "Discussion",
    "Discussion",
    "Discussion",
    "Supplement",
    "Methods/Supplement",
    "Methods/Supplement",
    "Results/Supplement",
    "Methods"
  ),
  How_Addressed = c(
    # 1. Title
    "'Development and external validation of a logistic regression model
    for predicting 10-year cardiovascular disease risk: a Framingham-NHANES
    study.' Identifies study type (development + external validation) and
    modelling approach.",

    # 2. Abstract
    "Structured abstract with: objective, study design, data sources,
    outcome (10-year CVD), predictors, sample sizes, C-statistic and
    calibration results for both development and validation cohorts.",

    # 3a. Background
    "Existing CVD risk models (Framingham, QRISK, SCORE2) and their
    limitations. Gap: need for updated validation in contemporary US
    population.",

    # 3b. Objectives
    "To develop a 10-year CVD risk prediction model using Framingham
    data and externally validate it in NHANES 2017-2020.",

    # 4a. Source of data
```

```
    "Development: Framingham Heart Study teaching dataset (riskCommunicator
    R package). Validation: NHANES 2017-2020 (nhanesA package). Both are
    publicly available.",

    # 4b. Dates
    "Framingham: original cohort with follow-up through 2005. NHANES:
    2017-2020 survey cycles.",

    # 5a. Eligibility
    "Adults aged 30-74, free of CVD at baseline, with complete data on
    key predictors. Exclude prior MI, stroke, heart failure.",

    # 5b. Treatments
    "Not applicable (prediction model, not treatment comparison).
    However, blood pressure treatment status is included as a predictor.",

    # 6a. Outcome definition
    "10-year cardiovascular event: composite of MI, coronary death,
    stroke, or heart failure requiring hospitalisation.",

    # 6b. Outcome timing
    "10-year follow-up from baseline examination. Censored at death
    from non-CVD causes or loss to follow-up.",

    # 6c. Blinding
    "Not applicable for retrospective analysis. Outcome ascertainment
    in Framingham was by adjudication committee blinded to risk factors.",

    # 7. Predictors
    "Pre-specified based on established Framingham model: age, sex, total
    cholesterol, HDL cholesterol, systolic blood pressure, blood pressure
    treatment status, smoking, diabetes. No data-driven selection.",
```

```
# 8. Sample size
"Sample size justified using Riley et al. (2020) criteria via
pmsampsize package: 8 predictors, anticipated outcome prevalence 10%,
target R-squared 0.15, requiring minimum ~800 events.",

# 9. Missing data
"Development cohort: <5% missing. Validation: up to 12% for some
laboratory values. Handled by multiple imputation (MICE, 50 datasets,
20 iterations). Complete-case sensitivity analysis performed.",

# 10a. Model development
"Logistic regression for 10-year risk. Pre-specified predictors,
no variable selection. Continuous predictors modelled with restricted
cubic splines (4 knots for age and SBP; 3 knots for cholesterol)
based on prior literature.",

# 10b. Model specification
"Full model equation provided in supplementary materials. Coefficients
table with 95% CIs in main text.",

# 10c. Predictor selection
"All predictors pre-specified from clinical knowledge. No stepwise
selection. Rationale provided for each predictor.",

# 10d. Performance measures
"Discrimination: C-statistic (Harrell's concordance) with 95% CI.
Calibration: calibration plot (observed vs predicted), calibration
slope, calibration-in-the-large. Both reported for development
(bootstrap-corrected) and external validation.",

# 11. Risk groups
"Patients categorised into low (<5%), moderate (5-10%), high (10-20%),
and very high (>20%) 10-year CVD risk groups. Classification tables
```

```
    provided.",

    # 12a. Internal validation
    "500 bootstrap resamples for optimism-corrected C-statistic and
    calibration slope. Apparent and corrected performance reported.",

    # 12b. External validation
    "Model applied to NHANES data without recalibration. C-statistic
    and calibration plot in external data. Recalibration of intercept
    also explored.",

    # 13. Fairness
    "C-statistic and calibration reported separately by sex (male/female)
    and by race/ethnicity (non-Hispanic White, non-Hispanic Black,
    Hispanic, Asian). Differences in performance >0.05 C-statistic
    flagged for discussion.",

    # 14. Participants
    "Flow diagram showing sample selection: patients screened, excluded
    (with reasons), and included in development and validation cohorts.
    Table 1 with baseline characteristics stratified by CVD status.",

    # 15. Model development
    "Full regression table with coefficients, standard errors, odds
    ratios, 95% CIs, and p-values. Presented as Table 2.",

    # 16. Model performance
    "Table 3: discrimination (C-statistic) and calibration metrics for
    development (apparent and corrected) and external validation.
    Figures: calibration plots (development and validation), decision
    curve analysis.",

    # 17. Model updating
```

```
"Recalibration of intercept in NHANES improved calibration-in-the-
large from X to Y. Full model updating was explored but not
recommended due to limited improvement.",

# 18. Interpretation
"Model performance compared with published Framingham, QRISK3, and
PCE models. Clinical utility assessed via decision curve analysis
across clinically relevant threshold probabilities (5-20%).",

# 19. Limitations
"Limitations discussed: (1) Framingham cohort may not represent
contemporary diverse populations; (2) NHANES validation limited by
shorter follow-up and self-reported outcomes; (3) Missing data
assumptions; (4) Ecological fallacy in subgroup performance
assessment.",

# 20. Implications
"Clinical implications for CVD risk assessment. Recommendation for
recalibration before use in non-US populations. Need for prospective
validation.",

# 21. Code and data
"R code provided via GitHub repository [URL]. Framingham data
available via riskCommunicator package. NHANES data publicly available.
Reproducible via renv for package management.",

# AI-1. Data preprocessing
"Not applicable (logistic regression, not ML/AI). However, data
cleaning steps, outlier handling, and transformation of predictors
(restricted cubic splines) described in supplement.",

# AI-2. Hyperparameter tuning
"Not applicable for logistic regression. Number of knots for RCS
```

```
    pre-specified based on Harrell's recommendations.",

    # AI-3. Model explainability
    "Not applicable (logistic regression is inherently interpretable).
    Coefficient interpretation provided. Nomogram included for clinical
    use.",

    # AI-4. Software
    "R version 4.4.1. Packages: rms (v6.8), survival (v3.7), mice
    (v3.16), gtsummary (v1.7), pmsampsize (v1.1), riskCommunicator
    (v1.0), nhanesA (v1.1). Full renv.lock file in repository."
  ),

  Applicable = c(
    "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes",
    "Partially (no treatment comparison)",
    "Yes", "Yes",
    "Partially (retrospective)",
    "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes",
    "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes",
    "No (not ML)", "No (not ML)", "No (not ML)", "Yes"
  ),

  stringsAsFactors = FALSE
)

# Print the checklist
cat("=" * 70, "\n")
cat("TRIPOD+AI Checklist - Capstone 1: CV Risk Prediction Model\n")
cat("=" * 70, "\n\n")

for (i in 1:nrow(checklist)) {
  cat("ITEM:", checklist$Item[i], "\n")
```

```r
  cat("  Section:", checklist$Section[i], "\n")
  cat("  Applicable:", checklist$Applicable[i], "\n")
  cat("  How addressed:", trimws(checklist$How_Addressed[i]), "\n\n")
}

# Summary
cat("\n=== SUMMARY ===\n")
n_applicable <- sum(checklist$Applicable == "Yes")
n_partial <- sum(grepl("Partially", checklist$Applicable))
n_na <- sum(checklist$Applicable == "No (not ML)")

cat("Items fully addressed:", n_applicable, "\n")
cat("Items partially applicable:", n_partial, "\n")
cat("Items not applicable (ML-specific):", n_na, "\n")
cat("Total items:", nrow(checklist), "\n")

cat("\nNote: AI-specific items (AI-1 through AI-3) are not applicable because\n")
cat("this capstone uses logistic regression, not machine learning. If a\n")
cat("gradient boosted model or neural network were used instead, these\n")
cat("items would need to be addressed with details on feature engineering,\n")
cat("hyperparameter search strategy, and SHAP/LIME explanations.\n")
```

D.15.4.2. Python

```python
# ============================================================================
# Chapter 19 - Exercise 4: TRIPOD+AI Checklist (Conceptual)
# Completed for Capstone 1: Cardiovascular Risk Prediction Model
# ============================================================================

# This exercise asks you to complete the TRIPOD+AI checklist for one of the
# capstone projects. Below is a completed checklist for Capstone 1:
# "Development and external validation of a cardiovascular risk prediction
```

```
# model using Framingham methodology, validated in NHANES data."

checklist = [
    {
        "item": "1. Title",
        "section": "Title page",
        "applicable": "Yes",
        "how": (
            "'Development and external validation of a logistic regression "
            "model for predicting 10-year cardiovascular disease risk: a "
            "Framingham-NHANES study.' Identifies study type (development + "
            "external validation) and modelling approach."
        )
    },
    {
        "item": "2. Abstract",
        "section": "Abstract",
        "applicable": "Yes",
        "how": (
            "Structured abstract with: objective, study design, data sources
            "outcome (10-year CVD), predictors, sample sizes, C-statistic and
            "calibration results for both development and validation cohorts
        )
    },
    {
        "item": "3a. Background/rationale",
        "section": "Introduction",
        "applicable": "Yes",
        "how": (
            "Existing CVD risk models (Framingham, QRISK, SCORE2) and their "
            "limitations. Gap: need for updated validation in contemporary "
            "US population."
        )
```

```
    },
    {
        "item": "3b. Objectives",
        "section": "Introduction",
        "applicable": "Yes",
        "how": (
            "To develop a 10-year CVD risk prediction model using Framingham "
            "data and externally validate it in NHANES 2017-2020."
        )
    },
    {
        "item": "4a. Source of data",
        "section": "Methods",
        "applicable": "Yes",
        "how": (
            "Development: Framingham Heart Study teaching dataset "
            "(riskCommunicator R package). Validation: NHANES 2017-2020 "
            "(nhanesA package). Both publicly available."
        )
    },
    {
        "item": "4b. Dates of study",
        "section": "Methods",
        "applicable": "Yes",
        "how": (
            "Framingham: original cohort with follow-up through 2005. "
            "NHANES: 2017-2020 survey cycles."
        )
    },
    {
        "item": "5a. Key eligibility criteria",
        "section": "Methods",
        "applicable": "Yes",
```

```
        "how": (
            "Adults aged 30-74, free of CVD at baseline, with complete data "
            "on key predictors. Exclude prior MI, stroke, heart failure."
        )
    },
    {
        "item": "5b. Treatments received",
        "section": "Methods",
        "applicable": "Partially",
        "how": (
            "Not applicable (prediction model, not treatment comparison). "
            "However, BP treatment status included as a predictor."
        )
    },
    {
        "item": "6a. Outcome definition",
        "section": "Methods",
        "applicable": "Yes",
        "how": (
            "10-year CVD event: composite of MI, coronary death, stroke, "
            "or heart failure requiring hospitalisation."
        )
    },
    {
        "item": "6b. Outcome timing",
        "section": "Methods",
        "applicable": "Yes",
        "how": (
            "10-year follow-up from baseline examination. Censored at "
            "death from non-CVD causes or loss to follow-up."
        )
    },
    {
```

```
        "item": "6c. Blinding of outcome",
        "section": "Methods",
        "applicable": "Partially",
        "how": (
            "Retrospective analysis. Framingham outcome adjudication was "
            "by committee blinded to risk factors."
        )
    },
    {
        "item": "7. Predictors",
        "section": "Methods",
        "applicable": "Yes",
        "how": (
            "Pre-specified: age, sex, total cholesterol, HDL cholesterol, "
            "systolic BP, BP treatment, smoking, diabetes. No data-driven "
            "selection."
        )
    },
    {
        "item": "8. Sample size",
        "section": "Methods",
        "applicable": "Yes",
        "how": (
            "Justified using Riley et al. (2020) criteria via pmsampsize: "
            "8 predictors, prevalence 10%, target R^2 0.15, requiring "
            "minimum ~800 events."
        )
    },
    {
        "item": "9. Missing data",
        "section": "Methods",
        "applicable": "Yes",
        "how": (
```

```
            "Development: <5% missing. Validation: up to 12%. Handled by "
            "MICE (50 datasets, 20 iterations). Complete-case sensitivity "
            "analysis performed."
        )
    },
    {
        "item": "10a. Model development",
        "section": "Methods",
        "applicable": "Yes",
        "how": (
            "Logistic regression for 10-year risk. Pre-specified predictors.
            "Continuous predictors with restricted cubic splines."
        )
    },
    {
        "item": "10b. Model specification",
        "section": "Methods",
        "applicable": "Yes",
        "how": "Full model equation in supplementary materials."
    },
    {
        "item": "10c. Predictor selection",
        "section": "Methods",
        "applicable": "Yes",
        "how": (
            "All predictors pre-specified from clinical knowledge. "
            "No stepwise selection."
        )
    },
    {
        "item": "10d. Performance measures",
        "section": "Methods",
        "applicable": "Yes",
```

```
        "how": (
            "Discrimination: C-statistic with 95% CI. Calibration: "
            "calibration plot, calibration slope, calibration-in-the-large."
        )
    },
    {
        "item": "11. Risk groups",
        "section": "Methods",
        "applicable": "Yes",
        "how": (
            "Low (<5%), moderate (5-10%), high (10-20%), very high (>20%) "
            "10-year CVD risk categories."
        )
    },
    {
        "item": "12a. Internal validation",
        "section": "Methods",
        "applicable": "Yes",
        "how": (
            "500 bootstrap resamples for optimism-corrected C-statistic "
            "and calibration slope."
        )
    },
    {
        "item": "12b. External validation",
        "section": "Methods",
        "applicable": "Yes",
        "how": (
            "Model applied to NHANES without recalibration. C-statistic "
            "and calibration reported. Recalibration of intercept explored."
        )
    },
    {
```

```
        "item": "13. Fairness assessment (AI-specific)",
        "section": "Methods",
        "applicable": "Yes",
        "how": (
            "Performance reported by sex and race/ethnicity. Differences "
            ">0.05 C-statistic flagged."
        )
    },
    {
        "item": "14. Results: participants",
        "section": "Results",
        "applicable": "Yes",
        "how": (
            "Flow diagram with sample selection. Table 1 stratified by "
            "CVD event status."
        )
    },
    {
        "item": "15. Results: model development",
        "section": "Results",
        "applicable": "Yes",
        "how": (
            "Full regression table with coefficients, ORs, 95% CIs, "
            "and p-values."
        )
    },
    {
        "item": "16. Results: model performance",
        "section": "Results",
        "applicable": "Yes",
        "how": (
            "Table of discrimination and calibration metrics. Calibration "
            "plots and decision curve analysis figures."
```

```
        )
    },
    {
        "item": "17. Results: model updating",
        "section": "Results",
        "applicable": "Yes",
        "how": (
            "Recalibration of intercept in NHANES. Full model updating "
            "explored but not recommended."
        )
    },
    {
        "item": "18. Discussion: interpretation",
        "section": "Discussion",
        "applicable": "Yes",
        "how": (
            "Comparison with Framingham, QRISK3, PCE models. Clinical "
            "utility via decision curve analysis."
        )
    },
    {
        "item": "19. Discussion: limitations",
        "section": "Discussion",
        "applicable": "Yes",
        "how": (
            "Framingham may not represent diverse populations. NHANES "
            "validation limited by follow-up. Missing data assumptions."
        )
    },
    {
        "item": "20. Discussion: implications",
        "section": "Discussion",
        "applicable": "Yes",
```

```
        "how": (
            "Clinical implications for CVD risk assessment. Need for "
            "recalibration in non-US populations."
        )
    },
    {
        "item": "21. Code and data",
        "section": "Supplement",
        "applicable": "Yes",
        "how": (
            "Code on GitHub. Data publicly available via R packages. "
            "Reproducible via renv/conda."
        )
    },
    {
        "item": "AI-1. Data preprocessing",
        "section": "Methods/Supplement",
        "applicable": "No (logistic regression)",
        "how": (
            "Not applicable for logistic regression. Data cleaning and "
            "spline transformations described in supplement."
        )
    },
    {
        "item": "AI-2. Hyperparameter tuning",
        "section": "Methods/Supplement",
        "applicable": "No (logistic regression)",
        "how": (
            "Not applicable. Number of RCS knots pre-specified based on "
            "Harrell's recommendations."
        )
    },
    {
```

```
        "item": "AI-3. Model explainability",
        "section": "Results/Supplement",
        "applicable": "No (logistic regression)",
        "how": (
            "Not applicable (logistic regression is inherently "
            "interpretable). Nomogram provided."
        )
    },
    {
        "item": "AI-4. Software and hardware",
        "section": "Methods",
        "applicable": "Yes",
        "how": (
            "Python 3.11 with scikit-learn, lifelines, statsmodels, "
            "tableone. Full requirements.txt in repository."
        )
    },
]

# --- Print the checklist ---
print("=" * 70)
print("TRIPOD+AI Checklist")
print("Capstone 1: Cardiovascular Risk Prediction Model")
print("=" * 70)

for entry in checklist:
    print(f"\nITEM: {entry['item']}")
    print(f"  Section: {entry['section']}")
    print(f"  Applicable: {entry['applicable']}")
    print(f"  How addressed: {entry['how']}")

# --- Summary ---
n_yes = sum(1 for e in checklist if e['applicable'] == 'Yes')
```

```
n_partial = sum(1 for e in checklist if 'Partially' in e['applicable'])
n_no = sum(1 for e in checklist if e['applicable'].startswith('No'))

print(f"\n{'=' * 70}")
print("SUMMARY")
print(f"{'=' * 70}")
print(f"Items fully addressed:              {n_yes}")
print(f"Items partially applicable:         {n_partial}")
print(f"Items not applicable (ML-specific):  {n_no}")
print(f"Total items:                        {len(checklist)}")

print("""
Note on AI-specific items:
AI-specific items (AI-1 through AI-3) are not applicable because this
capstone uses logistic regression, not machine learning. If a gradient
boosted model or neural network were used, these items would require:
  - AI-1: Feature engineering pipeline, normalization, encoding
  - AI-2: Hyperparameter search strategy (grid, random, Bayesian)
  - AI-3: SHAP values, partial dependence plots, or LIME explanations

The TRIPOD+AI checklist is available from the EQUATOR Network:
https://www.equator-network.org/reporting-guidelines/tripod-ai/

Key points for completing TRIPOD+AI:
1. NEVER report AUROC/C-statistic without calibration
2. Always include a calibration plot
3. Fairness assessment across demographic subgroups is now required
4. Code and data availability statements are mandatory
5. For AI/ML models, explainability methods must be described
""")
```

E. References